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A PRACTICAL GUIDE FOR CLINICIANS

Foundations of Clinical Research

From Clinical Question to Publication

Bernard Isekah Osang'ir
Vương Mỹ Lượng

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The teaching dataset used in this book, FLUID-ICU, is **simulated**: an invented cohort of 620 adults with sepsis in four intensive care units, generated by computer for teaching. It resembles real intensive-care data, but it describes no real patient, hospital or country, and its results illustrate methods, not clinical findings. The two published studies used as running examples are real papers, cited in the references.

The figures were drawn for this book by Neudata; the book was typeset with LuaLaTeX and biblatex.

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Preface

Every day, clinicians make decisions that research could improve. Should this patient receive more intravenous fluid, or less? Does a new turning schedule prevent pressure injuries? Is the delirium we see on the ward more common at night? Most doctors, nurses and pharmacists have such questions, and many have the data to answer them in their own units. Yet few have been taught how to turn a question from the bedside into a study, how to judge whether a published paper can be trusted, or how to write up a result so that it changes practice.

This book is written for them. It is for clinicians (doctors, residents, nurses, midwives, pharmacists and allied health professionals) who have **no previous training in research methods or statistics** and who want to read the medical literature critically, plan a sound study and see it through to publication. It is also intended for students of the health professions and for anyone who supervises clinical research and wants a shared vocabulary with the people they supervise.

What this book covers

The book follows the path of a research project, from the first question to the published paper, in four parts.

Part I, Asking and reading, starts with why research is needed at all when we already have clinical experience, and how research differs from audit and quality improvement. It then shows how to turn a clinical problem into a focused, answerable research question; how to search the literature systematically, manage references and cite honestly; and how to read and appraise a published paper efficiently.

Part II, Designing, explains the main study designs and what each can and cannot claim; the threats to validity (bias, confounding and reverse causation) and how good design reduces them; the measures used to describe disease frequency and association (risk, odds, risk ratio, odds ratio, risk difference, number needed to treat); the accuracy of diagnostic tests; and how systematic reviews and meta-analyses combine studies.

Part III, Data and analysis, covers collecting and managing data well, describing and presenting data honestly, the logic of statistical inference (confidence intervals, p-values, power and sample size), correlation and regression, survival analysis and prediction models, and how to plan an analysis and handle missing data. It ends with a full worked case study in which every method is applied to one dataset.

Part IV, Ethics, writing and publishing, deals with protocols, research ethics and informed consent, research integrity and authorship, how to write each section of a scientific paper, and how to choose a journal, respond to reviewers and get the work published.

The book assumes no mathematics beyond simple arithmetic. Formulas appear only where they make an idea clearer, and each is explained in words and worked through with numbers. The emphasis throughout is on **understanding and interpreting** methods rather than on calculation:

what a confidence interval means, why an odds ratio changed after adjustment, what an author is entitled to conclude from a given design.

How the chapters are organised

Every chapter can be read on its own, although the chapters build on each other. Each one opens with a short list of what you will learn and ends with a summary and a few annotated suggestions for further reading. Inside the chapters, four kinds of boxes help the reader:

- **Key point** boxes state an idea worth remembering;
- **Worked example** boxes take a concept through a full example, with the answer worked out;
- **Common mistake** boxes describe misconceptions and errors that appear again and again in proposals and published papers;
- **Note** boxes give definitions and side remarks.

Figures and tables are numbered by chapter. References are given in author–year form and listed in full at the end of the book. Appendix A is an English–Vietnamese glossary of the key terms, Appendix B collects the checklists and tools used throughout, and Appendix C is the data dictionary of the case-study dataset.

One running example

To make the methods concrete, one clinical topic runs through the whole book: **fluid therapy in sepsis and septic shock**. Intravenous fluids are given to almost every patient with septic shock, yet how much fluid is best remains genuinely uncertain. Sicker patients receive more fluid and also die more often, which makes the topic an ideal teacher of confounding, of the difference between association and causation, and of what randomised trials can add.

Two published studies answer the same clinical question with two different designs, and they are used as examples in many chapters:

- a **randomised controlled pilot trial** of a restrictive fluid strategy in septic shock (Boulet et al. 2024), which shows what randomisation can achieve and what a small trial cannot;
- a large **propensity-score-matched cohort study** of cumulative fluid balance and 28-day mortality in sepsis (Hyun et al. 2023), which shows the strengths and limits of observational data.

Reading the two side by side teaches more about study design than either alone: the trial may claim that its intervention caused (or did not cause) a difference, while the cohort may only claim an association.

The FLUID-ICU dataset

Part III uses a third example: **FLUID-ICU**, a realistic but entirely **simulated** dataset created for teaching. It describes an invented multicentre prospective cohort of 620 adults with sepsis admitted to four intensive care units, with their baseline characteristics, daily fluid input and output, cumulative fluid balance by day 3, kidney injury, ventilation, length of stay and survival to day 28. The data were generated by computer so that the classic lessons of real clinical data appear in them: data-entry errors that must be cleaned, a skewed outcome, confounding by severity of illness, missing laboratory values and an interaction worth discussing.

Because FLUID-ICU is simulated, its results illustrate methods and must never be read as clinical findings. Every time the book uses it, it says so. Its data dictionary is printed in Appendix C, and Chapter 16 analyses it from raw data to a draft paper.

How to use this book

Read Part I first: the ideas of a focused question, of fair comparison and of critical reading underpin everything else. After that the parts can be read in order or as needed. A reader planning a study may go straight from Part II to Chapter 15 and Part IV; a reader who mainly wants to understand published papers may concentrate on Chapters 4–9 and 12–14. Whatever your route, try to apply each chapter to a question from your own practice: methods are learned by using them.

We hope this book helps you ask better questions, read the evidence with confidence and contribute answers of your own.

Bernard Isekah Osang'ir and Vương Mỹ Lượng

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PART I

Asking and reading

CHAPTER 1

Why clinical research?

Every day on a ward or in an intensive care unit (ICU), clinicians make decisions that rest on a belief about what works: how much intravenous fluid to give, when to turn a patient, which dressing to use, whether to stop a drug. Some of those beliefs rest on good evidence. Many rest on habit, on what a senior colleague taught us, or on a handful of memorable patients. Clinical research is the set of methods we use to test those beliefs fairly, so that the next patient receives care that has been shown to help rather than care that merely feels right. This chapter explains why experience alone cannot answer clinical questions, how different kinds of research differ, how research differs from audit and quality improvement, and what the whole journey from a bedside doubt to a published, practice-changing result looks like. The rest of the book follows that journey step by step.

In this chapter you will learn

- why clinical experience is an excellent source of questions but a weak way of answering them;
- how chance behaves in small numbers, and why research needs enough patients and a fair comparison;
- how the "evidence pyramid" ranks study designs, and what it does and does not tell you;
- the difference between quantitative, qualitative and mixed-methods research, and between primary and secondary research;
- how research differs from clinical audit and quality improvement, and why the difference matters for ethics approval;
- the main stages of a research project, from question to impact on patients.

1.1 The running example: fluids in septic shock

Throughout this book we return to one clinical problem, so that each new idea can be attached to something familiar. Septic shock is a life-threatening response to infection in which blood pressure falls and organs are poorly perfused despite treatment; patients need vasopressor drugs and usually large volumes of intravenous fluid in the first hours. Fluids save lives early. But in the following days many patients keep receiving fluid (for resuscitation, to keep drips open, to dilute antibiotics, as nutrition) and accumulate a large positive fluid balance: more fluid in than out. Many clinicians suspect that this extra fluid causes harm through tissue oedema, kidney injury and difficulty weaning from the ventilator. Others worry that restricting fluid risks under-perfusion. Practice varies widely between units and even between consultants on the same unit.

Three sources of evidence about this problem appear again and again in the book:

- **Paper 1**, Boulet and colleagues (2024), a small randomised controlled trial (RCT) in three ICUs that tested a strategy of restricting "hidden" non-resuscitation fluids in adults with septic shock (Boulet et al. 2024).
- **Paper 2**, Hyun and colleagues (2023), a large observational cohort study from a multicentre sepsis registry that examined whether a positive fluid balance in the first three ICU days was associated with death at 28 days (Hyun et al. 2023).
- **FLUID-ICU**, a *simulated* cohort of ICU patients with sepsis created for teaching. It is not real data; it is used in later chapters so that every analysis can be shown in full.

Two large trials provide the wider context: CLASSIC, which randomised 1,554 adults with septic shock to restrictive or standard intravenous fluid (Meyhoff et al. 2022), and CLOVERS, which randomised 1,563 adults with sepsis-induced hypotension to a restrictive or liberal fluid strategy (National Heart, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Clinical Trials Network et al. 2023). Neither found a difference in mortality.

1.2 Why experience is not enough

1.2.1 Anecdote and evidence

Imagine a senior colleague who says: "I gave less fluid to my last three septic shock patients and they all did well." This is an honest observation, and it may be the start of a valuable idea. But it cannot tell us whether less fluid *works*. The figure below contrasts this kind of anecdote with evidence from a fair comparison.

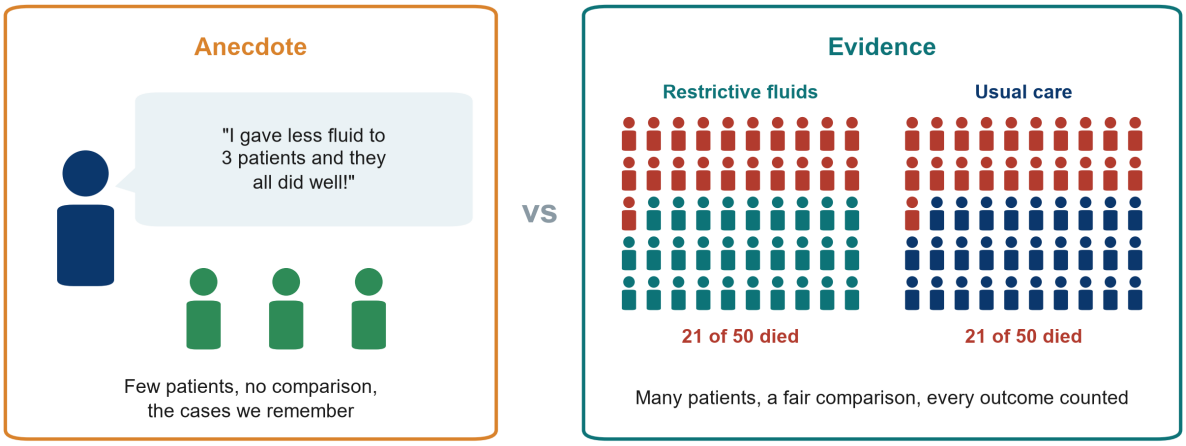


Figure 1.1: An anecdote (three remembered patients, no comparison group) contrasted with evidence from a fair comparison in which 50 patients receive each strategy and every death is counted; here 21 of 50 die in each group. The numbers are illustrative.

The anecdote has three problems. First, there is **no comparison**: we do not know what would have happened to those three patients with usual fluids. Many patients with septic shock survive whatever fluid strategy is used; in CLASSIC, about 58% of patients in both groups were alive at 90 days (Meyhoff et al. 2022). Three survivors are exactly what we would expect from either strategy. Second, **memory is selective**. Dramatic and recent cases stick in the mind; the patient who did badly after fluid restriction two years ago is easily forgotten, or explained away as "too sick anyway". Third, the patients were **chosen by the clinician**, who may have restricted fluid only in patients

who already looked as if they were improving. That choice, not the fluid, could explain the good outcome. Chapter 6 calls this confounding by indication.

On the right of the figure, the same question is answered differently: many patients, two groups treated differently, the same outcome counted in every patient. Real numbers from CLASSIC show the same pattern as the illustrative figure: 90-day mortality was 42.3% with restrictive fluid and 42.1% with standard care (Meyhoff et al. 2022). The three "successes" would very probably have done well with either approach.

Experience asks; fair comparisons answer

Clinical experience is the richest source of research questions we have. It is a poor way of answering them, because it lacks a comparison group, it is filtered by memory, and the patients who received a treatment were chosen for a reason. Research replaces these weaknesses with a planned comparison, measured in enough patients, with every outcome counted.

1.2.2 Chance in small numbers

Even if we compared two groups fairly, small groups would still mislead us, because chance alone produces large differences when numbers are small. The figure below shows a simulation. Imagine a disease in which the true risk of death is exactly 40%. In the top row, twenty wards each treat only five patients; in the middle row, groups of 50; in the bottom row, groups of 500. Each dot is the percentage of deaths that one group observed.

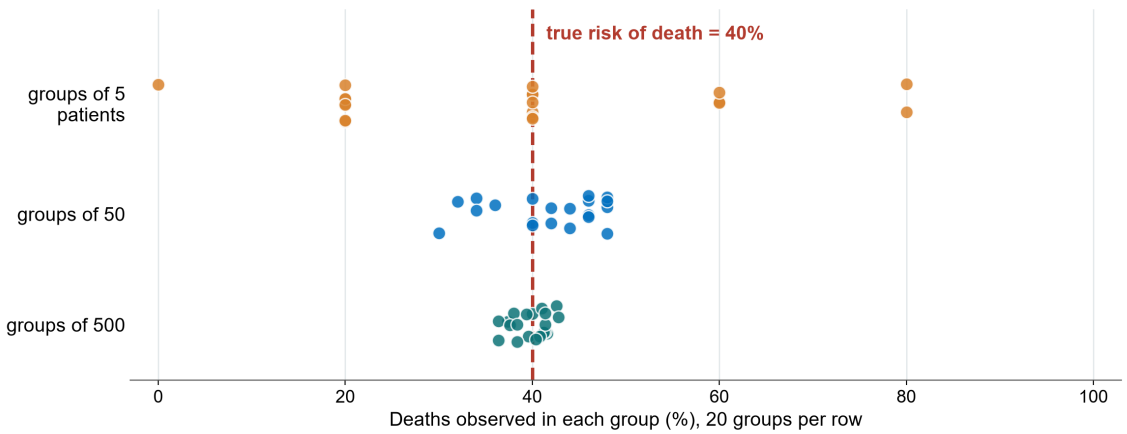


Figure 1.2: Simulated deaths in groups of 5, 50 and 500 patients when the true risk of death is 40%: small groups scatter between 0% and 80% by chance alone, while large groups cluster tightly around the true value.

With five patients, some wards see no deaths at all and others see four deaths out of five, although every ward is treating the same disease with the same true risk. Each ward would "learn" something different from its own experience. With 50 patients the results are closer to the truth but still scattered between about 30% and 50%. Only with 500 patients do the results settle tightly around 40%.

Worked example: how often does a ward see no deaths by chance?

Suppose the true risk of death is 40%, so the chance that any one patient survives is 60%, or 0.6. If a ward treats five patients and their outcomes are independent, the probability that all

five survive is

$$0.6 \times 0.6 \times 0.6 \times 0.6 \times 0.6 = 0.6^5 \approx 0.078$$

About 8% of wards, roughly one in thirteen, will see no deaths among five patients purely by chance. If twenty wards each try a new protocol on five patients, we should expect one or two of them to report “no deaths since we started”, even if the protocol does nothing at all. Those wards are likely to present their result enthusiastically at a meeting; the wards with three or four deaths are not.
The same calculation for the other extreme, four or five deaths out of five, gives about 9%. A dramatic result in either direction is common in small numbers.

This simple arithmetic is the reason research needs an adequate **sample size**, a number of patients large enough that a real difference between treatments can be distinguished from the play of chance. Chapters 12 and 15 show how sample size is calculated. For now the lesson is the one in the box.

! Common mistake: a big difference in a small group must be real

A big difference in a small group is exactly what chance produces. “We lost none of our last five patients after changing the protocol” is not evidence that the protocol works, because no deaths among five patients happens about 8% of the time even when the true risk is 40%.

1.2.3 Why opinion and practice drift

Two further reasons make research necessary. Clinicians who are equally experienced often disagree, and they cannot all be right. And practice changes over time for reasons that have little to do with evidence: fashion, a new product, an influential lecture. History contains many treatments that seemed obviously beneficial and were later shown to cause harm when finally tested in a fair comparison. Evidence-based medicine was proposed as a way to integrate individual clinical expertise with the best available external evidence from systematic research, and with the values of the patient (Sackett et al. 1996). Neither half is sufficient alone: evidence without expertise cannot be applied to the patient in front of us, and expertise without evidence quickly becomes out of date.

1.3 The evidence pyramid

Not all studies protect equally well against the errors described above. The **evidence pyramid** is a simple picture that ranks study designs by how well they protect against bias when the question is about the effect of a treatment.

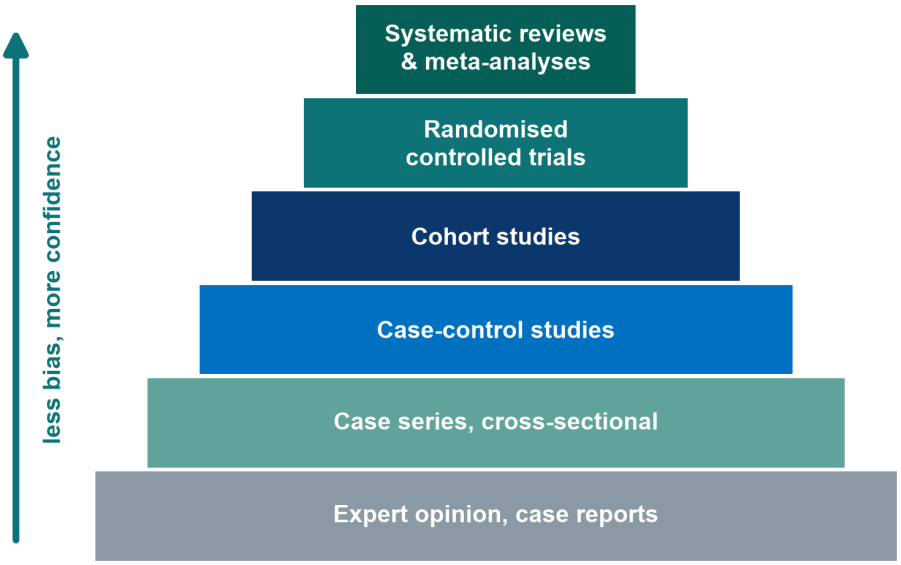


Figure 1.3: The evidence pyramid for questions about treatment effects: from expert opinion and case reports at the base, through case series and cross-sectional studies, case-control studies, cohort studies and randomised controlled trials, to systematic reviews and meta-analyses at the top.

Reading from the bottom:

- **Expert opinion and case reports** describe what one person believes or what happened to one patient. They are valuable for raising new questions (the first reports of a new disease or a rare adverse drug effect are case reports), but they have no comparison group.
- **Case series and cross-sectional studies** describe a group of patients, or measure many people at one moment. They tell us how common something is and who is affected, but they rarely allow a fair comparison of treatments.
- **Case-control studies** start from people who have an outcome (cases) and people who do not (controls), and look back at their past exposures.
- **Cohort studies** follow groups of people forward from an exposure to an outcome. Paper 2 is a cohort study (Hyun et al. 2023).
- **Randomised controlled trials** allocate patients to treatments by chance, which makes the groups comparable in both measured and unmeasured characteristics. Paper 1, CLASSIC and CLOVERS are randomised trials.
- **Systematic reviews and meta-analyses** search for, appraise and (where appropriate) statistically combine all the studies that answer one question.

Each of these designs is explained fully in Chapter 5, and systematic reviews in Chapter 9. Here we need only the logic of the pyramid: moving upwards, the design does more of the work of protecting against bias, so for a question such as “does restrictive fluid reduce deaths?” designs higher up give more confidence.

Worked example: ranking four sources on fluid restriction

Four sources answer the question “Do restrictive fluids reduce deaths in septic shock?” A. A senior colleague: “In twenty years, my dry patients did better.” B. A registry cohort of thousands of ICU patients with sepsis, similar to Paper 2. C. A trial that randomised 1,554 patients with septic shock (CLASSIC). D. A systematic review that pools every randomised trial of fluid restriction.

From weakest to strongest the ranking is A, B, C, D. Notice that B and C seem to disagree: Paper 2 found that a positive fluid balance on ICU days 2 and 3 was associated with higher 28-day mortality, whereas CLASSIC found no difference in mortality between restrictive and standard fluid. A likely explanation is that, in routine care, sicker patients receive more fluid. Their higher mortality is then caused partly by their illness, not by the fluid, and fluid looks harmful in a cohort even if it is not. Randomisation removes this problem because chance, not the clinician, decides who receives which strategy. Chapter 6 explains this in detail.

The pyramid is useful, but it is easily misused.

! Common mistake: the pyramid ranks papers

The pyramid ranks *designs*, not individual papers. A small, badly run trial can mislead more than a large, careful cohort study, and a systematic review of ten poor-quality trials is only as strong as the trials inside it. The quality of each study still has to be checked (Chapter 4), and frameworks such as GRADE rate the certainty of a whole body of evidence rather than trusting the design label alone (Guyatt et al. 2008).

! Common mistake: only randomised trials count

The pyramid applies to questions about the effects of treatments. For other questions it is the wrong tool. How common is delirium after hip-fracture surgery? A cross-sectional or cohort study answers that; a trial is unnecessary. Does smoking cause lung cancer? Randomising people to smoke would be unethical, so the evidence comes from observational studies. Why do nurses hesitate to stop intravenous fluids? That is a question for qualitative research, which sits outside the pyramid altogether. The right design is the one that fits the question.

1.4 Types of research

Two simple questions classify almost any study: *what kind of data does it collect?* and *where do the data come from?* The figure below shows both.



Figure 1.4: Types of research. Top row, by kind of data: quantitative research (numbers), qualitative research (words) and mixed methods (both). Bottom row, by source of data: primary research collects new data; secondary research reuses data that already exist.

1.4.1 Numbers or words?

Quantitative research counts and measures. It asks "how many?", "how much?" and "does X change Y?". Mortality at 28 days, cumulative fluid balance in millilitres per kilogram, and the proportion of patients who develop a pressure injury are quantitative outcomes. Most of this book is about quantitative research.

Qualitative research collects words, observations or images, through interviews, focus groups, observation of practice or analysis of documents. It asks "why?", "how?" and "what is it like?". If a unit introduces a fluid-restriction protocol and nobody follows it, counting fluid volumes will show *that* the protocol is ignored; interviewing nurses and doctors will show *why*: perhaps fear of hypotension, unclear responsibility for stopping a drip, or a belief that a patient "looks dry". Qualitative research has its own rigorous methods for sampling, collecting and analysing data, and its own reporting guideline, COREQ, for interviews and focus groups (Tong et al. 2007).

Mixed-methods research combines both in one study, for example a survey of fluid-prescribing practice followed by focus groups to understand the reasons behind the answers.

> **! Common mistake: qualitative research is just opinion**
>
> Qualitative research is not a collection of anecdotes. It uses systematic sampling, recorded and transcribed data, explicit analysis methods and checks on the researchers' own influence. It is the right tool whenever numbers cannot explain behaviour, experience or meaning.

1.4.2 New data or existing data?

Primary research collects new data for the purpose of the study: a trial, a prospective cohort, a survey, a set of interviews. Paper 1 is primary research; the investigators enrolled patients and measured their fluid balance specifically for the trial (Boulet et al. 2024).

Secondary research reuses data that already exist: medical records, registries, administrative databases, or published studies (a systematic review is secondary research on published studies). Paper 2 analysed data from a sepsis registry that had been set up for broader purposes (Hyun et al. 2023). Secondary research is often the most feasible kind of research for a busy department, because the data are already there. Its weakness is that the data were not collected with the research question in mind, so important variables may be missing or recorded inconsistently. Chapter 10 discusses data quality.

Worked example: classifying four projects

For each project, ask the two questions: numbers or words? new or existing data?

- Counting how many ICU patients develop pressure injuries this year: quantitative, primary (new counts are made).
- Interviewing families about visiting rules in the ICU: qualitative, primary.
- Re-analysing three years of discharge records to study readmissions: quantitative, secondary. The records were created for clinical care, not for this study. Reusing them is still research and still needs ethics review.
- A survey of hand-hygiene practice combined with focus groups of nurses: mixed methods, primary.

1.5 Research, audit and quality improvement

Many projects in hospitals look similar on the surface but are different activities with different purposes and different approval routes. The figure below compares the three most common.

	Research	Clinical audit	Quality improvement
Asks	What is the best practice?	Are we meeting the standard?	How do we get better, step by step?
Fluid example	Restrictive vs usual: which is better?	Is fluid balance recorded daily?	A daily checklist, tested in PDSA cycles
Results apply to	patients elsewhere (generalisable)	our unit	our unit
Ethics	**Ethics committee approval: always**	Local approval, usually no committee	Local approval, ethics review if new risk

Not sure which it is? Ask the ethics committee before you start. Journals ask for approval or a waiver.

Figure 1.5: Research, clinical audit and quality improvement compared: what each asks, an example from fluid management, to whom the results apply, and what approval each normally needs.

Research generates new knowledge that is intended to be generalisable: it asks *what is the best*

practice? "Is restrictive fluid better than usual fluid in septic shock?" is a research question, and its answer is meant to apply to patients in other hospitals too.

Clinical audit asks *are we meeting the standard?* It measures current practice against an agreed standard, such as a local protocol or a national guideline, and feeds the results back so that practice can change. "Is fluid balance recorded every day for every ICU patient, as our protocol requires?" is an audit question. The standard is already known; the audit checks whether it is met.

Quality improvement (QI) asks *how do we get better, step by step?* It changes a local process in small, repeated cycles, usually described as Plan-Do-Study-Act (PDSA) cycles, measuring the effect of each change. Introducing a daily fluid-balance checklist, testing it on one bay, refining it, and then spreading it is a QI project. QI has its own reporting guideline, SQUIRE (Ogrinc et al. 2016).

The distinction matters most for **approval**. Research always needs approval from an ethics committee before it starts (Chapter 17). Audit and QI usually need only local approval from the clinical governance or quality department. However, the label a team gives a project does not decide the route; what the project *does* decides it. If a project randomises patients, introduces a new intervention or an additional risk, or collects data beyond what is collected in routine care, it is research and needs ethics review, whatever it is called.

Worked example: research, audit or QI?

Table 1.1: Classifying five fluid-related projects. Rules differ between institutions, so local policy should always be checked.

Project	Type	Approval before starting
Checking whether fluid balance is charted daily, as the unit protocol requires	Audit	Local or clinical-governance approval
Randomising patients to restrictive or usual fluids	Research	Ethics committee approval and trial registration
Testing a new handover checklist in small PDSA cycles	Quality improvement	Local approval; ethics review if it adds risk
Interviewing nurses about why they hesitate to stop fluids	Research (qualitative)	Ethics committee approval
Reviewing three years of ICU records to see whether fluid balance is linked to mortality	Research (secondary, observational)	Ethics committee approval or a formal waiver of consent

The fourth row surprises many people. Interviewing colleagues may feel like "just talking", but systematic, recorded interviews that are analysed and reported are research: participants must give informed consent and their confidentiality must be protected.

! Common mistake: we can get approval later if we decide to publish

Publishing does not change what a project was. Journals ask for evidence of ethics approval, or a formal waiver, for any study involving patients or their data, and retrospective approval is often refused. Decide whether a project is research, audit or QI at the start, and when in doubt ask the ethics committee before collecting any data.

! Common mistake: audit and QI are lesser research

Audit and QI answer different questions from research, and they are essential to good care. Many good research questions start from an audit result: an audit that shows most septic shock patients are fluid-positive by day 3 is exactly the local evidence that justifies a research project (Chapter 3). Audit and QI projects can also be published, with the appropriate approvals.

1.6 The road from question to impact

Research is often imagined as "collecting data". In fact, data collection is only one stop on a long road, and it comes well after the most important decisions have been made. A typical clinical research project passes through eleven stages.

Table 1.2: The road from a clinical question to impact on patients.

Stage	What happens	Where in this book
1. Question	A clinical problem becomes a precise, answerable research question	Chapter 2
2. Design and protocol	The study design is chosen and the full plan is written down	Chapters 5–9, 17
3. Registration	Trials (and increasingly other studies) are registered publicly before the first participant is enrolled	Chapters 3, 17
4. Ethics approval	An ethics committee reviews and approves the protocol	Chapter 17
5. Data collection	Participants are recruited and data are recorded	Chapter 10
6. Analysis	The data are cleaned, described and analysed according to a pre-specified plan	Chapters 11–16
7. Writing	The study is written up as a paper	Chapter 19
8. Submission	The paper is sent to a suitable journal	Chapter 20
9. Peer review	Independent experts review it; the authors revise	Chapter 20
10. Publication	The paper becomes available to readers	Chapter 20
11. Impact on patients	The result changes guidelines, protocols or practice, and raises new questions	Chapters 4, 9

Paper 1 illustrates several stages. The trial was registered on ClinicalTrials.gov on 1 July 2021, before recruitment began in September 2021. It was approved by an ethics committee, patients were recruited until February 2023, and the paper was published in 2024 (Boulet et al. 2024). From first idea to publication, a clinical study commonly takes one to three years, and many good studies stall at the writing stage.

Two features of this road deserve emphasis. First, every stage depends on stage 1. A vague question leads to a weak protocol, an outcome that cannot be measured, an analysis that does not answer anything in particular, and a paper that is hard to publish. That is why the whole of Chapter 2 is devoted to the question. Second, the road does not end at publication. The point of clinical research is impact: a changed guideline, a revised protocol, a safer practice. And each answer raises new questions, so the road becomes a cycle: clinical problem, question, literature search, design and protocol, ethics approval, data collection, analysis and interpretation, publication and change in practice, and back to a new clinical problem.

! Common mistake: research starts with collecting data

Research starts with a question. Data collected before the question, design and protocol are fixed are usually incomplete, inconsistently recorded and unusable for the question that is eventually asked. Collecting data before ethics approval may also make the project impossible to publish.

1.6.1 Why getting it right matters

Poorly asked questions, unsuitable designs, unpublished results and badly reported studies waste a large share of the effort and money invested in clinical research (Chalmers and Glasziou 2009). They also waste the goodwill of patients who agreed to take part. Much of this waste is avoidable: it arises from decisions made at the start of a project (asking a question that has already been answered, choosing an outcome that does not matter to patients, enrolling too few participants to answer the question) and at the end (not publishing, or reporting selectively). The chapters that follow are designed to help you avoid these failures.

1.7 How the research papers in this book fit together

It is worth pausing to see how the sources of evidence on our running example fit the ideas in this chapter.

Table 1.3: The evidence on fluids in septic shock used in this book.

Source	Type of research	Design	Place on the pyramid	What it can tell us
A colleague's three patients	Anecdote	None	Base	A question worth asking
Paper 2 (Hyun 2023)	Quantitative, secondary (registry)	Propensity-matched cohort	Middle	Whether positive balance is *associated* with death
Paper 1 (Boulet 2024)	Quantitative, primary	Pilot randomised trial	Upper	Whether a restriction strategy *changes* fluid balance; too small for mortality
CLASSIC, CLOVERS	Quantitative, primary	Large randomised trials	Upper	Whether restrictive strategies change mortality
FLUID-ICU (simulated)	Quantitative, secondary	Cohort (teaching data)	Middle	Used to demonstrate analysis methods

No single source gives the whole answer. Learning to read each of them for what it can and cannot show is the purpose of Chapter 4, and the reason for the rest of the book.

Summary

- Clinical experience generates the best research questions, but it cannot answer them: it has no comparison group, it is filtered by memory, and treatments in routine care are chosen for reasons that also affect outcomes.
- Chance produces large differences in small numbers. With a true risk of death of 40%, about 8% of groups of five patients will see no deaths at all. Research needs a fair comparison in enough patients.
- The evidence pyramid ranks designs by how well they protect against bias for questions about treatment effects. It ranks designs, not individual papers, and it does not apply to questions about frequency, causes that cannot be randomised, or experiences.
- Quantitative research counts and measures; qualitative research explores reasons and experiences; mixed methods combine both. Primary research collects new data; secondary research reuses existing data.
- Research seeks generalisable knowledge and always needs ethics approval; audit compares practice with a standard; quality improvement changes local processes in cycles. What a project does, not what it is called, decides the approval route.
- A research project travels from question to design, registration, ethics approval, data collection, analysis, writing, submission, peer review, publication and impact. Everything depends on the first step, the question.

Further reading

- Hulley and colleagues, *Designing Clinical Research* (Hulley et al. 2013). The standard introductory text on planning clinical studies; its early chapters cover the anatomy and physiology of research and are an excellent companion to this chapter.
- Greenhalgh, *How to Read a Paper* (Greenhalgh 2019). A readable introduction to evidence-based practice, study types and critical appraisal, written for clinicians.
- Sackett and colleagues, “Evidence based medicine: what it is and what it isn’t” (Sackett et al. 1996). A two-page classic that defines evidence-based medicine as the integration of clinical expertise with the best external evidence.
- Chalmers and Glasziou, “Avoidable waste in the production and reporting of research evidence” (Chalmers and Glasziou 2009). Explains how much research effort is lost to poor questions, designs and reporting, and why it matters.

CHAPTER 2

From clinical problem to research question

"Are fluids bad for ICU patients?" is a question many clinicians have asked on a ward round. It is an important worry, but it is not yet a research question: it does not say which patients, which fluids, bad compared with what, bad in what way, or by when. No study can answer it as it stands. Turning a clinical worry into a precise, answerable question is the single most important step in any research project, because every later decision (the design, the sample size, the data to collect, the analysis, even the title of the final paper) follows from it. This chapter shows how to make that transformation using two simple tools, PICO and FINER, and how to turn the finished question into an aim, objectives and a hypothesis.

In this chapter you will learn

- where good research questions come from, and how to tell a research question from a question that already has an answer;
- how to write a question using PICO(T), and its variants PECO, P + O, and frameworks for diagnostic, prognostic and qualitative questions;
- the difference between descriptive and analytical questions, and how the question points to the design;
- how to test a question with the FINER criteria and rescue a question that fails;
- how to write an aim, measurable objectives and null and alternative hypotheses, and when no hypothesis is needed.

2.1 Where research questions come from

Good questions rarely come from a blank page. They come from daily clinical work, and most clinicians meet several every week without noticing. The figure below shows six common sources.

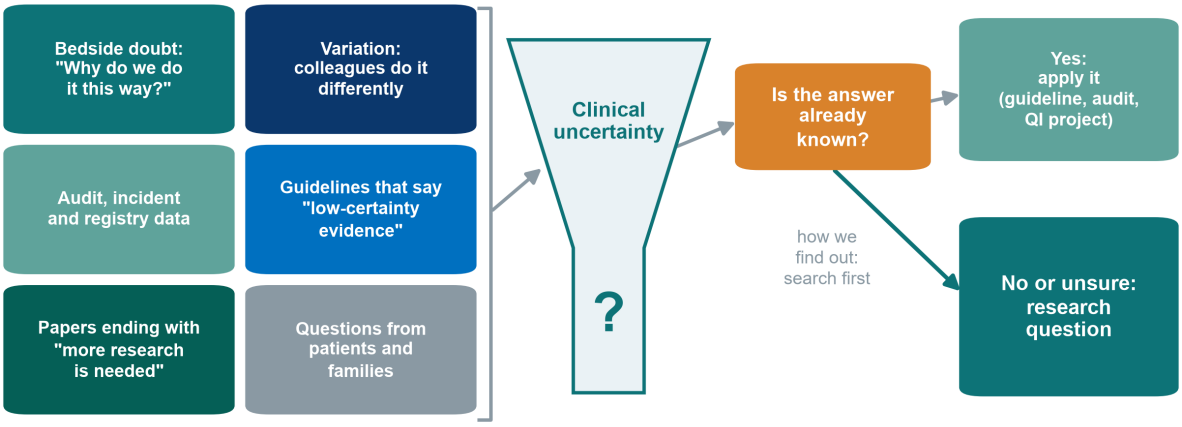


Figure 2.1: Six common sources of research questions feed into clinical uncertainty. Before treating it as research, ask whether the answer is already known: if it is, apply it through a guideline, audit or quality-improvement project; if it is not, or you are unsure, you have a research question.

- **Bedside doubt.** "Why do we do it this way?" A nurse who wonders why every patient has a maintenance drip running at the same rate has found a question.
- **Variation.** When colleagues do the same thing differently (sedation targets, fluid volumes, dressing changes, timing of mobilisation), at least one of them is probably not doing the best thing, or nobody knows what the best thing is.
- **Audit, incident and registry data.** An audit that shows most septic shock patients are fluid-positive by day 3, or a cluster of medication errors at ICU discharge, points to a problem worth studying.
- **Guidelines that admit uncertainty.** Clinical guidelines often state that a recommendation is based on "low-certainty evidence". Each such statement marks a gap.
- **Papers that end with "more research is needed".** The discussion section of a published study often names the next question.
- **Patients and families.** Their questions ("How long will I be confused after the operation?") are frequently the most relevant of all.

The orange decision box in the figure is the most important step. Before treating an uncertainty as a research question, find out whether the answer already exists. If it does, the task is not research but implementation: write or update a protocol, audit practice against it, or run a quality-improvement project (Chapter 1). If the answer is not known, or the evidence is weak or conflicting, you have a research question. The only way to know is to search the literature systematically, which is the subject of Chapter 3. In practice the question and the search develop together: a first draft of the question guides the search, and what the search finds reshapes the question.

2.2 PICO(T): the anatomy of a question

The most widely used tool for structuring a clinical question is **PICO** (Richardson et al. 1995), often extended to **PICOT**:

- **P, Population (or Patient or Problem):** who exactly is the question about?
- **I, Intervention:** what is being done, given or changed?
- **C, Comparison:** compared with what?
- **O, Outcome:** what result will be measured?
- **T, Time:** when will the outcome be measured?

The figure below applies PICO(T) to the running example of this book, fluids in septic shock.

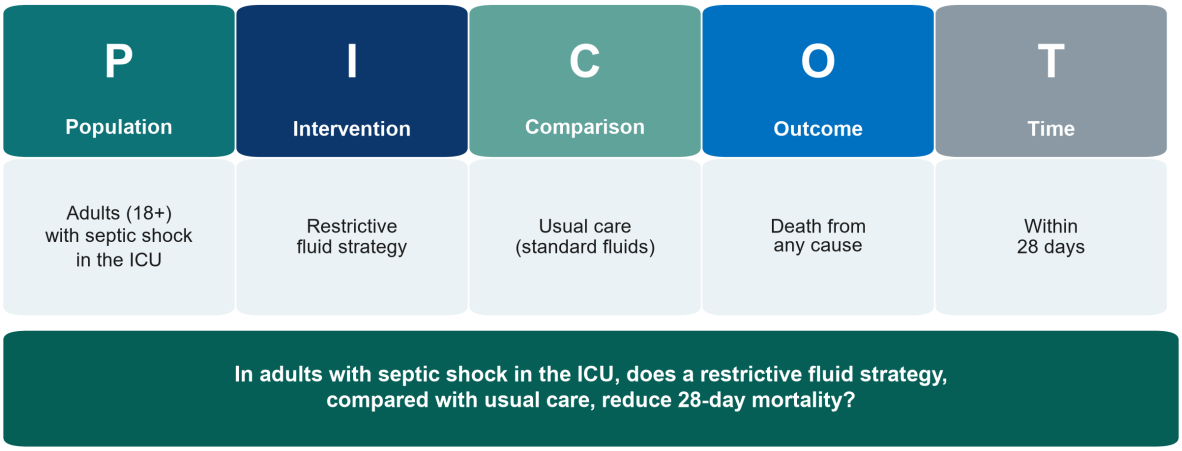


Figure 2.2: PICO(T) applied to the running example. P: adults with septic shock in the ICU; I: a restrictive fluid strategy; C: usual care with standard fluids; O: death from any cause; T: within 28 days. Together they form one question.

Read as a sentence, the boxes give:

> *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*

Each letter forces a decision that a vague question avoids.

Population. Which patients, exactly? "ICU patients" includes post-operative patients, trauma patients and patients with heart failure, whose fluid needs are very different. "Adults with septic shock" is narrower; a full protocol would go further, defining septic shock (for example by the Sepsis-3 criteria (Singer et al. 2016)), age limits, and the time window for enrolment. Paper 1 enrolled adults aged 18 to 85 within 24 hours of starting vasopressors, and excluded patients needing kidney replacement therapy, those with end-stage kidney disease and those with severe malnutrition (Boulet et al. 2024). A precise population makes the study reproducible and tells readers whether the results apply to their own patients.

Intervention. Restrictive *how*? Restricting resuscitation boluses (as CLASSIC did (Meyhoff et al. 2022)) is a different intervention from restricting maintenance fluids and drug diluents (as Paper 1

did (Boulet et al. 2024)). The intervention must be described well enough that someone else could deliver it.

Comparison. The comparison is often written as “usual care”, but usual care varies between hospitals and over time, and must itself be described. A comparison of “nothing” is rarely realistic.

Outcome. Which outcome, and only one as the main outcome. Death from any cause is objective and important to patients, but other outcomes are possible: days alive without life support, kidney injury needing dialysis, cumulative fluid balance. Each would make a different study. The choice of outcome is discussed further below.

Time. When is the outcome measured? Mortality at 28 days and at 90 days are both common; CLASSIC used 90 days (Meyhoff et al. 2022). The time point must be chosen in advance and justified, because an outcome measured at a time chosen after looking at the data is easily manipulated.

- **If you cannot fill all the boxes, the question is not ready**

  A research question is ready when every PICO(T) element can be written down specifically. An empty or vague box (no comparison, an outcome with no time point, “all patients”) shows exactly where more thinking is needed.

2.2.1 Variants of PICO for other kinds of question

PICO was designed for questions about treatments. Other kinds of question need slightly different letters, but the principle, being explicit about every element, is the same.

PECO for exposures. Many questions are about factors we cannot or should not assign: obesity, smoking, night shifts, a positive fluid balance that arises during routine care. Here the I becomes **E, Exposure**. Paper 2 asked a PECO question: in adults with sepsis admitted to the ICU (P), is a positive cumulative fluid balance on days 1 to 3 (E), compared with a negative balance (C), associated with death (O) at 28 days (T)? (Hyun et al. 2023)

P + O for "how many?" questions. Questions about frequency need only a population, an outcome and a time frame: "What proportion of adults aged 65 and over develop delirium in the first 7 days after hip-fracture surgery?" There is no intervention and no comparison.

Diagnostic questions. The new test being evaluated (the index test) takes the place of I, the reference standard takes the place of C, and the outcome is accuracy, measured as sensitivity, specificity and related quantities (Chapter 8).

Prognostic questions. "In patients with septic shock, which factors predict death within 28 days?" These are usually PECO questions with an emphasis on time (Chapter 14).

Qualitative questions. Questions about experiences and reasons often use **SPIDER**: Sample, Phenomenon of Interest, Design, Evaluation, Research type (Cooke et al. 2012). For example: in ICU nurses (S), what are the experiences of stopping maintenance fluids (PI), explored through semi-structured interviews (D), in terms of perceived barriers and facilitators (E), in a qualitative study (R)?

The figure below shows three worked examples from outside the ICU, one for each of the commonest question types.

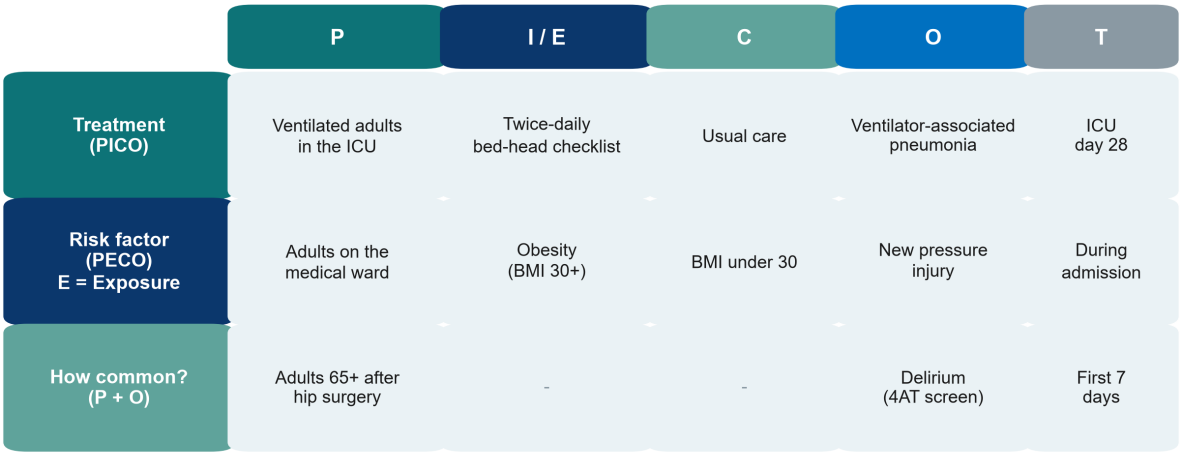


Figure 2.3: PICO(T) for three ward questions. A treatment question (PICO): in ventilated adults, a twice-daily bed-head checklist versus usual care, and ventilator-associated pneumonia by ICU day 28. A risk-factor question (PECO): on the medical ward, obesity versus BMI under 30, and a new pressure injury during admission. A frequency question (P + O): adults aged 65 and over after hip surgery, and delirium on the 4AT screen in the first 7 days.

Notice that the boxes you fill already reveal the type of question. A row with an assigned intervention and a comparison is a treatment question; a row with an exposure is a risk-factor question; a row with only P, O and T is a frequency question.

Worked example: spot the PICO

Consider the question: “In adults aged 65 and over after hip-fracture surgery, does a daily non-drug delirium-prevention bundle, compared with usual care, reduce delirium in the first 7 days?”

Table 2.1: The PICO(T) elements of the delirium question.

P	I	C	O	T
Adults 65+ after hip-fracture surgery	Daily non-drug delirium-prevention bundle	Usual care	Delirium, screened daily with a validated tool such as the 4AT or CAM	First 7 days after surgery

The commonest error is to label “hip fracture” as the outcome. The fracture has already happened; it is the reason patients are eligible, so it belongs to P. An outcome must be something that can happen *after* the intervention starts.
A follow-up question: how would delirium be measured? A validated screening tool, applied at the same time each day by trained staff, so that the outcome is measured identically in both groups.

2.2.2 Five worked scenarios

The following scenarios show how real clinical worries from different professions become PICO(T) questions. They are worth reading slowly, because each illustrates a typical difficulty.

Ventilator-associated pneumonia (treatment question). In adults ventilated for at least 48 hours in the ICU (P), does a twice-daily nurse checklist for head-of-bed elevation to 30–45 degrees (I), compared with usual care (C), reduce the incidence of ventilator-associated pneumonia (O) by ICU day 28 (T)? The difficulty here is the outcome: definitions of ventilator-associated pneumonia vary, so the study must state which definition it uses, and the condition is not very common, which affects feasibility.

Pressure injuries (descriptive first). A ward team worried about pressure injuries may want to test a repositioning schedule. But if nobody knows how often pressure injuries occur on the ward, the honest first question is descriptive: what is the incidence of hospital-acquired pressure injuries of stage 2 or worse (O) among adults admitted to our medical ward (P) over 6 months (T)? An analytical question, such as whether 2-hourly repositioning reduces new pressure injuries compared with 4-hourly repositioning, can follow once the size of the problem is known.

Stroke door-to-needle time (exposure question). In adults with acute ischaemic stroke who receive thrombolysis in our emergency department (P), is a pre-hospital ambulance alert (E), compared with no alert (C), associated with a shorter door-to-needle time (O), measured from arrival to the start of thrombolysis (T)? The outcome here is continuous (minutes), so the eventual hypothesis will be about a difference in average or median time.

Medication errors at ICU discharge (treatment question). In adults transferred from the ICU to a ward (P), does pharmacist-led medication reconciliation at transfer (I), compared with the usual process (C), reduce the proportion of patients with at least one medication discrepancy (O) within 48 hours (T)? The difficulty is definitional: “medication error” can mean any discrepancy or only one that causes harm, and the question must choose.

Delirium (treatment question). As in the worked example above.

2.3 Descriptive and analytical questions

Research questions fall into two broad families, and recognising the family tells you a great deal about the design.

Table 2.2: Descriptive and analytical questions, with examples from the running example and the designs that usually answer them.

Family	Typical wording	Running example	Usual designs
Descriptive	What? How many? Who?	"How many of our septic shock patients have a positive fluid balance on day 3?"	Case series, cross-sectional study
Analytical, observational	Is X associated with Y?	"Is a positive fluid balance associated with death?"	Cohort, case-control study
Analytical, experimental	Does X cause or change Y?	"Does restricting fluids reduce 28-day mortality?"	Randomised controlled trial

Descriptive questions describe a situation. They need no comparison group, and they do not test a hypothesis; they *estimate* something, such as a proportion or an average, with a measure of its precision (Chapter 12).

Analytical questions compare groups or test a possible cause. They come in two kinds. In **observational** studies the investigator watches what happens without assigning the exposure: Paper 2 observed which patients happened to have a positive fluid balance (Hyun et al. 2023). In **experimental** studies the investigator assigns the intervention, ideally at random: Paper 1 randomised patients to a restrictive or standard strategy (Boulet et al. 2024).

The verbs give it away. "How many" is descriptive; "is associated with" is observational; "does it reduce" implies an experiment. Getting the verb right at this stage prevents a serious error later: an observational study cannot support a causal claim, so a question phrased as "does X reduce Y?" answered with observational data will produce a paper whose conclusions overreach its design (Chapter 4).

> **! Common mistake: descriptive studies are not real research**
>
> Descriptive studies often come first, and they matter. They show that a problem exists, how large it is and whom it affects, and they provide the numbers needed to plan an analytical study. A well-done descriptive study of fluid balance in your own unit may be the most useful study you can do.

2.4 From vague to answerable

The figure below shows the transformation this chapter is about: the corridor question on the left, the PICO(T) question on the right.

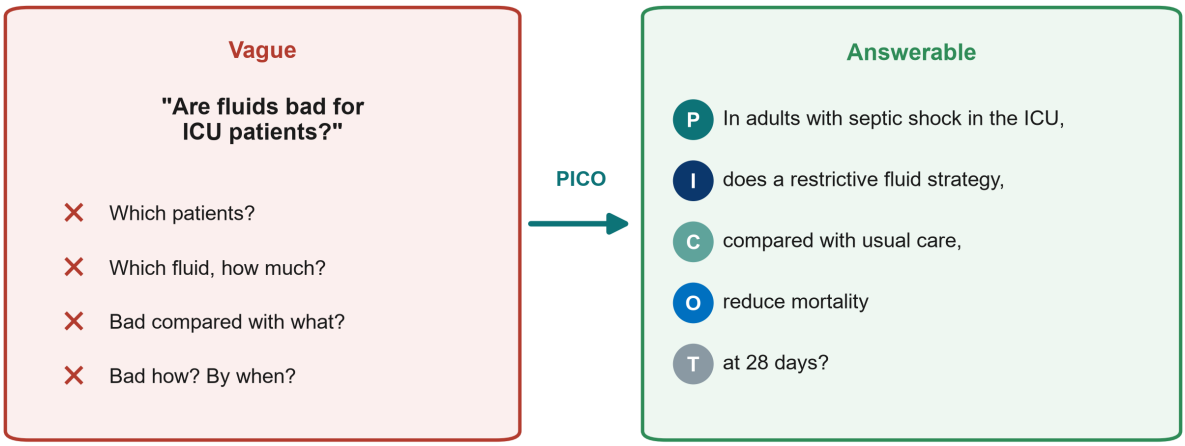


Figure 2.4: A vague question ("Are fluids bad for ICU patients?"), with its unanswered questions (which patients, which fluid, compared with what, bad how and by when), transformed by PICO into an answerable question about adults with septic shock, a restrictive strategy, usual care, mortality and 28 days.

A researchable question has three properties. It is **specific**: every PICO element is defined. It has **one primary outcome**: the single outcome on which the study will be judged, which also determines the sample size. Secondary outcomes can be added, but there must be one main one. And it **implies the design**: once the question is written, the appropriate design is usually obvious.

Worked example: making a corridor question answerable

Corridor question: "Is hand hygiene good on our unit?"
This could become a descriptive audit question ("What proportion of hand-hygiene opportunities are met by ICU staff, observed over one month?") or an analytical question. One analytical version:
In ICU staff (P), does a monthly feedback programme on hand-hygiene performance (I), compared with no feedback (C), increase hand-hygiene compliance (O) at 3 months (T)?
A second example from the ward: "Do night shifts cause medication errors?" Nurses cannot easily be randomised to night shifts, so this becomes a PECO question: among ward nurses (P), are night shifts (E), compared with day shifts (C), associated with more medication administration errors per 1,000 doses (O) over 6 months (T)?

! Common mistake: a broad question is more important

Broad questions produce vague answers. "How can we improve the care of septic patients?" cannot be answered by any single study. A narrow question that is answered well contributes more than a broad question answered badly.

! Common mistake: two outcomes in one question

"Does restrictive fluid reduce mortality and length of stay?" contains two outcomes. If one improves and the other does not, what is the answer? Choose one primary outcome; list the

others as secondary outcomes in the objectives.

2.4.1 Choosing the outcome

The choice of outcome deserves particular care, because it determines what the study can say. Outcomes can be thought of in two broad groups.

Patient-important outcomes are things patients themselves experience and care about: death, disability, quality of life, days alive and out of hospital, the need for dialysis. They are the outcomes that ultimately justify changing practice.

Process and surrogate outcomes are measurements that lie on the path between the intervention and a patient-important outcome: fluid balance, blood pressure, a laboratory value, protocol adherence. They change faster and more often than patient-important outcomes, so studies using them need fewer patients. But a change in a surrogate does not guarantee a change in what matters to patients.

Paper 1 chose cumulative fluid balance at day 5 as its primary outcome, a process outcome (Boulet et al. 2024). That was a sensible choice for a pilot trial, whose job is to ask "can we change fluid balance?" before a larger trial asks "does changing it save lives?". But it means Paper 1 cannot answer our running question about 28-day mortality, and a reader who assumes that it does will misread the paper.

2.5 Is the question worth asking? FINER

A question can be perfectly structured and still not be worth pursuing. The **FINER** criteria, proposed by Hulley and colleagues, are five tests every question should pass before time and money are invested in it (Hulley et al. 2013).

- **Feasible.** Are there enough eligible patients? Enough time, money, skills and equipment? Are the data obtainable? Feasibility is the commonest reason hospital studies fail.
- **Interesting.** Is the question interesting to you, to your team and to the people who will read the result? Research takes a long time, and enthusiasm sustains it.
- **Novel.** Will the study confirm, refute or extend what is known? Novel does not mean "never studied before". Repeating a study in a different setting, in a different population or with a better design can be novel, which is why the literature search matters.
- **Ethical.** Would an ethics committee approve it? Is there genuine uncertainty about which option is better (often called **equipoise**)? Can consent be obtained? Are the risks to participants acceptable?
- **Relevant.** Could the answer change clinical practice, health policy or future research?

2.5.1 FINER applied to the running example

The figure below applies FINER to the running question about restrictive fluids and 28-day mortality, with a traffic-light rating for each letter.

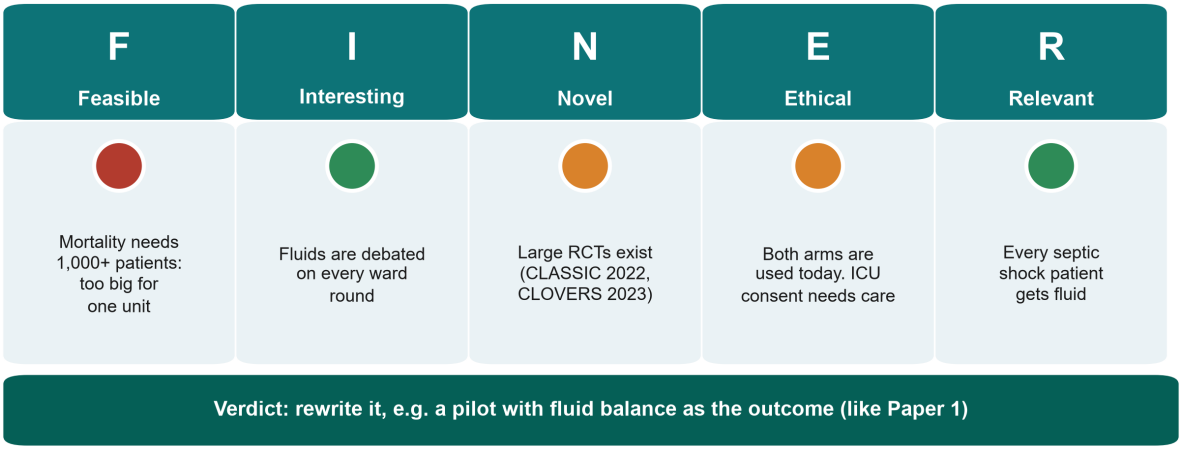


Figure 2.5: FINER check of the running question. Feasible is red (a mortality difference needs over a thousand patients, too many for one unit); Interesting is green; Novel is amber (large trials such as CLASSIC and CLOVERS already exist); Ethical is amber (both strategies are used today, but consent in the ICU needs care); Relevant is green. Verdict: rewrite the question, for example as a pilot with fluid balance as the outcome, as Paper 1 did.

Feasible is red. To detect a modest difference in mortality, a trial needs a very large number of patients, far more than one ICU can recruit in a reasonable time. The worked example below shows why.

Worked example: why a mortality question is not feasible for one unit

Suppose 28-day mortality with usual care is about 42%, and we hope restrictive fluid lowers it to 37%, an absolute reduction of 5 percentage points. A standard approximate formula for the number of patients needed per group, for 80% power and a two-sided significance level of 5%, is

$$n \approx \frac{(1.96 + 0.84)^2 \times [\, p_1(1 - p_1) + p_2(1 - p_2)\,]}{(p_1 - p_2)^2}$$

where p_1 and p_2 are the expected proportions dying in the two groups. Substituting $p_1 = 0.42$ and $p_2 = 0.37$:

$$n \approx \frac{7.84 \times (0.2436 + 0.2331)}{0.05^2} = \frac{7.84 \times 0.4767}{0.0025} \approx 1{,}495$$

About 1,500 patients per group, roughly 3,000 in total. An ICU that admits one eligible patient every two or three days would need decades. This is why CLASSIC (1,554 patients) and CLOVERS (1,563 patients) were run across many hospitals (Meyhoff et al. 2022; National Heart, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Clinical Trials Network et al. 2023). Chapters 12 and 15 explain power and sample size properly; the point here is that the arithmetic of feasibility should be checked before a question is adopted.

Interesting is green: fluid management is debated on every ward round.

Novel is amber: large, well-conducted trials already exist, and neither found a difference in mortality

(Meyhoff et al. 2022; National Heart, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Clinical Trials Network et al. 2023). Another small mortality trial would add little.

Ethical is amber: both strategies are used in current practice, so genuine uncertainty exists, but obtaining informed consent from critically ill patients is difficult and needs special arrangements, such as consent from a legal representative or deferred consent (Chapter 17).

Relevant is green: almost every patient with septic shock receives intravenous fluid.

2.5.2 Rescuing a question that fails FINER

A red or amber light does not mean the idea is bad. It means the question needs reshaping, which is normal and expected. There are several standard ways to rescue a question that fails on feasibility:

- **Change the outcome** to one that occurs more often or changes faster, such as cumulative fluid balance at day 5 rather than mortality. This is exactly what Paper 1 did (Boulet et al. 2024).
- **Make it a pilot or feasibility study**, whose aim is to test whether a larger trial is possible: can patients be recruited, will clinicians follow the protocol, can the outcome be measured? Pilot trials have their own reporting guidance (Eldridge et al. 2016).
- **Ask a descriptive question** about your own practice: what is the cumulative fluid balance of septic shock patients in our unit on days 1 to 5, and where does the fluid come from?
- **Join a multicentre collaboration** that can recruit enough patients.

Worked example: which FINER letter fails?

Table 2.3: Four questions, each failing a different FINER criterion.

Question	Main problem	Why
In all hospitalised adults in the country, does vitamin D reduce mortality over 10 years?	Feasible	Enormous, long and expensive; large vitamin D trials also already exist, so Novel is weak too
In septic shock, does withholding antibiotics for 24 hours reduce antimicrobial resistance?	Ethical	Withholding antibiotics in septic shock causes harm; there is no equipoise
Do our ICU nurses wash their hands? (already audited monthly for five years)	Novel	The answer is already measured; this is an audit question
In pneumonia, does the colour of the drug chart affect 28-day mortality?	Relevant (and Interesting)	No plausible mechanism and no impact on practice

2.6 Aims, objectives and hypotheses

A question tells us what we want to know. Aims and objectives tell us what we will do to find out, and the hypothesis states the answer the data will test. Together they are the question written as a plan, and they are the first part of a protocol that ethics committees and grant reviewers read. The figure below shows how the three relate.

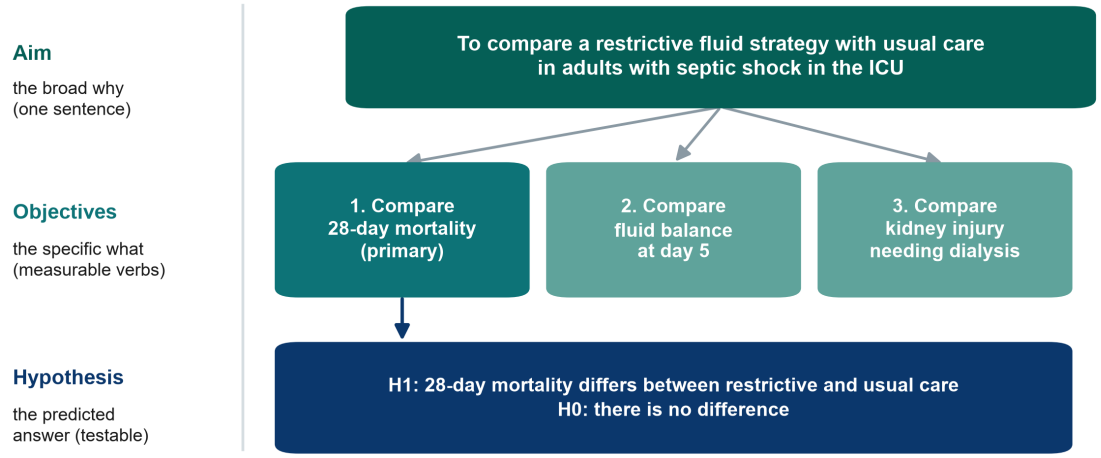


Figure 2.6: The aim (the broad purpose in one sentence) branches into specific, measurable objectives, one of which is marked primary; the primary objective leads to a testable hypothesis, stated as a null hypothesis (no difference) and an alternative hypothesis (a difference).

2.6.1 The aim

The **aim** is the broad purpose of the study in one sentence, usually beginning “To compare…”, “To describe…”, “To estimate…” or “To determine…”. For the running example: *To compare a restrictive fluid strategy with usual care in adults with septic shock in the ICU.* A study normally has one aim.

2.6.2 The objectives

Objectives are the specific, measurable steps that together achieve the aim. Each objective should name an outcome, a time point and, where relevant, a comparison, so that it maps directly onto one measurement and, later, one analysis. Two or three objectives are usually enough. One must be marked **primary**: the primary objective corresponds to the primary outcome, and it drives the sample size calculation (Chapter 15).

Objectives must start with a **measurable verb**. The figure below contrasts vague and measurable verbs.

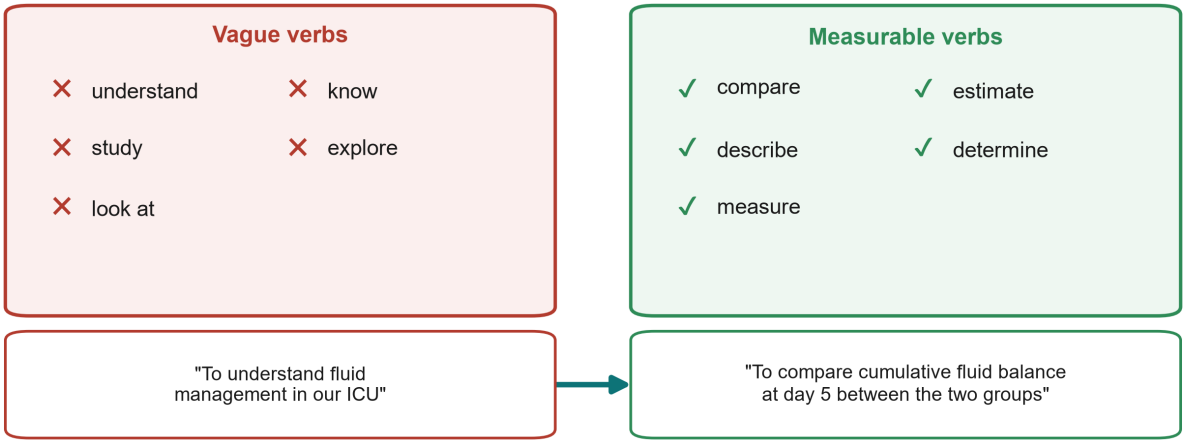


Figure 2.7: Vague verbs (understand, know, study, explore, look at) contrasted with measurable verbs (compare, estimate, describe, determine, measure). "To understand fluid management in our ICU" is rewritten as "To compare cumulative fluid balance at day 5 between the two groups".

"To understand fluid management in our ICU" cannot be finished or checked: when would we know we had understood it? "To compare cumulative fluid balance at day 5 between the two groups" tells the reader the outcome, the time point and the comparison, and it can clearly be completed. A useful test: if you cannot say how you will measure it, it is not yet an objective.

(Qualitative studies legitimately use verbs such as "explore", because their purpose is to understand experiences rather than to measure them. The rule about measurable verbs applies to quantitative objectives.)

! Common mistake: more objectives make a stronger proposal

A list of eight objectives suggests an unfocused study and invites a reviewer to ask which one the study is designed to answer. Two or three focused objectives, with one clearly primary, are stronger.

2.6.3 Hypotheses

A **hypothesis** is the predicted answer to the primary analytical objective, stated so that data can contradict it. Statistical testing works with two hypotheses, shown in the figure below.

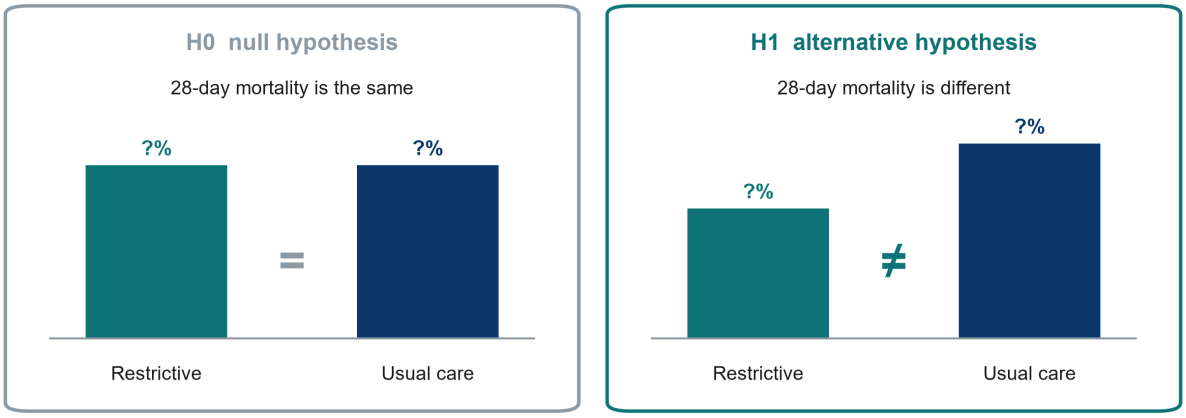


Figure 2.8: The null hypothesis H0 states that 28-day mortality is the same with restrictive fluid and usual care; the alternative hypothesis H1 states that it is different.

- The **null hypothesis (H0)** states that there is no difference or no association: *28-day mortality is the same with a restrictive fluid strategy and with usual care.*
- The **alternative hypothesis (H1)** states that there is a difference: *28-day mortality differs between the two strategies.*

The study then asks whether the data are compatible with H0. If the observed difference would be very unlikely if H0 were true, we **reject** H0. If not, we **fail to reject** it. We never "prove" H0.

A courtroom analogy helps. A court starts by assuming the defendant is not guilty (H0) and convicts only if the evidence is strong enough. A verdict of "not guilty" does not mean the defendant has been shown to be innocent; it means the evidence was insufficient. In the same way, "no statistically significant difference" does not mean "no difference". A small study may simply be too small to detect a real difference (Altman and Bland 1995). Paper 1, with 48 patients, was far too small to detect a difference in mortality, and was not designed to; its finding of similar mortality in the two groups is not evidence that the strategies are equally safe (Boulet et al. 2024). P-values and confidence intervals are explained in Chapter 12.

One-sided or two-sided? H1 is usually **two-sided** ("mortality differs"), even when the investigators expect benefit, because a new intervention can harm as well as help; fluid restriction could increase deaths from under-perfusion. A **one-sided** hypothesis ("mortality is lower") ignores the possibility of harm. It must be specified in advance and strongly justified, and reviewers are rightly suspicious of it. When in doubt, use a two-sided hypothesis.

Non-inferiority questions. Sometimes the aim is not to show that a new option is better, but that a cheaper, simpler or safer option is *not much worse*. These questions have their own hypotheses, framed around a pre-specified margin, and their own design (Chapter 5).

When no hypothesis is needed. Purely descriptive studies estimate rather than test. "To estimate the proportion of patients with septic shock who have a positive fluid balance on day 3" needs no hypothesis; it needs a precise estimate with a 95% confidence interval.

! Common mistake: every study needs a hypothesis

Descriptive studies have no hypothesis to test. Inventing one ("H1: more than 50% of patients have a positive balance") adds nothing and suggests the investigators have not understood their own design. State clearly that the study is descriptive and that its purpose is estimation.

2.7 Keeping it consistent

The question, the primary objective, the hypothesis and (later) the primary outcome in the protocol, the sample size calculation, the main analysis and the abstract of the final paper must all refer to the **same outcome measured at the same time**. Reviewers check this chain immediately, and inconsistency is one of the commonest weaknesses in research proposals.

Worked example: a consistent plan for the delirium question

Question. In adults aged 65 and over after hip-fracture surgery, does a daily non-drug delirium-prevention bundle, compared with usual care, reduce incident delirium in the first 7 postoperative days?
Aim. To compare a daily non-drug delirium-prevention bundle with usual care in adults aged 65 and over after hip-fracture surgery.
Objectives.

1. (Primary) To compare the incidence of delirium in the first 7 postoperative days between the two groups.
2. To compare length of hospital stay between the two groups.

Hypothesis. H0: the incidence of delirium in the first 7 days is the same in both groups. H1: it differs.
Check the chain: the question's outcome (delirium in 7 days) is the primary objective's outcome, which is the outcome in H0 and H1. The plan is consistent.
A descriptive version would need no hypothesis: *To estimate the proportion of patients aged 65 and over who develop delirium in the first 7 days after hip-fracture surgery.*
FINER watch-points. Ethical: patients who may become confused need careful consent procedures. Feasible: staff time to deliver the bundle every day.

Worked example: a feasible version of the running question

Because the mortality question fails on feasibility, a single ICU might instead plan a pilot trial similar to Paper 1.
Type of project. Research: patients are randomised and care is changed, so ethics committee approval and prospective trial registration are required.
PICO(T). P: adults with septic shock within 24 hours of starting vasopressors. I: a restrictive protocol for maintenance and drug-dilution fluids. C: usual care. O: cumulative fluid balance (ml/kg). T: day 5.
Question. In adults with septic shock in our ICU, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5?
FINER. Feasible: green to amber (a pilot of about 50 patients over 12 to 18 months). Interesting: green. Novel: amber to green (a different protocol in a new setting). Ethical: amber (consent

arrangements needed). Relevant: green.
Aim. To compare a restrictive fluid protocol with usual care in adults with septic shock in our ICU.
Objectives.

1. (Primary) To compare cumulative fluid balance (ml/kg) at day 5 between groups.
2. To estimate the recruitment rate and adherence to the protocol.
3. To describe 28-day mortality and kidney injury needing dialysis in each group (exploratory).

Hypothesis. H0: cumulative fluid balance at day 5 is the same in both groups. H1: it differs. Note objective 3: mortality is *described*, not tested, because the study is far too small to test it. Writing "to compare mortality" would promise something the study cannot deliver.

2.8 Common problems in research questions

The same faults appear again and again in draft questions. They are easy to fix once recognised.

Table 2.4: Common faults in research questions and how to fix them.

Fault	Example	Fix
No comparison	"Does a fluid protocol work?"	Name the comparison: usual care, a different protocol
Outcome with no time point	"…reduce mortality?"	"…reduce 28-day mortality?"
Two outcomes	"…mortality and length of stay?"	One primary outcome; others secondary
Population too broad	"In all patients…"	Define condition, age, setting, timing
Intervention undefined	"Does better care…"	Describe what is done, by whom, how often
Condition placed in O	"…reduce hip fracture?" in a post-operative study	The condition that brings patients in belongs to P
Causal verb for an exposure that cannot be assigned	"Does obesity cause pressure injuries?" studied observationally	"Is obesity associated with…"
Vague objective verb	"To understand delirium"	"To estimate the incidence of delirium…"

Summary

- Research questions come from daily practice: bedside doubts, variation between colleagues, audits and incidents, guidelines that admit uncertainty, papers that call for more research, and patients' own questions. Before treating an uncertainty as research, search to see whether the answer is already known.
- PICO(T) structures a question: Population, Intervention, Comparison, Outcome, Time. Exposure questions use PECO; frequency questions need only P, O and T; diagnostic, prognostic and qualitative questions have their own variants.

- Descriptive questions estimate; analytical questions compare, either by observing (association) or by intervening (cause). The verb of the question points to the design.
- A good question is specific, has one primary outcome measured at a stated time, and implies its design.
- FINER (Feasible, Interesting, Novel, Ethical, Relevant) tests whether a question is worth pursuing. Feasibility fails most often; questions can be rescued by changing the outcome, running a pilot, asking a descriptive question or collaborating.
- The aim is the broad purpose; objectives are specific and measurable, with one marked primary; the hypothesis (H0 and H1) is stated for the primary analytical objective. Descriptive studies need no hypothesis. Failing to reject H0 is not proof of no difference.
- The question, primary objective, hypothesis and primary outcome must all name the same outcome at the same time.

Further reading

- Hulley and colleagues, *Designing Clinical Research* (Hulley et al. 2013). The chapter on conceiving the research question is the origin of the FINER criteria and gives many further examples.
- Richardson and colleagues, “The well-built clinical question” (Richardson et al. 1995). The short paper that introduced the PICO structure for evidence-based practice.
- Cooke and colleagues, “Beyond PICO: the SPIDER tool for qualitative evidence synthesis” (Cooke et al. 2012). Explains how to frame questions about experiences and behaviour.
- Altman and Bland, “Absence of evidence is not evidence of absence” (Altman and Bland 1995). A one-page classic on why a non-significant result does not show that there is no difference.

CHAPTER 3

Finding, managing and citing the evidence

Once a clinical worry has become a research question (Chapter 2), the next task is to find out what is already known about it. A literature search decides whether the question is genuinely open, provides the background for a protocol or paper, shows how other investigators defined and measured the same outcome, and often reshapes the question itself. Searching well is a skill that can be learnt in an afternoon and improved for a lifetime. This chapter explains how to turn a PICO question into a documented search, how to use PubMed and other sources, how to store and cite what you find, how to use other people's work honestly, and how to turn a handful of key papers into the opening paragraphs of a research proposal.

In this chapter you will learn

- why a documented literature search is part of every research project;
- how to turn a PICO question into concept blocks joined with Boolean operators;
- how MeSH terms, free-text words, phrase searching, truncation, field tags and filters work in PubMed;
- when to use PubMed, Google Scholar, other databases, trial registries and grey literature, and why publication bias matters;
- how to manage references with a reference manager and choose a citation style;
- how to paraphrase without plagiarising, and the rules for citing honestly;
- how to summarise papers in a literature matrix and write a problem statement, working title and keywords.

3.1 Why search before you study

A literature search does four jobs in a research project.

1. **It checks novelty.** If the question has already been answered convincingly, the right response is to apply the answer (through a guideline, audit or quality-improvement project), not to repeat the study. If the evidence is conflicting, weak or comes from settings unlike yours, there is room for a new study.
2. **It provides the background.** Every protocol, grant application and paper opens with a short account of what is known and what is not. That account must be built from evidence, with citations.
3. **It shows how others did it.** Published studies reveal how the population was defined, how the intervention was delivered, which outcome was used and how it was measured, how many patients were recruited and what went wrong. Paper 1, for example, reports that only about 4% of screened patients were randomised (Boulet et al. 2024), a sobering figure for anyone planning a similar trial.
4. **It reshapes the question.** Searching often shows that the original question was too broad, already answered, or framed around the wrong outcome.

In clinical work we often look for evidence quickly: a search engine, a colleague, a point-of-care summary or a guideline. These are reasonable for patient care. A research proposal needs more: a **documented search**, recorded in enough detail that someone else could repeat it and get the same results.

3.2 The shape of a good search

The figure below shows how a good search narrows step by step, from the whole of a database to a handful of key papers.

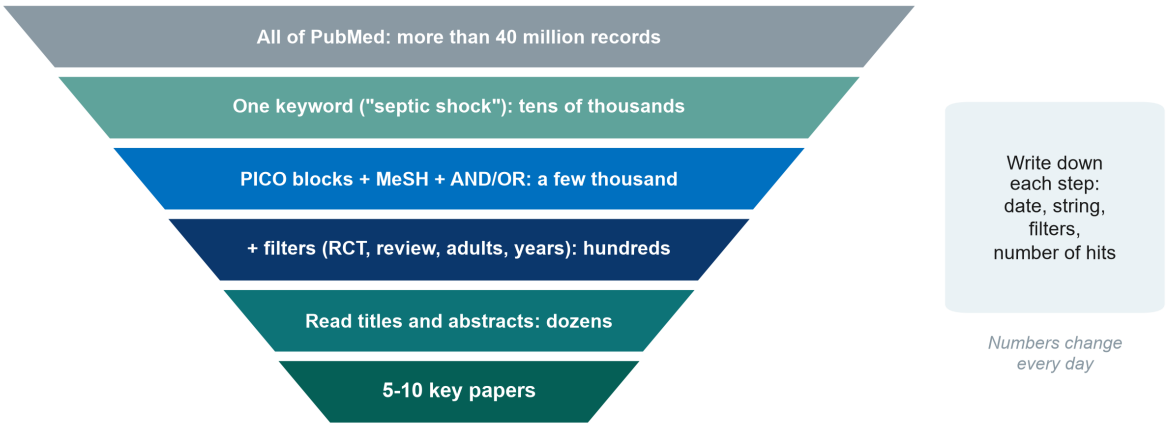


Figure 3.1: The search funnel: from all of PubMed (more than 35 million records), to one keyword (tens of thousands of hits), to structured PICO blocks with MeSH and Boolean operators (a few thousand), to filters for design, age and date (hundreds), to reading titles and abstracts (dozens), to 5 to 10 key papers. Each step should be written down: date, search string, filters and number of hits. The numbers are orders of magnitude and change every day.

Two features of the funnel are important. First, the last two steps are done by a human reader, not by the computer: no search string can judge relevance and quality. Second, every step is recorded. A search record contains:

- the database (for example PubMed);
- the date of the search;
- the exact search string;
- any filters applied, each with a reason;
- the number of hits at each stage;
- the papers selected, and why others were excluded.

Journals and reviewers increasingly ask for this information, and systematic reviews must report it in full (Chapter 9).

3.3 From PICO to search blocks

The most reliable way to build a search is to turn each PICO element into a **concept block**: a set of synonyms for one idea, combined so that a paper using any of them is found. The figure below shows the running question of this book, restrictive fluids in septic shock, translated into three blocks.

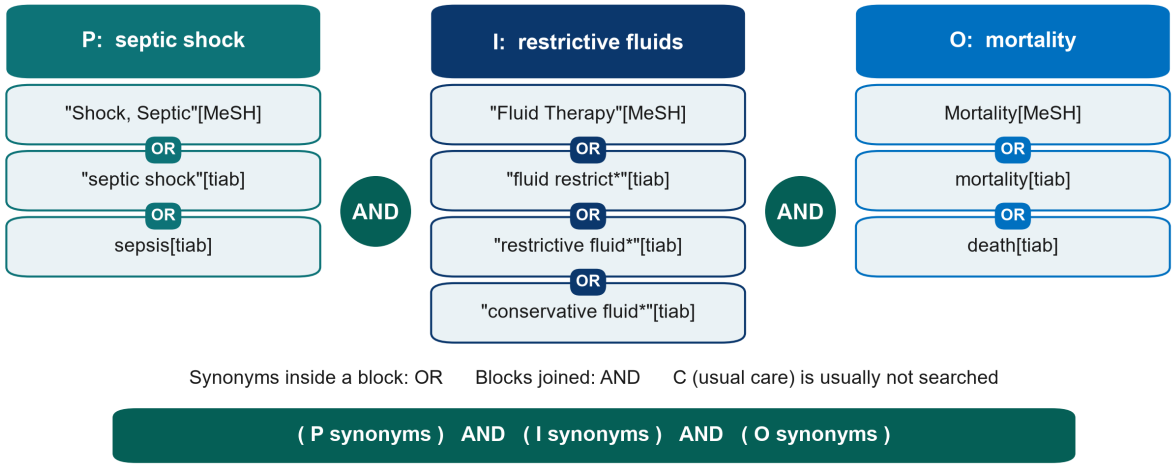


Figure 3.2: From PICO to search blocks. Population block: "Shock, Septic"[MeSH] OR "septic shock"[tiab] OR sepsis[tiab]. Intervention block: "Fluid Therapy"[MeSH] OR "fluid restrict*"[tiab] OR "restrictive fluid*"[tiab] OR "conservative fluid*"[tiab]. Outcome block: Mortality[MeSH] OR mortality[tiab] OR death[tiab]. Synonyms inside a block are joined with OR, blocks are joined with AND, and the comparison (usual care) is usually not searched.

The steps are:

1. **List the concepts.** Usually P, I (or E) and O. The comparison is normally left out, because "usual care" is described in too many different ways and adding it would lose relevant papers.
2. **List synonyms for each concept.** Authors describe the same idea in different words. For restrictive fluids: fluid restriction, restrictive fluid, conservative fluid, limited fluid, de-resuscitation, fluid removal, negative fluid balance. Spelling varies too (British and American

English), and PubMed indexes titles and abstracts in English, so synonyms must be in English whatever language the research will eventually be written in.

3. **Combine synonyms with OR** inside brackets, so that any synonym is enough.
4. **Combine blocks with AND**, so that every concept must be present.

Written out in full, the search for the running example is:

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*"[tiab]
     OR "restrictive fluid*"[tiab] OR "conservative fluid*"[tiab])
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Useful additions to the intervention block would be `de-resuscitation[tiab]`, `deresuscitation[tiab]`, `"fluid balance"[tiab]` and the MeSH heading `"Water-Electrolyte Balance"[Mesh]`.

The same method works for any topic. The figure below shows a ward question about pressure injuries.

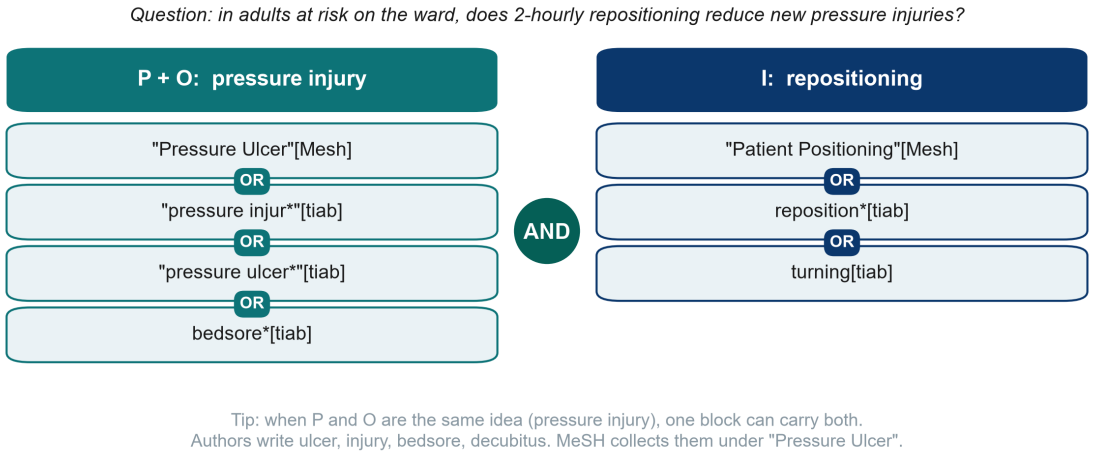


Figure 3.3: A ward search: in adults at risk on the ward, does 2-hourly repositioning reduce new pressure injuries? One block carries both P and O (pressure injury: “Pressure Ulcer”[Mesh], “pressure injur*”, “pressure ulcer*”, bedsore*), joined with AND to an intervention block (repositioning: “Patient Positioning”[Mesh], reposition*, turning). MeSH collects the different words authors use under “Pressure Ulcer”.

Here the population and the outcome share the same idea (patients at risk of pressure injury, and new pressure injury), so one block carries both. Authors write pressure ulcer, pressure injury, bedsore and decubitus ulcer; the block must catch them all.

Worked example: a search for the delirium question

Question (Chapter 2): in adults aged 65 and over after hip-fracture surgery, does a daily non-drug prevention bundle, compared with usual care, reduce delirium in the first 7 days?

```
(Delirium[Mesh] OR delirium[tiab] OR "acute confusion"[tiab])
AND ("Hip Fractures"[Mesh] OR "hip fracture*"[tiab])
AND (prevent*[tiab] OR multicomponent[tiab]
     OR "non-pharmacological"[tiab] OR bundle[tiab])
```

The three blocks are the outcome (delirium), the population (hip fracture) and the intervention

(non-drug prevention). Age is not searched as a word; if needed, it is handled with a filter. The comparison, usual care, is not searched.

! Common mistake: typing the whole question into the search box

Typing “does restrictive fluid reduce mortality in septic shock” into PubMed hands the logic of the search to the software. You lose control over synonyms and over how words are combined, and you cannot record or repeat exactly what was searched. Build blocks instead.

3.4 Boolean logic

The words AND, OR and NOT are called **Boolean operators**. The figure below shows what each does.

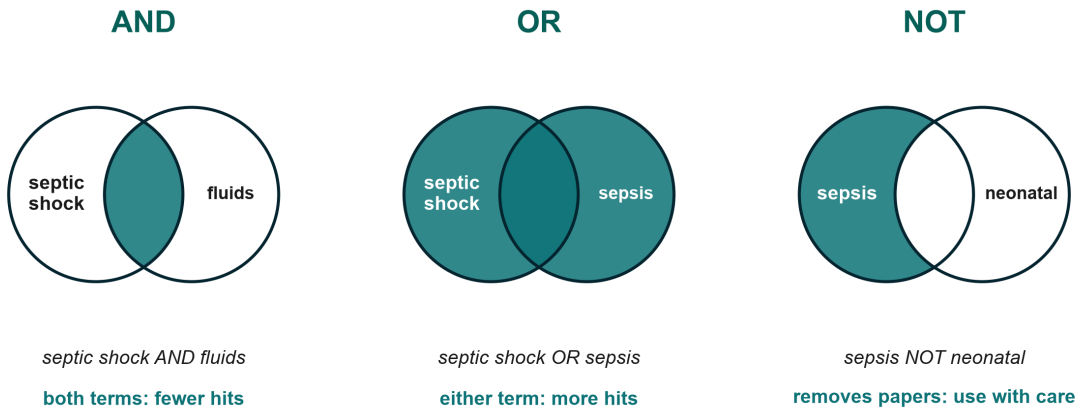


Figure 3.4: Boolean operators as overlapping circles. AND (septic shock AND fluids) finds only papers containing both terms: fewer hits. OR (septic shock OR sepsis) finds papers containing either term: more hits. NOT (sepsis NOT neonatal) removes every paper containing the second term: use with care.

- **AND narrows.** Only records containing both terms are returned. Adding another AND block always gives fewer hits.
- **OR widens.** Records containing either term are returned. OR is used for synonyms inside a block.
- **NOT removes.** Every record containing the second term is excluded. NOT is risky: `sepsis NOT neonatal` removes every paper that mentions neonates anywhere in the title or abstract, including good adult studies that mention neonates once in passing. Use NOT sparingly, and only with a reason you can write down.

Two practical rules: write operators in capital letters, and use brackets to control the order in which they are applied. `septic shock OR sepsis AND fluids` is ambiguous; `(septic shock OR sepsis) AND fluids` is not.

! Common mistake: AND inside a block

`("septic shock" AND sepsis)` finds only papers that use both words, and loses papers that use only one. Synonyms belong together with OR; AND belongs between blocks.

3.5 MeSH: the shared label behind many words

Authors describe the same condition in many ways. PubMed solves this with **MeSH** (Medical Subject Headings), the controlled vocabulary of the United States National Library of Medicine. Trained indexers read each article in MEDLINE, the main database inside PubMed, and tag it with MeSH headings, whatever words the authors used. The figure below shows how it works.

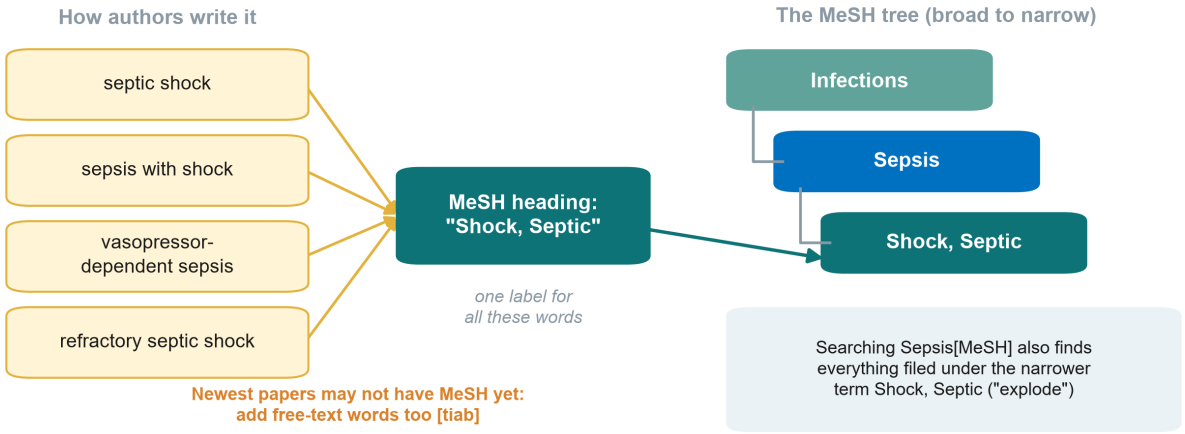


Figure 3.5: MeSH collects different author wordings (septic shock, sepsis with shock, vasopressor-dependent sepsis, refractory septic shock) under one heading, “Shock, Septic”. Headings sit in a tree from broad to narrow (Infections, then Sepsis, then Shock, Septic); searching Sepsis[MeSH] also finds everything filed under the narrower term (“explode”). The newest papers may not have MeSH yet, so free-text words should be added too.

Three properties of MeSH matter in practice:

- **One label for many words.** Searching `"Shock, Septic"[Mesh]` finds papers whether the authors wrote “septic shock”, “sepsis with shock” or “vasopressor-dependent sepsis”. For a nurse searching for pressure injuries, the heading “Pressure Ulcer” collects pressure ulcer, pressure injury, bedsore and decubitus ulcer.
- **A tree from broad to narrow.** MeSH headings are arranged hierarchically. By default, searching a heading also searches every narrower heading below it, which is called **exploding** the term: `Sepsis[Mesh]` includes `Shock, Septic`.
- **A delay.** Indexing takes time, so the newest records may not yet have MeSH headings. That is why every block should combine the MeSH heading with free-text words searched in the title and abstract.

The MeSH database can be browsed from the PubMed home page to find the right heading for a concept. The keywords listed under an abstract, and the MeSH terms shown for a relevant paper you have already found, are good sources of headings and synonyms.

3.6 Four tools that sharpen a search

Table 3.1: Four tools that make a PubMed search more precise.

Tool	Example	What it does
Quotation marks	`"septic shock"`	Searches the exact phrase, not two separate words
Truncation (*)	`restrict*`	Finds every ending: restrict, restricted, restriction, restrictive
Field tags	`[tiab]`, `[mh]`, `[pt]`	Restrict where PubMed looks: title and abstract, MeSH heading, publication type
Filters	Randomized Controlled Trial; Adult 19+; last 10 years	Narrow results by article type, age group, date, species, language

A few cautions apply. A truncation stem that is too short finds unwanted words: `flu*` finds influenza and fluoride as well as fluid. Every filter needs a reason that can be written down: “Randomized Controlled Trial and Systematic Review, because ours is a treatment question; adults, because our population is adult; last 10 years, to reflect current practice”. And after running any search, check PubMed’s **search details**, which show exactly how the software interpreted your string; automatic term mapping sometimes adds or changes terms.

! Common mistake: filtering to free full text

Restricting a search to papers with free full text throws away many of the most important studies, simply because their publisher charges for access. Find the key papers first; worry about access afterwards. Many institutions have library services that can obtain articles, and authors will often send a copy on request.

3.7 Where to search

3.7.1 PubMed and Google Scholar

The two most widely used free tools are compared in the figure below.

	PubMed	Google Scholar
What it covers	Biomedical journals, indexed by experts	Almost everything: theses, books, preprints, websites
MeSH (controlled vocabulary)	✓	✗
Filters (design, age, date)	✓ many	✗ few (date, reviews)
Transparent, repeatable	✓	✗ ranking is hidden
"Cited by" a key paper	partial	✓
Best for	your main, repeatable search	a quick check, grey literature, citations

Figure 3.6: PubMed and Google Scholar compared. PubMed covers biomedical journals indexed by experts, uses MeSH, has many filters and is transparent and repeatable; it is best for the main search. Google Scholar covers almost everything (theses, books, preprints, websites), has no MeSH and few filters, ranks results by a hidden algorithm, but is good at showing which papers cite a key paper; it is best for a quick check, grey literature and following citations.

PubMed is free, maintained by the United States National Library of Medicine, and contains more than 35 million records from biomedical journals. Its search is transparent: the same string run on the same day gives the same result, and the string can be reported. It should be the main database for a clinical research proposal.

Google Scholar is very broad, covering theses, books, preprints and conference papers as well as journals. Its most useful feature is “Cited by”, which follows citations *forward* in time: from a key paper published in 2022, it shows every later paper that cited it. Its weaknesses are a hidden ranking algorithm, few filters and results that change from month to month, so a Google Scholar search cannot be reliably documented or repeated. Use it to check that the main search has not missed something obvious and to follow citations.

Two simple techniques complement any database search. **Backward citation searching** means reading the reference lists of the key papers you have found. **Forward citation searching** means using “Cited by” (in Google Scholar or PubMed’s “Cited by” list) to find newer papers that cite them. Both often find papers the search string missed.

3.7.2 Other databases

Table 3.2: Sources beyond PubMed.

Source	What it contains	Access
Cochrane Library	Systematic reviews of interventions and a register of controlled trials	Many reviews free; check local access
Embase	Large biomedical database, strong on drugs and European journals	Subscription
CINAHL	Nursing and allied health literature	Subscription
Trial registries (ClinicalTrials.gov, WHO ICTRP)	Planned, ongoing and completed trials, including unpublished ones	Free
PubMed Clinical Queries	Ready-made filters for therapy, diagnosis, aetiology and prognosis questions	Free

For a treatment question, the **Cochrane Library** is a good place to start: a recent, high-quality systematic review may already answer the question, and its search strategy can be borrowed. Systematic reviews search several databases because no single one is complete (Higgins et al. 2019). For a single research proposal, PubMed plus a trial registry, supplemented by citation searching, is usually enough. National and regional databases can also matter, because some journals published in local languages are not indexed in PubMed.

3.7.3 Grey literature, registries and publication bias

Journal articles found in PubMed are only part of the evidence. The figure below uses the image of an iceberg.

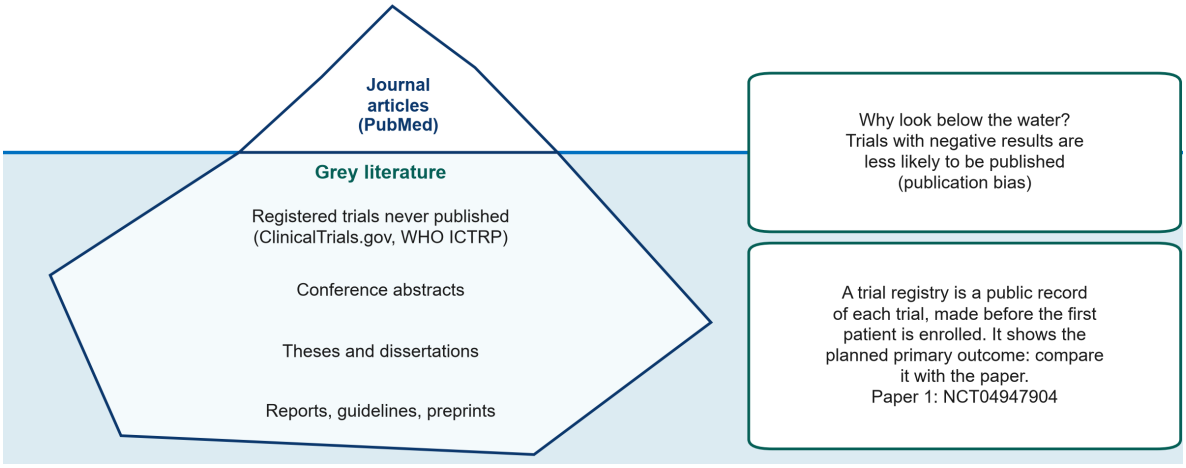


Figure 3.7: The evidence iceberg. Journal articles indexed in PubMed are the visible tip. Below the waterline lie registered trials that were never published, conference abstracts, theses and dissertations, reports, guidelines and preprints: the grey literature. Trials with negative results are less likely to be published (publication bias). Registries also show each trial's planned primary outcome, which can be compared with the published paper; Paper 1's registration number is NCT04947904.

Grey literature is material not published in conventional journals: conference abstracts, theses, reports from health agencies, guidelines and preprints. It matters because of **publication bias**: studies with positive, striking results are more likely to be published, published quickly and published in prominent journals than studies with negative or disappointing results. If only the published

studies are read, an intervention can look more effective than it really is.

Trial registries are a partial remedy. ClinicalTrials.gov and the World Health Organization International Clinical Trials Registry Platform (ICTRP) list trials that have been planned, are ongoing or have finished, including those whose results were never published. Since 2005, the International Committee of Medical Journal Editors (ICMJE) has required trials to be registered in a public registry before the first participant is enrolled as a condition for publication in member journals (De Angelis et al. 2004; International Committee of Medical Journal Editors 2024).

Registries have three uses for a reader and a researcher:

- **Finding unpublished and ongoing trials.** Before planning a trial, check that nobody is already running the same study.
- **Checking for outcome switching.** A registry entry records the *planned* primary outcome. Compare it with the primary outcome in the published paper. A study that quietly replaces a disappointing primary outcome with a more favourable secondary one has reported selectively, and this is a known and common problem (Chan et al. 2004).
- **Checking timing.** Was the trial registered before the first patient, or afterwards?

Paper 1 passes these checks: it was registered on ClinicalTrials.gov as NCT04947904 on 1 July 2021, before recruitment began in September 2021, and its primary outcome in the paper (cumulative fluid balance at day 5) is the one it was designed around (Boulet et al. 2024).

Preprints are papers posted online before peer review, for example on medRxiv. They make findings available quickly, but they have not been checked by independent reviewers. They may be cited, clearly labelled as preprints, and read with extra care.

3.8 Screening and selecting papers

A good search for the running question returns several hundred to a few thousand records. The next step is **screening**: reading titles, then abstracts, and deciding which papers are relevant. Useful questions at this stage are:

- Is the population similar to mine?
- Does it study the intervention or exposure I am interested in?
- Does it measure an outcome relevant to my question?
- What is the design, and how high on the evidence pyramid is it (Chapter 1)?
- Is it recent, or is it a landmark study that everyone cites?

From the dozens of relevant records, a proposal usually builds on **five to ten key papers**: the most recent high-quality systematic review if one exists, the largest and best randomised trials, the most relevant observational studies, the current guideline, and any study from a setting similar to yours. Chapter 4 explains how to read each of them efficiently.

For the running example, a sensible set of five key references would be: Paper 1, the pilot trial closest in design to a feasible local study (Boulet et al. 2024); Paper 2, the large matched cohort linking positive balance to mortality (Hyun et al. 2023); the CLASSIC trial (Meyhoff et al. 2022);

the CLOVERS trial (National Heart, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Clinical Trials Network et al. 2023); and the current Surviving Sepsis Campaign guideline, which shows where uncertainty remains (Evans et al. 2021).

3.9 Managing references

3.9.1 Reference managers

Every paper you find should be saved, with its full citation details, the moment you decide it is relevant. A **reference manager** is software that does this and much more. The workflow has five steps:

1. **Collect.** Tick relevant papers in PubMed, or save them from the journal website with a browser connector.
2. **Export.** In PubMed, choose “Send to: Citation manager”, which downloads a file containing the citation details.
3. **Import.** Open the file in the reference manager. Zotero and Mendeley are free; EndNote is common in universities but paid. Authors, title, journal, year, volume, pages and DOI arrive without being typed by hand.
4. **Cite.** While writing in a word processor, insert citations through the reference manager’s plug-in.
5. **List.** The reference manager builds the reference list automatically, in whatever style the journal requires.

The best reference manager is the one the whole team uses. The important habit is to start on day one.

! Common mistake: I will sort out the references at the end

Saving a reference takes seconds; rebuilding a list from memory at the end of a project takes hours and produces errors: wrong years, misspelt names, references cited in the text but missing from the list, and citations attached to the wrong claim. Wrong references are among the commonest errors in proposals and manuscripts.

3.9.2 Citation styles

Journals format citations and reference lists in different **styles**. The two most common in health research are shown below.

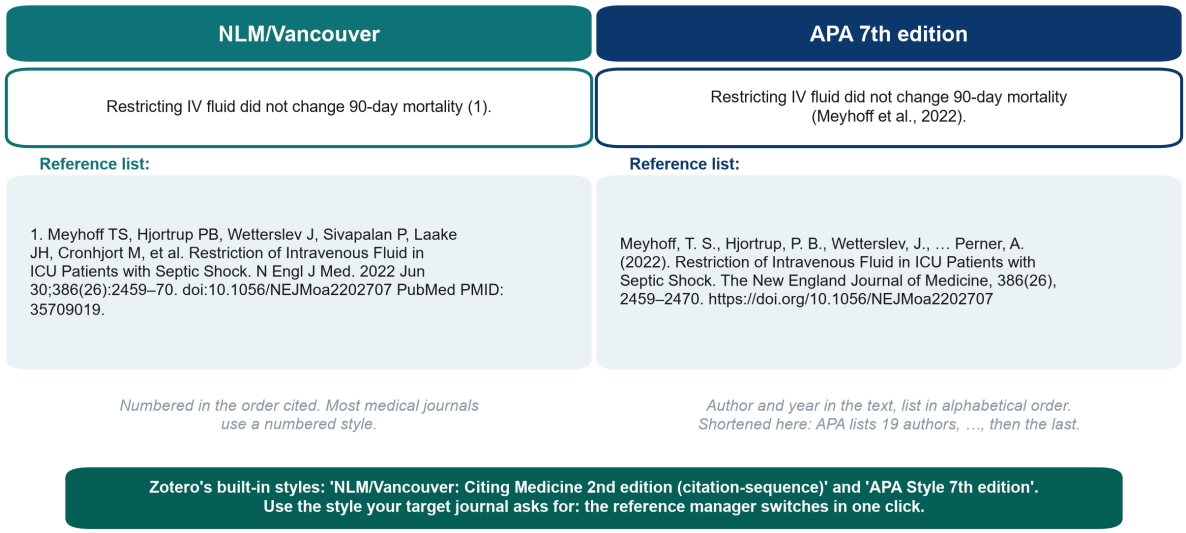


Figure 3.8: The same sentence and reference in the two citation styles built into Zotero. NLM/Vancouver: "Restricting IV fluid did not change 90-day mortality (1)", with a numbered reference list in the order cited; most medical journals use a numbered style. APA 7th edition: "(Meyhoff et al., 2022)" in the text, with an alphabetical reference list. Both reference lists are Zotero's own output for the CLASSIC trial. A reference manager switches between styles in one click.

NLM/Vancouver style (in Zotero: 'NLM/Vancouver: Citing Medicine 2nd edition (citation-sequence)') numbers references in the order they are first cited, as (1) in the text; it follows the ICMJE recommendations (International Committee of Medical Journal Editors 2024). Most medical journals use a numbered style, each with its own small variations, for example [1] or a superscript number. **APA style** (in Zotero: 'APA Style 7th edition') gives the author and year in the text and lists references alphabetically; it is common in nursing, psychology and the social sciences. Every journal states its required style in its instructions for authors. With a reference manager, changing style is a single click, so there is no reason to format references by hand.

3.10 Using other people's work honestly

A citation does two jobs: it credits the people whose work you are using, and it lets readers check your claim. Both depend on citing accurately and writing in your own words.

3.10.1 Paraphrasing and patchwriting

Medical writing rarely quotes sources word for word. Instead, writers **paraphrase**: they restate the source's meaning in their own words and cite it. The figure below shows the difference between a genuine paraphrase and its common imitation.

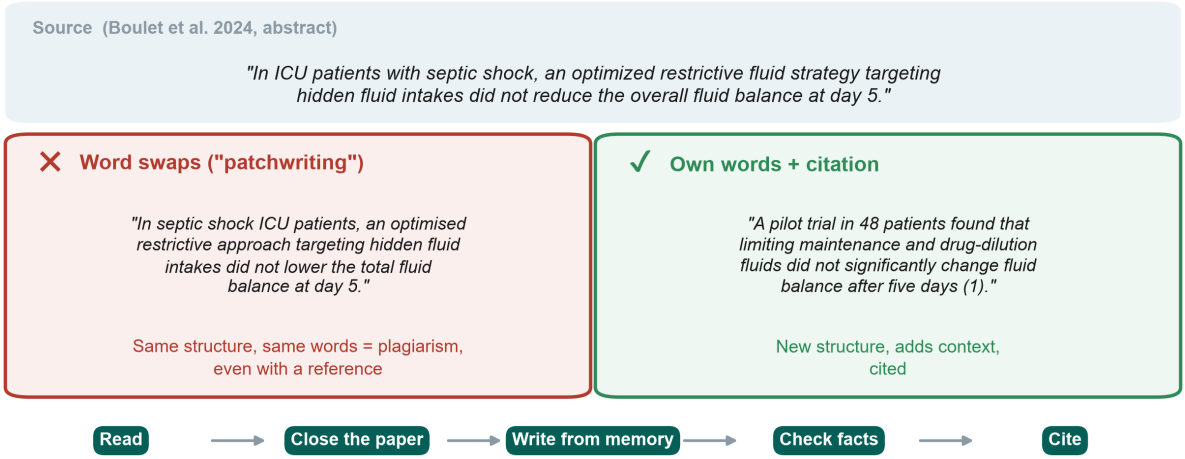


Figure 3.9: A source sentence from the abstract of Paper 1, a "patchwritten" version that swaps a few words but keeps the structure (plagiarism, even with a reference), and a genuine paraphrase with a new structure, added context (a pilot trial in 48 patients, limiting maintenance and drug-dilution fluids) and a citation. Method: read, close the paper, write from memory, check the facts, cite.

Patchwriting keeps the structure and most of the wording of the source and changes a few words. It is plagiarism *even when the source is cited*, because the words and their arrangement are still someone else's. Similarity-checking software used by journals detects it easily. A **genuine paraphrase** is written from understanding: read the source, close it, write the idea in your own words, then reopen the source to check that every fact is correct and nothing has been strengthened or weakened, and add the citation.

If the exact words matter, use quotation marks and a citation, but quotations are rare in medical writing.

Worked example: judging four paraphrases of Paper 2

Paper 2's conclusion reports that a positive cumulative fluid balance on the second and third ICU days was associated with higher 28-day mortality (Hyun et al. 2023). Four attempts to use this finding:
(a) The sentence copied with one word changed ("mortality" replaced by "death") and a citation added. **Patchwriting**: same structure, almost the same words. Plagiarism despite the citation.
(b) "In a large propensity-matched registry cohort, patients whose cumulative fluid balance was positive on ICU days 2 and 3 had higher 28-day mortality than those with a negative balance (hazard ratio 2.13 on day 2 and 1.56 on day 3) (2)." **Acceptable**: new structure, added context, the calibrated word "associated" is preserved in meaning, numbers given, source cited.
(c) "Giving more fluid in the first days of ICU doubles the risk of death (2)." **Distorts the source**: it turns an observational association into a causal claim, and "doubles" overstates the result (the hazard ratio was 2.13 for day 2 but 1.56 for day 3). A citation does not excuse

misrepresenting what a paper found.
(d) A well-written sentence in the writer's own words reporting the finding, with no citation.
Plagiarism of ideas: someone else's finding presented without credit.

Worked example: paraphrasing a secondary result of Paper 1

Paper 1 reports that, after 28 days, mortality and the number of days alive without life support were similar between the two groups (Boulet et al. 2024). A good paraphrase: "Mortality and days alive without organ support at day 28 did not differ between the groups, although the pilot trial was too small to detect such differences (1)."
The structure is new, the facts are unchanged, and the added context (the trial was too small) is accurate: with 48 patients the trial was not designed to detect differences in mortality. What a careful writer must *not* write is "restriction was shown to be safe", which claims far more than the data allow.

3.10.2 Four rules for citing safely

1. **Cite facts.** Numbers, findings and other people's ideas need a citation. Common knowledge ("sepsis is a life-threatening response to infection") does not.
2. **Cite sources you have read.** Cite the original study, not a review or textbook that quotes it, wherever possible. Citing a paper you have only seen quoted elsewhere repeats any error the intermediate author made.
3. **Use your own words.** Change the structure, not just the vocabulary; use quotation marks for the rare exact quotation.
4. **Check.** Every reference must say what you claim it says. Reopen the paper and confirm.

The summary of these rules is short: never cite a paper you have not read.

Worked example: OK or not OK?

Table 3.3: Four citing practices judged.

Situation	Verdict
A sentence copied from an abstract, two words changed, citation added	Not OK: patchwriting
One sentence quoted word for word, in quotation marks, with a citation	OK, though rarely needed in medical writing
A finding written in your own words, but the citation forgotten	Not OK: someone else's finding needs credit
A paper cited because a review quoted it, without reading the original	Not OK: you may repeat an error

3.10.3 Text recycling and AI tools

Two newer issues deserve a mention. **Text recycling**, sometimes called self-plagiarism, means reusing wording from your own published work. Some reuse of standard methods descriptions may be acceptable if cited and allowed by the journal; results and discussion text should never be reused. Check the journal's policy.

Artificial intelligence tools can reword text and suggest references, but they do not remove responsibility. Rewording a source with software is still patchwriting if the ideas and structure are someone else's. AI chat tools can also produce references that look entirely real but do not exist: every reference must be found in PubMed or on the journal's website before it is cited. Journals now require authors to disclose the use of AI tools. Chapter 18 discusses research integrity in more detail.

3.11 From reading to a problem statement

Searching and reading are not ends in themselves. The papers you find become the first page of a research proposal: the **problem statement** or background that explains why the study is needed. Two tools make this step manageable.

3.11.1 The literature matrix

A **literature matrix** summarises each key paper in one row, using the same columns for every paper. The figure below shows three rows for the running example.

Study	Design, size	Compared	Primary outcome: result	Key limitation
Boulet 2024 Crit Care	Pilot RCT, 3 ICUs, n = 48	Restrict hidden fluids vs standard	Fluid balance at day 5: no significant difference	Small, not powered for mortality
Hyun 2023 Ann Intensive Care	Matched cohort, registry, 20 hospitals	Positive vs negative fluid balance	28-day death higher if positive by day 2-3	Observational: confounding possible
Meyhoff 2022 N Engl J Med	Large RCT (CLASSIC), n = 1,554	Restrictive vs standard IV fluid	90-day death 42.3% vs 42.1%: no difference	Clinicians not blinded
Your paper 4	...	...	...	...

Figure 3.10: A literature matrix with one row per paper and the same columns for each: study, design and size, what was compared, primary outcome and result, and key limitation. Rows show Boulet 2024 (pilot RCT, 3 ICUs, n = 48; fluid balance at day 5, no significant difference; small and not powered for mortality), Hyun 2023 (matched registry cohort, 20 hospitals; 28-day death higher if positive by day 2 to 3; observational, confounding possible) and Meyhoff 2022 (CLASSIC, n = 1,554; 90-day death 42.3% vs 42.1%, no difference; clinicians not blinded). An empty row is ready for the next paper.

A matrix has three benefits. It forces you to extract the same information from every paper, which is the beginning of critical reading (Chapter 4). It makes differences between studies visible at a glance: in the example, the three papers measure different outcomes (fluid balance, 28-day mortality, 90-day mortality) at different times, which is easy to miss when the papers are read one by one. And the **key limitation** column is where the gap in knowledge, and therefore the

justification for a new study, comes from.

Write the matrix in your own words. A matrix filled with sentences copied from abstracts teaches nothing and invites patchwriting when the background is written.

3.11.2 The problem statement as a funnel

A problem statement moves from the general to the specific, like a funnel. The figure below shows its structure.

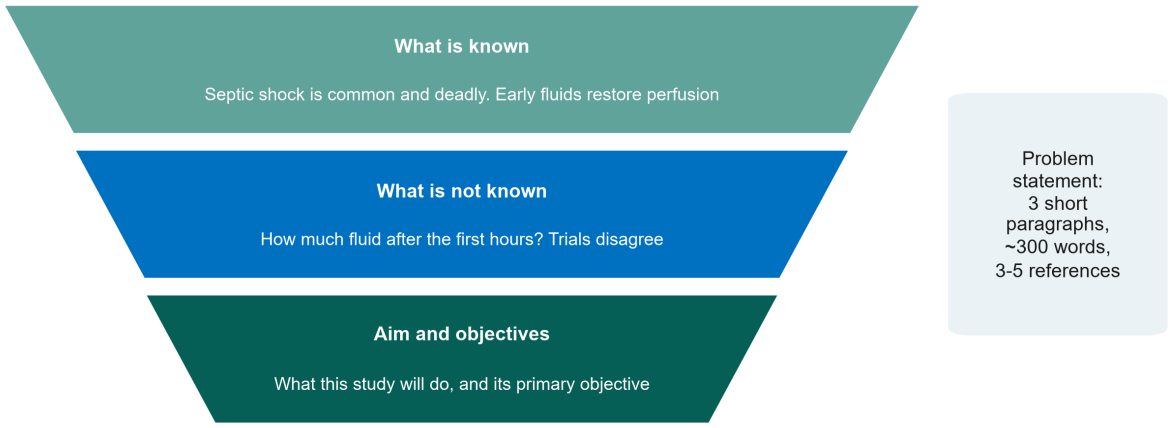


Figure 3.11: The problem statement as a funnel: what is known (septic shock is common and deadly; early fluids restore perfusion), what is not known (how much fluid after the first hours; trials disagree), and the aim and objectives (what this study will do, and its primary objective). About 300 words, three short paragraphs, 3 to 5 references.

- **Paragraph 1: what is known.** The size and importance of the problem, and current practice, with references.
- **Paragraph 2: what is not known.** The gap: conflicting studies, no studies in patients like yours, important outcomes not measured, designs too weak to answer the question. The limitation column of the literature matrix feeds this paragraph.
- **Paragraph 3: aim and objectives.** What the study will do and its primary objective (and any secondary objectives), stated in one flow, as published protocols and papers do: “In this pilot randomised trial, we aim to assess whether ... Our primary objective is to ...”. The aim should follow naturally from the gap. Your own data or results never belong in the Introduction: a local figure may appear only if it has been published, with its reference.

The same funnel shapes the Introduction of a published paper (Chapter 19). Paper 1’s Introduction is a good example: it moves from the frequency and mortality of sepsis, through evidence that a positive fluid balance is associated with harm and that most fluid is not resuscitation fluid, to the observation that earlier trials restricted mainly resuscitation fluid, and ends by stating the trial’s primary objective, all in well under 500 words (Boulet et al. 2024).

The figure below shows a model problem statement for the feasible version of the running question.

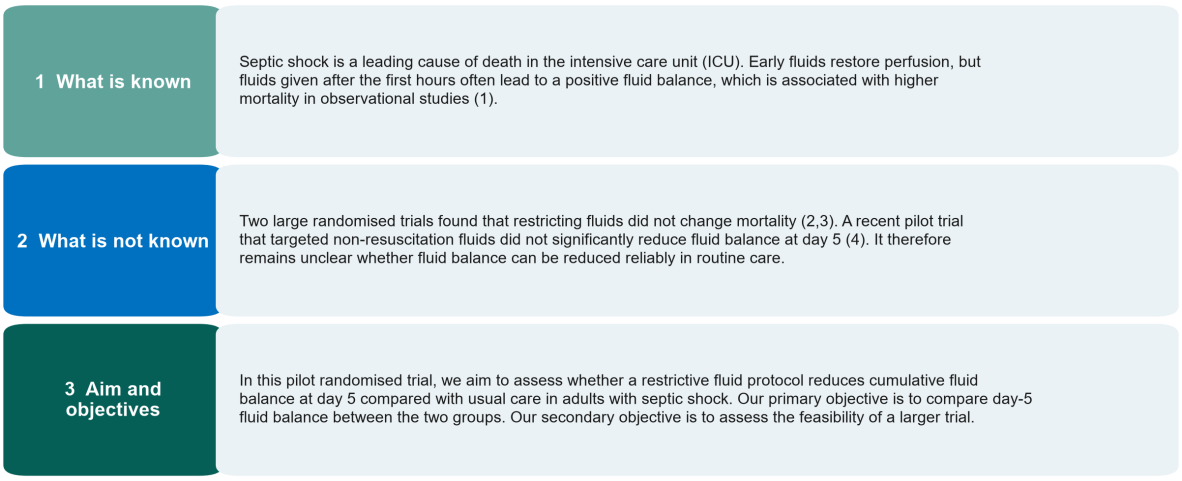


Figure 3.12: A model three-paragraph problem statement. What is known: septic shock is a leading cause of death in the ICU; early fluids restore perfusion, but fluids given after the first hours often lead to a positive fluid balance, which is associated with higher mortality in observational studies (1). What is not known: two large randomised trials found that restricting fluids did not change mortality (2,3), and a recent pilot trial that targeted non-resuscitation fluids did not significantly reduce fluid balance at day 5 (4); it remains unclear whether fluid balance can be reduced reliably in routine care. Aim and objectives: in this pilot randomised trial, we aim to assess whether a restrictive fluid protocol reduces cumulative fluid balance at day 5 compared with usual care in adults with septic shock; the primary objective is to compare day-5 fluid balance between the groups, and the secondary objective is to assess the feasibility of a larger trial. References are numbered in the order first cited.

Notice three features. Every factual sentence carries a citation, numbered in the order first cited. The second paragraph ends with an explicit statement of the gap. And the third paragraph states the aim and objectives, which follow directly from that gap; it contains no data of its own.

! Common mistake: the background must review everything ever written

A problem statement for a proposal is not a literature review. About 300 words in three paragraphs, citing three to five key references, is usually enough. Length does not impress reviewers; a clearly stated gap does.

! Common mistake: a literature dump with no gap

A background that lists study after study (“Smith found… Jones found… Lee found…”) without saying what is still unknown leaves the reader asking why the new study is needed. Every paragraph should move the reader towards the gap, and the gap towards the question.

3.11.3 Working title and keywords

A good working title can be built directly from the PICO question, as the figure below shows.

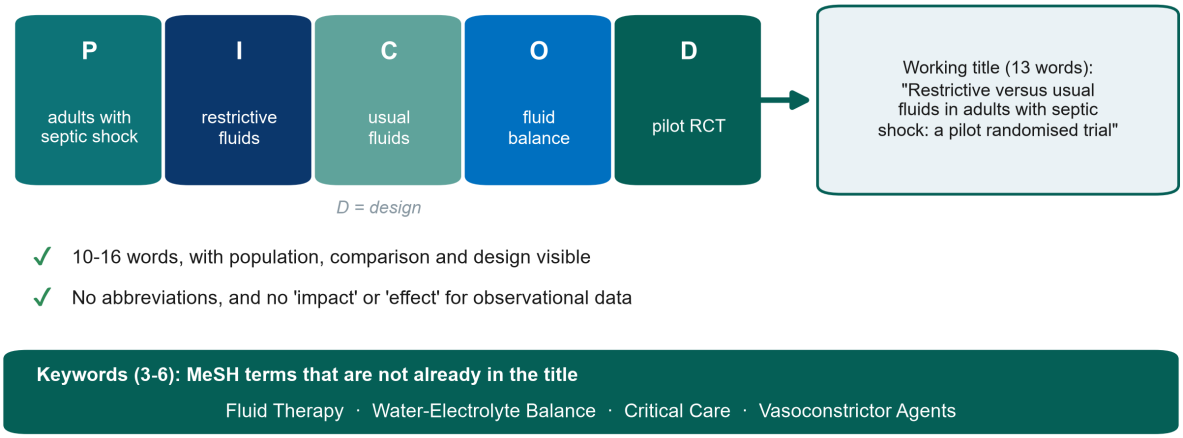


Figure 3.13: Building a working title from PICO plus design (D): adults with septic shock (P), restrictive fluids (I), usual fluids (C), fluid balance (O), pilot RCT (D), giving the 13-word title "Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial". A good title has 10 to 16 words with the population, comparison and design visible, no abbreviations, and no "impact" or "effect" for observational data. Keywords (3 to 6) are MeSH terms not already in the title: Fluid Therapy, Water-Electrolyte Balance, Critical Care, Vasoconstrictor Agents.

A good title:

- names the population, the intervention and comparison (or exposure), and the design;
- is about 10 to 16 words long;
- avoids abbreviations;
- avoids promotional words ("novel", "first") and causal words ("impact", "effect of") when the design is observational.

"Fluids in the ICU" is a poor title: it names no population, no comparison and no design. "Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial" (13 words) tells the reader exactly what the study is. Compare Paper 2's title, which begins "Impact of a cumulative positive fluid balance…" (Hyun et al. 2023): the design is a cohort study, which can show association but not impact, so the title promises more than the design can deliver (Chapter 4 returns to this).

Keywords, usually three to six, help the paper be found. The best keywords are MeSH terms that are *not* already in the title, because words in the title are already searchable; keywords that add new terms widen the range of searches that will find the paper.

Summary

- A documented literature search checks novelty, provides the background, shows how others designed similar studies and often reshapes the question. Record the database, date, string, filters and hits.
- Turn the PICO question into concept blocks (usually P, I and O, not C). Combine synonyms with OR inside a block and blocks with AND. Use NOT rarely.

- MeSH headings collect many author wordings under one label and include narrower terms; always add free-text words searched in the title and abstract, because new records may not yet be indexed. Phrase searching, truncation, field tags and justified filters sharpen the search.
- Use PubMed for the main, repeatable search; Google Scholar and citation searching to check and to follow citations; the Cochrane Library for existing reviews; and trial registries for unpublished and ongoing trials. Publication bias means the published literature can look more positive than the truth; registries reveal planned primary outcomes and outcome switching.
- Save every reference in a reference manager from the start, and let it format citations in the style the journal requires.
- Paraphrase from understanding, change the structure, check the facts and cite. Patchwriting is plagiarism even with a citation. Never cite a paper you have not read, and verify every reference, especially any suggested by AI tools.
- Summarise key papers in a literature matrix; write the problem statement as a funnel (known, not known, why here) ending in the question; build the working title from PICO plus design, and choose keywords that add MeSH terms not in the title.

Further reading

- Greenhalgh, *How to Read a Paper* (Greenhalgh 2019). Its early chapters give a practical, clinician-friendly guide to searching the literature.
- Higgins and colleagues (editors), *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins et al. 2019). The chapter on searching for and selecting studies is the reference standard for comprehensive searching, far beyond what a single proposal needs, but invaluable when in doubt.
- Chan and colleagues, “Empirical evidence for selective reporting of outcomes in randomized trials” (Chan et al. 2004). Shows, by comparing protocols with publications, how often outcomes are changed or omitted, and why registries matter.
- International Committee of Medical Journal Editors, *Recommendations for the Conduct, Reporting, Editing, and Publication of Scholarly Work in Medical Journals* (International Committee of Medical Journal Editors 2024). The source of the trial registration requirement and of many conventions for references and citation.

CHAPTER 4

Reading and appraising a paper

A literature search (Chapter 3) produces a pile of papers. Busy clinicians cannot read every word of every one, and they do not need to. What they need is a routine: a way of reading the right parts in the right order, so that within a few minutes they know what a paper asked, what it found and whether it deserves closer attention, and then a structured way of judging whether its findings can be trusted and used. The first activity is *reading*; the second is *critical appraisal*. This chapter teaches both. It explains the standard structure of a research paper, a fifteen-minute reading routine, what a good abstract contains, how to recognise spin, and a ten-question checklist for appraisal. Throughout, the two papers on fluids in septic shock that run through this book are read and appraised in full, so that every step is shown with real examples.

In this chapter you will learn

- the IMRaD structure of an original research paper and what each section is for;
- a fifteen-minute reading order that tells you quickly whether a paper is relevant and what it found;
- what a structured abstract should contain, and how to judge a results sentence: how big, how sure, by when;
- what spin is, its common forms, and how to check whether the words of a paper match its data and design;
- how to appraise a paper with a ten-question checklist, and how this relates to reporting guidelines and formal risk-of-bias tools;
- how the checklist applies to a randomised trial (Paper 1) and a matched cohort study (Paper 2), and what each design may claim.

4.1 The two papers used in this chapter

Both papers address the clinical question at the heart of this book: should we give less intravenous fluid to adults with septic shock or sepsis after the first hours of resuscitation?

Paper 1 is Boulet and colleagues (2024), published in *Critical Care* (Boulet et al. 2024). It is a pilot randomised controlled trial in three ICUs. Fifty adults with septic shock were randomised either to an "optimised restrictive strategy", which reduced maintenance fluids, drug-dilution volumes and nutrition volumes as far as possible for 7 days while leaving resuscitation fluid to the clinicians, or to standard care. The primary outcome was cumulative fluid balance in millilitres per kilogram of body weight after 5 days.

Paper 2 is Hyun and colleagues (2023), published in *Annals of Intensive Care* (Hyun et al. 2023). It is an analysis of a multicentre sepsis registry from 20 hospitals. From 11,981 registered patients, 2,516 who stayed at least three days in the ICU were analysed in three propensity-score-matched cohorts, comparing patients whose cumulative fluid balance was positive or negative at the end of ICU day 1, day 2 and day 3. The primary outcome was death at 28 days.

The two papers are deliberately different. One is a small experiment; the other is a large observation of routine care. Reading them side by side shows that the design of a study determines what it is entitled to claim.

4.2 The anatomy of a research paper: IMRaD

Almost every original research paper in clinical medicine follows the same structure, known by the initials of its four main sections: **Introduction, Methods, Results and Discussion (IMRaD)**, preceded by a title and abstract. The figure below shows its shape.

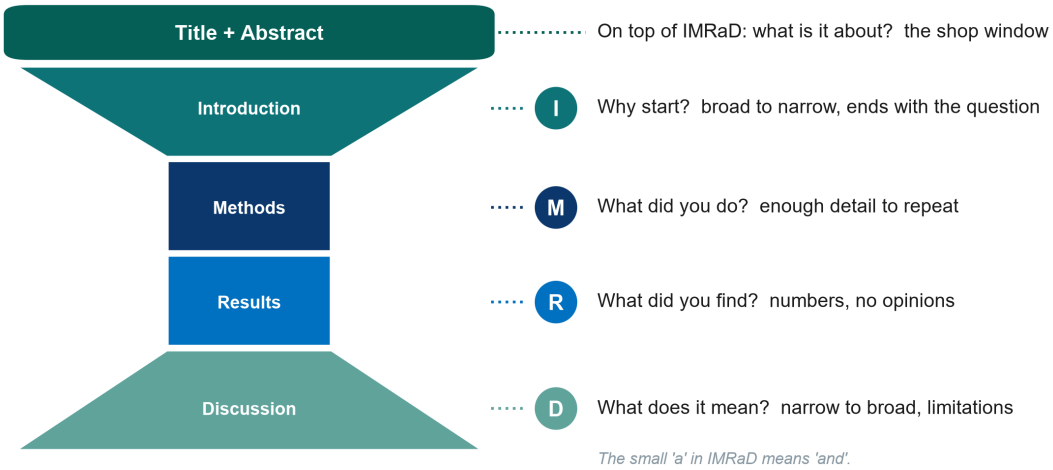


Figure 4.1: The IMRaD hourglass. The title and abstract are the shop window: what is the paper about? The Introduction narrows from the broad problem to the specific question. Methods (what was done, in enough detail to repeat it) and Results (what was found, numbers without opinions) are narrow and factual. The Discussion widens again to what the findings mean, including limitations.

The hourglass image captures the logic of the structure:

- The **Title and Abstract** form the shop window: they tell the reader what the paper is about and what it found, in a few hundred words.

- The **Introduction** starts broad (the clinical problem and its importance), narrows through what is known and not known, and ends with the specific question or objective. The question is usually in the last paragraph of the Introduction. This is the same funnel used to write a problem statement (Chapter 3).
- The **Methods** describe what was done: the design, setting, participants, intervention or exposure, outcomes, sample size and statistical analysis, in enough detail that another team could repeat the study.
- The **Results** report what was found: who was included, how the groups compared at the start, and the outcomes, as numbers, without interpretation.
- The **Discussion** widens again: what the findings mean, how they compare with other studies, the study's strengths and limitations, and the implications for practice and research.

Knowing the shape tells you where to look. The question is at the end of the Introduction. The primary outcome is defined in the Methods. The main number is in the Results and the abstract. Who was included is in the Methods (eligibility criteria) and the Results (flow chart and the first table). The authors' interpretation, and their acknowledged limitations, are in the Discussion.

Around the four sections sit other important parts: the **declarations** (ethics approval, consent, funding, conflicts of interest, author contributions, data availability), the **references**, and **supplementary material**, which in many journals now contains essential details such as full eligibility criteria, the protocol or a flow chart.

! Common mistake: the Discussion tells me what was found

The Discussion is the authors' interpretation of their findings, written by people who naturally hope their study was worthwhile. The Results, and especially the tables and figures, are the evidence. Read the numbers before you read what the authors say about them.

4.3 A fifteen-minute reading order

Papers are not novels, and reading them from front to back is slow and often unnecessary. The figure below shows a reading order that takes about fifteen minutes for a paper you decide to read properly, and less than four minutes for one you decide to reject.

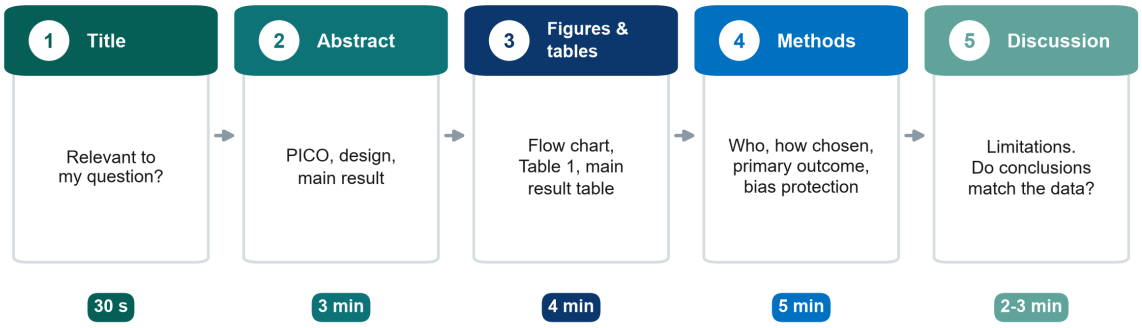


Figure 4.2: A five-step reading order. 1, Title (30 seconds): relevant to my question? 2, Abstract (3 minutes): PICO, design, main result. 3, Figures and tables (4 minutes): flow chart, Table 1, main result table. 4, Methods (5 minutes): who, how chosen, primary outcome, protection against bias. 5, Discussion (2 to 3 minutes): limitations; do the conclusions match the data? Stop after step 1 or 2 if the paper does not answer your question.

Step 1: Title (30 seconds). Is the paper about my question? Does it name a population, an intervention or exposure and a design that are relevant? Many papers can be set aside here.

Step 2: Abstract (3 minutes). Can I write the PICO, the design and the main result? Most papers are rejected at this step, correctly, because they study a different population, intervention or outcome.

Step 3: Figures and tables (4 minutes). Three items carry most of the information. The **flow chart** shows how many people were screened, included, allocated and analysed, and who was lost along the way. **Table 1** describes the participants at the start of the study and shows whether the groups being compared were similar. The **main results table** (or figure) shows the primary outcome in each group and the estimate of the difference.

Step 4: Methods (5 minutes). Who exactly was studied and how were they chosen? What was the primary outcome, and how and when was it measured? How did the design protect against bias: randomisation, concealment of allocation, blinding, matching, statistical adjustment?

Step 5: Discussion (2 to 3 minutes). What limitations do the authors acknowledge? Do their conclusions match their data and their design?

The Introduction, and the Discussion in full, are read only for papers you will cite or appraise in depth. This order also guards against being led by the authors' framing: by the time you reach the Discussion, you already know what the numbers say.

> **! Common mistake: reading only the abstract is enough**
>
> The abstract is the most-read part of any paper and the least complete. It rarely mentions limitations, it cannot describe the population in detail, and it is where spin is most common

(Boutron et al. 2010). The abstract tells you whether to read further; it is not a substitute for doing so before you change practice or cite a finding.

4.4 Reading the abstract

4.4.1 What a structured abstract contains

Most clinical journals require a **structured abstract** with labelled headings. The figure below lists what each part should contain.

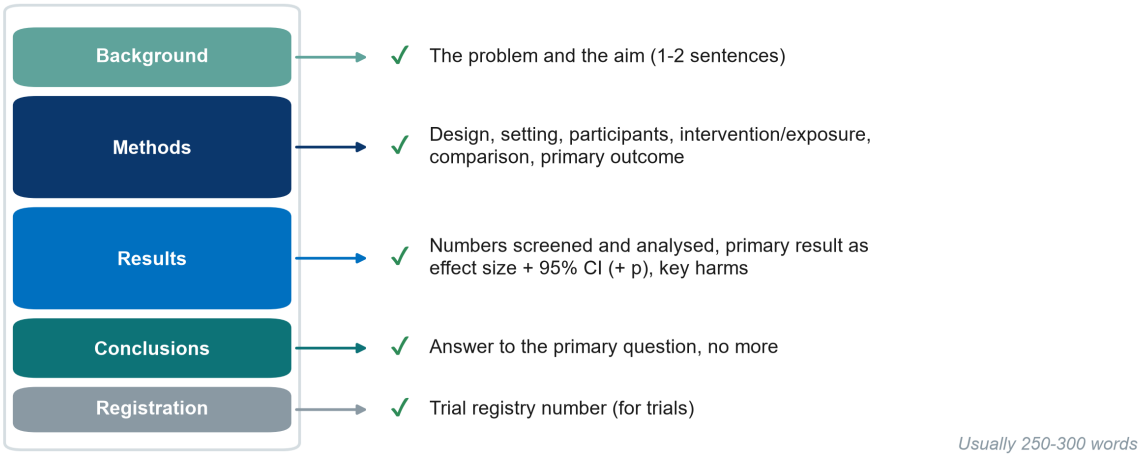


Figure 4.3: The parts of a structured abstract. Background: the problem and the aim in one or two sentences. Methods: design, setting, participants, intervention or exposure, comparison and the primary outcome. Results: numbers screened and analysed; the primary result as an effect size with its 95% confidence interval (and p-value); key harms. Conclusions: the answer to the primary question, no more. Registration: the trial registry number. Usually 250 to 300 words.

A useful test of an abstract is whether you can write the PICO and the main result from it without opening the paper. The Methods should name the design and the primary outcome. The Results should give the numbers included and analysed and the **primary result as an effect size with its 95% confidence interval**, not just a p-value or the word “significant”. The Conclusions should answer the primary question and claim no more. For trials, the abstract should include the registration number. Reporting guidelines specify the items an abstract should contain, for example CONSORT for Abstracts for randomised trials (Hopewell et al. 2008).

4.4.2 How big, how sure, by when?

The figure below contrasts two ways of reporting the same kind of result.

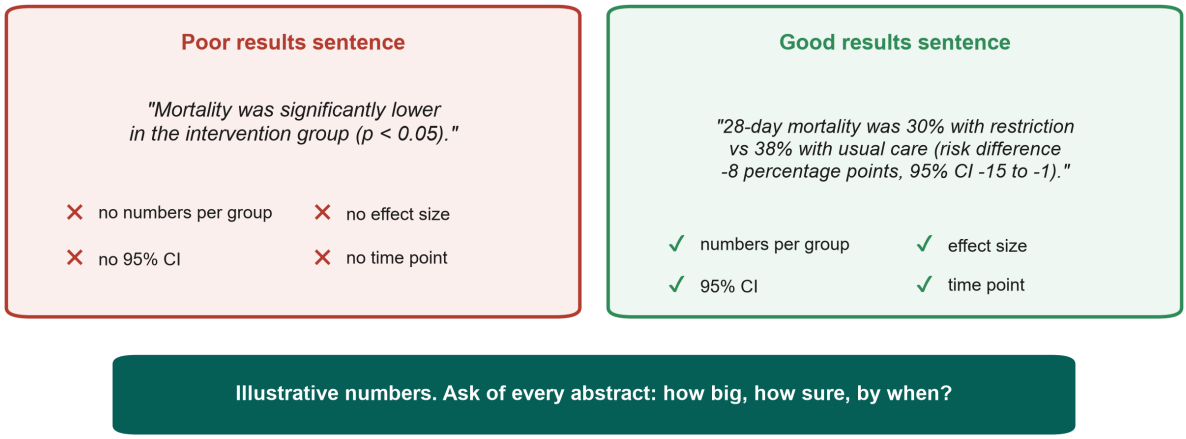


Figure 4.4: A poor results sentence ("Mortality was significantly lower in the intervention group ($p < 0.05$)") lacks numbers per group, an effect size, a 95% confidence interval and a time point. A good results sentence ("28-day mortality was 30% with restriction vs 38% with usual care (risk difference -8 percentage points, 95% CI -15 to -1)") has all four. The numbers are illustrative.

The poor sentence tells us only that a statistical test crossed a threshold. It does not say how many patients died in each group, how large the difference was, how precisely it was estimated, or when it was measured. The good sentence answers three questions that should be asked of every result:

- **How big?** The size of the effect: here a risk difference of 8 percentage points (30% compared with 38%). Whether a difference matters clinically depends on its size, which a p-value does not show.
- **How sure?** The **95% confidence interval (CI)**: here, from 15 points lower to 1 point lower. Informally, the confidence interval is the range of true effects that are reasonably compatible with the data. A narrow interval means the effect is estimated precisely; a wide interval means there is much uncertainty. Chapter 12 explains confidence intervals properly.
- **By when?** The time point: 28 days.

One more piece of interpretation is needed to read the papers in this chapter. For a **difference** between groups (such as a difference in fluid balance or a risk difference), "no effect" corresponds to a value of 0. For a **ratio** (such as a relative risk, odds ratio or hazard ratio), "no effect" corresponds to a value of 1. If the 95% confidence interval includes the no-effect value, the data are compatible with no effect as well as with the effects at either end of the interval. Measures of effect are explained in Chapter 7, and hazard ratios in Chapter 14.

4.4.3 Worked example: reading the abstract of Paper 1

Worked example: the abstract of Paper 1, line by line

Reading the abstract of Paper 1 with the questions above gives the grid below.

Design	Pilot RCT, single-blind (patients and statisticians), 3 ICUs
P	Adults 18-85 y, septic shock, first 24 h of vasopressors
I vs C	Restrict maintenance and drug-dilution fluids (7 days) vs standard
Primary outcome	Cumulative fluid balance at day 5 (ml/kg)
Numbers	1,201 screened, 50 randomised, 48 analysed (24 per group)
Main result	**Median 6.9 vs 35.0 ml/kg. Difference 13.2 (95% CI -15.2 to 41.6), p = 0.42**
Conclusion	Did not reduce fluid balance at day 5 (matches the result)
Registration	ClinicalTrials.gov NCT04947904

Figure 4.5: A reading grid for the abstract of Paper 1. Design: pilot RCT, single-blind (patients and statisticians), 3 ICUs. P: adults 18 to 85 years with septic shock, in the first 24 hours of vasopressors. I vs C: restricting maintenance and drug-dilution fluids for 7 days vs standard care. Primary outcome: cumulative fluid balance at day 5 (ml/kg). Numbers: 1,201 screened, 50 randomised, 48 analysed (24 per group). Main result: median 6.9 vs 35.0 ml/kg; difference 13.2 (95% CI -15.2 to 41.6), p = 0.42. Conclusion: did not reduce fluid balance at day 5, which matches the result. Registration: ClinicalTrials.gov NCT04947904.

Five points stand out.

1. **It is a pilot trial with 48 analysed patients.** Pilot trials are small by design; they test whether a strategy can be delivered and a larger trial run.
2. **The primary outcome is cumulative fluid balance at day 5, not mortality.** This is the commonest error readers make with this paper. Mortality was a secondary outcome.
3. **The confidence interval includes zero.** The estimated difference was 13.2 ml/kg, but the 95% CI runs from −15.2 to 41.6 ml/kg, so the data are compatible with the restrictive strategy lowering the balance by a clinically useful amount, with no difference, and even with a slightly higher balance. There is no evidence of a difference, and much uncertainty.
4. **The conclusion matches the result.** The abstract concludes that the strategy did not reduce overall fluid balance at day 5. That is a plain statement of a negative result, without spin: good practice.
5. **The trial was registered** before recruitment began.

A careful reader also notices something odd: the medians (6.9 and 35.0 ml/kg) differ by 28.1 ml/kg, yet the reported difference is 13.2 ml/kg. The difference was estimated by a different method from the medians, and the paper does not explain how. This is not a reason to dismiss the trial, but it is a question a reviewer would ask (it returns under question 7 of the checklist below).

Finally, the link to our running question: this trial does **not** answer "does restrictive fluid reduce 28-day mortality?". It answers a feasibility-type question about fluid balance. That is exactly the feasible rewrite of the running question described in Chapter 2.

The same reading applied to Paper 2's abstract gives: design, a propensity-score-matched cohort

from a prospective multicentre registry; P, adults with sepsis in the ICU; E versus C, a positive versus a negative cumulative fluid balance at the end of ICU day 1, 2 or 3; primary outcome, 28-day mortality; numbers, 11,981 registered, 2,516 included in matching, 483, 373 and 392 matched pairs for days 1, 2 and 3; main result, hazard ratios for death of 1.17 (95% CI 0.85 to 1.60) for day 1, 2.13 (1.48 to 3.06) for day 2 and 1.56 (1.10 to 2.22) for day 3 (Hyun et al. 2023). The abstract's conclusion uses the word "associated", which matches an observational design. But the title begins with the word "Impact", which does not. That contrast introduces the next topic.

4.5 Spin: when the words go beyond the data

Spin is reporting that makes results look more favourable, or more certain, than the data justify. It is not usually fabrication; the numbers are correct, but the words around them lead the reader to a stronger conclusion than the numbers support. Spin is common. In a study of randomised trials whose primary outcome was not statistically significant, Boutron and colleagues found spin in the abstracts or main texts of a large proportion of reports (Boutron et al. 2010). Peer review reduces spin but does not remove it.

The figure below shows four common forms, two of them taken from the papers in this chapter.

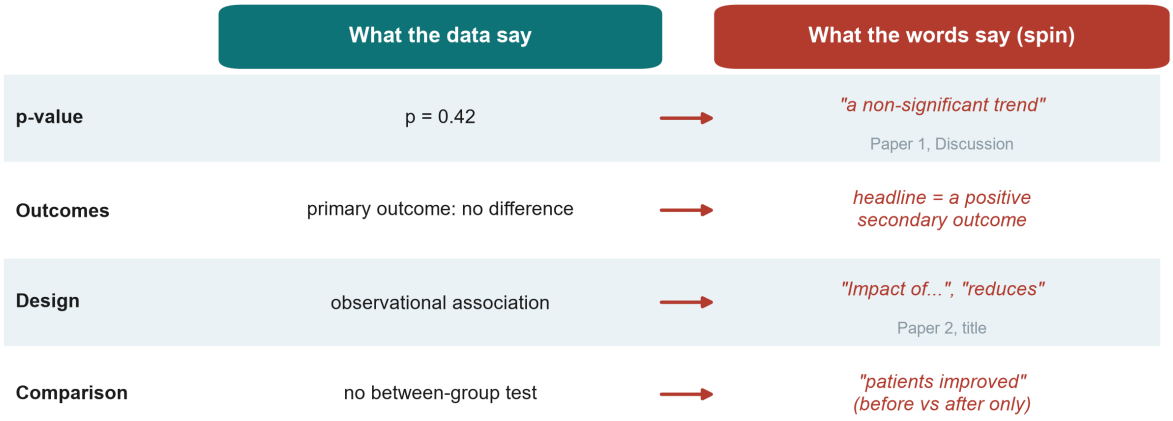


Figure 4.6: Four common forms of spin. P-value: the data show p = 0.42, but the words say "a non-significant trend" (Paper 1, Discussion). Outcomes: the primary outcome shows no difference, but the headline is a positive secondary outcome. Design: the data show an observational association, but the words say "Impact of…" or "reduces" (Paper 2, title). Comparison: there is no between-group test, but the words say "patients improved" (before versus after only).

Turning a null result into a "trend". A p-value of 0.42 means that a difference at least as large as the one observed would arise by chance more than four times in ten even if there were no true difference. It is no evidence of a difference. Calling it a "non-significant trend" invites the reader to see an effect that the data do not show. The honest phrase is "no significant difference; the confidence interval was wide". Paper 1's Discussion uses "non-significant trend" for p = 0.42 when arguing for a larger trial, although its abstract conclusion, to its credit, has no spin (Boulet et al. 2024).

Switching the spotlight to a secondary outcome. When the primary outcome shows nothing, the abstract leads with a positive secondary outcome, or with a subgroup in which the treatment appeared to work. Secondary outcomes and subgroups are numerous, and some will be "significant" by chance.

Causal words for observational data. Words such as “impact”, “effect”, “reduces”, “prevents” and “leads to” imply that one thing causes another. Only a design that controls who receives the exposure, ideally by randomisation, can support such claims. An observational study can show that an exposure is *associated* with an outcome. Paper 2’s title begins “Impact of a cumulative positive fluid balance…”, although its abstract conclusion correctly says “associated” (Hyun et al. 2023).

Before-after claims without a comparison. “Mortality fell from 30% to 22% after the new protocol” ignores everything else that changed over the same period: other treatments, case mix, staffing, seasonal variation. Without a concurrent comparison group, the fall cannot be attributed to the protocol.

4.5.1 Matching the verb to the design

The strength of a verb is a claim. The figure below arranges common verbs as a ladder, from the weakest claim to the strongest, with the design needed to climb each rung.

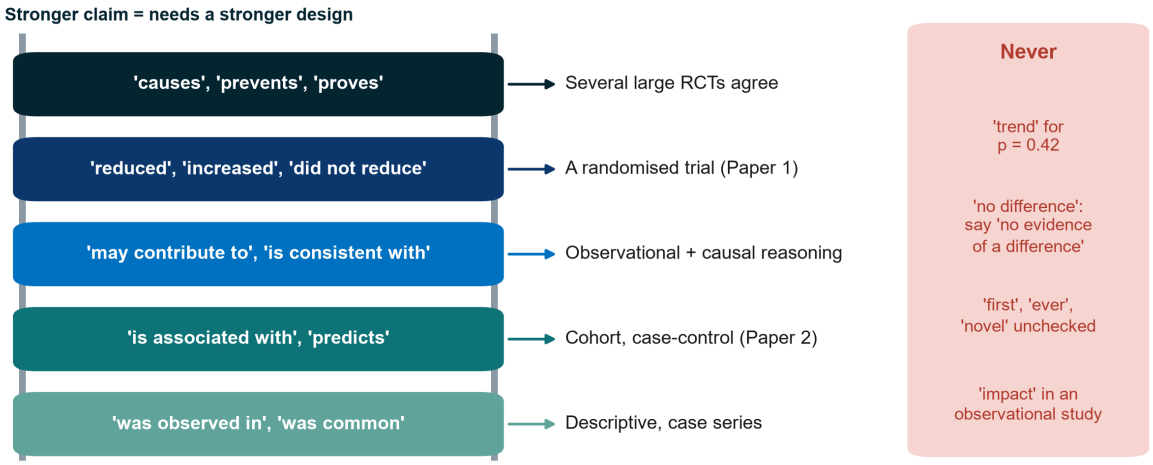


Figure 4.7: A ladder of verbs: a stronger claim needs a stronger design. “Was observed in”, “was common”: descriptive studies and case series. “Is associated with”, “predicts”: cohort and case-control studies (Paper 2). “May contribute to”, “is consistent with”: observational evidence plus causal reasoning. “Reduced”, “increased”, “did not reduce”: a randomised trial (Paper 1). “Causes”, “prevents”, “proves”: several large trials that agree. Never: “trend” for p = 0.42; “no difference” when the study was too small (say “no evidence of a difference”); “first”, “ever” or “novel” without checking; “impact” in an observational study.

Reading with this ladder in mind is one of the most useful habits a clinician can develop. When you read a conclusion, identify the verb, identify the design, and check whether the design is high enough on the ladder to support the verb.

Worked example: spot the spin and rewrite it

Table 4.1: Four examples of spin with calibrated rewrites.

Sentence	The spin	A calibrated rewrite
A cohort study: "Early feeding reduces pneumonia in stroke patients."	A causal verb for an observational design	"Early feeding was associated with a lower incidence of pneumonia."
Primary outcome p = 0.31: "a promising trend towards benefit."	A null result described as a trend	"There was no significant difference; the confidence interval was wide."
Primary outcome showed no difference; the abstract headline is "improved patient satisfaction".	A secondary outcome promoted to the headline	Lead with the primary result, then report secondary outcomes as secondary.
"Mortality fell from 30% to 22% after the new protocol" (no control group).	Before-after change attributed to the intervention	"Mortality was lower after implementation; without a control group we cannot attribute this to the protocol."

Numbers first, then words

Read the numbers (the effect size, its confidence interval and the time point) before reading what the authors say about them. Then check whether the words match the numbers and the design. Absence of evidence is not evidence of absence (Altman and Bland 1995): a non-significant result from a small study means "we don't know yet", not "there is no effect", and certainly not "a trend".

4.6 From reading to appraising

Reading tells you what a paper says. **Critical appraisal** asks whether you should believe it and whether it applies to your patients. Three separate questions are involved:

- **Internal validity.** Within this study, is the result likely to be true, or could it be explained by bias, confounding or chance? Bias and confounding are explained fully in Chapter 6.
- **The size and precision of the result.** How big is the effect, and how certain is the estimate?
- **External validity, or applicability.** Would the result hold for my patients, in my setting?

Several kinds of tool exist, and it helps to know how they relate.

Reporting guidelines are checklists of what a paper should *report*, so that readers can judge it. Each design has its own: CONSORT for randomised trials (Schulz et al. 2010), STROBE for observational studies (Elm et al. 2007), COREQ for qualitative interviews (Tong et al. 2007), and others described in Chapter 19. A paper can report everything well and still be a weak study; a reporting guideline checks transparency, not quality. But a paper that fails to report key items cannot be appraised properly, and missing items are themselves warning signs.

Formal risk-of-bias tools are structured instruments used in systematic reviews to judge the internal validity of each study in a standard way: for example RoB 2 for randomised trials (Sterne

et al. 2019) and ROBINS-I for non-randomised studies of interventions (Sterne et al. 2016). They are explained in Chapter 9.

Appraisal checklists for everyday reading are shorter and broader, combining questions about validity, results and applicability. The ten-question checklist below is a simplified tool of this kind, suitable for any clinician reading a trial or cohort study. It is not a substitute for a formal risk-of-bias assessment, but it covers the issues that matter most and can be completed in about half an hour.

4.7 The ten-question appraisal checklist

Table 4.2: The ten-question appraisal checklist.

#	Topic	Question
1	Question	Can you write the study's PICO in one line?
2	Design	What is the design, and is it the right one for the question?
3	Participants	Who was included, and who was left out? Would your patients have been included?
4	Comparison groups	Were the groups similar at the start (Table 1)? How was this achieved: randomisation, matching or adjustment?
5	Bias	Who knew which group patients were in (blinding)? Could that change the outcome?
6	Outcome	Was there one pre-specified primary outcome, measured the same way in both groups?
7	Numbers	What is the main result as an effect size with a 95% CI? Is the CI narrow or wide?
8	Words versus design	Do the conclusions match the design ("associated" versus "caused"; no "trend" for p = 0.42)?
9	Limitations	Are the main limitations stated honestly? Name one the authors missed.
10	Use	Would this change practice in your unit? Why, or why not?

Answer each question **Yes, No or Can't tell**, and write down **where in the paper** you found the answer (page, table or figure). "Can't tell" is a perfectly good answer when the paper does not report something; indeed, a long list of "Can't tell" answers is itself informative. The page reference matters because it forces you to find the evidence rather than rely on an impression, and it lets you, or a colleague, check your judgement later.

The questions follow the reading order. Questions 1 to 3 concern the question, design and participants (abstract, Introduction, Methods, flow chart). Questions 4 to 6 concern how the groups were made comparable, blinding and the outcome (Methods, Table 1). Question 7 concerns the result (Results, main table). Questions 8 to 10 concern interpretation and use (abstract conclusion, Discussion, your own judgement).

The rest of this chapter applies the checklist to both papers. Page numbers refer to the journal page numbers printed on each paper.

4.8 Appraising Paper 1: a pilot randomised trial

Q1. Question: can you write the PICO in one line? Yes (abstract, page 1; end of the Background, page 3). P: adults aged 18 to 85 with septic shock, within 24 hours of starting vasopressors, in three ICUs. I: an optimised restrictive strategy reducing maintenance fluids, drug-dilution volumes and artificial nutrition volumes as far as possible for 7 days, with resuscitation fluid left to the clinicians. C: standard fluid strategy. O: cumulative fluid balance (ml/kg) at day 5, with mortality, organ-support-free days, length of stay and adverse events as secondary outcomes. The primary outcome is a process or surrogate outcome, not a patient-important one (Chapter 2): the trial asks "can we change fluid balance?" before "does it save lives?".

Q2. Design: is it the right one? Yes, with a caveat (title, page 1; trial design, page 3). A multicentre, single-blind, parallel-group, randomised controlled pilot trial is the right design to test whether a strategy changes something. The caveat concerns the sample size (page 4). Pilot trials usually aim to test feasibility, but this one calculated its size to detect a difference in efficacy: assuming a balance of 70 ml/kg in the control group and 40 ml/kg with restriction, with a standard deviation of 35 ml/kg, 44 patients were needed, increased to 50 to allow for losses. In fact the control group's balance was about 35 ml/kg, half the assumed value, so the trial faced a much smaller potential difference than it was designed to detect.

Q3. Participants: who was included, who was left out? Partly; Can't tell for many of our patients (patients, page 3; flow chart, Figure 1, page 5; limitations, page 10). Eligible patients had suspected or confirmed infection, a lactate of at least 2 mmol/L and a vasopressor infusion, and were within 24 hours of starting vasopressors; patients needing kidney replacement therapy within 24 hours, with end-stage kidney disease, or with severe malnutrition were excluded. Of 1,201 patients screened, 50 were randomised, about 4%. The authors acknowledge that the strict criteria excluded some of the most severe patients and that baseline lactate was low. A 4% inclusion rate means the trial population was highly selected; check the flow chart before applying the results to your sickest patients.

Q4. Comparison groups: similar at the start? Mostly yes, by randomisation (randomisation, page 3; Table 1, page 6). Allocation used a central, computer-generated sequence in a 1:1 ratio with permuted blocks of 2. Table 1 shows broadly similar groups for 24 patients each (median age 69.5 versus 67.0 years; similar severity scores and sources of infection). But baseline creatinine was considerably higher in the control group (median 170.5 versus 94.5 µmol/L). With only 24 patients per group, chance imbalances like this are expected even with perfect randomisation. Two reporting errors are worth noticing: the text on page 5 states that creatinine was higher in the intervention group, contradicting Table 1 and the Discussion, and Table 1 gives sodium and potassium in micromoles rather than millimoles per litre. On the positive side, Table 1 correctly contains no p-values: testing for baseline differences in a randomised trial is pointless, because any differences are by definition due to chance.

Q5. Bias: who knew which group patients were in? Yes, and it could change the outcome (page 3; limitations, page 8). Allocation was concealed from patients and statisticians but not from clinicians or investigators. This matters particularly here because the unblinded clinicians prescribe the fluids that *make up* the primary outcome. Two consequences follow. First, clinicians in the control group, aware of the trial and perhaps convinced of the value of restriction, may also have restricted fluids. The authors themselves suggest such **contamination**, which pushes the result towards no difference. Second, with **permuted blocks of 2** and unblinded clinicians, once the first patient in each block has been allocated, the next allocation can be predicted. That threatens allocation concealment (Chapter 6), and the authors do not discuss it.

Q6. Outcome: one pre-specified primary outcome, measured the same way? Yes (outcomes, page 4; registration, page 1). There is one primary outcome, clearly separated from secondary and safety outcomes, and the trial was registered before the first patient was enrolled. Fluid doses were calculated using admission weight in both groups. Could be better: about forty secondary comparisons are reported without any comment on the risk of chance findings from multiple testing.

Q7. Numbers: effect size with a 95% CI? Yes; the CI is wide (primary outcome, page 6; Table 3). Median cumulative balance at day 5 was 6.9 versus 35.0 ml/kg; the reported difference was 13.2 ml/kg (95% CI −15.2 to 41.6), p = 0.42. The interval includes 0 (no difference) and also includes 30 ml/kg (the difference the trial was designed to detect), so the result is **inconclusive**. For mortality, 6 of 24 patients died by day 28 in each group: a relative risk of 1.0 with a 95% CI from 0.38 to 2.66, compatible with restriction reducing deaths by more than half or more than doubling them. The trial tells us nothing about mortality. Two things a careful reader notices: the summary statistics are medians but the estimate is described as an absolute or mean difference, and the test used (Mann-Whitney) does not correspond to the estimate, so estimate, summary and test do not match. The one clear effect was on fluids used to keep veins open, reduced by about 20 ml/kg at day 5 (95% CI 6.4 to 33.6).

Q8. Words versus design: do the conclusions match? Partly; the abstract yes, the Discussion no. The abstract and the formal Conclusions state plainly that the strategy did not reduce fluid balance at day 5 (pages 1 and 11). As an RCT, the trial is entitled to a causal claim, and its honest causal claim is negative. But the Discussion (page 10) describes p = 0.42 as a "non-significant trend" in fluid balance reduction; suggests that the strategy would be suitable for all septic shock patients "regardless of severity", a generalisation from 24 patients after the sickest were excluded; and claims that the protocol achieved the lowest day-5 fluid balance "ever published" in septic patients, an unchecked superlative for a strategy that did not significantly change the balance.

Q9. Limitations: stated honestly? Yes; name one missed (pages 8 to 10). The authors' limitations section is honest and specific: contamination of the control group, unblinded clinicians, worse renal function in the control group, exclusion of the most severe patients, and the lack of effect of the nutrition component, each with a mechanism and often with a suggested remedy, such as a stepped-wedge cluster design. Limitations they missed include: permuted blocks of 2 with unblinded clinicians making allocation predictable; two randomised patients excluded from an analysis described as intention-to-treat, which is strictly a *modified* intention-to-treat analysis; sample-size assumptions that were not met; and the many secondary outcomes without comment on multiplicity.

Q10. Use: would this change practice? Not by itself. A 48-patient pilot with an inconclusive surrogate primary outcome cannot change practice or tell us about mortality. It does show that "keep-vein-open" fluids can be reduced substantially without obvious harm in this sample. That is a reasonable trigger for a local audit of hidden fluids (Chapter 1), and a basis for planning a larger trial.

Overall verdict. A well-designed pilot with an honest answer to its own question. Its strengths are structural: a textbook Introduction funnel, one pre-specified and registered primary outcome, a flow chart, specific limitations paired with remedies, complete declarations, and an abstract conclusion that matches the result. Its weaknesses are mostly in presentation (mismatched estimates, a text-table contradiction, a unit error, a "non-significant trend") and could have been fixed without new data. The paper does not name the reporting guideline it followed; CONSORT is the appropriate one (Schulz et al. 2010).

4.9 Appraising Paper 2: a propensity-matched cohort

Q1. Question: can you write the PICO in one line? Yes (abstract, page 1; end of the Background, page 2). P: adults with sepsis (Sepsis-3 criteria (Singer et al. 2016)) admitted to the ICU in 20 hospitals and staying at least three days. E: a positive cumulative fluid balance from ICU admission to midnight of day 1, 2 or 3. C: a negative cumulative balance over the same period. O: 28-day mortality; secondary outcomes were ICU discharge and freedom from mechanical ventilation. In a cohort the "I" is an **exposure** that occurred during routine care; nobody assigned it.

Q2. Design: is it the right one? Yes for an association; No for a causal "impact" (title, page 1; Methods, pages 2 to 3). This is an analysis of a prospective multicentre registry, with three propensity-score-matched cohorts. Fluid balance cannot easily be randomised as such, because it is partly a consequence of how ill a patient is, so a cohort is a reasonable way to study whether it is associated with death. It cannot show that changing a patient's balance would change their risk of death.

Q3. Participants: who was included, who was left out? Partly (data source and patient selection, page 2; Results, page 3). The registry included adults with sepsis from emergency departments and wards. But only patients admitted to the ICU who stayed at least three days and had fluid data were analysed: 2,516 of 11,981, about 21%. The flow chart is only in the supplement. Patients who died or left the ICU within three days, that is, the sickest and the mildest, are missing by design. The matched patients were also less severely ill than the whole cohort, receiving less resuscitation, fewer steroids, less ventilation and fewer vasopressors (page 4). The authors note that all patients came from one country and one ethnic group.

Q4. Comparison groups: similar at the start? Partly, by propensity-score matching (Methods, pages 2 to 3; Table 1, pages 5 to 6). A **propensity score** is each patient's estimated probability of having the exposure (here, a positive balance), calculated from measured characteristics (Rosenbaum and Rubin 1983). Patients with and without the exposure but with similar scores are then matched, so that the matched groups resemble each other on those characteristics. Here the score included age, sex, severity (SOFA score), septic shock, source of referral, comorbidities, site of infection, additional treatments, pathogens and fluid output; matching was 1:1, and balance was judged by a **standardised mean difference** (SMD) below 0.1. The text says almost all patients were closely matched, but Table 1 shows several SMDs above the authors' own threshold, including lactate (0.132 in the day-2 cohort and 0.182 in the day-3 cohort), which is a key marker of severity and was higher in the positive-balance groups. The fundamental limitation is that matching can only balance what was **measured**. Randomisation balances measured and unmeasured factors on average; matching cannot.

Q5. Bias: who knew which group patients were in? Not applicable as "blinding"; the main bias is confounding. In an observational study everyone knows what fluid was given. Death is an objective outcome, so bias in measuring it is small. The real problem is **confounding by indication**: clinicians give more fluid to patients who look sicker, and sicker patients are more likely to die. A positive fluid balance may therefore be partly a *marker* of severity rather than a cause of death (Chapter 6). The authors do not name this problem. A further, subtler issue is that fluid was recorded per calendar day, so "day 1" lasted a few hours for some patients and nearly 24 hours for others.

Q6. Outcome: one pre-specified primary outcome, measured the same way? Yes for one outcome measured the same way; Can't tell for pre-specified (page 3). The primary outcome, 28-day mortality, was analysed with Cox proportional hazards models accounting for clustering,

and secondary outcomes with competing-risk models, which is good practice. But no registered protocol or pre-specified analysis plan is mentioned, and the three cohorts (days 1, 2 and 3) amount to three primary comparisons.

Q7. Numbers: effect size with a 95% CI? Yes; the CIs are moderately narrow (abstract, page 1; outcomes, page 4). The effect measure is the **hazard ratio (HR)**, which compares the rate of death over time between the groups (Chapter 14). Day 1: HR 1.17 (95% CI 0.85 to 1.60), $p = 0.354$, no evidence of an association. Day 2: HR 2.13 (1.48 to 3.06), $p < 0.001$. Day 3: HR 1.56 (1.10 to 2.22), $p = 0.012$. For days 2 and 3 the intervals exclude 1, so the data are not compatible with no association; a hazard ratio of 2.13 means that, at any time during follow-up, the rate of death in the positive-balance group was about twice that in the negative-balance group. The day-2 and day-3 intervals overlap widely, and no test compares them, which matters for question 8.

Q8. Words versus design: do the conclusions match? No. The abstract conclusion uses "associated", correctly (page 1). But the title and the stated aim use "impact", causal wording for an observational design. The Discussion (page 4) says that a positive balance on the first day "did not impact" mortality and that "our results show" an effect of cumulative balance. Later (page 7) it describes the matched design as allowing "the most reliable comparison" and the study as having "a low selection bias", although 79% of the registry was excluded and matching cannot handle unmeasured confounders. And the Conclusions describe day 2 as associated with "the highest risk", ranking hazard ratios from three different matched cohorts, made up of different patients, without any formal comparison.

Q9. Limitations: stated honestly? Partly; name one missed (pages 7 to 8). The authors acknowledge possible unmeasured confounders, daily rather than hourly data, missing data on diuretics and a single ethnic group. Missed limitations include: **survivor bias** from requiring at least three ICU days, which excludes early deaths and early discharges; confounding by indication; residual imbalance in lactate after matching; and matched patients who were less sick than the full cohort, limiting generalisability. There is also no sample-size statement.

Q10. Use: would this change practice? Not on its own. The study supports what many guidelines already suggest, to review fluid balance daily after the resuscitation phase and avoid unnecessary accumulation (Evans et al. 2021), but it cannot show that actively achieving a negative balance would save lives. The authors themselves call for randomised trials (page 7).

Overall verdict. A large, carefully analysed registry study with genuine strengths: a Table 1 that reports SMDs and footnotes missing data, a clear timeline figure, competing-risk analysis for secondary outcomes, and transparent statements on the funder's role and the waiver of consent. Its weaknesses are in its claims: causal language, unacknowledged survivor bias and confounding by indication, overstated reliability, and ranking risks across cohorts. The paper does not name the reporting guideline it followed; STROBE is the appropriate one (Elm et al. 2007).

	Paper 1: Boulet 2024 (RCT)	Paper 2: Hyun 2023 (cohort)

4.10 The two papers side by side

Table 4.3: Papers 1 and 2 compared using the ten-question checklist.

	Paper 1: Boulet 2024 (RCT)	Paper 2: Hyun 2023 (cohort)
Design	Multicentre (3 ICUs), single-blind, parallel, 1:1 randomised pilot trial	Prospective multicentre registry (20 hospitals), three 1:1 propensity-matched cohorts
Size	1,201 screened, 50 randomised, 48 analysed	11,981 in registry, 2,516 analysed; 483, 373 and 392 matched pairs
Question	Does restricting non-resuscitation fluids reduce day-5 fluid balance?	Is a positive balance on ICU days 1 to 3 associated with 28-day mortality?
Groups made comparable by	Randomisation (central, computer-generated, blocks of 2)	Propensity-score matching on measured variables
Who knew the group	Clinicians and investigators; patients and statisticians blinded	Everyone (observational)
Primary outcome	Cumulative fluid balance (ml/kg) at day 5	28-day mortality
Main result	13.2 ml/kg (95% CI −15.2 to 41.6), p = 0.42	Day 1 HR 1.17 (0.85–1.60); day 2 HR 2.13 (1.48–3.06); day 3 HR 1.56 (1.10–2.22)
Biggest threats to validity	Small sample; unblinded clinicians control the outcome; contamination; blocks of 2	Confounding by indication; survivor bias; residual imbalance in lactate
Wording problems	“Non-significant trend” for p = 0.42; “ever published”; “regardless of severity”	“Impact” in title and aim; “did not impact”, “show”; “most reliable”; “highest risk”
Strongest features	Introduction funnel; one registered primary outcome; specific limitations with remedies	Table 1 with SMDs; competing-risk analysis; funder-role and consent statements
Reporting guideline (not named in the paper)	CONSORT	STROBE
May claim	The strategy did not reduce day-5 balance in this small pilot (uncertain)	A positive balance on days 2 and 3 is associated with higher 28-day mortality

The figure below summarises the most important row of the table: what each design is entitled to say.

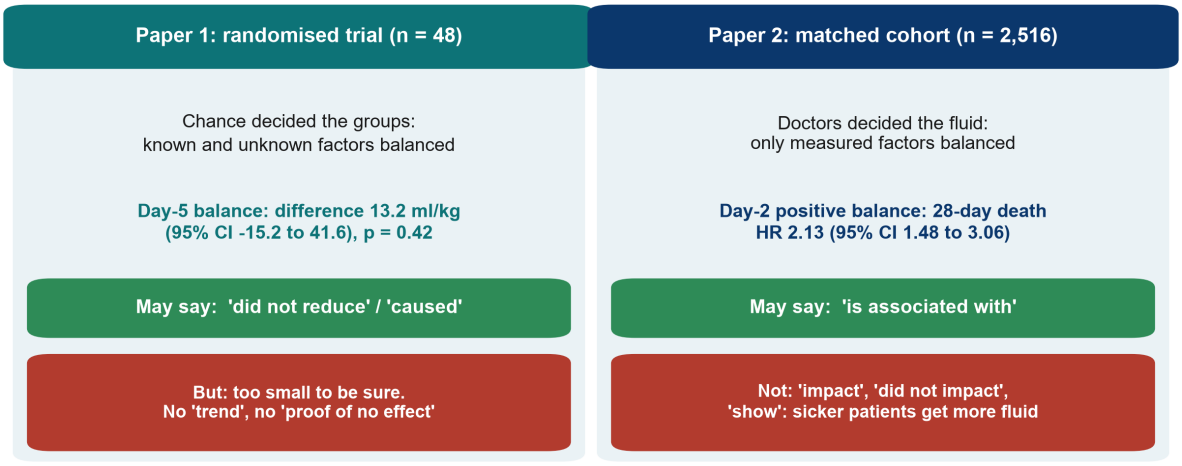


Figure 4.8: What each design may claim. Paper 1, a randomised trial (n = 48): chance decided the groups, so known and unknown factors are balanced; day-5 balance difference 13.2 ml/kg (95% CI -15.2 to 41.6), p = 0.42; it may say "did not reduce" or "caused", but it is too small to be sure, so no "trend" and no "proof of no effect". Paper 2, a matched cohort (n = 2,516): doctors decided the fluid, so only measured factors are balanced; day-2 positive balance and 28-day death HR 2.13 (95% CI 1.48 to 3.06); it may say "is associated with", but not "impact", "did not impact" or "show", because sicker patients get more fluid.

Which paper is stronger evidence? It depends on the question. For **causation**, the randomised trial is the stronger design, but this one is a small pilot that cannot answer the mortality question. For the **real-world association** between fluid balance and death in a large population, the cohort tells us more, but the association needs a trial to show whether it is causal. The large trials of restrictive strategies, CLASSIC and CLOVERS, which found no difference in mortality (Meyhoff et al. 2022; National Heart, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Clinical Trials Network et al. 2023), suggest that at least part of the association in Paper 2 reflects confounding by severity rather than harm from the fluid itself. Neither paper settles the question alone, which is why evidence is synthesised across studies (Chapter 9).

Four messages from appraising the two papers

1. **The design sets the verb.** Randomisation earns "caused" or "did not reduce"; observation earns "is associated with".
2. **A wide confidence interval means "we don't know yet".** P = 0.42 is not a trend and not proof of no effect; a 48-patient pilot cannot answer a mortality question.
3. **Randomisation balances what we cannot measure; matching balances only what we measured.** In studies of treatment intensity, sicker patients receive more treatment, so confounding by indication always threatens observational comparisons.
4. **Published is not the same as correct.** Both papers were peer-reviewed and published in well-regarded journals, and both contain fixable errors. Neither names its reporting guideline. Appraisal is a clinical skill, not an act of disrespect.

4.11 Common mistakes in appraisal

! Common mistake: it is a randomised trial, so there is no bias

Randomisation protects against confounding at the start of a trial. It does not protect against bias that arises afterwards: unblinded clinicians who change their behaviour, predictable allocation, losses to follow-up, or outcomes measured differently in the two groups. In Paper 1, the clinicians who knew the allocation also prescribed the fluids that made up the outcome.

! Common mistake: propensity matching makes a cohort as good as a trial

Propensity scores balance the variables that went into them, and only those. Severity of illness that was never measured, or was measured crudely, remains unbalanced. Always ask which variables were in the score, whether balance was achieved in Table 1, and what important factor might be missing.

! Common mistake: no significant difference means the treatments are equal

"The restrictive strategy did not significantly reduce fluid balance, so it does not work" ignores the width of the confidence interval. Paper 1's interval includes both no difference and the difference the investigators hoped to find. The correct reading is that the trial could not tell (Altman and Bland 1995).

! Common mistake: so both papers are rubbish

Appraisal is balanced. Both papers have real strengths worth copying: Paper 1's Introduction, registered primary outcome and honest limitations; Paper 2's transparent Table 1 and competing-risk analysis. The aim of appraisal is to know how much weight a study can bear, not to dismiss it.

! Common mistake: I can't understand the statistics, so I can't appraise it

Most of the checklist is about judgement, not mathematics. If a statistical method is unfamiliar, answer "Can't tell" and move on; questions 8 and 9 (do the words match the design, and what limitations were missed) can be answered by any careful clinical reader. Later chapters explain the methods, and each one you learn will make appraisal easier.

Summary

- Original research papers follow the IMRaD structure: Introduction (from the broad problem to the question), Methods (what was done), Results (what was found) and Discussion (what it means), with a title, abstract, declarations and supplementary material. The Results are the evidence; the Discussion is the authors' interpretation.
- Read in the order title, abstract, figures and tables (flow chart, Table 1, main results), Methods, then Discussion, stopping early if the paper does not answer your question. About fifteen minutes is enough to decide relevance and extract the main result.
- A good abstract lets you write the PICO, design and main result. For every result ask: how big (effect size), how sure (95% CI), by when (time point)? For a difference, no effect is 0; for a ratio, no effect is 1.
- Spin makes results look stronger than the data allow: "trends" for null results, secondary outcomes promoted to headlines, causal words for observational designs, and before-after claims without a comparison. Read the numbers first, then check whether the verb matches the design.
- Appraisal asks whether a result is valid, how large and precise it is, and whether it applies to your patients. Reporting guidelines (CONSORT, STROBE) check transparency; formal risk-of-bias tools (RoB 2, ROBINS-I) assess validity in systematic reviews; the ten-question checklist is a practical everyday tool. Answer each question Yes, No or Can't tell, with a page reference.
- Paper 1, a small pilot RCT, may claim that its strategy did not reduce day-5 fluid balance, with much uncertainty; it says nothing about mortality. Paper 2, a large matched cohort, may claim that a positive balance on days 2 and 3 is associated with higher mortality, but not that fluid caused the deaths. Neither alone would change practice.

Further reading

- Greenhalgh, *How to Read a Paper* (Greenhalgh 2019). The classic clinician's guide to reading and appraising papers of every design, with a chapter on each.
- Boutron and colleagues, "Reporting and interpretation of randomized controlled trials with statistically nonsignificant results for primary outcomes" (Boutron et al. 2010). The study that defined and measured spin in trial reports; essential for understanding why the words of a paper must be checked against its numbers.
- Schulz and colleagues, the CONSORT 2010 Statement (Schulz et al. 2010), and von Elm and colleagues, the STROBE Statement (Elm et al. 2007). Reading the checklist for each design shows what a well-reported trial or observational study should contain, and is one of the fastest ways to learn appraisal.
- Altman and Bland, "Absence of evidence is not evidence of absence" (Altman and Bland 1995). One page that every reader of a negative trial should know.

PART II

Designing

CHAPTER 5

Study designs

A clinical question can be answered in many ways. You can look at the patients on your ward today and count how many have the problem. You can start from patients who already have a complication and look back into their records for what might have caused it. You can follow a group of patients forward from the day they received a treatment and count what happens to them. Or you can decide, by the toss of a coin, which patients receive which treatment and then follow them forward. Each of these is a *study design*, and the choice between them is the most consequential decision in a research project. A good question answered with the wrong design produces a weak study; the same question answered with the right design can change practice. Above all, the design decides what you are allowed to claim at the end. This chapter explains the main designs used in clinical research, shows how to recognise each one from a short description, and explains how to choose a design that fits a question and is feasible in a real clinical setting.

In this chapter you will learn

- How to name the exposure, the outcome and the comparison group in any clinical question.
- The difference between descriptive and analytical designs, and between observational and experimental designs.
- How cross-sectional, case-control, cohort and randomised controlled designs work, drawn as timelines, with their strengths and weaknesses.
- How to name the design of a published study with three simple questions.
- What variants of the randomised trial exist (crossover, cluster, factorial; superiority and non-inferiority) and when each is used.
- What pilot studies, interrupted time series and qualitative studies are for.
- How to match a design to a question, write inclusion and exclusion criteria, and define outcomes precisely.
- Why a randomised trial may say “caused” while a cohort study should say “associated with”.

5.1 One question, two designs

Throughout this part of the book we follow one clinical question. Patients with septic shock receive large volumes of intravenous fluid: fluid to restore blood pressure (resuscitation fluid), but also maintenance fluid, fluid to dilute drugs, fluid to keep lines open and fluid in nutrition. Many clinicians suspect that the fluid that accumulates in the days after resuscitation is harmful. So we may ask: *in adults with septic shock, does a restrictive fluid strategy, compared with usual care, reduce mortality at 28 days?*

Two published studies approached this question from different directions, and we will use them repeatedly. The first, by Boulet and colleagues (Paper 1 in this book), is a randomised controlled trial (Boulet et al. 2024). Fifty adults with septic shock in three intensive care units (ICUs) were allocated by a central computer, by chance, either to an "optimised restrictive strategy" that cut non-resuscitation fluids for seven days, or to usual care. Two patients in the usual-care group withdrew consent, so 48 were analysed. The primary outcome was the cumulative fluid balance at day 5, not mortality. The second study, by Hyun and colleagues (Paper 2), is a cohort study (Hyun et al. 2023). It used a registry of 11,981 patients with sepsis from 20 hospitals, selected 2,516 adults who stayed at least three days in the ICU and had fluid data, and compared patients whose cumulative fluid balance was positive with those whose balance was negative. The outcome was death by day 28.

The figure below sets the two studies side by side. Both are about fluids and both are well conducted. Yet they can tell us quite different things, because in one study the investigators *assigned* the fluid strategy and in the other they only *observed* what clinicians did.

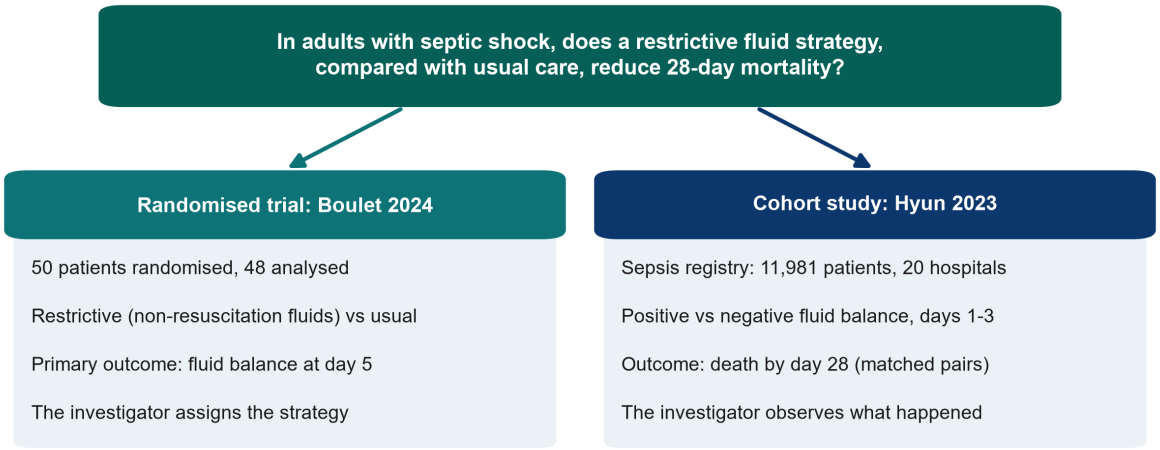


Figure 5.1: The same fluid question answered by a randomised trial (Boulet 2024), in which the investigators assigned the strategy, and by a cohort study (Hyun 2023), in which they observed what happened.

Notice a detail that is easy to miss: Boulet's trial was a pilot trial, designed to measure fluid balance, and it recorded only 6 deaths in each group of 24. Hyun's cohort measured the outcome we care about, 28-day mortality, in thousands of patients, but could not decide who received more fluid. Neither study answers our question completely. By the end of this chapter you will be able to say exactly what each can and cannot claim, and why.

5.2 The pieces of a question: exposure, outcome and comparison

Before choosing any design, you must be able to name two things in your question. The **exposure** is what comes first: a treatment, a strategy, a risk factor, a behaviour or a characteristic of the patient. The **outcome** is what we count later: death, infection, a pressure injury, a laboratory value, a score. The **comparison group** is the people without the exposure, or with a different exposure, often usual care. Once these three pieces are named, most design decisions become straightforward.

The word "exposure" can sound as if it must mean something harmful, like exposure to smoke or radiation. It does not. In research language a treatment is an exposure too; so is an air mattress, a nursing protocol or a time of day. The figure below gives four examples from intensive care and ward nursing.

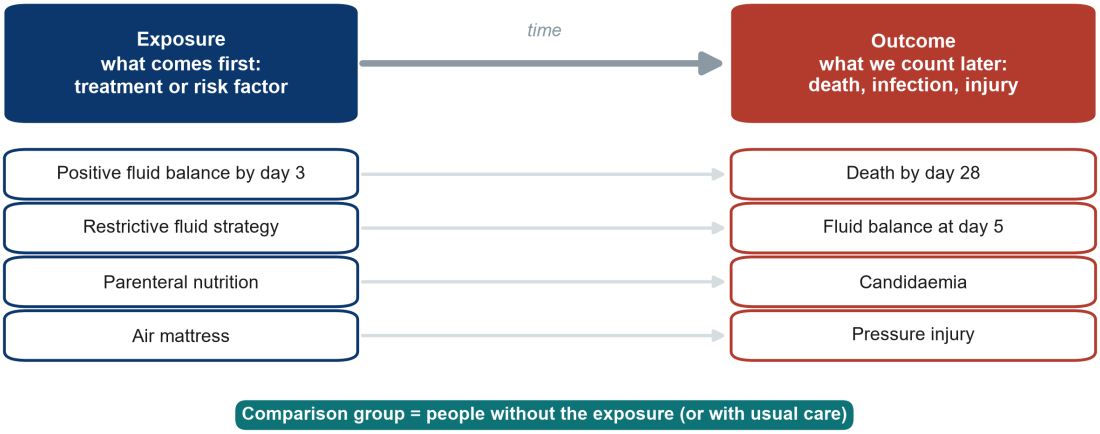


Figure 5.2: Exposure is what comes first and outcome is what is counted later; the comparison group is people without the exposure or with usual care.

The same variable can play different roles in different studies. In Boulet's trial, the fluid balance at day 5 is the *outcome*: the trial asks whether the restrictive strategy changes it. In Hyun's cohort, the fluid balance over the first three ICU days is the *exposure*: the study asks whether it predicts death. Neither use is wrong; what matters is that you know which role each variable plays in the study you are reading or planning.

Worked example: naming the pieces

Question: "Does early mobilisation reduce delirium in ventilated ICU patients?"

- Exposure: early mobilisation (for example, sitting out of bed within 72 hours of intubation).
- Outcome: delirium, assessed with a validated screening tool, during the ICU stay.
- Comparison: usual care, in which mobilisation starts later.

Question: "Is parenteral nutrition a risk factor for candidaemia in the ICU?"

- Exposure: parenteral nutrition in the preceding 14 days.
- Outcome: candidaemia (a positive blood culture for *Candida*).
- Comparison: ICU patients who did not receive parenteral nutrition.

Writing these three lines takes a minute and prevents most design errors that follow.

5.3 The family tree of study designs

Study designs can be arranged in a simple family tree, shown below (Grimes and Schulz 2002a). The first split is between **descriptive** studies, which describe what is there without formally comparing groups, and **analytical** studies, which compare groups to test whether an exposure is associated with an outcome. Analytical studies split again. In **observational** studies the investigator watches what happens in normal care, without influencing who receives the exposure. In **experimental** studies the investigator decides who receives what. When the decision is made by chance, the study is a **randomised controlled trial** (RCT).

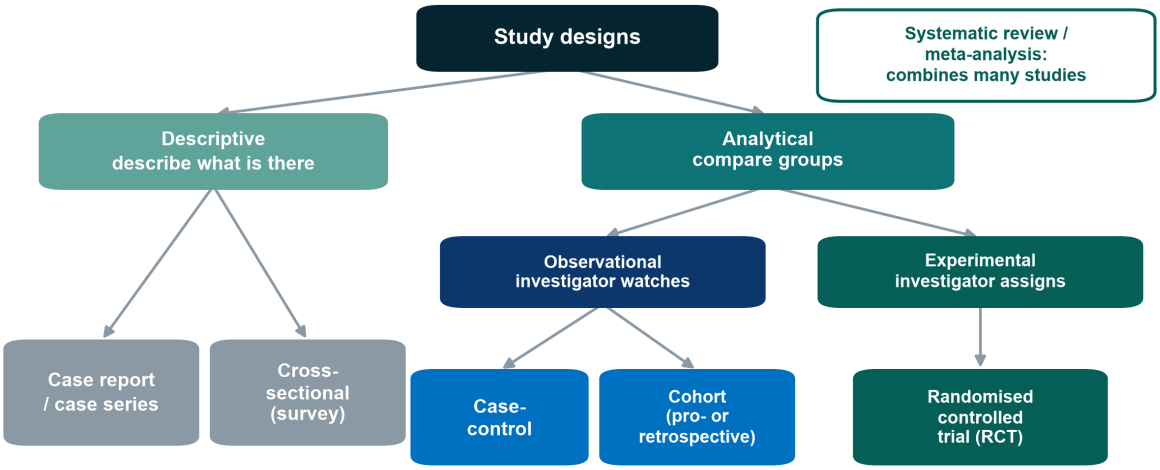


Figure 5.3: The family tree of study designs: descriptive designs describe, analytical designs compare groups, and analytical designs are either observational (the investigator watches) or experimental (the investigator assigns).

Systematic reviews and meta-analyses sit outside the tree because they are studies of studies: they combine the results of many studies, of any of the designs below them. They are the subject of Chapter 9.

A common misunderstanding is that "observational" means "bad". It does not. Many important questions can only be answered observationally: questions about harm (we cannot randomise people to smoke), about prognosis (we cannot assign a diagnosis), about rare outcomes and about what happens in routine practice. The right design is the one that fits the question; there is no design that is best for everything.

5.4 Descriptive designs

5.4.1 Case reports and case series

A **case report** describes one patient in detail; a **case series** describes a handful of patients who share an unusual feature. They have no comparison group, so they cannot test whether an exposure causes an outcome. But they matter. Case reports and series have raised the first alarm about many new diseases and drug harms, and they generate the hypotheses that other designs then test. A report of five patients who developed severe hypernatraemia within a week of a new fluid protocol being introduced on a unit is a case series: it raises a reasonable suspicion about the protocol, but it cannot tell us whether hypernatraemia became more common, because there is nothing to compare with.

5.4.2 Cross-sectional studies

A **cross-sectional study** takes a snapshot of a population at one moment. Everyone in the population (or a sample of it) is examined once, and the exposure and the outcome are recorded at the same visit. The figure below contrasts this snapshot with a case series.

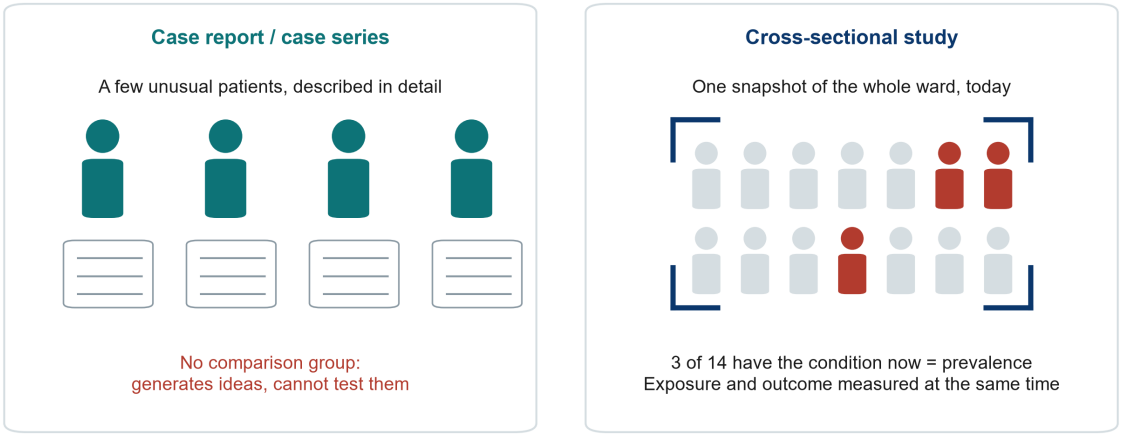


Figure 5.4: A case series describes a few unusual patients without a comparison group; a cross-sectional study takes one snapshot of everyone and measures how many have the condition now (the prevalence).

A cross-sectional study measures **prevalence**: how many people have a condition *now* (Chapter 7 explains prevalence and the related idea of incidence). It is the natural design for questions such as "how many of our inpatients have a pressure injury?" or "how many ICU patients have delirium today?". Cross-sectional studies are quick and cheap, and for questions about how common something is they are exactly the right tool.

When a cross-sectional study compares groups, for example pressure injuries on standard versus air mattresses, it becomes an *analytical* cross-sectional study. Its key weakness is then time order. Because exposure and outcome are measured at the same moment, you usually cannot tell which came first.

Worked example: a one-day point-prevalence survey

A hospital wants to know how common pressure injuries are, and whether they are linked to mattress type.

1. *Choose a day.* All inpatients on all wards are included on one weekday.
2. *Examine everyone the same way.* Two trained nurses examine each patient's skin using one standard form. Using the same method for everybody protects against measurement error (Chapter 6).
3. *Record both exposure and outcome at the same visit.* Pressure injury (yes or no) and the type of mattress the patient is lying on.
4. *Count.* Suppose 37 of 412 patients have a pressure injury: the prevalence is $37/412 = 9\%$.

Suppose further that injuries are more common on standard mattresses (9%) than on pressure-relieving ones (4%). Does this show that pressure-relieving mattresses prevent injury? Not on its own. Patients who already have an injury are often *moved onto* a pressure-relieving

mattress, so the exposure may be a consequence of the outcome rather than its cause. This is reverse causation, which Chapter 6 discusses. The survey can honestly say that mattress type and pressure injury are *associated*; it cannot say which came first.

5.5 Case-control studies: start from the outcome and look back

In a **case-control study**, the investigator starts from the outcome. **Cases** are people who already have the outcome; **controls** are similar people who do not. The investigator then looks back in time, in records or by interview, to compare how often cases and controls were exposed. The arrow of the investigation runs backwards, from outcome to exposure, as the figure shows.

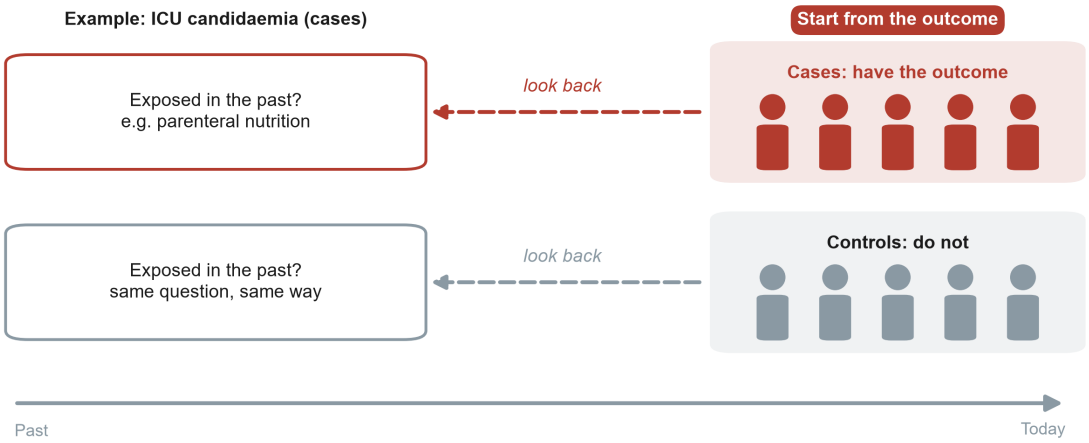


Figure 5.5: A case-control study starts from people who have the outcome (cases) and people who do not (controls) and looks back to compare their past exposures, asking the same question in the same way for both groups.

The case-control design is efficient when the outcome is **rare** or takes a long time to develop. If only 1 in 200 ICU patients develops candidaemia, following thousands of patients forward to find a few dozen cases would be slow and expensive. Starting from the cases that have already occurred is far quicker. Case-control studies also played a historic role: the early studies linking cigarette smoking with lung cancer were case-control studies (Doll and Hill 1950).

The weaknesses are equally important. First, the exposure is measured after the outcome is known, which invites **recall bias**: cases, or their families, may search their memories harder than controls do. Second, choosing good controls is difficult. Controls must come from the same population that produced the cases, so that if a control had developed the outcome, he or she would have become a case in the study. Healthy volunteers are usually poor controls for hospital cases. Third, because the investigator decides how many cases and how many controls to include, the proportion of people with the outcome in the study is set by design, not by nature. That means a case-control study cannot estimate the *risk* of the outcome; it can only estimate an **odds ratio**, a measure explained in Chapter 7.

Worked example: parenteral nutrition and ICU candidaemia

Question: is parenteral nutrition a risk factor for candidaemia in the ICU?

1. *Find cases.* The microbiology records identify 40 ICU patients with candidaemia over a defined period.

2. *Find controls.* Choose 80 ICU patients *without* candidaemia from the same units and the same months (two controls per case). They might be matched on age and month of admission.
3. *Look back.* Search the records of both groups, in the same way, for parenteral nutrition in the 14 days before the culture date (or an equivalent date for controls).
4. *Compare.* Suppose 24 of the 40 cases (60%) and 16 of the 80 controls (20%) had received parenteral nutrition.
5. *Report.* The result is expressed as an odds ratio, which here equals 6.0 (the calculation is worked through in Chapter 7).

Why can we not report the risk of candidaemia from this study? Because we chose to include one case for every two controls. Among the 40 exposed patients in the study, 24 are cases, so a naive "risk" would be 60%. Had we chosen four controls per case instead of two, the same calculation would give a much smaller figure, although nothing about the patients had changed. The proportion with the outcome is our choice, not a property of the population.

Choosing good controls

Three rules help. *Same source:* controls come from the population that produced the cases (same hospital, units and period). *Same measurement:* exposure is measured in exactly the same way for cases and controls, ideally by someone who does not know which is which. *Matching with care:* matching on a few strong factors, such as age or month of admission, can make the comparison more efficient, but never match on the exposure itself or on things closely linked to it (over-matching), because that hides the very effect you are looking for.

5.6 Cohort studies: start from the exposure and follow forward

In a **cohort study**, people are grouped by their **exposure** and then followed forward in time to see who develops the outcome. Time runs in the natural direction: exposure first, outcome later. The figure below shows the logic using fluid balance.

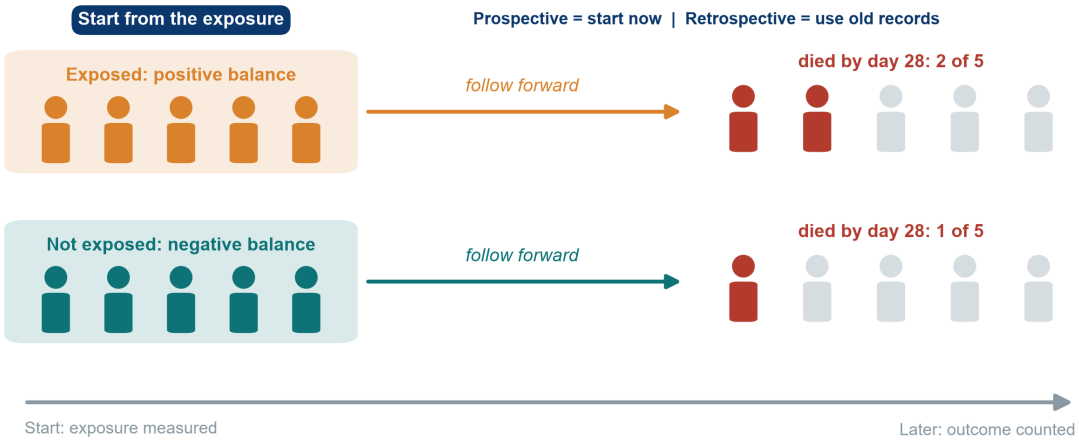


Figure 5.6: A cohort study starts from the exposure (here, positive or negative fluid balance) and follows people forward to count the outcome (death by day 28); it can be prospective or use old records (retrospective).

A **prospective** cohort starts now and follows people into the future. A **retrospective** cohort

(sometimes called a historical cohort) uses records that already exist, but the logic is the same: exposure is defined first and the outcome is counted afterwards. “Retrospective” does not mean “poor”: the quality of a retrospective cohort depends on how well the original data were recorded. Hyun’s study is a good example of the middle ground. Its data were collected prospectively, as part of an ongoing registry, but the question about fluid balance was asked later (Hyun et al. 2023). What matters for the design is that fluid balance (the exposure) was measured before death (the outcome) was counted.

Cohort studies have important strengths. The time order is clear, so reverse causation is less of a problem than in cross-sectional studies. They can study several outcomes of one exposure, and they give real **risks** and **incidence** (Chapter 7). Their weaknesses are that prospective cohorts are slow and costly, that they are inefficient for rare outcomes, and, above all, that they are open to **confounding**. The investigator did not choose who was exposed. In Hyun’s study, the treating clinicians decided how much fluid each patient received, often because of how sick the patient looked. Sicker patients receive more fluid and also die more often, so part of any association between fluid and death may be due to severity of illness. Chapter 6 is devoted to this problem.

Worked example: the Hyun cohort, step by step

Question: is a positive cumulative fluid balance in the first ICU days associated with death by day 28?

1. *Define the cohort.* Adults with sepsis from a registry covering 20 hospitals who were admitted to the ICU and stayed there for at least three days, with fluid data recorded. Of 11,981 registry patients, 2,516 met these conditions.
2. *Measure the exposure.* Cumulative fluid balance (input minus output) from ICU admission to the first, second and third midnight, classified as positive or negative.
3. *Follow forward.* Record who died by day 28.
4. *Compare like with like.* Patients with a positive balance were matched one-to-one with patients with a negative balance who had a similar *propensity score*, a summary of many measured characteristics such as age, sex, SOFA score, septic shock, comorbidities, site of infection and fluid output. This produced 483, 373 and 392 matched pairs for the day-1, day-2 and day-3 analyses.
5. *Report.* A positive balance by day 1 showed no clear difference in mortality (hazard ratio 1.17, 95% confidence interval 0.85 to 1.60). A positive balance by day 2 (hazard ratio 2.13, 1.48 to 3.06) and by day 3 (1.56, 1.10 to 2.22) was associated with higher mortality.

The hazard ratio is explained in Chapters 7 and 14. For now, note what the study can say: a positive balance on days 2 and 3 was *associated* with higher mortality. It cannot say that the fluid *caused* the deaths, because clinicians, not chance, decided who received more fluid.

! Common mistake: calling every study that uses old records a case-control study

Both case-control studies and retrospective cohorts use existing records, so they are easily confused. The difference lies in how people are *selected*. If people are chosen because of their **outcome** (they developed candidaemia, or did not), it is a case-control study. If they are chosen and grouped by their **exposure** (they did or did not receive a benzodiazepine

infusion), and the outcome is counted afterwards, it is a cohort study, however old the records are.

5.7 Randomised controlled trials: chance decides

A **randomised controlled trial** is a cohort study in which the investigator assigns the exposure, and does so by chance. Eligible patients are enrolled, randomised to one of the groups, and then followed forward to the outcome. The figure below shows Boulet's trial drawn on a timeline.

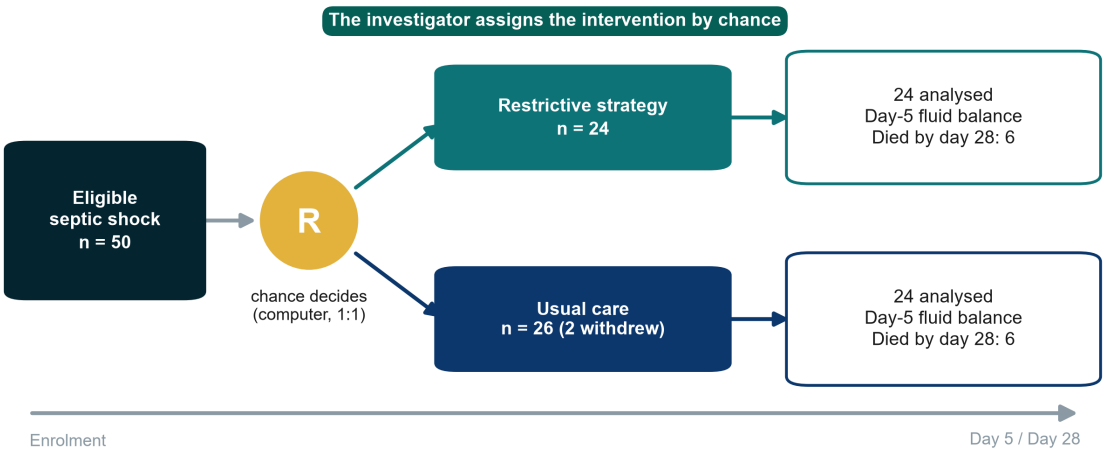


Figure 5.7: In a randomised controlled trial chance decides who receives which intervention; in Boulet 2024, 50 patients were randomised by computer and 24 per group were analysed.

Why go to the trouble of randomising? Because when chance decides who receives the intervention, the groups are alike at the start, on average, in every respect: in the factors we measured, such as age and severity, *and in the factors we did not or could not measure*, such as frailty, genetic make-up or a clinician's bedside impression. Any difference in outcome that is larger than chance would produce can then be attributed to the intervention. No other design offers this protection against unmeasured confounding, which is why RCTs are the preferred design for questions about whether a treatment works. Chapter 6 explains randomisation, allocation concealment and blinding in detail.

Worked example: the Boulet trial, step by step

Question: does restricting non-resuscitation fluids reduce the cumulative fluid balance at day 5 in septic shock?

1. *Eligibility.* Adults aged 18 to 85 with septic shock, within 24 hours of starting a vasopressor. Excluded: acute kidney injury needing renal replacement therapy within 24 hours, end-stage chronic kidney disease, and severe malnutrition (body mass index below 18).
2. *Consent.* From the patient or a legal surrogate, before randomisation or as soon as possible afterwards.
3. *Randomise.* A centralised, computer-generated sequence allocated patients 1:1 in small blocks, so that nobody could predict the next allocation.
4. *Treat.* For seven days, the restrictive group had maintenance fluids, drug dilutions and artificial nutrition minimised; resuscitation fluids were left to clinical judgement in both

groups.

5. *Measure.* Primary outcome: cumulative fluid balance (ml/kg) at day 5. Secondary outcomes included death by day 28.

Step 3 is what makes this an *experiment* rather than a cohort: the investigators, through chance, assigned the exposure. The result was honest and null: the median day-5 balance was 6.9 versus 35.0 ml/kg, a difference whose confidence interval ran from -15.2 to 41.6 ml/kg ($p = 0.42$), and 6 of 24 patients died in each group (Boulet et al. 2024).

Randomisation is necessary for a causal claim but not sufficient. A small, unblinded or poorly conducted RCT can still mislead. In Boulet's trial, clinicians knew the allocation (only patients and statisticians were blinded), and the authors themselves suspected that clinicians gave less fluid than usual to the control group too, narrowing the difference between groups. With 24 patients per arm and 6 deaths in each, the trial tells us almost nothing about mortality. It was never designed to.

Why not randomise everything? Some exposures cannot ethically be assigned (we cannot randomise people to a harmful exposure). Some outcomes are rare or take years to appear. Trials are expensive and slow, and they often enrol highly selected patients (Boulet screened 1,201 patients to randomise 50), so their results may not apply to everyone. These are reasons why observational designs remain essential.

5.8 Naming a design in three questions

Titles and abstracts often contain words such as "prospective", "registry" or "retrospective" that do not, on their own, identify a design. A more reliable approach is to ask three questions, shown as a decision tree below.

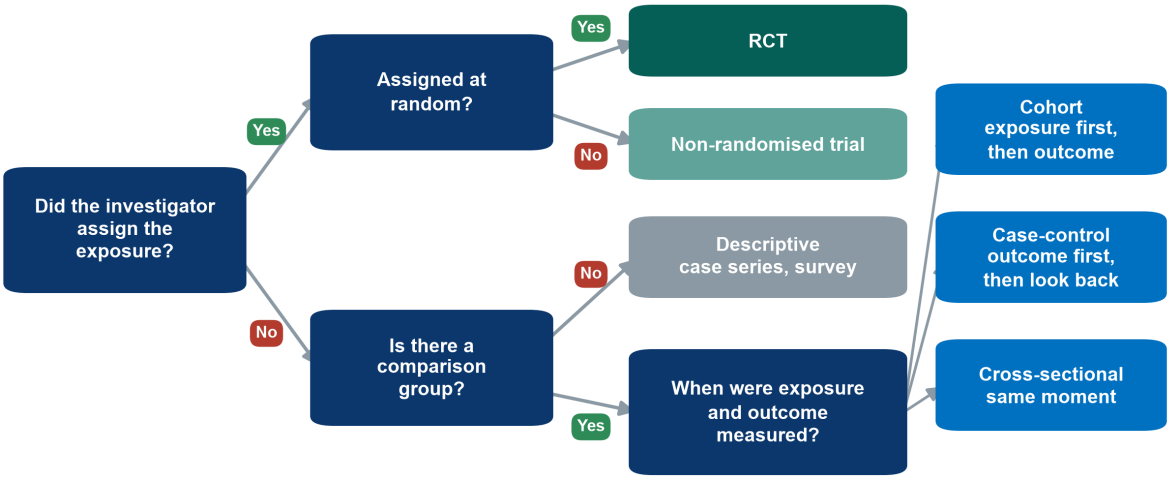


Figure 5.8: Three questions name most designs: did the investigator assign the exposure, was it assigned at random, and, if not assigned, when were exposure and outcome measured?

1. **Did the investigator assign the exposure?** If yes, the study is experimental. *Was it assigned at random?* If yes, it is an RCT; if not, it is a non-randomised (quasi-experimental) study.
2. **If the exposure was not assigned, is there a comparison group?** If not, the study is

descriptive (a case series or a survey of prevalence).

3. **If there is a comparison group, in which direction does time run?** Exposure first, then outcome: a cohort. Outcome first, then look back: a case-control study. Both at the same moment: a cross-sectional study.

Applied to our papers: in Boulet, the investigators assigned the strategy (yes), at random (yes), so it is an RCT. In Hyun, nothing was assigned (no); there is a comparison group (positive versus negative balance); exposure was measured in days 1 to 3 and the outcome counted by day 28, so it is a cohort.

Worked example: naming designs from short abstracts

Each description below is followed by the design and the phrase that proves it.

1. *"Two hundred and forty ventilated adults were randomly allocated by a central computer to chlorhexidine oral care or standard oral care; ventilator-associated pneumonia was recorded over 28 days by a blinded assessor."* RCT: "randomly allocated by a central computer".
2. *"On one day, two nurses examined all 412 inpatients for pressure injuries and recorded each patient's mattress."* Cross-sectional: "on one day", exposure and outcome recorded together.
3. *"Forty ICU patients who developed candidaemia were compared with 80 from the same units who did not; records were searched for earlier parenteral nutrition."* Case-control: selected by outcome, then "searched the records ... before".
4. *"All 600 adults given thrombolysis for stroke were classified on arrival by door-to-needle time and followed up for functional independence at 90 days."* Prospective cohort: "classified on arrival ... followed up".
5. *"From electronic records for 2019 to 2022, 1,150 ICU patients were divided into those who did and did not receive a benzodiazepine infusion for at least 3 days, and delirium later in the stay was compared."* Retrospective cohort: old records, but people were grouped by *exposure* and the outcome came later. This is the description most often misnamed as case-control.
6. *"We describe five patients who developed severe hypernatraemia after a new fluid protocol."* Case series: no comparison group.
7. *"Central-line infection rates were compared for six months before and six months after an insertion-bundle programme that the research team introduced on all beds on the same day."* Non-randomised intervention study (a before-after study): the investigators assigned the intervention, but not at random.
8. *"Twelve wards were randomly allocated, ward by ward, to a hand-hygiene feedback programme or usual practice, and infections were counted in all their patients."* Cluster randomised trial: randomised, but the unit of randomisation is the ward.

! Common mistake: two groups make a trial

A study that compares patients who received early mobilisation (because the physiotherapist decided so) with patients who did not, and follows both groups to discharge, has two groups and follows them forward. It is still a cohort study, not a trial, because chance did not decide who was mobilised. Always ask *who decided* the exposure.

5.9 Strengths, weaknesses and the evidence pyramid

Every design trades something away. The table below summarises the four basic analytical designs.

Table 5.1: Strengths and weaknesses of the four basic analytical designs.

Design	Main strengths	Main weaknesses	Main measure
Cross-sectional	Quick, cheap; gives prevalence	Cannot show which came first	Prevalence; prevalence ratio
Case-control	Efficient for rare outcomes and long latency; few patients	Recall bias; choosing controls is hard; no risks	Odds ratio
Cohort	Clear time order; several outcomes; risk and incidence can be calculated	Slow if prospective; poor for rare outcomes; confounding	Risk, risk ratio, hazard ratio
RCT	Balances known and unknown confounders; strongest for cause	Costly; ethical limits; selected patients	Risk ratio, risk difference, mean difference

You may have met the **evidence pyramid**, which ranks designs from expert opinion and case reports at the bottom, through cross-sectional, case-control and cohort studies, to RCTs and then systematic reviews at the top. The figure below places our two papers on it.

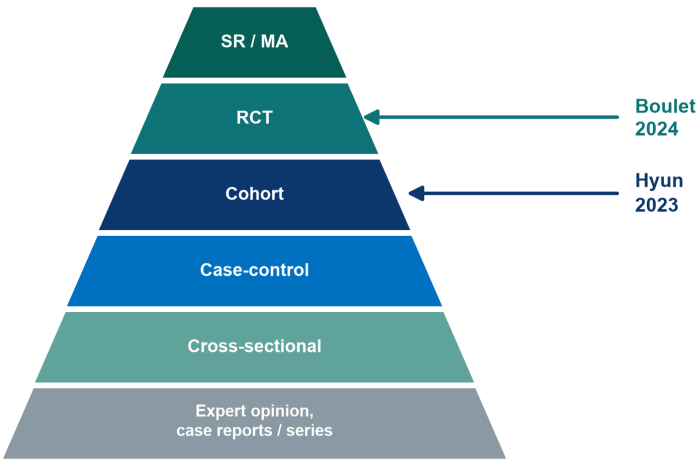


Figure 5.9: The evidence pyramid ranks designs by their typical protection against bias for questions about treatment effects; Boulet 2024 sits at the level of RCTs and Hyun 2023 at the level of cohort studies.

The pyramid is a useful rule of thumb, but it must be read with care. Higher levels are, on average, less prone to bias *for questions about the effects of treatments*. The pyramid does not rank designs for questions about prevalence, prognosis or rare harms, for which a cross-sectional study, a cohort or a case-control study is the right tool. And it describes designs, not individual studies: a careful

multicentre cohort can be more informative than a small or badly run trial. For our mortality question, Boulet sits higher, but a 48-patient pilot trial with 12 deaths in total cannot overrule a large cohort on mortality; the two studies answer different parts of the question.

5.10 What each design may claim

The design determines the language of the conclusion. When chance decided who received the intervention, a difference in outcome can be attributed to the intervention: an RCT may use causal words such as “reduced”, “caused” or “prevented”. When the exposure was not assigned, the most a study can honestly say is that the exposure was “associated with” the outcome. The figure below applies this rule to our two papers.

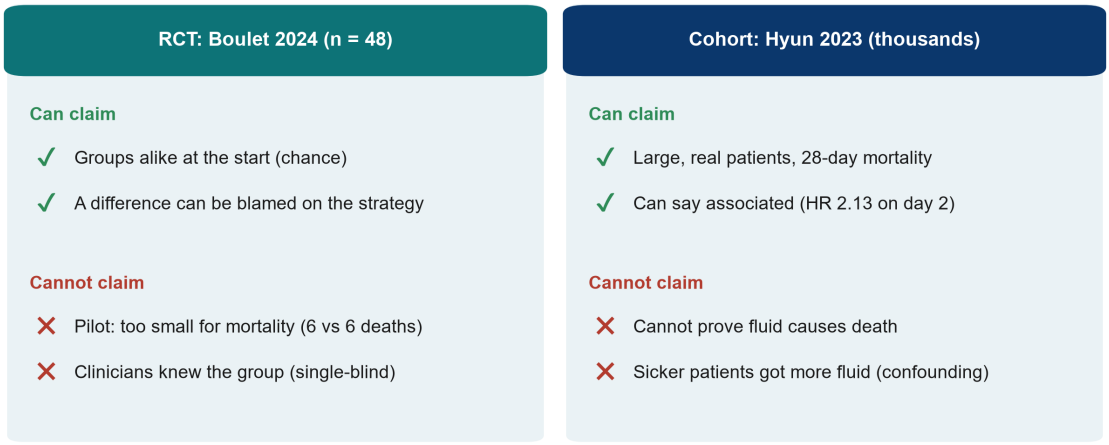


Figure 5.10: What each paper may claim: the RCT can attribute differences to the strategy but is too small for mortality; the cohort has a large sample and can report an association but cannot prove that fluid causes death.

Boulet’s honest conclusion is that the restrictive strategy “did not reduce the overall fluid balance at day 5” (Boulet et al. 2024): a causal statement about a null result, which is permitted. Hyun’s abstract correctly says that a positive balance on days 2 and 3 “was associated with” higher mortality (Hyun et al. 2023). Its title, however, speaks of the “impact” of fluid balance, a word that implies more than a cohort can show. Checking that the words match the design is one of the most useful habits a reader can develop (Chapter 4 and Chapter 19 return to this).

• The design decides the claim

Randomised designs may say “caused” (if they are well conducted and large enough). Observational designs should say “associated with”, however large they are and however carefully they were adjusted or matched. Adjustment and matching reduce confounding by the factors that were measured; they do not turn an association into a proven cause.

5.11 Variants of the randomised trial

Most trials are **parallel-group** trials: each patient is randomised to one treatment and stays in that group, as in Boulet's trial. Three variants solve particular problems; the figure below shows all four layouts.

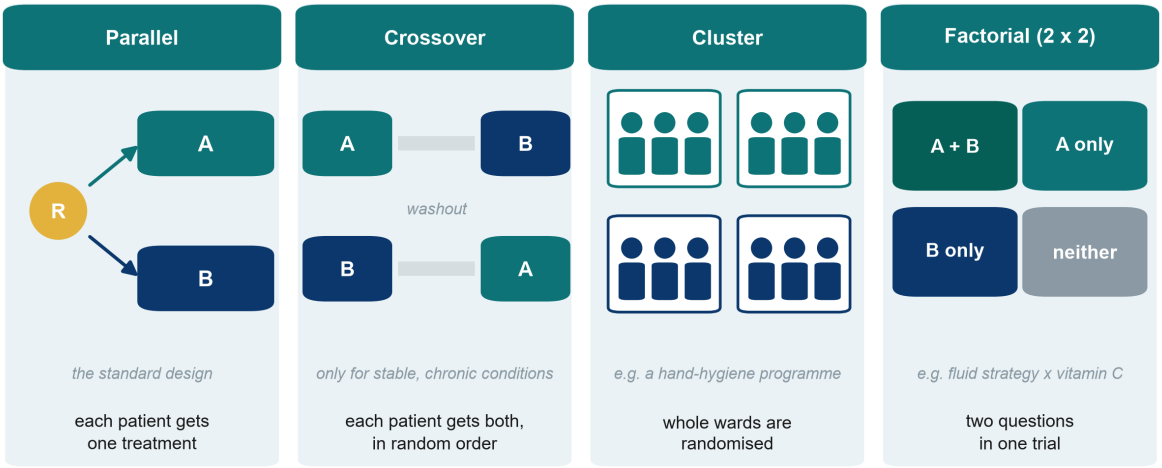


Figure 5.11: Four trial layouts: parallel (each patient gets one treatment), crossover (each patient gets both in random order), cluster (whole wards are randomised) and factorial (two questions in one trial).

Crossover trials. Each patient receives both treatments, one after the other, in a random order, usually with a *washout* period in between. Because each patient acts as his or her own control, far fewer patients are needed. The design only works for stable, chronic conditions in which the treatment effect wears off, such as inhalers in chronic obstructive pulmonary disease. It is useless in acute illness such as septic shock: the patient on day 7 is not the same patient as on day 1.

Cluster randomised trials. Whole groups (wards, ICUs, hospitals or general practices) are randomised rather than individual patients. This is the right design when an intervention works at the level of a unit, such as a hand-hygiene programme, a sepsis bundle or a handover checklist, or when patients in the same ward would "contaminate" one another (a nurse cannot easily follow a new protocol for half of her patients and the old one for the rest). Patients in the same cluster resemble one another, so a cluster trial needs more patients than an individually randomised trial, and the analysis must take the clustering into account. A cluster trial is a proper randomised trial, not a weaker one; it has its own reporting guidance (Campbell et al. 2012).

Factorial trials. Two questions are answered in one trial by randomising each patient twice, for example to fluid strategy (restrictive or usual) and to vitamin C (yes or no), giving four groups. This is efficient as long as the two treatments do not interact, that is, as long as the effect of one does not depend on whether the patient also receives the other.

5.12 Superiority, equivalence and non-inferiority

Most trials are **superiority** trials: they ask whether a new treatment is *better* than the comparator. Sometimes that is not the question. A new treatment may be cheaper, safer, simpler or require less fluid, and we would adopt it as long as it is *not unacceptably worse*. Such a question calls for a **non-inferiority** trial. A third type, the **equivalence** trial, asks whether two treatments are about the *same*, within a margin on either side; it is common for generic drugs.

The key ingredient of a non-inferiority trial is the **margin**: the largest loss of effectiveness that we would accept in exchange for the new treatment's other advantages. The margin is a clinical judgement, made by clinicians (and ideally patients) with the statistician, and it must be fixed in the protocol before any data are seen (Piaggio et al. 2012). The result is then judged by where the whole 95% confidence interval (Chapter 12) lies relative to zero and to the margin. The figure below shows four possible results for a hypothetical trial with a margin of 5 percentage points of mortality.

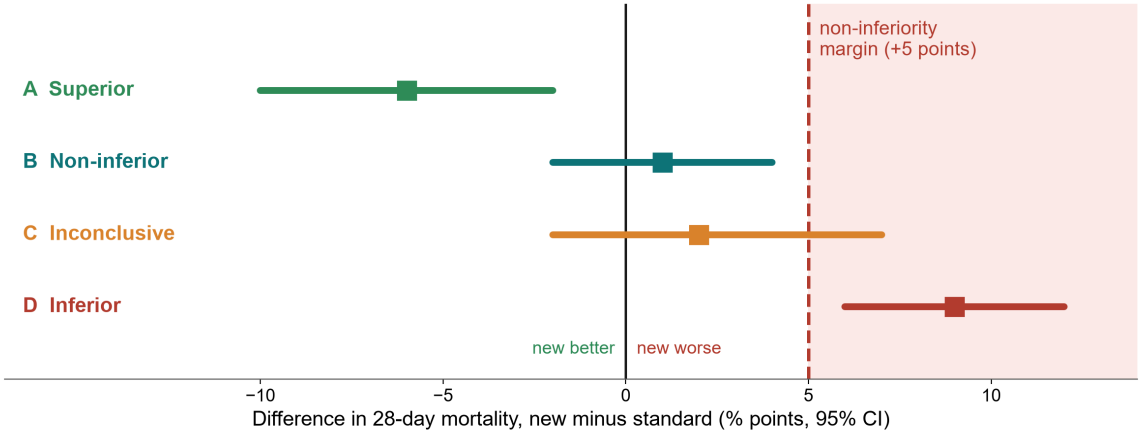


Figure 5.12: Reading a non-inferiority trial: if the whole confidence interval lies below zero the new treatment is superior, if it lies below the margin it is non-inferior, if it crosses the margin the result is inconclusive, and if it lies above the margin the new treatment is inferior.

- **A, superior:** the whole interval lies below zero; the new treatment is better.
- **B, non-inferior:** the interval crosses zero but stays entirely below the margin; we cannot say it is better, but we can say it is not unacceptably worse.
- **C, inconclusive:** the interval crosses the margin; the new treatment might be unacceptably worse.
- **D, inferior:** the whole interval lies above the margin.

! Common mistake: a non-significant superiority trial shows the treatments are equivalent

Boulet's primary result had $p = 0.42$. This does not show that the restrictive strategy is "as good as" usual care. A superiority trial that fails to show a difference has simply failed to show a difference; the absence of evidence is not evidence of absence (Altman and Bland 1995). To claim that two treatments are similar, the trial must be designed for it, with a margin fixed in advance and enough patients to make the confidence interval narrow.

5.13 Pilot and feasibility studies

Boulet's trial calls itself a pilot, and its real value lies in the numbers shown on the left of the figure below: 1,201 patients screened, 50 randomised (about 4%), 48 analysed (96%), at a rate of roughly three patients a month across three ICUs.

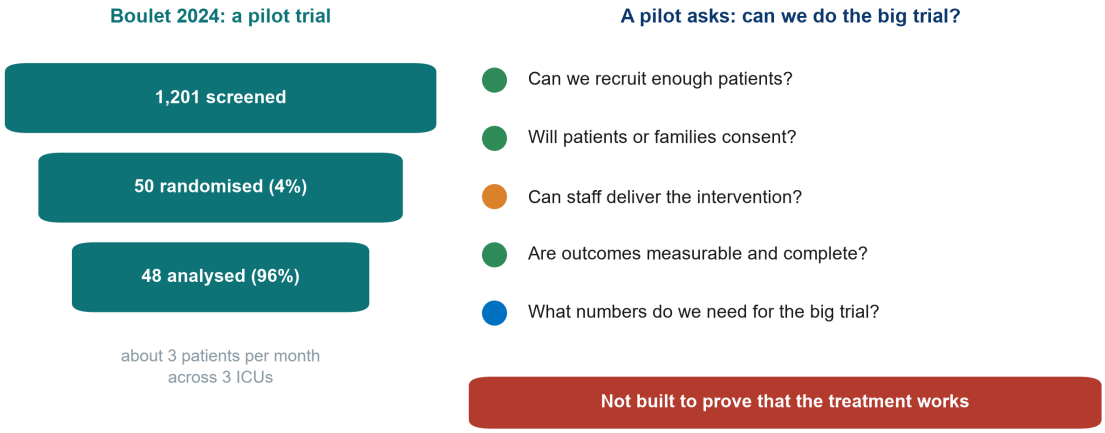


Figure 5.13: A pilot study rehearses the full trial: Boulet 2024 screened 1,201 patients to randomise 50, and a pilot asks whether recruitment, consent, delivery and measurement are feasible rather than whether the treatment works.

A **pilot** or **feasibility** study tests the machinery of a future large trial. Can enough patients be recruited? Will patients or families consent? Can staff deliver the intervention as intended? Can the outcomes be measured completely? And what numbers (event rates, variability) are needed to calculate the sample size of the main trial? A pilot should have its own feasibility objectives, with success criteria such as "recruit at least two patients per site per month" or "complete outcome data for more than 90% of patients" (Eldridge et al. 2016).

What a pilot should not do is claim effectiveness. It is too small, and it was not designed for that purpose. From Boulet's numbers we can also draw a practical lesson: at three patients a month, a 1,000-patient trial in the same three ICUs would take decades. The main trial would need many more sites or broader eligibility criteria. This is exactly the kind of knowledge a pilot exists to provide.

! Common mistake: renaming a small negative study as a pilot

A study that was planned to test effectiveness and found nothing is not a pilot. A pilot is planned as a pilot from the start, with feasibility objectives stated in the protocol.

5.14 Quasi-experimental designs for quality improvement

Many projects in hospitals evaluate a change that was introduced for everyone at once: a new bundle, a protocol, a checklist or a policy. There is no randomisation and often no parallel control group. Such designs are called **quasi-experimental**.

The simplest is the **before-after study**, which compares the average rate before the change with the average after it. It is weak. Rates often fall anyway: because of the seasons, because new staff arrive, because another project started in the same month, or because the project was launched after an unusually bad month that would have been followed by better ones in any case (regression to the mean).

The **interrupted time series** (ITS) is much stronger (Bernal et al. 2017). It uses many measurements at regular intervals before and after the change, for example 24 monthly infection rates on each side. The pre-change data define a *trend*, which is projected forward to show what would have been expected without the change. The post-change data are then compared with this expected line, looking for a change in **level** (an immediate drop) and in **slope** (a steeper decline). The figure below shows simulated data for a central-line bundle.

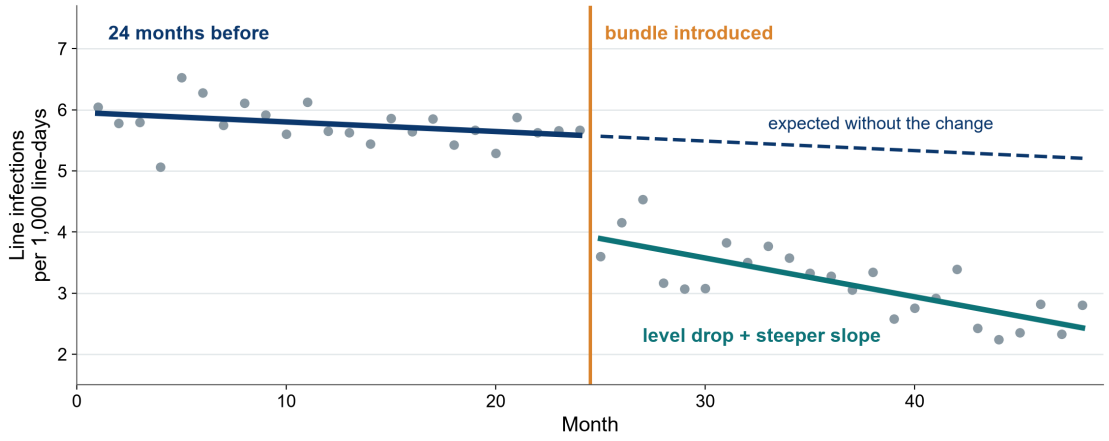


Figure 5.14: An interrupted time series of simulated monthly line-infection rates: after the bundle was introduced, rates fell below the line expected from the earlier trend, with a drop in level and a steeper slope.

An ITS becomes more convincing with more time points (at least eight to twelve on each side, with the same definition every month), with a **control series** (a ward or hospital that did not make the change, measured over the same months), and with a written diary of other changes that happened during the study period, such as staff turnover, new policies or audits. Quality-improvement projects have their own reporting guideline, SQUIRE 2.0 (Ogrinc et al. 2016).

! Common mistake: infections fell after our bundle, so the bundle worked

Only if the fall is larger than the trend that was already present, and larger than what happened elsewhere over the same months. Most hospitals introduce several changes in any given year. Ask what else changed in the same month.

5.15 Qualitative designs

Some questions are not about how many or how much but about *why*, *how* and *what it is like*. Why do nurses not chart urine output at night? How do families experience being asked to consent for a relative who cannot speak? What stops early mobilisation from happening on a busy unit? Numbers cannot answer these questions; words can. **Qualitative research** studies them systematically. The figure below shows its three main methods.

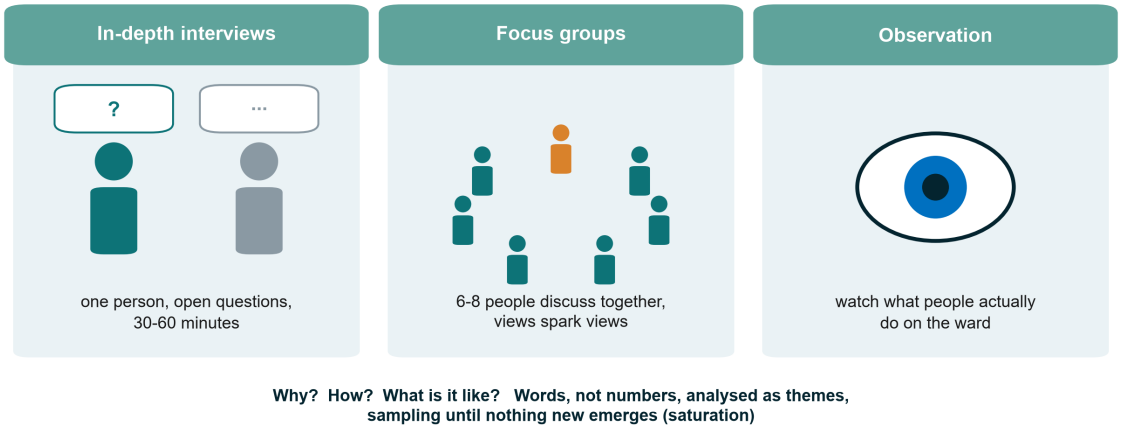


Figure 5.15: The three main qualitative methods are in-depth interviews, focus groups and observation; they answer why, how and what-it-is-like questions with words analysed as themes.

In-depth interviews explore one person's views with open questions, usually for 30 to 60 minutes. **Focus groups** bring together six to eight people, whose discussion reveals shared views and disagreements. **Observation** records what people actually do, which may differ from what they say they do. Participants are chosen by **purposive sampling**, that is, because they know about the topic, not at random, and recruitment continues until new interviews no longer produce new themes (**saturation**). Transcripts are analysed by identifying and refining themes. Qualitative studies have their own reporting guidelines, such as COREQ (Tong et al. 2007) and SRQR (O'Brien et al. 2014).

Qualitative research is not anecdote. Done properly it is systematic, transparent and peer-reviewed. It often works best alongside numbers in **mixed-methods** studies: a trial, for example, may include interviews with staff to understand why the intervention was or was not delivered as planned.

5.16 Matching the design to the question

The practical summary of this chapter is a simple mapping from the type of question to the design that usually answers it best, shown below.

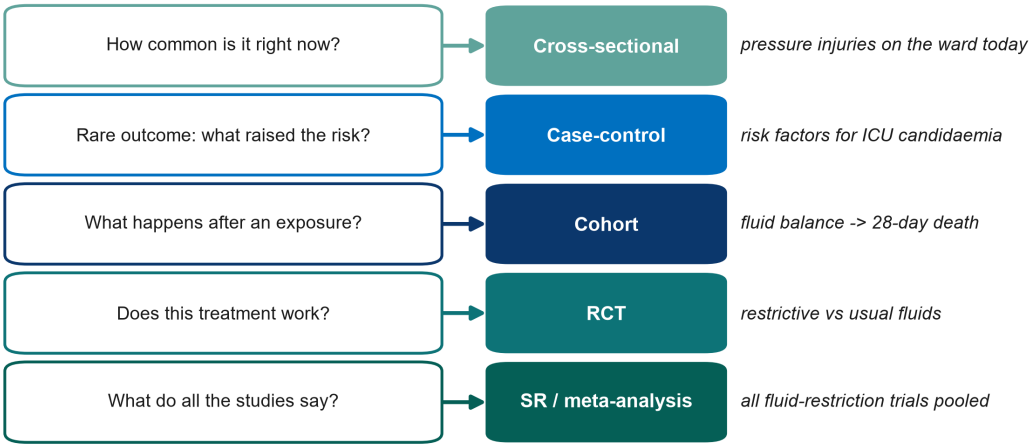


Figure 5.16: Match the design to the question: how common now (cross-sectional), what raised the risk of a rare outcome (case-control), what happens after an exposure (cohort), does a treatment work (RCT), and what do all the studies say (systematic review).

Two further rows complete the picture: if a change has *already been made* and routine data exist, consider an interrupted time series; if the question is *why* or *how*, consider a qualitative design.

The word that matters most in practice is **feasible**. The ideal design for "does restriction save lives?" is a large, multicentre RCT. A clinician with one ICU, no funding and six months may honestly choose a retrospective cohort, state its limitations, use the language of association, and produce a useful, publishable study that estimates the size of the association and informs a future trial. A well-conducted, honestly reported observational study is far more valuable than an underpowered trial that cannot answer its own question.

Worked example: choosing feasible designs

1. *"How many inpatients have a pressure injury today, and is it more common on standard mattresses?"* Cross-sectional (point-prevalence survey). Reason: "how many today" is prevalence; one visit per patient. Main weakness: cannot tell whether the mattress came first.
2. *"Does daily chlorhexidine bathing reduce bloodstream infections? It will be introduced ward by ward, and nurses cannot bathe only half their patients."* Cluster randomised trial, randomising wards. Weakness: needs more patients and an analysis that accounts for clustering; with few wards, chance imbalance is likely.
3. *"A sepsis bundle was introduced three years ago; we have monthly mortality for three years before and two after. Did mortality fall?"* Interrupted time series. Weakness: other changes at the same time; strengthen with a control series.
4. *"Why do nurses not report near-miss medication errors?"* Qualitative interviews or focus groups. Weakness: findings are not generalisable as numbers; quality depends on the interviewer's skill.
5. *"What are the risk factors for Stevens-Johnson syndrome in hospital patients?"* Case-

control: a rare outcome. Weakness: recall and record bias; only an odds ratio.

6. *"A 5-day antibiotic course is cheaper than the 10-day course; we only need to show it is not worse for cure."* Non-inferiority RCT with a margin fixed in advance. Weakness: requires a justified margin; poor adherence can make the arms look falsely similar.

! Common mistake: choosing an RCT for every question

Before choosing a trial, ask whether the exposure could ethically and practically be assigned. Risk factors, harms and prognosis usually cannot be randomised; for these questions an observational design is not a compromise but the correct choice.

5.17 Who is in the study: inclusion and exclusion criteria

Once the design is chosen, the study population must be defined. The figure below shows the process as a funnel, using an observational version of the fluid question.



Figure 5.17: From the source population to the study population: inclusion criteria describe who the question is about, and exclusion criteria remove eligible people for a stated reason.

The **source population** is everyone who could be studied, for example all adults admitted to one ICU over three years. **Inclusion criteria** are the P of the PICO question (Chapter 2) made measurable: age 18 or over, septic shock by a stated definition, a vasopressor started within 24 hours. **Exclusion criteria** remove people who meet the inclusion criteria but must be left out for a stated reason, usually safety or the impossibility of measuring the outcome: chronic dialysis (because fluid balance means something different), or a decision for comfort care only.

Good criteria follow a few rules. Each exclusion has a reason. Exclusions are not simply the opposite of an inclusion ("age under 18" is not an exclusion criterion; such patients are simply not included). Criteria can be checked from records on day 1. And every additional exclusion makes the study population cleaner but narrower, so that the result applies to fewer patients. This trade-off between internal and external validity is discussed in Chapter 6. Boulet's exclusions (renal replacement therapy within 24 hours, end-stage kidney disease, severe malnutrition), together with the upper age limit of 85, mean that its results apply most clearly to patients like those enrolled.

Worked example: design, setting and population for an observational fluid study

Design and justification. A retrospective cohort. The question concerns the effect of fluid balance on death; an RCT would be ideal but is not feasible in one ICU without funding. Existing charts record daily fluid input and output, severity scores and outcomes, so a cohort can estimate the association between a positive balance and death and provide effect sizes for a future trial. The study will say "associated with", not "caused".
Setting. The adult mixed ICU of one teaching hospital; consecutive admissions over three calendar years; data from the electronic chart and the ICU register.
Inclusion. Age 18 or over; septic shock (Sepsis-3) on ICU admission; vasopressor started within 24 hours of admission; ICU stay of at least 72 hours (so that a three-day balance can be calculated).
Exclusion. Chronic dialysis (fluid balance is managed differently); transfer from another ICU after more than 24 hours there (early fluid data are missing); a treatment-limitation decision within 24 hours (the outcome reflects that decision rather than fluid); missing fluid charts for days 1 to 3.
Note that requiring a 72-hour stay excludes patients who died early. This is a deliberate choice with a cost, which Chapter 6 explains as selection bias.

5.18 Defining the outcome

A study is designed and sized around one **primary outcome**: the single result that will answer the question. A primary outcome is not a word such as "mortality" or "improved outcomes"; it is a measurement with three parts, shown below.

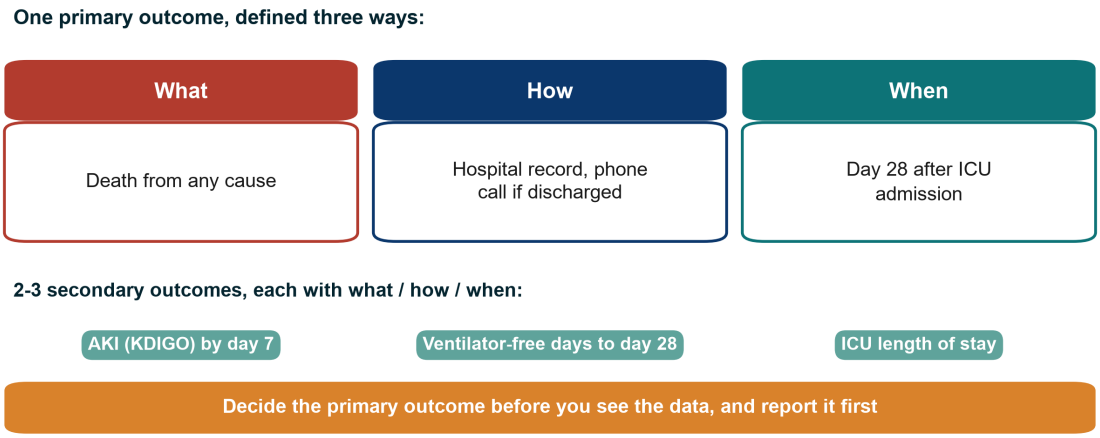


Figure 5.18: One primary outcome defined three ways, by what is measured, how and when, with two or three secondary outcomes defined in the same way and the primary outcome decided before the data are seen.

What is measured (death from any cause), **how** it is measured (the hospital record, or a telephone call if the patient has been discharged) and **when** (day 28 after ICU admission). Boulet's primary outcome, the cumulative fluid balance in ml/kg at day 5, is a good example of all three parts. Secondary outcomes, such as acute kidney injury by day 7 or ventilator-free days to day 28, are defined in the same way. The primary outcome must be chosen before the data are seen and reported first. Having several "primary" outcomes multiplies the chances of finding something by chance and invites selective reporting, a form of spin (Boutron et al. 2010).

5.19 Reporting guidelines for each design

Each design has a reporting guideline: a checklist of what a paper must report so that readers can judge it. You will meet them again in Chapters 9 and 19. CONSORT covers randomised trials (Schulz et al. 2010), with extensions for cluster (Campbell et al. 2012), non-inferiority (Piaggio et al. 2012) and pilot trials (Eldridge et al. 2016). STROBE covers cohort, case-control and cross-sectional studies (Elm et al. 2007). STARD covers diagnostic accuracy studies (Chapter 8), PRISMA covers systematic reviews (Chapter 9), SQUIRE covers quality-improvement work, and COREQ and SRQR cover qualitative research. Reading the relevant guideline before you design a study is one of the best ways to avoid omissions that cannot be repaired later.

Summary

The design of a study is chosen to fit its question, and it determines what the study may claim. Start every question by naming the exposure, the outcome and the comparison group. Descriptive designs (case series, cross-sectional surveys) describe and raise hypotheses; analytical designs compare groups. Three questions name most designs: was the exposure assigned, was it assigned at random, and which way does time run? Cross-sectional studies take a snapshot and give prevalence but cannot show which came first. Case-control studies select by outcome and look back; they suit rare outcomes and give odds ratios. Cohort studies select by exposure and follow forward; they give risks but are open to confounding. Randomised trials let chance assign the exposure, balancing measured and unmeasured factors, and may therefore make causal claims, provided they are well conducted and large enough. Trial variants (crossover, cluster, factorial; non-inferiority with a margin fixed in advance) solve particular problems. Pilot studies test feasibility, not effectiveness; interrupted time series are the design of choice for evaluating changes already made; qualitative designs answer why and how. Inclusion criteria describe the P of the question, each exclusion needs a reason, and the primary outcome is defined by what, how and when. Our two running papers illustrate the central lesson: Boulet's randomised pilot may attribute differences to the strategy but cannot speak about mortality, while Hyun's large cohort can speak about mortality but only in the language of association.

Further reading

- Hulley and colleagues, *Designing Clinical Research* (Hulley et al. 2013). The standard practical textbook on choosing and planning designs, with clear chapters on cohort, case-control, cross-sectional and trial designs.
- Grimes and Schulz, "An overview of clinical research: the lay of the land" (Grimes and Schulz 2002a). A short, readable paper that sets out the family tree of designs used in this chapter; the rest of the same *Lancet* series covers each design in turn.
- Rothman, Greenland and Lash, *Modern Epidemiology* (Rothman et al. 2008). The authoritative reference for the theory behind cohort and case-control designs, for readers who want depth.
- Bernal, Cummins and Gasparrini, "Interrupted time series regression for the evaluation of public health interventions: a tutorial" (Bernal et al. 2017). An accessible introduction to the strongest design for evaluating changes already introduced.
- Eldridge and colleagues, the CONSORT extension for pilot and feasibility trials (Eldridge et al. 2016). Explains what a pilot is for and how to report it.

CHAPTER 6

Bias, confounding and causation

Every study result invites two questions. The first is: *can we trust it?* The second is: *how big is it?* This chapter is about the first question; Chapter 7 is about the second. The order matters. A large, precise effect is worthless if the study that produced it was distorted, and a beautifully narrow confidence interval around a biased estimate simply makes us confident in the wrong answer. Clinicians meet this problem every day. Patients who receive more fluid in the ICU die more often. Patients seen by the dietitian lose more weight. Patients on more antibiotics have worse outcomes. In none of these examples is the treatment necessarily harmful; the association may arise from who receives the treatment, how outcomes were measured, or which patients ended up in the study. This chapter explains the main ways in which a study can be misled, how good design protects against them, and how to reason, carefully and honestly, from an association to a cause.

In this chapter you will learn

- The difference between random error and bias, and why a larger sample fixes only the first.
- How selection bias and information bias (recall, observer and measurement bias) arise, with examples from the ward and the ICU.
- What confounding is, why confounding by indication is so common in clinical data, and how stratification reveals it.
- How to tell confounding from effect modification.
- How randomisation, allocation concealment, blinding, restriction, matching and adjustment protect against bias, and why only randomisation deals with unmeasured confounders.
- The difference between internal and external validity.
- How reverse causation can masquerade as an effect.
- How to use the Bradford Hill viewpoints to judge whether an association is likely to be causal.

6.1 The running example

This chapter uses the fluid question introduced in Chapter 5. In septic shock, does a positive fluid balance in the first days of ICU care increase the risk of death? Two published studies will help us. Boulet and colleagues randomised 50 adults with septic shock to a restrictive strategy for non-resuscitation fluids or to usual care; 48 were analysed, and the primary outcome was fluid balance at day 5 (Boulet et al. 2024). Hyun and colleagues analysed a registry of 11,981 patients with sepsis from 20 hospitals, selected 2,516 who stayed at least three days in the ICU, matched patients with a positive fluid balance to similar patients with a negative balance, and compared death by day 28 (Hyun et al. 2023). The first study is at risk mainly of random error, because it is small; the second is at risk mainly of bias, because it is observational. Later in this book, the simulated FLUID-ICU cohort, created for teaching, raises the same issues in a dataset you can analyse yourself.

6.2 Random error and bias

Imagine throwing darts at a target whose centre is the true answer to a research question. The figure below shows three possible patterns.

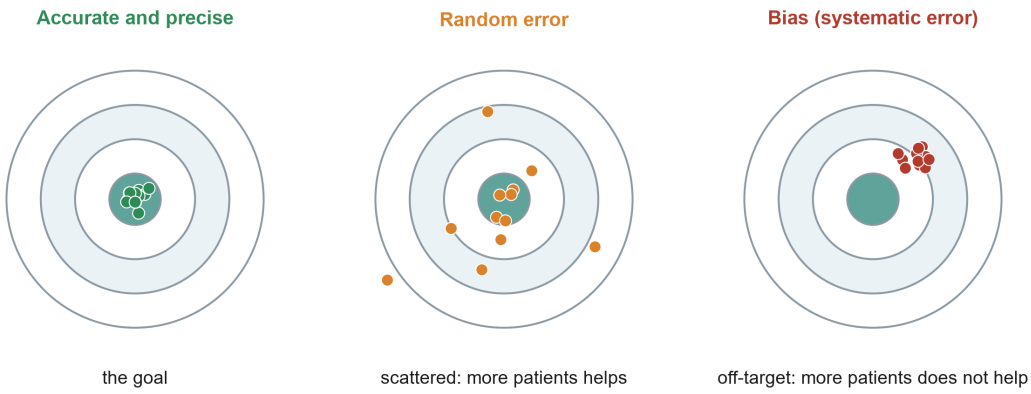


Figure 6.1: Three dartboards: accurate and precise results cluster on the centre; random error scatters results around the centre and shrinks with more patients; bias groups results tightly in the wrong place and is not fixed by more patients.

On the left, the darts are close to the centre and close to one another: the result is both accurate and precise. In the middle, the darts are scattered, but they are scattered *around* the centre. This is **random error**: the play of chance that makes any one sample differ from the population it came from. Random error has two reassuring properties. It does not push results in any particular direction, and it shrinks as the sample grows: with more patients, the darts draw together. Confidence intervals and p-values (Chapter 12) are tools for describing random error.

On the right, the darts are tightly grouped, but in the wrong place. This is **bias**, or systematic error: a distortion that pushes the result consistently in one direction. Bias is not reduced by a larger sample. A study of 100,000 patients that is biased will give a very precise wrong answer, with a narrow confidence interval that tells you nothing about the bias (Sackett 1979).

For our two papers, the main threat differs. Boulet's 48 patients make it vulnerable to random error: with 6 deaths in each group, the true effect on mortality could be anything from substantial benefit to substantial harm. Hyun's thousands of patients make random error small, but as an

observational study it is vulnerable to bias, especially confounding.

> **Size fixes random error, not bias**
>
> A bigger study gives a narrower confidence interval, but the interval is centred wherever the study's methods put it. A narrow interval says nothing about whether the study was biased. Judge the methods first, then the precision.

Epidemiologists group the many named biases into three families (Rothman et al. 2008; Grimes and Schulz 2002b). **Selection bias** concerns *who* got into the study, or stayed in it. **Information bias** concerns *how* exposures and outcomes were measured. **Confounding** concerns the *mixing* of the effect of the exposure with the effect of other factors. We take each in turn.

6.3 Selection bias: who got into the study?

Selection bias arises when the people who enter or remain in a study differ systematically from the people we want to learn about, in a way that is related to both exposure and outcome. The result may then be correct for the people studied but wrong, or at least different, for the population of interest.

Hyun's study gives a clear example. Only patients who remained in the ICU for at least three days and had fluid data were analysed: 2,516 of the 11,981 registry patients. The figure below shows what such a filter does.

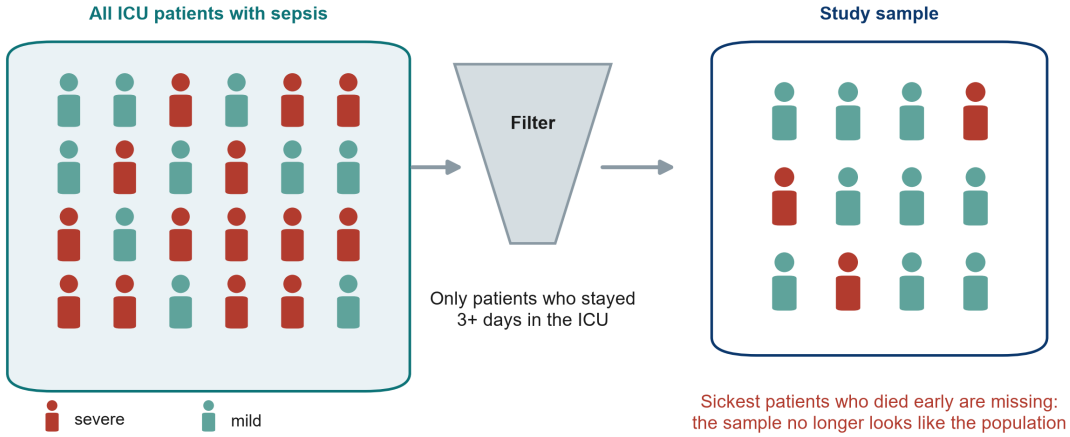


Figure 6.2: Restricting a study to patients who stayed three or more days in the ICU filters out the sickest patients who died early, so the study sample no longer looks like the population of all ICU patients with sepsis.

The sickest patients, who died in the first days, are missing, and so are the mildest, who left the ICU early. The result therefore applies to "patients with sepsis who are still in the ICU on day 3", not to all patients with sepsis. The restriction was necessary (a three-day fluid balance cannot be calculated for a patient who died on day 1), but it changes what the result means.

Selection bias has many common sources:

- **Volunteers and responders.** People who agree to take part, or who return a questionnaire, are often healthier and more motivated than those who do not.
- **Refusal of consent.** In trials, patients or families who refuse may differ from those who agree.

- **Loss to follow-up.** If patients who do badly are more likely to be lost, the outcome among those who remain looks better than it is.
- **Referral filters.** A tertiary centre sees the most complex cases; results may not apply to district hospitals.
- **Survivor selection.** Studying only patients who survived to a certain point excludes those who did not.

Selection bias is not limited to surveys. Randomised trials are also selective: Boulet screened 1,201 patients to randomise 50 (about 4%), and its exclusion criteria (age over 85, need for renal replacement therapy, severe malnutrition) define a particular group. Within a properly randomised trial, selection does not bias the comparison between groups, because both groups come from the same selected population. But it limits to whom the result applies, a point we return to under external validity.

Worked example: a follow-up clinic study

A study of quality of life after intensive care recruits only the survivors who attend the hospital's follow-up clinic six months after discharge. What is the problem, and how could it be reduced?
Problem. Survivors who come to the clinic are probably healthier, more mobile and more motivated than those who do not. Those who are too unwell to travel, who live far away, or who have died are missing. The study will paint too optimistic a picture of life after intensive care. This is selection bias, including loss to follow-up.
Remedies. Plan follow-up from the start for every survivor; contact all of them by telephone or post; record how many could not be reached and compare their baseline characteristics with those of the responders; report the numbers clearly in a flow diagram.

6.4 Information bias: how things were measured

Information bias (also called measurement bias) arises when the data on exposure or outcome are wrong in a systematic way. The figure below illustrates three common forms.

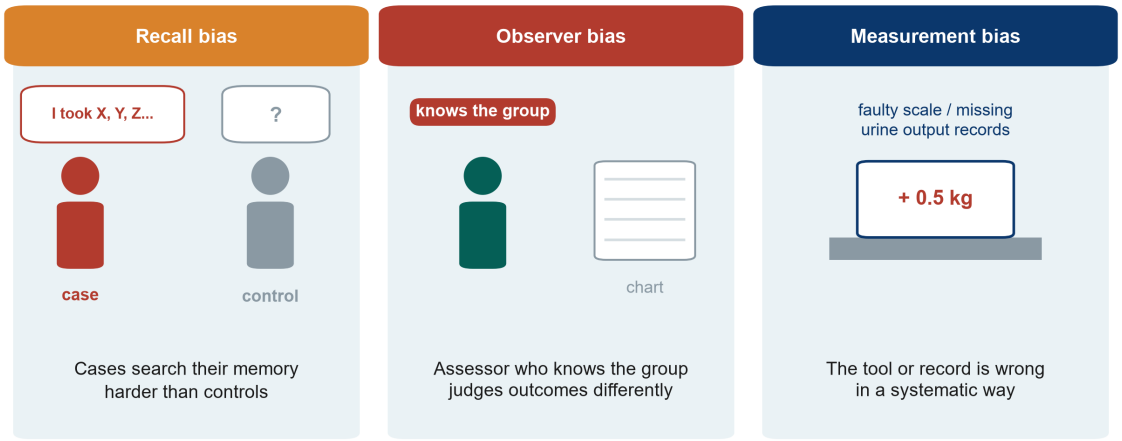


Figure 6.3: Three forms of information bias: recall bias, in which cases search their memory harder than controls; observer bias, in which an assessor who knows the group judges outcomes differently; and measurement bias, in which a tool or record is systematically wrong.

Recall bias. In a case-control study, cases (or their families) search their memories harder than

controls. Mothers of babies born with a birth defect tend to remember medicines taken in pregnancy in more detail than mothers of healthy babies, so exposure appears more common among cases even if it is not. The remedy is to use exposure information recorded *before* the outcome was known (prescription or pharmacy records), to ask cases and controls exactly the same structured questions, and to keep interviewers unaware of who is a case.

Observer bias. An assessor who knows which group a patient is in may judge a subjective outcome differently. Whether a patient is "ready for discharge", whether delirium is present or how much pain a patient has are all judgements that can be influenced, often unconsciously, by expectations. In Boulet's trial the clinicians knew the allocation; for an objective outcome such as measured fluid balance this matters less than it would for a judgement-based outcome. Remedies include an assessor who is blinded to the group, objective and predefined outcome criteria, and standard definitions with training.

Measurement bias. The instrument or the record itself is wrong in a systematic way: a scale that always reads 0.5 kg too high, or urine output that is under-recorded on night shifts. If the error affects all groups equally, it mostly blurs the comparison. If it affects one group more than another (for example, if the faulty scale is used on one ward only), it distorts the comparison.

This last point has a technical name. When measurement errors are equally likely in all groups, the misclassification is **non-differential**; it usually weakens an association towards no effect (making real effects harder to find). When errors differ between groups, the misclassification is **differential**, and it can push the result in either direction, creating effects that do not exist or hiding ones that do (Rothman et al. 2008).

> **! Common mistake: hard outcomes are immune to information bias**
>
> Death is the most reliable outcome there is, but the *timing* of death, its cause and whether it occurred before or after a cut-off such as day 28 can all be misrecorded, especially after discharge. "Death from any cause by day 28, from the hospital record or a telephone call" is much more reliable than "death attributable to sepsis".

6.5 Confounding

6.5.1 What a confounder is

Confounding is the mixing of the effect of the exposure with the effect of a third factor. Consider the fluid example. Clinicians give more fluid to patients who look sicker: those with deeper shock, higher lactate, more organ failure. Those same patients are also more likely to die, whatever fluid they receive. So, when we compare patients who received more fluid with patients who received less, the high-fluid group dies more often, even if the fluid itself does nothing. The figure below shows the triangle of relationships.

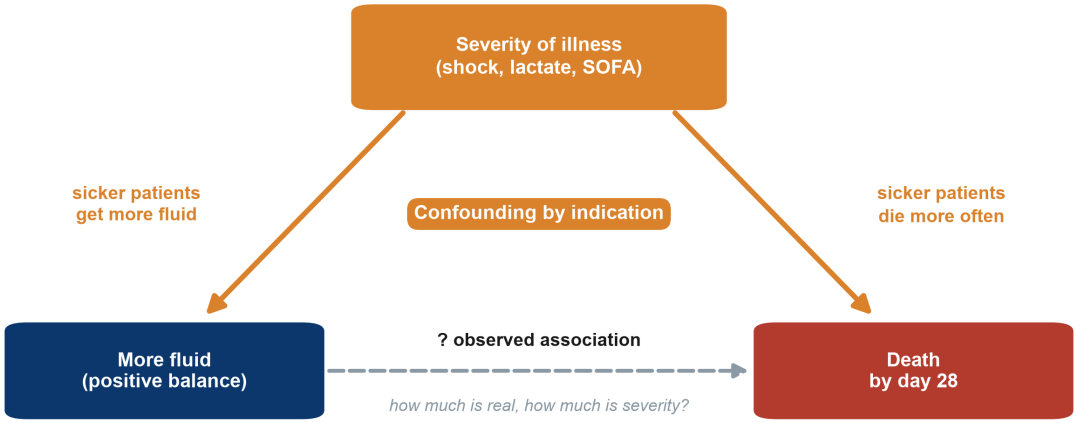


Figure 6.4: Confounding by indication: severity of illness leads clinicians to give more fluid and also independently causes death, creating an observed association between fluid and death that may be partly or wholly due to severity.

A variable is a **confounder** when three conditions hold (Rothman et al. 2008):

1. It is associated with the exposure (sicker patients receive more fluid).
2. It is a cause of the outcome, or a marker of a cause, independently of the exposure (sicker patients die more often).
3. It is not on the causal pathway between exposure and outcome (severity of illness is not a consequence of the fluid).

The third condition matters. If fluid caused kidney injury and kidney injury caused death, kidney injury would be a step on the pathway (a *mediator*), not a confounder, and adjusting for it would remove part of the very effect we are trying to measure.

6.5.2 Confounding by indication

The fluid example is a special case called **confounding by indication**: the reason for giving a treatment is itself linked to the outcome. It is the single most common problem in observational studies of treatments, because in clinical practice treatments are not given at random; they are given to the patients who seem to need them. The useful question to ask of any observational comparison of treatments is: *why did these patients receive the exposure?* The reason is often the confounder.

Worked examples: find the confounder

In each case, a third factor drives both the exposure and the outcome.

- *Antibiotics and death.* Patients who receive more antibiotics die more often. Confounder: severity of infection. Sicker patients receive more drugs.
- *Dietitian and weight loss.* Patients seen by the dietitian lose more weight. Confounder: malnutrition at admission, which is the reason for the referral.
- *Vasopressors and arrhythmia.* Patients on vasopressors have more arrhythmias. Confounder: depth of shock. Here the picture is mixed, because vasopressors can genuinely cause arrhythmias *and* deeper shock drives both. Real associations are often a mixture of a true effect and confounding, which is why we need randomisation or careful adjustment to separate them.
- *Flu vaccination and death.* People who choose to be vaccinated die less often over the following winter. Confounder: general health and health-seeking behaviour (the "healthy-user" effect). People who are frail or bed-bound are less likely to attend for vaccination and more likely to die.

6.5.3 Confounding in numbers

Confounding can be revealed, and removed, by comparing like with like. The figure below shows hypothetical data for 200 ICU patients, 100 who received high volumes of fluid and 100 who received low volumes.

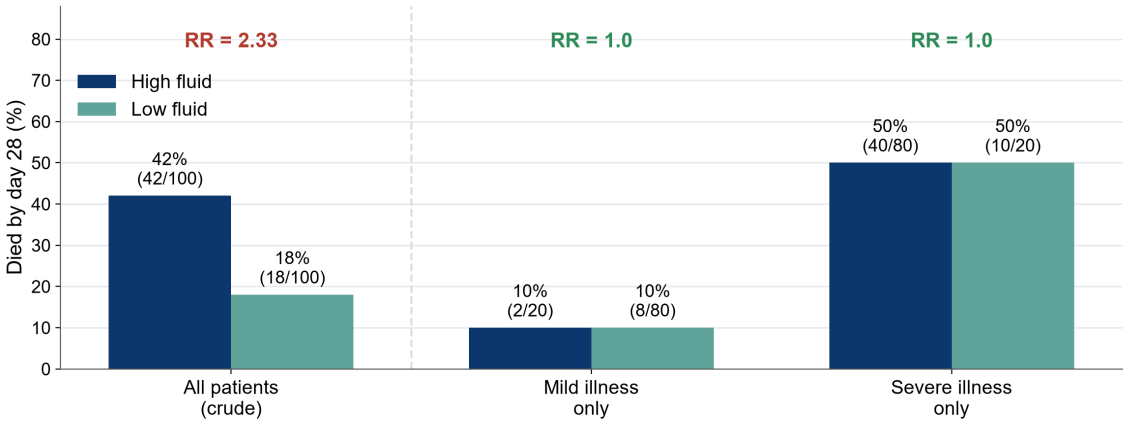


Figure 6.5: Hypothetical data showing confounding: in all patients together, 42% of high-fluid patients and 18% of low-fluid patients died (risk ratio 2.33), but within mild illness (10% versus 10%) and within severe illness (50% versus 50%) the risk ratio is 1.0.

Worked example: an association that disappears

All patients together (the crude comparison). Of 100 high-fluid patients, 42 died (42%). Of 100 low-fluid patients, 18 died (18%). The risk ratio (Chapter 7) is $0.42/0.18 = 2.33$. It looks as if high fluid volumes more than double the risk of death.
Now split the patients by severity of illness.

Table 6.1: Hypothetical fluid and mortality data, stratified by severity of illness.

	High fluid: died / total	Low fluid: died / total	Risk ratio
Mild illness	2 / 20 (10%)	8 / 80 (10%)	1.0
Severe illness	40 / 80 (50%)	10 / 20 (50%)	1.0
All patients (crude)	42 / 100 (42%)	18 / 100 (18%)	2.33

Within each level of severity, fluid makes no difference at all. Why did the crude comparison suggest harm? Because 80 of the 100 high-fluid patients were severely ill, compared with only 20 of the 100 low-fluid patients. Severity, not fluid, explained the whole of the crude difference. Check the arithmetic: high-fluid deaths are $2 + 40 = 42$; low-fluid deaths are $8 + 10 = 18$. This is what "adjusting for severity" does: it compares patients of similar severity with one another. Stratification, as here, is the simplest form; regression models (Chapter 13) and propensity scores do the same thing for many confounders at once.

Real data are rarely so tidy. Usually adjustment shrinks an association rather than abolishing it, and the question becomes how much of the remaining association is real. The answer depends on whether every important confounder was measured, and measured well. That is never guaranteed in observational data, which is why adjusted results from observational studies are still described as associations.

! Common mistake: a confounder is any variable that differs between groups

Groups differ in many ways that do not matter. A difference only confounds the comparison if the variable is linked to the exposure *and* affects the outcome, and is not a consequence of the exposure. Conversely, a variable can be an important confounder even if the difference between groups looks modest, if its effect on the outcome is strong.

6.6 Confounding or effect modification?

When we split data into strata, two very different things can happen. The figure below shows both.

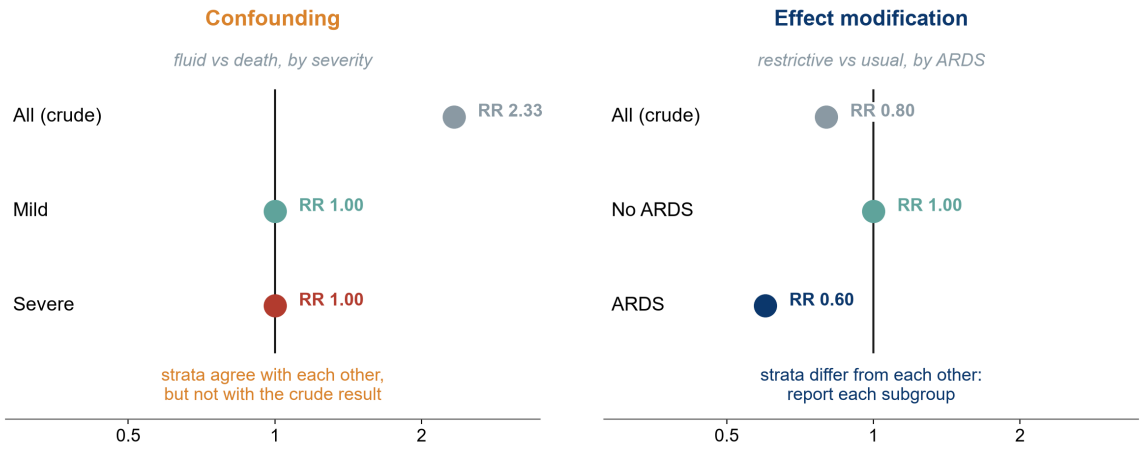


Figure 6.6: Confounding versus effect modification: with confounding, the strata agree with each other but differ from the crude result; with effect modification, the strata differ from each other and each subgroup should be reported.

On the left is our confounding example. The strata *agree with each other* (risk ratio 1.0 in both) but *differ from the crude result* (2.33). The third factor, severity, created a false association. The response is to adjust for it and report the adjusted estimate.

On the right is a hypothetical example of **effect modification**, also called **interaction**. Suppose a restrictive fluid strategy reduced mortality in patients with acute respiratory distress syndrome (ARDS), with a risk ratio of 0.60, but made no difference in patients without ARDS (risk ratio 1.00). The overall risk ratio of 0.80 is an average of two genuinely different effects. Here the strata *differ from each other*. This is not a bias to be removed; it is a finding about who benefits, and each subgroup should be reported. Averaging it away would hide useful information.

• A rule of thumb

Strata similar to each other but different from the crude result: confounding (remove it by adjusting). Strata different from each other: effect modification (report it by subgroup).

Worked example: confounding, effect modification or neither?

1. *Vasopressor X versus Y and death.* Clinicians chose X mainly for patients with high lactate. Overall risk ratio 1.80; lactate below 4 mmol/L: 1.00; lactate 4 or above: 1.00. **Confounding** by lactate (by indication): the strata agree with each other but not with the crude result. Report the lactate-adjusted estimate, about 1.0.
2. *Early mobilisation and delirium.* Overall risk ratio 0.70; age under 65: 0.50; age 65 or over: 0.95. **Effect modification** by age: the strata differ from each other. Report each age group, ideally as a pre-specified subgroup with a formal test for interaction.
3. *Hand-hygiene programme and infection.* Overall 0.80; day shift 0.80; night shift 0.79. **Neither**: the strata agree with each other and with the crude result. Report the overall 0.80.

Subgroup findings need caution. If a study looks at twenty subgroups, one is likely to show a

"significant" difference by chance alone. Credible effect modification is usually pre-specified, limited to a few subgroups with a plausible reason, supported by a formal test for interaction, and consistent across studies (Wang et al. 2007). Chapter 13 shows how interaction is represented in a regression model.

6.7 How design fights bias

Bias is far easier to prevent at the design stage than to repair at the analysis stage. The main tools are summarised in the table below.

Table 6.2: Tools that protect against bias, the stage at which they act and their limits.

Tool	Stage	What it protects against	Limits
Randomisation	Design	Confounding by measured *and unmeasured* factors	Balance is on average; small trials can be unbalanced by chance
Allocation concealment	Design	Selection bias at enrolment in a trial	Always possible; must be planned
Blinding	Design	Biased care and biased outcome assessment	Often impossible for clinicians in ICU trials
Restriction	Design	Confounding by the restricted factor	Narrows to whom results apply
Matching	Design or analysis	Confounding by matched factors	Measured factors only
Stratification, regression, propensity scores	Analysis	Confounding by measured factors	Measured factors only; residual confounding remains

6.7.1 Randomisation

When chance decides who receives the intervention, each patient's characteristics, known or unknown, are as likely to end up in one group as in the other. The figure below shows 24 patients, of whom 10 are severely ill (a known confounder) and 6 carry a hidden factor that nobody measured.

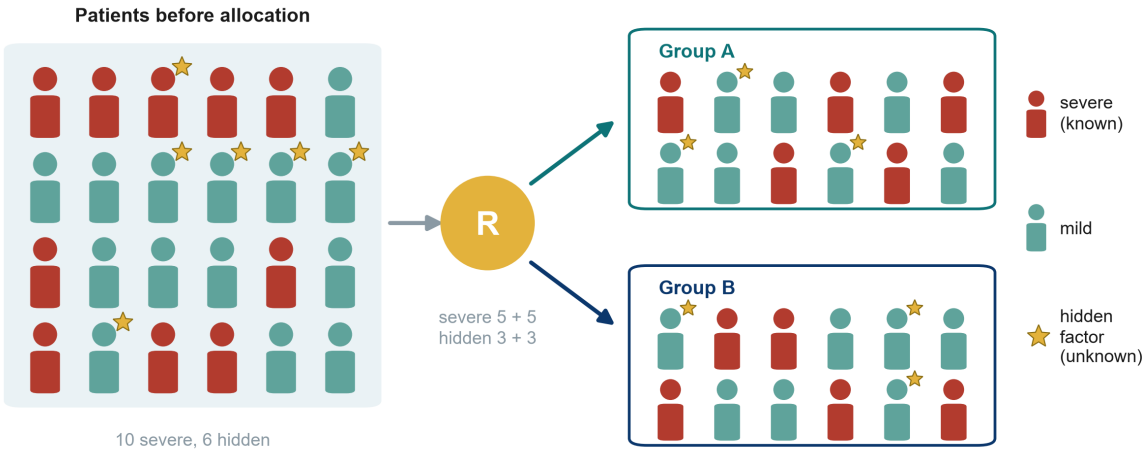


Figure 6.7: Randomisation sends patients to group A or group B by chance, so both groups receive a similar mix of severe patients (a known factor) and of a hidden factor that nobody measured.

After randomisation each group contains five severely ill patients and three with the hidden factor. In real trials the balance is rarely this exact. Randomisation balances groups *on average*; in small trials, chance imbalances are common. In Boulet's trial, baseline serum creatinine was higher in one group, which is not surprising with 24 patients per arm, and the authors discussed it as a possible explanation for some differences. Larger trials balance better, which is one reason (besides precision) why size matters. The table of baseline characteristics, usually Table 1 of a trial report, lets readers check the balance.

6.7.2 Allocation concealment and blinding

Two protections are often confused. The figure below separates them in time.

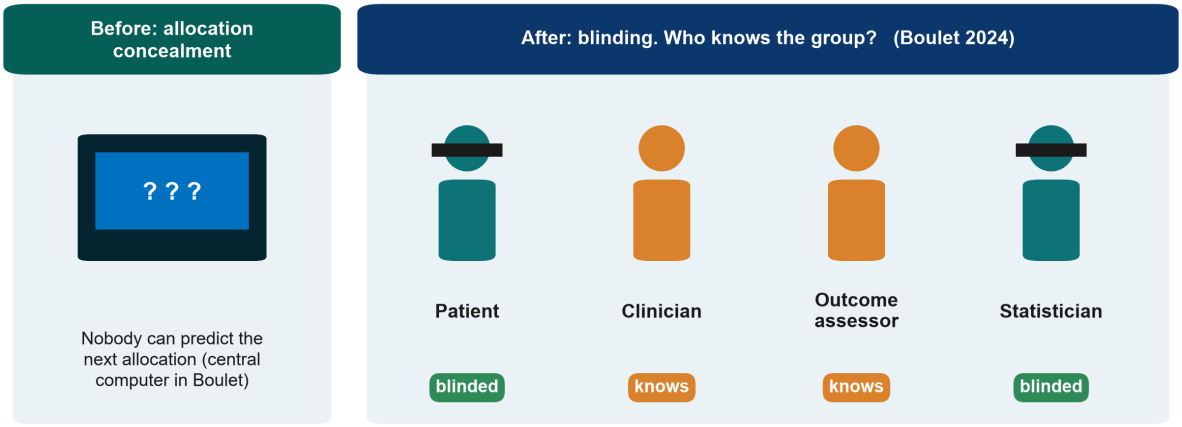


Figure 6.8: Allocation concealment happens before randomisation so that nobody can predict the next allocation; blinding happens afterwards and concerns who knows each patient's group. In Boulet 2024 patients and statisticians were blinded, while clinicians and outcome assessors knew the group.

Allocation concealment happens *before* and at randomisation: the person enrolling a patient cannot know or predict the next allocation. Without it, a clinician who suspects that the next envelope says "new treatment" might steer a very sick patient away from the trial or delay enrolment, and the groups would no longer be comparable. Central, computer-based randomisation, as in Boulet's trial, conceals allocation well; envelopes left on the desk do not. Concealment is always possible and always required (Schulz and Grimes 2002a).

Blinding (masking) happens *after* randomisation and concerns who knows each patient's group: patients, clinicians, outcome assessors, data analysts. Blinding protects against differences in care (clinicians may treat patients differently if they know the group) and against biased outcome assessment. In Boulet's trial, patients and statisticians were blinded, but clinicians and investigators were not: a "single-blind" trial. Terms such as "single-blind" and "double-blind" have no fixed meaning, so good reports state exactly who was blinded (Schulz and Grimes 2002b). In fluid trials and many ICU trials, blinding the treating clinicians is impossible. The next best steps are to blind the outcome assessor and to choose an objective outcome.

! Common mistake: unblinded means useless

Lack of blinding matters most for subjective outcomes assessed by people who know the group. For an objective outcome such as death, or fluid balance measured from charts, its effect is much smaller. Concealment of allocation, by contrast, is never optional.

6.7.3 Restriction, matching and adjustment

In observational studies the investigator cannot randomise, but can still reduce confounding. **Restriction** limits the study to one level of a confounder (for example, only patients with septic shock), so that it cannot vary between groups. **Matching** pairs each exposed patient with an unexposed patient who is similar on chosen characteristics. **Adjustment** uses statistics in the analysis (stratification, regression models or propensity scores) to compare like with like.

A **propensity score** is each patient's estimated probability of receiving the exposure, calculated from his or her measured characteristics (Austin 2011). Patients with similar scores are similar, on average, across all the characteristics that went into the score. Hyun and colleagues used this approach: the probability of a positive fluid balance was modelled from age, sex, SOFA score, septic shock, referring facility, comorbidities, site of infection, adjunct interventions, pathogen and fluid output, and each positive-balance patient was matched to a negative-balance patient with a similar score (Hyun et al. 2023). The figure below shows the idea and its limit.

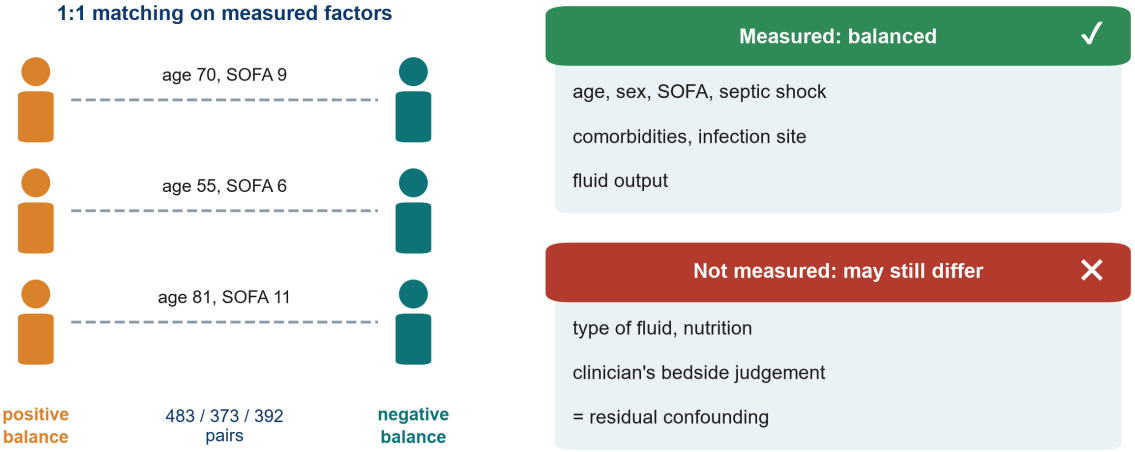


Figure 6.9: Propensity-score matching pairs patients on measured factors such as age, sex, SOFA score, septic shock, comorbidities, infection site and fluid output, but factors that were not measured, such as fluid type, nutrition and the clinician's bedside judgement, may still differ: residual confounding.

After matching, the measured characteristics were well balanced. But the authors themselves note that information on the type of fluid and nutritional support was not available, and that diuretic use was not recorded. The clinician's bedside judgement of how sick a patient looks, which drives fluid prescribing and also predicts death, is never fully captured by any score. Whatever confounding remains after adjustment is called **residual confounding**.

• Only randomisation protects against confounders nobody measured

Restriction, matching, stratification, regression and propensity scores all work only on factors that were measured, and measured well. Randomisation is the only tool that also balances factors that were never recorded. This is why a cohort, however carefully matched, says "associated with", and why the question "does restriction save lives?" still needs a large randomised trial.

! Common mistake: propensity-score matching is as good as randomisation

Propensity-score matching is a clever way of balancing many measured factors at once. It is still matching on measured factors. Read the matched baseline table, and then ask what is *not* in it.

6.8 Internal and external validity

Two different questions should be asked of every study, in a fixed order. The figure below sets them out.

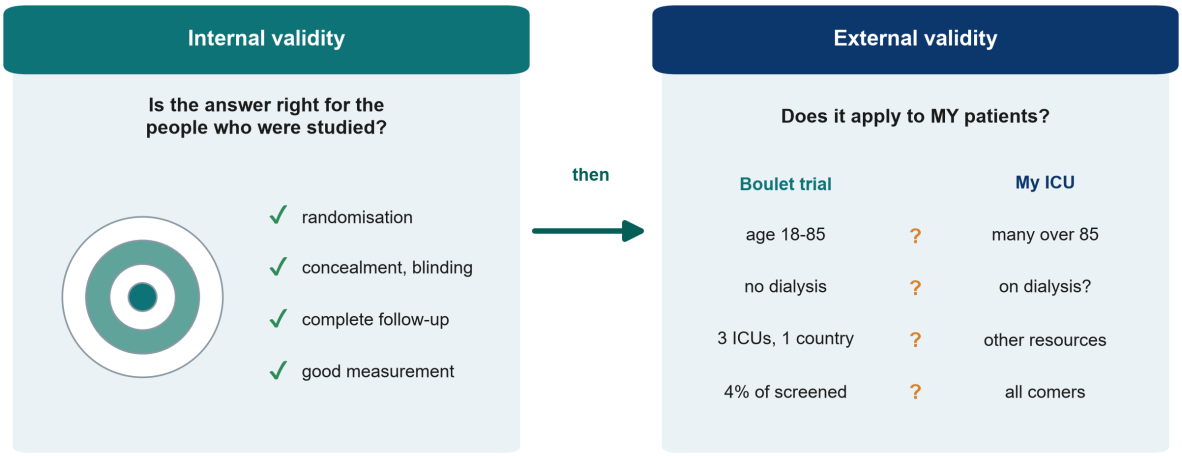


Figure 6.10: Internal validity asks whether the answer is right for the people who were studied; external validity then asks whether it applies to my patients, comparing the trial population with one's own.

Internal validity asks: *is the answer correct for the people who were in the study?* It is protected by randomisation, concealment, blinding, complete follow-up and good measurement, the tools described above. **External validity** (generalisability) asks: *does the answer apply to my patients, in my setting?* (Rothwell 2005) To judge it, compare the study population with your own. Boulet's trial enrolled adults aged 18 to 85, without the need for dialysis or severe malnutrition, within 24 hours of starting vasopressors, in three ICUs in one country, and only about 4% of the patients screened were randomised. An ICU that cares for many patients over 85 or on chronic dialysis must ask whether the result transfers.

Internal validity comes first. A biased result generalises nowhere. But a perfectly valid trial in highly selected patients may still not help a particular unit. The trade-off appears at the design stage: every additional exclusion criterion makes a study cleaner (higher internal validity) and narrower (lower external validity).

6.9 Reverse causation

Before claiming that X causes Y, make sure that Y did not cause X. In **reverse causation**, the supposed outcome, or an early stage of it, is actually the cause of the supposed exposure. The figure below shows how this could happen with fluids.

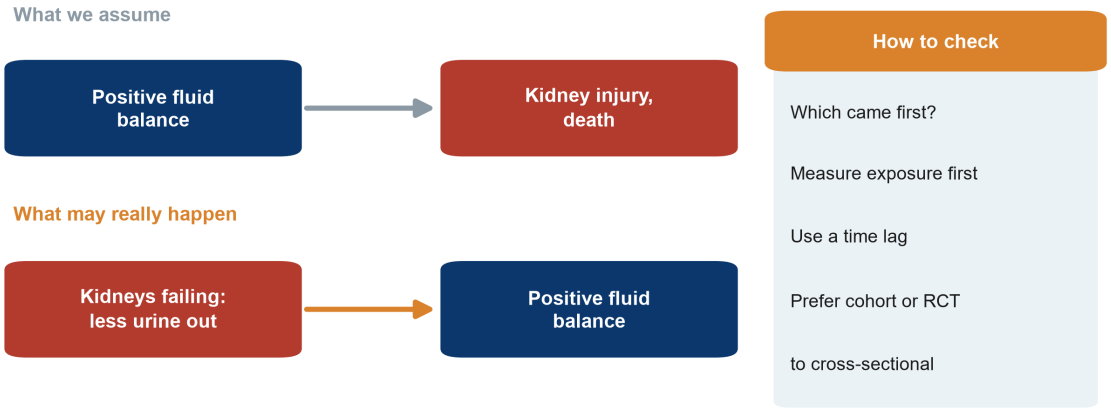


Figure 6.11: Reverse causation: we may assume that a positive fluid balance causes kidney injury and death, but failing kidneys produce less urine, which itself makes the fluid balance positive.

We may assume that a positive fluid balance leads to kidney injury and death. But failing kidneys produce less urine, and less urine output means a more positive fluid balance. The arrow may run, at least partly, the other way. Hyun and colleagues anticipated this by including fluid output in their propensity score, so that patients were compared with others who had similar urine and drain output (Hyun et al. 2023). Other examples are familiar: people with early, undiagnosed illness stop exercising, so inactivity looks harmful; deteriorating patients receive more tests and more drugs, so tests and drugs look dangerous; delirious patients receive more antipsychotics, so antipsychotics appear to cause delirium (they may also genuinely contribute).

The defences are to make sure the exposure was measured *before* the outcome began, to introduce a time lag between exposure and outcome, and to prefer cohort or trial designs to cross-sectional ones. Reverse causation can affect cohorts as well as cross-sectional studies, whenever the exposure is measured while the outcome is already developing.

6.10 From association to causation

How, then, do we decide whether an association reflects a cause? There is no single test. In 1965 the medical statistician Austin Bradford Hill proposed nine "viewpoints" from which an association can be examined before deciding that the most likely interpretation is causation (Hill 1965). The figure below summarises them.

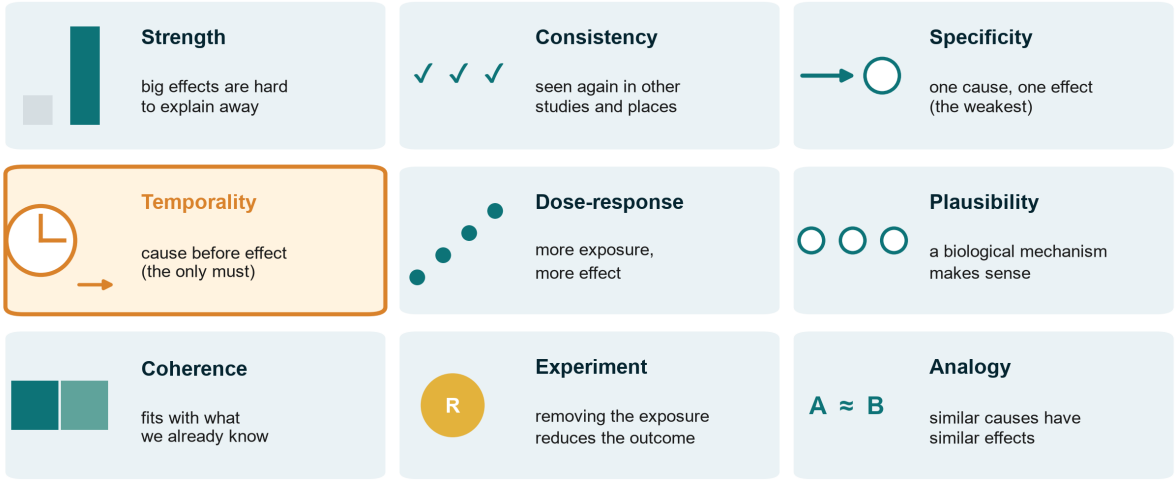


Figure 6.12: The nine Bradford Hill viewpoints for judging causation: strength, consistency, specificity, temporality (the only essential one), dose-response, plausibility, coherence, experiment and analogy.

- **Strength.** A large effect is harder to explain away by confounding than a small one.
- **Consistency.** The association is seen again in different studies, places and populations.
- **Specificity.** One cause leads to one effect. This is the weakest viewpoint, since many causes have many effects (smoking causes much more than lung cancer).
- **Temporality.** The cause comes before the effect. This is the only viewpoint that is essential.
- **Biological gradient (dose-response).** More exposure leads to more effect.
- **Plausibility.** A biological mechanism makes sense.
- **Coherence.** The association fits with what is already known about the disease.
- **Experiment.** Removing or reducing the exposure reduces the outcome. A randomised trial is the strongest form of this evidence.
- **Analogy.** Similar causes are known to have similar effects.

Hill was explicit that these are not criteria to be scored or a checklist that "proves" causation. They are clues that raise or lower our confidence. A causal judgement is built from many pieces of evidence, from different designs, and it remains open to revision.

Worked example: is a positive fluid balance a cause of death in sepsis?

Apply the viewpoints to Hyun's findings and the wider evidence.

- *Strength:* moderate. Hazard ratios of about 1.6 to 2.1 for days 2 and 3 are not so large that confounding could not explain them.

- *Consistency:* broadly yes; other cohorts cited by the authors report similar associations, although findings on the first ICU day disagree.
- *Temporality:* fluid balance was measured before death, but reverse causation (low urine output making the balance positive) is plausible.
- *Dose-response:* the authors report a stepwise increase in the hazard of death with increasing cumulative balance on days 2 and 3, which supports a causal reading.
- *Plausibility:* yes; excess fluid can worsen tissue oedema and organ dysfunction.
- *Experiment:* this is the weak point. Randomised trials of fluid restriction, including Boulet's small pilot and larger trials cited by both papers, have not shown a mortality benefit.

Judgement: a causal effect of fluid accumulation on death is possible and plausible, but not proven. The honest summary is the one Hyun's abstract uses: a positive balance on days 2 and 3 *was associated with* higher mortality.

! Common mistake: if most Bradford Hill viewpoints are met, causation is proven

The viewpoints are not votes. An association can satisfy most of them and still be confounded, and a causal effect can fail several (specificity in particular). Temporality is the only one that must hold.

6.11 Putting it together: spotting the bias

When reading any study, ask the three family questions in turn. Who got in, and who was lost (selection)? How were exposure and outcome measured, and could the measurement differ between groups (information)? Why did people receive the exposure, and could that reason also affect the outcome (confounding)? Then ask whether the result could be reverse causation, whether the strata suggest effect modification, and whether the result applies to your patients.

Worked example: spot the bias and propose a remedy

1. *Mothers of babies with a birth defect recall medicines in pregnancy better than other mothers.* Recall bias (information bias). Remedy: use prescription records made before the birth; ask cases and controls the same structured questions; blind interviewers.
2. *In an unblinded early-mobilisation trial, the nurse who knows each patient's group decides whether the patient is "ready for discharge".* Observer bias (information bias). Remedy: a blinded assessor; objective, predefined discharge criteria; or a harder outcome such as actual days to discharge.
3. *A survey of ICU survivors' quality of life recruits only those who attend the follow-up clinic.* Selection bias, including loss to follow-up. Remedy: contact every survivor and report non-responders.
4. *In a registry, patients with higher lactate received more fluid and also died more often; the authors conclude that fluid increases mortality.* Confounding by indication. Remedy: randomise; or measure severity at baseline (lactate, SOFA, vasopressor dose) and restrict, match or adjust; and conclude "associated with", not "increases".

When you plan your own study, the same questions become a design checklist. For an observational version of the fluid question, a well-designed protocol would name confounding by severity as the main threat and record SOFA score, lactate, vasopressor dose, age and comorbidity at admission for adjustment or matching; acknowledge that requiring a 72-hour ICU stay excludes early deaths and report how many were excluded; and reduce information bias in fluid charts with predefined definitions and double data entry for a sample of records. "We will be careful" is not a mitigation; each bias needs a concrete step. Increasingly, observational studies of treatments are also designed by explicitly describing the randomised trial they are trying to imitate (a "target trial"), which forces clear decisions about eligibility, time zero and the comparison (Hernán and Robins 2016).

Summary

Random error scatters results around the truth and shrinks with larger samples; bias pushes results systematically away from the truth and is untouched by sample size. Selection bias concerns who enters and stays in a study; information bias concerns how exposures and outcomes are measured, through recall, observer and measurement errors that may be non-differential or differential. Confounding mixes the effect of the exposure with that of a third factor linked to both exposure and outcome; confounding by indication, in which sicker patients receive more treatment and also do worse, is the commonest form in clinical data. Stratification reveals confounding when strata agree with each other but not with the crude result; effect modification is present when strata differ from each other, and it should be reported rather than removed. Randomisation with concealed allocation is the only protection against unmeasured confounders; blinding protects care and outcome assessment; restriction, matching and adjustment, including propensity scores, deal only with measured factors and leave residual confounding. Internal validity comes before external validity. Reverse causation must be excluded by ensuring that exposure precedes outcome. Finally, causation is a judgement, informed by the Bradford Hill viewpoints, of which only temporality is essential. For our running example, the evidence supports the statement that a positive fluid balance on days 2 and 3 is associated with higher mortality, and leaves open whether it causes it.

Further reading

- Bradford Hill, “The environment and disease: association or causation?” (Hill 1965). The original nine viewpoints, still worth reading in full for their wisdom and caution.
- Grimes and Schulz, “Bias and causal associations in observational research” (Grimes and Schulz 2002b). A concise and practical guide to selection bias, information bias and confounding.
- Rothman, Greenland and Lash, *Modern Epidemiology* (Rothman et al. 2008). The definitive reference on confounding, effect modification and misclassification.
- Schulz and Grimes, “Allocation concealment in randomised trials” (Schulz and Grimes 2002a) and “Blinding in randomised trials” (Schulz and Grimes 2002b). Two short papers that make clear the difference between concealment and blinding.
- Austin, “An introduction to propensity score methods” (Austin 2011). A readable introduction to the method used by Hyun and colleagues.

CHAPTER 7

Measuring disease frequency and association

Chapter 6 asked whether a study result can be trusted. This chapter asks the second question: *how big is it?* To answer it we need a small set of measures. Some describe how common a condition is: how many patients have a pressure injury today, how many develop one each week. Others describe how strongly an exposure is linked to an outcome: how much more likely death is with one fluid strategy than another, or how many patients must be treated for one to benefit. These measures appear in the abstract of almost every clinical paper, and they are frequently misread, sometimes by the authors themselves. The good news is that nearly all of them come from one simple tool, the two-by-two table, and the only arithmetic required is division. This chapter builds the measures one at a time, with worked examples, and explains how to choose between them and how to report them honestly.

In this chapter you will learn

- The difference between prevalence and incidence, and between a risk and a rate.
- The difference between risk and odds.
- How to lay out a two-by-two table so that every measure is calculated the same way.
- How to calculate and interpret the risk ratio, risk difference, number needed to treat and odds ratio.
- Why the odds ratio is close to the risk ratio only when the outcome is rare, and why case-control studies can only give an odds ratio.
- What hazard ratios and mean differences are, and where you will meet them.
- Why results should be reported in both relative and absolute terms.

7.1 The running example

We continue with the question introduced in Chapter 5: in adults with septic shock, does a restrictive fluid strategy reduce mortality? The randomised pilot trial by Boulet and colleagues reported 6 deaths among 24 patients in each group by day 28 (Boulet et al. 2024). The registry cohort by Hyun and colleagues reported hazard ratios for death by day 28 comparing positive with negative fluid balance (Hyun et al. 2023). Because a pilot trial with 12 deaths is too small to illustrate the measures clearly, much of this chapter also uses a *hypothetical* larger trial of the same question, with 100 patients per group. Its numbers are invented for teaching and are labelled as such wherever they appear. Ward examples (pressure injuries, central-line infections, candidaemia) complete the set.

7.2 How common is it? Prevalence and incidence

7.2.1 The bathtub

Two measures of disease frequency are easily confused. The figure below uses a bathtub to separate them.

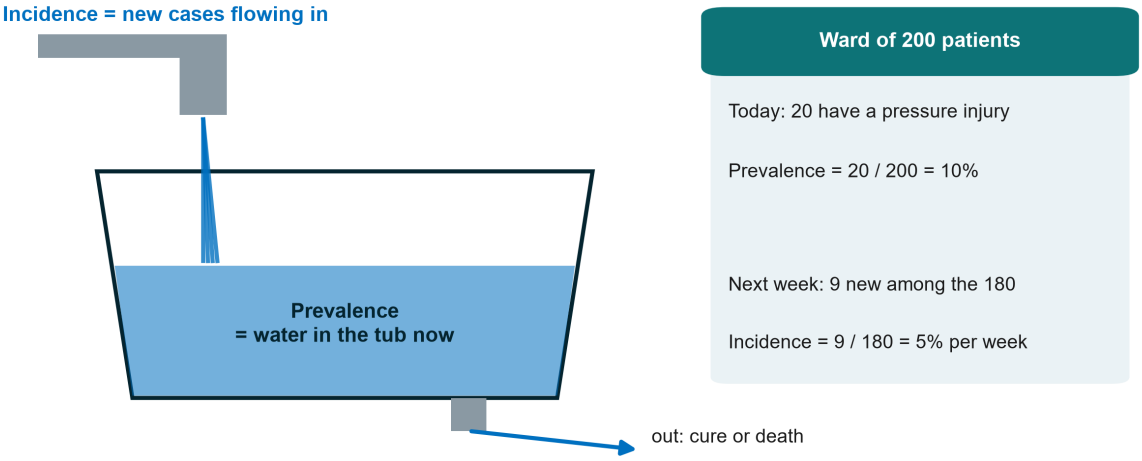


Figure 7.1: Incidence is the flow of new cases into the bathtub; prevalence is the water in the tub at a given moment; cure or death drains water out. On a ward of 200 patients, 20 have a pressure injury today (prevalence 10%) and 9 of the 180 without one develop a new injury over a week (incidence 5% per week).

Prevalence is the water in the tub now: the proportion of a population that *has* the condition at a given time.

$$\text{prevalence} = \frac{\text{number of people with the condition at a given time}}{\text{number of people in the population at that time}}$$

Incidence is the flow from the tap: the occurrence of *new* cases over a period of time, among people who did not have the condition at the start. Water leaves the tub through the drain when patients recover or die.

Worked example: pressure injuries on a ward

A ward has 200 patients. On Monday, 20 of them have a pressure injury.
Prevalence on Monday $= 20/200 = 10\%$.
Over the following week, 9 *new* pressure injuries develop. Who could have developed a new injury? Only the 180 patients who did not already have one: they are the population **at risk**.
Incidence over one week $= 9/180 = 5\%$ per week.
Two details make this calculation correct. The denominator for incidence excludes people who already have the condition, and the result always carries a time period ("per week"). An incidence without a time period is meaningless.

Prevalence depends on both how often new cases arise and how long they last. When a condition is fairly rare and the situation is stable, an approximate relationship holds:

$$\text{prevalence} \approx \text{incidence rate} \times \text{average duration}$$

This explains some patterns that otherwise seem puzzling. A long-lasting condition such as

diabetes can have a high prevalence despite a modest incidence. A short-lived condition such as the common cold, or a brief episode of delirium, can have a high incidence but a low prevalence at any moment. And if we get better at keeping people alive with a chronic disease without preventing it, prevalence *rises*: the drain slows while the tap runs as before. A rising prevalence is not always bad news.

Which measure suits which design? A cross-sectional study, a snapshot, measures prevalence. A cohort study, which follows people forward, measures incidence. A case-control study measures neither, because the investigator chooses how many people with and without the outcome to include (Chapter 5).

7.2.2 Risk and rate

Incidence comes in two forms, which papers often call by the same name.

Cumulative incidence, also called **risk**, is the proportion of people at risk who develop the outcome over a stated period: 9 of 180 patients in one week, or 20 of 100 patients dying by day 28. It is a proportion between 0 and 1 (or 0% and 100%) and it always needs a time frame.

An **incidence rate** divides the number of new cases by the total amount of time that people were observed and at risk, called **person-time**. Rates are needed when people are followed for different lengths of time, as when patients enter and leave a unit. Infection-control teams use them routinely.

Worked example: an incidence rate of central-line infections

Over one year, an ICU records 12 central-line-associated bloodstream infections. Counting each day that each patient had a central line in place gives a total of 4,000 line-days.

$$\text{incidence rate} = \frac{12 \text{ infections}}{4{,}000 \text{ line-days}} = 0.003 \text{ per line-day} = 3 \text{ per } 1{,}000 \text{ line-days}$$

A unit that inserts more lines, or keeps them longer, will have more infections simply because it has more line-days. The rate takes that into account, which is why quality-improvement reports express line infections “per 1,000 line-days”.

7.3 Risk and odds

Before comparing groups we need one more distinction. The figure below shows a *hypothetical* trial of our fluid question with 100 patients in each group.

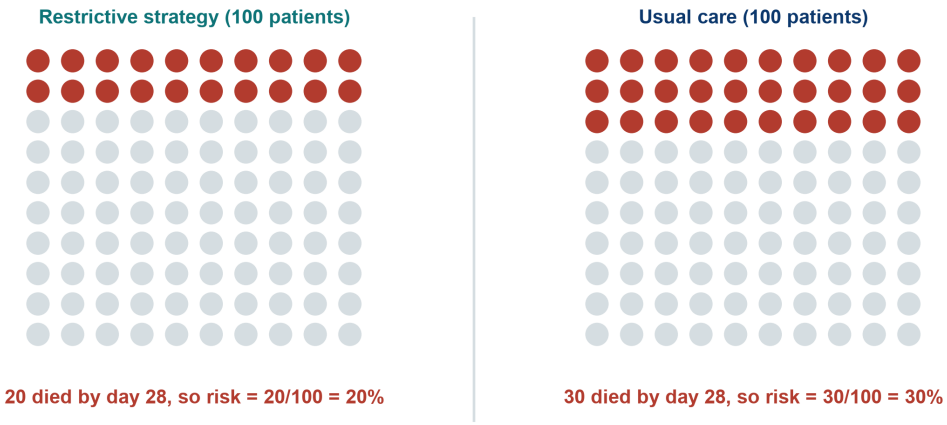


Figure 7.2: Hypothetical trial with 100 patients per group: 20 of 100 died with the restrictive strategy (risk 20%) and 30 of 100 with usual care (risk 30%).

The **risk** of death in the restrictive group is the number who died divided by everyone in the group: $20/100 = 0.20$, or 20%. The **odds** of death compare those who died with those who did not: 20 to 80, or $20/80 = 0.25$. In the usual-care group the risk is $30/100 = 0.30$ and the odds are $30/70 = 0.43$.

Risk is the measure clinicians and patients understand intuitively: “out of everyone like you, 20 in 100 died”. Odds are the language of betting (“one to four against”) and of some statistical methods, including logistic regression (Chapter 13) and case-control studies. When the outcome is rare, risk and odds are almost the same (1 in 100 versus 1 to 99); when it is common they diverge (50 in 100 is a risk of 0.5 but odds of 1.0). This divergence explains much of the confusion around the odds ratio, which we meet below.

Pictures like the one above, called **icon arrays**, are also one of the best ways to explain risk to patients and families. They make the denominator visible. A headline of “10 more deaths” means little until you know “out of how many?”.

7.4 The two-by-two table

Almost every measure of association for a yes/no outcome comes from four numbers arranged in a **two-by-two table**. The figure below shows the template and the hypothetical trial.

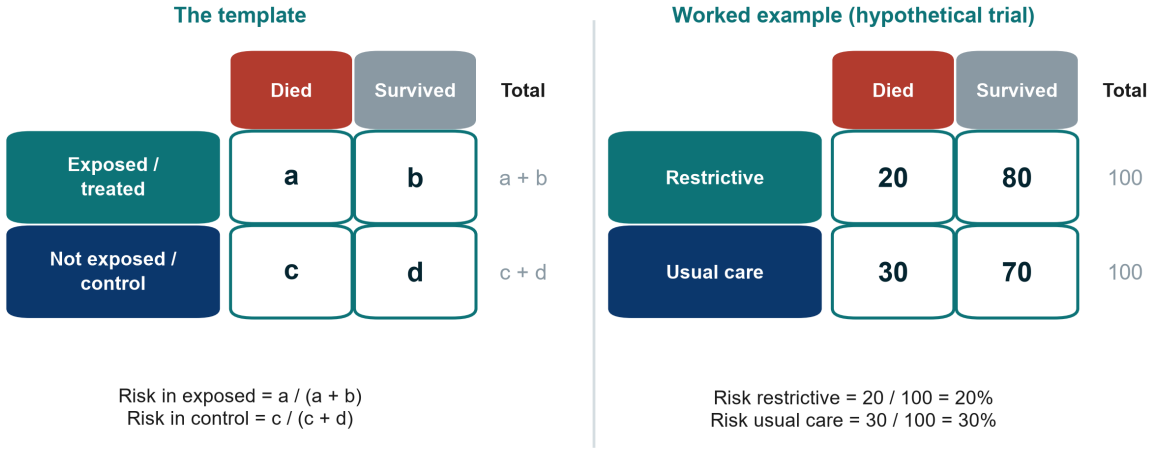


Figure 7.3: The two-by-two table template places exposure in rows (exposed on top) and outcome in columns (outcome yes on the left), labelling the cells a, b, c and d; in the hypothetical trial a = 20, b = 80, c = 30 and d = 70.

Adopt one layout and keep it for life: **exposure (or treatment) in the rows, with the exposed or treated group on top; outcome in the columns, with "outcome yes" on the left**. The four cells are then always labelled the same way.

Table 7.1: The two-by-two table layout used throughout this book.

	Outcome: yes	Outcome: no	Total
Exposed / treated	a	b	$a+b$
Unexposed / control	c	d	$c+d$

With this layout, the risk in the exposed group is $a/(a+b)$ and the risk in the unexposed group is $c/(c+d)$. In the hypothetical trial, $a = 20$, $b = 80$, $c = 30$ and $d = 70$. In Boulet's real trial, for death by day 28, $a = 6$, $b = 18$, $c = 6$ and $d = 18$.

! Common mistake: drawing the table the other way round

Most errors in these calculations come from the layout, not the arithmetic. If outcome is placed in the rows, or "outcome no" on the left, every formula below gives the wrong answer. Before calculating anything, check that the exposed group is on top and that a counts exposed people *with* the outcome.

7.5 Four measures from one table

7.5.1 The risk ratio

The **risk ratio** (RR), also called the **relative risk**, divides the risk in the exposed group by the risk in the unexposed group:

$$RR = \frac{a/(a+b)}{c/(c+d)}$$

In the hypothetical trial, $RR = 0.20/0.30 = 0.67$. The risk of death with the restrictive strategy is two-thirds of the risk with usual care, or one-third lower.

An RR of 1 means no difference. An RR below 1 means the exposure is associated with less of the outcome (protective, if the outcome is bad); above 1 means more of the outcome. The **relative risk reduction** (RRR) is $1 - RR$: here $1 - 0.67 = 0.33$, a 33% relative reduction. Papers and advertisements love this figure because it sounds impressive.

7.5.2 The risk difference

The **risk difference** (RD) subtracts one risk from the other:

$$RD = \frac{a}{a+b} - \frac{c}{c+d}$$

In the hypothetical trial, $RD = 0.20 - 0.30 = -0.10$: ten fewer deaths per 100 patients treated, or 10 percentage points. When the treatment reduces a bad outcome, the absolute value of the RD is called the **absolute risk reduction** (ARR), here 10%. An RD of 0 means no difference.

Note the units. A risk difference is measured in **percentage points**, not per cent. A fall from 30% to 20% is a fall of 10 percentage points, but a 33% relative fall. Mixing up the two is a common source of exaggeration.

7.5.3 The number needed to treat

The **number needed to treat** (NNT) turns the risk difference into a number of patients:

$$NNT = \frac{1}{|RD|}$$

with the RD written as a proportion (0.10, not 10). In the hypothetical trial, $NNT = 1/0.10 = 10$: on average, 10 patients must receive the restrictive strategy for one death to be prevented over 28 days (Laupacis et al. 1988). The figure below shows the idea with 10 patients per group.

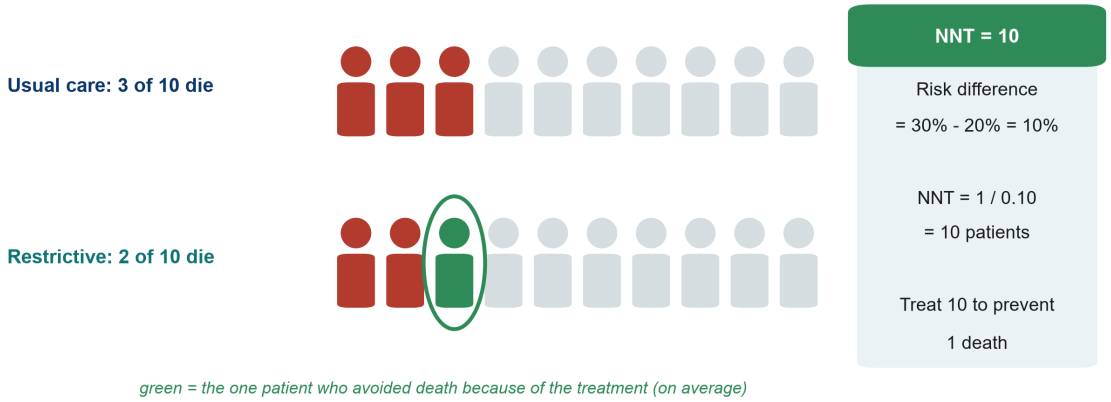


Figure 7.4: The number needed to treat in pictures: with usual care 3 of 10 die and with the restrictive strategy 2 of 10 die, so treating 10 patients prevents one death on average (NNT = 1/0.10 = 10).

The green patient is the one who, on average, avoided death because of the treatment. The other nine either survived anyway or died anyway; they bore the burden of the treatment without its benefit. This is why the NNT is so useful in conversations with patients, managers and pharmacists: it shows the effort needed for one good outcome.

When an exposure *increases* a bad outcome, the same calculation gives the **number needed to harm** (NNH). An NNT must always be given with its time frame (here, 28 days) and is an average for a group, not a prediction for an individual. A smaller NNT means a more effective treatment, but it must always be weighed against harms and costs.

7.5.4 The odds ratio

The **odds ratio** (OR) divides the odds of the outcome in the exposed group by the odds in the unexposed group. With the standard layout it simplifies to the “cross-product”:

$$OR = \frac{a/b}{c/d} = \frac{a \times d}{b \times c}$$

In the hypothetical trial, $OR = (20 \times 70)/(80 \times 30) = 1{,}400/2{,}400 = 0.58$. It is further from 1 than the risk ratio (0.67). The next section explains why.

Worked example: all four measures from one trial

A hypothetical trial: 20 of 100 patients die with the restrictive strategy and 30 of 100 with usual care.

Table 7.2: Hypothetical fluid trial used for the worked calculations.

	Died	Survived	Total
Restrictive	20	80	100
Usual care	30	70	100

- Risks: $20/100 = 20\%$ and $30/100 = 30\%$.
- $RR = 0.20/0.30 = 0.67$: the risk of death is one-third lower.
- $RD = 0.20 - 0.30 = -0.10$: 10 fewer deaths per 100 patients.
- $NNT = 1/0.10 = 10$: treat 10 patients to prevent one death by day 28.
- $OR = (20 \times 70)/(80 \times 30) = 0.58$.

In words for a colleague: "With the restrictive strategy, 20% of patients died by day 28 compared with 30% with usual care: about one death prevented for every 10 patients treated."

7.6 Relative and absolute effects

"One-third lower" and "ten fewer per hundred" describe the same result. Why insist on both? Because the relative effect alone hides how much difference a treatment makes, and that depends on how common the outcome is to begin with, the **baseline risk**.

Worked example: the same relative effect at three baseline risks

Suppose the restrictive strategy always lowers the risk of death by one-third (RR = 0.67).

Table 7.3: The same risk ratio gives very different absolute benefits at different baseline risks.

Baseline risk (usual care)	Risk with restriction	Risk difference	NNT
30%	20%	10 percentage points	10
15%	10%	5 percentage points	20
3%	2%	1 percentage point	100

Take the middle row as a check calculation. With 200 patients per group, 20 die with the new treatment and 30 with usual care. Risks: $20/200 = 10\%$ and $30/200 = 15\%$. $RR = 0.10/0.15 = 0.67$, as before. $RD = 0.10 - 0.15 = -0.05$. $NNT = 1/0.05 = 20$. $OR = (20 \times 170)/(180 \times 30) = 3{,}400/5{,}400 = 0.63$.
The relative effect is identical in all three rows, yet in high-risk patients one death is prevented for every 10 treated, while in low-risk patients 100 must be treated. Whether the treatment is worth its costs and side-effects may differ completely between the two groups.

Report effects twice

Ratios (RR, OR) tell you how *strong* an association is. Differences (RD, NNT) tell you how many *patients* are affected. A complete report of a yes/no outcome gives both, each with a confidence interval, as the CONSORT statement recommends for trials (Schulz et al. 2010).

! Common mistake: a relative effect presented alone

"The new drug halves the risk of stroke" may mean a fall from 2% to 1% (NNT 100) or from 40% to 20% (NNT 5). Relative effects sound larger and are favoured in press releases; patients and policymakers need the absolute effect to make decisions.

7.7 When is the odds ratio close to the risk ratio?

Odds ratios appear constantly in clinical papers, because case-control studies and logistic regression (Chapter 13) produce them. Many readers interpret an OR of 2 as "twice the risk". That reading is safe only when the outcome is rare. The figure below shows why.

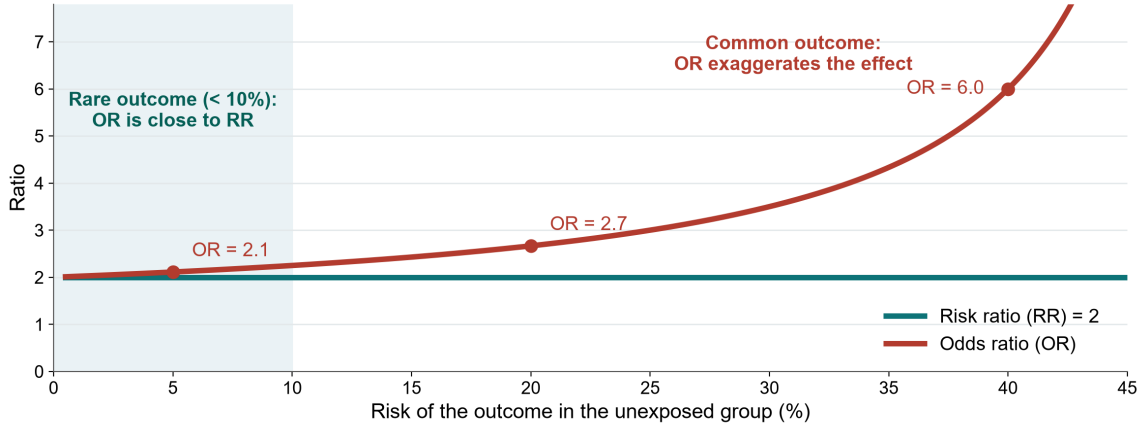


Figure 7.5: When the exposure truly doubles the risk (risk ratio 2), the odds ratio is close to 2 when the outcome is rare (about 2.1 at a baseline risk of 5%) but grows rapidly as the outcome becomes common (2.7 at 20% and 6.0 at 40%).

In the figure, the exposure truly doubles the risk at every baseline level (RR = 2, the flat green line). The odds ratio for the same data is shown in red. When the outcome is rare, in the shaded region below about 10%, the OR is practically the same as the RR: a baseline risk of 5% doubled to 10% gives $OR = (0.10/0.90)/(0.05/0.95) = 2.1$. As the outcome becomes more common, the two diverge: a baseline risk of 20% doubled to 40% gives $OR = 2.7$, and 40% doubled to 80% gives $OR = 6.0$. The odds ratio always lies further from 1 than the risk ratio.

This matters in intensive care, where mortality of 20% to 40% is common. In our hypothetical trial the OR (0.58) suggests a larger benefit than the RR (0.67). Neither is wrong, but they answer different questions, and an OR for a common outcome must not be read as "times the risk".

! Common mistake: an odds ratio of 2 means twice the risk

Only when the outcome is rare. For common outcomes, the odds ratio exaggerates the risk ratio. In trials and cohorts, prefer the risk ratio and risk difference; when a paper reports only an OR for a common outcome, interpret it with care.

7.8 The odds ratio in case-control studies

In a case-control study the investigator chooses how many cases and how many controls to include. The proportion of people with the outcome in the study is therefore fixed by design, and the "risks" calculated from its rows are meaningless. The odds ratio, however, remains valid, because it does not depend on the ratio of cases to controls. This is why the odds ratio is *the* measure of association in case-control studies, and why, when the outcome is rare, it can be read as an approximate risk ratio.

Worked example: parenteral nutrition and candidaemia

Investigators selected 40 ICU patients with candidaemia (cases) and 80 without it (controls) and looked back for parenteral nutrition. Among cases, 24 had received it and 16 had not; among controls, 16 had and 64 had not.

In a case-control study the columns are the case and control groups; the cross-product formula is unchanged.

Table 7.4: Case-control study of parenteral nutrition and ICU candidaemia (hypothetical).

	Cases (candidaemia)	Controls (no candidaemia)
Parenteral nutrition: yes	$a = 24$	$b = 16$
Parenteral nutrition: no	$c = 16$	$d = 64$

$$OR = \frac{24 \times 64}{16 \times 16} = \frac{1{,}536}{256} = 6.0$$

Why no risk, risk ratio or NNT? The investigators chose to include 40 cases and 80 controls. A spreadsheet will happily calculate "risks" of $24/40 = 60\%$ among exposed and $16/80 = 20\%$ among unexposed patients, but these numbers describe the sampling scheme, not the patients. Had the investigators chosen four controls per case, they would change completely.

In words for a colleague: "ICU patients with candidaemia had about six times the odds of having received parenteral nutrition compared with similar ICU patients without candidaemia. Because candidaemia is rare, this is roughly a six-fold relative risk. As an observational study, it cannot show that nutrition caused the infection: sicker patients are more likely to receive parenteral nutrition."

7.9 A second trial, worked through

Worked example: a pressure-relieving mattress trial

A trial randomised 400 adults on a medical ward to a new pressure-relieving mattress (200) or a standard mattress (200). A new pressure injury developed in 10 patients on the new mattress and 20 on the standard one.

Table 7.5: Hypothetical mattress trial.

	Pressure injury: yes	Pressure injury: no	Total
New mattress	10	190	200
Standard mattress	20	180	200

- Risks: $10/200 = 0.05$ (5%) and $20/200 = 0.10$ (10%).
- $RR = 0.05/0.10 = 0.50$.
- $RD = 0.05 - 0.10 = -0.05$: five percentage points fewer.
- $NNT = 1/0.05 = 20$.
- $OR = (10 \times 180)/(190 \times 20) = 1{,}800/3{,}800 = 0.47$.

The OR (0.47) is close to the RR (0.50) because the outcome is uncommon (5% to 10%).
In words: "With the new mattress, pressure injuries were halved, from 10% to 5%: about one injury prevented for every 20 patients who use it."
Common slips: using 30 or 400 as a denominator; writing the RD as 5 rather than 0.05 and then calculating an NNT of 0.2; and drawing the table with outcome in the rows.

7.10 Measures for time-to-event and continuous outcomes

7.10.1 Hazard ratios

Hyun and colleagues did not report a risk ratio. They reported a **hazard ratio** (HR): for the day-2 cohort, $HR = 2.13$ (95% confidence interval 1.48 to 3.06) (Hyun et al. 2023). A hazard is, informally, the rate at which the outcome occurs among those still at risk at each moment; the hazard ratio compares these instantaneous rates between groups across the whole follow-up period. It uses the *timing* of each death, not only whether it happened by day 28, and it handles patients who are followed for different lengths of time. Hazard ratios come from Cox regression (Cox 1972), which is explained, with Kaplan-Meier curves, in Chapter 14. For reading purposes, an HR is interpreted much like a risk ratio: 1 means no difference, 2.13 means that the rate of death at any time was about twice as high among patients with a positive fluid balance. Like a risk ratio, it is a relative measure, and it says nothing on its own about how many patients are affected.

7.10.2 Mean differences

Not every outcome is a yes/no event. Boulet's primary outcome was a measurement: cumulative fluid balance in ml/kg at day 5. For such **continuous** outcomes, the natural measure of effect is the **mean difference** (or, when the data are skewed, a difference in medians): the average value in one group minus the average in the other, in the original units. Boulet and colleagues reported median balances of 6.9 ml/kg (restrictive) and 35.0 ml/kg (control), with an absolute difference of 13.2 ml/kg and a 95% confidence interval from -15.2 to 41.6 ml/kg (Boulet et al. 2024). The interval includes zero and is wide, so the trial could not exclude no difference. Chapter 11 explains means, medians and their spread; Chapter 12 explains confidence intervals.

7.11 Reading measures of effect in published papers

When you meet a measure of effect in a paper, run through a short checklist.

1. *Which measure is it?* Risk ratio, odds ratio, hazard ratio, risk difference, mean difference? Each answers a slightly different question.
2. *What is the comparison and the direction?* Which group is in the numerator? An RR of 0.67 for "restrictive versus usual" is an RR of 1.5 for "usual versus restrictive".
3. *Over what time?* A risk or NNT without a time frame is incomplete.
4. *How precise is it?* Look for the 95% confidence interval. In Boulet's trial, the risk ratio for death by day 28 was 1.0 with an interval from 0.38 to 2.66: compatible with anything from a 62% reduction to a near tripling of risk. With 12 deaths, the trial cannot speak about mortality.
5. *Is there an absolute measure?* If only relative effects are given, find the baseline risk and work out the risk difference and NNT yourself.
6. *Could bias explain it?* Precision is no defence against bias (Chapter 6). An observational odds ratio of 6.0 may still be confounded.

Choosing the measure for your own study

For a randomised trial or cohort with a yes/no outcome at a fixed time, report risks in each group, the risk ratio and the risk difference, with confidence intervals, and the NNT if it helps interpretation. For a case-control study, report the odds ratio. For outcomes where the time to the event matters and follow-up varies, use Kaplan-Meier curves and hazard ratios. For continuous outcomes, report the mean (or median) in each group and the difference with its confidence interval. Fix the choice in the protocol and statistical analysis plan before the data are seen (Chapter 15).

Summary

Prevalence is the proportion of people who have a condition now; incidence is the occurrence of new cases over time among those at risk, expressed either as a risk (a proportion over a stated period) or as a rate (per unit of person-time). Risk divides those with the outcome by everyone in the group; odds divide those with the outcome by those without it. Laid out in a two-by-two table with exposure in the rows and outcome in the columns, four numbers give the risk ratio $[a/(a+b)]/[c/(c+d)]$, the risk difference $a/(a+b) - c/(c+d)$, the number needed to treat $1/|RD|$ and the odds ratio $(a \times d)/(b \times c)$. Ratios describe the strength of an association; differences describe how many patients are affected, and the same relative effect gives very different absolute effects at different baseline risks. The odds ratio is close to the risk ratio only when the outcome is rare, and it is the only valid measure in case-control studies. Hazard ratios compare event rates over time and mean differences compare continuous outcomes. Every measure needs its time frame and its confidence interval, and every one remains vulnerable to the biases described in Chapter 6.

Further reading

- Kirkwood and Sterne, *Essential Medical Statistics* (Kirkwood and Sterne 2003). Clear chapters on risks, odds, rates and their comparison, with worked examples.
- Laupacis, Sackett and Roberts, “An assessment of clinically useful measures of the consequences of treatment” (Laupacis et al. 1988). The paper that introduced the number needed to treat.
- Rothman, Greenland and Lash, *Modern Epidemiology* (Rothman et al. 2008). The authoritative account of incidence, prevalence and measures of association, including the relationship between them.
- Altman, *Practical Statistics for Medical Research* (Altman 1991). A classic, approachable text covering the analysis of two-by-two tables and the interpretation of relative and absolute effects.

CHAPTER 8

Diagnostic test accuracy

Much of clinical work is diagnosis, and much of diagnosis relies on tests: a lactate level, a blood culture, a chest radiograph, a delirium screening tool, a bedside ultrasound, a risk score at triage. No test is perfect. Some patients with the disease test negative, and some without it test positive. To use a test well, a clinician needs to know how often it is right and, more importantly, what a particular result means for the patient in front of them. These are different questions, and confusing them is one of the commonest errors in medicine. This chapter explains how the accuracy of a test is measured, why the meaning of a positive result depends on how common the disease is, how tests that give a number rather than a yes or no are judged, and how studies of test accuracy are designed, reported and appraised.

In this chapter you will learn

- How to lay out a diagnostic two-by-two table and calculate sensitivity and specificity.
- Why positive and negative predictive values depend on prevalence, and why the same test means different things in different settings.
- How likelihood ratios turn a pre-test probability into a post-test probability.
- How the ROC curve and the area under it describe a test that gives a continuous result, and how a cut-off is chosen.
- How diagnostic accuracy studies are designed, which biases threaten them, and how STARD and QUADAS-2 are used to report and appraise them.

8.1 Tests, reference standards and the two-by-two table

A diagnostic accuracy study compares the result of the test being evaluated, the **index test**, with the best available method of establishing whether the disease is truly present, the **reference standard** (sometimes called the “gold standard”, although few references are perfect). For example, a point-of-care lactate meter used at the bedside might be the index test, with the laboratory lactate as the reference standard for detecting hyperlactataemia. A rapid sepsis screening tool at triage might be the index test, with a later expert diagnosis based on all available information as the reference.

Every patient falls into one of four cells:

Table 8.1: The diagnostic two-by-two table.

	Disease present (reference)	Disease absent (reference)	Total
Test positive	True positive (TP)	False positive (FP)	TP + FP
Test negative	False negative (FN)	True negative (TN)	FN + TN
Total	TP + FN	FP + TN	All

This is the same tool as the two-by-two table of Chapter 7, laid out with the test result in the rows and the true disease status in the columns. Every measure of accuracy in this chapter comes from these four numbers.

Diagnostic questions run through the fluid example followed in this part of the book. The randomised trial by Boulet and colleagues defined septic shock partly by a plasma lactate of 2 mmol/L or more (Boulet et al. 2024), so the accuracy of the lactate measurement decides who is eligible. Hyun and colleagues identified sepsis with the Sepsis-3 consensus definitions (Hyun et al. 2023), themselves built on a score, and any screening tool used to find such patients early has its own accuracy. And the bedside decision at the heart of fluid therapy, whether a patient will respond to more fluid, is itself a diagnostic question, answered by tests whose accuracy can be measured exactly as this chapter describes.

8.2 Sensitivity and specificity

Consider 100 patients, of whom 20 truly have a disease according to the reference standard and 80 do not. The figure below shows how a test performs in each group.

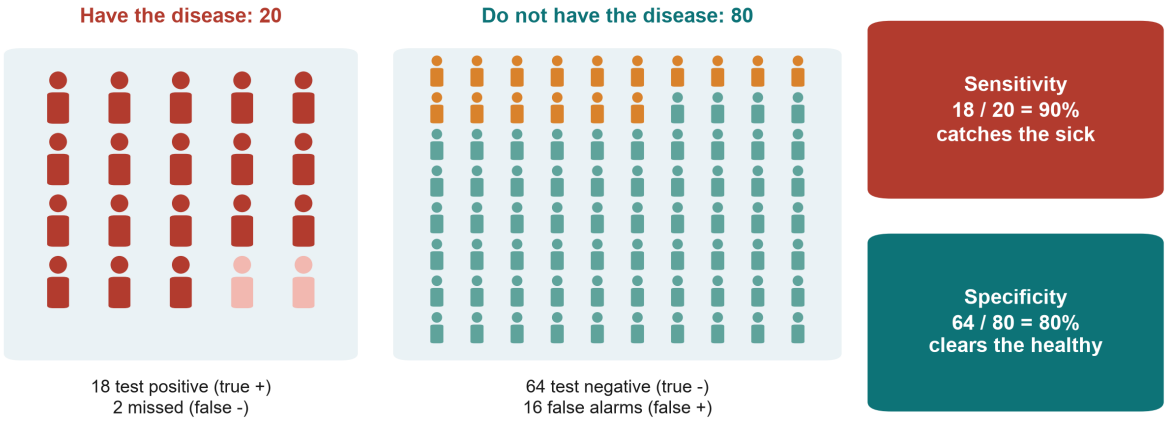


Figure 8.1: Sensitivity and specificity: of 20 patients with the disease the test detects 18 (sensitivity 90%), and of 80 without the disease it correctly clears 64 (specificity 80%), raising 16 false alarms.

Sensitivity is the proportion of people *with* the disease who test positive:

$$\text{sensitivity} = \frac{TP}{TP + FN} = \frac{18}{20} = 90\%$$

Specificity is the proportion of people *without* the disease who test negative:

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{64}{80} = 80\%$$

The two numbers answer questions that start from the truth: *if a patient has the disease, how likely is the test to detect it?* and *if a patient does not have the disease, how likely is the test to clear them?* (Altman and Bland 1994a)

A highly sensitive test rarely misses disease, so a **negative** result from it is reassuring and helps to *rule out* the disease. A highly specific test rarely raises false alarms, so a **positive** result helps to *rule in* the disease. Medical students learn two mnemonics for this: *SnNout* (a Sensitive test, when Negative, rules out) and *SpPin* (a Specific test, when Positive, rules in). They are useful reminders, provided the test is very sensitive or very specific; for moderate values, likelihood ratios (below) give a more precise answer.

Which matters more depends on the purpose of the test. A screening tool for sepsis at the emergency department door should above all not miss cases, so sensitivity comes first, and some false alarms are accepted. A test used to confirm a diagnosis before a risky or expensive treatment should above all not be falsely positive, so specificity comes first.

! Common mistake: a 90% sensitive test means a positive patient has a 90% chance of disease

Sensitivity starts from people who *have* the disease. A clinician holding a positive result needs the reverse: of people who *test positive*, how many have the disease? That is the positive predictive value, and in the example above it is only 53%, not 90%.

8.3 Predictive values

At the bedside the question is different. We do not know whether the patient has the disease; we know the test result. The **predictive values** answer the clinician's question.

The **positive predictive value** (PPV) is the proportion of people who test positive who truly have the disease:

$$PPV = \frac{TP}{TP + FP}$$

The **negative predictive value** (NPV) is the proportion of people who test negative who truly do not have the disease:

$$NPV = \frac{TN}{TN + FN}$$

In the 100 patients above, 34 tested positive (18 true positives and 16 false positives), so $PPV = 18/34 = 53\%$. Of the 66 who tested negative, 64 were truly negative, so $NPV = 64/66 = 97\%$ (Altman and Bland 1994b).

8.3.1 Predictive values depend on prevalence

Sensitivity and specificity are properties of the test (at least approximately, as we shall see). Predictive values are properties of the test *in a particular population*, because they depend on how common the disease is there. The figure below applies the same test, with sensitivity 90% and specificity 80%, to 500 people in two settings.

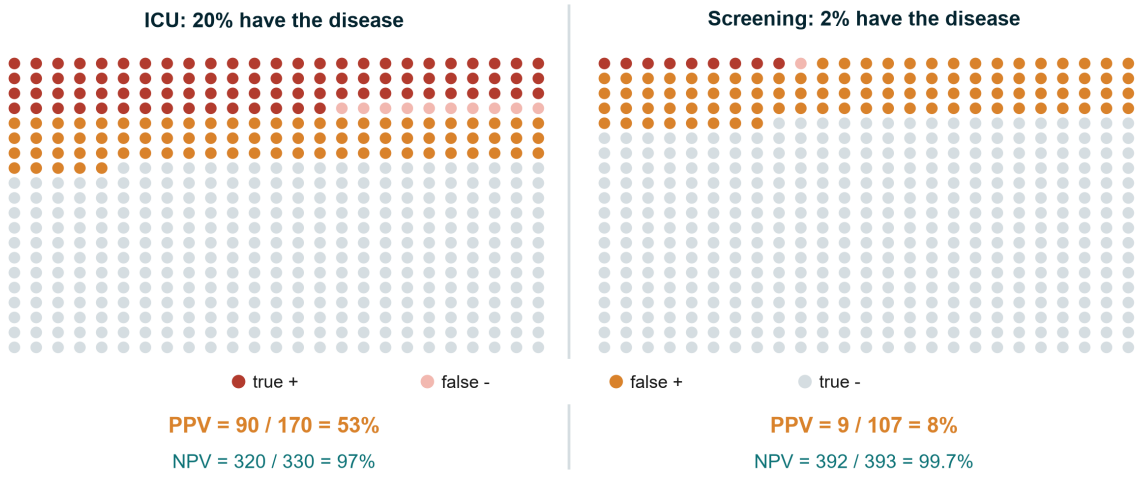


Figure 8.2: The same test (sensitivity 90%, specificity 80%) gives a positive predictive value of 53% in an ICU where 20% have the disease, but only 8% in a screening population where 2% have it, because false positives from the many healthy people swamp the few true positives.

Worked example: the same test in the ICU and in screening

Setting 1: an ICU where 20% of 500 patients have the disease.

- With disease: $0.20 \times 500 = 100$. True positives $= 0.90 \times 100 = 90$; false negatives $= 10$.

- Without disease: 400. True negatives $= 0.80 \times 400 = 320$; false positives $= 80$.

Table 8.2: Same test in an ICU population with 20% prevalence.

	Disease present	Disease absent	Total
Test positive	90	80	170
Test negative	10	320	330
Total	100	400	500

$PPV = 90/170 = 53\%$ and $NPV = 320/330 = 97\%$.
Setting 2: screening, where 2% of 500 people have the disease.

- With disease: 10. True positives $= 9$; false negatives $= 1$.
- Without disease: 490. True negatives $= 0.80 \times 490 = 392$; false positives $= 98$.

Table 8.3: Same test in a screening population with 2% prevalence.

	Disease present	Disease absent	Total
Test positive	9	98	107
Test negative	1	392	393
Total	10	490	500

$PPV = 9/107 = 8.4\%$ and $NPV = 392/393 = 99.7\%$.
In screening, more than nine out of ten positive results are false alarms. Nothing about the test changed: the 20% false-positive rate, applied to the many healthy people, simply produces far more false positives (98) than there are true cases to find (10). For completeness, at a prevalence of 50% the same test gives $PPV = 225/275 = 82\%$ and $NPV = 200/225 = 89\%$.

The lesson is fundamental. **Never borrow a predictive value from a study done in a different setting.** A test that performs well in a specialist unit, where the disease is common, may produce mostly false positives when used in a general population, generating anxiety, further investigations and harm. This is why screening programmes usually follow a positive first test with a second, more specific test before acting.

Worked example: adding a confirmatory test

In the screening setting above, 107 people tested positive, of whom 9 had the disease. Suppose all 107 then receive a second test with sensitivity 90% and specificity 95%, and assume that its errors are independent of the first test's.

- Of the 9 with disease, $0.90 \times 9 \approx 8.1$ test positive again.
- Of the 98 without disease, $0.05 \times 98 \approx 4.9$ test positive again.

The PPV after both tests is $8.1/(8.1 + 4.9) \approx 62\%$, compared with 8% after the first test alone. The price is a small loss of sensitivity: about one of the nine true cases is now missed. Sequential testing trades sensitivity for specificity.

8.4 Likelihood ratios: from pre-test to post-test probability

Predictive values change with every new setting, which makes them awkward to carry around. **Likelihood ratios** offer a more portable way to express how much a test result should change our suspicion (Deeks and Altman 2004).

The **positive likelihood ratio** (LR+) is how much more likely a positive result is in someone with the disease than in someone without it:

$$LR+ = \frac{\text{sensitivity}}{1 - \text{specificity}}$$

The **negative likelihood ratio** (LR−) is how much more likely a negative result is in someone with the disease than in someone without it:

$$LR- = \frac{1 - \text{sensitivity}}{\text{specificity}}$$

For our test, $LR+ = 0.90/0.20 = 4.5$ and $LR- = 0.10/0.80 = 0.125$. A positive result is 4.5 times as likely in a person with the disease; a negative result is about eight times *less* likely in a person with the disease. Because likelihood ratios are built from sensitivity and specificity, they do not depend on prevalence, and they travel between settings better than predictive values.

To use a likelihood ratio, start with the **pre-test probability**: your estimate of the chance of disease before the test, based on the setting and the patient (the prevalence among similar patients is a reasonable starting point). Convert it to odds, multiply by the likelihood ratio, and convert back:

$$\text{pre-test odds} = \frac{p}{1-p}, \quad \text{post-test odds} = \text{pre-test odds} \times LR, \quad \text{post-test probability} = \frac{\text{odds}}{1 + \text{odds}}$$

Worked example: updating suspicion with a likelihood ratio

ICU patient, pre-test probability 20%, test positive. Pre-test odds $= 0.20/0.80 = 0.25$. Post-test odds $= 0.25 \times 4.5 = 1.125$. Post-test probability $= 1.125/2.125 = 53\%$, the same as the PPV calculated from the table.
ICU patient, pre-test probability 20%, test negative. Post-test odds $= 0.25 \times 0.125 = 0.031$. Post-test probability $= 0.031/1.031 \approx 3\%$, which matches $1 - NPV$.
Screening, pre-test probability 2%, test positive. Pre-test odds $= 0.02/0.98 = 0.0204$. Post-test odds $= 0.0204 \times 4.5 = 0.092$. Post-test probability $\approx 8.4\%$.
The likelihood ratio is the same in both settings; the starting point differs, and so does the conclusion.

A rough guide helps in practice. Likelihood ratios above 10 or below 0.1 usually change clinical decisions substantially. Values between 5 and 10, or between 0.1 and 0.2, change them moderately. Values close to 1 change very little: such a test result hardly moves the probability of disease. A test with $LR+ = 4.5$, like ours, is useful but not decisive on its own. Graphical tools called nomograms let clinicians read off the post-test probability without calculation.

8.5 Tests with a continuous result: the ROC curve

Many tests give a number rather than a yes or no: lactate, procalcitonin, troponin, an early-warning score. To turn a number into a decision we need a **cut-off** (threshold): results above it count as positive. Every possible cut-off gives its own pair of sensitivity and specificity. A low cut-off calls many people positive: it catches almost every case (high sensitivity) but raises many false alarms (low specificity). A high cut-off does the reverse.

The **receiver operating characteristic** (ROC) curve plots all these pairs at once, with sensitivity on the vertical axis and the false-positive rate (1 − specificity) on the horizontal axis. The figure below shows the ROC curve of a simulated test.

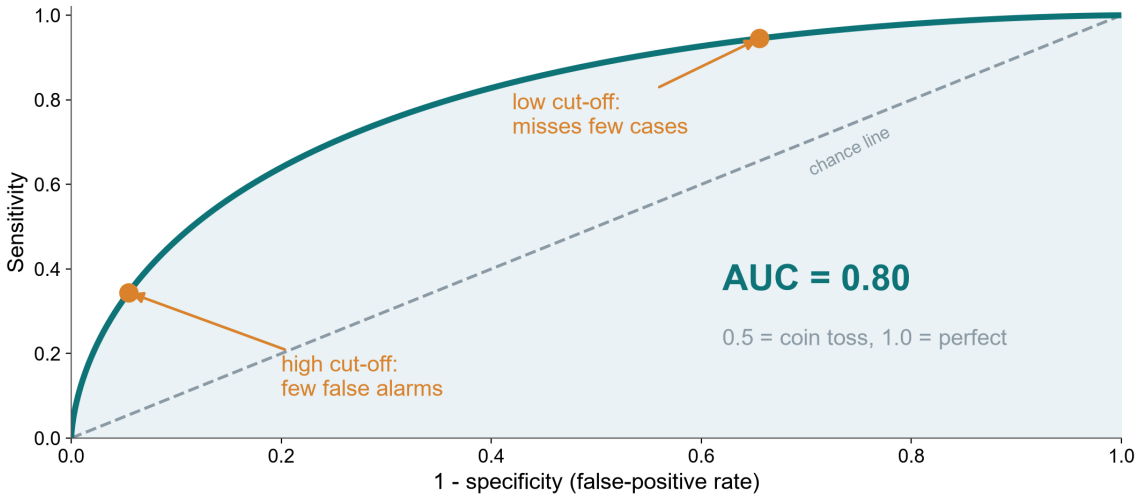


Figure 8.3: A receiver operating characteristic curve for a simulated test: each point is one cut-off, low cut-offs miss few cases but raise many false alarms, high cut-offs do the opposite, and the area under the curve is 0.80, compared with 0.5 for the chance line.

The diagonal dashed line is what a useless test, no better than tossing a coin, would produce. The further the curve bows towards the top-left corner, the better the test separates people with and without the disease. The **area under the ROC curve** (AUC), also called the **c-statistic**, summarises this ability, known as *discrimination*, in one number (Hanley and McNeil 1982). An AUC of 0.5 means no discrimination; 1.0 means perfect discrimination. The AUC has an intuitive meaning: it is the probability that a randomly chosen person with the disease has a higher test value than a randomly chosen person without it. The simulated test in the figure has an AUC of 0.80. As a loose guide, values around 0.7 to 0.8 are often described as acceptable and above 0.8 as good, but the AUC must always be judged against the clinical use.

The AUC describes the whole curve; it does not tell you which cut-off to use. That choice depends on the consequences of the two kinds of error. For ruling out sepsis at triage, where a missed case may be fatal and a false alarm costs an extra blood test, a low cut-off with high sensitivity is appropriate. For selecting patients for an invasive procedure, a high cut-off with high specificity may be preferred. A cut-off chosen to look best in the same data in which it was found will usually perform less well in new patients, so cut-offs should be pre-specified or validated in separate data. The same AUC reappears in Chapter 14 as a measure of the discrimination of prediction models, where it is joined by *calibration*, the agreement between predicted and observed risks.

! Common mistake: a good test is good everywhere

Sensitivity and specificity themselves can change between settings, because the mix of patients changes. A test evaluated in very sick patients compared with healthy volunteers looks much better than it will in the real, mixed population of patients in whom the diagnosis is genuinely uncertain. This is called *spectrum bias*, discussed below.

8.6 Designing a diagnostic accuracy study

The numbers in a diagnostic two-by-two table are only as good as the study that produced them. A good diagnostic accuracy study is, in design terms, a cross-sectional study (Chapter 5): a series of patients in whom the diagnosis is uncertain receive both the index test and the reference standard at about the same time, and the results are cross-tabulated. Several decisions determine whether the result can be trusted.

Who is tested. The patients should be those in whom the test will be used in practice, ideally a consecutive or random series of patients with the clinical problem, for example all adults presenting to the emergency department with suspected infection. A **case-control** diagnostic design, which compares patients known to have severe disease with healthy volunteers, exaggerates accuracy because both extremes are easy to tell apart. This is **spectrum bias**.

The reference standard. It should be the best available method of classifying the disease, applied in the same way to everyone. If only patients with a positive index test receive the reference standard (for example, only those with a positive screen undergo the definitive investigation), the study suffers **partial verification bias**: false negatives are never discovered and sensitivity is overestimated. If different patients receive different reference standards depending on their index result, the study suffers **differential verification bias**. If the index test result forms part of the reference standard, for example if the lactate under study is used by the expert panel to decide the final diagnosis, the study suffers **incorporation bias**, which inflates accuracy.

Blinding. Those who interpret the index test should not know the reference result, and vice versa. A radiologist who knows the final diagnosis will "see" more in an image.

Timing and flow. Index test and reference standard should be done close together, before treatment changes the disease. All enrolled patients should be accounted for, including those with indeterminate or missing results, which are often quietly dropped.

Worked example: planning a study of point-of-care lactate

Question: how accurate is a bedside point-of-care lactate meter for detecting hyperlactataemia (laboratory lactate of 2 mmol/L or more) in adults with suspected sepsis?

- *Participants:* consecutive adults with suspected sepsis in whom lactate is clinically indicated, in the emergency department and ICU.
- *Index test:* point-of-care lactate on a capillary or arterial sample, read by the bedside nurse, with the cut-off fixed in the protocol.
- *Reference standard:* laboratory lactate on a sample drawn at the same time, analysed by laboratory staff unaware of the bedside result.
- *Flow:* every enrolled patient has both tests; failed or unreadable bedside results are recorded and reported, not discarded.

- *Analysis:* sensitivity, specificity, predictive values (in this population) and likelihood ratios, each with a 95% confidence interval; a flow diagram of all patients.
- *Applicability:* results apply to adults with suspected sepsis in similar settings; whether they apply to children, or to patients on dialysis, is a separate question.

8.7 Reporting and appraising diagnostic studies

Diagnostic accuracy studies have their own reporting guideline, **STARD** (Standards for Reporting of Diagnostic Accuracy Studies). Its 2015 version contains 30 items and asks for a flow diagram showing how many patients were eligible, received each test, and ended up in each cell of the two-by-two table (Bossuyt et al. 2015).

Their quality is appraised with **QUADAS-2**, a tool recommended for systematic reviews of diagnostic accuracy (Whiting et al. 2011). It examines four domains:

Table 8.4: The four QUADAS-2 domains, with risk-of-bias and applicability questions.

Domain	Risk of bias question (summary)	Applicability question (summary)
Patient selection	Was a consecutive or random sample enrolled? Was a case-control design avoided? Were inappropriate exclusions avoided?	Do the patients match the review question?
Index test	Was it interpreted without knowledge of the reference standard? Was any threshold pre-specified?	Does the test, its conduct or its interpretation match the question?
Reference standard	Is it likely to classify the condition correctly? Was it interpreted without knowledge of the index test?	Does the target condition as defined by the reference match the question?
Flow and timing	Was the interval between tests appropriate? Did all patients receive the same reference standard? Were all patients included in the analysis?	(Not assessed)

Each domain is judged at low, high or unclear risk of bias. QUADAS-2 is distinctive in also asking about **applicability**: whether the study, even if unbiased, answers the reader's question. This mirrors the distinction between internal and external validity in Chapter 6. Chapter 9 places QUADAS-2 alongside the risk-of-bias tools for other designs.

! Common mistake: a diagnostic study is just a two-by-two table

The arithmetic is easy; the design is where studies succeed or fail. Who was tested, whether everyone received the same reference standard, and whether results were read blind matter as much as the numbers. A sensitivity of 95% from a study of severe cases and healthy volunteers, read without blinding, should carry little weight.

8.8 How precise is an estimate of accuracy?

Sensitivity and specificity are proportions estimated from a sample, so, like any estimate, they carry random error (Chapters 6 and 12). The precision of a sensitivity depends on the number of people *with the disease*, not on the total sample size. In our example, sensitivity was estimated from only 20 diseased patients: 18 of 20, or 90%. An exact 95% confidence interval for 18 out of 20 runs from about 68% to 99%. The true sensitivity could be as low as two-thirds. Specificity, estimated from 80 patients without disease (64 of 80, 80%), is more precise, with an interval of roughly 70% to 88%.

This has practical consequences. A diagnostic study of 200 patients in which only 15 have the disease cannot estimate sensitivity precisely, however large it looks. When planning such a study, the sample size is driven by the expected number of people with the target condition, and therefore by its prevalence among those enrolled. When reading one, look for the confidence intervals around sensitivity and specificity, not only the point estimates. STARD asks authors to report them (Bossuyt et al. 2015).

8.9 Accuracy is not the same as benefit

A more accurate test is not automatically a better test for patients. A test improves outcomes only if its result changes what clinicians do, and if the changed management helps. Consider a new biomarker that detects sepsis more accurately than current practice. If clinicians already start antibiotics in every patient with suspected sepsis, a better test may change nothing, except cost. If the test leads to earlier treatment of patients who would otherwise have been missed, it may save lives; if it leads to treating many false positives, it may cause harm through unnecessary antibiotics and investigations.

The question “does using this test improve patient outcomes?” is therefore a question about an *intervention*, the test-and-treat strategy as a whole, and it is best answered by a randomised trial comparing patients managed with and without the test (Chapter 5). Diagnostic accuracy studies are an essential first step, and they tell us whether a test *can* help; trials of test-and-treat strategies tell us whether it *does*.

Screening and diagnosis are different settings

Screening applies a test to people without symptoms to find disease early; the prevalence is low, so false positives are common and the harms of follow-up investigations fall on many healthy people. *Diagnosis* applies a test to people with symptoms or signs; the prevalence is higher, and the same test yields far more useful positive results. The worked example of the ICU and the screening setting earlier in this chapter shows the arithmetic behind this difference. A test validated for diagnosis should never be assumed to work equally well for screening.

8.10 Reading a diagnostic accuracy paper

When you read a paper that reports a test’s accuracy, ask, in order:

1. *Were the patients like mine*, with genuine diagnostic uncertainty, recruited consecutively?
2. *Was the reference standard appropriate*, applied to everyone, and interpreted blind to the index test?

3. *Was the cut-off pre-specified*, or chosen after looking at the data?
4. *Are sensitivity and specificity given with confidence intervals?* Small studies give wide intervals.
5. *What does a result mean for my patient?* Combine the likelihood ratios with the pre-test probability in your own setting, rather than borrowing the paper's predictive values.

Summary

A diagnostic accuracy study compares an index test with a reference standard in a two-by-two table of true positives, false positives, false negatives and true negatives. Sensitivity is the proportion of people with the disease who test positive; specificity is the proportion without it who test negative. A highly sensitive negative test helps rule out; a highly specific positive test helps rule in. Predictive values answer the clinician's question (what does this result mean?), but they depend on prevalence: the same test with 90% sensitivity and 80% specificity has a positive predictive value of 53% in an ICU where 20% have the disease and 8% in a screening population where 2% have it. Likelihood ratios ($LR+ = 4.5$ and $LR- = 0.125$ for this test) do not depend on prevalence and convert a pre-test probability into a post-test probability. For tests with a continuous result, the ROC curve shows the trade-off between sensitivity and specificity across cut-offs, and the AUC summarises discrimination, from 0.5 (chance) to 1.0 (perfect); the choice of cut-off depends on the costs of each kind of error. Good diagnostic studies recruit patients with genuine uncertainty, apply one reference standard to all, read results blind and account for every patient; STARD guides their reporting and QUADAS-2 their appraisal.

Further reading

- Altman and Bland, "Diagnostic tests 1: sensitivity and specificity" and "Diagnostic tests 2: predictive values" (Altman and Bland 1994a; Altman and Bland 1994b). Two one-page classics that explain the core measures with clinical examples.
- Deeks and Altman, "Diagnostic tests 4: likelihood ratios" (Deeks and Altman 2004). A short, clear account of likelihood ratios and how to use them.
- Hanley and McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve" (Hanley and McNeil 1982). The standard reference on the AUC, for readers who want the underlying ideas.
- Whiting and colleagues, "QUADAS-2" (Whiting et al. 2011), and Bossuyt and colleagues, "STARD 2015" (Bossuyt et al. 2015). The appraisal tool and the reporting guideline for diagnostic accuracy studies.

CHAPTER 9

Systematic reviews, meta-analysis and risk of bias

One study is rarely the answer. Studies of the same question differ in their patients, their methods and their results, and any single study can be misleading through chance or bias. When a clinician, a guideline panel or a researcher planning a new study wants to know what the evidence says, they need a summary of *all* the relevant studies, gathered and judged in a way that is transparent and reproducible. That is what a systematic review provides. When the studies are similar enough, their numbers can be combined in a meta-analysis to give a single, more precise estimate. Both are only as good as the studies they include, so they depend on a careful, structured judgement of each study's risk of bias. This chapter explains how systematic reviews are done, how to read the forest plot at the heart of a meta-analysis, what heterogeneity means, how risk-of-bias tools work and which tool fits which design, and how to judge whether the conclusions of a review, or of any paper, are supported by its data.

In this chapter you will learn

- The difference between a narrative review, a systematic review and a meta-analysis.
- The steps of a systematic review, from protocol to report, and how to read a PRISMA flow diagram.
- How a meta-analysis weights and pools studies, and how to read a forest plot step by step.
- What heterogeneity is, how I^2 is interpreted, and the difference between fixed-effect and random-effects models.
- How publication bias distorts reviews and how it is investigated.
- How risk-of-bias tools work, which tool suits which design (RoB 2, ROBINS-I, Newcastle-Ottawa or ROBINS-E, QUADAS-2, AMSTAR 2 and others), and how they differ from reporting guidelines.
- How to judge whether the conclusions of a paper are supported by its design and its numbers.

9.1 The running example

The question followed throughout this part of the book is whether restricting fluids benefits adults with septic shock. Individual studies have approached it in different ways. Boulet and colleagues ran a small randomised pilot trial of a strategy restricting non-resuscitation fluids, in which 6 of 24 patients died in each group (Boulet et al. 2024). Hyun and colleagues analysed a registry cohort and found that a positive fluid balance on ICU days 2 and 3 was associated with higher mortality (Hyun et al. 2023). Both papers cite larger randomised trials and earlier meta-analyses of fluid restriction that did not show a clear effect on mortality. A clinician trying to make sense of this literature needs a method for gathering and weighing all of it. In this chapter we use a set of six *hypothetical* trials of restrictive versus usual fluids, invented for teaching, to show how such a method works.

9.2 Studies of studies

A **narrative review** is an expert's summary of a field. It can be insightful, but the author chooses which studies to discuss and how to weigh them, and readers cannot check what was left out. Two experts can write narrative reviews of the same question and reach opposite conclusions, each citing the studies that support their view.

A **systematic review** answers a focused question using explicit, pre-specified and reproducible methods to find, select, appraise and summarise all the relevant studies. Its method is written down in advance, so that another team following the same protocol should find the same studies. A **meta-analysis** is the statistical part: pooling the numerical results of the included studies into a single estimate. The figure below shows the relationship.

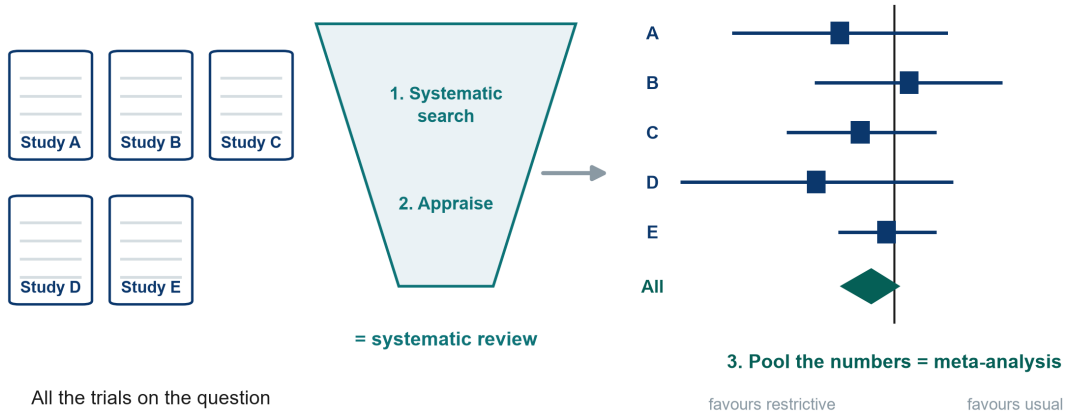


Figure 9.1: A systematic review gathers all the studies on a question through a systematic search and appraises each one; a meta-analysis then pools their numbers into a single estimate, shown as the diamond at the bottom of a forest plot.

Every meta-analysis should sit within a systematic review, but not every systematic review contains a meta-analysis. When studies are too different in their patients, interventions or outcomes, pooling them would produce a meaningless average, and the review instead summarises them in structured tables and text.

Systematic reviews sit at the top of the evidence pyramid described in Chapter 5, but only because they summarise the levels below. A meta-analysis of small, biased trials does not become reliable

by being pooled; it becomes *precisely* wrong. “Garbage in, garbage out” is the shortest summary of why the appraisal of each included study matters.

9.3 The steps of a systematic review

A systematic review follows a sequence of steps, each documented (Higgins et al. 2019).

1. A focused question. Usually framed with PICO (Chapter 2): for example, “In adults with septic shock (P), does a restrictive fluid strategy (I), compared with usual care (C), reduce mortality at 28 to 90 days (O)?” The review also specifies which study designs it will include, here randomised trials.

2. A protocol, registered in advance. The protocol states the question, eligibility criteria, search strategy, outcomes and methods of analysis before the work begins, and is registered publicly (for example in PROSPERO, the international register of systematic review protocols). Registration protects against the temptation to change the question or outcomes after seeing which ones give interesting results.

3. A comprehensive search. Several bibliographic databases are searched with a strategy combining synonyms with OR and concepts with AND (Chapter 3). Trial registers and other “grey” sources are also searched, because studies with negative or disappointing results are less likely to be published in journals. Reference lists of included studies are checked.

4. Selection by two reviewers. Titles and abstracts, then full texts, are screened against the eligibility criteria by two reviewers working independently, with disagreements resolved by discussion or a third reviewer. Reasons for excluding full texts are recorded.

5. Data extraction. Two reviewers extract the characteristics and results of each study onto a standard form.

6. Risk-of-bias assessment. Each included study is appraised with a tool suited to its design (discussed later in this chapter).

7. Synthesis. Results are combined, in a meta-analysis if appropriate, with investigation of heterogeneity and sensitivity analyses.

8. Certainty of the evidence. Many reviews rate how confident we can be in each pooled result, most often with the GRADE approach. GRADE starts randomised evidence at high certainty and rates it down for risk of bias, inconsistency (heterogeneity), indirectness (the studies do not quite match the question), imprecision (wide confidence intervals) and publication bias, ending at high, moderate, low or very low certainty (Guyatt et al. 2008).

9. Reporting. The review is reported following PRISMA 2020, which includes a 27-item checklist and a flow diagram (Page et al. 2021).

9.3.1 The PRISMA flow diagram

The PRISMA flow diagram shows how the studies in a review were found, screened and selected, with the number excluded at each stage and the reasons. The figure below gives an illustrative example.

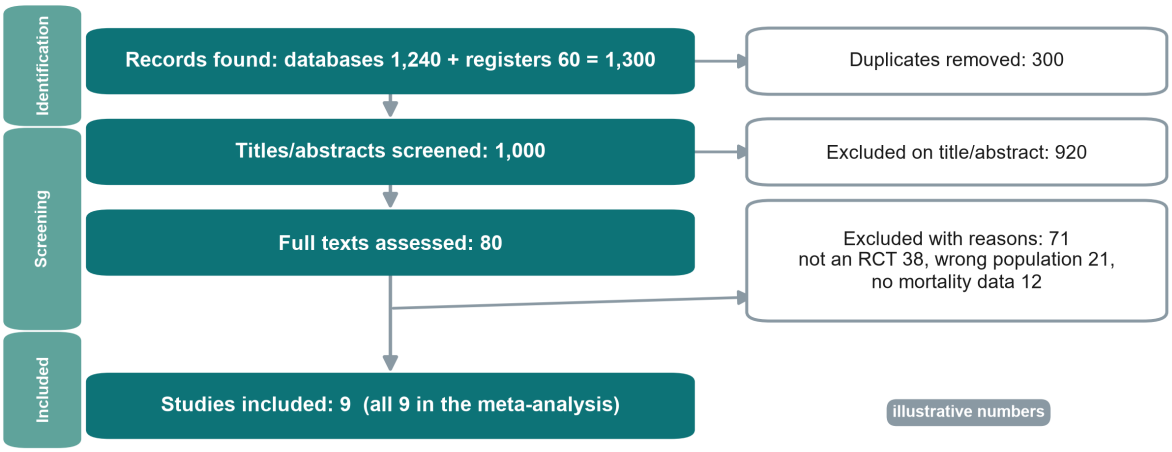


Figure 9.2: A PRISMA flow diagram with illustrative numbers: 1,300 records were identified, 300 duplicates removed, 1,000 titles and abstracts screened, 80 full texts assessed, 71 excluded with reasons, and 9 studies included in the meta-analysis.

Worked example: reading a PRISMA flow diagram

Identification. The search found 1,240 records in databases and 60 in trial registers, 1,300 in all. After removing 300 duplicates, 1,000 records remained.
Screening. Two reviewers screened 1,000 titles and abstracts and excluded 920 as clearly irrelevant. They read 80 full texts and excluded 71, with reasons: 38 were not randomised trials, 21 studied the wrong population and 12 reported no mortality data. Check that the reasons add up: $38 + 21 + 12 = 71$.
Included. Nine trials were included, and all nine contributed to the meta-analysis.
What should a reader look for? That trial registers were searched, which reduces the risk of missing unpublished trials; that reasons for exclusion are given and add up; and that the number of studies in the meta-analysis matches the number included, or that any difference is explained. A larger number of included studies does not make a review better; clear, appropriate inclusion criteria do.

9.4 Meta-analysis: pooling the numbers

9.4.1 Weighting the studies

A meta-analysis is not a simple average of study results. Large studies with many events estimate the effect more precisely than small ones, so they should count for more. The standard approach, **inverse-variance weighting**, gives each study a weight proportional to the inverse of the variance (the square of the standard error) of its estimate. A study whose result is twice as precise counts for more than a study whose result is uncertain. For ratio measures such as the risk ratio, the calculation is done on the logarithmic scale and the pooled result transformed back.

9.4.2 Reading a forest plot

The results of a meta-analysis are displayed in a **forest plot**. The figure below shows one for six hypothetical trials of restrictive versus usual fluids, with death as the outcome.

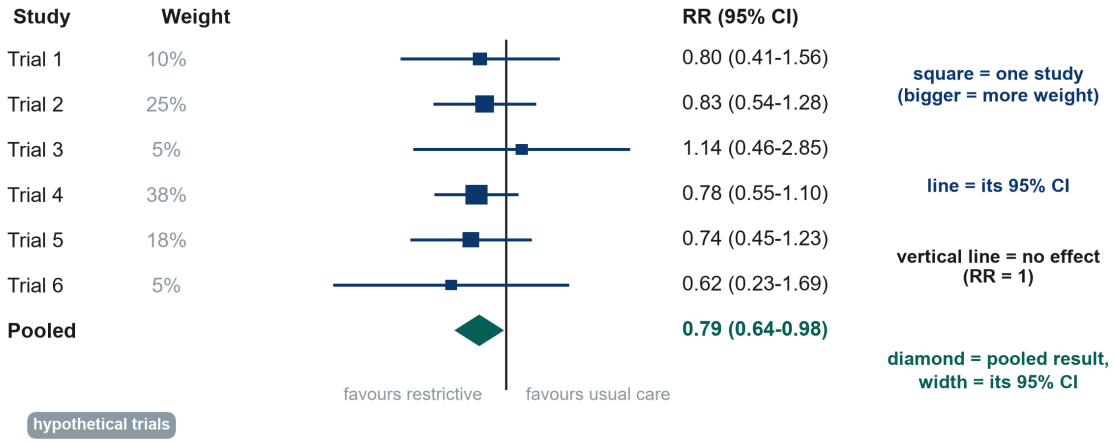


Figure 9.3: A forest plot of six hypothetical trials: each square is one trial's risk ratio, sized by its weight, each horizontal line its 95% confidence interval, the vertical line marks no effect (risk ratio 1), and the diamond shows the pooled risk ratio of 0.79 (95% CI 0.64 to 0.98).

A forest plot is read in four steps.

1. **Find the vertical line of no effect.** For a ratio measure it is at 1. Here, results to the left favour the restrictive strategy (fewer deaths) and results to the right favour usual care. The plot's labels tell you which side is which; always check them.
2. **Read each square.** Each square is one study's point estimate. Its size reflects the study's weight in the meta-analysis.
3. **Read each horizontal line.** It is the study's 95% confidence interval. If the line crosses the line of no effect, that study on its own is compatible with no difference.
4. **Read the diamond.** It is the pooled estimate. Its centre is the pooled point estimate and its width is the pooled 95% confidence interval.

Worked example: reading the forest plot of six fluid trials

The raw data and results behind the plot are:

Table 9.1: Six hypothetical trials of restrictive versus usual fluids, pooled by fixed-effect inverse-variance meta-analysis.

Trial	Restrictive: died / total	Usual care: died / total	RR (95% CI)	Weight
Trial 1	12 / 60	15 / 60	0.80 (0.41 to 1.56)	10%
Trial 2	30 / 150	36 / 150	0.83 (0.54 to 1.28)	25%
Trial 3	8 / 40	7 / 40	1.14 (0.46 to 2.85)	5%
Trial 4	45 / 250	58 / 250	0.78 (0.55 to 1.10)	38%
Trial 5	20 / 100	27 / 100	0.74 (0.45 to 1.23)	18%
Trial 6	5 / 30	8 / 30	0.62 (0.23 to 1.69)	5%
Pooled			**0.79 (0.64 to 0.98)**	$I^2 = 0\%$

Which trial carries the most weight? Trial 4 (38%): it is the largest and has the most deaths

(103), so its confidence interval is the narrowest. Trial 3 and Trial 6, with few deaths, carry 5% each.
Does any single trial show a clear effect? No. Every individual confidence interval crosses 1. Taken one at a time, each trial is inconclusive.
Does the diamond cross 1? No. The pooled risk ratio is 0.79, with a 95% confidence interval from 0.64 to 0.98. In words: "Across these six trials, the restrictive strategy was associated with about a 21% lower relative risk of death (RR 0.79, 95% CI 0.64 to 0.98); the upper limit is close to no effect."
Should Trial 3, which points the other way (RR 1.14), worry us? Not much. It is small, its confidence interval is very wide and overlaps all the others, and the spread of results is compatible with chance ($I^2 = 0\%$, explained below).

This example shows the power of meta-analysis: several trials that are individually inconclusive can, when pooled, give a precise answer. It also shows why a single inconclusive trial should never be described as showing "no effect": it shows uncertainty, and the effect may become clear when the evidence accumulates (Altman and Bland 1995).

9.5 Heterogeneity: do the studies agree?

Pooling assumes that the studies are estimating something similar. When their results differ more than chance would explain, they are said to show **heterogeneity**. The figure below contrasts two sets of hypothetical trials.

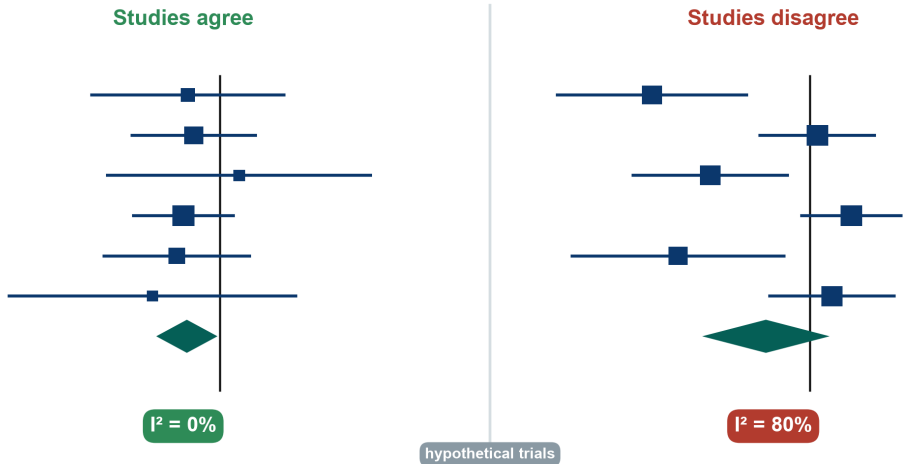


Figure 9.4: Two sets of hypothetical trials: on the left the studies agree, their confidence intervals overlap and I^2 is 0%; on the right they disagree, some favouring each side, I^2 is 80%, and the pooled diamond is wide and crosses the line of no effect.

On the left, the six trials from our worked example point the same way and their confidence intervals overlap; the variation among them is compatible with chance. On the right, six other hypothetical trials disagree: some clearly favour restriction, others point the other way, and several confidence intervals do not overlap. Their random-effects pooled estimate is a risk ratio of 0.74 with a confidence interval from 0.47 to 1.15, wide and crossing 1.

Heterogeneity is quantified with the I^2 **statistic**, which estimates the percentage of the variation between study results that reflects real differences rather than chance (Higgins et al. 2003). It is derived from Cochran's Q test of heterogeneity, with $k - 1$ degrees of freedom for k studies:

$$I^2 = \frac{Q - (k-1)}{Q} \times 100\%$$

(set to 0 when Q is smaller than $k - 1$). The Cochrane Handbook offers a rough guide, with deliberately overlapping ranges: 0% to 40% might not be important; 30% to 60% may represent moderate heterogeneity; 50% to 90% substantial; 75% to 100% considerable (Higgins et al. 2019). The overlap is intentional: I^2 should be judged together with the plot itself and the size and direction of the effects.

When heterogeneity is high, the most useful question is *why*. Fluid-restriction trials might disagree because they enrolled different patients (septic shock versus major surgery), restricted different fluids (resuscitation fluids versus maintenance and dilution fluids, as in Boulet's trial), started at different times, used different definitions of the outcome, or differed in their risk of bias. A single pooled number may hide these differences. Reviews explore them with **subgroup analyses** (for example, pooling trials in septic shock separately from trials in surgery) and **sensitivity analyses** (for example, repeating the meta-analysis without the trials at high risk of bias). Like subgroups within a trial (Chapter 6), subgroup analyses in reviews should be few, planned in the protocol and interpreted cautiously.

9.5.1 Fixed-effect and random-effects models

A **fixed-effect** meta-analysis assumes that all the studies estimate one true effect, and that they differ only by chance. A **random-effects** meta-analysis assumes that the true effect itself varies from study to study (because patients, interventions and settings differ) and estimates the average of these true effects. When studies are heterogeneous, a random-effects model gives relatively more weight to small studies and produces a wider confidence interval, reflecting the extra uncertainty.

> **! Common mistake: a random-effects model fixes heterogeneity**
>
> Switching to a random-effects model widens the confidence interval; it explains nothing. With substantial heterogeneity, the right response is to investigate its causes and to ask whether a single pooled estimate makes sense at all.

9.6 Publication bias and small-study effects

A systematic review can only include studies that it finds. Studies with positive, "significant" results are more likely to be published, published quickly, published in English and cited, than studies with null results. If a review finds only the published studies, its pooled estimate will overstate the benefit of the intervention. This is **publication bias**, one of a family of reporting biases that also includes selective reporting of favourable outcomes within studies.

Reviews defend against it by searching trial registers and grey literature (Chapter 3), and by looking for its signature in the data. A **funnel plot** plots each study's effect estimate against a measure of its precision. Large, precise studies cluster near the true effect at the top; small studies scatter more widely at the bottom. Without bias, the scatter is symmetrical, like an inverted funnel. If small studies with unfavourable results are missing, one corner of the funnel is empty and the plot is asymmetrical. Statistical tests of asymmetry exist (Egger et al. 1997), but they are unreliable with fewer than about ten studies, and asymmetry can have other causes, such as genuine differences between small and large studies. Prospective registration of trials, now required by journals (Chapter 17), is

the most powerful remedy.

9.7 Risk of bias: appraising each study

9.7.1 What risk-of-bias tools do

A risk-of-bias tool is a structured way of asking: *could the design or conduct of this study have distorted its result?* Rather than a general impression of "quality", it breaks the question into **domains**, each corresponding to a way in which bias can arise (Chapter 6), and asks a series of signalling questions within each. The reviewer then judges each domain, typically as low risk, some concerns or high risk, and reaches an overall judgement.

Three principles guide their use.

1. **Choose the tool that matches the design.** Each design has its characteristic biases, and each tool is built around them.
2. **Appraise the result, not the whole paper.** The same study may be at low risk of bias for an objective outcome such as death and at high risk for a subjective outcome assessed by unblinded staff. Appraise each outcome you care about.
3. **Judge by domain; do not add up scores.** A total score hides *which* domain failed, and one serious flaw can invalidate a study that scores well elsewhere.

9.7.2 Which tool for which design

The figure below matches each common design with its risk-of-bias tool and its reporting guideline.

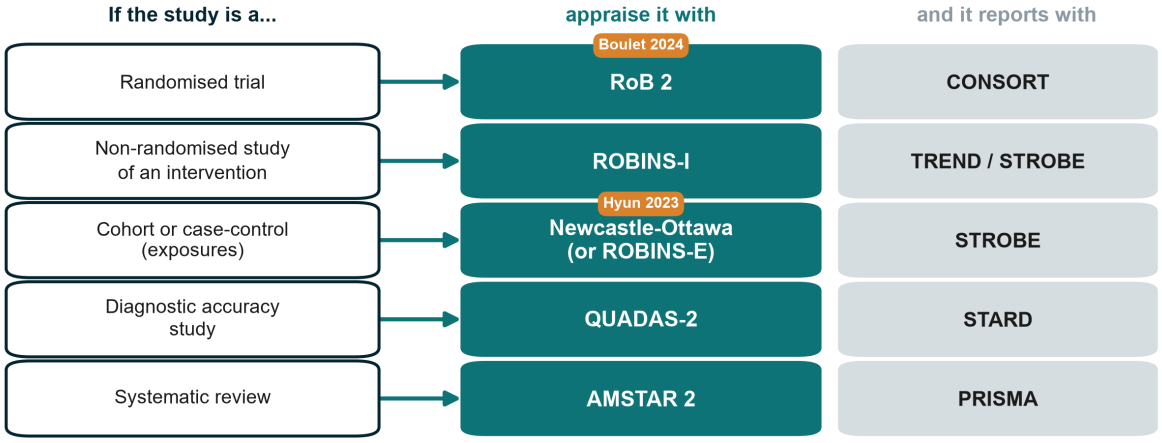


Figure 9.5: Choosing an appraisal tool by design: randomised trials are appraised with RoB 2 (reported with CONSORT), non-randomised studies of interventions with ROBINS-I, cohort and case-control studies of exposures with the Newcastle-Ottawa Scale or ROBINS-E (STROBE), diagnostic accuracy studies with QUADAS-2 (STARD) and systematic reviews with AMSTAR 2 (PRISMA).

Table 9.2: Risk-of-bias and critical appraisal tools by study design.

Study design	Tool	What it examines
Randomised controlled trial	RoB 2	Randomisation process; deviations from intended interventions; missing outcome data; measurement of the outcome; selection of the reported result
Non-randomised study of an intervention	ROBINS-I	Confounding; selection of participants; classification of interventions; deviations; missing data; outcome measurement; selection of the reported result, all judged against an ideal “target trial”
Cohort or case-control study of an exposure	Newcastle-Ottawa Scale, or ROBINS-E	Selection of the groups; comparability (confounding); ascertainment of outcome or exposure (NOS); ROBINS-I-style domains for exposures (ROBINS-E)
Diagnostic accuracy study	QUADAS-2	Patient selection; index test; reference standard; flow and timing; plus applicability
Systematic review	AMSTAR 2	16 items, including a registered protocol, a comprehensive search, risk-of-bias assessment of included studies, appropriate meta-analysis and publication bias
Qualitative study	CASP Qualitative Checklist	Aims; appropriateness of method and design; recruitment; data collection; researcher relationship (reflexivity); ethics; rigour of analysis; findings; value

Other tools cover other designs: PROBAST for prediction model studies (Chapter 14), and the JBI critical appraisal checklists for cross-sectional studies and case series. All of these tools are freely

available from their developers, who publish guidance documents explaining each question.

9.7.3 RoB 2 for randomised trials

RoB 2, the Cochrane tool for randomised trials, examines five domains (Sterne et al. 2019):

- **D1, the randomisation process:** was the allocation sequence random and concealed, and are baseline differences compatible with chance?
- **D2, deviations from intended interventions:** were participants and carers aware of the allocation, and did this lead to deviations that could affect the outcome?
- **D3, missing outcome data:** were outcome data available for nearly all participants, and if not, could missingness depend on the true outcome?
- **D4, measurement of the outcome:** could assessors' knowledge of the allocation have influenced the measurement?
- **D5, selection of the reported result:** was the result reported in line with a pre-specified plan, or chosen from several measurements or analyses?

Reviews often display RoB 2 judgements in a "traffic-light" plot, as in the figure below.

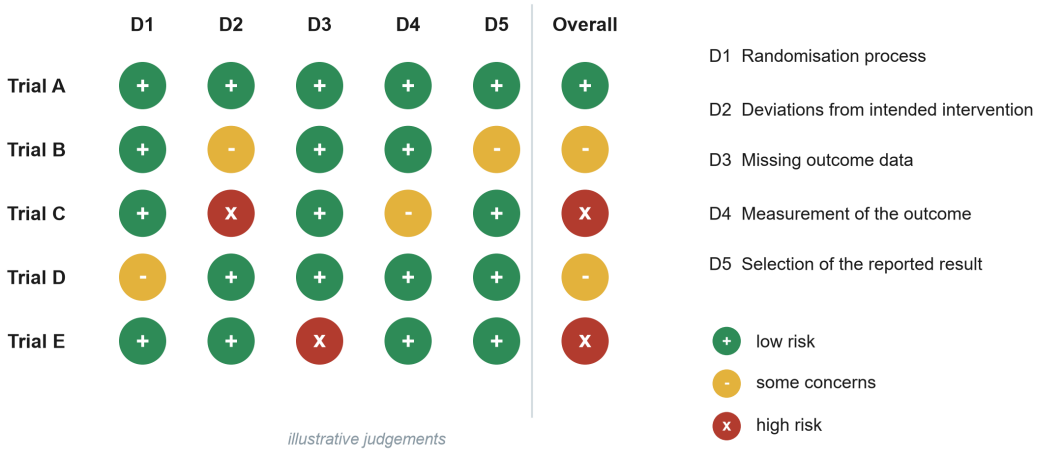


Figure 9.6: A traffic-light plot of illustrative RoB 2 judgements for five trials across five domains; the overall judgement is driven by the worst domain, so Trial C is at high risk because of deviations from intended interventions and Trial E because of missing outcome data.

Each row is a trial and each column a domain, with green for low risk, yellow for some concerns and red for high risk. The overall judgement is driven by the worst domain: Trial C is at high risk overall because of domain 2, and Trial E because of domain 3. Readers can see at a glance which trials drive a pooled result and whether excluding the high-risk ones changes it.

Worked example: RoB 2 applied informally to Boulet 2024

Consider the primary outcome, cumulative fluid balance at day 5.

- *D1, randomisation:* likely **low risk**. A centralised, computer-generated sequence in small permuted blocks concealed the allocation. Baseline characteristics were balanced except serum creatinine, a difference compatible with chance in a trial of this size.

- *D2, deviations from intended interventions:* **some concerns**. Clinicians knew the allocation, and the authors suggest that clinicians in the participating units may have restricted fluids in the control group too, which would narrow the difference between arms. (This would bias towards no effect.)
- *D3, missing outcome data:* likely **low risk**. Two of 50 randomised patients, both in the control group, withdrew consent; 48 were analysed.
- *D4, measurement of the outcome:* likely **low risk** for an objective, chart-derived measure such as fluid balance; the risk would be higher for a subjective outcome.
- *D5, selection of the reported result:* likely **low risk**. The protocol, including the statistical analysis plan, was registered before recruitment, and the pre-specified primary outcome is reported first.

Overall: some concerns, driven by domain 2. This is an informal illustration; a formal assessment works through every signalling question in the RoB 2 guidance.

! Common mistake: unblinded means high risk of bias

Lack of blinding does not automatically make a trial high risk. RoB 2 asks whether awareness of the allocation led to deviations that affected the outcome (D2) or influenced how the outcome was measured (D4). For an objective outcome with small deviations, an open trial can be at low risk.

9.7.4 ROBINS-I, the Newcastle-Ottawa Scale and ROBINS-E for observational studies

ROBINS-I (Risk Of Bias In Non-randomised Studies of Interventions) appraises studies that compare interventions without randomisation, such as before-after studies of a care bundle or cohorts comparing two treatments (Sterne et al. 2016). Its central idea is the **target trial**: the reviewer first describes the randomised trial that the study is trying to imitate, then judges how far the study departs from it in seven domains, beginning with confounding. Because an observational study is compared with an ideal trial, even a good one rarely reaches "low risk" overall; "moderate" is often the best realistic judgement.

For cohort and case-control studies of *exposures* (risk factors rather than chosen treatments), the **Newcastle-Ottawa Scale** (NOS) is still widely used (Wells et al. n.d.). It awards stars in three areas: selection of the groups, their comparability (control of confounding) and ascertainment of the outcome (in cohorts) or the exposure (in case-control studies). Its simplicity is attractive, but adding stars produces exactly the kind of total score that hides which domain failed. **ROBINS-E**, a newer and more rigorous tool built on the ROBINS-I approach, is designed for exposure studies (Higgins et al. 2024).

Worked example: appraising Hyun 2023 by domain

Using the questions common to these tools, the main concerns for the day-2 result (hazard ratio 2.13) are:

- *Confounding:* serious concern. Propensity-score matching balanced many measured factors, including SOFA score, septic shock and fluid output. But fluid type, nutrition,

diuretic use and clinicians' bedside judgement were not recorded, and confounding by indication is likely (Chapter 6).

- *Selection of participants:* some concern. Only patients who remained in the ICU for at least three days and had fluid data were analysed (2,516 of 11,981), which excludes early deaths and early discharges.
- *Classification of the exposure:* moderate concern. Fluid input and output were recorded by hospital day, so the first "day" varied in length between patients, as the authors acknowledge.
- *Measurement of the outcome:* low concern. Death by day 28 is objective.
- *Missing data and reporting:* the analysis was not registered in advance; reviewers would look for evidence that the three time windows were chosen before the results were seen.

The usual tool for this study is the Newcastle-Ottawa Scale or ROBINS-E. Some reviewers would choose ROBINS-I, arguing that fluid balance is partly the result of treatment decisions; either choice should be justified.

9.7.5 QUADAS-2 and AMSTAR 2

QUADAS-2, for diagnostic accuracy studies, was described in Chapter 8: four domains (patient selection, index test, reference standard, flow and timing), each judged for risk of bias and, for the first three, for applicability (Whiting et al. 2011).

AMSTAR 2 appraises systematic reviews themselves (Shea et al. 2017). Its 16 items ask, among other things, whether the protocol was registered before the review began, whether the search was comprehensive, whether study selection and data extraction were done in duplicate, whether risk of bias in the included studies was assessed and taken into account when interpreting the results, whether the meta-analytic methods were appropriate, and whether publication bias was investigated. Certain items are designated critical, and the overall confidence in the review is rated as high, moderate, low or critically low. A reader faced with two reviews reaching different conclusions can use AMSTAR 2 to decide which deserves more weight.

9.7.6 Reporting guidelines are not risk-of-bias tools

The right-hand column of the tool chooser figure lists the **reporting guidelines**: CONSORT for trials (Schulz et al. 2010), STROBE for observational studies (Elm et al. 2007), STARD for diagnostic accuracy studies (Bossuyt et al. 2015), and PRISMA for systematic reviews (Page et al. 2021). They tell authors what to *report*, so that readers can judge the study. Risk-of-bias tools judge how *trustworthy* the result is. A study can be impeccably reported and still at high risk of bias; indeed, good reporting is what allows the bias to be seen. Compliance with a reporting checklist should never be presented as a quality score.

Worked example: choosing the tool

- Boulet 2024, a randomised trial of restrictive versus usual fluids: **RoB 2**.
- Hyun 2023, a registry cohort comparing positive with negative fluid balance: **Newcastle-Ottawa Scale or ROBINS-E**.

- A before-after study of a central-line insertion bundle on one unit: **ROBINS-I** (a non-randomised study of an intervention).
- A study comparing point-of-care lactate with the laboratory reference for detecting hyperlactataemia: **QUADAS-2**.
- A systematic review and meta-analysis of fluid-restriction trials: **AMSTAR 2**.
- A cluster-randomised trial of a hand-hygiene programme on 12 wards: **RoB 2**, using its version for cluster-randomised trials.

9.8 Are the conclusions supported by the data?

The final question to ask of any study, single or synthesised, is whether its conclusions follow from its design and its numbers. Three checks cover most problems.

The words match the design. "Caused", "reduced" and "prevented" need randomisation (Chapter 5). Observational studies should say "associated with". A pilot trial should conclude about feasibility, not effectiveness. Boulet's conclusion, that the strategy "did not reduce the overall fluid balance at day 5", matches its design and its result (Boulet et al. 2024). Hyun's abstract correctly says "associated", but its title speaks of the "impact" of fluid balance, a stronger word than a cohort supports (Hyun et al. 2023). Words such as "impact", "effect", "reduces" or "prevents" in the title or conclusions of an observational study are a warning sign.

The numbers match the words. The main result should be given as an effect size with a 95% confidence interval, relative and absolute where the outcome is binary (Chapter 7). A non-significant result should be described as showing no clear difference, not as "a trend towards benefit". Boulet's discussion refers to a "non-significant trend" in fluid balance reduction; with $p = 0.42$ and a confidence interval from -15.2 to 41.6 ml/kg, the honest description is that the trial did not show a clear difference.

Honest limits, no spin. The pre-specified primary outcome should be reported first, and the conclusion should rest on it rather than on a favourable secondary outcome or subgroup. The main limitations should be stated. Presenting a study as more favourable than its results justify is called **spin**, and it is common in abstracts of trials with non-significant primary outcomes (Boutron et al. 2010). Both of our running papers, to their credit, list their main limitations: Boulet's unrepresentative control group and lack of clinician blinding, and Hyun's unmeasured factors and daily recording of fluids.

Read the conclusion last

Read the methods and results first, then decide what you would conclude, and only then read the authors' conclusion. If it claims more than the design and the numbers support, trust your own reading.

Summary

A systematic review answers a focused question with explicit, reproducible methods: a registered protocol, a comprehensive search including trial registers, duplicate screening and extraction, risk-of-bias assessment and a transparent synthesis, reported with PRISMA 2020 and its flow diagram. A meta-analysis pools the results, weighting each study by the inverse of its variance. A forest plot is read in four steps: the line of no effect, the squares (point estimates sized by weight), the horizontal lines (confidence intervals) and the diamond (the pooled estimate and its interval); several inconclusive trials can pool to a conclusive answer, as in the hypothetical example with a pooled risk ratio of 0.79 (0.64 to 0.98). Heterogeneity, measured by I^2, asks whether studies agree; when it is high, investigate why rather than relying on a random-effects model to fix it. Publication bias inflates pooled effects and is countered by searching registers and examining funnel plots. Risk-of-bias tools judge each result domain by domain with a tool matched to the design: RoB 2 for trials, ROBINS-I for non-randomised studies of interventions, Newcastle-Ottawa or ROBINS-E for exposure studies, QUADAS-2 for diagnostic studies and AMSTAR 2 for reviews. Reporting guidelines check what was reported, not whether the result is trustworthy. Finally, check that a paper's words match its design and its numbers, and read its conclusion last.

Further reading

- Higgins and colleagues (eds), *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins et al. 2019). The definitive, freely available guide to every step of a systematic review and meta-analysis.
- Page and colleagues, "The PRISMA 2020 statement" (Page et al. 2021). The reporting guideline and flow diagram for systematic reviews; useful as a checklist when reading a review.
- Sterne and colleagues, "RoB 2" (Sterne et al. 2019), and Sterne and colleagues, "ROBINS-I" (Sterne et al. 2016). The papers describing the two main Cochrane risk-of-bias tools, with their rationale.
- Higgins, Thompson, Deeks and Altman, "Measuring inconsistency in meta-analyses" (Higgins et al. 2003). The short paper that introduced I^2 and explains how to interpret it.
- Shea and colleagues, "AMSTAR 2" (Shea et al. 2017). The tool for judging the trustworthiness of a systematic review.

PART III

Data and analysis

CHAPTER 10

Collecting and managing data

Every clinical study ends in a table of numbers, and every number in that table began life as a mark on a form, a value typed into a tablet, or a result copied from a laboratory system. Statistics can only work with what it is given. If the box for fluid balance was left blank, if one hospital reported lactate in a different unit, or if the same patient was entered twice, no analysis performed months later can recover the truth. This chapter is about the unglamorous but decisive part of research: deciding what to collect, designing the form that collects it, describing each variable so that anyone can understand the dataset, checking the data as they arrive, keeping them safe, and cleaning them in a way that can be traced and repeated. It ends with a worked example of a messy dataset being cleaned, step by step.

In this chapter you will learn

- Why the quality of the data sets an upper limit on the quality of the answer.
- How to classify a variable as continuous, count, binary, nominal, ordinal or time-to-event, and why the type matters.
- How to write case report form (CRF) items that have only one possible reading.
- What a data dictionary is and how to write one.
- The four main types of validation check, and how double data entry and electronic data capture protect data quality.
- The principles of safe data management: coded identifiers, audit trails, versions, backups and data-sharing plans.
- Why missing data can bias a study and how to prevent them.
- How a raw dataset is cleaned with a script and a cleaning log, illustrated with the simulated FLUID-ICU cohort.

10.1 Bad data cannot be rescued by good statistics

Most problems in clinical research are not in the analysis; they are in the data. A statistician cannot fix a blank box, a missing unit or an impossible value six months after the patient has gone home. The old phrase “garbage in, garbage out” is blunt but accurate: the quality of the answer is capped by the quality of the form.

Many clinicians have had the experience of being asked to “help with the data” after a study has finished, only to find that half the forms are incomplete, that different nurses interpreted the same question in different ways, and that the spreadsheet contains names, phone numbers and results side by side. The work of this chapter is to prevent that situation. It costs a few days at the start of a study and saves months at the end, and it often decides whether the study can be published at all.

Throughout the chapter we follow one variable from the bedside to the analysis file: **cumulative fluid balance**, the total fluid a patient has received minus the total fluid lost, over a defined period. It is the central variable of the running example of this book, fluid management in septic shock. Paper 1 (Boulet et al. 2024), a pilot randomised trial of a restrictive fluid strategy, used cumulative fluid balance at day 5 as its primary outcome (Boulet et al. 2024); Paper 2 (Hyun et al. 2023), a cohort study, used a positive cumulative balance during the first three ICU days as its exposure (Hyun et al. 2023). We will also use **FLUID-ICU**, a cohort of 620 adults with sepsis in four intensive care units. FLUID-ICU is **simulated**: it was generated by computer for teaching, and contains no real patient, hospital or country. It was built to behave like real ICU data, including the problems real data have.

10.2 What type of variable is it?

The most basic decision in data collection is what *type* of variable you are collecting. The type decides how you write the question on the form, how you will summarise it (Chapter 11), and which statistical test or model you will use (Chapters 12 to 14). It is worth getting into the habit of labelling the type of every variable in a study.

Every variable is either a **number you measure** or a **category you assign**. Each of these then splits once more, as the figure below shows.

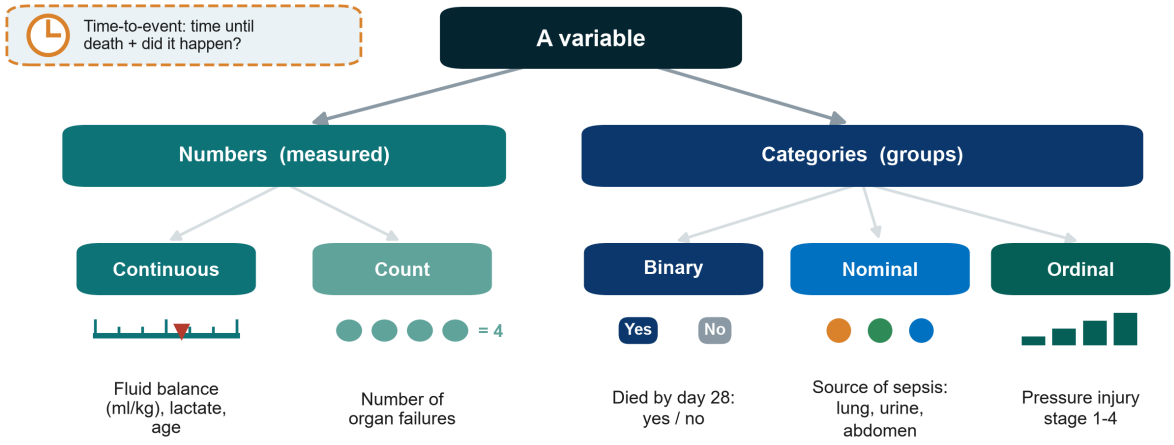


Figure 10.1: A variable is either a number (continuous or count) or a category (binary, nominal or ordinal); time-to-event variables combine how long with whether the event happened.

Numerical variables come in two kinds:

- **Continuous** variables can take any value on a scale, within the limits of measurement: age, weight, lactate, mean arterial pressure, cumulative fluid balance in ml/kg. Fluid balance can be negative or positive.
- **Count** variables are whole numbers: the number of organ failures, the number of vasopressors, the number of days on a ventilator. Counts cannot be negative and are often skewed, with many small values and a few large ones.

Categorical variables come in three kinds:

- **Binary** variables have exactly two categories: died by day 28 (yes or no), mechanical ventilation on admission (yes or no), sex.
- **Nominal** variables have several categories with no natural order: source of infection (lung, urine, abdomen, skin, other), blood group.
- **Ordinal** variables have several categories in a natural order, but the steps between them are not necessarily equal: pressure-injury stage 1 to 4, pain rated none, mild, moderate or severe, the stage of acute kidney injury. Clinical scores such as the Glasgow Coma Scale (3 to 15) or the SOFA score (0 to 24) are ordinal by nature, although they are often analysed as if they were numbers. That is acceptable if it is stated, but the reader should know that the difference between a GCS of 3 and 4 is not clinically the same as the difference between 14 and 15.

A sixth type deserves special mention: **time-to-event** variables. "Time to death within 28 days" or "time from ICU admission to extubation" combine two pieces of information: *how long* until the event, and *whether* the event was observed at all. Some patients are never extubated because they die or are transferred; some are still alive when follow-up ends. These patients are **censored**: we know they had not had the event by a certain time, but not what happened afterwards. Time-to-event data need their own methods (Kaplan-Meier curves and Cox regression), which are described in Chapter 14. On the form, a time-to-event variable is collected as two items: the dates (of admission and of the event or last contact) and the status (event or no event).

Worked example: classifying ICU variables

Consider ten variables that might appear in a study of septic shock.

Table 10.1: Ten ICU variables classified by type.

Variable	Type	Comment
Age (years)	Continuous	Keep the exact number, or better, collect the date of birth
Sex	Binary	Coding it 1 and 2 does not make it numerical
Glasgow Coma Scale (3 to 15)	Ordinal	Ordered scores with unequal steps; often analysed as a number
Source of sepsis	Nominal	Several categories, no order
Number of vasopressors (0 to 3)	Count	Whole numbers
Cumulative fluid balance (ml/kg)	Continuous	Can be negative or positive
Alive at day 28	Binary	Yes or no
Time to death, censored at day 28	Time-to-event	How long, and whether death occurred
Pain: none, mild, moderate, severe	Ordinal	Ordered categories
Blood group	Nominal	No order

The three classic errors are to call the Glasgow Coma Scale continuous, to call "time to death" continuous (which ignores censoring), and to call the number of vasopressors continuous.

! Common mistake: coded as a number means numerical

Categories are often stored as numbers in a database: sex as 1 = male and 2 = female, source of infection as 1 = lung, 2 = urine, and so on. This does not make them numerical. The "mean sex" of 1.4 or the "mean infection source" of 2.3 means nothing. The type of a variable comes from what it measures, not from how it is stored.

A practical rule follows from this classification: **collect continuous variables as numbers whenever possible**. You can always group exact ages into bands such as 60 to 69 years later, but you can never recover the exact age from a box ticked "60 to 69". The same applies to lactate: record the value in mmol/L, not a tick box for "lactate high", and decide on the cut-off during the analysis.

10.3 Planning what to collect

Before designing any form, write a short **variables table**: a list of every variable the study needs, its role, how it will be measured and its type. Each variable should have a reason to be there. In practice there are only four good reasons:

1. it is the **outcome** (primary or secondary);
2. it is the **exposure** or intervention;
3. it is a **confounder** or effect modifier that the analysis must take into account (Chapter 6);
4. it **describes the population**, so that readers can judge to whom the results apply.

"Nice to know" items make forms long, and long forms are left incomplete. The table below shows a variables table for a study like FLUID-ICU.

Table 10.2: A variables table for a cohort study of fluid balance and mortality.

Role	Variable	How measured	Type
Primary outcome	Death by day 28	Hospital records or phone call at day 28	Binary (and time-to-event)
Exposure	Cumulative fluid balance, ICU days 1 to 3	Fluid charts, ml/kg of admission weight	Continuous (also used as positive or negative)
Confounders	Age, SOFA score, lactate, vasopressor use, mechanical ventilation, chronic kidney disease	CRF at ICU admission	Continuous, ordinal score, continuous, binary, binary, binary
Effect modifier	Chronic heart failure	CRF at ICU admission	Binary
Description	Sex, source of infection, comorbidities	CRF at ICU admission	Binary, nominal, binary

Notice what is *not* in the confounder row: acute kidney injury and renal replacement therapy. They occur after the exposure and may be consequences of it, so they are outcomes or mediators, not confounders. Deciding this before the data are collected, and writing it down in the protocol and statistical analysis plan, is part of good design (Chapters 6 and 15).

10.4 Designing the case report form

The **case report form (CRF)** is the document, on paper or on screen, on which the data for each participant are recorded. A CRF item is good when two different people, filling it in for the same patient from the same records, would write the same answer. That simple test catches most problems.

10.4.1 What makes a CRF item bad or good

The first figure below shows four poorly written items; the second shows improved versions.

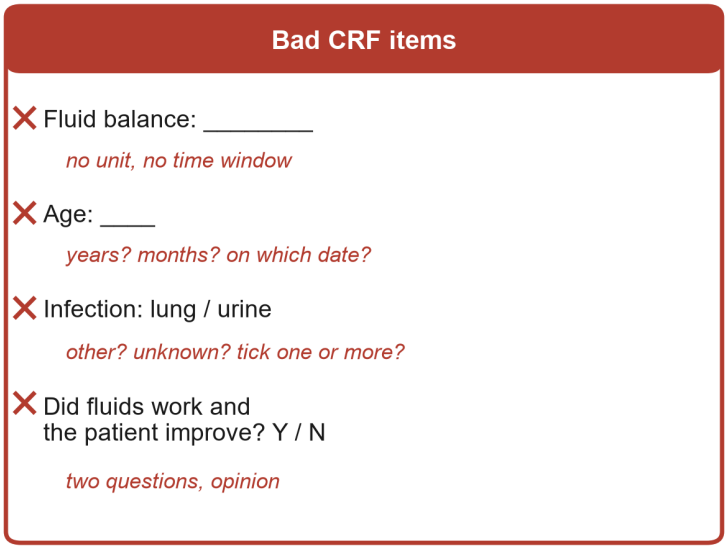


Figure 10.2: Four badly written CRF items: no unit or time window, an unclear age, incomplete infection options, and two opinions combined in one question.

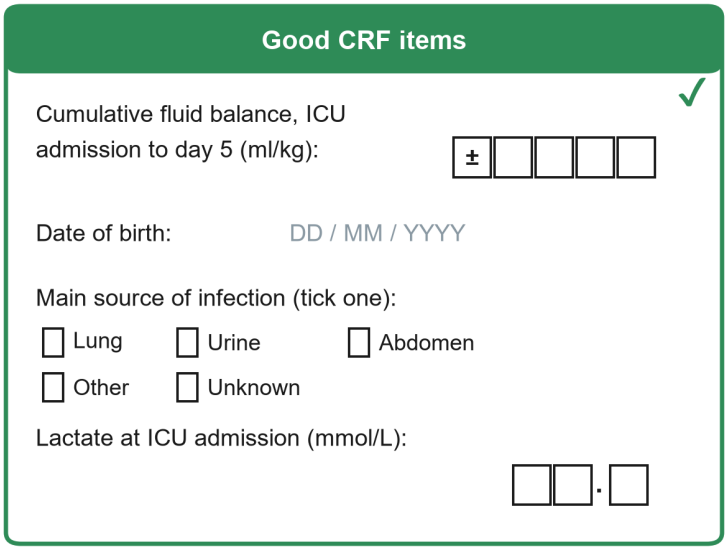


Figure 10.3: The same items rewritten: unit and time window inside the question, digit boxes with a sign, a fixed date format, one tick and an "unknown" option.

Take each bad item in turn.

- **"Fluid balance: __"** has no unit (ml? litres? ml/kg?) and no time window (today? since admission? cumulative to day 5?). In a fluid study this single box may be the primary outcome of the whole trial. The good version reads "Cumulative fluid balance, ICU admission to day 5 (ml/kg)", with digit boxes and a plus or minus sign, because fluid balance can be negative.
- **"Age: __"** leaves open whether the answer is in years or months and on which date it was calculated. Better: collect the date of birth and the date of admission, and let the computer calculate age.

- **"Infection: lung / urine"** offers incomplete options (no "other", no "unknown") and does not say whether more than one box may be ticked. The good version offers lung, urine, abdomen, other and unknown, and says "tick ONE" for the main source.
- **"Did fluids work and the patient improve? Y / N"** is two questions in one (a *double-barrelled* question), and both halves are opinions. Replace it with objective measurements: lactate, mean arterial pressure, vasopressor dose.

One more item deserves attention: what does a blank box mean? It might mean the test was not done, that the result was forgotten, or that the value was zero. A good CRF removes this ambiguity by giving an explicit option for "not done" and for "unknown", so that **a blank always means "forgotten"** and can be chased.

10.4.2 Worked example: from a question to one CRF item

Worked example: building the CRF item for cumulative fluid balance

Suppose the exposure of a study is "cumulative fluid balance". Writing the CRF item requires five decisions, made before the first patient is enrolled.

1. **What?** Fluid in minus fluid out. Fluid in includes crystalloid boluses, colloids and albumin, maintenance fluids, drug diluents and line flushes, blood products, and enteral and oral intake. Fluid out includes urine, drains, gastric losses and net fluid removal by renal replacement therapy. Decide whether insensible losses (sweat, breathing) are included; FLUID-ICU did not include them. Write the decision in the data dictionary.
2. **When?** From ICU admission to the end of day 3 (the FLUID-ICU definition), or to the end of day 5 (Paper 1). Day 1 must be defined too: for example, from ICU admission to the first midnight, then midnight to midnight. A different time window gives a different variable.
3. **Unit?** Millilitres per kilogram of admission body weight, so that patients of different sizes can be compared. This decision has a consequence: the CRF must now also contain **body weight at admission**, and whether it was measured or estimated.
4. **Format?** A signed number with one box per digit, because the balance can be negative.
5. **Missing?** A "not recorded" box and a space for the reason, so that a blank always means "forgotten".

In practice, the best CRF does not ask the nurse to calculate the balance at all. It records the raw daily totals of input and output in millilitres for each day, and the database calculates the daily balance, the cumulative balance and the balance per kilogram. Raw values can be checked against the fluid chart; a hand calculation cannot.

10.4.3 Five rules for a good CRF

Most CRF problems are covered by five rules.

1. **Only what you need.** Every item must answer an objective, adjust for a confounder or describe the population. Short forms get completed.
2. **One question, one answer.** Avoid double-barrelled questions. Prefer a measured value to an opinion.
3. **Units and timing in the question.** Write the unit (ml/kg, mmol/L) and the time point ("at ICU admission", "worst value in the first 24 hours", "day 5") inside the question itself.
4. **Closed answers.** Coded tick boxes are faster to fill in and to analyse than free text. Always include "other (specify)" and "unknown" or "not done".
5. **Pilot it.** Give the form to two colleagues to fill in for the same three to five patients, and compare. Every disagreement points to an item that needs rewriting.

A further set of practical details turns a list of questions into a usable form:

- Identify participants by a **study ID** only, never by name or medical record number, and print the ID on every page.
- Use a fixed date format (for example DD/MM/YYYY) and 24-hour times (hh:mm).
- Record **raw values** and let the database do calculations such as age, body mass index or cumulative balance.
- For laboratory values, choose **one unit per test for the whole study** and print it on the form. Creatinine and bilirubin, for example, are reported in µmol/L in some laboratories and mg/dL in others.
- Give completion instructions: write in capitals, one digit per box; to correct an error, draw a single line through it, write the correct value beside it, and add initials and the date. Never use correction fluid or overwrite.
- List the **source documents** (ICU chart, fluid balance chart, laboratory system, medical notes) so that every value can be verified.
- Put a **version number and date** on every page, and design the form before data collection starts. Changing a form halfway through a study creates two different datasets.

Worked example: rewriting four poor CRF items

Table 10.3: Four poor CRF items and improved versions.

Poor item	Problem	Improved item
Fluid given: __	No unit, no time window, no definition	Total IV fluid received from ICU admission to 24 h (ml), including crystalloids, colloids and drug diluents: [ ][ ][ ][ ][ ] ml; ☐ not recorded
Weight: __	No unit; measured or estimated? When?	Body weight at ICU admission (kg): [ ][ ][ ].[ ]; ☐ measured ☐ estimated ☐ not available
Comorbidities: _____	Free text, impossible to analyse	Comorbidities present at admission (tick ALL that apply): ☐ diabetes ☐ chronic kidney disease ☐ heart failure ☐ COPD ☐ cirrhosis ☐ cancer ☐ none ☐ unknown
Lactate high? Y / N	Opinion, loses the value, no time point	Serum lactate at ICU admission (mmol/L): [ ][ ].[ ]; ☐ not done

Each improved item passes four checks: the unit is in the question, the time point is in the question, there is only one possible reading, and there is a way to record “unknown” or “not done”. Note also the lactate item: collecting the number is better than collecting “high, yes or no”, because the definition of “high” can then be chosen, and changed, in the analysis.

10.5 The data dictionary

A **data dictionary** (also called a codebook) is the instruction manual of a dataset. It has one row for every variable and explains, at a minimum, the short variable name, what it means, its type, its unit or codes, and the values it is allowed to take. The figure below shows a few rows for a study of fluids in septic shock.

Variable name	Label (meaning)	Type	Unit / codes	Allowed values
fb_d5	Cumulative fluid balance, day 5	Continuous	ml/kg	-100 to 300
age	Age at ICU admission	Continuous	years	18 - 110
sex	Sex	Binary	1 = male, 2 = female	1, 2
src	Main source of infection	Nominal	1 lung, 2 urine, 3 abdomen, 4 other, 9 unknown	1 - 4, 9
died28	Died by day 28	Binary	0 = alive, 1 = died	0, 1

Figure 10.4: Example rows of a data dictionary: each variable has a short name, a label, a type, a unit or codes, and an allowed range.

Read one row aloud: `fb_d5` is the cumulative fluid balance at day 5, a continuous variable measured in ml/kg, allowed to lie between -100 and 300. That allowed range is not decoration; it becomes an automatic check on the data, as we will see in the next section. Note also the code 9 for "unknown" in the source-of-infection row: missing and unknown are written down, not left blank.

Why does a data dictionary matter so much? In six months nobody will remember whether sex 1 meant male or female, or whether `fb_d5` was in millilitres or ml/kg. A statistician, a co-author, a reviewer, or the investigator herself a year later must be able to understand every column without phoning anyone. Writing the dictionary *before* data collection also forces the decisions about units, codes and time windows that make the CRF clear.

Good conventions for a data dictionary are:

- **Variable names**: short, lower case, no spaces or special characters, never reused (`fb_d5`, not "Fluid balance day 5"). Such names work in every statistical program.
- **Codes**: the same codes everywhere (for example 1 = yes, 0 = no), defined in the dictionary.
- **Missing-value codes**: if special codes are used for missing values (for example -88 = not done, -99 = unknown), they must be defined in the dictionary and used consistently, and they must be converted to "missing" before any analysis. A code of 999 that is mistaken for a real lactate value is a disaster, as the FLUID-ICU example below shows.
- **Derived variables**: variables calculated from others (age from dates, cumulative balance from daily inputs and outputs, "any acute kidney injury" from the stage) are marked as derived, with the formula.
- **Link to the CRF**: a column giving the CRF item number makes it easy to trace each variable back to its source.

The table below shows selected rows of the FLUID-ICU data dictionary.

Table 10.4: Selected rows of the FLUID-ICU data dictionary (simulated study).

Variable	Label	Type	Units or codes	Allowed range	Notes
`age`	Age	Numeric (integer)	Years	18 to 110	
`weight_kg`	Body weight on admission	Numeric	kg, 1 decimal	30 to 250	Used to compute balance in ml/kg
`lactate_mmol_l`	Blood lactate on admission	Numeric	mmol/L, 1 decimal	0.1 to 30	About 15% missing
`fluid_in_d1_ml`	Total fluid input, ICU day 1	Numeric (integer)	ml	0 to 20 000	All IV and enteral fluid, including drugs and nutrition
`cum_fluid_balance_d3_ml_kg`	Cumulative fluid balance, ICU days 1 to 3	Numeric (derived)	ml/kg, 1 decimal	-150 to 250	(Fluid in days 1 to 3 minus fluid out days 1 to 3) / weight
`positive_balance_d3`	Positive cumulative balance at day 3	Binary (derived)	No / Yes	No, Yes	Yes if balance > 0 ml/kg
`aki_stage`	Acute kidney injury, maximum KDIGO stage, days 1 to 7	Ordinal	0 = none, 1, 2, 3	0 to 3	
`died_28d`	Death from any cause by day 28	Binary	No / Yes	No, Yes	Primary outcome
`time_to_event_28d`	Time from ICU admission to death or censoring	Numeric (integer)	Days	1 to 28	28 if alive at day 28

10.6 Validation: let the rules catch the errors

Data-entry errors are inevitable. The aim is to catch them early, while the patient's chart is still on the ward, rather than a year later when it is archived. Four simple types of check catch most errors, as the figure below shows.

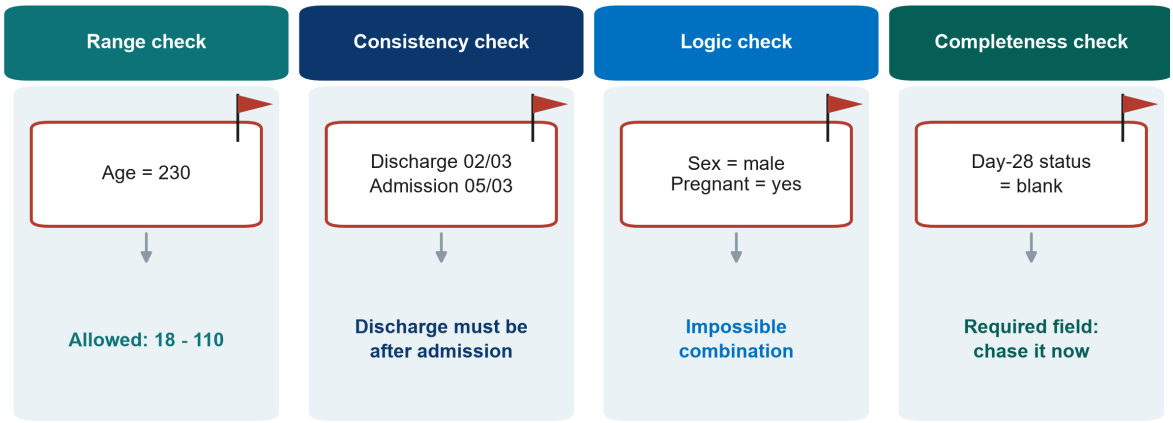


Figure 10.5: Four types of validation check: range (an age of 230), consistency (discharge before admission), logic (a male patient recorded as pregnant) and completeness (a required field left blank).

- A **range check** flags a value outside the allowed range in the data dictionary. An age of 230

was probably 23 or 30, but we cannot know which.

- A **consistency check** flags two values that cannot both be true: a discharge date before the admission date, or a date of death after the day-28 visit at which the patient was recorded as alive.
- A **logic check** flags an impossible combination of answers: a male patient recorded as pregnant; a "date of death" filled in for a patient whose vital status is "alive".
- A **completeness check** flags a required field that is empty, so that it can be chased now.

Electronic forms can run these checks at the moment of entry and refuse to save an impossible value. On paper, a study coordinator can run the same checks weekly on the entered data. Either way, the checks come directly from the data dictionary, which is another reason to write the dictionary first.

10.7 From bedside to database: data management in practice

Good data need a safe route from the patient to the analysis file. The figure below shows the two common routes.

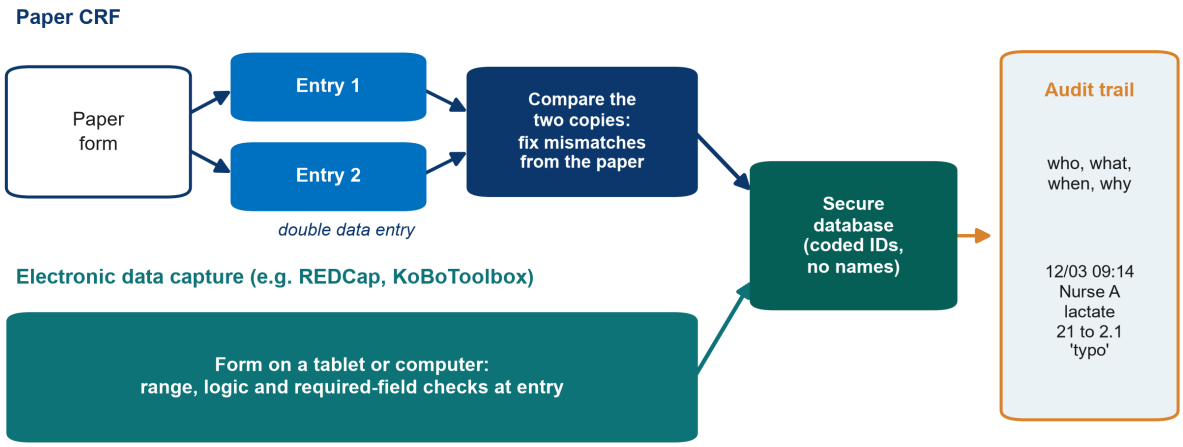


Figure 10.6: Two routes from patient to database: a paper CRF entered twice by different people and compared (double data entry), or electronic data capture with checks at entry; both feed a secure database with coded IDs and an audit trail.

The paper route. Data are written on a paper CRF and then typed into a computer. The safest approach is **double data entry**: two different people type the same forms independently, a program compares the two copies, and every mismatch is resolved by looking at the paper form. This catches most typing errors, such as a lactate of 21 instead of 2.1. When resources are limited, a common compromise is to double-enter a random sample of forms (for example 10%) and the key variables of all forms, and to report the error rate found.

Electronic data capture (EDC). The form is completed directly on a tablet or computer. Free and widely used systems such as REDCap (Harris et al. 2009) and KoBoToolbox allow the range, logic and required-field checks from the data dictionary to run at entry, so impossible values are blocked before they are saved. EDC also removes the transcription step from paper to computer.

Whichever route is used, five principles of data management apply.

- **Coded identifiers.** The database holds study IDs, not names. The key linking study IDs to names and medical record numbers (sometimes called the *linking log* or *enrolment log*) is kept separately, locked and accessible only to named staff. This protects confidentiality and is usually required by the ethics committee (Chapter 17).
- **Audit trail.** A good system records who changed what, when and why: "12/03, 09:14, nurse A changed lactate from 21 to 2.1, reason: typing error". Values are never overwritten silently. Monitors, ethics committees and journals may ask to see this trail, and the principles of Good Clinical Practice expect it (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use 2025).
- **Versions.** The raw data file, as exported or received, is set to read-only and never edited. Every cleaning step produces a new, dated version (for example `data_2026-03-12_v2`) with a short record of what changed. Any result can then be traced back to the raw data.
- **Backups.** A common rule is "3-2-1": three copies, on two different media, one stored off-site and encrypted. A backup that has never been restored is only a hope; test it.
- **Data sharing.** Many journals now require a data-sharing statement (Chapter 19), for example: "De-identified participant data and the data dictionary will be available from the corresponding author on reasonable request after publication, subject to ethics approval." If sharing is not mentioned in the protocol and consent form, it may not be allowed later. Data that are well documented, with a clear dictionary and stable identifiers, are much easier to share and reuse, an idea summarised in the FAIR principles (findable, accessible, interoperable, reusable) (Wilkinson et al. 2016).

! Common mistake: the spreadsheet that does everything

A single spreadsheet containing patient names, phone numbers, hospital numbers and results, edited by several people and emailed back and forth, breaks almost every rule in this section: no coded IDs, no audit trail, no version control, no protection if a laptop is lost. If your laptop were stolen tonight, would your study survive? If the answer is uncomfortable, fix the data pipeline before the next patient is enrolled.

10.8 Missing data: why they matter and how to prevent them

Missing data do not simply reduce the number of patients. They can give the **wrong answer**. The figure below explains why.

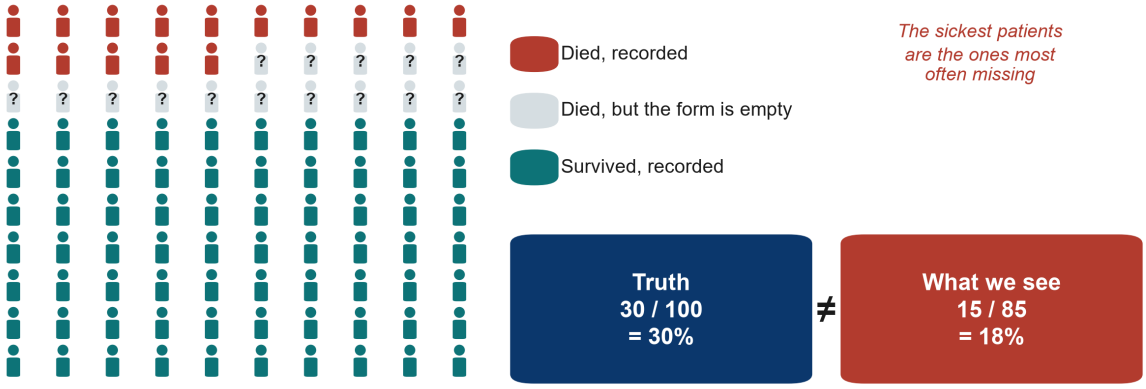


Figure 10.7: Of 100 patients, 30 truly died, but 15 of the deaths have an empty outcome box; among the 85 recorded patients mortality appears to be 18% instead of 30%.

Imagine 100 patients, of whom 30 truly died. For 15 of the deaths the outcome box is empty: the patient was transferred, died at night, or the form was never completed. Among the 85 patients with a recorded outcome we see 15 deaths, a mortality of 18%. The truth was 30%. The estimate is less precise, and it is also biased, because whether the outcome is missing is linked to the outcome itself. The sickest and most chaotic patients are the ones whose forms stay empty.

The common reassurance "we lost only 15% of patients" therefore misses the point. **Who** is missing matters more than **how many**. Chapter 15 names the mechanisms by which data go missing (missing completely at random, missing at random, and missing not at random) and explains statistical methods such as multiple imputation (Sterne et al. 2009). But all statistical fixes are second best. Prevention is the only reliable cure, and it rests on five habits.

1. **Fewer items.** Every extra item increases the chance that something is left blank.
2. **A clear form.** Mark required fields; offer "not done" and "unknown" options so that a blank always means "forgotten".
3. **Training.** A short briefing for every nurse or doctor who will collect data, with the data dictionary in hand, so that everyone applies the same definitions.
4. **Weekly checks.** A simple count of empty boxes per variable and per patient. If one item is always empty, the item is the problem, not the staff.
5. **Chase early.** Fix gaps while the chart is still on the ward. Plan follow-up for patients transferred out: in ICU studies the day-28 outcome is the variable most often missing, so decide in advance who will check it, and how (hospital records, a phone call).

10.9 Cleaning data: a worked case with FLUID-ICU

Even with a good form, data arrive messy. This section follows the simulated FLUID-ICU cohort from the raw file to the analysis-ready file, to show what real problems look like and how they are fixed in a way that anyone can check.

10.9.1 The study

FLUID-ICU (simulated) is a multicentre prospective cohort of adults with sepsis admitted to four intensive care units, labelled ICU A to D, between January 2023 and December 2024. It asks an observational question: **is a positive cumulative fluid balance by day 3 associated with death by day 28?** The exposure is the cumulative fluid balance over ICU days 1 to 3 (all fluid in minus all fluid out), divided by admission weight, in ml/kg. The primary outcome is death from any cause by day 28; the time to death is also recorded, with survivors censored at day 28. Other variables include demographics, comorbidities, the source of infection, severity scores (SOFA and APACHE II), lactate, vasopressor use, mechanical ventilation, acute kidney injury, renal replacement therapy, ventilator-free days and ICU length of stay.

After cleaning, the dataset contains **620 patients** (ICU A 191, ICU B 164, ICU C 147, ICU D 118). Twenty-eight-day mortality was 179 of 620 (28.9%). Of the 618 patients with a computable balance, 373 (60.4%) had a positive balance by day 3.

10.9.2 What the four ICUs actually sent

The raw file arrived with 623 rows and 29 columns. The figure below summarises the main problems found and how each was resolved.

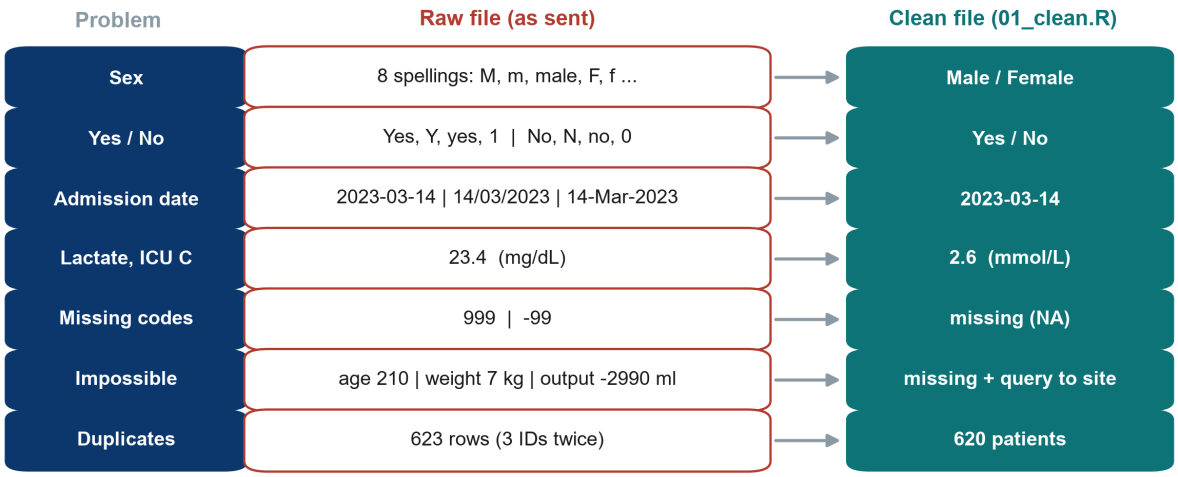


Figure 10.8: Problems in the raw FLUID-ICU file (simulated) and their clean versions: inconsistent spellings, mixed date formats, lactate in the wrong unit, missing-value codes, impossible values and duplicate records.

The full cleaning log records twelve steps. The most instructive are listed in the table below.

Table 10.5: Main steps of the FLUID-ICU cleaning log (simulated data).

Problem found	How many	What was done	Why
Duplicate records: the same patient entered twice	3 IDs (623 rows for 620 patients)	Checked that both copies were identical in every column, then kept one	Each patient must be counted once
ICU name typed in different ways (`icu a`, `ICU-A`, `ICUA`)	14 rows	Standardised to ICU A, B, C, D	Otherwise software sees 11 "different" ICUs
Admission date in three formats (`2023-03-14`, `14/03/2023`, `14-Mar-2023`)	365, 164 and 94 rows	Each format read explicitly and converted to YYYY-MM-DD	Is 03/04 the 3rd of April or the 4th of March?
Sex coded nine ways (`Male`, `M`, `m`, `male`, `F`, `f`, ...)	200 rows	Recoded to Female / Male	One variable, one coding
Yes/No variables coded `Yes`, `Y`, `yes`, `1` (one ICU sent 1/0)	8 variables, about 1,900 cells	Recoded to No / Yes; checked that no value was lost	`1` and `Yes` must mean the same thing
Missing-value codes used as numbers: lactate `999` and `-99`, APACHE II `-99`, MAP `999`	Lactate 21, APACHE II 6, MAP 4	Set to missing	A lactate of 999 would wreck the mean
ICU C reported lactate in mg/dL; the others in mmol/L (ICU C median 23.8 vs about 2.5 elsewhere)	126 values	Divided by 9.01 (1 mmol/L = 9.01 mg/dL); ICU C median became 2.65; unit confirmed with the site laboratory	One unit for one variable
Impossible values: an age of 210, a weight of 7 kg, a day-2 fluid output of -2990 ml	3 values	Set to missing; query sent to the site; patient kept in the study	The true value cannot be guessed; deleting the patient would bias the sample
Derived variables needed	3 new variables	Cumulative balance in ml/kg, positive balance (> 0 ml/kg), any AKI (KDIGO stage 1 or more) calculated by the script	Derived values are calculated, never typed by hand

Each of these problems would have broken the analysis silently. A lactate of 999 inflates the mean. "Y" and "Yes" become two different categories. Values in mg/dL make one ICU look like a lactate catastrophe, roughly nine times higher than the others. The unit problem is particularly instructive: it was found only by **summarising the variable separately for each site** and noticing that one median was nine times the others. Summarising key variables by site, by month and by data collector is one of the most powerful cleaning tools available.

Every one of these problems would also have been prevented, or caught at entry, by the measures described earlier in this chapter: an electronic CRF with coded answers, a fixed date format, one printed unit per laboratory test, range checks from the data dictionary, and a unique patient ID checked at enrolment.

10.9.3 Principles of reproducible cleaning

The FLUID-ICU cleaning follows four rules that apply to any study.

1. **The raw file is never edited.** It is kept exactly as received, as the reference for every later question.
2. **Changes are made by a script, not by hand.** A short program (in R, Stata, SPSS syntax, Python or similar) reads the raw file and applies each correction, one block per problem. Because it is a script, anyone can re-run it and obtain exactly the same clean file: the cleaning is **reproducible**. In FLUID-ICU the last step of the script rebuilds the clean file and confirms that it is identical, value for value, to the reference version.
3. **Every change is written down in a cleaning log.** Each row of the log states the problem found, how many records were affected, what was done and why. Where a value is queried with a site, the query is logged too (patient, variable, original value, status).
4. **Impossible values are set to missing and queried; patients are not deleted.** The patient with an age of 210 still has a valid outcome, fluid balance and severity score. Deleting the whole record for a typing error loses information and can bias the sample, because errors are not always random. We never guess the true value: 210 might have been 21, 71 or something else.

! Common mistake: fixing errors by hand in a spreadsheet

Correcting values directly in a spreadsheet is fast, but it is invisible and cannot be repeated. Nobody, including the person who did it, can later tell which values were changed, from what, or why. If the raw data are updated (for example, after a site answers a query), all the manual corrections must be repeated. A script and a cleaning log make every change traceable and repeatable.

10.9.4 What remains after cleaning: the flow of patients

After cleaning, what is left missing is *real* missing data: values that were never measured or that could not be trusted. In FLUID-ICU, lactate was missing for 93 of 620 patients (15.0%), APACHE II for 6 (1.0%), mean arterial pressure for 4 (0.6%), and age, weight and day-2 fluid output for one patient each (the impossible values set to missing). Every other variable was complete.

These gaps determine how many patients can enter each analysis, which is shown in the flow diagram below.

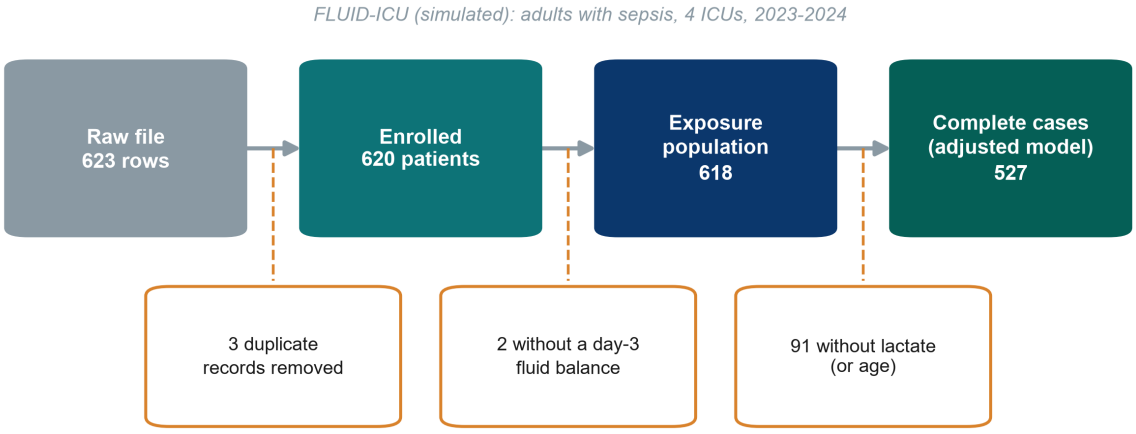


Figure 10.9: Flow of patients in FLUID-ICU (simulated): 623 raw rows, 620 enrolled patients after removing 3 duplicates, 618 with a day-3 fluid balance, and 527 complete cases for the adjusted model.

The raw file had 623 rows; removing three duplicates left 620 enrolled patients. Two patients had no computable day-3 fluid balance because their weight or one fluid output had been set to missing, leaving 618 in the exposure population, used for describing and comparing the groups. A further 91 had no lactate (or no age), leaving 527 complete cases for the adjusted analysis. That is 85% of the enrolled patients. Whether the 15% left out differ from the rest is an important question: in FLUID-ICU, patients without a lactate measurement were less severely ill (median SOFA 5 vs 8) and less likely to die (19.4% vs 30.6%). Chapter 15 explains what this means for the analysis, and Chapter 16 follows the whole FLUID-ICU analysis from start to finish.

A flow diagram like this one appears in almost every published cohort study and trial, usually as the first figure of the Results. Reporting guidelines ask for it: STROBE asks authors to report the numbers of individuals at each stage of the study and the reasons for non-participation (Elm et al. 2007), and CONSORT asks for a participant flow diagram in trials (Schulz et al. 2010).

10.10 How data collection appears in published papers

Readers rarely see the CRF or the cleaning log, but a well-written Methods section tells them what they need. Look for:

- **Exact definitions** of the outcome and exposure, including units and time windows. Paper 1, for instance, reports its primary outcome as cumulative fluid balance at day 5 in ml/kg of admission body weight (Boulet et al. 2024).
- **How and by whom data were collected**: prospectively on a dedicated CRF, or retrospectively from records; by trained research staff or by routine clinical staff.

- **Data quality measures**: electronic data capture with range checks, double entry, source-data verification, monitoring.
- **Handling of data problems**: "Data were cleaned with a script; all changes are documented in a cleaning log. Implausible values were set to missing and queried with the sites."
- **Missing data**: the number missing for each variable, often in Table 1 or a supplementary table, and the participant flow diagram.

When appraising a paper (Chapter 4), the absence of this information is itself informative. A retrospective study that does not say how variables were defined or extracted, or that reports no missing data at all for an ICU cohort, deserves a careful second look.

Summary

Good analysis starts with good data. The type of each variable (continuous, count, binary, nominal, ordinal or time-to-event) decides how it is collected, summarised and analysed, and continuous variables should be collected as exact numbers. A good CRF item has only one possible reading: the unit and time window are in the question, answers are closed and coded, and "not done" and "unknown" are explicit options. The data dictionary documents every variable, its codes and its allowed range, and those ranges become the validation checks that catch errors at entry. Safe data management uses coded IDs with a separate locked linking log, an audit trail, read-only raw files with dated versions, tested backups and a planned data-sharing statement. Missing data are rarely random, so prevention matters more than any statistical fix. Finally, cleaning is done with a script and documented in a cleaning log, never by hand: the simulated FLUID-ICU example showed duplicates, inconsistent codes, mixed date formats, a unit error found by summarising by site, missing-value codes and impossible values, each fixed in a way that anyone can trace and repeat.

Further reading

- Hulley and colleagues, *Designing Clinical Research* (Hulley et al. 2013): the chapters on planning measurements and on data management are a practical, clinician-friendly guide to CRFs, data dictionaries and quality control.
- Harris et al. on REDCap (Harris et al. 2009): describes the design of a widely used electronic data capture system and the metadata-driven approach (the data dictionary drives the form) that underlies most modern EDC tools.
- ICH E6(R3) Guideline for Good Clinical Practice (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use 2025): the international standard for the conduct of clinical trials, including expectations for data integrity, audit trails and record keeping.
- Wilkinson et al., the FAIR Guiding Principles (Wilkinson et al. 2016): a short paper on making research data findable, accessible, interoperable and reusable, useful when planning documentation and sharing.
- The STROBE statement (Elm et al. 2007): shows what readers expect a cohort or case-control study to report about variables, data sources, missing data and participant flow.

CHAPTER 11

Describing and presenting data

Before comparing groups, describe them. Every clinical paper begins its Results by telling the reader who the patients were: how old, how sick, how many had each comorbidity, how long they stayed. These descriptions look simple, and they are, but they are also where many papers first go wrong: a mean reported for a variable that no typical patient resembles, a bar chart that hides a single extreme patient, a table with four decimal places and no units. This chapter explains how to summarise each type of variable honestly, how to choose a graph, how to build and read a "Table 1", and how to present tables and figures that a reader can understand without the text. Description is not a preliminary chore before the "real" analysis: done well, it often reveals the most important feature of the data, as it does in the simulated FLUID-ICU cohort used throughout this chapter.

In this chapter you will learn

- How to summarise a categorical variable with counts and percentages.
- What the mean and the median are, how to calculate them, and why they differ for skewed data.
- How to recognise the shape of a distribution and why it decides the summary you report.
- What the standard deviation and the interquartile range measure, and why you should never mix the pairs.
- How to choose a graph from the type of variable, and how to avoid misleading figures.
- How to read a Table 1 with two questions in mind: do these patients look like mine, and were the groups alike at the start?
- How to lay out a publication-quality table and write descriptive sentences for a Results section.

11.1 Why describe before you compare

Description answers two questions that come before any test or model. The first is **who were the patients?** A reader in another hospital wants to know whether the results apply to her own patients, and she can only judge that if the study describes its population clearly. The second is **what do the data look like?** Is a variable symmetric or skewed? Are there extreme values? Are there impossible values that escaped cleaning? Is one group sicker than the other at the start? The answers decide which summary to report, which test to use and, sometimes, whether a simple comparison can be trusted at all.

The running example of this book is fluid management in septic shock. Paper 1 (Boulet et al. 2024), a pilot randomised trial of 48 patients, compared an optimised restrictive fluid strategy with usual

care, with cumulative fluid balance at day 5 as its primary outcome (Boulet et al. 2024). Paper 2 (Hyun et al. 2023), a propensity-matched cohort study, examined whether a positive cumulative fluid balance during the first three ICU days was associated with mortality (Hyun et al. 2023). We also use **FLUID-ICU**, a **simulated** cohort of 620 adults with sepsis in four ICUs, created for teaching. Its data behave like real ICU data, but no real patient, hospital or country is involved. Chapter 10 described how the FLUID-ICU data were collected and cleaned; this chapter describes them.

The variable types introduced in Chapter 10 (continuous, count, binary, nominal, ordinal and time-to-event) organise everything that follows. Categories are summarised with counts and percentages; numbers need a measure of centre and a measure of spread, chosen according to the shape of the data. Time-to-event variables have their own summaries, the Kaplan-Meier curve and the median survival time, described in Chapter 14.

11.2 Describing categories: counts and percentages

A binary, nominal or ordinal variable is summarised by the **number** of patients in each category and the **percentage** that number represents, written "n (%)". In FLUID-ICU, for example, 376 of 620 patients were male, written "376 (60.6%)", and 179 of 620 died by day 28: "179 (28.9%)".

Three points need care.

- **State the denominator.** A percentage without its denominator cannot be checked or interpreted. When some values are missing, the denominator changes: if lactate is missing for some patients, "lactate above 4 mmol/L" must be calculated among those with a lactate value, and the number missing reported. Writing "n/N (%)", for instance "134/373 (35.9%)", is the clearest form.
- **Report percentages, not proportions, and choose sensible decimals.** "0.375" is harder to read than "9 (38%)". With small groups, decimals in percentages suggest a precision that does not exist: in a group of 24 patients, one patient is about 4%, so "37.5%" adds nothing to "38%". With several hundred patients, one decimal is reasonable.
- **For ordinal variables, give every category.** For pressure-injury stage or acute kidney injury stage, report n (%) for each stage. The median stage is also acceptable. A "mean stage of 2.3" is not, because the steps between stages are not equal.

The natural graph for a categorical variable is a **bar chart** of counts or percentages, ideally with the bars sorted from largest to smallest for nominal variables, and in their natural order for ordinal ones.

11.3 Describing numbers: centre and spread

A numerical variable needs two numbers: a **centre** (where is the typical patient?) and a **spread** (how different are patients from each other?). There are two families of summaries: the **mean with the standard deviation**, and the **median with the interquartile range**. The choice between them depends on one thing: the **shape** of the data.

11.3.1 The mean and the median

The **mean** is the arithmetic average: add up all the values and divide by the number of values. For n values $x_1, x_2, \ldots, x_n$,

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

The **median** is the middle value when all the values are lined up from smallest to largest. Half the patients have a value below the median and half above. With an odd number of patients, the median is the middle one; with an even number, it is halfway between the two middle ones.

Worked example: mean and median by hand

Seven patients stayed in the ICU for 1, 2, 2, 3, 4, 6 and 24 days.

1. **Line up** the values from shortest to longest stay. They are already in order: 1, 2, 2, 3, 4, 6, 24.
2. **Median.** The middle patient of seven is the fourth: **3 days**. Three patients stayed less, three stayed longer.
3. **Mean.** Add them up: 1 + 2 + 2 + 3 + 4 + 6 + 24 = 42. Divide by 7: **6 days**.
4. **Compare.** Six of the seven patients stayed 6 days or less. The mean of 6 days is pulled upwards by the single 24-day stay; the median describes the typical patient.

Now suppose the long-stay patient had stayed 60 days instead of 24. The median does not change: it is still 3 days. The mean jumps to 78 / 7 = 11.1 days, longer than six of the seven patients actually stayed. One extreme value moves the mean a lot and the median not at all. The median is said to be **robust** to extreme values.

11.3.2 Shape: symmetric or skewed

The **distribution** of a variable is the pattern of how often each value occurs, best seen in a **histogram**: the range of values is divided into intervals, and a bar shows how many patients fall into each interval. Two shapes cover most clinical data, as the figure below shows.

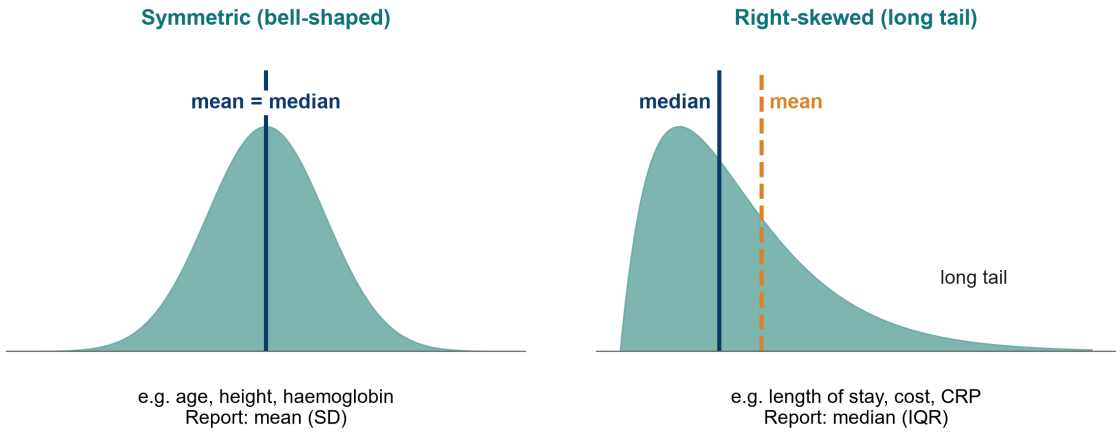


Figure 11.1: Two common shapes of clinical data: symmetric and bell-shaped, where the mean equals the median (report mean and SD), and right-skewed with a long tail, where the mean is pulled above the median (report median and IQR).

- **Symmetric, bell-shaped** distributions (often called “normal” or Gaussian) cluster around a typical value, with values falling away evenly on both sides. The mean and median are in the same place. Many measurements of body size and of physiology that the body regulates tightly look like this: age in adult patients, height, haemoglobin, systolic blood pressure.
- **Right-skewed** distributions have a long tail of high values. They arise for quantities that cannot go below zero but can go very high: lengths of stay, times, costs, counts, and many laboratory values such as C-reactive protein, troponin or lactate. The long tail pulls the mean to the right of the median.

Other shapes exist but are less common. **Left-skewed** distributions have a ceiling, such as oxygen saturation or gestational age at birth. **Bimodal** distributions, with two peaks, usually mean that two different groups have been mixed, such as children and adults on the same ward. For any shape that is not clearly symmetric, show the histogram and report the median.

The figure below shows the most famous skewed variable in hospital medicine: length of stay.

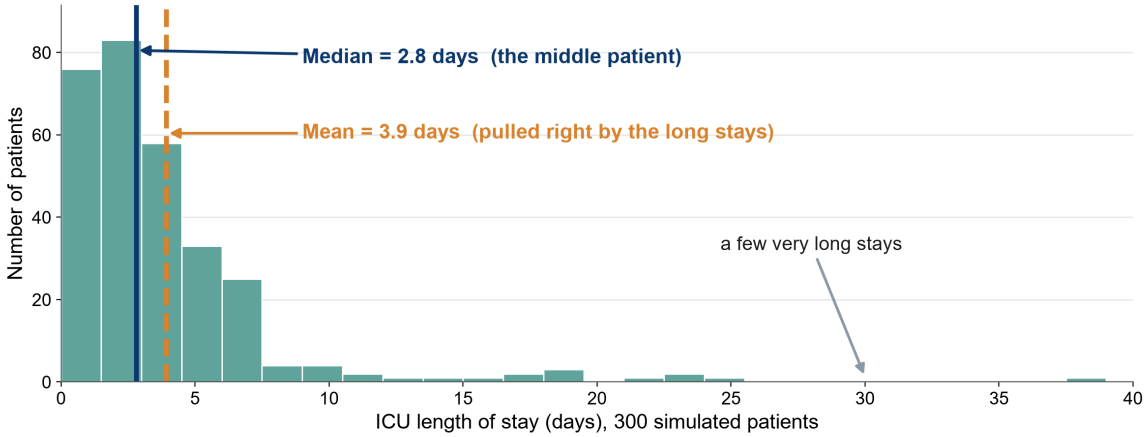


Figure 11.2: ICU length of stay in 300 simulated patients is right-skewed: the median of 2.8 days describes the middle patient, while a few very long stays pull the mean to 3.9 days.

Most of these patients stayed between one and five days, but a few stayed three to six weeks. Those few long stays pull the mean to 3.9 days while the median stays with the typical patient at 2.8 days. If a manager says "the average ICU stay is 3.9 days", most patients actually leave earlier than that. For skewed data the mean describes no real patient.

Two quick checks help when no histogram is available. First, **if the mean and the median are far apart, the data are skewed**; report the median. Second, for a variable that cannot be negative, **if the standard deviation is larger than about half the mean, the distribution is probably skewed**, because a symmetric distribution with that spread would have to extend below zero.

11.3.3 Spread: the standard deviation and the interquartile range

Knowing the centre is not enough. Two ICUs can both have a median stay of 3 days, one with almost every patient staying 2 to 4 days and the other with stays ranging from a few hours to two months. The spread tells you how different patients are from each other.

The **standard deviation (SD)** goes with the mean. It is, roughly, the typical distance of a patient's value from the mean. It is calculated by taking each value's distance from the mean, squaring it (so that distances above and below count equally), averaging the squares (dividing by $n - 1$ rather than n, for technical reasons), and taking the square root to return to the original units:

$$\text{SD} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}.$$

Worked example: a standard deviation by hand

Five patients are aged 50, 55, 60, 65 and 70 years. The mean is 300 / 5 = 60 years.
The distances from the mean are -10, -5, 0, 5 and 10. Their squares are 100, 25, 0, 25 and 100, which add up to 250. Dividing by $n - 1 = 4$ gives 62.5 (this is the *variance*). The square root of 62.5 is about 7.9.
So the ages are summarised as **mean 60 (SD 7.9) years**: the typical patient is about 8 years away from the mean age. In practice a computer does this calculation; what matters is the interpretation.

For a bell-shaped distribution the SD has a useful property: about two thirds of patients (68%) lie

within one SD of the mean, and about 95% lie within two SDs (more precisely, 1.96 SDs). In the figure below, adult ICU patients have a mean age of about 61 years with an SD of 12, so roughly two thirds are aged between about 49 and 73, and about 95% between about 37 and 85. This rule only works for symmetric data: for a skewed variable, "mean minus two SDs" can give an impossible negative length of stay.

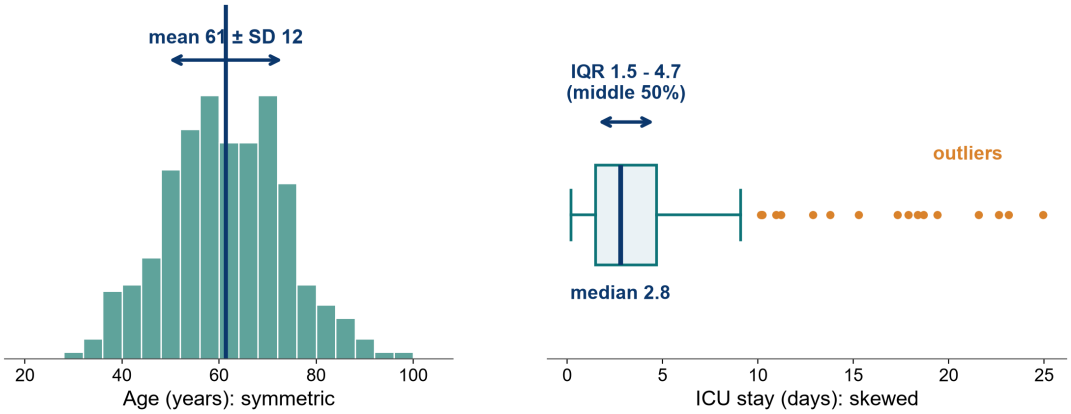


Figure 11.3: Left: age is roughly symmetric and is summarised by its mean and SD. Right: ICU stay is skewed and is summarised by a boxplot showing the median (2.8 days), the interquartile range (1.5 to 4.7 days) and outliers.

The **interquartile range (IQR)** goes with the median. Line the patients up again and divide them into four equal groups. The value one quarter of the way along is the **first quartile** (the 25th percentile); the median is the second quartile; the value three quarters of the way along is the **third quartile** (the 75th percentile). The IQR runs from the first to the third quartile and contains the middle 50% of patients. Because it ignores the top and bottom quarters, it is not affected by a few extreme values.

The right-hand panel of the figure above shows a **boxplot**, the standard picture of a median and IQR. The thick line is the median; the box runs from the 25th to the 75th percentile; the whiskers extend to the most extreme values that are not unusually far from the box; and points beyond the whiskers are plotted individually as **outliers**. "Median 2.8 days (IQR 1.5 to 4.7)" reads: half the patients stayed less than 2.8 days, and the middle half stayed between 1.5 and 4.7 days.

In the seven-patient example above (1, 2, 2, 3, 4, 6, 24 days), the lower half of the values below the median is 1, 2, 2, with middle value 2, and the upper half above the median is 4, 6, 24, with middle value 6. So the summary is "median 3 days (IQR 2 to 6)". Software packages use slightly different rules for quartiles, so small differences between programs are normal.

The IQR is usually written as the two quartiles, "IQR 1.5 to 4.7", rather than as a single difference ("IQR 3.2"), because the two values show the reader where the middle half of the patients lies. The **range** (minimum to maximum) is sometimes added, but on its own it is a poor measure of spread, because it depends entirely on the two most extreme patients.

Paper 1 offers an instructive example. The trial reported the cumulative fluid balance at day 5 as medians: 6.9 ml/kg (IQR -13.7 to 52.1) in the restrictive group and 35.0 ml/kg (IQR -7.9 to 40.2) in the usual-care group (Boulet et al. 2024). Medians were used because fluid balance in small groups is skewed by a few patients with very large balances. Notice the negative lower quartiles: at least a quarter of the patients in each group had a negative balance at day 5.

11.3.4 Which summary for which variable?

The rules can now be gathered in one place.

Table 11.1: Choosing a summary and a graph from the type and shape of a variable.

Variable type and shape	Summary	Graph
Continuous, roughly symmetric (age, haemoglobin)	Mean (SD)	Histogram
Continuous or count, skewed (length of stay, CRP, lactate, days on a ventilator)	Median (IQR)	Histogram or boxplot
Clinical score (SOFA, APACHE II, Glasgow Coma Scale)	Median (IQR)	Boxplot or bar chart
Binary (died, ventilated)	n (%)	Bar chart (often unnecessary)
Nominal (source of infection)	n (%) per category	Bar chart, sorted
Ordinal (pressure-injury stage)	n (%) per category, or median	Bar chart in natural order
A number compared between groups	Mean (SD) or median (IQR) per group	Side-by-side boxplots with the individual points
Two numbers together	Correlation (Chapter 13)	Scatter plot

! Common mistake: mixing the pairs

Report **mean (SD)** or **median (IQR)**, never "mean (IQR)" or "median (SD)". Each centre has its matching measure of spread. Also avoid the notation "mean ± SD" without saying what follows the ± sign: readers cannot tell whether it is a standard deviation, a standard error or a confidence interval. Writing "64 (SD 11)" or "64 (11)" with a clear label in the row is unambiguous.

11.3.5 The standard deviation is not the standard error

One more distinction prevents a very common error. The **standard deviation** describes how much **patients** differ from each other. The **standard error (SE)** describes how precisely the **mean** has been estimated; it equals the SD divided by the square root of the number of patients, $\text{SE} = \text{SD}/\sqrt{n}$, so it is always smaller than the SD and shrinks as the study grows. The SD belongs in descriptive tables; the SE, and the confidence interval built from it, belong in the comparison of groups and are explained in Chapter 12 (Altman and Bland 2005). Some authors show the SE in descriptive figures precisely because it is smaller and makes the data look tidier. Always ask which one you are looking at.

11.4 Describing FLUID-ICU

The rules above can now be applied to the simulated FLUID-ICU cohort. Of 620 enrolled patients, the mean age was 62.6 years (SD 15.2), 376 (60.6%) were male, the median SOFA score on admission was 7 (IQR 5 to 9), and 179 (28.9%) died by day 28. Age is roughly symmetric, so the mean and SD are appropriate; the SOFA score is a clinical score with a skewed distribution, so the median and IQR are used.

The exposure. It is always worth looking at the exposure before using it. The figure below shows the cumulative fluid balance at day 3 for the 618 patients in whom it could be calculated.

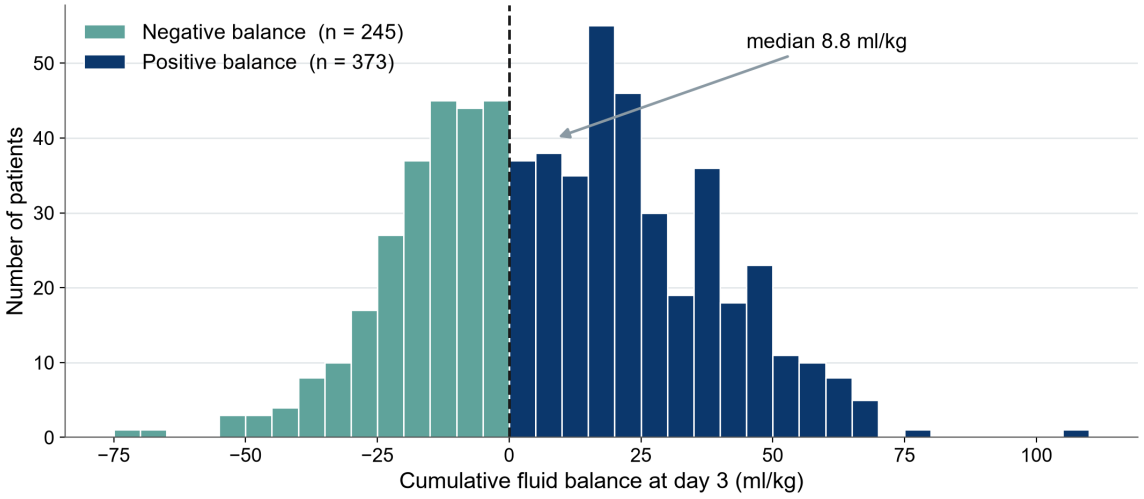


Figure 11.4: Cumulative fluid balance at day 3 in FLUID-ICU (simulated, 618 patients): a wide distribution with a median of 8.8 ml/kg, split at 0 ml/kg into negative (245 patients) and positive (373 patients) balance.

The balance ranges widely, from about -75 to more than 100 ml/kg, with a median of 8.8 ml/kg (IQR -10.3 to 26.2). The middle of the distribution is roughly symmetric, but there are long tails, so the median and IQR are a safe summary; the mean and SD would also be acceptable if stated. For the main analysis the continuous variable was cut at 0 ml/kg into two groups: 373 of 618 patients (60.4%) had a positive balance and 245 a negative one. Cutting a continuous variable loses information, but the groups are easier to explain, and the choice was fixed in the statistical analysis plan before the data were analysed (Chapter 15). Other analyses kept the balance continuous.

A skewed outcome. ICU length of stay in FLUID-ICU behaves exactly like the textbook example.

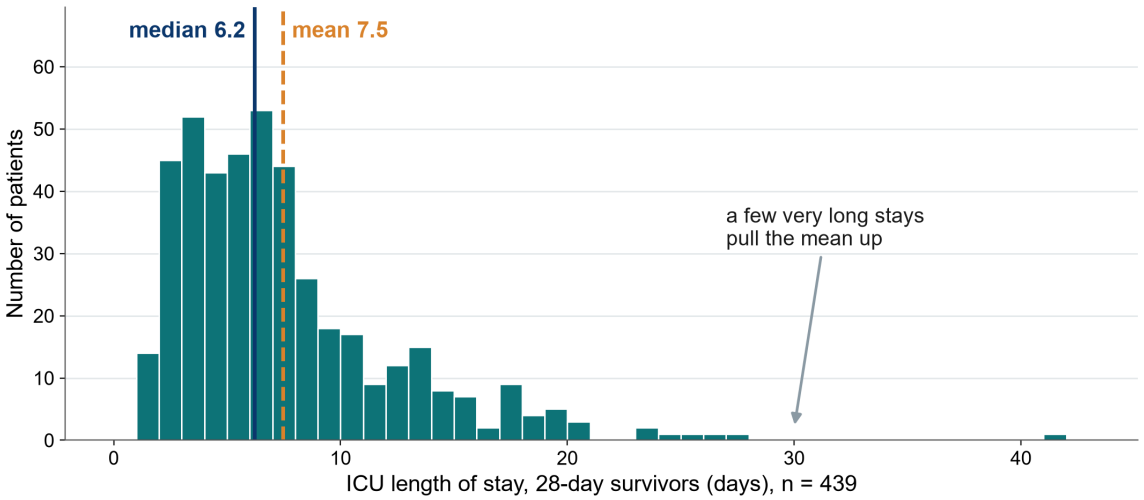


Figure 11.5: ICU length of stay among the 439 survivors at day 28 in FLUID-ICU (simulated) is right-skewed: median 6.2 days, mean 7.5 days, pulled up by a few very long stays.

Among the 439 patients alive at day 28, a few very long stays pull the mean (7.5 days) above the median (6.2 days). Across all 620 patients the mean was 7.3 days and the median 6.0. The median (IQR) is the right summary, and the comparison of groups will need a method suited to skewed data (Chapter 12).

11.5 Choosing a graph

Graphs show what summaries hide: the shape of the data, the spread, the outliers and the overlap between groups. Four graphs cover most of what is needed to describe clinical data, and the choice follows from the variable type, not from the software menu.

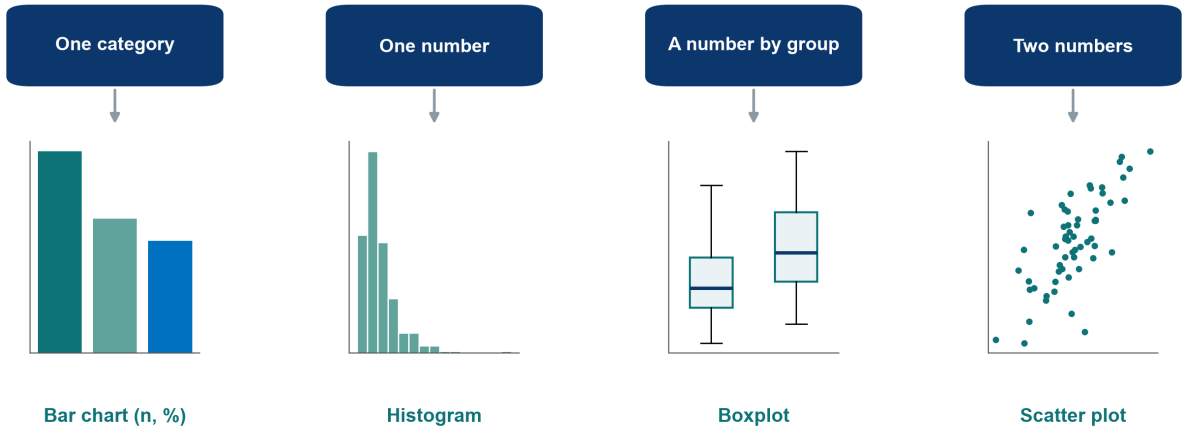


Figure 11.6: Four graphs for describing data: a bar chart for one categorical variable, a histogram for one numerical variable, side-by-side boxplots for a number compared between groups, and a scatter plot for two numerical variables.

- **One categorical variable**: a bar chart of counts or percentages, such as the source of sepsis.
- **One numerical variable**: a histogram, to see the shape, such as length of stay.

- **A number compared between groups**: side-by-side boxplots, such as fluid balance in a restrictive and a usual-care group, ideally with the individual patients shown as dots.
- **Two numerical variables**: a scatter plot, such as fluid balance against lactate.

Time-to-event outcomes have their own graph, the Kaplan-Meier survival curve, introduced in Chapter 14.

11.5.1 Show the data: the dynamite plot and its alternative

The most common poor figure in clinical journals is the **dynamite plot**: a bar showing the mean of each group with a thin error bar on top, like a stick of dynamite. It shows two numbers per group and hides everything else. The two figures below show the same 48 simulated patients drawn in two ways.

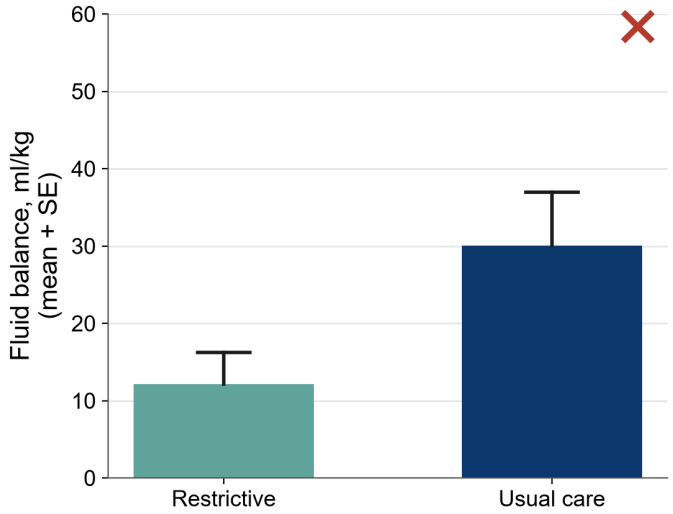


Figure 11.7: A dynamite plot: bars of mean fluid balance with standard-error bars suggest that the usual-care group is much higher.

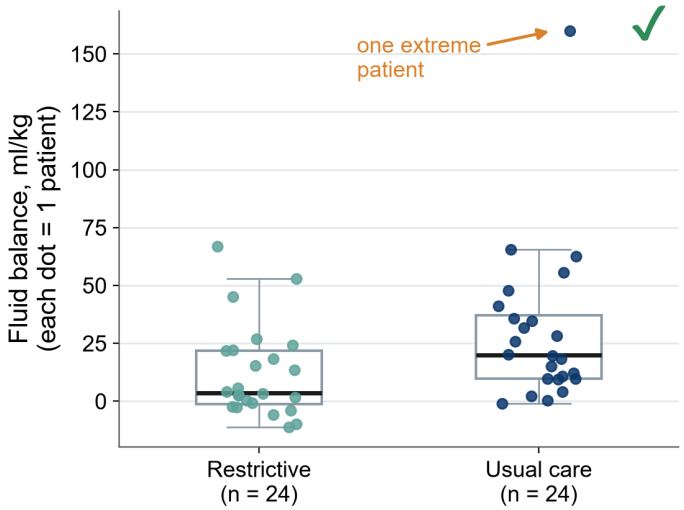


Figure 11.8: The same 48 patients shown as individual dots over boxplots: the data are skewed, the groups overlap a lot, and one extreme patient at 160 ml/kg pulls the usual-care mean upwards.

From the bars alone, most readers would conclude that the usual-care group clearly had a higher fluid balance. The dots-plus-box version tells a different story. One extreme patient, at 160 ml/kg, drags the usual-care mean upwards. The data are skewed, so the mean is the wrong summary in the first place. And the two groups overlap a great deal: many restrictive-group patients had higher balances than many usual-care patients. Note also that the error bars in the dynamite plot are standard errors, which make the spread look small. Whenever error bars appear, the caption must say whether they show the SD, the SE or a 95% confidence interval.

The rule that follows is simple: **with fewer than about 100 patients per group, show every patient.** Many journals now discourage bar charts for continuous data for exactly these reasons (Weissgerber et al. 2015).

The same principle applies to FLUID-ICU. The figure below shows the admission SOFA score in the two fluid-balance groups, with every patient as a dot over a boxplot.

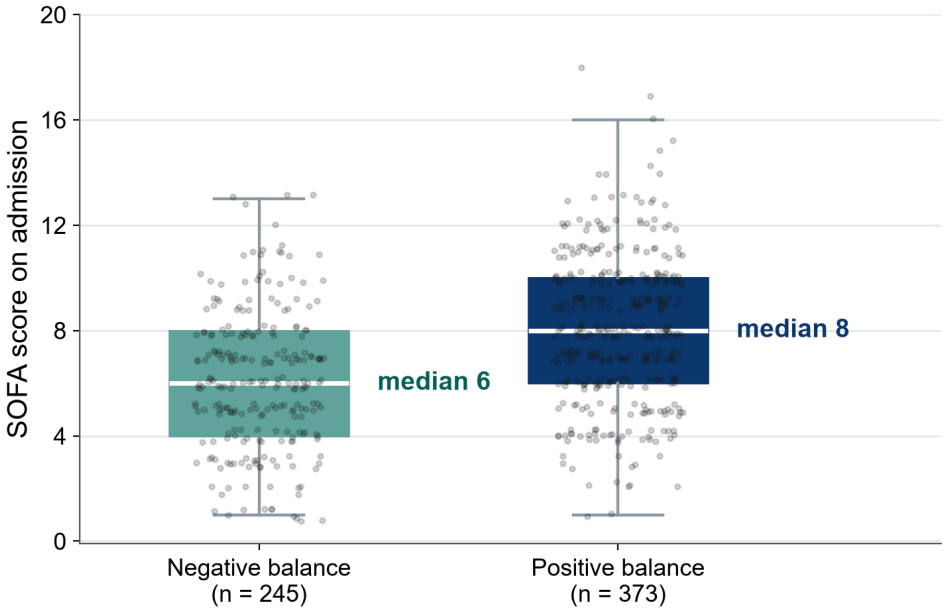


Figure 11.9: SOFA score on admission by fluid-balance group in FLUID-ICU (simulated): median 6 in the negative-balance group and 8 in the positive-balance group; every dot is one patient.

The reader sees both the overlap and the shift: the median SOFA score was 6 (IQR 4 to 8) in the negative-balance group and 8 (IQR 6 to 10) in the positive-balance group. A bar of the mean SOFA score would have hidden the spread and the overlap. This single picture turns out to be one of the most important in the FLUID-ICU analysis, for reasons explained in the section on Table 1.

11.5.2 Three more ways graphs mislead

Three other distortions appear regularly in posters, reports, dashboards and papers. The figure below shows each one (top row) with a better version (bottom row).

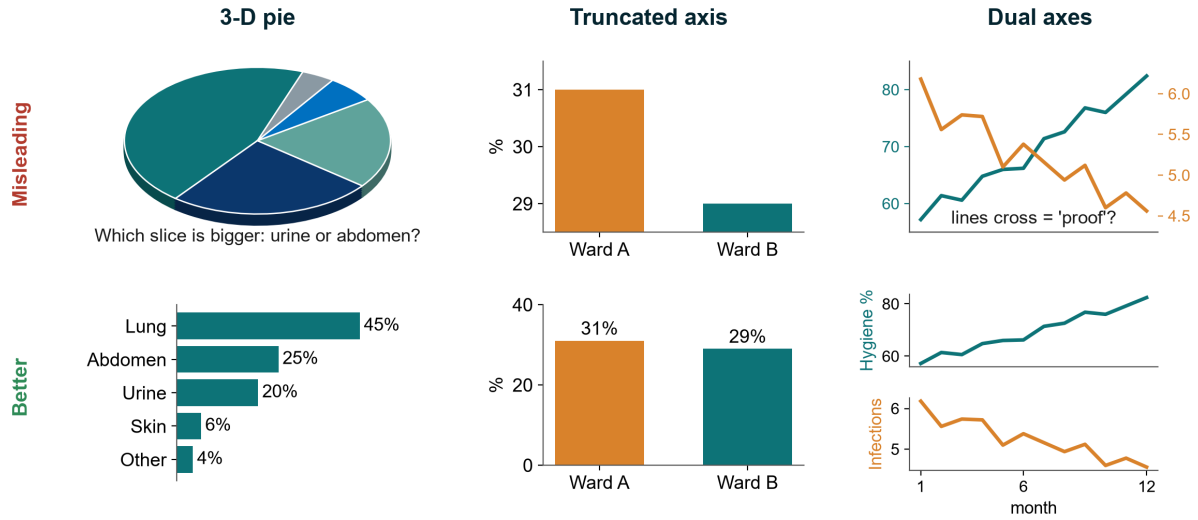


Figure 11.10: Three misleading graphs and their fixes: a 3-D pie chart (better: a sorted bar chart), a truncated axis that makes 31% vs 29% look like a three-fold difference (better: bars starting at zero), and dual axes that suggest a relationship (better: two panels sharing the time axis).

- **3-D pie charts.** The tilt makes slices at the front look larger, and the eye compares angles poorly anyway. Which slice is bigger, urine or abdomen? The sorted bar chart answers instantly: 20% versus 25%.
- **Truncated axes.** The y-axis starts at 28.5%, so a two-point difference in mortality (31% vs 29%) looks like a tripling. Bars must start at zero, because their length represents the value. Line charts may zoom in, but should say so.
- **Dual axes.** Two different y-axes, each scaled by the author, can make any two lines cross or move together wherever the author wishes, suggesting here that “hand hygiene went up, so infections went down”. Better: two panels sharing the time axis. And remember that two trends moving together over time is correlation, not causation (Chapter 6).

Rules for honest figures

- Choose the graph from the variable type.
- Show the individual data when groups are small.
- Start bar charts at zero; avoid 3-D effects and dual axes.
- Label axes with units, and give the number of patients in each group.
- Say what every error bar represents: SD, SE or 95% confidence interval.
- Write a caption that lets the figure stand alone.

11.6 Table 1: who were the patients?

Almost every clinical paper contains a table describing the participants at baseline, usually by group. Because it is so often the first table, it is universally known as **Table 1**. The figure below shows a typical example for a small trial like Paper 1, with illustrative numbers.

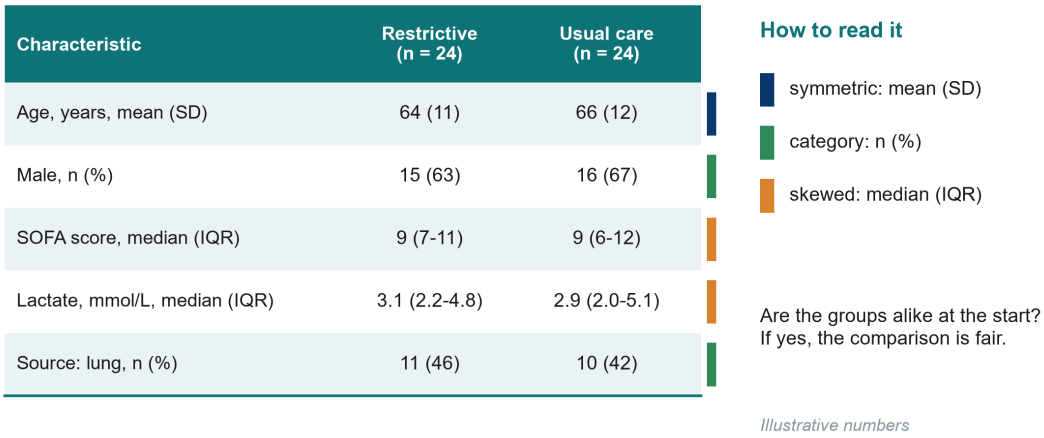


Figure 11.11: A typical Table 1 for a two-arm trial (illustrative numbers): symmetric variables as mean (SD), skewed variables as median (IQR), and categories as n (%).

Each row uses the summary that matches the variable: age as mean (SD) because it is roughly symmetric, SOFA score and lactate as median (IQR) because they are skewed, and sex and source of infection as n (%). The number of patients in each group appears at the top of each column.

Read every Table 1 with two questions in mind.

1. **Do these patients look like mine?** This is a question of generalisability, or external validity. If the trial patients were younger, less sick or free of the comorbidities common in your unit, the results may not apply to your patients.
2. **Were the groups alike at the start?** This is a question of fair comparison, or internal validity. If one group was sicker before the intervention or exposure, any difference in outcome may reflect that, not the intervention.

The meaning of differences in Table 1 depends on the study design.

- **In a randomised trial**, the groups were formed by chance, so any baseline differences are due to chance by definition. Small differences are expected, and testing them answers a question whose answer is already known. For this reason CONSORT advises **against p-values in the baseline table of a trial** (Schulz et al. 2010). A large imbalance in an important prognostic factor still matters, but its importance is judged by its size and clinical relevance, not by a p-value.
- **In a cohort study**, the groups were formed by what happened to the patients, often by clinical decisions. Large baseline differences are a warning sign of **confounding** (Chapter 6). That is why Paper 2 matched patients on a propensity score before comparing mortality (Hyun et al. 2023).

Even in observational studies, many authors and analysis plans leave p-values out of Table 1: the table describes the groups, and the size of a difference is more informative than whether it is “significant”.

11.6.1 FLUID-ICU's Table 1

The figure below shows the main rows of the FLUID-ICU Table 1 (simulated data), comparing patients with a negative and a positive fluid balance at day 3.

Characteristic	Overall n = 618	Negative balance n = 245	Positive balance n = 373
Age, years, mean (SD)	62.6 (15.2)	64.6 (15.0)	61.4 (15.2)
Male sex, n (%)	375 (61%)	143 (58%)	232 (62%)
Heart failure, n (%)	97 (16%)	54 (22%)	43 (12%)
SOFA score, median (IQR)	**7 (5, 9)**	**6 (4, 8)**	**8 (6, 10)**
Lactate, mmol/L, median (IQR)	**2.6 (1.7, 3.7)**	**2.2 (1.4, 3.0)**	**2.9 (1.9, 4.0)**
Vasopressor, n (%)	**266 (43%)**	**61 (25%)**	**205 (55%)**
Mechanical ventilation, n (%)	**313 (51%)**	**97 (40%)**	**216 (58%)**

Figure 11.12: FLUID-ICU Table 1 (simulated): patients with a positive fluid balance had higher SOFA scores and lactate, and more often received vasopressors and mechanical ventilation, than patients with a negative balance.

Apply the two questions. The patients look like a typical general ICU sepsis population: mean age about 62 years, around 60% men, median SOFA 7. But the groups were **not** alike at the start.

- **Age** was similar: mean 64.6 years in the negative-balance group and 61.4 in the positive-balance group.
- **Severity** was not similar. The median SOFA score was 6 (IQR 4 to 8) versus 8 (IQR 6 to 10). Vasopressors were given on admission to 61 of 245 patients (24.9%) versus 205 of 373 (55.0%). Mechanical ventilation was used in 97 of 245 (39.6%) versus 216 of 373 (57.9%). Median lactate was 2.2 (IQR 1.4 to 3.0) versus 2.9 (IQR 1.9 to 4.0) mmol/L.
- **Heart failure** went the other way: 54 of 245 (22.0%) in the negative-balance group versus 43 of 373 (11.5%) in the positive-balance group. Clinicians give less fluid to patients with heart failure.

The interpretation is clinically obvious once it is pointed out: patients with a positive balance were **sicker before the fluid was given**. Sicker patients receive more fluid *and* are more likely to die. A crude comparison of mortality between the two groups will therefore mix the effect of fluid with the effect of severity: this is **confounding by indication** (Chapter 6). From Table 1 alone we can predict that the crude association between a positive balance and death will be larger than an association adjusted for severity. Chapter 13 shows that this is exactly what happens, and Chapter 16 works through the whole analysis.

! Common mistake: reading Table 1 as a result

"The positive-balance group had more vasopressor use and higher SOFA scores, so fluid is harmful" is a misreading. Table 1 describes the patients at baseline, before (or at the moment of) exposure. Differences in Table 1 are not effects of the exposure; in a cohort study, they

are the confounders that the analysis must take into account.

11.6.2 Publication-quality tables

A table in a paper must stand alone: a reader who has not read the text should understand every row and column. The two figures below show the same baseline data laid out badly and well.

Variable	Group A	Group B	p
Age	63.4567±11.2	65.1±12.03	0.6123
Sex	15	16	0.7642
SOFA	8.9±3.1	9.21±3.4	0.73
Lactate	3.1	2.94	0.8
Died	0.375	0.4583	0.555

No n, no units, no %, mixed decimals, unexplained abbreviations, p-values in an RCT

Figure 11.13: A poorly laid out baseline table: vague group names, no number of patients, no units, inconsistent decimals, a proportion instead of a percentage, unexplained abbreviations and p-values in a trial.

Characteristic	Restrictive (n = 24)	Usual care (n = 24)
Age, years, mean (SD)	63 (11)	65 (12)
Male, n (%)	15 (63)	16 (67)
SOFA score, median (IQR)	9 (7-11)	9 (6-12)
Lactate, mmol/L, median (IQR)	3.1 (2.2-4.8)	2.9 (2.0-5.1)

SD, standard deviation; IQR, interquartile range; SOFA, Sequential Organ Failure Assessment. Illustrative data.

Figure 11.14: The same table laid out well: group names with n in the column headers, unit and summary in each row label, consistent decimals, and footnotes explaining abbreviations.

Compare the two versions row by row.

- **Column headers.** “Group A” and “Group B” tell the reader nothing. Write the group names and the number of patients in each: “Restrictive (n = 24)”.

- **Row labels.** State the unit and the summary used: “Age, years, mean (SD)”; “Lactate, mmol/L, median (IQR)”.
- **Numbers.** Use sensible, consistent decimals: age in whole years, lactate to one decimal, percentages without decimals for small groups. “63.4567” suggests a precision that does not exist.
- **Categories** as n (%), never as a proportion such as 0.375; and for a binary variable such as sex, say which category is counted (“Male, n (%)”).
- **Footnotes** explain every abbreviation (SD, IQR, SOFA) and, if any p-values or tests are shown, which test was used.
- **Layout.** No vertical lines and few horizontal ones. Journals re-typeset tables anyway, but reviewers notice messy ones.
- **Missing data.** Report the number of patients with missing values for each variable, either in the row label (“Lactate, mmol/L, median (IQR); missing 91”) or in a footnote.
- **No p-values** in the baseline table of a randomised trial.

11.7 Writing the description

Descriptive results are usually the first paragraph of a Results section. They report the flow of participants and then the main characteristics, using the same summaries as the table, without repeating every number. For FLUID-ICU, two sentences capture the essentials:

> *Of 620 enrolled patients, 618 had a computable day-3 fluid balance, of whom 373 (60.4%) had a positive balance. Patients with a positive balance were more severely ill at admission (median SOFA 8 vs 6; vasopressor use 55.0% vs 24.9%; mechanical ventilation 57.9% vs 39.6%).*

Notice what these sentences do: they give the denominators, use medians for the skewed score and percentages for the binary variables, highlight the one feature of the table that matters for interpretation (the difference in severity), and avoid any language of “significance”. Reporting guidance such as STROBE asks authors to give the characteristics of participants and the number with missing data for each variable (Elm et al. 2007); CONSORT asks for a table of baseline characteristics for each group (Schulz et al. 2010). Chapter 19 returns to how descriptive results fit into the structure of a paper.

Summary

Every analysis starts with a description of the patients and of the data. Categorical variables are summarised as n (%) with a clear denominator. Numerical variables need a centre and a spread, chosen according to the shape of the data: mean (SD) for roughly symmetric variables, median (IQR) for skewed ones such as length of stay, lactate and clinical scores. One extreme value moves the mean a lot and the median not at all; if the mean and median are far apart, report the median. The SD describes how patients differ; the standard error describes the precision of a mean and belongs to inference. Choose graphs from the variable type, show individual patients when groups are small, and avoid dynamite plots, 3-D pies, truncated axes and dual axes. Read Table 1 with

two questions: do these patients look like mine, and were the groups alike at the start? In the simulated FLUID-ICU cohort, Table 1 and the SOFA dot plot revealed that patients with a positive fluid balance were sicker before the fluid was given, a clear warning of confounding by indication that later chapters must address.

Further reading

- Altman, *Practical Statistics for Medical Research* (Altman 1991): the chapters on describing data and on presentation remain among the clearest explanations of means, medians, standard deviations and good tables.
- Kirkwood and Sterne, *Essential Medical Statistics* (Kirkwood and Sterne 2003): a concise, rigorous treatment of summaries, distributions and the normal distribution, written for medical researchers.
- Weissgerber et al. 2015 (Weissgerber et al. 2015): a short, persuasive paper showing how bar graphs of continuous data hide the distribution, with practical alternatives such as dot plots.
- Altman and Bland, “Standard deviations and standard errors” (Altman and Bland 2005): a one-page explanation of the difference between the SD and the SE, and when to use each.

CHAPTER 12

From sample to population: statistical inference

Nobody can study every adult with septic shock in the world. We study a sample, perhaps 48 patients in a pilot trial or 620 in a cohort, and hope to learn something about all the patients like them. Statistical inference is the set of tools that lets us do this honestly: it tells us how far the answer from our sample might be from the truth, and how surprised we should be by a difference if, in reality, there were none. This chapter introduces those tools without assuming any previous knowledge of statistics. It explains sampling and sampling error, confidence intervals, p-values and the ways they are commonly misread, the two kinds of error a study can make, statistical power and how to calculate a sample size, how to choose a statistical test, why paired data need paired tests, and why running many tests produces false alarms. The aim is not to make you a statistician but to make you a confident reader and a careful planner: someone who can read "HR 2.13 (95% CI 1.48 to 3.06)" correctly, and who knows how many patients a study needs before it starts.

In this chapter you will learn

- The difference between a population and a sample, and what sampling error and the standard error are.
- How a 95% confidence interval is built, what "95%" really means, and how to read an interval in three steps.
- What a p-value is, what it is not, and five common misinterpretations.
- What type I and type II errors are, and what statistical power means.
- The four ingredients of a sample-size calculation, the standard formulas, and how to apply them to worked examples.
- How to choose between the t-test, Mann-Whitney U test, chi-square test and Fisher's exact test, and their paired versions.
- Why paired designs are more powerful, and why multiple comparisons produce false-positive findings.

12.1 Populations, samples and sampling error

The **population** is everyone we want to learn about: for example, all adults with septic shock. We never see all of them. The **sample** is the patients we actually study. From the sample we calculate an **estimate**, such as a mortality, a mean fluid balance or a difference between two groups, and we want to use it to say something about the corresponding **true value** in the population.

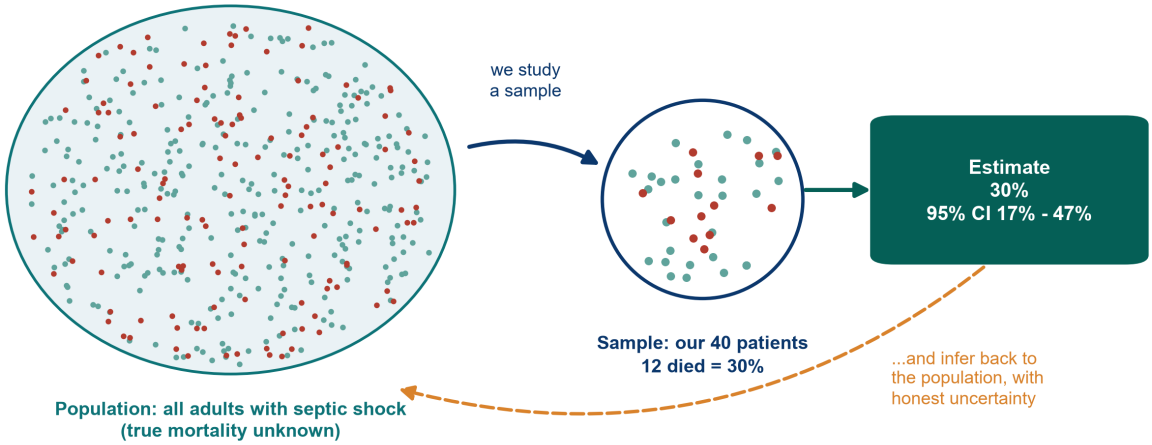


Figure 12.1: We study a sample of 40 patients from the population of all adults with septic shock; 12 die (30%), and the 95% confidence interval of 17% to 47% expresses how far the true mortality might be from this estimate.

The figure above shows the idea. In our sample of 40 patients, 12 died: an estimated mortality of 30%. But a different sample of 40 patients from the same population would almost certainly give a slightly different figure, perhaps 25% or 35%. This variation from sample to sample is called **sampling error**. It is not a mistake by anyone; it is simply chance, the luck of which patients happened to be in the sample.

For the estimate to tell us about the population, two conditions must hold.

1. **The sample must be representative.** If patients were selected in a way that is related to the outcome, for instance only patients who survived long enough to be transferred to a tertiary ICU, the estimate will be wrong however large the sample. This is **selection bias** (Chapter 6), a systematic error that statistics cannot fix. Ideally the sample is a **random sample** of the population. In clinical research that is rarely possible, and studies usually enrol **consecutive** eligible patients over a period of time, which is the next best thing. A **convenience** sample (whoever was easy to recruit) is the most vulnerable to bias.
2. **The sample must be large enough.** Even a perfectly representative sample gives an imprecise estimate if it is small. This is a question of **precision**, and it is what statistical inference measures.

Bias and imprecision are different things, and it helps to keep them apart. Bias is like a bathroom scale that always reads 2 kg too heavy: weighing yourself more often does not help. Imprecision is like a scale that wobbles randomly around the right value: weighing yourself several times and averaging does help. Larger samples reduce imprecision, never bias.

12.1.1 The standard error

How much would an estimate vary from sample to sample? Imagine repeating the same study many times, each time with a new sample of the same size, and plotting all the estimates. They would form a distribution, called the **sampling distribution**, centred on the true value. A remarkable result of probability theory (the *central limit theorem*) says that, for reasonably large samples, this distribution is approximately bell-shaped even when the data themselves are skewed.

The spread of the sampling distribution is called the **standard error (SE)**. It measures the precision of an estimate. For a mean,

$$\mathrm{SE}(\bar{x}) = \frac{\mathrm{SD}}{\sqrt{n}},$$

and for a proportion p estimated from n patients,

$$\mathrm{SE}(p) = \sqrt{\frac{p(1-p)}{n}}.$$

Two features of these formulas are worth remembering. First, the standard error gets smaller as the sample grows, but only with the square root of n: to halve the standard error you need **four times** as many patients. Second, the standard error depends on the variability between patients (the SD): the noisier the data, the less precise the estimate.

! Common mistake: confusing the SD and the SE

The standard deviation describes how much **patients** differ from one another; it does not shrink as the study grows. The standard error describes how precisely the **estimate** is known; it shrinks as the study grows. Descriptive tables report the SD (Chapter 11). Comparisons and estimates report confidence intervals, built from the SE. A figure with "error bars" must say which one it shows (Altman and Bland 2005).

12.2 Confidence intervals

12.2.1 Building a 95% confidence interval

Because the sampling distribution is approximately bell-shaped, about 95% of estimates fall within 1.96 standard errors of the true value. Turning this around gives the most widely used tool in medical statistics, the **95% confidence interval (CI)**:

$$95\%\ \mathrm{CI} = \text{estimate} \pm 1.96 \times \mathrm{SE}.$$

The number 1.96 comes from the normal distribution; for a 90% interval it would be 1.645 and for a 99% interval 2.576.

Worked example: a confidence interval for a proportion

In the sample of 40 patients shown earlier, 12 died: $p = 12/40 = 0.30$.
The standard error is $\sqrt{0.30 \times 0.70/40} = \sqrt{0.00525} = 0.072$.
The 95% CI is $0.30 \pm 1.96 \times 0.072 = 0.30 \pm 0.14$, that is, from about **16% to 44%**.

The figure gives 17% to 47%, slightly different, because with small samples and proportions far from 50% the simple formula is not accurate, and software uses more exact methods. The message is the same either way: with only 40 patients, the true mortality could plausibly be anywhere from the mid-teens to the mid-forties. If the study had included 400 patients with the same 30% mortality, the standard error would have been $\sqrt{0.21/400} = 0.023$ and the interval about 25% to 35%: roughly three times narrower ($\sqrt{10} \approx 3.2$).

The same logic applies to differences between groups. For the difference between two proportions p_1 and p_2 observed in groups of size n_1 and n_2, the standard error is

$$\text{SE}(p_1 - p_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}.$$

Worked example: the mortality difference in FLUID-ICU

In the simulated FLUID-ICU cohort, 134 of 373 patients with a positive fluid balance by day 3 died by day 28 (35.9%), compared with 45 of 245 patients with a negative balance (18.4%).

- **Risk difference:** $134/373 - 45/245 = 0.3592 - 0.1837 = 0.1756$, or 17.6 percentage points (the risk ratio is $35.9/18.4 = 1.96$; Chapter 7).
- **Standard error:** $\sqrt{0.359 \times 0.641/373 + 0.184 \times 0.816/245} = \sqrt{0.000617 + 0.000612} = 0.035$.
- **95% CI:** $0.176 \pm 1.96 \times 0.035 = 0.176 \pm 0.069$, that is, **10.7 to 24.4 percentage points**.

The data are compatible with a true difference in mortality of anywhere from about 11 to about 24 percentage points, and the interval is far from zero. This is an unadjusted comparison: Chapter 11 showed that patients with a positive balance were sicker at baseline, so part of this difference reflects severity rather than fluid. The interval quantifies chance; it says nothing about confounding.

12.2.2 What “95%” means: the dance of the confidence intervals

The figure below is the single most useful picture for understanding confidence intervals.

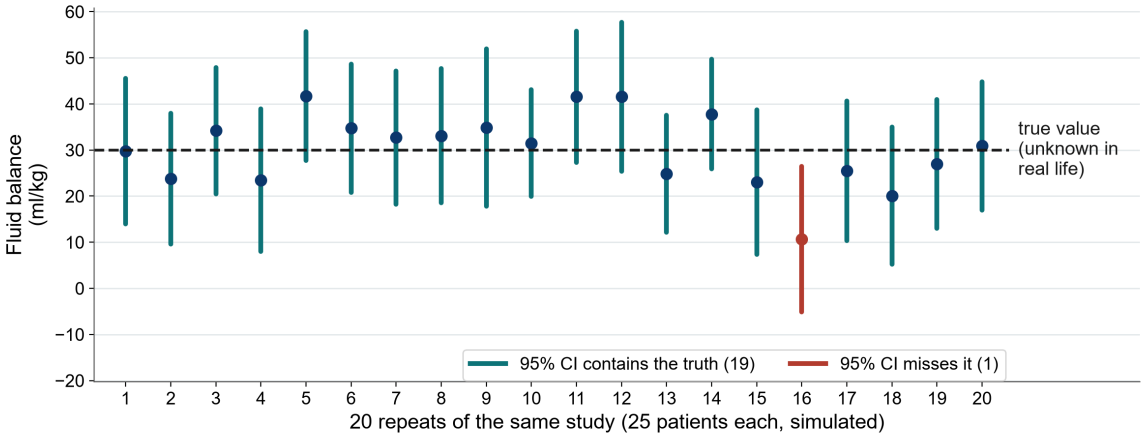


Figure 12.2: Twenty simulated repeats of the same study (25 patients each) from a population whose true mean fluid balance is 30 ml/kg: each study gives a different estimate and 95% CI; 19 of the 20 intervals contain the true value and one misses it.

The same study was simulated 20 times, each time with 25 patients drawn from a population in which the true mean fluid balance is 30 ml/kg (the dashed line). Each study gives a different estimate (the dot) and a different 95% CI (the vertical line). The intervals “dance” from study to study. Nineteen of the twenty contain the true value; one, shown in red, misses it.

That is what “95%” means: the *method* of building the interval captures the true value in about 95% of studies. In real life you run one study, see one interval, and cannot see the dashed line. You do not know whether yours is one of the lucky nineteen or the unlucky one. Strictly speaking, therefore, it is not correct to say that “there is a 95% probability that the true value lies in this particular interval”; the 95% refers to the long-run behaviour of the method. For practical purposes, a safe and useful reading is:

> *A 95% confidence interval is the range of values for the true effect that are reasonably compatible with our data.*

A narrow interval means a precise estimate; a wide interval means an imprecise one. Intervals get narrower when there are more patients and when patients vary less.

12.2.3 Reading a confidence interval in three steps

Every confidence interval in a paper can be read in three steps.

1. **The estimate.** What is the best single guess, and is it clinically important?
2. **The null value.** Where is “no effect”? For a **difference** (in means, risks or proportions), no effect is **0**. For a **ratio** (risk ratio, odds ratio, hazard ratio), no effect is **1**. Does the interval include the null value?
3. **The width.** How wide is the interval? Do both ends lead to the same clinical conclusion?

The figure below applies these steps to the two papers of the running example.

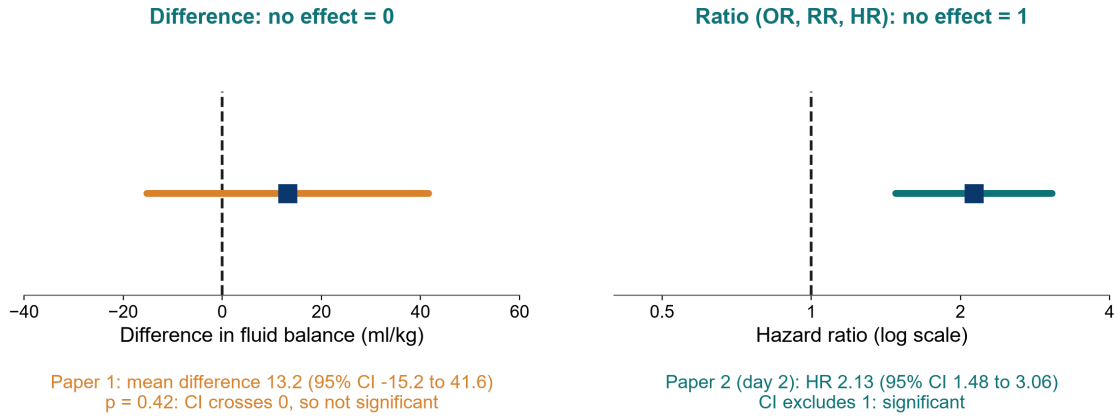


Figure 12.3: Two confidence intervals: Paper 1's mean difference in fluid balance crosses 0 (no effect for a difference) and is very wide, while Paper 2's hazard ratio lies entirely above 1 (no effect for a ratio).

Paper 1 (Boulet et al. 2024) was a pilot randomised trial of 48 patients with septic shock comparing an optimised restrictive fluid strategy with usual care (Boulet et al. 2024). The median cumulative fluid balance at day 5 was 6.9 ml/kg in the restrictive group and 35.0 ml/kg in the control group; the trial reported the difference as 13.2 ml/kg (95% CI -15.2 to 41.6), p = 0.42. Step 1: the estimate favours the restrictive strategy by 13 ml/kg. Step 2: this is a difference, so the null value is 0, and the interval includes 0: no difference is compatible with the data. Step 3: the interval is very wide, from 15 ml/kg in favour of usual care to 42 ml/kg in favour of the restrictive strategy. The honest conclusion is that this small pilot cannot tell us whether the strategy reduces fluid balance. (The difference of 13.2 is not 35.0 minus 6.9, because it is not a difference between the medians; medians describe skewed data, and differences between means or other estimators need not match them, as Chapter 11 explained.)

Paper 2 (Hyun et al. 2023) was a propensity-matched cohort study of patients with sepsis (Hyun et al. 2023). For patients with a positive versus a negative cumulative fluid balance on day 2, the hazard ratio for 28-day death was 2.13 (95% CI 1.48 to 3.06), $p < 0.001$. Step 1: the hazard of death was about twice as high. Step 2: this is a ratio, so the null value is 1, and the whole interval lies above 1. Step 3: the interval is fairly wide, but even its lower end (1.48) represents a substantial increase. The data are compatible with a true hazard ratio from about 1.5 to about 3.1. (Because this is a cohort, "associated with" is the right wording, not "caused"; Chapter 6 explains why.) Ratio scales are best drawn on a logarithmic axis, as in the figure, so that a halving (0.5) and a doubling (2) are the same distance from 1.

12.3 Hypothesis tests and p-values

12.3.1 The null hypothesis and the p-value

A statistical test starts from a deliberately sceptical assumption, the **null hypothesis** (H0): there is no real difference between the groups, or no real association. The test then asks: *if the null hypothesis were true, how surprising would our data be?*

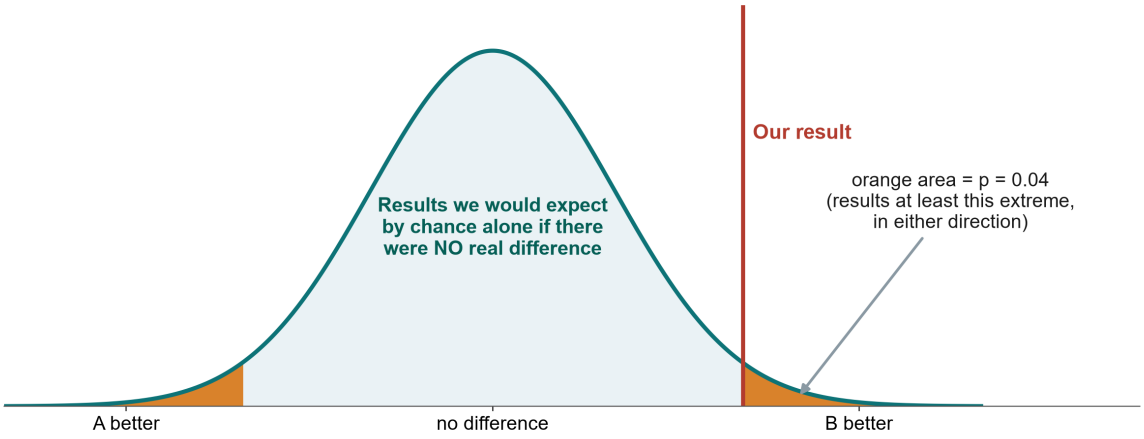


Figure 12.4: The p-value in one picture: if there were no real difference, chance alone would produce results scattered around "no difference" (the bell curve); the p-value is the orange area, the proportion of results at least as extreme as ours in either direction.

The figure above shows the idea. Even if there were no real difference, chance alone would produce results that scatter around "no difference", forming the bell-shaped curve. Now place our result on the picture. The **p-value** is the orange area: the proportion of results at least as extreme as ours, in either direction, that chance alone would produce if there were no real difference.

- **p = 0.04** means that if there were truly no difference, results this extreme or more would occur in about 4 of every 100 studies. That is fairly surprising, so we have some evidence against the null hypothesis.
- **p = 0.42** (Paper 1) means that results like these would be common even with no real difference. They are not surprising, so there is no evidence against the null hypothesis. But that is not evidence *for* the null hypothesis either.

By convention, $p < 0.05$ is called "statistically significant". The convention is useful as a shared yardstick, but it is not a law of nature: $p = 0.049$ and $p = 0.051$ represent practically the same strength of evidence. The FLUID-ICU analysis illustrates this neatly. The adjusted odds ratio for death with a positive balance was 1.62 (95% CI 1.02 to 2.59), $p = 0.044$, while the adjusted hazard ratio from a survival model of the same patients was 1.46 (95% CI 1.00 to 2.13), $p = 0.053$. One p-value falls just below 0.05 and the other just above, but the two analyses tell the same story: an association in the same direction, of similar size, with overlapping intervals whose lower ends lie close to "no effect".

Tests are usually **two-sided**: they count extreme results in either direction, because a new treatment could turn out better or worse than the old one. One-sided tests are rarely justified in clinical research and should be pre-specified with a good reason.

12.3.2 Confidence intervals and p-values tell a consistent story

For the same analysis, a 95% CI and a two-sided p-value are linked: **if the 95% CI excludes the null value, $p < 0.05$; if it includes the null value, $p > 0.05$.** Paper 1's interval includes 0 and its p-value is 0.42; Paper 2's interval excludes 1 and its p-value is below 0.001. The confidence interval therefore contains everything the p-value tells you about "significance", plus the size of the effect and the precision with which it is known. For this reason, journals and statisticians have long urged authors to report effect sizes with confidence intervals rather than p-values alone (Gardner and Altman 1986).

12.3.3 Five common misinterpretations

Each of the following statements appears regularly in published papers. All are wrong (Greenland et al. 2016; Wasserstein and Lazar 2016).

! Misinterpretation 1: the p-value is the probability that there is no effect

"p = 0.03 means there is a 3% chance that the null hypothesis is true." False. The p-value is calculated *assuming* the null hypothesis is true; it cannot also be the probability that it is true. It describes how surprising the data are under that assumption. For the same reason, avoid saying "the p-value is the probability that the result is due to chance".

! Misinterpretation 2: a small p-value means a big effect

"$p < 0.001$, so the effect is large." False. With thousands of patients, a clinically trivial difference, such as a 1 mmHg fall in blood pressure, can have a tiny p-value. Statistical significance is not clinical importance. Judge the size of the effect from the estimate and its confidence interval, and ask whether it matters to patients.

! Misinterpretation 3: $p > 0.05$ means no effect

"p = 0.42, so the two fluid strategies are equivalent." False. Paper 1 is the perfect example: its confidence interval (-15.2 to 41.6 ml/kg) includes both no difference and a large difference. The study simply cannot tell. **Absence of evidence is not evidence of absence** (Altman and Bland 1995). Showing that two treatments are equivalent requires a study designed for that purpose, with a pre-specified margin and a confidence interval that lies entirely within it.

! Misinterpretation 4: p = 0.08 is a trend

"p = 0.08 shows a trend towards benefit." False. A p-value does not move towards anything; it is a single number from a single analysis. If more patients were recruited, the p-value could go up just as easily as down. Report the estimate and its confidence interval, and call the result what it is: not statistically significant at the 5% level, with an imprecise estimate.

! Misinterpretation 5: the p-value is enough

"Results: p = 0.02." Not enough. The reader needs the size of the effect and its uncertainty. Paper 2 reports it well: HR 2.13 (95% CI 1.48 to 3.06), $p < 0.001$. Report the effect size with its 95% CI; the p-value is a supporting actor, not the result.

The American Statistical Association summarised these points in a short statement that is worth

reading in full: p-values can indicate how incompatible the data are with a statistical model, but they do not measure the probability that a hypothesis is true, do not measure the size or importance of an effect, and should not be the sole basis of a scientific conclusion (Wasserstein and Lazar 2016).

12.4 Type I and type II errors, and power

A statistical test is like a fire alarm. Two things can go wrong, as the figure below shows.

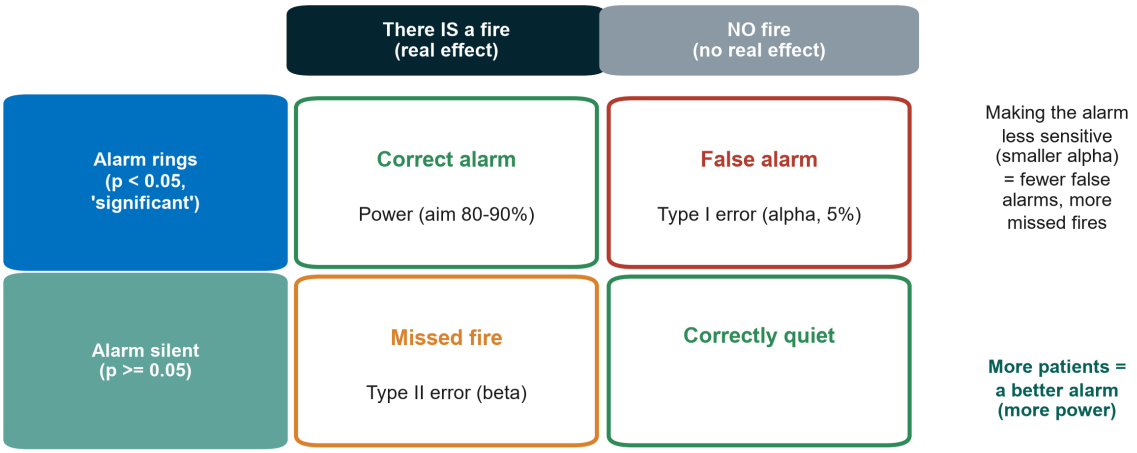


Figure 12.5: A test is like a fire alarm: a type I error is a false alarm (the alarm rings with no fire, risk alpha), a type II error is a missed fire (the alarm stays silent although there is a fire, risk beta), and power is the chance of the alarm ringing when there really is a fire.

- **Type I error (false alarm).** The alarm rings but there is no fire: we conclude that a treatment works, or that an association exists, when in truth it does not. The risk of a type I error is called **alpha** (α). By convention it is set at 5%, which is the same thing as using $p < 0.05$ as the threshold for significance.
- **Type II error (missed fire).** There is a fire but the alarm stays silent: the treatment truly works, but the study fails to show it. The risk of a type II error is called **beta** (β).
- **Power** is $1 - \beta$: the probability that the alarm rings when there really is a fire, that is, the probability that the study detects a real effect of a given size. Studies usually aim for a power of 80% or 90%.

There is a trade-off. Making the alarm less sensitive (a smaller alpha, such as 1%) gives fewer false alarms but more missed fires, unless you buy a better alarm. In research, a better alarm means **more patients**, or less noisy measurements.

Which error is worse depends on the question. For a new, expensive drug with serious side effects, a false alarm could lead to widespread harmful prescribing, so a small alpha is appropriate. For a cheap, safe intervention, missing a real benefit may be the bigger loss. There is no single right answer, which is why alpha and power are chosen and justified when the study is designed.

A small “negative” study may simply be a missed fire. Paper 1 was a pilot designed to detect a large difference in fluid balance; a smaller real difference could easily have been missed. This is another reason never to read $p > 0.05$ as proof of no effect. Note also that **power is a property of the design, before the data are collected**. Calculating “post hoc power” after a non-significant

result adds nothing: the confidence interval already shows which effects the study can and cannot rule out.

12.5 Sample size: how many patients?

Every protocol must justify its number of participants: too few, and the study cannot answer its question (and exposes patients to research for nothing); too many, and resources are wasted and patients are needlessly exposed to an inferior treatment. Ethics committees, funders and journal reviewers all expect a sample-size calculation (Chapters 15 and 17).

12.5.1 The four ingredients

A sample-size calculation for comparing two groups needs four ingredients.

1. **The significance level, alpha.** Usually 5%, two-sided.
2. **The power,** $1 - \beta$**.** Usually 80% or 90%.
3. **The effect you want to detect.** The difference between groups that the study should be able to find: a difference in means (for example 70 versus 40 ml/kg) or in proportions (for example 40% versus 30% mortality). This is the hardest ingredient. Take it from previous studies, pilot data, or, best of all, from the **minimal clinically important difference**: the smallest difference that would change practice. Never choose it just to make the number small.
4. **The variability.** For a numerical outcome, the standard deviation, from previous studies or pilot data. For a binary outcome, the variability is determined by the proportions themselves.

Then add an allowance for **drop-outs and losses** (often 10% to 20%): if a fraction d of patients is expected to be lost, recruit $n/(1-d)$.

The figure below shows how the ingredients interact for a numerical outcome.

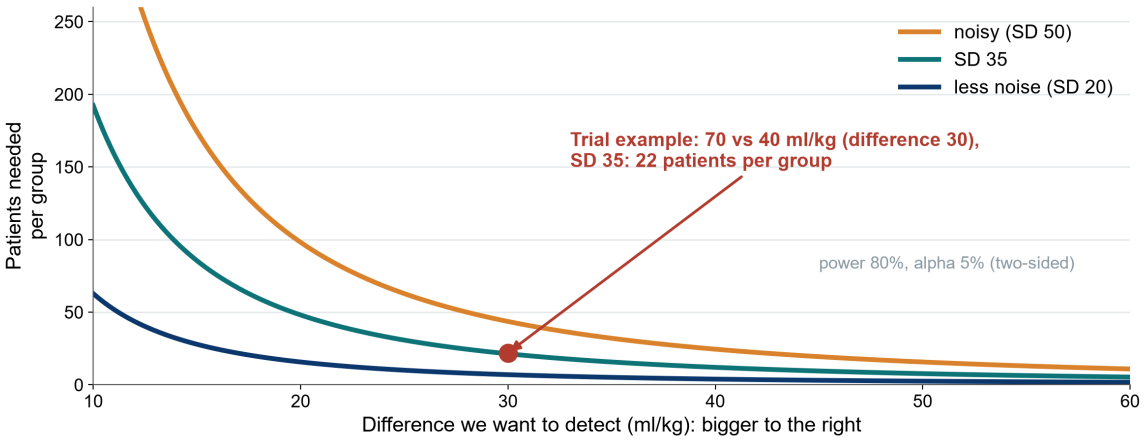


Figure 12.6: Patients needed per group to detect a difference in fluid balance, for three levels of variability (SD 20, 35 and 50 ml/kg), with 80% power and two-sided alpha of 5%: bigger differences and less noise need fewer patients; a 30 ml/kg difference with SD 35 needs 22 patients per group.

You need fewer patients when the difference you want to detect is larger, and when patients vary less. For a 30 ml/kg difference with an SD of 35 ml/kg, 22 patients per group are needed. Halve

the difference to 15 ml/kg and you need about four times as many (86 per group). With noisier data (SD 50), every point on the curve moves up.

12.5.2 The formulas

For a **numerical outcome compared between two groups of equal size**, the number of patients needed in *each* group is

$$n = \frac{2\,(z_{1-\alpha/2} + z_{1-\beta})^2\,\sigma^2}{\Delta^2},$$

where Δ is the difference in means to be detected, σ is the standard deviation, and the z values come from the normal distribution: $z_{1-\alpha/2} = 1.96$ for a two-sided alpha of 5%, and $z_{1-\beta} = 0.84$ for 80% power or 1.28 for 90% power. With alpha 5% and 80% power, $(1.96 + 0.84)^2 \approx 7.85$, so the formula becomes approximately $n \approx 15.7\,\sigma^2/\Delta^2$, often rounded to "Lehr's rule": $n \approx 16/(\Delta/\sigma)^2$ per group (Lehr 1992). The ratio Δ/σ, the difference measured in standard deviations, is called the **standardised effect size**: a difference of one SD needs about 16 patients per group, half an SD about 64.

For a **binary outcome compared between two groups of equal size**, with expected proportions p_1 and p_2, the usual formula (without continuity correction) is

$$n = \frac{\left[\, z_{1-\alpha/2}\sqrt{2\,\bar{p}(1-\bar{p})} + z_{1-\beta}\sqrt{p_1(1-p_1) + p_2(1-p_2)}\,\right]^2}{(p_1 - p_2)^2},$$

where $\bar{p} = (p_1 + p_2)/2$ is the average of the two proportions (Fleiss et al. 2003).

For **estimating a single proportion** with a given precision (for example in a prevalence survey), the number needed is

$$n = \frac{z_{1-\alpha/2}^2\; p(1-p)}{d^2},$$

where p is the expected proportion and d is the acceptable margin of error, the half-width of the 95% CI.

Always **round up** to the next whole patient. In practice the calculation is done with software or an online calculator, but working through the formulas once shows exactly what drives the answer. Different programs use slightly different formulas (with or without a continuity correction, for example), so small differences between calculators are normal.

Worked example 1: Paper 1's sample size (two means)

Paper 1 planned to detect a difference in cumulative fluid balance at day 5 of 70 versus 40 ml/kg ($\Delta = 30$), assuming an SD of 35 ml/kg, with 80% power and a two-sided alpha of 5% (Boulet et al. 2024).

$$n = \frac{2 \times (1.96 + 0.84)^2 \times 35^2}{30^2} = \frac{2 \times 7.84 \times 1225}{900} = 21.3,$$

which rounds up to **22 patients per group**, 44 in total. (Lehr's rule gives the same: the

standardised effect is $30/35 = 0.857$, and $16/0.857^2 = 21.8$.) The authors then increased the number by 10% to allow for deaths between day 1 and day 5, giving 50 patients in total, 25 per group ($22/0.9 = 24.4$, rounded up to 25).
Notice that this calculation was for **fluid balance**, the trial's primary outcome. The trial was never designed to detect a difference in mortality, which is why its mortality results cannot be interpreted as evidence of no effect.

Worked example 2: a mortality trial (two proportions)

Suppose you plan a trial of the restrictive strategy with 28-day mortality as the primary outcome. From published data you expect 40% mortality with usual care, and you consider a reduction to 30% the smallest clinically important effect. Alpha is 5% two-sided and power 80%.

- $p_1 = 0.40$, $p_2 = 0.30$, $\bar{p} = 0.35$.
- First term: $1.96 \times \sqrt{2 \times 0.35 \times 0.65} = 1.96 \times 0.675 = 1.322$.
- Second term: $0.84 \times \sqrt{0.40 \times 0.60 + 0.30 \times 0.70} = 0.84 \times \sqrt{0.45} = 0.84 \times 0.671 = 0.565$.
- $n = (1.322 + 0.565)^2 / 0.10^2 = 3.56 / 0.01 = 356$ patients per group.

With an allowance of 10% for loss to follow-up, recruit $356/0.9 = 395.6$, rounded up to **396 per group, 792 in total**. (With a continuity correction, some calculators give about 376 per group before the allowance; both are acceptable if the method is stated.)
Now change one ingredient at a time and watch the number move:

- **Power 90%** instead of 80% ($z_{1-\beta} = 1.28$): about 477 per group.
- **A smaller effect**, 40% versus 35%: more than 1,400 per group.

The lesson is striking. Paper 1 needed 50 patients to study fluid balance; a trial powered to detect a 10-point fall in mortality needs nearly 800. The choice of primary outcome changes everything.
A sample-size statement for a protocol would read: "Assuming 28-day mortality of 40% with usual care and 30% with the restrictive strategy, 356 patients per group give 80% power at a two-sided alpha of 5%. Allowing for 10% loss to follow-up, we will recruit 396 patients per group (792 in total)."

Worked example 3: estimating a proportion with given precision

An audit aims to estimate the proportion of patients with septic shock who have a positive fluid balance on day 3. You expect about 30% and want the 95% CI to be no wider than ±5 percentage points.

$$n = \frac{1.96^2 \times 0.30 \times 0.70}{0.05^2} = \frac{3.8416 \times 0.21}{0.0025} = 322.7,$$

so 323 patients are needed; allowing 10% for incomplete records, $323/0.9 = 358.9$, so include 359 patients. If you had no idea of the expected proportion, using $p = 0.5$ gives the largest possible answer: $1.96^2 \times 0.25 / 0.05^2 = 384.2$, rounded up to the familiar figure of

385.

12.5.3 Sample size in observational studies

Many observational studies, including cohort studies based on all eligible patients over a period, do not choose their size; it is fixed by how many patients are available. A sample-size or power statement is still needed, but it is turned around: given the expected number of patients, what effect can the study detect? The FLUID-ICU statistical analysis plan (simulated) gives an example. It expected about 600 patients, roughly 240 with a negative and 360 with a positive balance, and a mortality of about 20% in the negative-balance group. With these numbers and a two-sided 5% significance level, the study would have about 80% power to detect an odds ratio of 1.7 (20% versus 30% mortality) but only about 55% power for an odds ratio of 1.5. The plan therefore stated honestly that the study was adequately powered for a moderate association only, and that the confidence interval would show the precision achieved.

Studies that adjust for several variables in a regression model have an additional requirement: enough outcome events for the number of variables in the model. A common rule of thumb is about ten events per variable (Peduzzi et al. 1996); Chapters 13 and 15 explain why.

! Common mistakes in sample-size calculations

- **No calculation at all**: “we included all patients admitted last year”. Reviewers will ask what the study could detect.
- **An optimistic effect** chosen to make the number small. If the true effect is smaller, the study will be underpowered and probably “negative”.
- **A calculation for the wrong outcome**: powering the study for a secondary outcome, or for a surrogate, and then drawing conclusions about mortality.
- **Forgetting drop-outs**, or ignoring the design: cluster-randomised trials, survival outcomes, equivalence and non-inferiority trials, and multivariable models all need their own methods. Ask a statistician.
- **Not stating the inputs**: a sample-size statement must give all four ingredients and where they came from, so that a reader can repeat the calculation.

12.6 Choosing a statistical test

Once the data are collected, how do you choose the test to compare two groups? Two questions, and sometimes a third, decide it, as the decision chart below shows.

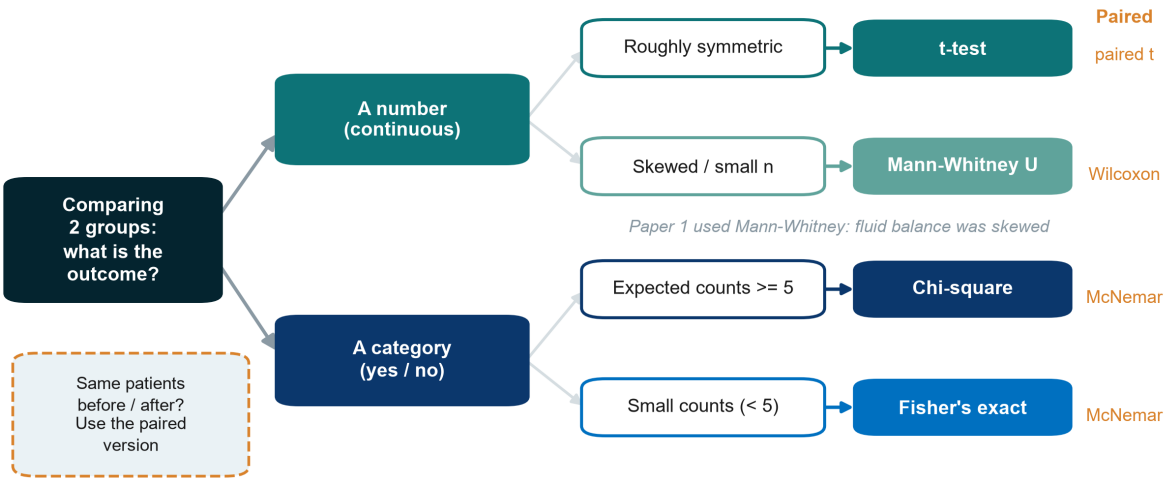


Figure 12.7: A decision chart for comparing two groups: for a numerical outcome, use the t-test if roughly symmetric or the Mann-Whitney U test if skewed or small; for a categorical outcome, use the chi-square test if expected counts are 5 or more or Fisher's exact test if smaller; for the same patients measured twice, use the paired versions.

Question 1: what type of outcome is it? A number (continuous or count) or a category (such as yes or no)?

Question 2a, for numbers: is the outcome roughly symmetric?

- If yes, use the **t-test** (Student's t-test for two independent groups). It compares the **means** of the two groups and gives the difference in means with its 95% CI.
- If the outcome is skewed, or the sample is very small, use the **Mann-Whitney U test** (also called the Wilcoxon rank-sum test). It replaces the values by their ranks (1st, 2nd, 3rd...) and asks whether one group tends to have higher values than the other. It is usually reported with the median (IQR) in each group. An estimate of the size of the shift between groups, with a confidence interval, can be obtained with the **Hodges-Lehmann** method.

Tests like the t-test, which rely on assumptions about the shape of the data, are called **parametric**; tests based on ranks, like the Mann-Whitney test, are called **non-parametric**. With large samples the t-test is fairly robust to moderate skew, because the sampling distribution of a mean becomes bell-shaped; but for strongly skewed outcomes, especially in small studies, the mean is the wrong summary in the first place, and a rank-based method or a transformation (for example analysing the logarithm of length of stay) is more appropriate.

Question 2b, for categories: are the expected counts large enough?

- If every cell of the table would be expected to contain 5 or more patients under the null hypothesis, use the **chi-square test**.
- If some expected counts are small (below 5), use **Fisher's exact test**, which calculates the p-value exactly rather than relying on an approximation.

For a binary outcome, the test is only half of the answer: report the effect as a risk difference or risk ratio (or odds ratio) with its 95% CI (Chapter 7).

Question 3: are the data paired? If the same patients are measured twice, or patients are deliberately matched in pairs, use the paired version of the test: the **paired t-test** for symmetric differences, the **Wilcoxon signed-rank test** for skewed differences, and **McNemar's test** for a binary outcome. The next section explains why.

More than two groups. To compare a number across three or more groups, the extensions are **analysis of variance (ANOVA)** for symmetric outcomes and the **Kruskal-Wallis test** for skewed ones; the chi-square test extends directly to larger tables. These are best planned with a statistician.

Table 12.1: Choosing a test to compare two groups.

Outcome	Two independent groups	Same patients twice, or matched pairs
Number, roughly symmetric	t-test (difference in means, 95% CI)	Paired t-test
Number, skewed or small sample	Mann-Whitney U test (medians; Hodges-Lehmann difference)	Wilcoxon signed-rank test
Binary, large expected counts	Chi-square test (risk difference or ratio, 95% CI)	McNemar's test
Binary, small expected counts (< 5)	Fisher's exact test	McNemar's test (exact version)

Worked example: choosing tests for four scenarios

1. **ICU length of stay (skewed) in patients with a positive versus a negative fluid balance**: a number, skewed, two independent groups, so the **Mann-Whitney U test**.
2. **28-day death (yes or no) in 300 versus 300 patients**: binary outcome, large groups, so the **chi-square test**, with the risk difference and its 95% CI.
3. **Lactate at ICU admission and six hours later in the same 40 patients**: the same patients measured twice, so the **paired t-test**, or the Wilcoxon signed-rank test if the changes are skewed.
4. **Dialysis (yes or no) in 2 of 24 versus 5 of 24 patients**: binary outcome with very small counts, so **Fisher's exact test**.

Paper 1 made the same kind of choice: because cumulative fluid balance was skewed, its primary outcome was compared between groups with a Mann-Whitney test, consistent with reporting medians (Boulet et al. 2024).

12.6.1 The tests applied to FLUID-ICU

The table below shows four unadjusted comparisons between the two fluid-balance groups in the simulated FLUID-ICU cohort, each with the test chosen from the decision chart.

Outcome (type)	Negative (n = 245)	Positive (n = 373)	Test	Result
28-day death (yes/no)	45 (18.4%)	134 (35.9%)	Chi-square	RD 17.6 points (10.7 to 24.4), p < 0.001
AKI (yes/no)	71 (29.0%)	230 (61.7%)	Chi-square	p < 0.001
Ventilator-free days (0-28, skewed)	median 28 (22 to 28)	median 20 (0 to 28)	Mann-Whitney	p < 0.001
ICU stay, survivors (days, skewed)	4.8 (3.3 to 7.1) n = 200	7.5 (5.2 to 11.2) n = 239	Mann-Whitney	difference 2.5 days (1.8 to 3.2), p < 0.001

n (%) or median (IQR). Unadjusted comparisons. Source: 03_compare.R, key_results.md (simulated data)

Figure 12.8: Four outcomes compared between negative and positive fluid-balance groups in FLUID-ICU (simulated, unadjusted): death and AKI with the chi-square test, ventilator-free days and ICU stay in survivors with the Mann-Whitney test.

- **Death by day 28** (binary): 134 of 373 (35.9%) versus 45 of 245 (18.4%); risk difference 17.6 percentage points (95% CI 10.7 to 24.4); chi-square $p < 0.001$.
- **Acute kidney injury** (binary): 230 of 373 (61.7%) versus 71 of 245 (29.0%); chi-square $p < 0.001$.
- **Ventilator-free days to day 28**, a count from 0 to 28 that piles up at both ends (0 for patients who died or stayed ventilated, 28 for those never ventilated): median 20 (IQR 0 to 28) versus 28 (IQR 22 to 28); Mann-Whitney $p < 0.001$.
- **ICU length of stay in survivors** (skewed): median 7.5 (IQR 5.2 to 11.2) versus 4.8 (IQR 3.3 to 7.1) days; Mann-Whitney $p < 0.001$; Hodges-Lehmann difference 2.5 days (95% CI 1.8 to 3.2).

For contrast, a t-test applied to the skewed ICU stay gives a difference in means of 2.7 days: not wildly different here, but describing the means of a distribution that no typical patient resembles. With smaller samples or more extreme skew, the wrong test can give a misleading answer.

Every outcome is worse in the positive-balance group. But all four comparisons are **unadjusted**, and the positive-balance group was sicker at baseline (Chapter 11). These differences mix the effect of fluid with the effect of severity. The tests correctly quantify the role of chance; they cannot remove confounding. That requires the regression methods of Chapter 13.

Worked example: writing up a comparison

A good Results sentence reports the numbers in each group, the effect with its confidence interval, and avoids causal language:

> *Twenty-eight-day mortality was 35.9% (134/373) in patients with a positive fluid balance and 18.4% (45/245) in those with a negative balance (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4).*

Compare this with “Positive fluid balance significantly increased mortality ($p < 0.001$)”, which gives no effect size, no interval, no denominators, and claims causation from an unadjusted cohort comparison.

12.7 Paired or unpaired?

The decision chart asked whether the data are paired. The figure below shows why this matters.

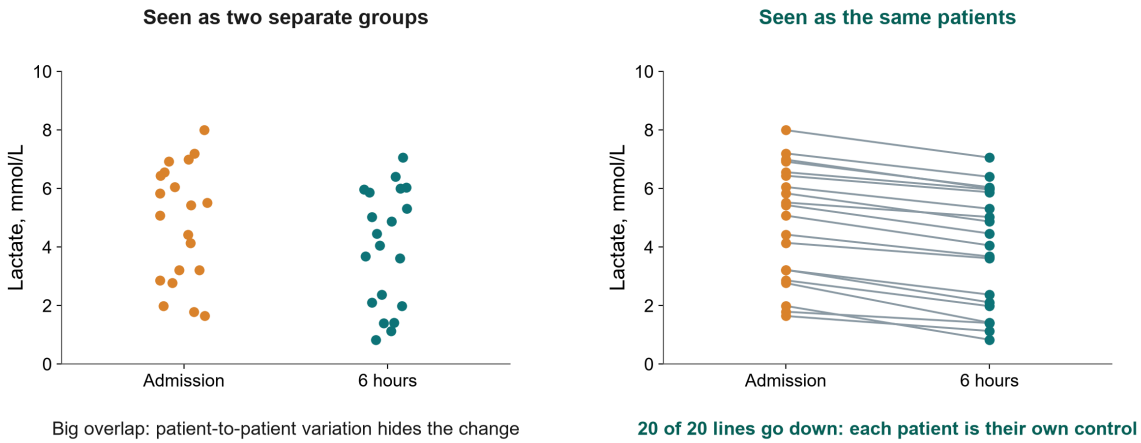


Figure 12.9: Lactate in 20 simulated patients at ICU admission and six hours later: drawn as two separate groups (left) the values overlap heavily, but joining each patient’s two values (right) shows that all 20 lines go down.

The same simulated lactate values, measured in 20 patients at ICU admission and six hours later, are drawn twice. On the left they are shown as two separate clouds: the before and after values overlap heavily. Patients differ a great deal from one another (some start at 2 mmol/L, some at 8), and this between-patient variation hides the change. On the right, each patient’s two values are joined by a line. Every one of the 20 lines goes down.

In a paired design **each patient is their own control**, so the variation between patients disappears from the comparison. A paired test analyses the *differences* within each patient, which vary much less than the raw values, and it therefore has much more power to detect a real change. Analysing paired data with an unpaired test throws this advantage away and can miss a real effect. The opposite error, treating independent groups as if they were paired, is also wrong, and can produce spurious precision.

Common paired designs include:

- **before-and-after measurements** in the same patients (pain scores before and after a new analgesia protocol, lactate at admission and at six hours);
- **crossover trials**, in which each patient receives both treatments in turn;
- **matched designs**, in which each patient is paired with a similar patient in the other group. Paper 2 matched patients one-to-one on a propensity score and analysed the matched pairs with methods that took the pairing into account (Hyun et al. 2023).

12.8 Multiple comparisons: the jelly-bean problem

Every test carries a risk of a false alarm, usually 5%. Run many tests and false alarms become almost certain. The figure below illustrates the problem, in a version of a well-known cartoon.

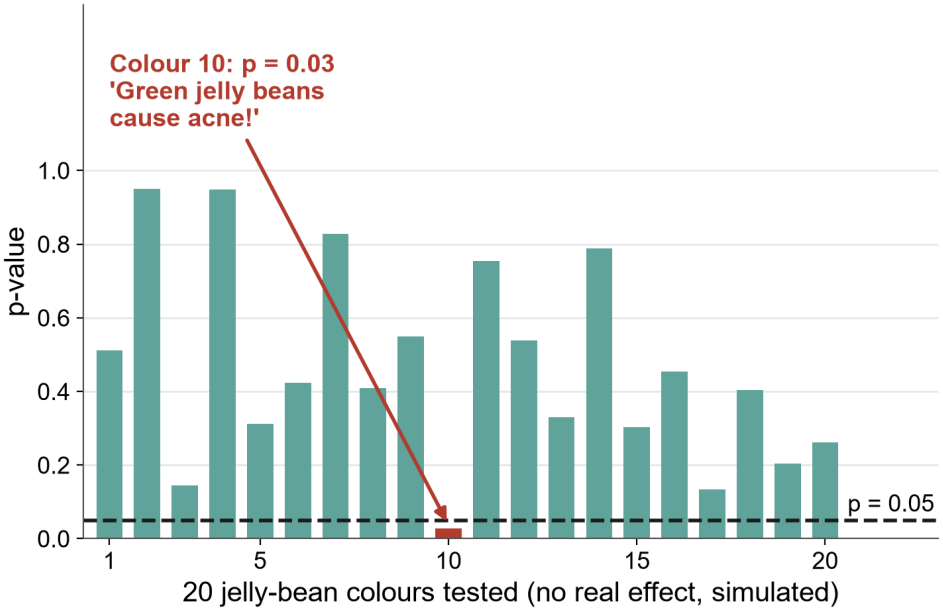


Figure 12.10: Twenty jelly-bean colours tested for an effect on acne when there is no real effect (simulated): one colour gives p = 0.03 by chance alone, producing the headline "Green jelly beans cause acne!".

Researchers test whether jelly beans cause acne. Nothing is found overall, so they test 20 colours separately. There is no real effect anywhere (the data are simulated), yet one colour comes out at p = 0.03, and the headline reads "Green jelly beans cause acne!".

The arithmetic is simple. If each test has a 5% chance of a false alarm, the chance that a single test is *not* a false alarm is 95%. With 20 independent tests and no real effects, the chance that none of them gives a false alarm is $0.95^{20} = 0.36$, so the chance of **at least one p < 0.05 is about 64%**.

Clinical versions of the jelly-bean problem are everywhere: ten secondary outcomes, eight subgroups, five time points and several laboratory values, all tested, with only the "significant" ones reported. Subgroup analyses are especially treacherous: when a treatment appears to work in women but not in men, or in one age group but not another, the difference is often a chance finding.

Four remedies help.

1. **Pre-specify one primary outcome** in the protocol and statistical analysis plan (Chapter 15). The study's main conclusion rests on that outcome.
2. **Label everything else as secondary or exploratory**, to be confirmed in further studies.
3. **Adjust the threshold** when many tests address the same question. The simplest method, the **Bonferroni correction**, divides the significance level by the number of tests: for 20 tests, $0.05/20 = 0.0025$ (Bland and Altman 1995). It is conservative (it makes real effects harder to detect), and other methods exist, but it makes the point clearly.
4. **Report every test you ran**, not just the winners.

For subgroups, the right question is not "is the effect significant in one subgroup and not in the other?" but "does the effect *differ* between subgroups?", which is answered by a test of **interaction** (Chapter 13). Interactions should be few, pre-specified and clinically plausible. FLUID-ICU (simulated) pre-specified a single one: whether the association between a positive balance and death differed in patients with heart failure, who tolerate fluid less well. The adjusted odds ratio was 4.39 (95% CI 1.65 to 12.38) in the 90 patients with heart failure and 1.29 (95% CI 0.77 to 2.19) in the 437 without, with p = 0.031 for interaction. Because it was pre-specified and plausible, it deserves attention; because the subgroup was small and the interval very wide, the analysis plan correctly treated it as evidence needing confirmation, not a definite finding.

! Common mistake: the highlighted subgroup

A paper reports twelve subgroup analyses and highlights the one with p = 0.04. How convinced should you be? Not very. With twelve tests, at least one $p < 0.05$ would be expected about 46% of the time even if the treatment had exactly the same effect in everyone ($1 - 0.95^{12} = 0.46$). Ask whether the subgroup was pre-specified, whether a test of interaction was reported, and whether the finding is biologically plausible and replicated elsewhere.

12.9 How inference appears in published papers

When reading the Methods and Results of a paper, check the following.

- **Is there a sample-size calculation**, with all four ingredients and their sources? Was the study powered for the primary outcome, and is that the outcome on which the conclusions rest?
- **Are the tests appropriate** for the type and shape of each outcome, and for paired or matched data? Paper 1, for example, states that continuous variables were compared with a t-test if normally distributed or a Mann-Whitney test otherwise, and categorical variables with the chi-square or Fisher's exact test (Boulet et al. 2024).
- **Are effects reported with confidence intervals**, not p-values alone? Is the interval read correctly in the discussion, or is a wide interval including the null value described as "no effect"?
- **Is the language honest?** Look out for "trend" for $p > 0.05$, "proves" or "causes" in observational studies, and conclusions drawn from secondary outcomes or subgroups.
- **How many tests were run?** Is there one pre-specified primary outcome? Are subgroup findings supported by interaction tests?

Reporting guidelines such as CONSORT and STROBE ask authors to present estimates with their precision, such as a 95% confidence interval, and to report the sample-size calculation (for trials) or how the study size was arrived at (for observational studies) (Schulz et al. 2010; Elm et al. 2007).

Summary

We study samples to learn about populations. Estimates vary from sample to sample by chance (sampling error), and the standard error measures that variation; it shrinks with the square root of the number of patients. Bias, by contrast, is not reduced by larger samples. A 95% confidence interval, estimate ± 1.96 SE, is the range of true values reasonably compatible with the data; read it in three steps: the estimate, whether it includes the null value (0 for differences, 1 for ratios), and its width. A p-value is the probability of results at least as extreme as those observed if there were no real effect; it is not the probability that the null hypothesis is true, not a measure of effect size, and $p > 0.05$ is not proof of no effect. Report effects with confidence intervals. Type I errors are false alarms (risk alpha), type II errors are missed effects (risk beta), and power is the chance of detecting a real effect. Sample size depends on alpha, power, the effect to be detected and the variability, plus an allowance for losses: Paper 1 needed 22 patients per group for fluid balance, whereas a trial to detect a fall in mortality from 40% to 30% needs 356 per group before allowing for losses. Choose tests from the outcome type, its shape or expected counts, and whether the data are paired. Finally, many tests produce false alarms: pre-specify the primary outcome and the few subgroups that matter, label the rest as exploratory, and report everything.

Further reading

- Altman, *Practical Statistics for Medical Research* (Altman 1991) and Kirkwood and Sterne, *Essential Medical Statistics* (Kirkwood and Sterne 2003): two standard textbooks covering sampling, confidence intervals, hypothesis tests, sample size and the choice of test in more depth, with medical examples throughout.
- Greenland et al. 2016 (Greenland et al. 2016): a thorough and readable guide to 25 common misinterpretations of p-values, confidence intervals and power.
- Wasserstein and Lazar 2016 (Wasserstein and Lazar 2016): the American Statistical Association's short statement on the context, process and purpose of p-values.
- Altman and Bland 1995, "Absence of evidence is not evidence of absence" (Altman and Bland 1995): a one-page classic on why non-significant results from small studies must not be read as proof of no effect.
- Gardner and Altman 1986 (Gardner and Altman 1986): the influential argument for reporting confidence intervals rather than relying on p-values.

CHAPTER 13

Relationships: correlation and regression

Chapter 12 showed how to compare two groups: a mortality rate in one group against another, a median length of stay in one group against another. Much of clinical research asks a richer question. How does one quantity change as another changes? Is the association between a treatment and an outcome still there once we account for the fact that the patients who received the treatment were different to begin with? Does the effect of an exposure depend on who the patient is? These questions are answered by correlation and, above all, by regression models. This chapter explains what these tools do, how to read their output in a published paper, and the mistakes that most often lead readers and authors astray. You do not need to run a model to read one, and the aim here is that by the end of the chapter you can read any regression table with confidence.

In this chapter you will learn

- What a correlation coefficient measures, and why correlation is neither agreement nor causation
- How linear regression describes a numerical outcome with an intercept, a slope and R^2
- Why a yes/no outcome needs logistic regression, and what an odds ratio from a model means
- How the type of outcome determines the model and the "no effect" value (0 or 1)
- What "adjusted" means, and how a model compares like with like
- How to read a regression table and a forest plot, row by row
- What interaction (effect modification) is and how to test it properly
- Why too many variables for too few events leads to overfitting, and why stepwise selection misleads

13.1 The running example

Throughout this book we follow one clinical problem: how much intravenous fluid to give adults with sepsis and septic shock. Two published studies anchor the example. Paper 1 is a randomised pilot trial by Boulet and colleagues, which compared a restrictive fluid strategy with usual care in 50 patients with septic shock (Boulet et al. 2024). Paper 2 is a large cohort study by Hyun and colleagues, which used data from a national sepsis registry to ask whether a positive cumulative fluid balance in the first three ICU days was associated with 28-day mortality (Hyun et al. 2023). Alongside them we use **FLUID-ICU**, a simulated cohort of 620 adults with sepsis in four ICUs. FLUID-ICU is not real: it was generated by a computer program for teaching, so that the classic lessons of data analysis appear in it. Its main question is whether a positive cumulative fluid balance by day 3 is associated with death by day 28. Chapter 16 analyses it from start to finish; this chapter borrows a few of its results to show what models do.

Paper 2 and FLUID-ICU share a central difficulty. In everyday practice, sicker patients receive more fluid, and sicker patients die more often. A crude comparison of patients with more fluid against patients with less fluid therefore mixes the effect of fluid with the effect of severity. This is **confounding by indication** (Chapter 6), and regression models are one of the main tools for dealing with it.

13.2 Correlation: how closely two numbers move together

The simplest relationship is between two numerical variables measured on the same patients: age and systolic blood pressure, weight and creatinine clearance, fluid given on day 1 and ICU length of stay. The natural first step is a **scatter plot**, with one variable on each axis and one dot per patient. The **correlation coefficient**, written r (Pearson's correlation), summarises how closely the dots lie along a straight line.

r runs from -1 to $+1$. A value of $+1$ means the points lie exactly on a rising straight line; -1 means they lie exactly on a falling line; 0 means there is no straight-line association at all. The figure below shows five clouds of points with their correlation coefficients.

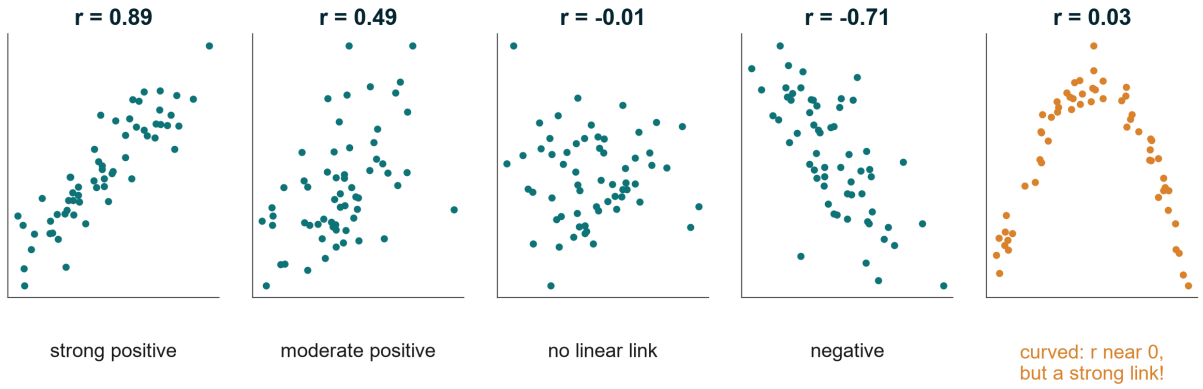


Figure 13.1: Five scatter plots with their correlation coefficients: strong positive (r = 0.89), moderate positive (0.49), no linear link (-0.01), negative (-0.71) and a strong curved relationship that nevertheless has r close to 0.

Read the panels from left to right. With $r = 0.89$ the points form a tight rising band. With $r = 0.49$ there is a clear upward trend but a lot of scatter: knowing one value helps predict the other, but only roughly. With r near 0 there is no straight-line trend. With $r = -0.71$ one variable falls as the other rises. The last panel is the trap. Its r is almost exactly 0, yet the relationship is very strong: it is simply curved, like an inverted U. Think of a drug with an optimal dose, or the U-shaped relationship between blood pressure and mortality. Because r measures only *straight-line* association, a correlation coefficient without a scatter plot can hide the most important feature of the data. Always look at the plot.

A rough guide, used in many textbooks, calls an r around 0.1 weak, around 0.3 moderate and 0.5 or more strong (Altman 1991). Treat this as a guide, not a rule: the clinical meaning matters more than the label. The square of r gives the proportion of the variation in one variable that is "explained" by a straight-line relationship with the other. An r of 0.5 corresponds to $r^2 = 0.25$, so three quarters of the variation is left unexplained.

A correlation also has a p-value, which tests the null hypothesis that the true correlation is zero. With large samples this p-value can be tiny for a correlation that is clinically useless: in 5,000 patients, $r = 0.05$ is "statistically significant" but explains a quarter of one per cent of the variation. As always (Chapter 12), the size of the estimate and its confidence interval matter more than the p-value.

Pearson's r assumes that both variables are numerical and that the relationship is roughly linear, without extreme outliers. When the data are skewed, include outliers, or are ordered scores (such

as a pain score or a SOFA score), the **Spearman rank correlation** is used instead. It correlates the ranks of the values rather than the values themselves and measures whether one variable tends to increase as the other increases, whether or not in a straight line.

13.2.1 Correlation is not agreement, and not causation

Correlation is often misused in two ways, shown side by side in the figure below.

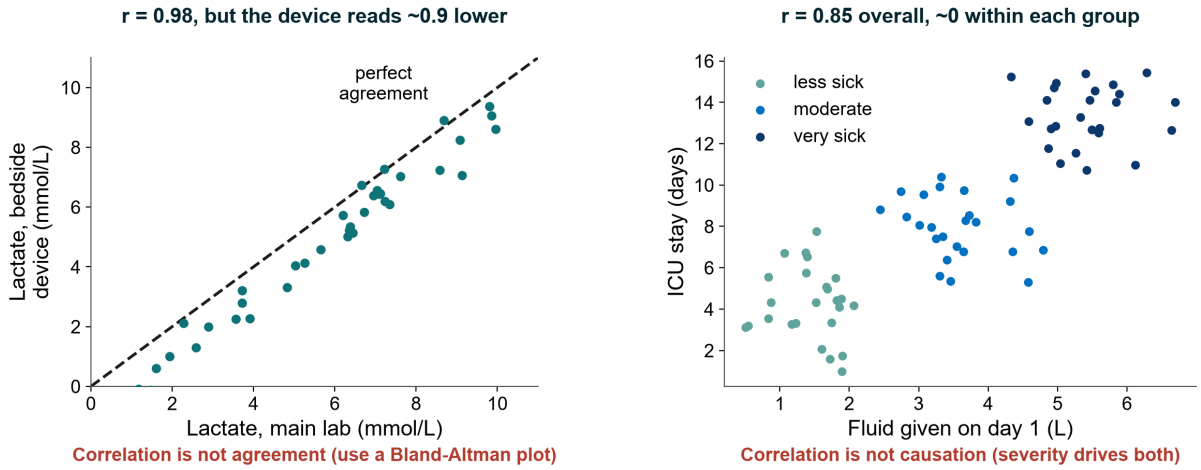


Figure 13.2: Left: a bedside lactate device and the main laboratory are highly correlated (r = 0.98) but the device reads about 0.9 mmol/L lower, so they do not agree. Right: fluid on day 1 and ICU stay are strongly correlated overall (r = 0.85) but not at all within each severity group, because severity drives both.

On the left, a bedside lactate device is compared with the main laboratory analyser in simulated patients. The correlation is 0.98, and many papers would describe this as “excellent agreement”. Yet every point lies below the line of perfect agreement: the device reads about 0.9 mmol/L lower than the laboratory, across the whole range. Correlation measures whether two sets of values rise and fall together, not whether they are the same. Two thermometers, one of which always reads two degrees too high, correlate perfectly. To compare two methods of measurement, the right tool is the **Bland–Altman plot**: for each patient, plot the difference between the two methods against their average, and report the mean difference (the bias) and the 95% limits of agreement (the range within which most differences fall) (Bland and Altman 1986). The clinician then decides whether differences of that size are acceptable.

On the right, the amount of fluid given on day 1 is strongly correlated with ICU length of stay ($r = 0.85$). It is tempting to conclude that fluid prolongs ICU stay. But the points fall into three clusters, coloured by severity, and within each cluster there is no relationship at all. Severity drives both the fluid given and the length of stay. The overall correlation is entirely produced by a third variable: confounding again. A correlation, however strong, cannot by itself tell us that one variable causes the other.

! Common mistake: reading r as agreement or as cause

“The new point-of-care haemoglobin meter correlated very well with the laboratory (r = 0.95), so it can replace the laboratory test.” Not on this evidence. A high correlation is compatible with a large, systematic bias. Ask for a Bland–Altman analysis with the bias and limits of agreement. Equally, “fluid correlated with length of stay (r = 0.85), so fluid prolongs ICU stay”

ignores the obvious confounder, illness severity.

13.3 Linear regression: the best straight line

Correlation tells us how closely two variables are related, but not *how much* one changes when the other changes. That is the job of **regression**. When the outcome is a number (a blood pressure, an oxygenation ratio, a number of ventilator-free days), the basic tool is **linear regression**, which fits the straight line that best describes how the average outcome changes with the exposure:

$$\text{outcome} = a + b \times \text{exposure}$$

The **intercept** a is the expected outcome when the exposure equals zero. The **slope** b, usually called the **regression coefficient** (written B or β in tables), is the change in the average outcome for each one-unit increase in the exposure. “Best” has a precise meaning: the line chosen is the one that makes the squared vertical distances between the points and the line as small as possible (the method of least squares).

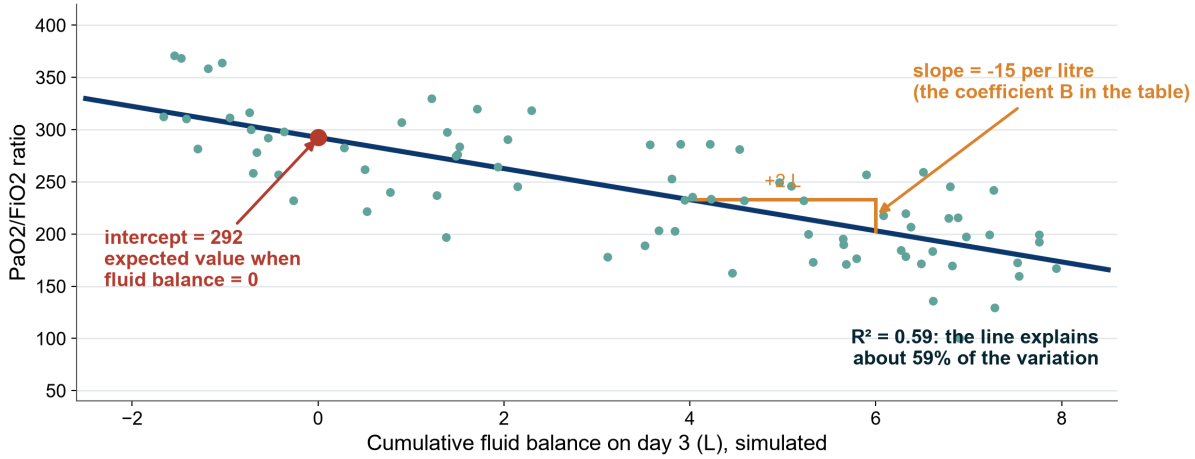


Figure 13.3: Linear regression of the PaO2/FiO2 ratio on cumulative fluid balance at day 3 in simulated patients: intercept 292 (expected value at zero balance), slope -15 per litre of positive balance, and R^2 = 0.59.

The figure above shows a simulated example. The outcome is the PaO_2/FiO_2 ratio, a measure of oxygenation; the exposure is the cumulative fluid balance at day 3 in litres. The fitted line has an intercept of 292: a patient with a fluid balance of zero is expected to have a ratio of about 292. Its slope is -15 per litre: each extra litre of positive balance is associated with a ratio about 15 points lower. The orange step shows this concretely: moving from 4 to 6 litres (two extra litres) lowers the expected ratio by about 30 points.

Worked example: reading a linear regression coefficient

A table reports, for the PaO_2/FiO_2 ratio, a coefficient for fluid balance of $B = -15$ per litre (95% CI -19 to -11).

- **The comparison:** one litre more of cumulative balance versus one litre less.

- **The effect:** the expected ratio is 15 points lower for each extra litre.
- **The precision:** values between 11 and 19 points lower per litre are compatible with the data.
- **The null value:** for a difference, "no effect" is 0. The interval does not include 0, so the association is statistically significant at the 5% level.
- **A prediction:** a patient with a balance of 4 litres has an expected ratio of $292 - 15 \times 4 = 232$.

In plain language: "Each additional litre of cumulative fluid balance was associated with a PaO_2/FiO_2 ratio about 15 points lower (95% CI 11 to 19)." The data in this example are simulated; it is not a claim about real oxygenation.

The figure also reports **R^2** (R-squared), here 0.59. It is the proportion of the variation in the outcome that the line explains: 59%. The other 41% is due to everything the model does not include, from lung disease to measurement error. R^2 is useful for describing how well a model fits, but a low R^2 does not make an association unimportant: the effect of a single exposure on a complex clinical outcome will rarely explain most of its variation.

Linear regression rests on a few assumptions. The relationship should be roughly linear; the scatter of the points around the line should be similar along its length; and, for the confidence intervals and p-values to be accurate in small samples, the residuals (the vertical distances from the points to the line) should be roughly normally distributed. With large samples the last assumption matters less, but strongly skewed outcomes such as length of stay are often analysed on a logarithmic scale instead, so that the result becomes a ratio of geometric means rather than a difference (Chapter 16 shows an example).

! Common mistake: extrapolating beyond the data

The line in the figure was fitted to fluid balances between about -2 and $+8$ litres. Applied to a balance of 20 litres it predicts a negative PaO_2/FiO_2 ratio, which is impossible. A regression line describes the data it was fitted to; outside that range it is guesswork.

13.4 Logistic regression: an S-curve for yes/no outcomes

Many of the most important clinical outcomes are binary: died or survived, developed acute kidney injury or not, readmitted or not. A straight line does not work for these outcomes, because a probability cannot fall below 0% or rise above 100%, and a straight line eventually crosses both limits. **Logistic regression** solves this by modelling not the probability itself but the logarithm of the **odds** of the outcome (Chapter 7 defined odds as the probability that the event happens divided by the probability that it does not):

$$\log(\text{odds of outcome}) = a + b \times \text{exposure}$$

On the log-odds scale the relationship is a straight line; translated back into probabilities it becomes an S-shaped curve that stays between 0 and 1. The coefficient b is the change in log-odds per unit of exposure, which nobody finds intuitive. Its exponential, e^b, is much more useful: it is the **odds ratio (OR)** for a one-unit increase in the exposure. This is the number reported in almost

every logistic regression table.

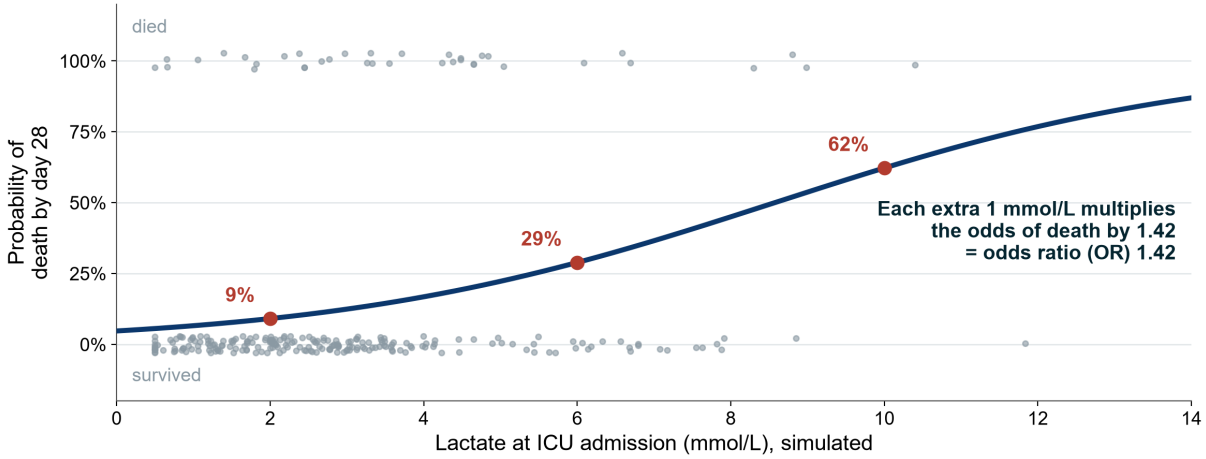


Figure 13.4: A logistic regression of 28-day death on admission lactate in simulated patients: each grey dot is a patient (top row died, bottom row survived); the blue S-curve gives the probability of death, about 9% at 2 mmol/L, 29% at 6 and 62% at 10, with an odds ratio of 1.42 per mmol/L.

In the figure each grey dot is a simulated patient, placed at the top if they died and at the bottom if they survived. The blue curve is the fitted probability of death at each lactate level. The model reports an odds ratio of 1.42 per mmol/L: each extra 1 mmol/L of lactate multiplies the odds of death by 1.42.

Worked example: one odds ratio, three different risk changes

Start at a lactate of 2 mmol/L, where the predicted risk of death is about 9%. The odds are $0.09/0.91 \approx 0.099$.
Four mmol/L higher (lactate 6) the odds are multiplied by $1.42^4 \approx 4.07$, giving odds of about 0.40, which is a probability of $0.40/1.40 \approx 29\%$.
Another four mmol/L higher (lactate 10) the odds are multiplied by 4.07 again, to about 1.64, a probability of $1.64/2.64 \approx 62\%$.
The odds ratio is the same along the whole curve, but the change in *risk* is not: going from 2 to 6 mmol/L adds about 20 percentage points, and going from 6 to 10 adds about 33. Where a patient sits on the S-curve determines how much the same odds ratio changes their absolute risk.

Because the odds ratio is constant along the curve, it is a convenient summary, and logistic regression allows several variables to be included at once (multivariable logistic regression), which we come to shortly. But it has a well-known weakness as a communication tool. When the outcome is rare (say under 10%), the odds ratio is close to the risk ratio and can be read loosely as "times the risk". When the outcome is common, as death is in intensive care, the odds ratio is further from 1 than the risk ratio.

Worked example: odds ratio versus risk ratio in FLUID-ICU

In the simulated FLUID-ICU cohort, 134 of 373 patients with a positive fluid balance died by day 28 (35.9%), against 45 of 245 with a negative balance (18.4%).

- Risk ratio: $35.9/18.4 = 1.96$.

- Odds of death with a positive balance: $134/239 = 0.561$. With a negative balance: $45/200 = 0.225$.
- Odds ratio: $0.561/0.225 = 2.49$.

The same data give a risk ratio of 1.96 and an odds ratio of 2.49. Writing "a positive balance was associated with a 2.5-fold risk of death" would overstate the association. The odds ratio is not wrong; it is simply a ratio of odds, and should be described as such.

! Common mistake: 'OR 1.8 means 80% more deaths'

An odds ratio of 1.8 means 80% higher *odds*. When the outcome is common, the increase in risk is smaller. Write "1.8 times the odds" or "80% higher odds", and if readers need absolute risks, report them separately (from the raw data or from the model's predicted probabilities).

13.5 Which model? It depends on the outcome

Just as the type of outcome determines which statistical test to use (Chapter 12), it determines which regression model to use and what number the model reports. The figure below sets out the three models met most often in clinical papers.

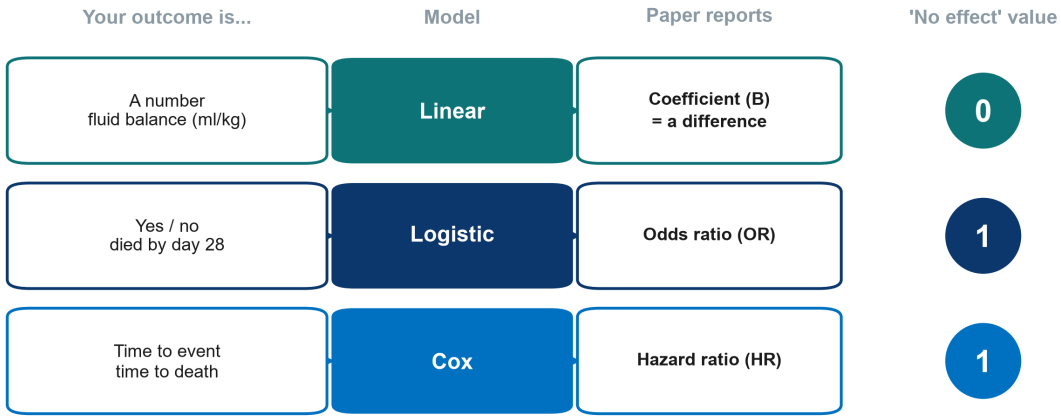


Figure 13.5: The outcome decides the model: a numerical outcome goes with linear regression, which reports a coefficient (a difference, no effect = 0); a yes/no outcome goes with logistic regression, which reports an odds ratio (no effect = 1); a time-to-event outcome goes with Cox regression, which reports a hazard ratio (no effect = 1).

- A **numerical outcome** (fluid balance in ml/kg, a blood pressure) goes with **linear regression**. The result is a coefficient: a *difference* in the outcome per unit of the exposure. No effect is **0**.
- A **yes/no outcome** (died by day 28) goes with **logistic regression**. The result is an **odds ratio**. No effect is **1**.
- A **time-to-event outcome** (time to death, with patients still alive at the end of follow-up "censored") goes with **Cox regression**. The result is a **hazard ratio (HR)**. No effect is **1**. Chapter 14 explains survival analysis in detail; Paper 2 used a Cox model (Hyun et al. 2023).

Why 0 for one and 1 for the others? A difference of 0 means the two groups are the same; a ratio of 1 also means they are the same (two divided by two is one). Before reading any confidence interval,

find the null value. A 95% confidence interval that excludes the null corresponds to $p < 0.05$, and one that includes it corresponds to $p \geq 0.05$. Forest plots of ratios are usually drawn on a logarithmic axis, on which halving (0.5) and doubling (2) are the same distance from 1; this is why such plots look symmetrical around 1.

Other models exist for other outcome types: Poisson or negative binomial regression for counts and rates, ordinal regression for ordered categories such as AKI stages, and log-binomial or modified Poisson models when authors want to report risk ratios directly instead of odds ratios. A statistician may also choose a different model for a skewed numerical outcome. The reading principle is always the same: what is compared, how big is the effect, how wide is the interval, and where is the null?

! Common mistake: looking for 0 in a table of ratios

"The 95% CI for the odds ratio was 0.85 to 1.60; since it does not include 0, the result is significant." Wrong: for a ratio the null value is 1, and this interval includes 1. The data are compatible with no association.

13.6 What "adjusted" means: compare like with like

The most important use of regression in observational research is **adjustment** for confounders. Chapter 6 introduced confounding: a third factor, associated with both the exposure and the outcome, that distorts the comparison. The intuition of adjustment is best seen without any model at all, by splitting the patients into groups that are alike on the confounder. This is **stratification**.

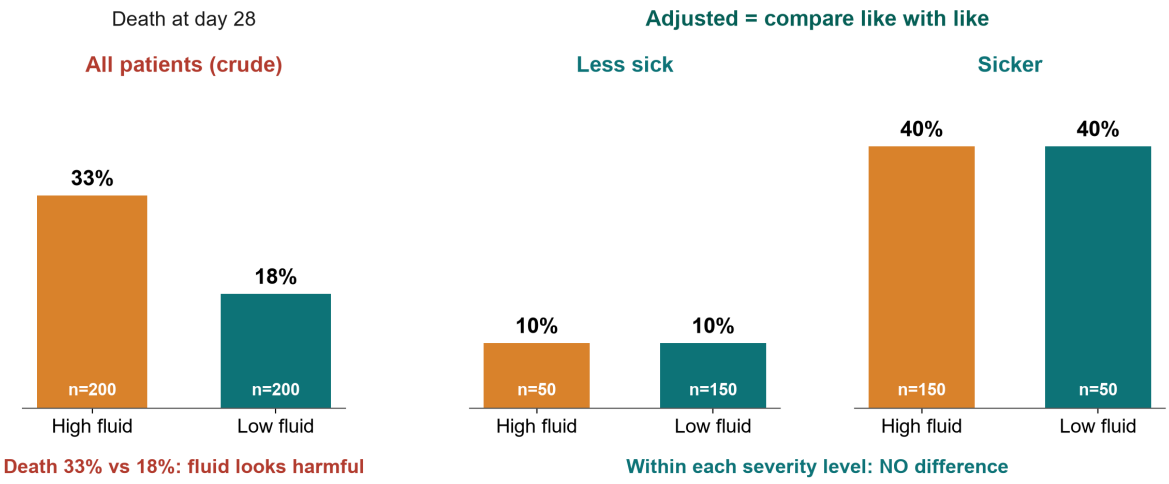


Figure 13.6: Simulated example of stratification: crudely, death is 33% with high fluid and 18% with low fluid; split by severity, death is 10% versus 10% in less sick patients and 40% versus 40% in sicker patients, so within each severity level there is no difference.

On the left of the figure are all 400 simulated patients. Those who received a lot of fluid died more often (33% versus 18%): fluid looks harmful. On the right the same patients are split by severity. Among less sick patients, 10% died whether they received high or low fluid. Among sicker patients, 40% died either way. Within each level of severity there is no difference at all. The crude difference arose because sicker patients were both more likely to receive more fluid (150 of the 200 sicker patients) and more likely to die. Severity, not fluid, produced the crude association. This is confounding by indication made visible.

Stratification works well for one confounder with two levels. With six confounders, several of them

numerical, the strata become too many and too small. A **multivariable regression model** does the same job for many confounders at once. Including the exposure and the confounders together in one model gives, for the exposure, an estimate of its association with the outcome *at fixed values of the other variables*: in effect, a comparison of patients who are alike on everything else in the model. This is what "adjusted for age, SOFA score and lactate" means. The estimate from a model with only the exposure is called the **crude** (or unadjusted) estimate; the estimate from the model with confounders is the **adjusted** estimate.

Worked example: crude and adjusted odds ratios in FLUID-ICU

In the simulated FLUID-ICU cohort, the crude odds ratio for death by day 28 with a positive versus negative fluid balance was 2.49 (95% CI 1.70 to 3.70; 618 patients). The analysis plan named six confounders in advance: age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease. After adjustment for these, the odds ratio fell to 1.62 (95% CI 1.02 to 2.59; 527 patients with complete data).
Part of the crude association was severity: sicker patients received more fluid and died more often. What remains is a smaller, less precise association whose lower confidence limit is close to 1. Chapter 16 walks through this analysis in full.

Three warnings go with every adjusted estimate.

First, a model can adjust only for confounders that were **measured**, and measured well. SOFA score and lactate capture severity imperfectly; whatever severity they miss remains as **residual confounding**. Only randomisation protects against confounders that nobody measured, which is why Paper 1 is a trial and why the strongest claim a cohort can make is one of association. Paper 2 used propensity-score matching rather than regression adjustment, but the limitation is the same: patients can be matched only on what was recorded (Hyun et al. 2023).

Second, **not every variable should be adjusted for**. A confounder is a cause of both exposure and outcome that comes *before* the exposure. A variable that lies on the causal pathway between exposure and outcome (a **mediator**) must not be adjusted for if the aim is the total effect of the exposure: adjusting for it removes part of the very effect we want to estimate. In FLUID-ICU, acute kidney injury occurs after the fluid has been given and may be one way in which fluid harms patients, so the analysis plan deliberately does not adjust for it. Choosing confounders requires clinical knowledge of how the variables are related, ideally drawn as a simple causal diagram, not a statistical rule (Hernán et al. 2002; Rothman et al. 2008).

Third, the other rows of an adjusted model are not answers to the research question. In the FLUID-ICU model, the odds ratio for vasopressor use is adjusted for fluid balance, which may lie on its causal pathway; it does not estimate the effect of vasopressors on death. Interpreting every row of a multivariable table as if it were a separate causal finding is called the **"Table 2 fallacy"** (Westreich and Greenland 2013). Only the exposure row answers the question the model was built for.

• Adjusting = comparing like with like

An adjusted estimate compares patients who are alike on the variables in the model. It removes confounding by *measured* factors, not by unmeasured ones, and it is still an association. Confounders are chosen in advance from clinical knowledge; mediators and consequences of the exposure are left out.

13.7 Reading a regression table, row by row

Regression results are usually presented as a table with one row per variable. Reading such a table is a skill every clinician can learn, and it uses the same four questions for every row.

Outcome: death by day 28. Multivariable logistic regression (mock data)

Variable	Adjusted OR	95% CI	p-value
Positive fluid balance (yes vs no)	**1.80**	**1.20 - 2.70**	**0.004**
Age (per 10 years)	1.25	1.10 - 1.42	< 0.001
SOFA score (per 1 point)	1.15	1.08 - 1.23	< 0.001
Female (vs male)	0.92	0.65 - 1.30	0.64
Lactate (per 1 mmol/L)	1.10	0.98 - 1.24	0.11

Covariates are the confounders the authors 'adjusted for'

Row 1: after adjusting for age, SOFA, sex, lactate, odds of death are 1.8 times

CI 1.20-2.70 does not include 1, so p < 0.05

Red CI crosses 1: compatible with no effect

Figure 13.7: A mock multivariable logistic regression table for death by day 28: positive fluid balance OR 1.80 (1.20 to 2.70, p = 0.004), age per 10 years 1.25, SOFA per point 1.15, female sex 0.92 (0.65 to 1.30) and lactate per mmol/L 1.10 (0.98 to 1.24); the last two intervals include 1.

1. **What is compared?** Each row states a comparison: “positive fluid balance (yes vs no)”, “age (per 10 years)”, “female (vs male)”. Check the reference group and the unit. An odds ratio “per 10 years” is very different from one “per year”.
2. **How big is the effect?** In the first row, an odds ratio of 1.80: patients with a positive fluid balance had 1.8 times the odds of death, adjusted for the other variables in the table.
3. **How precise is it?** The 95% CI is 1.20 to 2.70: the data are compatible with anything from a 20% to a 170% increase in the odds.
4. **Does the interval include the null?** For a ratio, the null is 1. The interval 1.20 to 2.70 excludes 1, so $p < 0.05$ (here $p = 0.004$). For female sex (0.65 to 1.30) and lactate (0.98 to 1.24) the intervals include 1: the data are compatible with no association.

Rows for numerical variables need one more step. An odds ratio of 1.15 per SOFA point means that each extra point multiplies the odds by 1.15. Over five points the odds are multiplied by $1.15^5 \approx 2.0$, a doubling. A modest-looking odds ratio per unit can describe a large effect across the range of the variable, and a large-looking one can describe a trivial effect if the unit is tiny.

Worked example: a regression table on hospital-acquired delirium

A mock cohort of 600 adults in a medical ICU reports a multivariable logistic regression for hospital-acquired delirium, adjusted for all variables in the table and for APACHE II score:

Table 13.1: Mock multivariable logistic regression for delirium in 600 ICU patients.

Variable	Adjusted OR (95% CI)	p
Benzodiazepine infusion (yes vs no)	2.10 (1.40 to 3.20)	< 0.001
Age (per 10 years)	1.30 (1.12 to 1.51)	< 0.001
Mechanical ventilation (yes vs no)	1.60 (0.95 to 2.70)	0.08
Female sex (vs male)	0.85 (0.60 to 1.20)	0.36
Pre-existing dementia (yes vs no)	3.40 (1.90 to 6.10)	< 0.001
Sleep protocol (yes vs no)	0.62 (0.41 to 0.94)	0.02

The outcome is binary, so the model is logistic and the null value is 1. The intervals for mechanical ventilation and female sex include 1. For the main exposure: “After adjustment for age, sex, ventilation, dementia, the sleep protocol and illness severity, patients who received a benzodiazepine infusion had about twice the odds of delirium (OR 2.10, 95% CI 1.40 to 3.20).” A patient 20 years older has $1.30 \times 1.30 \approx 1.69$ times the odds, all else equal.

Suppose the authors conclude: “Benzodiazepines cause delirium. Mechanical ventilation showed a trend towards more delirium (p = 0.08). Our sleep protocol reduces delirium by 38%.” There are three problems. “Cause” and “reduces” are causal words from an observational study, in which agitated or sicker patients are more likely to receive benzodiazepines (confounding by indication). A “trend” for $p = 0.08$ describes an interval (0.95 to 2.70) that includes 1. And “reduces delirium by 38%” confuses a 38% reduction in *odds* (OR 0.62) with a reduction in risk, and implies that a non-randomised protocol caused the difference.

13.8 Forest plots

A **forest plot** shows several estimates with their confidence intervals against a vertical line at the null value. Each estimate is a square or dot; each horizontal line is its 95% CI. Forest plots are used for the rows of a regression model, for subgroups, for several time points and, most famously, for the studies in a meta-analysis (Chapter 9). The reading rules are those of the regression table, applied visually: find the null line, see which side each estimate falls, and see whether its interval crosses the line.

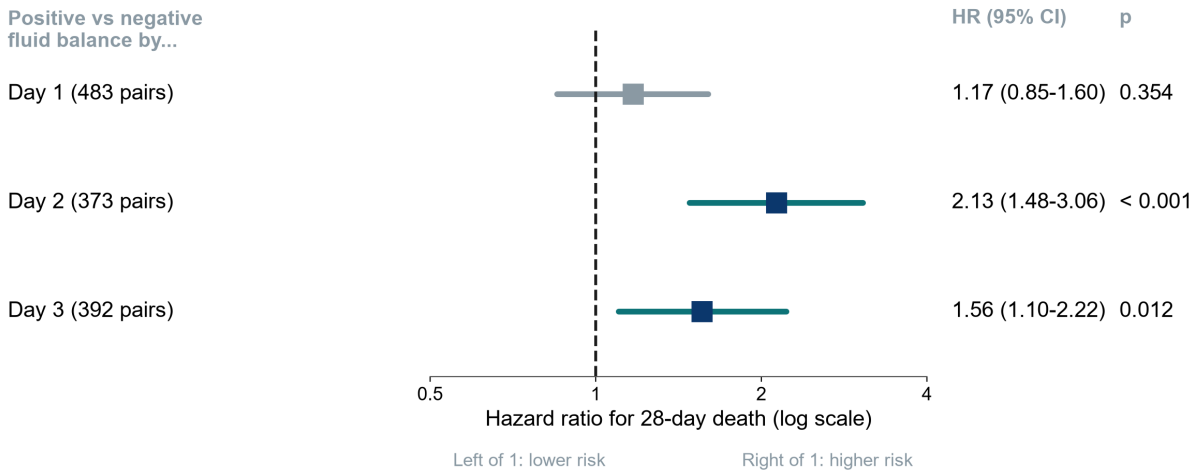


Figure 13.8: Forest plot from Paper 2: hazard ratios for 28-day death with a positive versus negative cumulative fluid balance in three propensity-matched cohorts. Day 1: 1.17 (0.85 to 1.60); day 2: 2.13 (1.48 to 3.06); day 3: 1.56 (1.10 to 2.22).

The figure shows the main results of Paper 2 (Hyun et al. 2023). For a positive balance on day 1 (483 matched pairs) the hazard ratio was 1.17, and the interval (0.85 to 1.60) crosses 1: no clear association. For a positive balance by day 2 (373 pairs) the hazard ratio was 2.13 (1.48 to 3.06), entirely above 1: about double the hazard of death. By day 3 (392 pairs) it was 1.56 (1.10 to 2.22), above 1 but with a lower limit close to it. The authors' interpretation is that a positive balance that persists beyond the first day is associated with higher mortality. Note the word "associated": even after matching, sicker patients may still have received more fluid.

The width of each line also tells a story. Intervals are wider when there are fewer patients and events, or more variability. In a forest plot of subgroups, small subgroups always have wide intervals, which is one reason why apparent differences between subgroups must be tested rather than eyeballed (next section).

13.9 Interaction: does the effect differ between groups?

Adjustment assumes that the exposure has the same effect in every kind of patient. Sometimes that is not true. **Interaction**, also called **effect modification** (Chapter 6), means that the association between exposure and outcome differs according to a third variable.

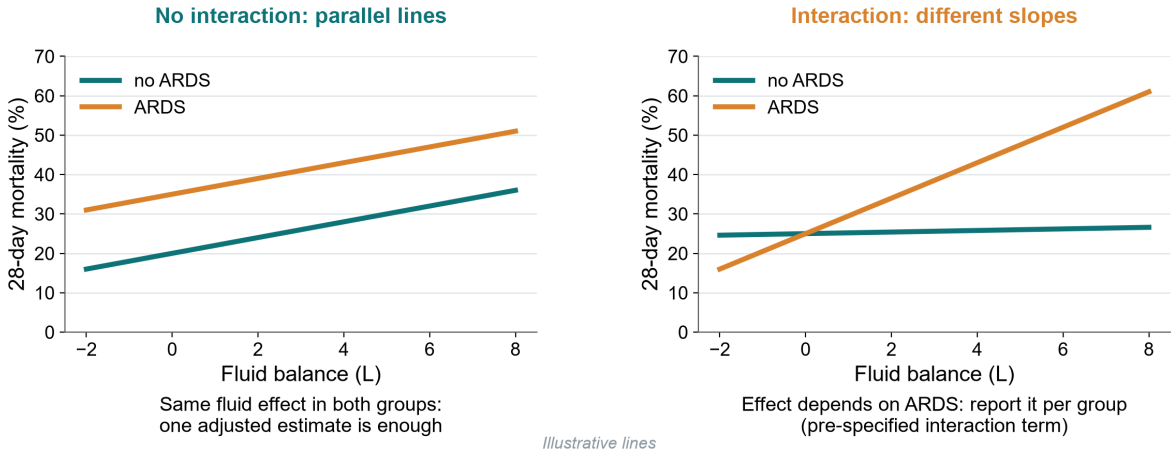


Figure 13.9: Illustrative lines: on the left (no interaction) mortality rises with fluid balance at the same rate with and without ARDS, so the lines are parallel; on the right (interaction) fluid balance hardly matters without ARDS but matters a lot with ARDS.

On the left of the figure, mortality rises with fluid balance at the same rate in patients with and without acute respiratory distress syndrome (ARDS); patients with ARDS simply start at a higher level. The lines are parallel, there is no interaction, and a single adjusted estimate describes everyone. On the right the slopes differ: fluid balance hardly matters without ARDS but matters a great deal with ARDS. A single "average" effect would describe nobody well. The right approach is to include an **interaction term** (fluid × ARDS) in the model, test it, and, if the interaction is important, report the effect separately in each group.

Confounding and interaction are different things. Confounding is a bias to be removed: the crude estimate is distorted by a third variable. Interaction is a real feature of the world to be described: the effect genuinely differs between groups. "The drug works in women but not in men" is a claim of interaction.

Claims of interaction are among the least reliable findings in clinical research, and a few rules protect against false ones (Wang et al. 2007; Altman and Bland 2003):

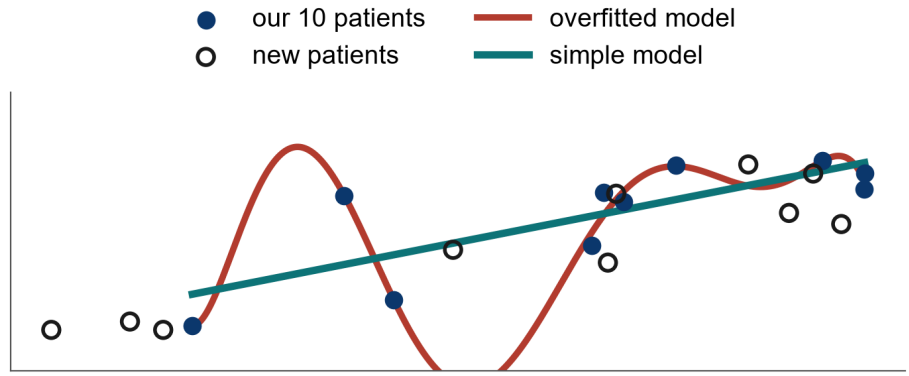
- **Pre-specify** a small number of interactions, each with a clinical reason and an expected direction, in the analysis plan. Testing twenty subgroups after seeing the data will produce at least one "significant" difference by chance (the multiple comparisons problem of Chapter 12).
- **Test the interaction itself.** Finding $p < 0.05$ in one subgroup and $p > 0.05$ in the other does *not* show that the effects differ. The right question is whether the difference between the two subgroup estimates is larger than chance would produce, which is what the test for interaction answers.
- **Remember the power.** Tests for interaction have much less power than tests of the main effect, and subgroups are often small.
- **Interpret cautiously.** Even a pre-specified interaction with a small p-value is usually a hypothesis needing confirmation in another study.

Worked example: the heart-failure interaction in FLUID-ICU

The FLUID-ICU analysis plan named one effect modifier in advance: chronic heart failure, because patients with heart failure tolerate fluid less well. In the simulated data, the adjusted odds ratio for death with a positive balance was 4.39 (95% CI 1.65 to 12.38) in the 90 patients with heart failure and 1.29 (0.77 to 2.19) in the 437 without. The test for interaction gave $p = 0.031$. The finding is plausible and pre-specified, but it rests on 90 patients and has a very wide interval. It should be reported as evidence of effect modification that needs confirmation, not as a definite finding.

13.10 Overfitting: too many variables for too few events

A regression model learns from the data it is given. With many variables and few events it can learn the noise in *this* dataset as well as the signal, fitting the patients in hand closely and failing on the next ones. This is **overfitting**.



Rule of thumb: about 10 events per variable
80 deaths: no more than about 8 variables in the model

Figure 13.10: Ten patients (filled dots) fitted by an overfitted model (red curve) that passes through all of them and by a simple model (teal line); new patients (hollow dots) are predicted much better by the simple line. Rule of thumb: about 10 events per variable.

In the figure the red curve passes through all ten filled points, but the hollow points (new patients from the same population) fall far from it. The simple teal line misses some of the original points but predicts the new ones much better. The red curve has memorised its data.

The practical consequence in clinical papers is a limit on how many variables a logistic or Cox model can contain. A widely used rule of thumb is about **10 events per variable** (EPV), counting *events*, not patients: deaths, not admissions (Peduzzi et al. 1996). A study with 80 deaths can support a model with no more than about eight variables, with each category of a categorical variable beyond the first counted separately. For a binary outcome, the "events" are whichever of the two outcomes is less common. The rule is approximate, and more recent work shows that the number needed depends on the situation; prediction models in particular generally need more

data, and their sample size should be calculated rather than assumed (Chapter 14) (Riley et al. 2020; Harrell 2015).

Worked example: events per variable in FLUID-ICU

The FLUID-ICU primary model includes the exposure and six confounders: seven terms. In the complete-case model there were 161 deaths among 527 patients. That gives $161/7 \approx 23$ events per variable, comfortably above 10. If the same model had been fitted to a study with 40 deaths, it would have had fewer than 6 events per variable and its estimates would be unstable.

13.10.1 Why stepwise selection misleads

Faced with many candidate variables, authors often let the software choose them, adding or dropping variables according to their p-values (**stepwise selection**), or screen variables by including only those with $p < 0.05$ in univariable analyses. These procedures look objective, but they have serious problems (Harrell 2015). They multiply the multiple-comparisons problem, because dozens of tests are run behind the scenes. They tend to keep variables that are associated with the outcome by chance and to drop real confounders that happen to be weakly associated in this sample. The effects of the variables that survive are overestimated and their confidence intervals are too narrow, because the reported analysis ignores the selection that preceded it. And a different sample from the same population would often select different variables.

For causal questions (is the exposure associated with the outcome, after accounting for confounding?), the better approach is to **choose confounders in advance** from clinical knowledge and the literature, with the help of a causal diagram, write them in the statistical analysis plan (Chapter 15), and stay within the events-per-variable budget. A variable that confounds the association should be in the model whether or not its own p-value is small.

! Common mistake: 15 variables, 40 events, stepwise selection

A study with 40 deaths reports a model with 15 variables chosen by stepwise selection and highlights three “independent predictors”. With fewer than 3 events per variable and data-driven selection, the model is almost certainly overfitted: the effect sizes are inflated, the confidence intervals too narrow, and the “predictors” may not replicate.

13.11 How regression appears in published papers

A well-reported regression analysis, as recommended by the STROBE statement for observational studies, tells the reader the following (Elm et al. 2007):

- The outcome and its type, and which model was used for it.
- How each exposure and confounder was entered (the units of numerical variables, the reference category of categorical ones, and whether any continuous variable was cut into categories, and where).
- Which confounders were adjusted for, and why they were chosen (ideally pre-specified, with a rationale).
- Both the **crude and the adjusted estimates**, with 95% confidence intervals, so the reader can see how much adjustment changed the answer (STROBE item 16a).
- The number of patients and events in each model; when patients are dropped because of missing data, how many and how this was handled (Chapter 15).
- Any interactions tested, and whether they were pre-specified.
- How model assumptions were checked.

When reading, look for these elements. A methods section that says only "variables with $p < 0.1$ in univariable analysis were entered into a multivariable model" or "data were analysed with statistical software" should prompt questions. Note, too, whether continuous variables were cut into categories: splitting a continuous measurement in two (for example, positive versus negative fluid balance) throws away information and reduces power, although it can make results easier to communicate (Altman and Royston 2006). FLUID-ICU's analysis plan uses the binary version for its primary question and the continuous version (per 10 ml/kg) for others, a choice fixed before the analysis.

Summary

- A scatter plot and the correlation coefficient r describe how closely two numerical variables follow a straight line. Always look at the plot: curved relationships can have r near zero. Correlation is not agreement (use a Bland–Altman plot) and not causation.
- Linear regression describes a numerical outcome with an intercept and a slope; the slope is the change in the outcome per unit of exposure (null value 0), and R^2 is the proportion of variation explained.
- Logistic regression models the log-odds of a binary outcome and reports odds ratios (null value 1). The odds ratio is constant along the S-curve, but the change in risk is not; with common outcomes the OR is further from 1 than the risk ratio.
- The outcome type decides the model: linear (difference), logistic (OR), Cox (HR).
- Adjustment compares like with like on measured confounders. It does not remove unmeasured confounding, and mediators should not be adjusted for. Only the exposure row answers the research question.
- Read every row of a regression table and forest plot with four questions: what is compared, how big, how precise, and does the interval include the null?

- Interaction means the effect differs between groups. Pre-specify few interactions, test the interaction itself, and treat findings as hypotheses.
- Keep to roughly 10 events per variable, choose confounders in advance, and avoid stepwise selection.

Further reading

- Altman DG. *Practical Statistics for Medical Research* (Altman 1991). The chapters on correlation and regression remain one of the clearest introductions for clinicians.
- Kirkwood BR, Sterne JAC. *Essential Medical Statistics* (Kirkwood and Sterne 2003). Covers linear and logistic regression and confounding with medical examples and a gentle level of mathematics.
- Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement (Bland and Altman 1986). The paper that introduced the Bland–Altman plot and explained why correlation is the wrong tool for agreement.
- Westreich D, Greenland S. The Table 2 fallacy (Westreich and Greenland 2013). A short, readable explanation of why only the exposure row of a multivariable model answers the research question.
- Harrell FE. *Regression Modeling Strategies* (Harrell 2015). An advanced but practical reference on overfitting, variable selection and how to build models that hold up.

CHAPTER 14

Time-to-event analysis and prediction models

Two kinds of question come up again and again in clinical research, and both need methods beyond those of Chapter 13. The first is about *time*: not only whether patients die, relapse or are readmitted, but when, in studies where follow-up ends before everyone has had the event. The second is about *individual patients*: not whether an exposure is associated with an outcome on average, but how likely this particular patient is to have the outcome, so that a clinician can act on it. The first question is answered by survival analysis, with Kaplan–Meier curves, the log-rank test and the Cox model. The second is answered by prediction models, including those built with machine learning. This chapter explains both, with the emphasis on reading them critically in published papers.

In this chapter you will learn

- What censoring is, and why censored patients must be kept, not dropped
- How a Kaplan–Meier curve is built and how to read one, including the numbers at risk
- What the log-rank test does and does not tell you
- What a hazard ratio from a Cox model means, and what the proportional-hazards assumption is
- How prediction differs from explanation (causal association)
- How prediction models are judged: discrimination, calibration, internal and external validation, clinical usefulness
- How to appraise a machine-learning or "AI" prediction tool, and the role of TRIPOD+AI

14.1 The running example

The clinical thread of this book is fluid management in adults with sepsis. Paper 1 is a randomised pilot trial of a restrictive fluid strategy in septic shock (Boulet et al. 2024). Paper 2 is a propensity-matched cohort study from a national sepsis registry, which asked whether a positive cumulative fluid balance during the first three ICU days was associated with 28-day mortality; it presented its results as Kaplan–Meier curves and hazard ratios from Cox models (Hyun et al. 2023). FLUID-ICU is a simulated cohort of 620 adults with sepsis in four ICUs, generated by computer for teaching; it is not real patient data. Its primary outcome is death by day 28, with the time from ICU admission to death recorded for every patient. Chapter 16 analyses it in full.

14.2 Part 1: Time-to-event analysis

14.2.1 Why time matters

Suppose we want to compare survival after septic shock in two groups. The simplest approach is to count the proportion who died by day 28. This is perfectly valid if every patient is followed for the full 28 days. But it ignores *when* deaths happen: a strategy that delays death from day 3 to day 20 looks no different from one with no effect at all. And in many studies, not every patient is followed for the same length of time. Patients are recruited over months or years, so those recruited late are followed for less time; some move away, are transferred, or stop answering the telephone. Methods for **time-to-event data** (also called **survival analysis**, although the event need not be death) use both whether and when the event happened, and they handle unequal follow-up correctly (Clark et al. 2003).

14.2.2 Censoring

The key idea is **censoring**. A patient is censored when follow-up stops before the event has been observed. We know they were event-free up to that moment, but not what happened afterwards.

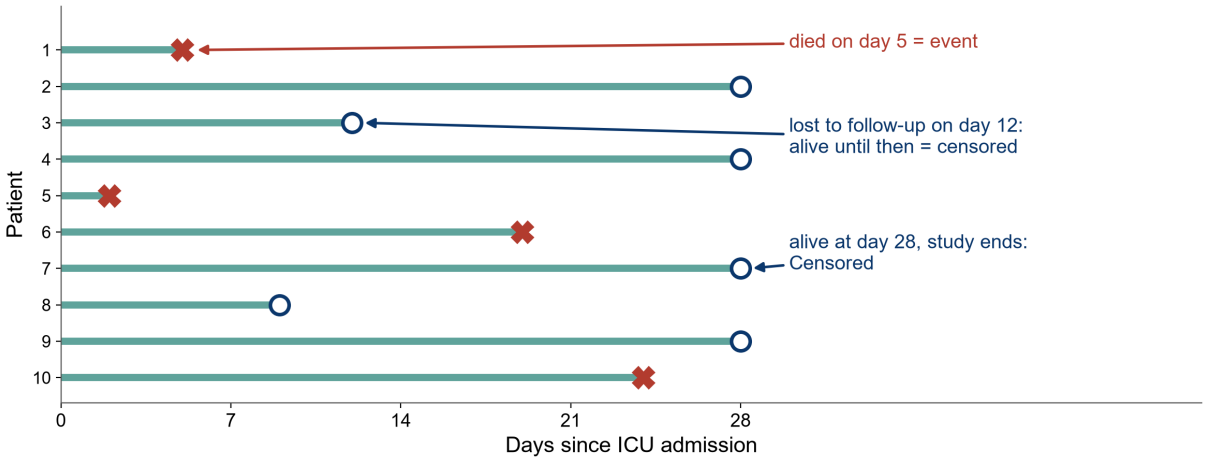


Figure 14.1: Ten patients followed from ICU admission: a red cross is a death (the event); a circle is a censored patient, either lost to follow-up (patient 3 on day 12) or alive when the study's follow-up ended at day 28.

In the figure each horizontal line is one patient, followed from ICU admission. A red cross is the event (death). A circle marks a censored patient. Patient 3 was lost to follow-up on day 12: we know they were alive for 12 days, but not afterwards. Patients 2, 4, 7 and 9 were alive at day 28, when follow-up ended by design; they are censored at day 28.

Censored does **not** mean excluded. Patient 3 tells us something useful: they survived at least 12 days. Survival methods use that information. The two tempting alternatives are both wrong. Dropping patient 3 from the analysis wastes information and, if the patients lost to follow-up differ from the others, biases the result. Counting patient 3 as a survivor to day 28 assumes something we do not know.

Survival methods rest on one important assumption: **censoring is non-informative**. The patients who are censored at a given time must have the same future risk as those who remain under observation. Being lost to follow-up because one moved house is usually harmless. Being "lost"

because one was transferred to another hospital for palliative care is not: those patients were about to die, and treating them as censored makes survival look better than it was. The logic is exactly the same as for missing data (Chapter 15), and the remedy is the same: minimise loss to follow-up by design, record why it happens, and check how sensitive the results are to it.

Worked example: event or censored?

A patient is discharged home from the ward on day 9 and cannot be contacted again before day 28. They are **censored at day 9**: alive on day 9, status unknown afterwards. A patient who dies on day 15 has the event at day 15. A patient seen alive at the day-28 follow-up call is censored at day 28. In FLUID-ICU, every survivor was followed to day 28 and is censored there; the variable `time_to_event_28d` holds the day of death or 28, and `died_28d` says which.

14.2.3 The Kaplan–Meier curve

The **Kaplan–Meier (KM) method** estimates the proportion of patients still event-free at each point in time, using censored patients correctly (Kaplan and Meier 1958; Bland and Altman 1998). Its logic is simple. Survival to any time is built up as a product of the probabilities of surviving each successive event time, given survival up to it. At each time when a death occurs, the probability of surviving that moment is

$$1 - \frac{\text{number of deaths at that time}}{\text{number at risk just before that time}}$$

and the KM estimate of survival is the running product of these probabilities. The **number at risk** is the number of patients still being followed and still alive just before that time. A censored patient contributes to the number at risk up to the moment they are censored, and then leaves the risk set without causing a step in the curve.

Worked example: a Kaplan–Meier estimate by hand

Take ten patients like those in the censoring figure: deaths on days 2, 5, 19 and 24; censored on days 9 and 12 (lost to follow-up); the remaining four censored alive at day 28.

Table 14.1: Kaplan–Meier calculation for ten patients with two censored before day 28.

Day	At risk	Deaths	Probability of surviving that day	Cumulative survival
2	10	1	9/10 = 0.900	0.900
5	9	1	8/9 = 0.889	0.900 × 0.889 = 0.800
9	8	0 (1 censored)	1	0.800
12	7	0 (1 censored)	1	0.800
19	6	1	5/6 = 0.833	0.800 × 0.833 = 0.667
24	5	1	4/5 = 0.800	0.667 × 0.800 = 0.533

The estimated survival at day 28 is 53%. Compare the two naive approaches. Counting the censored patients as survivors gives 4 deaths in 10, or 60% survival: too optimistic, because two patients' fates after days 9 and 12 are unknown. Dropping them gives 4 deaths in 8, or 50%: too pessimistic, because it ignores that they survived at least 9 and 12 days. The Kaplan–Meier estimate uses exactly the information available.

14.2.4 How to read a Kaplan–Meier curve

A Kaplan–Meier figure plots the cumulative survival (from 100% downwards) against time, as a staircase, usually with one curve per group. Four things need to be read on every such figure.

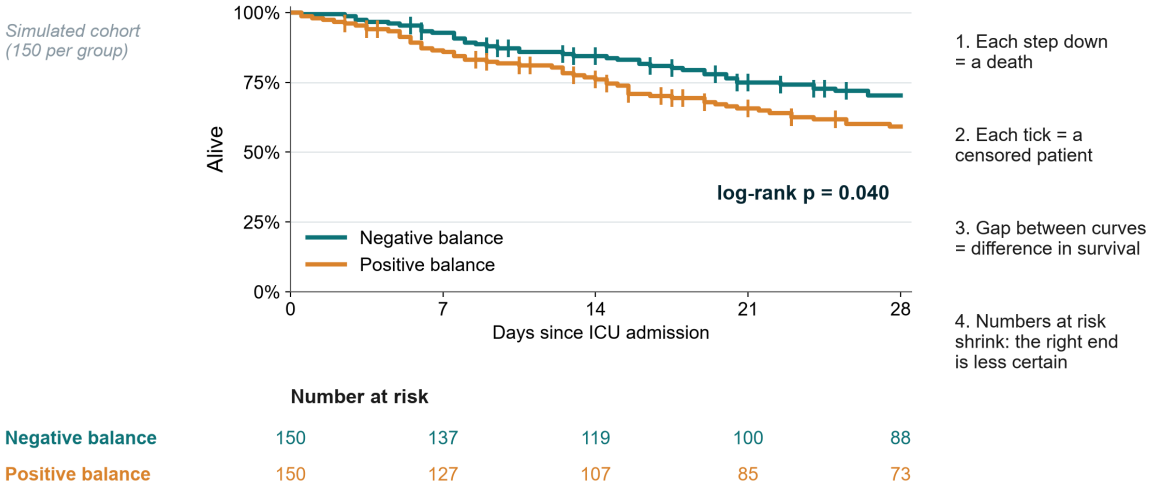


Figure 14.2: A simulated Kaplan–Meier figure with 150 patients per group: each step down is a death, each small vertical tick a censored patient, the gap between the curves is the difference in survival (about 70% versus 59% at day 28, log-rank p = 0.040), and the numbers at risk below the plot shrink over time.

1. **Each step down is one or more events.** Steep sections show periods when many patients die.
2. **Each small vertical tick is a censored patient.** Many ticks mean much censoring, which should prompt questions about who was lost and why.
3. **The vertical gap between the curves** at a given time is the difference in survival at that time. In this simulated cohort, about 70% of patients with a negative balance and 59% with a positive balance were alive at day 28.
4. **The numbers at risk** below the plot show how many patients were still being followed at each time point. As they shrink, the right-hand end of each curve rests on fewer patients and becomes less certain. A long flat tail supported by a handful of patients is a common source of overinterpretation.

A Kaplan–Meier figure without numbers at risk hides the last of these, and reporting guidelines and reviewers expect them. Good figures also show confidence bands around each curve, or at least the confidence interval for survival at the key time point.

The KM curve can also give the **median survival time**, the time at which the curve crosses 50%. If more than half the patients are still alive at the end of follow-up, as in most ICU studies with 28-day follow-up, the median cannot be estimated, and authors report survival at a fixed time instead (for example, survival at day 28).

14.2.5 The log-rank test

To compare two or more KM curves, the most common test is the **log-rank test** (Bland and Altman 2004). At each event time it compares the number of deaths observed in each group with the number that would be expected if the groups had the same survival, and sums these comparisons over the whole follow-up. It gives a p-value for the null hypothesis that the survival curves are the same.

The log-rank test has three limitations worth knowing. It gives a p-value but **no measure of the size** of the difference. It **cannot adjust for confounders**, so in an observational study it compares the groups crudely, with all the confounding that implies. And it is most powerful when one group has a consistently higher event rate throughout follow-up; when curves cross, it can miss real differences.

14.2.6 The Cox proportional-hazards model

The **Cox model** is to survival data what logistic regression is to binary outcomes (Cox 1972). It estimates a **hazard ratio (HR)** with a 95% confidence interval, and it can include several variables, so the hazard ratio can be adjusted for confounders.

The **hazard** is the instantaneous event rate: among patients still alive at a given moment, how quickly are they dying at that moment? The Cox model assumes that the hazard in one group is a constant multiple of the hazard in the other at every point in time:

$$h(t) = h_0(t) \times e^{bx}$$

Here $h_0(t)$ is the baseline hazard, which can have any shape over time and does not need to be estimated, and e^b is the hazard ratio for a one-unit increase in x. A hazard ratio of 2 means that, at any moment during follow-up, the death rate in the exposed group is twice that in the unexposed group. The null value is 1.

A hazard ratio is not quite a risk ratio. “HR 2” does not mean “twice as many deaths by day 28”, although when events are uncommon the two are close. When events are common, the risk ratio at a fixed time is closer to 1 than the hazard ratio, because once most patients in both groups have died, a doubled rate cannot double the proportion dead.

Worked example: reading a hazard ratio from Paper 2

Paper 2 compared 28-day mortality between propensity-matched patients with a positive and a negative cumulative fluid balance (Hyun et al. 2023). For the day-2 cohort (373 matched pairs) the hazard ratio was 2.13 (95% CI 1.48 to 3.06), $p < 0.001$.

- **Comparison:** positive versus negative cumulative fluid balance by day 2, in matched patients.
- **Effect:** at any moment during the 28 days, the death rate was about twice as high with a positive balance.
- **Precision:** the data are compatible with anything from about 1.5 to about 3 times the rate.
- **Null:** the interval excludes 1.
- **Interpretation:** an association, in a cohort, after matching on recorded variables. Residual

confounding by unrecorded severity remains possible.

14.2.7 The proportional-hazards assumption

The Cox model assumes that the hazard ratio is the same throughout follow-up: **proportional hazards**. If an exposure doubles the death rate in the first week and has no effect afterwards, a single hazard ratio averages these into one misleading number. In the extreme, if one group does worse early and better late, the curves cross and a single hazard ratio near 1 can conceal two opposite effects.

The assumption can be checked in several ways. Look at the KM curves: curves that cross, or that separate early and then converge strongly, are a warning. Statistical tests based on **Schoenfeld residuals** test whether the hazard ratio changes over time (Schoenfeld 1982); a small p-value suggests it does. When the assumption fails, analysts can report hazard ratios for separate time periods, include a time interaction in the model, or use other summaries such as the difference in **restricted mean survival time** (the average event-free time up to a fixed horizon). Good papers say how they checked the assumption.

Worked example: the Cox model in FLUID-ICU

In the simulated FLUID-ICU cohort, Kaplan–Meier survival at day 28 was 81.6% with a negative fluid balance and 64.1% with a positive balance (log-rank $p < 0.001$). The curves separated in the first days and then ran roughly parallel. The Cox model gave a crude hazard ratio of 2.17 (95% CI 1.55 to 3.05) and, adjusted for the six pre-specified confounders, 1.46 (95% CI 1.00 to 2.13), $p = 0.053$. The global test of proportional hazards gave $p = 0.10$: no strong evidence against the assumption.

The adjusted logistic model for the same question gave an odds ratio of 1.62 (95% CI 1.02 to 2.59), $p = 0.044$. One p-value is just below 0.05 and the other just above, but the two analyses tell the same story: a modest, imprecise association whose lower confidence limit is close to 1. Whether p lands on one side of 0.05 or the other does not change the interpretation.

14.2.8 Other issues in time-to-event analysis

Three further issues appear often enough in published papers to be worth recognising.

Competing risks. Sometimes another event prevents the event of interest from happening. A patient who dies cannot later be readmitted or develop dialysis dependence. Treating death as ordinary censoring in an analysis of readmission assumes that the dead patients would have behaved like the living ones, which is false, and overestimates the probability of readmission. Methods for competing risks (cumulative incidence functions, and models such as the Fine–Gray model) handle this properly. When an outcome other than death is analysed in a population with high mortality, look for how death was handled. Paper 2 did exactly this for its secondary outcomes (time to discharge alive from the ICU and time to freedom from invasive ventilation), using a Fine–Gray model with death before day 28 as the competing event (Hyun et al. 2023).

Immortal time. Exposure must be defined using information available at the start of follow-up ("time zero"). If patients are classified by something that happens later (say, a positive fluid balance by day 3) but follow-up starts at admission, patients who die before day 3 have had less time to accumulate fluid, or cannot be classified at all. The time before the exposure is defined is "immortal" for the patients who are classified as exposed, and it distorts the comparison. Paper 2 restricted its analysis to patients who stayed in the ICU for at least three days, which avoids this problem at the price of excluding the earliest deaths (Hyun et al. 2023); FLUID-ICU lists it as a limitation (Chapter 16).

Clustered data. When patients are matched in pairs, or treated in the same ICU, their outcomes are not independent, and analyses should account for this. Paper 2 used a clustered Cox model on its matched pairs and checked the proportional-hazards assumption by inspecting Schoenfeld residuals (Hyun et al. 2023).

Three tools that go with every Kaplan–Meier curve

The **Kaplan–Meier curve** shows survival over time and uses censored patients correctly. The **log-rank test** asks whether the curves differ, with a p-value but no effect size and no adjustment. The **Cox model** gives a hazard ratio with a confidence interval, can adjust for confounders, and assumes the hazard ratio is constant over time.

14.3 Part 2: Prediction models and machine learning

14.3.1 Explaining versus predicting

Everything so far in this chapter and the last has been about **explanation**: is an exposure associated with an outcome, after accounting for confounders? The analyst chooses the variables to answer a causal question, and the result of interest is one coefficient, for the exposure.

Prediction is a different task. A prediction model (also called a risk score, prognostic model or clinical prediction rule) combines several characteristics of a patient to estimate their individual probability of an outcome: the risk of death in the ICU, of deterioration on the ward, of a complication after surgery (Steyerberg 2019). Familiar examples include the SOFA and APACHE II scores, early warning scores used on wards, sepsis alerts embedded in electronic records, and many tools built with machine learning.

The two tasks follow different rules. For prediction, a variable earns its place by improving predictions, whether or not it is a cause. Being admitted at night may predict death well because it marks

sicker or less well-staffed admissions; it is not a cause to be treated. Conversely, a coefficient in a prediction model should not be interpreted as a causal effect. The first question to ask of any prediction model is: **what is it for?** Who is it applied to, at what moment, and which clinical decision will the prediction change?

14.3.2 Discrimination and calibration

A prediction model is judged on two main properties, illustrated in the figure below.

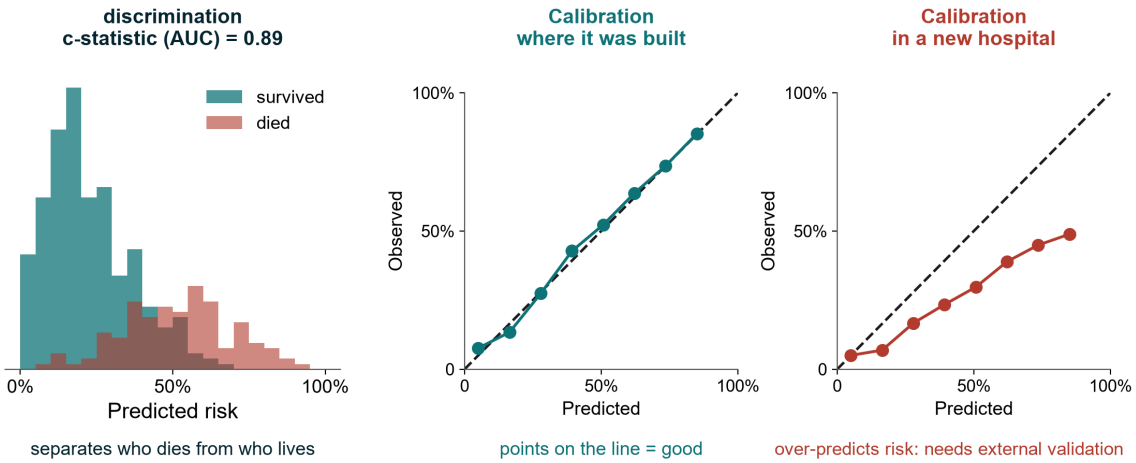


Figure 14.3: Left: discrimination; the predicted risks of patients who died are mostly higher than those of survivors (c-statistic 0.89, simulated). Middle: calibration where the model was built; predicted and observed risks lie on the diagonal. Right: calibration in a new hospital; the model over-predicts risk, so the points fall below the diagonal.

Discrimination is the ability of the model to separate patients who will have the outcome from those who will not: does it give higher predicted risks to patients who die than to those who survive? It is measured by the **c-statistic**, which for a binary outcome equals the **area under the ROC curve (AUC)** met in Chapter 8. The c-statistic is the probability that, for a randomly chosen patient who died and a randomly chosen patient who survived, the model gives the higher risk to the one who died. A value of 0.5 is no better than tossing a coin; 1.0 is perfect separation. In the left panel of the figure the two distributions of predicted risk overlap only partly, and the c-statistic is 0.89.

Calibration is the agreement between predicted and observed risks: when the model predicts a 30% risk, do about 30% of such patients have the outcome? It is shown in a **calibration plot**, with predicted risk on the horizontal axis and observed risk on the vertical axis, usually after grouping patients by their predicted risk. A well-calibrated model lies on the diagonal, as in the middle panel. Calibration is often neglected in published papers but matters greatly for clinical use: a clinician who tells a family that the risk of death is 60% needs that number to be right, not merely higher than another patient's (Van Calster et al. 2019).

The two properties are independent. A model can rank patients perfectly and still be badly calibrated, for example if every predicted risk is twice the true risk. The right panel of the figure shows exactly this: the same model applied in a new hospital still separates patients, but it **over-predicts** risk systematically; observed mortality is about half of what it predicts. Case mix, treatments, measurement practices and baseline risk all differ between settings. A high AUC alone is never enough.

14.3.3 Validation: does it work in new patients?

Any model fits its own development data better than it will fit new patients, for the reasons of overfitting described in Chapter 13. Performance must therefore be checked in patients who were not used to build the model.

- **Internal validation** estimates how much the model is overfitted using the development data themselves, by techniques such as **bootstrapping** (refitting the model in many resamples of the data) or **cross-validation** (repeatedly building the model on part of the data and testing it on the rest). Simply splitting the data once into "training" and "test" sets is common but wasteful in small datasets.
- **External validation** tests the model in a different population: another time period, another hospital, another country. It is the real test of whether a model can be used elsewhere, and it often reveals poorer discrimination and, especially, miscalibration (Steyerberg 2019).

When a model is miscalibrated in a new setting, it can sometimes be **updated** (recalibrated) using local data rather than discarded.

The number of patients needed to develop a reliable model is larger than many authors assume. The old rule of 10 events per candidate variable is often insufficient for prediction; methods now exist to calculate the required sample size from the expected outcome rate, the number of candidate predictors and the anticipated model performance (Riley et al. 2020).

Worked example: questions for a vendor

A vendor says its sepsis alert has "an AUC of 0.92 in our development data". The right questions are: How was it validated internally, and was the 0.92 corrected for overfitting? Has it been externally validated in a setting like ours, by people independent of the developers? What does its calibration look like? How many alerts will it fire per true case, at the threshold we would use? Has its use been shown to improve patient outcomes? When one widely implemented proprietary sepsis model was externally validated in hospitalised patients, its discrimination was considerably poorer than the developers had reported (c-statistic about 0.63), and it missed many patients with sepsis while generating frequent alerts (Wong et al. 2021).

14.3.4 Is the model useful?

Good discrimination and calibration show that a model predicts well. They do not show that using it helps patients. A prediction becomes useful only when it changes a decision for the better, and every decision has a threshold: above what predicted risk will we escalate care, start a treatment, or call the outreach team? At any threshold there is a trade-off between true alerts and false alarms, and false alarms have costs (alarm fatigue, unnecessary investigations). **Decision curve analysis** summarises the net benefit of using a model across a range of thresholds, compared with treating everyone or no one (Vickers and Elkin 2006). The final test is an **impact study**, ideally a randomised trial, comparing outcomes when clinicians use the model with outcomes when they do not. Very few published prediction models ever reach this stage.

14.3.5 Machine learning and "AI" prediction tools

Machine learning (ML) refers to flexible algorithms, such as random forests, gradient boosting and neural networks, that learn patterns from data with fewer assumptions than traditional regression. Many tools described as "artificial intelligence" in clinical medicine are prediction models built in this way. ML can capture complex non-linear relationships and interactions and can use unusual data such as images, waveforms or free text. But it is judged by exactly the same standards as any other prediction model, and it brings some risks of its own:

- **It needs more data.** Flexible methods overfit easily and generally need many more events than a regression model; in modest clinical datasets they often perform no better than logistic regression.
- **Validation is essential.** Check how internal validation was done, and look for external validation in a different time or place.
- **Data problems are magnified.** How were missing values handled? Do the training data reflect the population where the tool will be used? A model trained on data from one population may perform poorly, or unfairly, in groups under-represented in its training data.
- **Leakage.** If information that would not be available at the moment of prediction (for example, a treatment given *because* the patient was deteriorating) creeps into the predictors, the model looks excellent in development and fails in practice.
- **Transparency.** Are the code and model available, and can the predictions be explained to clinicians?

A systematic review of prediction models for COVID-19 published early in the pandemic found hundreds of models, almost all at high risk of bias because of small samples, overfitting, poor handling of missing data and lack of external validation, and recommended that none be used in practice (Wynants et al. 2020). The episode is a lesson in how quickly prediction models are published and how rarely they are properly validated.

14.3.6 Reporting and appraising prediction models

The **TRIPOD+AI** statement is the reporting guideline for studies that develop or validate clinical prediction models, whether they use regression or machine learning methods (Collins et al. 2024). It updates the original TRIPOD statement and asks authors to report, among other things, the source of data and setting, the outcome and predictors and when they were measured, the sample size and how it was chosen, how missing data were handled, the modelling method and how overfitting was addressed, discrimination *and* calibration with confidence intervals, internal and external validation, fairness across groups, and the availability of the model and code. It is found, like the other reporting guidelines, on the EQUATOR Network website (Chapter 19).

For appraising the risk of bias and applicability of a prediction model study, the **PROBAST** tool asks structured questions about participants, predictors, outcome and analysis (Wolff et al. 2019). Its analysis domain is where most published models fail: too few events, inappropriate handling of missing data, selection of predictors by univariable p-values, and no correction for overfitting.

An AI score is a prediction model

Ask four questions of any prediction tool, however it was built. **What is it for** (which patients, which moment, which decision)? **Was there enough data**, and was overfitting addressed by internal validation? **Does it work**: discrimination, calibration, and above all external validation in a setting like yours? **Is it reported and tested properly**: TRIPOD+AI reporting, fairness, and evidence that using it improves patient outcomes, not just accuracy?

! Common mistake: accuracy taken for benefit

"The early warning score has excellent accuracy (AUC 0.90), so it should be used on every ward." Accuracy is not benefit. Without calibration in the local setting, a chosen threshold with an acceptable alert burden, and ideally an impact study showing better patient outcomes, a score is not ready for routine use.

14.4 How these methods appear in published papers

In a paper using time-to-event methods, look for: a clear definition of time zero and of the event; how long patients were followed and how many were censored, and why; Kaplan–Meier curves with numbers at risk; the log-rank p-value if used; crude and adjusted hazard ratios with 95% confidence intervals and the variables adjusted for; how the proportional-hazards assumption was checked; and, where relevant, how competing risks and clustering were handled. STROBE asks for many of these items for observational studies (Elm et al. 2007).

In a prediction model paper, look for the elements of TRIPOD+AI: the intended use of the model, the number of events relative to candidate predictors, how predictors were selected, how missing data were handled, discrimination and calibration with confidence intervals, the type of validation, and whether the model is presented in a form that others can use and test.

Summary

- Time-to-event data record whether and when an event happened. Censored patients (follow-up ended without the event) contribute information up to their censoring time and must not be dropped; censoring should be unrelated to the outcome.
- The Kaplan–Meier curve estimates survival over time. Read the steps (events), the ticks (censoring), the gap between curves, and the numbers at risk.
- The log-rank test compares whole curves but gives no effect size and no adjustment. The Cox model gives an adjusted hazard ratio (null value 1), assuming the hazard ratio is constant over time (proportional hazards), which should be checked.
- Prediction models estimate individual risk; they are judged by discrimination (c-statistic/AUC), calibration (predicted versus observed risk), internal and especially external validation, and ultimately by whether their use improves patient outcomes.
- Machine learning tools are prediction models: they need more data, rigorous validation and transparent reporting, using TRIPOD+AI, and they can be appraised with PROBAST.

Further reading

- Clark TG, Bradburn MJ, Love SB, Altman DG. Survival analysis part I: basic concepts and first analyses (Clark et al. 2003). A clear introduction to censoring, Kaplan–Meier curves and the log-rank test, the first of a well-known series.
- Bland JM, Altman DG. Survival probabilities (the Kaplan–Meier method) and The logrank test (Bland and Altman 1998; Bland and Altman 2004). Two one-page "Statistics Notes" that explain each method with a worked example.
- Steyerberg EW. *Clinical Prediction Models* (Steyerberg 2019). The standard textbook on developing, validating and updating prediction models.
- Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement (Collins et al. 2024). The reporting guideline for prediction models using regression or machine learning; useful as a checklist for reading as well as writing.
- Van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW. Calibration: the Achilles heel of predictive analytics (Van Calster et al. 2019). Explains why calibration matters as much as discrimination.

CHAPTER 15

Planning the analysis and handling missing data

The previous chapters described the statistical tools that turn data into answers. This chapter is about two things that decide whether those answers can be trusted. The first is the **statistical analysis plan**: a document, written before anyone looks at the results, that fixes how the data will be analysed. The second is **missing data**: the gaps that every clinical dataset contains, which can quietly bias the answer if they are ignored. The two belong together, because the most important decisions about missing data, like all analytic decisions, should be made in advance and written down. A good analysis is planned before, adjusted honestly, tested for robustness and reported transparently.

In this chapter you will learn

- Why the analysis should be planned and written down before the data are analysed
- What a statistical analysis plan (SAP) contains, section by section, using the FLUID-ICU plan as a model
- Why missing data matter: who is missing matters more than how many
- The three missing-data mechanisms (MCAR, MAR and MNAR) and what each implies
- What complete-case analysis does, and when it is biased
- Why simple fixes such as mean imputation and "last value carried forward" mislead
- How multiple imputation works, when it is valid and how results are pooled
- How to plan sensitivity analyses and report missing data

15.1 The running example

This book follows one clinical problem throughout: fluid management in adults with sepsis. Paper 1 is a randomised pilot trial comparing a restrictive fluid strategy with usual care in septic shock (Boulet et al. 2024); Paper 2 is a propensity-matched cohort study of cumulative fluid balance and 28-day mortality (Hyun et al. 2023). **FLUID-ICU** is a simulated cohort of 620 adults with sepsis in four ICUs, created by computer for teaching; it describes no real patients. Its primary question is whether a positive cumulative fluid balance by day 3 is associated with 28-day mortality. It comes with a written statistical analysis plan, and its admission lactate is missing for 15% of patients, so it illustrates both halves of this chapter. Chapter 16 presents its full analysis.

15.2 Part 1: The statistical analysis plan

15.2.1 Why plan the analysis first?

Every dataset can be analysed in many reasonable ways. Which confounders to adjust for, whether to treat fluid balance as continuous or to cut it at zero, which patients to include, which outcome is primary, how to handle the patients without a lactate value, which subgroups to examine: each choice changes the answer a little, and sometimes a lot. If the choices are made after looking at the results, it becomes almost impossible to resist, consciously or not, the analysis that gives the most interesting answer or the smallest p-value. Trying many analyses and reporting the one that "works" has been given many names (data dredging, fishing, p-hacking, the garden of forking paths), but the effect is always the same: false-positive findings and exaggerated effects. Chapter 12 described why running many tests produces chance findings; an unplanned analysis is a hidden version of the same problem.

A **statistical analysis plan (SAP)** prevents this by fixing the main analytic decisions before the data are seen. In a randomised trial the SAP is finalised before the database is locked and the allocation is unblinded, and international guidance expects it (International Conference on Harmonisation 1998; Gamble et al. 2017). In an observational study the principle is the same: the plan should be written, dated and ideally signed before the analysis of outcome data begins. The SAP protects the authors as much as the readers. When reviewers ask why these six confounders and not others, the answer is "because we said so, in writing, before we saw the results".

Planning in advance does not forbid anything else. Unplanned analyses can be very informative. They must simply be **labelled** as exploratory or post hoc, so that readers can weigh them accordingly. And plans can change for good reasons: a variable turns out to be unusable, or an assumption fails. Changes are made openly, with a new version number, a date and a reason, and reported in the paper.

- **Plan, then analyse**

  The SAP records what will be done before anyone knows what the data show. Pre-specified analyses carry their full weight; analyses chosen after seeing the data are exploratory and must be labelled as such. Any change to the plan is dated, justified and reported.

Section	What it says	FLUID-ICU (simulated)

15.2.2 What a SAP contains

Published guidance on the content of SAPs for clinical trials lists the items a plan should contain (Gamble et al. 2017), and the same structure works well for observational studies. The FLUID-ICU plan follows fourteen sections, which make a convenient checklist.

Table 15.1: The fourteen sections of a statistical analysis plan, with the simulated FLUID-ICU plan as an example.

Section	What it says	FLUID-ICU (simulated)
1. Study summary and objectives	Design, setting, primary and secondary objectives	Multicentre prospective cohort, 4 ICUs, 2 years; primary objective: association of positive balance at day 3 with 28-day mortality
2. Outcomes	Each outcome, its type, definition and time point; the exposure	28-day mortality (binary), time to death, AKI, ventilator-free days, ICU stay in survivors
3. Analysis populations	Who is included in which analysis	Enrolled; exposure population; complete cases; survivors
4. Descriptive analysis	Table 1 and its summaries	By fluid-balance group; mean (SD) or median (IQR); no p-values
5. Primary analysis	Method, effect measure, adjustment, significance level	Multivariable logistic regression; OR with 95% CI; crude OR and risk difference also reported
6. Secondary analyses	One row per secondary objective	Logistic, linear and log-linear models; Kaplan–Meier and Cox as supportive
7. Confounders	Which, why, and what is not adjusted for	Age, SOFA, lactate, vasopressor, ventilation, CKD; AKI not adjusted (mediator)
8. Subgroups and interaction	Pre-specified effect modifiers and expected direction	Heart failure only
9. Multiple comparisons	How multiplicity is handled	One primary outcome; the rest exploratory
10. Missing data	Expected missingness, main handling, sensitivity analysis	Complete case for the main models; multiple imputation (m = 20) as sensitivity, and as main analysis where lactate is the exposure
11. Sample size	The calculation or the justification	About 600 patients; about 80% power for an OR of 1.7
12. Software	Programs and versions	R, with named packages; scripted and reproducible
13. Table and figure shells	Empty tables showing exactly what will be reported	Table 1, Table 2, Kaplan–Meier, forest plots
14. Version history and sign-off	Dates, changes and reasons, signatures	Version 1.0 signed before database lock

A few of these sections deserve a closer look, because they are where plans most often fall short.

Outcomes and exposure. Each outcome is defined precisely: what is measured, how, when, and what type of variable it is (Chapter 11), because the type determines the analysis (Chapters 12 and 13). FLUID-ICU defines its exposure as the cumulative fluid balance over ICU days 1 to 3 divided by admission weight, analysed as positive (above 0 ml/kg) versus negative for the primary question and as a continuous variable (per 10 ml/kg) for others. Writing this down in advance prevents the temptation to try several cut-points.

Analysis populations. Who is in each analysis? In a trial, the plan defines the intention-to-treat and per-protocol populations (Chapter 5). In FLUID-ICU there are four: all 620 enrolled patients; the 618 with a computable fluid balance; the complete cases used in the adjusted models; and the 28-day survivors, used for the length-of-stay question. A flow diagram reports the number at each step.

Primary analysis. The single most important section. It names the outcome, the population, the model, the effect measure (with its confidence interval) and the adjustment, and states the significance level. The FLUID-ICU plan adds a sentence worth copying: "The estimate and its CI are the result; the p-value supports it."

Confounders. Chosen in advance from clinical knowledge and the literature, ideally with a simple causal diagram (Chapter 6), within the events-per-variable budget (Chapter 13). The plan also says what will *not* be adjusted for and why. FLUID-ICU names age, four markers of severity (SOFA score, lactate, vasopressor use, mechanical ventilation) and chronic kidney disease; it excludes APACHE II because it overlaps with SOFA and age, and excludes AKI and renal replacement therapy because they occur after the exposure and may be mediators. With about 160 to 180 deaths and seven model terms, there are more than 20 events per term.

Subgroups. A small number of pre-specified effect modifiers, each with a reason and an expected direction, tested with an interaction term (Chapter 13). FLUID-ICU names one, heart failure, and states in advance that "a p for interaction below 0.05 will be described as evidence of effect modification needing confirmation, not as a definite finding."

Multiple comparisons. Which result is primary, and how the rest will be interpreted. FLUID-ICU tests one primary outcome at the 5% level and reports the secondary outcomes and the subgroup analysis with confidence intervals as exploratory, without formal adjustment.

Missing data. Discussed in Part 2 of this chapter. The plan states which variables are expected to be missing and why, how missingness will be assessed, the main approach, and the sensitivity analysis.

Sample size. For a trial, the calculation with its four inputs (Chapter 12). For a cohort that enrols every eligible patient, a statement of the expected numbers and the precision or power they give. FLUID-ICU expected about 600 patients and calculated about 80% power to detect an odds ratio of 1.7 (20% versus 30% mortality) but only about 55% power for an odds ratio of 1.5: honest information about what the study can and cannot detect.

Table shells. Empty versions of the tables and figures that will be reported, with row and column headings but no numbers. They force decisions about exactly what will be shown and make it obvious afterwards if something was dropped or added.

Worked example: a plain-language analysis plan for a trial

A team plans a trial of a restrictive fluid strategy against usual care in septic shock, with 28-day mortality as the primary outcome. A plan that would satisfy a reviewer, in plain language:

Sample size. “Assuming 28-day mortality of 40% with usual care and 30% with the restrictive strategy, 356 patients per group give 80% power at a two-sided alpha of 5%; allowing 10% loss to follow-up, we will recruit 396 per group.” (356 divided by 0.9 is 395.6, rounded up.)
Analysis. “Baseline characteristics will be described with mean (SD), median (IQR) or n (%), as appropriate. The primary outcome will be compared between groups in the intention-to-treat population with a chi-square test, and reported as a risk ratio and risk difference with 95% confidence intervals.”
Missing data. “We expect 28-day status to be missing for fewer than 5% of patients, mainly those transferred to other hospitals; research staff will contact the receiving hospital or the patient’s family. The primary analysis will use patients with known status; sensitivity analyses will assume that all patients with missing status died, and that all survived.”
A plan that says only “data will be analysed with statistical software; a p-value below 0.05 will be considered significant” says nothing about any of the decisions that matter.

15.2.3 Where the plan lives

For a trial, the SAP is usually a separate document, referenced in the protocol (SPIRIT, Chapter 17) and often published or posted with the trial registration. For an observational study it may be part of the protocol, a separate document, or a registered analysis plan. Whatever its form, it should be dated, version-controlled and available to reviewers. When the paper is written, the methods section summarises it, and any deviations are reported (Chapter 19). The FLUID-ICU plan is a model of the form: fourteen short sections, with tables rather than paragraphs where possible, signed before the database was locked.

15.3 Part 2: Missing data

15.3.1 Why missing data matter

Every clinical dataset has gaps. A test was not ordered, a form was left blank, a patient was transferred and never contacted, a sample was lost, an impossible value was set to missing during cleaning (Chapter 10). Most statistical software deals with gaps silently, by dropping every patient with a missing value in any variable used in the analysis. Many researchers think of this as a minor loss of precision. It can be much worse than that.

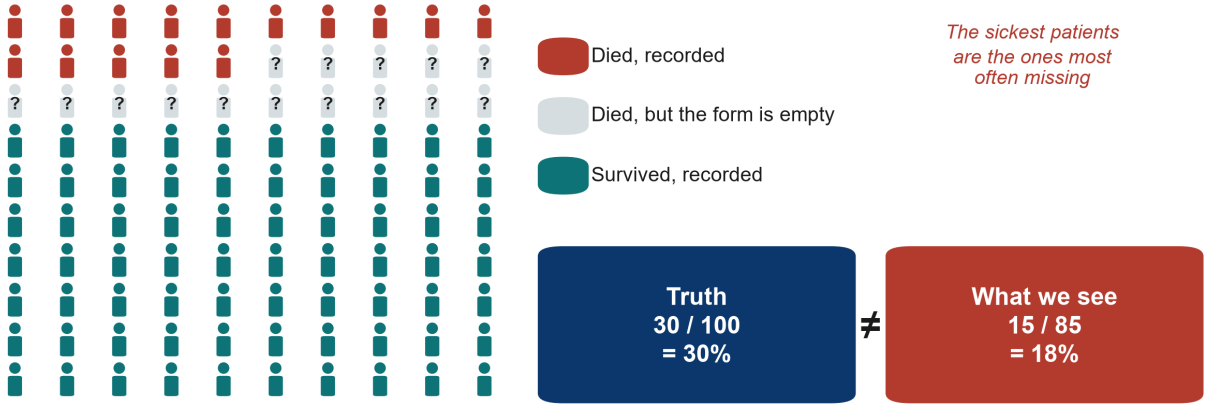


Figure 15.1: One hundred patients, of whom 30 truly died; 15 of the deaths have an empty outcome form, so the 85 recorded patients show 15 deaths (18%) when the truth is 30%.

The figure shows 100 patients, of whom 30 died. But 15 of the deaths have an empty outcome box: the patients were transferred, died at night, or the form was never completed. Among the 85 recorded patients there are 15 deaths, a mortality of 18%. The truth was 30%. Only 15% of the data are missing, yet the estimate is wrong by almost half, because missingness is linked to the outcome itself: the sickest, most chaotic patients are the ones whose forms stay empty.

The lesson is that **who is missing matters more than how many**. Ten per cent missing can seriously bias a result if the missing patients are different; thirty per cent missing may do little harm if they are a random subset. The questions to ask are always why the data are missing and whether the patients with missing values differ from the others.

15.3.2 Prevention first

Statistical methods for missing data are always second best to not having missing data. Most gaps can be prevented by design and by habit (Chapter 10 discusses data collection in detail):

- **Collect fewer items.** Every extra item on a form increases the chance that something is left blank. Collect what the analysis plan needs.
- **Design clear forms.** Mark required fields, and provide “not done” and “unknown” options, so that a blank always means “forgotten” rather than “not measured”.
- **Train every data collector** with the data dictionary in hand, so that everyone uses the same definitions.
- **Check completeness regularly.** A weekly count of empty fields per variable and per site shows problems while they can still be fixed. If one item is always empty, the item is the problem.

- **Chase early, especially the outcome.** The primary outcome at the end of follow-up is the most valuable item and often the most frequently missing. Plan in advance who will check it, and how, for patients who are transferred or discharged.
- **Record why data are missing.** Knowing the reason is what later allows the right analysis to be chosen.

15.3.3 Three mechanisms: MCAR, MAR and MNAR

Statisticians classify missing data by the mechanism that produced the gaps (Little and Rubin 2002). The three mechanisms have awkward names, but the ideas are simple, and they decide how dangerous the gaps are and what can be done about them.

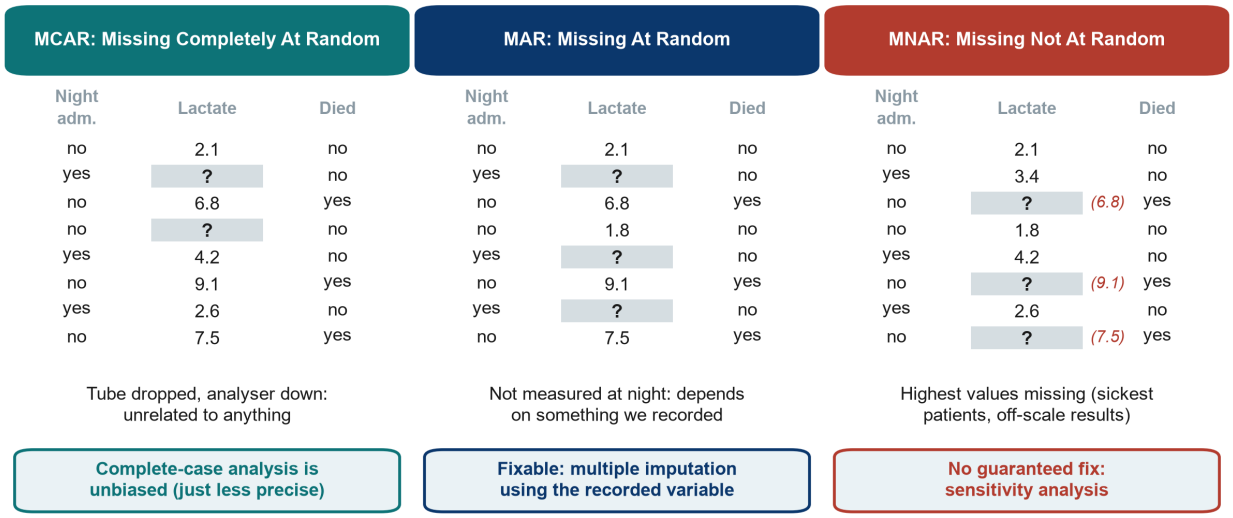


Figure 15.2: The same eight patients with some lactate values missing under three mechanisms: missing completely at random (a tube dropped), missing at random (not measured at night, which is recorded), and missing not at random (the highest values are the ones missing).

Missing completely at random (MCAR). The probability of a value being missing has nothing to do with the patient: a sample tube was dropped, an analyser broke down, a courier box was lost. The patients with complete data are a random subset of all patients. An analysis restricted to them is **unbiased**, just less precise. True MCAR is rare in clinical data.

Missing at random (MAR). The probability of being missing depends on other variables that *were recorded*, but not on the missing value itself once those variables are taken into account. In the figure, lactate was not measured for night admissions, and the time of admission is in the dataset. The patients with missing lactate may differ from the others (night admissions may be sicker), so a complete-case analysis can be biased, but because the reason for missingness is recorded, methods such as multiple imputation can use it to correct the bias. The name is confusing, because MAR data are not missing "at random" in the everyday sense; "missing at random given what we recorded" is a better description.

Missing not at random (MNAR). The probability of being missing depends on the unseen value itself, even after accounting for everything recorded. In the figure the missing lactates are the highest ones: the sickest patients, or values off the analyser's scale. No statistical method can fully correct this from the data alone, because the information needed to do so is precisely what is missing. The best we can do is test how much it might matter, with sensitivity analyses.

One point matters above all: **the data alone cannot prove which mechanism applies.** We can show that data are *not* MCAR, by finding that patients with and without a value differ on recorded characteristics. But we can never prove from the data that they are MAR rather than MNAR, because that would require knowing the values we do not have. Clinical knowledge of why values are missing is essential, which is why the reasons should be recorded at the time.

Worked example: naming the mechanism

For each scenario, the most likely mechanism and its consequence for a complete-case analysis:

Table 15.2: Missing-data mechanisms in six scenarios.

Scenario	Mechanism	Consequence
Twelve lactate samples lost when a courier box was dropped	MCAR	Unbiased, less precise
Weight missing more often for night admissions; admission time recorded for all	MAR	Possibly biased; multiple imputation using admission time can correct it
Pain scores missing for the patients in most pain, who could not complete the questionnaire	MNAR	Biased (pain looks better than it is); sensitivity analysis
Day-28 status missing for patients transferred because they were deteriorating	MNAR (at best MAR if the reason for transfer is recorded)	Mortality underestimated; chase the outcome; sensitivity analysis
An early form version omitted the smoking question for one month	MCAR, unless early patients differed	Usually unbiased; check whether early patients differed
Older patients less likely to return a 3-month quality-of-life questionnaire; age recorded	MAR	Biased towards younger patients; imputation including age

15.3.4 Complete-case analysis

Complete-case analysis (also called listwise deletion) uses only the patients with no missing values in any of the variables in the analysis. It is simple, transparent and the default in most statistical software, which is why it is so common, often without the authors realising they have chosen it.

Its problems are two. First, it **loses patients**, and the losses add up across variables: a model with ten variables, each missing in 3% of patients, may lose a quarter of the sample. Second, it is **biased unless the complete cases are representative**, which is guaranteed only under MCAR. (There are some exceptions: for example, in a regression model, complete-case analysis is unbiased when missingness in a covariate does not depend on the outcome. But these are technical conditions that are difficult to verify.)

Complete-case analysis is a reasonable main analysis when the proportion of incomplete cases is small (a few per cent), when the patients with missing data do not differ much from the others, or as one of the analyses compared in a sensitivity analysis. It should never be chosen by default, and the number of patients it excludes should always be reported.

Worked example: who is missing lactate in FLUID-ICU?

In the simulated FLUID-ICU cohort, admission lactate is missing for 93 of 620 patients (15.0%). Other variables are missing for 1% of patients or fewer. The patients without lactate were clearly different: their median SOFA score was 5 (IQR 4 to 6), against 8 (6 to 10) in patients with lactate, fewer were on vasopressors (15% against 48%), more came from one ICU without a point-of-care analyser, and fewer died (18 of 93, 19.4%, against 161 of 527, 30.6%). Lactate was missing mostly in less sick patients, presumably because clinicians judged it unnecessary.

So the data are **not MCAR**: a complete-case analysis drops mostly the less sick. The missingness is explained by recorded variables (severity, blood pressure, vasopressor use, ICU), which makes **MAR plausible**. That is the situation in which multiple imputation works well. Chapter 16 shows the comparison.

15.3.5 Simple fixes that mislead

Several quick ways of "filling in" missing values are still common, and all of them are problematic (Sterne et al. 2009).

- **Mean (or median) imputation** replaces each missing value with the average of the observed values. It keeps every patient but pretends we know the missing values exactly, so it understates variability, makes standard errors and confidence intervals too narrow, weakens associations, and is biased whenever the data are not MCAR. In FLUID-ICU, the patients missing lactate were less sick and probably had lower lactate than average; giving them the average would be wrong.
- **Last observation carried forward (LOCF)**, common in trials with repeated measurements, assumes that a patient who dropped out would have stayed exactly as they were at their last visit. That is rarely plausible, and it can bias results in either direction.
- **The missing-indicator method** adds a category "missing" to a variable, or an indicator variable for missingness, and keeps everyone in the model. It is generally biased when used for confounders in observational studies.
- **Single regression imputation** predicts each missing value from the other variables and then treats the predictions as if they were real. It is better than the mean, but it still ignores the uncertainty about the missing values, so the confidence intervals are too narrow.

The common flaw is that each method treats a guess as a measurement. The method described next keeps the guesses but is honest about how uncertain they are.

15.3.6 Multiple imputation

Multiple imputation (MI), developed by Rubin, is now the standard approach when data are plausibly MAR (Rubin 1987; Sterne et al. 2009). Its idea fits in three steps, shown in the figure below.

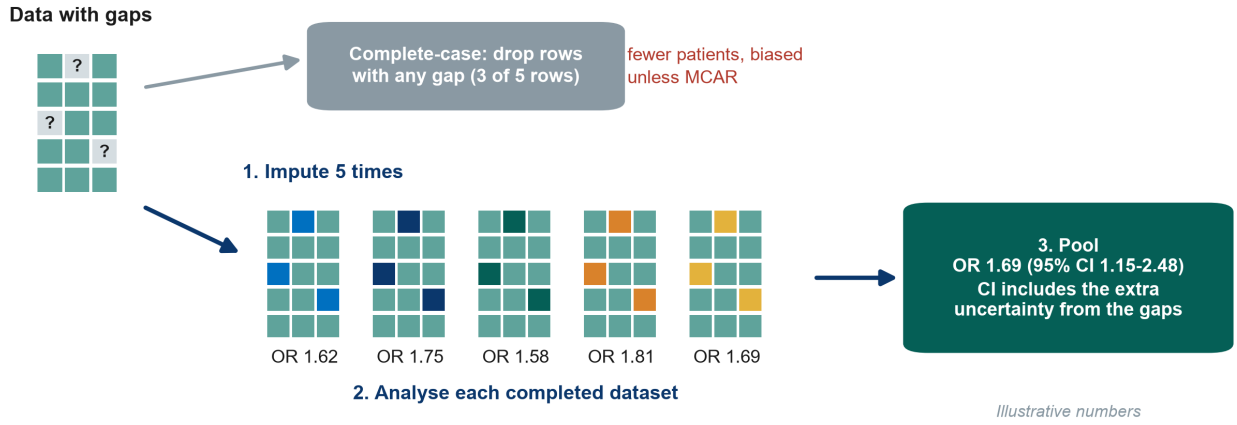


Figure 15.3: Complete-case analysis drops every row with a gap. Multiple imputation instead (1) fills each gap several times with plausible values, (2) analyses each completed dataset, and (3) pools the results into one estimate whose confidence interval includes the extra uncertainty from the gaps (illustrative numbers).

1. **Impute.** Each missing value is filled in several times, creating several complete datasets. Each filled-in value is drawn from a prediction based on the patient's other recorded variables, with a random element added so that the imputed values vary from one dataset to the next as much as our uncertainty about them warrants.
2. **Analyse.** Each completed dataset is analysed in the usual way, with the same model. Each gives a slightly different estimate (in the figure, odds ratios from 1.58 to 1.81).
3. **Pool.** The results are combined using **Rubin's rules**. The pooled estimate is the average of the estimates. Its variance combines the average variance within each dataset (W, the ordinary sampling uncertainty) and the variance between the datasets (B, the extra uncertainty because we do not know the missing values):

$$\text{total variance} = W + \left(1 + \frac{1}{m}\right) B$$

where m is the number of imputed datasets. The more the imputed datasets disagree, the wider the pooled confidence interval. This is what makes multiple imputation honest: it uses all the patients, but it does not pretend to know what it does not know.

In practice, most software uses **multiple imputation by chained equations (MICE)**, also called fully conditional specification, which imputes each incomplete variable in turn from all the others, with a model suited to its type: for example, predictive mean matching for continuous variables and logistic regression for binary ones (White et al. 2011; Buuren 2018).

Multiple imputation is not "making up data". It is valid under conditions that should be checked and reported:

- **The data should be plausibly MAR.** MI uses the recorded variables to predict the missing ones; if missingness depends on the unseen values themselves (MNAR), it will not remove the bias.
- **The imputation model must include the outcome.** This surprises many people, but leaving the outcome out of the imputation model weakens the association between the imputed variable and the outcome and biases it towards zero.
- **The imputation model should include every variable in the analysis model**, plus variables that predict missingness or the missing values ("auxiliary variables"), even if they are not in the analysis model.
- **Enough imputations.** Older advice of 5 imputations is often too few. A common rule of thumb is that the number of imputations should be at least the percentage of incomplete cases: 20 or more is typical (White et al. 2011).
- **Imputed values should be plausible.** Compare the distribution of imputed and observed values; imputed values should differ only in ways that make sense.

Worked example: the FLUID-ICU imputation

The FLUID-ICU plan specifies multiple imputation by chained equations with 20 imputed datasets, predictive mean matching for continuous variables and logistic regression for binary ones. The imputation model includes the outcome (death by day 28), all covariates of the analysis models, and the predictors of missingness: SOFA, APACHE II, mean arterial pressure, vasopressor use, ventilation, ICU, source of infection and fluid-balance group. Results are pooled with Rubin's rules.

For the question of lactate and 28-day death, adjusted for age, sex, diabetes, CKD and heart failure, the odds ratio per 1 mmol/L was 1.39 (95% CI 1.24 to 1.57) in the 527 complete cases and 1.45 (1.29 to 1.64) with multiple imputation in all 620 patients. The median lactate was 2.6 mmol/L in the complete cases and 2.4 mmol/L after imputation: the imputed values were lower, as expected, because the patients missing lactate were less sick. The conclusion is the same with both methods; the size of the association changes modestly. These data are simulated.

15.3.7 Sensitivity analyses

Because the mechanism can never be proven, every important analysis with missing data should be accompanied by at least one **sensitivity analysis**: the same question answered under different, reasonable assumptions. If the conclusion holds across them, it is robust; if it changes, readers need to know.

- **Complete case versus multiple imputation.** The commonest pair. Similar results are reassuring; different results show that the handling of missing data matters and must be discussed.
- **Best case and worst case** for a missing binary outcome: repeat the analysis assuming that all patients with missing outcome survived, and again assuming that all died (or, in a trial, the most and least favourable outcome for each arm). If the conclusion survives both extremes, missing outcomes cannot explain it.
- **Departures from MAR** ("delta adjustment" or **tipping-point analysis**): impute under MAR, then shift the imputed values systematically (for example, assume missing patients had lactate

1 mmol/L higher than predicted, or a 10% higher risk of death) and find how large the shift must be to change the conclusion. If only an implausibly large shift changes it, the result is robust to MNAR of plausible size.

Worked example: a robust primary result

The FLUID-ICU primary model (positive fluid balance and 28-day death, adjusted for six confounders) was fitted in the 527 complete cases as the main analysis: OR 1.62 (95% CI 1.02 to 2.59). With multiple imputation in all 620 patients, as a pre-specified sensitivity analysis, the OR was 1.70 (1.09 to 2.64). The conclusion does not depend on how the missing lactate values were handled.

15.3.8 Reporting missing data

Missing data are a result, and they are reported like one. The STROBE statement for observational studies asks authors to explain how missing data were addressed (item 12c) and to report the number of participants with missing data for each variable of interest (item 14b) (Elm et al. 2007); CONSORT asks for the same in trials, with a flow diagram of losses and exclusions. Guidance on reporting analyses that use multiple imputation adds the details of the imputation model (Sterne et al. 2009). A good paper tells the reader:

- **How much** is missing for each variable, in Table 1 or a footnote, and for the outcome, a flow diagram showing who was lost and why.
- **Who** is missing: a comparison of patients with and without complete data on baseline characteristics and outcomes.
- **How** missing data were handled in the main analysis, and why that approach was chosen.
- **For multiple imputation**: the number of imputations, the variables in the imputation model, the methods used, and the software.
- **Sensitivity analyses** and whether they changed the conclusion.
- **That all of this was planned**, in the SAP.

Worked example: a methods sentence on missing data

"Admission lactate was missing for 93 of 620 patients (15.0%), more often in less severely ill patients. The primary analysis used complete cases (n = 527). As a pre-specified sensitivity analysis, missing values were imputed by multiple imputation by chained equations (20 datasets), with an imputation model including the outcome, all covariates and predictors of missingness, and estimates were combined using Rubin's rules. Results were similar (Supplementary Table S1)."

! Common mistake: 'patients with missing data were excluded'

A paper excludes 30% of patients because of missing data, analyses the rest, and says no more. A reader should ask: who was excluded, were they different from those analysed, and was there a sensitivity analysis using imputation or best- and worst-case assumptions? "Missing data were excluded" is not a missing-data plan.

Missing data: report, handle, test

Prevent what you can. Report how much is missing and who is missing. Think about why it is missing (MCAR, MAR, MNAR), using clinical knowledge. Choose the main handling in advance: complete case when missingness is small and plausibly harmless, multiple imputation when MAR is plausible. Test the answer with at least one sensitivity analysis.

Summary

- A statistical analysis plan written before the analysis fixes the outcome definitions, populations, primary and secondary analyses, confounders, subgroups, multiplicity, missing data, sample size and table shells. It protects against data-driven analysis; changes are dated, justified and reported.
- Missing data can bias results, not just reduce precision. Who is missing matters more than how many. Prevention is the best remedy.
- MCAR: missingness unrelated to anything; complete-case analysis is unbiased. MAR: missingness explained by recorded variables; multiple imputation can correct the bias. MNAR: missingness depends on the unseen values; no method fully fixes it, so test it with sensitivity analyses. The data cannot prove MAR.
- Mean imputation, last observation carried forward and missing indicators give falsely precise and often biased results.
- Multiple imputation fills each gap several times, analyses each dataset and pools the results with Rubin's rules, so that the confidence interval reflects the uncertainty from the missing values. The imputation model must include the outcome and the variables that predict missingness.
- Report the amount and pattern of missing data, the handling method and a sensitivity analysis.

Further reading

- Gamble C, Krishan A, Stocken D, et al. Guidelines for the content of statistical analysis plans in clinical trials (Gamble et al. 2017). The consensus list of what a SAP should contain; easily adapted to observational studies.
- Sterne JAC, White IR, Carlin JB, et al. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls (Sterne et al. 2009). The best short introduction to multiple imputation for clinicians, with reporting recommendations.
- White IR, Royston P, Wood AM. Multiple imputation using chained equations: issues and guidance for practice (White et al. 2011). Practical advice on building imputation models and choosing the number of imputations.
- van Buuren S. *Flexible Imputation of Missing Data* (Buuren 2018). A thorough and readable book by the author of the most widely used imputation software, freely available online.
- Little RJA, Rubin DB. *Statistical Analysis with Missing Data* (Little and Rubin 2002). The classic reference on missing-data mechanisms and methods, for readers who want the theory.

CHAPTER 16

A worked case study: the FLUID-ICU cohort

The chapters of Part III introduced, one at a time, the steps that take a study from raw data to an answer: collecting and cleaning data (Chapter 10), describing them (Chapter 11), estimating with confidence intervals and testing (Chapter 12), modelling (Chapters 13 and 14), and planning the analysis and handling missing data (Chapter 15). In a real study these steps are not separate. Each decision depends on the ones before it, and mistakes made early cannot be repaired later. This chapter follows a single study, the FLUID-ICU cohort, from the files the participating ICUs sent to the results that would go into a manuscript. The primary question is analysed in depth; four secondary questions are analysed more briefly, each chosen because it teaches a different lesson. The chapter ends with how the results would be reported.

FLUID-ICU is simulated

FLUID-ICU is an invented study. Every value was produced by a computer program with a fixed random seed. It describes no real patient, hospital, ICU or country, and its results say nothing about whether fluid harms real patients. The data were built so that the classic lessons of analysis appear in them: confounding by indication, missing data that are not missing at random, skewed outcomes and an interaction worth discussing. Clinical ranges were chosen to resemble those in published studies of fluids in sepsis (Boulet et al. 2024; Hyun et al. 2023). Any write-up of these data must say clearly that they are simulated. All numbers in this chapter come from the study's analysis scripts and results summaries.

In this chapter you will learn

- How a realistic dataset moves from raw files through cleaning to an analysis-ready file, with an audit trail
- How the statistical analysis plan shapes every analysis that follows
- How to read Table 1 for warning signs of confounding
- How confounding by indication inflates a crude odds ratio, and what adjustment does to it
- How Kaplan–Meier curves, the log-rank test and a Cox model support the primary analysis
- How to present and interpret a pre-specified interaction
- How complete-case analysis and multiple imputation compare when data are not missing completely at random
- How to analyse binary, continuous, bounded and skewed outcomes, and how to report all of this

16.1 The study

FLUID-ICU is a simulated multicentre prospective cohort study. Consecutive adults (18 years or older) admitted with sepsis to four ICUs, called ICU A to ICU D, were enrolled over two years, from January 2023 to December 2024. Patients received usual care; the study only recorded data. In total 620 patients were enrolled: 191 in ICU A, 164 in ICU B, 147 in ICU C and 118 in ICU D.

The **exposure** is the cumulative fluid balance at day 3: all fluid given over ICU days 1 to 3, minus all fluid output over the same days, divided by admission weight, in ml/kg. A balance above 0 ml/kg is called a positive balance.

The **primary question** is: *in adults with sepsis in the ICU, is a positive cumulative fluid balance by day 3 associated with higher 28-day mortality?* Written in PECO form (Chapter 2): the population is adults with sepsis in the ICU; the exposure is a positive cumulative balance by day 3; the comparator is a negative or zero balance; the outcome is death from any cause by day 28 (a binary outcome), with time to death also recorded.

Four **secondary questions** use the same data:

Table 16.1: The five FLUID-ICU research questions.

Question	Exposure	Outcome and type	Main method
Q1 (primary)	Positive vs negative balance	Death by day 28 (binary; also time to death)	Logistic regression, crude and adjusted; Kaplan–Meier, Cox; interaction with heart failure
Q2	Balance per 10 ml/kg	Acute kidney injury (binary)	Logistic regression
Q3	Balance per 10 ml/kg	Ventilator-free days to day 28 (count 0–28, piled at both ends)	Mann–Whitney; linear regression
Q4	Admission lactate per 1 mmol/L	Death by day 28 (binary); effect of missing lactate	Complete case vs multiple imputation
Q5	Positive vs negative balance	ICU length of stay in survivors (continuous, right-skewed)	Median (IQR); Mann–Whitney; regression on the log scale

FLUID-ICU asks the observational version of the question that Paper 1 asked experimentally. Paper 1 randomised patients to a restrictive fluid strategy (Boulet et al. 2024); Paper 2, like FLUID-ICU, compared patients who happened to accumulate more or less fluid (Hyun et al. 2023). A cohort can include many more patients treated in everyday conditions, but it can show only association, never causation, because the patients who accumulate fluid differ from those who do not. That difference is the central theme of this chapter.

16.2 From raw data to an analysis file

16.2.1 What the ICUs sent

The raw file, as received from the four ICUs, had 623 rows and 29 columns, and it contained the kinds of problems found in almost every real multicentre dataset. The cleaning was done entirely by a script; the raw file was never edited by hand; and every change was recorded in a cleaning log (Chapter 10). The main problems and what was done about them are summarised below.

Table 16.2: Main problems found in the raw FLUID-ICU file and how they were handled (cleaning log).

Problem found	How many	What was done
Column names typed like a spreadsheet (`Weight (kg)`, `MAP (mmHg)`)	29 columns	Converted to short lower-case names
The same patient entered twice	3 patients (623 rows for 620 patients)	Checked the copies were identical, kept one
ICU name typed in several ways (`icu a`, `ICU-A`, `ICUA`)	14 rows	Standardised to ICU A to ICU D
Dates in three formats (`2023-03-14`, `14/03/2023`, `14-Mar-2023`)	365 / 164 / 94 rows	Each format read explicitly and converted to one
Sex coded nine ways; yes/no variables coded as Yes, Y, yes, 1 and so on	200 rows; about 1,900 cells	Recoded to one coding per variable
Missing-value codes used as numbers (lactate 999 and −99, APACHE II −99, MAP 999)	21 + 6 + 4 values	Set to missing
One ICU reported lactate in mg/dL instead of mmol/L	126 values	Divided by 9.01; confirmed with the site laboratory
Impossible values: age 210, weight 7 kg, a negative fluid output	3 values	Set to missing; query sent to the site; patient kept
Derived variables not in the raw file	3	Fluid balance, positive balance and any AKI computed by the script

Three of these problems carry lessons that go beyond this dataset. The **unit problem** in one ICU was invisible in the raw data: its median lactate was 23.8 against about 2.5 elsewhere, and it was found only by summarising lactate by ICU. Left uncorrected, it would have made that ICU look like a lactate catastrophe and wrecked any analysis of lactate. The **missing-value codes** would have done similar damage: a lactate of 999 inflates every mean it enters. And the **impossible values** were set to missing rather than deleted or guessed: we cannot know whether an age of 210 should have been 21, 71 or something else, and deleting the patient would throw away an otherwise valid record and could bias the sample.

Because the cleaning is a script, it is **reproducible**: running it again on the raw file produces exactly the same analysis file, value for value. Anyone can check every change, and a reviewer can trace every number in the paper back to the raw data.

```r
# Two of the cleaning steps (from 01_clean.R), each one block of the script
raw <- read_csv("fluid_icu_raw.csv",
                col_types = cols(.default = col_character()))  # read everything as
                text first

dat <- raw |> clean_names() |> distinct()     # tidy names; drop the 3 exact
duplicates
```

```
# ICU C reported lactate in mg/dL: 1 mmol/L = 9.01 mg/dL
dat <- dat |>
  mutate(lactate_mmol_l = if_else(site == "ICU C",
                                  round(lactate_mmol_l / 9.01, 1), lactate_mmol_l))
```

16.2.2 What remained missing, and the flow of patients

After cleaning, the remaining gaps were genuine missing values. Admission lactate was missing for 93 of 620 patients (15.0%), because it had not been measured; APACHE II was missing for 6, mean arterial pressure for 4, and age, weight and one fluid output for 1 each (the impossible values). Every other variable was complete.

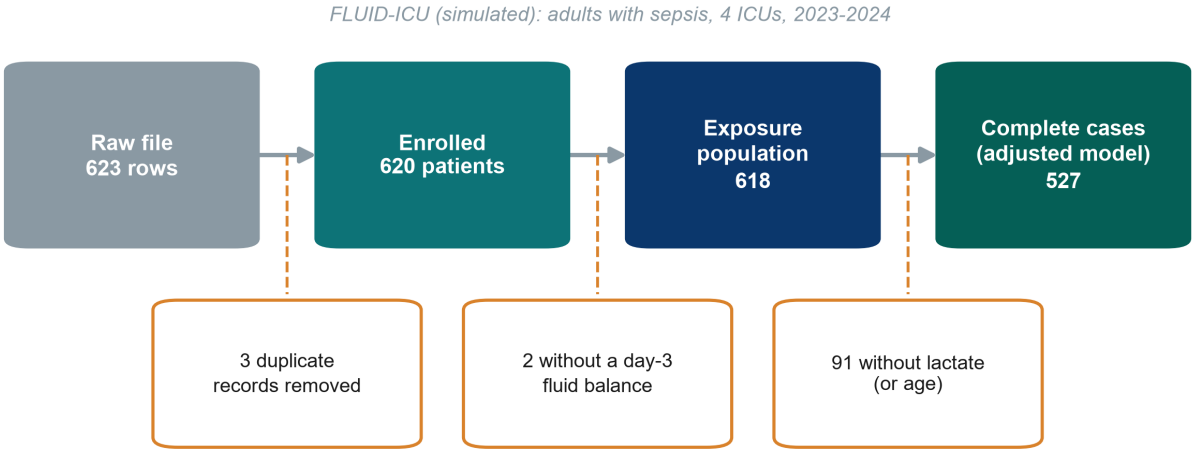


Figure 16.1: Flow of patients in the simulated FLUID-ICU cohort: 623 rows in the raw file; 3 duplicates removed, giving 620 enrolled patients; 2 without a day-3 fluid balance, giving 618 in the exposure population; 91 without lactate (or age), giving 527 complete cases for the adjusted model.

The flow diagram above is the first figure of any observational paper (STROBE item 13) (Elm et al. 2007). From 620 enrolled patients, two had no computable fluid balance at day 3 (because their weight or one fluid output had been set to missing), leaving **618 in the exposure population**. A further 91 lacked lactate or age, leaving **527 complete cases** for the adjusted models. The adjusted analysis therefore uses 85% of the enrolled patients, and the obvious question, whether the 91 excluded patients were different, is answered later in this chapter.

16.3 The plan came first

Before the database was locked, the study team wrote and signed a statistical analysis plan (Chapter 15). Everything in the rest of this chapter was decided there, before anyone saw an outcome:

- **Primary analysis:** 28-day mortality by multivariable logistic regression; odds ratio for positive versus negative balance with 95% confidence interval; the crude odds ratio and the risk difference also reported.
- **Confounders:** age; severity (SOFA score, lactate, vasopressor use and mechanical ventilation on admission); chronic kidney disease. Chosen from clinical knowledge with a simple causal

diagram: sicker patients receive more fluid and die more often. Variables measured after day 3, such as acute kidney injury and dialysis, were deliberately *not* adjusted for, because they are outcomes or possible mediators.

- **Supportive analyses:** Kaplan–Meier curves with numbers at risk and the log-rank test; a Cox model, crude and adjusted for the same confounders, with a check of proportional hazards.
- **One pre-specified effect modifier:** heart failure, because patients with heart failure tolerate fluid less well; expected direction a stronger association in heart failure.
- **Missing data:** complete-case analysis for the main models, with multiple imputation (20 datasets) as a sensitivity analysis; multiple imputation as the main analysis for Q4, where lactate is the exposure.
- **Sample size:** about 600 patients expected, giving about 80% power for an odds ratio of 1.7 and about 55% for an odds ratio of 1.5. The study was powered for a moderate association only.

16.4 Question 1 in depth: positive fluid balance and 28-day mortality

16.4.1 Step 1: Who was studied? Reading Table 1

Before comparing outcomes, describe the groups. Table 1 shows the baseline characteristics of the 618 patients in the exposure population, by fluid-balance group. Following the analysis plan, it uses mean (SD) for roughly symmetric variables, median (IQR) for skewed ones and scores, and n (%) for categories, reports the number missing, and contains **no p-values**: it describes the groups, it does not test them (Chapter 11).

Table 16.3: Baseline characteristics by cumulative fluid balance at day 3 (simulated FLUID-ICU data; selected rows).

Characteristic	Overall (n = 618)	Negative balance (n = 245)	Positive balance (n = 373)
Age, years, mean (SD)	62.6 (15.2)	64.6 (15.0)	61.4 (15.2)
Male sex, n (%)	375 (61%)	143 (58%)	232 (62%)
Diabetes, n (%)	158 (26%)	70 (29%)	88 (24%)
Chronic kidney disease, n (%)	64 (10%)	28 (11%)	36 (9.7%)
Heart failure, n (%)	97 (16%)	54 (22%)	43 (12%)
SOFA score, median (IQR)	7 (5, 9)	6 (4, 8)	8 (6, 10)
APACHE II score, median (IQR)	19 (15, 24)	17 (12, 21)	20 (17, 25)
Lactate, mmol/L, median (IQR)	2.6 (1.7, 3.7)	2.2 (1.4, 3.0)	2.9 (1.9, 4.0)
Lactate missing, n	91	50	41
Mean arterial pressure, mmHg, median (IQR)	73 (66, 80)	76 (70, 84)	71 (65, 78)
Vasopressor on admission, n (%)	266 (43%)	61 (25%)	205 (55%)
Mechanical ventilation, n (%)	313 (51%)	97 (40%)	216 (58%)

Positive balance was common: 373 of 618 patients (60.4%) had a positive cumulative balance by day 3, and the median balance was 8.8 ml/kg (IQR −10.3 to 26.2).

Read Table 1 with two questions. Do these patients look like the patients I treat? And were the groups alike at the start? Age and sex are similar. **Severity is not.** Patients with a positive balance

had a higher median SOFA score (8 against 6), were more than twice as likely to be on vasopressors (55% against 25%), were more often ventilated (58% against 40%), had higher lactate (median 2.9 against 2.2 mmol/L) and lower blood pressure. They were sicker *before* the fluid was given. Heart failure, in contrast, was more common in the negative-balance group (22% against 12%): clinicians give less fluid to patients with heart failure.

This is **confounding by indication** (Chapter 6) in plain view. Sicker patients receive more fluid, because they are in shock, and sicker patients die more often, because they are sicker. Any crude comparison of mortality between the groups will mix the effect of fluid with the effect of severity, and we can predict from Table 1 alone that the crude association will be larger than the adjusted one.

16.4.2 Step 2: The crude comparison

The outcome is binary and the groups are large, so the crude comparison is a 2×2 table with a chi-square test, reported as an effect with its confidence interval (Chapter 12).

Table 16.4: Twenty-eight-day mortality by fluid-balance group (unadjusted, simulated data).

	Died by day 28	Survived	Total	Risk
Positive balance	134	239	373	35.9%
Negative balance	45	200	245	18.4%

Overall, 179 of 620 enrolled patients (28.9%) died by day 28. Among the 618 with a fluid balance, mortality was 35.9% with a positive balance and 18.4% with a negative balance.

- **Risk difference:** 17.6 percentage points (95% CI 10.7 to 24.4). The interval is far from 0.
- **Risk ratio:** 1.96: about twice the risk.
- **Odds ratio:** $(134/239)/(45/200) = 2.49$, larger than the risk ratio because death is common (Chapter 13).
- **Chi-square test:** $p < 0.001$.

A results sentence would read: “Twenty-eight-day mortality was 35.9% (134/373) in patients with a positive and 18.4% (45/245) in patients with a negative fluid balance (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4).” The sentence is correct, but Table 1 has already told us that it cannot be interpreted as the effect of fluid.

16.4.3 Step 3: Crude versus adjusted: confounding by indication made visible

The pre-specified primary analysis is a multivariable logistic regression including the exposure and the six confounders. The full model is shown below, with the crude (one-variable) odds ratios alongside.

Table 16.5: Logistic regression for death by day 28: crude and adjusted odds ratios (adjusted model n = 527; simulated data).

Variable	Crude OR (95% CI)	p	Adjusted OR (95% CI)	p
Positive fluid balance (vs negative)	2.49 (1.70 to 3.70)	<0.001	1.62 (1.02 to 2.59)	0.044
Age, per year	1.02 (1.01 to 1.03)	<0.001	1.02 (1.01 to 1.04)	0.002
SOFA score, per point	1.26 (1.18 to 1.35)	<0.001	1.10 (1.01 to 1.21)	0.039
Lactate, per mmol/L	1.38 (1.23 to 1.55)	<0.001	1.15 (1.00 to 1.32)	0.059
Vasopressor	3.31 (2.31 to 4.78)	<0.001	1.86 (1.20 to 2.88)	0.005
Mechanical ventilation	2.98 (2.07 to 4.34)	<0.001	1.69 (1.11 to 2.60)	0.016
Chronic kidney disease	2.58 (1.52 to 4.36)	<0.001	1.85 (1.00 to 3.44)	0.050

One subtlety must be handled before comparing the crude and adjusted odds ratios for the exposure. The crude odds ratio of 2.49 uses all 618 patients; the adjusted model can use only the 527 with complete data. Part of any difference could simply reflect the different patients. The fair comparison is with the crude odds ratio *in the same 527 patients*, which is 2.22 (95% CI 1.48 to 3.39). The figure below shows all three.

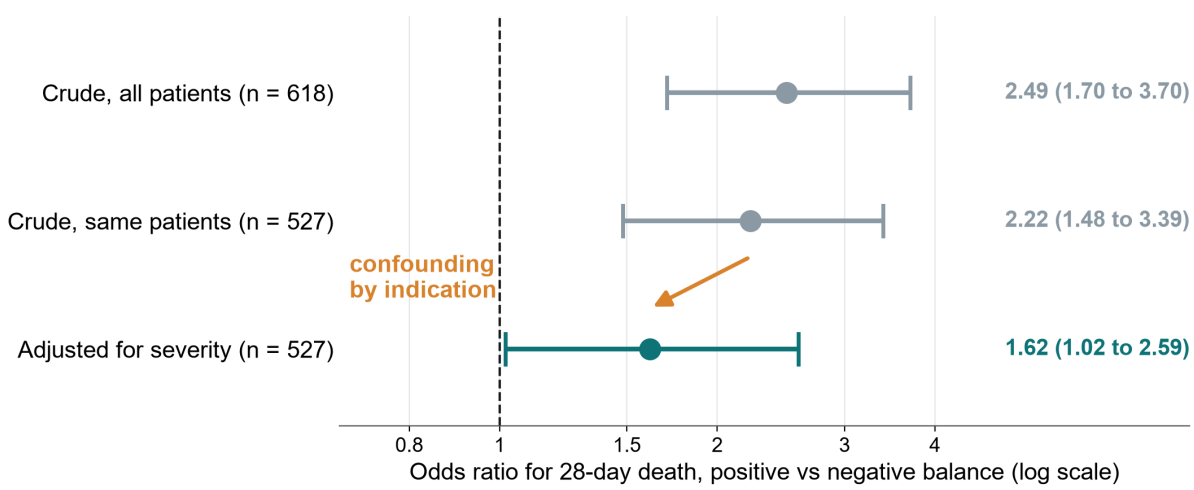


Figure 16.2: Odds ratio for 28-day death, positive versus negative balance (log scale): crude in all 618 patients 2.49 (1.70 to 3.70); crude in the same 527 complete cases 2.22 (1.48 to 3.39); adjusted for severity in the 527, 1.62 (1.02 to 2.59). The shift from crude to adjusted is the effect of confounding by indication.

Adjustment for the six confounders reduced the odds ratio from 2.22 to **1.62 (95% CI 1.02 to 2.59)**, $p = 0.044$. A large part of the crude association was severity: patients who were sicker received more fluid and died more often. This is the stratification picture of Chapter 13 done by a model, for six confounders at once.

What remains is a smaller and less precise association. The confidence interval is wider than the crude one, because there are fewer patients and because the confounders are correlated with the exposure, and its lower limit is close to 1. The data are compatible with a modest harm

associated with positive balance, and also with almost no association at all. With 161 deaths in the complete-case model and seven terms, there are about 23 events per variable, so the model is not overfitted.

The forest plot below shows the adjusted model as a picture.

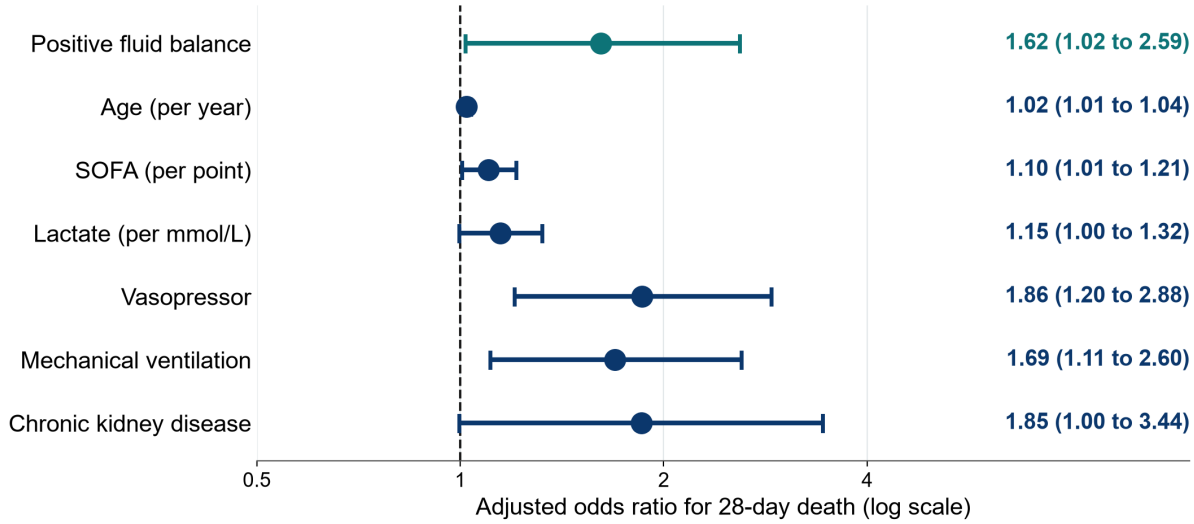


Figure 16.3: Adjusted odds ratios for 28-day death from the primary model (n = 527, log scale): positive fluid balance 1.62 (1.02 to 2.59); age per year 1.02; SOFA per point 1.10; lactate per mmol/L 1.15 (1.00 to 1.32); vasopressor 1.86; mechanical ventilation 1.69; chronic kidney disease 1.85 (1.00 to 3.44).

Only the first row answers the research question. The other rows are the confounders; they are in the model to adjust, not to be interpreted (the Table 2 fallacy, Chapter 13) (Westreich and Greenland 2013). Note the units: an odds ratio of 1.02 per year of age looks trivial, but over 20 years it multiplies the odds by about $1.02^{20} \approx 1.5$.

> **! Common mistake: adjusting for everything in the dataset**
>
> Why not also adjust for acute kidney injury, which is strongly associated with death? Because AKI in FLUID-ICU is measured over days 1 to 7, after or during the fluid exposure, and fluid overload may be one route by which a positive balance harms patients. Adjusting for a mediator removes part of the association we are trying to estimate. The analysis plan said so in advance.

```r
# Crude and adjusted logistic regression (from 04_models.R)
icu <- readRDS("fluid_icu_clean.rds") |> filter(!is.na(positive_balance_d3))   # 618
patients

m_crude <- glm(died_28d ~ positive_balance_d3, family = binomial, data = icu)
m_adj   <- glm(died_28d ~ positive_balance_d3 + age + sofa + lactate_mmol_l +
                 vasopressor + mech_vent + ckd, family = binomial, data = icu)

nobs(m_adj)                                          # 527: rows with missing
values are dropped silently
tidy(m_adj, exponentiate = TRUE, conf.int = TRUE)    # odds ratios with 95% CIs
```

The comment on `nobs()` deserves emphasis. R, like most software, drops every patient with a missing value in any model variable without warning. Always check how many patients a model

actually used.

16.4.4 Step 4: Survival analysis: Kaplan–Meier and Cox

The supportive analyses use the time from ICU admission to death, with survivors censored at day 28 (Chapter 14). Because every survivor in FLUID-ICU was followed to day 28, there is no censoring before day 28, which keeps the analysis simple.

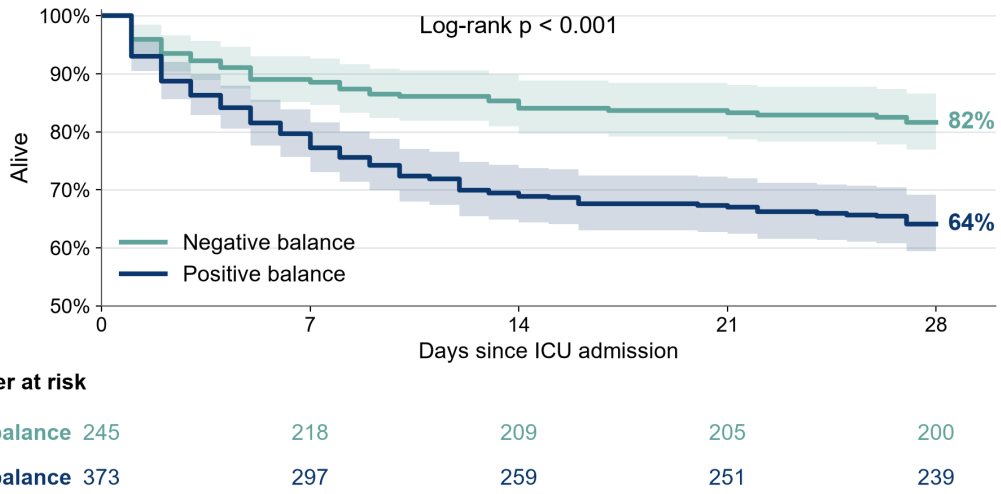


Figure 16.4: Kaplan–Meier survival to day 28 by fluid-balance group, with 95% confidence bands: survival at day 28 was 82% with a negative and 64% with a positive balance (log-rank p < 0.001); numbers at risk fell from 245 to 200 and from 373 to 239.

Read the figure with the four steps of Chapter 14. Each step down is a death; most deaths happen in the first week. The gap at day 28 is the difference in survival: **81.6% with a negative balance against 64.1% with a positive balance**. The numbers at risk under the plot fall from 245 to 200 and from 373 to 239: these are the patients still alive, since nobody was censored early. The log-rank test gives $p < 0.001$ (chi-square 21.4): the curves differ. But the log-rank test is unadjusted and carries exactly the same confounding as the crude odds ratio.

The curves separate in the first days and then run roughly parallel, which is reassuring for the proportional-hazards assumption of the Cox model.

Table 16.6: Cox regression for time to death within 28 days, positive versus negative balance (simulated data).

Cox model	n	Deaths	HR (95% CI)	p
Crude	618	179	2.17 (1.55 to 3.05)	<0.001
Adjusted (pre-specified confounders)	527	161	1.46 (1.00 to 2.13)	0.053

The Cox model tells the same story in hazard ratios. The crude hazard ratio of 2.17 shrinks to **1.46 (95% CI 1.00 to 2.13)** after adjustment. The global test of proportional hazards, based on Schoenfeld residuals, gave $p = 0.10$: no strong evidence against the assumption.

Now compare the two adjusted analyses. The logistic model gives an odds ratio of 1.62 with $p = 0.044$; the Cox model gives a hazard ratio of 1.46 with $p = 0.053$. One is “significant” and the other is not. A reader fixated on 0.05 would conclude that the two analyses disagree. They do

not. Both estimates point in the same direction, are of similar size, and have confidence intervals that overlap almost entirely and whose lower limits sit at about 1. Together they describe a modest, imprecise association. The correct reading is to report both estimates with their intervals, not to let one side of an arbitrary line decide whether the finding exists (Chapter 12) (Wasserstein and Lazar 2016).

```r
# Kaplan-Meier, log-rank and Cox (from 04_models.R)
km  <- survfit(Surv(time_to_event_28d, died_28d == "Yes") ~ positive_balance_d3,
data = icu)
summary(km, times = 28)                                    # survival at day 28 in
each group
survdiff(Surv(time_to_event_28d, died_28d == "Yes") ~ positive_balance_d3, data =
icu)  # log-rank

cox_adj <- coxph(Surv(time_to_event_28d, died_28d == "Yes") ~ positive_balance_d3 +
age +
                   sofa + lactate_mmol_l + vasopressor + mech_vent + ckd, data =
                   icu)
cox.zph(cox_adj)                                           # proportional-hazards
check
```

16.4.5 Step 5: The pre-specified interaction with heart failure

The analysis plan named one effect modifier in advance: heart failure. The reasoning was clinical: patients with heart failure tolerate fluid less well, so a positive balance might be more harmful in them. The interaction was tested by adding a positive-balance × heart-failure term to the primary model.

Table 16.7: Positive fluid balance and 28-day death by heart-failure status (simulated data). P for interaction 0.031 (Wald test; likelihood-ratio test 0.028). Adjusted ORs from the complete-case model (n = 437 without and 90 with heart failure).

Subgroup	Deaths, negative balance	Deaths, positive balance	Adjusted OR (95% CI)
No heart failure	34/191 (18%)	108/330 (33%)	1.29 (0.77 to 2.19)
Heart failure	11/54 (20%)	26/43 (60%)	4.39 (1.65 to 12.38)

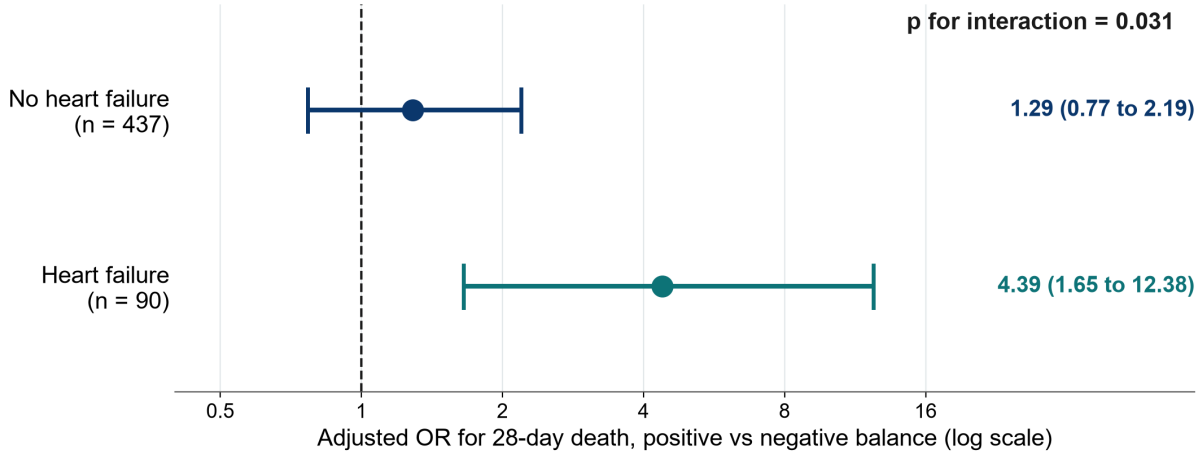


Figure 16.5: Adjusted odds ratio for 28-day death with a positive versus negative balance, by heart-failure status (log scale): no heart failure (n = 437) 1.29 (0.77 to 2.19); heart failure (n = 90) 4.39 (1.65 to 12.38); p for interaction 0.031.

In the 90 patients with heart failure in the adjusted model, the odds ratio was 4.39 (95% CI 1.65 to 12.38); in the 437 without, 1.29 (0.77 to 2.19). The test for interaction gave $p = 0.031$. Notice the logic of the test: it does not compare "significant in one subgroup" with "not significant in the other", which would be the classic error (Chapter 13) (Altman and Bland 2003). It asks directly whether the two odds ratios differ more than chance would explain.

How should this be interpreted? In favour of taking it seriously: it was pre-specified, with a clinical rationale and the expected direction, and it was the only subgroup tested. Against over-interpreting it: it rests on only 90 patients with heart failure in the adjusted model (37 of the 97 patients with heart failure in the exposure population died); the confidence interval for the heart-failure subgroup runs from 1.65 to 12.38, compatible with anything from a modest to an enormous association; and the analysis plan itself said in advance that an interaction p-value below 0.05 would be "evidence of effect modification needing confirmation, not a definite finding". The honest report is exactly that.

```R
# Interaction term and likelihood-ratio test (from 04_models.R)
```

```
m_int   <- glm(died_28d ~ positive_balance_d3 * heart_failure + age + sofa +
lactate_mmol_l +
                 vasopressor + mech_vent + ckd, family = binomial, data = icu)
m_noint <- update(m_int, . ~ . - positive_balance_d3:heart_failure)
anova(m_noint, m_int, test = "LRT")       # p for interaction (likelihood ratio)
```

16.4.6 Step 6: Missing lactate: complete case versus multiple imputation

The adjusted model excluded 91 patients, almost all because their lactate had not been measured. Were they different? Chapter 15 gave the answer in summary; here are the details. Across all 620 enrolled patients, lactate was missing for 93.

Table 16.8: Patients with and without an admission lactate (simulated data).

Characteristic	Lactate measured (n = 527)	Lactate missing (n = 93)
SOFA, median (IQR)	8 (6 to 10)	5 (4 to 6)
MAP, mmHg, median (IQR)	72 (65 to 78)	80 (75 to 86)
Vasopressor, n (%)	253 (48%)	14 (15%)
From ICU D, n (%)	89 (17%)	29 (31%)
Died by day 28, n (%)	161 (31%)	18 (19%)

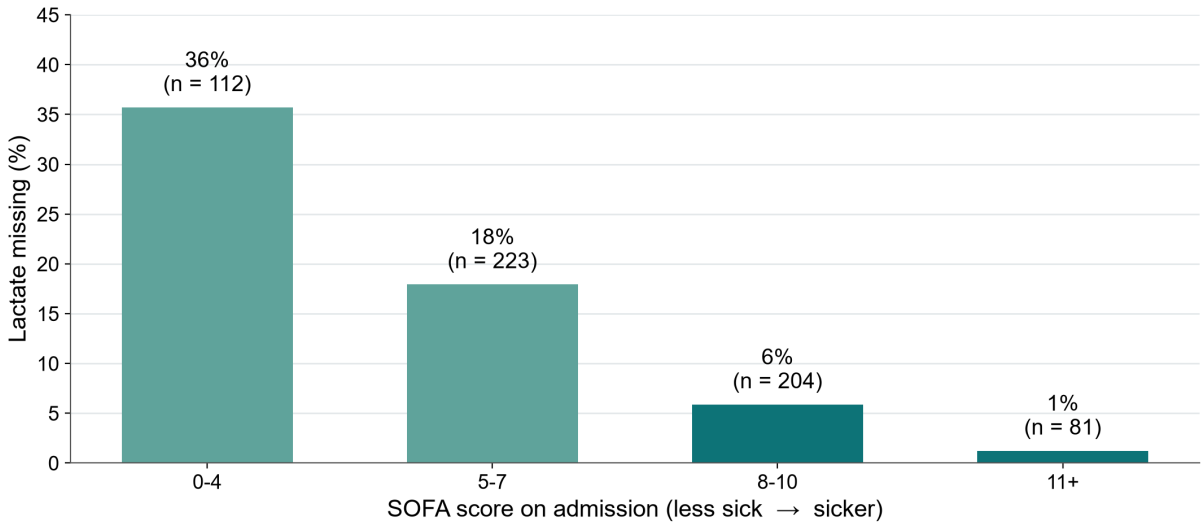


Figure 16.6: Percentage of patients with missing admission lactate by SOFA score on admission: 36% of 112 patients with SOFA 0 to 4, 18% of 223 with SOFA 5 to 7, 6% of 204 with SOFA 8 to 10 and 1% of 81 with SOFA 11 or more.

The pattern is unmistakable. Lactate was missing in 36% of the least sick patients (SOFA 0 to 4) and in 1% of the sickest (SOFA 11 or more). Patients without lactate had lower SOFA scores, higher blood pressure, fewer vasopressors, came more often from one ICU, and died less often (19.4% against 30.6%). Clinically this makes sense: lactate was not measured when the clinician judged the patient to be well enough, or where there was no point-of-care analyser.

So lactate is **not missing completely at random**, and the complete-case analysis drops a group of mostly less sick patients. But the missingness is well explained by variables that were recorded (severity, blood pressure, vasopressors, ICU). **Missing at random is plausible**, which is the

condition under which multiple imputation is valid (Chapter 15). It cannot be proven: if, among patients with the same SOFA score and blood pressure, clinicians skipped lactate precisely when they guessed it would be normal, the data would be missing not at random to some degree.

As the analysis plan specified, the primary model was repeated with multiple imputation by chained equations: 20 imputed datasets, predictive mean matching for continuous variables and logistic regression for binary ones, with an imputation model that included the outcome, all model covariates and the predictors of missingness. Estimates were pooled with Rubin's rules (Rubin 1987; White et al. 2011).

Table 16.9: Sensitivity analyses for the primary association between positive fluid balance and 28-day death (simulated data).

Analysis	n	OR (95% CI)	p
Crude, all patients	618	2.49 (1.70 to 3.70)	<0.001
Crude, complete cases	527	2.22 (1.48 to 3.39)	<0.001
Adjusted, complete cases (main analysis)	527	1.62 (1.02 to 2.59)	0.044
Adjusted, multiple imputation (m = 20)	620	1.70 (1.09 to 2.64)	0.020

With multiple imputation, using all 620 patients, the adjusted odds ratio was 1.70 (95% CI 1.09 to 2.64), close to the complete-case estimate of 1.62. The conclusion does not depend on how missing lactate was handled: the primary result is **robust** to this choice.

```r
# Multiple imputation with mice (from 05_missing.R)
imp_vars <- icu |> select(died_28d, lactate_mmol_l, age, sex, sofa, apache_ii,
map_mmhg,
                          vasopressor, mech_vent, ckd, diabetes, heart_failure,
                          site,
                          positive_balance_d3, infection_source)   # outcome +
                          predictors of missingness
imp <- mice(imp_vars, m = 20, seed = 2024, printFlag = FALSE)    # pmm for numbers,
logreg for yes/no
fit <- with(imp, glm(died_28d ~ positive_balance_d3 + age + sofa + lactate_mmol_l +
                       vasopressor + mech_vent + ckd, family = binomial))
summary(pool(fit), conf.int = TRUE, exponentiate = TRUE)          # Rubin's rules
```

16.4.7 Step 7: Putting Question 1 together

Bringing the steps together, the answer to the primary question, in plain language, is:

- Patients with a positive fluid balance by day 3 died about twice as often as those with a negative balance (36% against 18%). But they were also much sicker at the start.
- After adjustment for pre-specified markers of severity, age and chronic kidney disease, the association became much smaller (odds ratio 2.49 crude, 2.22 in the same patients, 1.62 adjusted). Much of the crude difference was confounding by indication.
- A modest association remained. Its confidence interval only just excludes 1 in the logistic model and just touches 1 in the Cox model. The evidence is weak to moderate, not strong.
- The association appeared stronger in patients with heart failure, a pre-specified hypothesis based on only 90 patients that needs confirmation.

- The result did not depend on how missing lactate was handled.

What the results do **not** show is equally important. They do not show that fluid *causes* death: this is an observational study, and the right words are "associated with", not "caused", "increased" or "led to". They do not tell clinicians to restrict fluid: only a randomised trial, such as the larger trials that pilot studies like Paper 1 are designed to prepare, can test a fluid strategy (Boulet et al. 2024). And $p = 0.044$ is not proof; an interval from 1.02 to 2.59 is compatible with a tiny or a large association.

The limitations that a discussion section must address follow directly from the analysis:

- **Residual confounding.** SOFA score and lactate measure severity imperfectly. Unmeasured aspects of severity could explain the remaining association.
- **Reverse causation.** Patients who are deteriorating, in refractory shock or with falling urine output, accumulate fluid *because* they are dying; the arrow may point from impending death to positive balance.
- **Immortal time.** Fluid balance is measured over days 1 to 3; patients who died early had less time to accumulate fluid, which can distort the comparison (Chapter 14).
- **Missing lactate.** The complete-case model lost 91 patients, mostly less sick; multiple imputation gave a similar answer.
- **Measurement.** Insensible losses are not recorded, and dichotomising the balance at zero throws away information (Question 2 uses the continuous version).

16.5 Questions 2 to 5: four further lessons

The secondary questions use the same data, the same confounders and the same complete-case approach for adjusted models (except Q4). Each is presented briefly, with the lesson it teaches. All are secondary and exploratory, as the analysis plan specified.

16.5.1 Question 2: fluid balance and acute kidney injury: a dose–response gradient

Acute kidney injury (KDIGO stage 1 or higher, the maximum stage over ICU days 1 to 7) occurred in 302 of 620 patients (48.7%); 79 (12.7%) received renal replacement therapy. AKI occurred in 61.7% (230/373) of patients with a positive balance and 29.0% (71/245) with a negative balance.

Here the exposure is kept continuous, as the plan specified, and the figure below shows why that is informative.

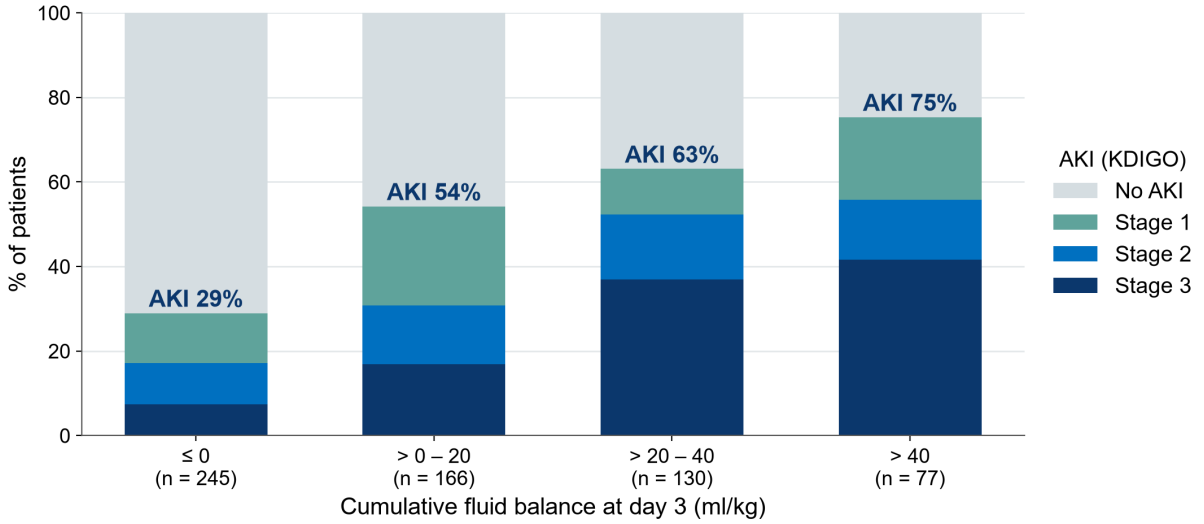


Figure 16.7: Maximum KDIGO acute kidney injury stage by category of cumulative fluid balance at day 3: any AKI in 29% of 245 patients with a balance of 0 ml/kg or less, 54% of 166 with above 0 to 20, 63% of 130 with above 20 to 40, and 75% of 77 above 40 ml/kg, with stage 3 increasing most.

AKI rose steadily across categories of fluid balance: 29%, 54%, 63% and 75%. A gradient of this kind (dose–response) is one of the Bradford Hill considerations for causation (Chapter 6). In logistic regression with the balance per 10 ml/kg, the crude odds ratio was 1.37 (95% CI 1.27 to 1.47) and the adjusted odds ratio **1.27 (95% CI 1.16 to 1.40)**, $p < 0.001$ (n = 527). Each extra 10 ml/kg of balance (about 650 ml for a 65 kg adult) was associated with about 27% higher odds of AKI. Treated as a binary exposure, the adjusted odds ratio for positive versus negative balance was 2.70 (95% CI 1.76 to 4.15); for AKI stage 2–3, the adjusted odds ratio per 10 ml/kg was 1.23 (1.12 to 1.35).

The lesson: the gradient is striking, but **reverse causation** is a serious alternative explanation. Patients who develop AKI pass less urine, and lower urine output makes the balance more positive. The exposure (days 1 to 3) and the outcome (days 1 to 7) overlap in time, so we do not know which came first. Fluid also dilutes creatinine, which may cause AKI to be under-diagnosed in patients with large positive balances. And because AKI is common (about 49%), the odds ratios overstate the corresponding risk ratios. A causal reading is not justified.

16.5.2 Question 3: fluid balance and ventilator-free days: an awkward outcome

Ventilator-free days to day 28 count the days a patient was alive and free of invasive ventilation. A patient who dies scores 0, however long they lived; a patient never ventilated who survives scores 28. The result is an outcome piled up at both ends.

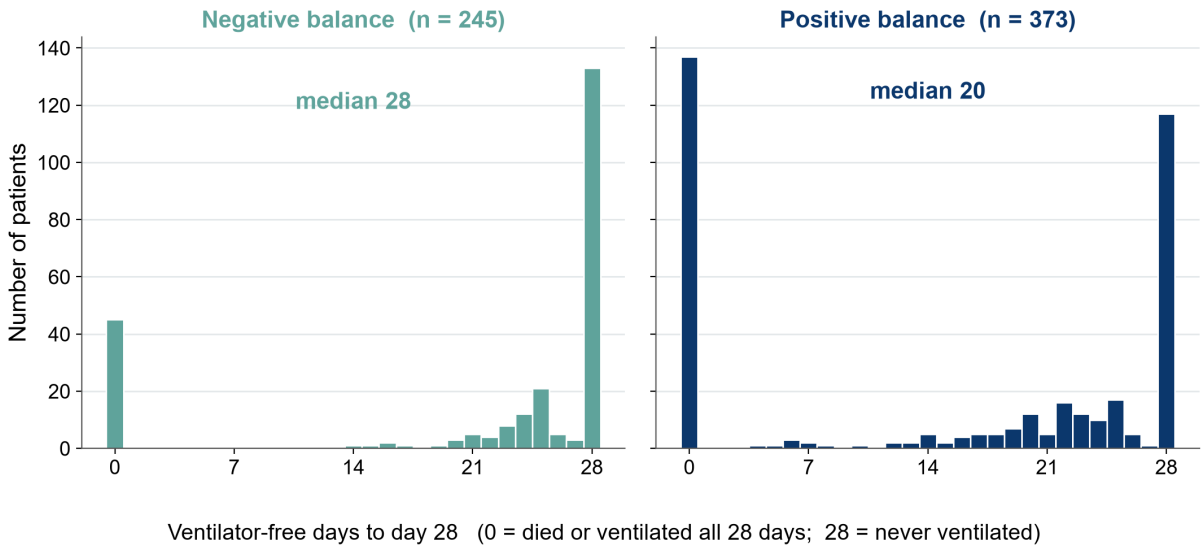


Figure 16.8: Distribution of ventilator-free days to day 28 by fluid-balance group: in the negative-balance group (n = 245) most patients are at 28 (median 28), with a smaller pile at 0; in the positive-balance group (n = 373) large piles sit at both 0 and 28 (median 20).

The figure shows two spikes and little in between: nothing like a bell curve. The median was 20 days (IQR 0 to 28) with a positive balance and 28 (22 to 28) with a negative balance (Mann–Whitney $p < 0.001$). A score of 0 was recorded for 37% of patients with a positive balance and 18% with a negative balance. Linear regression gave, per 10 ml/kg of balance, a crude difference of −1.48 days (95% CI −1.84 to −1.13) and an adjusted difference of **−0.61 days (95% CI −1.00 to −0.21)**, $p = 0.003$; for positive versus negative balance the adjusted difference was −2.78 days (−4.76 to −0.80).

The lesson: always look at the distribution before choosing the summary and the model. The median and the Mann–Whitney test describe this outcome honestly; the linear model gives an easily understood average difference, but its assumptions are strained by the two spikes, and a sensitivity analysis with a method suited to the shape (for example, an ordinal model) would strengthen the conclusion. Ventilator-free days are also a **composite** of death and duration of ventilation: a score of 0 can mean early death or 28 days on the ventilator, and the paper must say which drives the result. A difference in mean ventilator-free days does not mean “each patient was ventilated 0.6 days longer”.

16.5.3 Question 4: lactate, mortality and missing data

Question 4 asks whether admission lactate is associated with 28-day death, and whether the answer changes when the missing values are imputed. It is a prognostic-factor question, and since lactate is the exposure, its missing values matter directly; the plan therefore made multiple imputation the main analysis.

Non-survivors had higher admission lactate than survivors (median 3.1, IQR 2.2 to 4.5, against 2.3, IQR 1.6 to 3.4 mmol/L). Mortality rose across lactate categories: 19% below 2 mmol/L, 31% from 2 to 4, and 50% above 4. Patients with lactate missing had a mortality of 19%, like the lowest category, consistent with their lower severity.

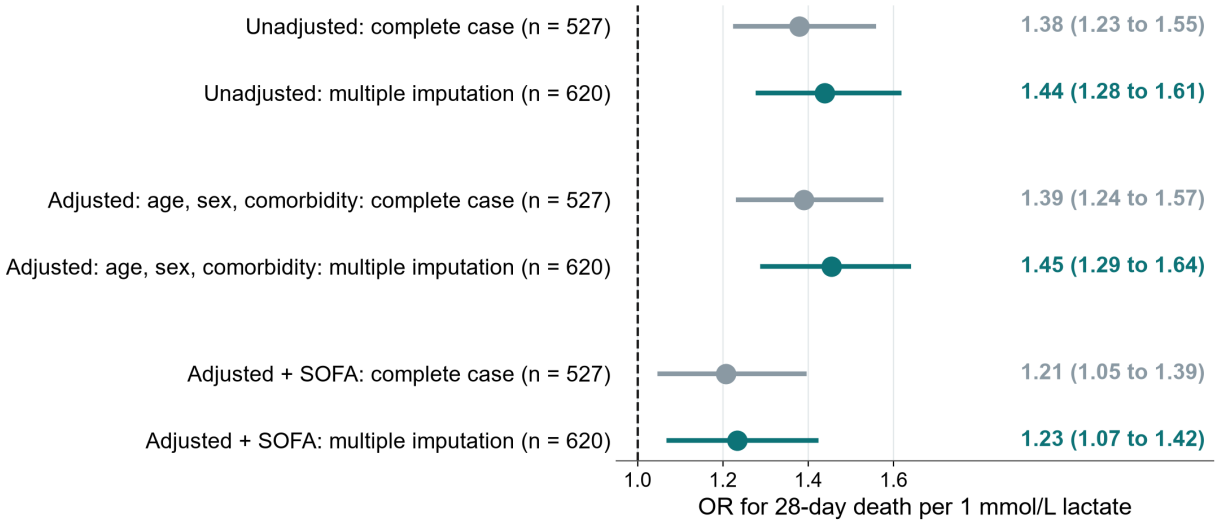


Figure 16.9: Odds ratio for 28-day death per 1 mmol/L of lactate, complete case (n = 527) versus multiple imputation (n = 620): unadjusted 1.38 versus 1.44; adjusted for age, sex and comorbidity 1.39 (1.24 to 1.57) versus 1.45 (1.29 to 1.64); with SOFA added 1.21 (1.05 to 1.39) versus 1.23 (1.07 to 1.42).

Adjusted for age, sex, diabetes, chronic kidney disease and heart failure, the odds ratio per 1 mmol/L was **1.45 (95% CI 1.29 to 1.64)** with multiple imputation in all 620 patients, and 1.39 (1.24 to 1.57) in the 527 complete cases. After adding the SOFA score, the odds ratios fell to 1.23 (1.07 to 1.42) and 1.21 (1.05 to 1.39). The median lactate was 2.6 mmol/L in the complete cases and 2.4 mmol/L after imputation: the imputed values were lower, as they should be for less sick patients.

The lessons: first, the complete cases were a sicker, selected group, and the complete-case estimate was modestly smaller than the imputed one; the conclusion was the same, and reporting both lets readers see that it is robust. Second, much of lactate's association with death overlaps with overall severity: adding SOFA reduced the odds ratio from about 1.4 to about 1.2, so lactate is partly a marker of how sick the patient is. It does not follow that lowering lactate would save lives. Third, multiple imputation is valid here only because missingness depends on recorded variables; if lactate had been missing because of its unseen value (MNAR), no method could fully correct it. Finally, the odds ratio per 1 mmol/L assumes a straight-line relationship on the log-odds scale; the categories let readers check that assumption.

16.5.4 Question 5: fluid balance and ICU stay in survivors: a skewed outcome

Question 5 asks whether, among patients who survived to day 28, a positive balance is associated with a longer ICU stay. Of the 618 patients with a fluid balance, 439 survived (239 with a positive and 200 with a negative balance).

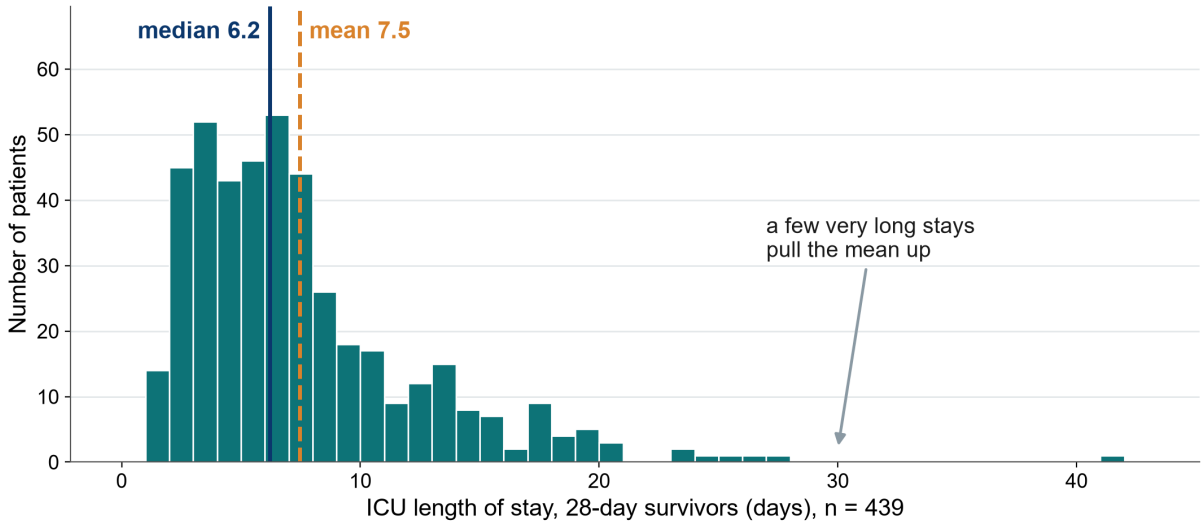


Figure 16.10: ICU length of stay in 439 survivors: a right-skewed histogram in which a few very long stays pull the mean (7.5 days) above the median (6.2 days).

ICU stay is **right-skewed**: most survivors stayed under 10 days, a few stayed much longer, and the long tail pulls the mean (7.5 days) above the median (6.2 days). The appropriate summaries are the median and IQR (Chapter 11): 7.5 days (IQR 5.2 to 11.2) with a positive balance against 4.8 days (3.3 to 7.1) with a negative balance (Mann–Whitney $p < 0.001$; Hodges–Lehmann difference 2.5 days, 95% CI 1.8 to 3.2). A t-test on the raw values would report a difference in means of 2.7 days, the wrong summary for this shape.

For adjustment, the stay was analysed on the logarithmic scale with linear regression. Back-transformed, the coefficient becomes a **ratio of geometric means**: crude 1.50 (95% CI 1.34 to 1.68) and adjusted **1.24 (95% CI 1.10 to 1.39)**, $p < 0.001$ (n = 366 survivors with complete data). A ratio of 1.24 means that the typical stay was about 24% longer with a positive balance, after adjustment. Per 10 ml/kg of balance the ratio was 1.06 (1.03 to 1.08).

The lessons: for a skewed outcome, report the median and IQR, use a rank test or analyse the logarithm and present a ratio. And beware of the population. Restricting the analysis to survivors **conditions on an outcome** that is itself associated with the exposure: death shortens ICU stay, and more patients with a positive balance died. The survivors in the two groups are therefore not comparable in the way the full cohort was, a form of selection bias. An alternative is an outcome such as ICU-free days, which counts death as zero. Reverse causation is again possible: patients who stay longer have more time and more reasons to receive fluid. And ICU discharge depends on bed pressure and local practice, not only on the patient.

16.5.5 The pattern across all five questions

Table 16.10: Crude and adjusted estimates for the fluid-balance questions (simulated data; 95% CIs).

Question	Crude estimate	Adjusted estimate
Q1 Death, OR positive vs negative	2.49 (1.70 to 3.70)	1.62 (1.02 to 2.59)
Q2 AKI, OR per 10 ml/kg	1.37 (1.27 to 1.47)	1.27 (1.16 to 1.40)
Q3 Ventilator-free days, difference per 10 ml/kg	−1.48 (−1.84 to −1.13)	−0.61 (−1.00 to −0.21)
Q5 ICU stay in survivors, ratio of geometric means	1.50 (1.34 to 1.68)	1.24 (1.10 to 1.39)

In every question, the adjusted estimate is closer to "no effect" than the crude one. This is the fingerprint of confounding by indication: in every comparison, the patients who accumulated more fluid were sicker, and part of every crude association was severity. It is also a reminder that adjustment can remove only the confounding that was measured.

16.6 Reporting the results

An analysis is finished only when it is reported clearly enough that a reader could check it. FLUID-ICU is an observational cohort, so the STROBE statement is the reporting guideline (Elm et al. 2007). The analysis above maps onto STROBE items as follows: design and setting (items 4–5), participants and eligibility (item 6), variables with the exposure, outcomes, confounders and effect modifier defined (item 7), statistical methods including confounding, interaction, missing data and sensitivity analyses (item 12), participant numbers and the flow diagram (item 13), descriptive data with missing values per variable (item 14), outcome data (item 15), crude and adjusted estimates with confidence intervals (item 16), and the subgroup and sensitivity analyses (item 17). Chapter 19 describes how to write each section of the paper; here are examples of the key passages for Question 1.

Worked example: the statistical methods paragraph

"The analysis followed a statistical analysis plan finalised before database lock. Baseline characteristics were summarised by fluid-balance group as mean (SD), median (IQR) or n (%). The primary analysis used multivariable logistic regression to estimate the association between a positive cumulative fluid balance at day 3 and death by day 28, adjusted for age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease, chosen a priori as confounders; acute kidney injury and renal replacement therapy were not adjusted for because they may lie on the causal pathway. Kaplan–Meier curves were compared with the log-rank test, and Cox regression was used to estimate crude and adjusted hazard ratios; proportional hazards were checked with Schoenfeld residuals. Effect modification by heart failure, pre-specified, was tested with an interaction term. The primary analysis used complete cases; as a pre-specified sensitivity analysis, missing values were imputed by multiple imputation by chained equations (20 datasets), and estimates were combined with Rubin's rules. Analyses used R. The data are simulated and were generated for teaching purposes."

Worked example: the results paragraph

"Of 620 enrolled patients, 618 had a computable day-3 fluid balance, of whom 373 (60.4%) had a positive balance. Patients with a positive balance were more severely ill at admission (median SOFA 8 vs 6; vasopressor use 55.0% vs 24.9%; mechanical ventilation 57.9% vs 39.6%). Twenty-eight-day mortality was 35.9% (134/373) with a positive and 18.4% (45/245) with a negative balance (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4; crude OR 2.49, 95% CI 1.70 to 3.70). After adjustment for pre-specified confounders (n = 527), positive balance remained associated with higher mortality (OR 1.62, 95% CI 1.02 to 2.59); the adjusted hazard ratio was 1.46 (95% CI 1.00 to 2.13). Results were similar with multiple imputation (OR 1.70, 95% CI 1.09 to 2.64; n = 620). The association appeared stronger in patients with heart failure (OR 4.39, 95% CI 1.65 to 12.38) than without (OR 1.29, 95% CI 0.77 to 2.19; p for interaction 0.031)."

Worked example: a structured abstract

Background: The association between early positive fluid balance and mortality in sepsis is confounded by illness severity. **Methods:** Simulated multicentre prospective cohort (teaching dataset) of 620 adults with sepsis in four ICUs. Exposure: positive cumulative fluid balance at day 3. Outcome: death by day 28. Multivariable logistic regression adjusted for pre-specified confounders; Cox regression; multiple imputation for missing lactate. **Results:** Mortality was 35.9% with a positive and 18.4% with a negative balance (crude OR 2.49, 95% CI 1.70 to 3.70). After adjustment, the association was smaller (OR 1.62, 95% CI 1.02 to 2.59; adjusted HR 1.46, 95% CI 1.00 to 2.13) and robust to multiple imputation (OR 1.70, 95% CI 1.09 to 2.64). A pre-specified interaction suggested a stronger association in heart failure (p for interaction 0.031). **Conclusions:** In these simulated data, a positive fluid balance by day 3 was associated with modestly higher 28-day mortality after adjustment for severity; much of the crude association reflected confounding by indication. Causation cannot be inferred; randomised trials are needed.

Notice what these passages do and avoid. They give the effect and its confidence interval before any p-value. They give the crude and adjusted estimates, so the reader can see how much adjustment mattered. They state the number of patients in each analysis. They say "associated with", never "caused" or "increased". They report the interaction as it was planned, without overstating it. They mention the sensitivity analysis and its result. And they say, every time, that the data are simulated.

! Common mistakes when reporting an analysis like this

- Reporting only the adjusted odds ratio, so readers cannot see the confounding.
- "Positive fluid balance increased mortality by 62%": causal language, and an odds ratio described as a risk.
- "The Cox model showed no significant association (p = 0.053)": dismissing a result that agrees with the primary analysis because of an arbitrary threshold.
- "Fluid was harmful in heart failure but not in other patients": comparing subgroup p-values instead of reporting the interaction test, and overstating an exploratory finding.
- Not saying how many patients the adjusted model lost, or who they were.
- Presenting simulated teaching data as real research.

Summary

- FLUID-ICU is a simulated cohort of 620 adults with sepsis, built to teach the analysis of observational data. Its raw data contained duplicates, inconsistent codes, a unit error, missing-value codes and impossible values, all fixed by a reproducible script and recorded in a cleaning log.
- The statistical analysis plan, written before the analysis, fixed the outcome, the model, the confounders, the single interaction and the missing-data strategy.
- Table 1 showed that patients with a positive balance were sicker at baseline: a warning of confounding by indication.
- Crude mortality was 35.9% against 18.4% (OR 2.49). Adjustment for six pre-specified confounders reduced the odds ratio to 1.62 (95% CI 1.02 to 2.59); the Cox model gave an adjusted HR of 1.46 (1.00 to 2.13). Both describe a modest, imprecise association, not a causal effect.
- A pre-specified interaction suggested a stronger association in heart failure ($p = 0.031$), based on 90 patients and requiring confirmation.
- Lactate was missing in 15%, mostly in less sick patients (not MCAR, MAR plausible); multiple imputation gave results similar to complete-case analysis.
- The secondary questions taught dose–response and reverse causation (AKI), awkward bounded outcomes (ventilator-free days), missing data in the exposure (lactate) and skewed outcomes with survivor selection (ICU stay). In every question the adjusted estimate was closer to no effect than the crude one.
- Reporting follows STROBE: crude and adjusted estimates with confidence intervals, numbers in each analysis, sensitivity analyses, cautious language, and a clear statement that the data are simulated.

Further reading

- von Elm E, Altman DG, Egger M, et al. The STROBE statement (Elm et al. 2007). The reporting guideline used to structure the FLUID-ICU write-up; its explanation and elaboration document gives examples of good reporting for each item.
- Hyun D-g, Ahn JH, Huh JW, et al. Impact of a cumulative positive fluid balance during the first three ICU days in patients with sepsis (Hyun et al. 2023). A real cohort study of the same question, using propensity-score matching and clustered Cox models; worth reading alongside this chapter.
- Boulet N, Quenot J-P, Serrand C, et al. Impact on fluid balance of an optimized restrictive strategy targeting non-resuscitative fluids (Boulet et al. 2024). The randomised pilot trial that addresses the causal version of the question.
- Sterne JAC, White IR, Carlin JB, et al. Multiple imputation for missing data in epidemiological and clinical research (Sterne et al. 2009). Explains the methods and reporting standards used for the missing-lactate analyses.

PART IV

Ethics, writing and publishing

CHAPTER 17

Protocols, ethics and informed consent

Everything in the earlier parts of this book (the question, the design, the data collection plan, the analysis) can be done well on paper and still lead to a study that should never have been run. Research asks people, often people who are ill and frightened, to accept some burden or risk for the sake of knowledge that will mostly help others. That request is only acceptable when the study is worth doing, is planned in advance, has been checked by people independent of the researchers, and respects each participant's right to decide. This chapter is about the protective framework that surrounds every study before the first data point is collected: the written protocol, the international ethical codes, the research ethics committee, informed consent (including the hard case of patients who cannot decide for themselves), registration, confidentiality and a frank check of feasibility.

The running example is the management of fluids in septic shock. Two published studies are used throughout: Paper 1, a small randomised pilot trial of a restrictive strategy for non-resuscitative fluids (Boulet et al. 2024), and Paper 2, a large propensity-matched cohort from a national sepsis registry (Hyun et al. 2023). Where a study of our own is needed, we use FLUID-ICU, a cohort of 620 adults with sepsis in four intensive care units that is **simulated for teaching**: no real patients were involved, which is why it never needed ethics approval, but we will ask what its paperwork would look like if it had been real.

In this chapter you will learn

- why research needs protections beyond those of ordinary clinical care, and what makes a study ethical;
- what a study protocol contains, why it is written and approved before the first participant, and how it is changed;
- the core principles of the Declaration of Helsinki (2024 revision) and of Good Clinical Practice, ICH E6(R3);
- how a research ethics committee reviews a study, and what it needs from you;
- when a study must be registered, and where;
- what makes consent valid, and how consent works when a patient lacks capacity (representative consent, deferred consent and waivers);
- how to protect confidentiality by design, and how to judge whether a study is feasible.

17.1 Why research needs special protection

In clinical care, every decision is meant to serve the patient in front of you. In research, at least part of what is done serves a different goal: producing generalisable knowledge. A patient in a trial is allocated by chance rather than by the clinician's judgement; a patient in a cohort may have extra blood drawn, extra questions asked or their records read by people who are not treating them. Even when the burden is small, the purpose has shifted, and that shift is the reason research needs its own safeguards.

The history of medicine shows why these safeguards were not left to the good intentions of individual researchers. Experiments on prisoners during the Second World War led to the Nuremberg Code in 1947, which made voluntary consent the first condition of human experimentation. Later scandals in which patients were deliberately left untreated, or exposed to harm without their knowledge, showed that consent alone was not enough: studies also needed independent review, a favourable balance of risk and benefit, and fair selection of participants. The modern framework grew from these lessons.

A useful summary of what makes a study ethical comes from Emanuel and colleagues, who proposed seven requirements (Emanuel et al. 2000). A study must have **social or scientific value** (it asks a question worth answering); **scientific validity** (it is designed well enough to answer it, because a badly designed study exposes people to burden for nothing); **fair subject selection** (participants are chosen for scientific reasons, not because they are convenient or powerless); a **favourable risk-benefit ratio**; **independent review**; **informed consent**; and **respect for enrolled participants** (privacy, the right to withdraw, information about new risks and about results). Behind these lie the familiar principles of biomedical ethics: respect for autonomy, beneficence, non-maleficence and justice (Beauchamp and Childress 2019).

- **A poorly designed study is an unethical study**

  Scientific quality and ethics are not separate topics. A trial too small to answer its question, an outcome that cannot be measured reliably or an analysis chosen after seeing the data all waste the time and goodwill of participants. Everything in Parts I to III of this book is, in that sense, also an ethical requirement.

17.2 The protocol: the study's rule book

A **protocol** is the complete written plan of a study, written and approved before the first participant is enrolled, and then followed. It is the document the ethics committee reviews, the reference that the study team works from at three o'clock in the morning, and, once published or registered, the record that lets readers check that the researchers did what they planned.

For randomised trials, the international guidance on what a protocol should contain is SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials), first published in 2013 and updated as SPIRIT 2025 (Chan et al. 2013; Chan et al. 2025). Observational studies have no equivalent checklist with the same authority, but the same logic applies, and STROBE tells you what the final paper will need to report, so it is wise to plan for those items from the start (Elm et al. 2007). A protocol can be thought of in four blocks.

Table 17.1: The four blocks of a study protocol, following the logic of SPIRIT.

Block	What it contains
Who and what	Full title naming the design, version number and date, registration number, sponsor, funder, investigators and their roles, sites, ethics committee and approval number
Why	Background and rationale (what is known, what is not, why this study now and here); the research question in PICO form; the primary objective and hypothesis, and a few secondary objectives
How	Design; setting; eligibility criteria; screening and recruitment; intervention or exposure; outcomes with definitions and time points (only one primary outcome); schedule of assessments; sample size with every assumption and its source; allocation and blinding (trials); data collection and management; statistical methods
Protect and share	Monitoring and safety reporting; ethics and consent procedure; confidentiality and data protection; dissemination plan, authorship rules and data sharing; funding and conflicts of interest; appendices (information sheet, consent form, case report form, data dictionary, budget)

Two features of a good protocol deserve emphasis. First, it is **specific**. "Data will be collected on fluid balance" is not a protocol; "cumulative fluid balance at day 5 (ml/kg) is the sum of daily input minus output from the ICU fluid chart, divided by the admission weight" is. The protocol of Paper 1 had to define exactly which fluids were "non-resuscitative" (maintenance fluids, drug dilution, nutrition) and which were left to the clinicians, because that distinction *was* the intervention (Boulet et al. 2024). Second, a protocol is **version-controlled**. Once approved, it is not quietly edited. If something must change (a new exclusion criterion, an extra outcome, a longer recruitment period) the change is written as a formal **amendment**, given a new version number and date, and approved by the ethics committee before it is applied. A version history table at the front of the protocol records every change.

The protocol is closely linked to the **statistical analysis plan** (SAP) described in Chapter 15. The SAP gives the analysis in more detail than the protocol (exact models, covariates, handling of missing data, sensitivity analyses) and, like the protocol, it should be finalised before anyone sees outcome data by group. Paper 1 registered its trial at ClinicalTrials.gov (NCT04947904) on 1 July 2021, before its first patient in September 2021, so readers can compare what was planned with what was reported (Boulet et al. 2024).

! Common mistake: writing the protocol after the study has started

A protocol written after recruitment begins is not a protocol; it is a description. It cannot protect against outcome switching or post hoc analysis choices, an ethics committee cannot approve it retrospectively, and journals will notice the dates. Write it first, get it approved, then start.

17.3 Two rule books: the Declaration of Helsinki and Good Clinical Practice

Two international documents shape almost every ethics committee's expectations. They have different jobs: the Declaration of Helsinki says *why* and *what* we must protect; Good Clinical Practice says *how* to run a trial so that participants are protected and the data can be trusted.

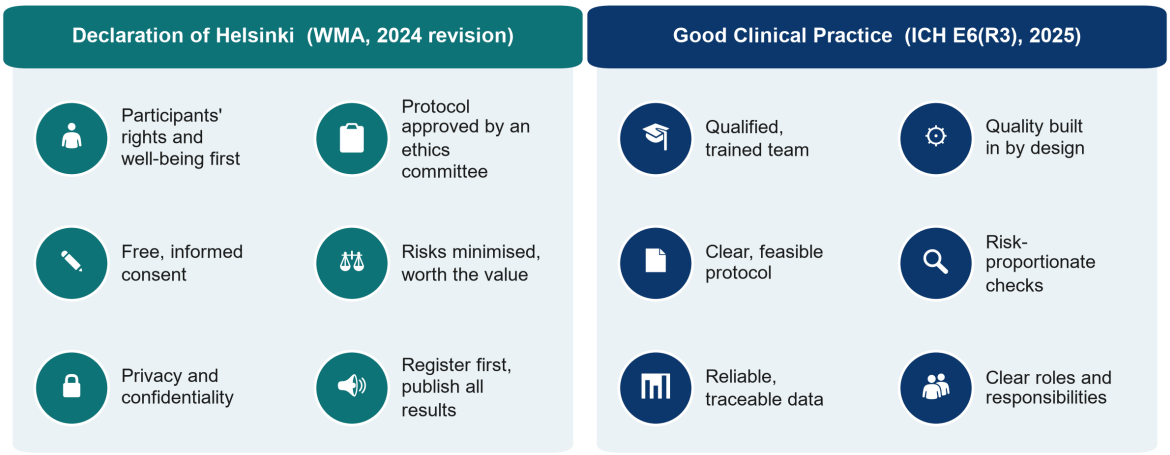


Figure 17.1: The Declaration of Helsinki (left) sets out the ethical principles that protect research participants; Good Clinical Practice, ICH E6(R3) (right), sets out how a trial is run so that it is safe and its data are reliable.

The **Declaration of Helsinki** is issued by the World Medical Association and was most recently revised in October 2024 (World Medical Association 2024). Its central principle is that the rights, safety and well-being of the individual participant take precedence over the goal of generating new knowledge. From that principle follow its main requirements, shown on the left of the figure above: a protocol reviewed and approved by an independent research ethics committee before the study starts; free and informed consent; risks that are minimised and justified by the importance of the question; protection of privacy and confidentiality; and registration of every study involving human participants before the first participant is recruited, with **all** results made publicly available, including negative and inconclusive ones. The 2024 revision speaks of "participants" rather than "subjects", strengthens the protection of vulnerable groups and of people who cannot consent, and puts more emphasis on scientific integrity and on fairness between the communities that bear the burdens of research and those that benefit from it. It explicitly allows research with people who cannot consent, under the conditions discussed later in this chapter.

Good Clinical Practice (GCP) is the international quality standard for clinical trials, maintained by the International Council for Harmonisation (ICH). Its third revision, ICH E6(R3), was adopted in January 2025 (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use 2025). The revision moves away from box-ticking towards **quality by design** and **risk-proportionate** oversight: identify what really matters for participant safety and for the reliability of the results (for example, correct consent, correct allocation, accurate measurement of

the primary outcome) and concentrate effort there. Its principles, shown on the right of the figure, include a qualified and trained team, a clear and feasible protocol, checks proportionate to the risks, reliable and traceable data, and clearly defined roles and responsibilities. Investigators on a trial are usually required to have completed GCP training recently.

A common belief is that GCP is "only for drug companies". It is not. Paper 1, an academic trial funded by an internal hospital grant, states that its analyses were performed in accordance with ICH Good Clinical Practice guidelines, and that the trial was monitored by the coordinating centre (Boulet et al. 2024). And although GCP is written for trials, its spirit (trained people, a written plan, traceable data, honest records) is good practice for every study, including a cohort study like FLUID-ICU.

Most countries also have their own laws on research with human participants, on medicines trials and on data protection. These add to the international principles; they do not replace them. Every protocol should state that the study will follow the Declaration of Helsinki, GCP and the applicable national regulations, and the investigators should know what those regulations require.

17.4 The research ethics committee

A **research ethics committee** (REC) is an independent group that reviews research involving people before it starts and oversees it while it runs. Depending on the country it may be called an institutional review board (IRB) or an independent ethics committee (IEC). Its members typically include clinicians, scientists, at least one lay member and often a lawyer or ethicist. The committee's job is to protect participants: it asks whether the question is worth answering, whether the design can answer it, whether risks are minimised and justified, whether consent is properly planned, and whether privacy is protected.



Figure 17.2: The journey through ethics review: draft the documents, obtain scientific or departmental review, submit to the committee, answer its questions, receive the approval letter, register if required, and only then begin. No data are collected before the approval letter.

The figure above follows a study through the process. Most institutions first require a **scientific or departmental review**, which checks the design and the resources before the ethics committee sees the study. The application dossier then goes to the committee. It typically includes the protocol, the participant information sheet and consent form, the case report form (CRF), investigators' CVs and evidence of their GCP training, a budget, any insurance documents required for interventional studies, and the committee's own application form. A completed example application, for an obser-

vational fluid-balance cohort, answers each of the committee's questions in specific, measurable terms and points out its own design problem (patients who die before day 5 cannot have a 5-day balance) along with how it will be handled; that kind of self-critical precision is what committees look for.

Committees usually distinguish three **levels of review**. Some low-risk studies (for example, analyses of fully anonymised existing data) may be declared **exempt** from full review; studies of minimal risk may receive an **expedited** review by one or two members; trials and anything with more than minimal risk go to the **full committee**. The committee decides which category applies, not the researcher. Questions and requests for revision are normal, and a realistic plan allows one to three months from first submission to approval.

Approval is not the end of the committee's role. During the study, the investigators submit amendments for approval before applying them, report serious adverse events within the required time, send progress reports (often annually) and send a final report when the study ends.

Research, audit or quality improvement?

Not every data collection is research. A **clinical audit** compares current practice with an agreed standard (for example, "were 90% of patients with sepsis given antibiotics within one hour?"); **quality improvement** tests changes to local care. Both are usually approved through the hospital's quality structures rather than the research ethics committee. **Research** aims to produce new, generalisable knowledge. The boundary can be blurred, and the intention to publish does not by itself turn an audit into research; but when in doubt, ask the ethics committee, which can confirm in writing that its review is not needed. That letter will also be useful when a journal asks.

! Common mistake: assuming a retrospective chart review needs no ethics review

Studies of existing records feel harmless, but they involve access to confidential information by people outside the treating team. The committee must review them. It may well grant a waiver of individual consent, as it did for Paper 2 (Hyun et al. 2023), but that is the committee's decision. Equally, "we will collect a few patients now and apply for approval later" is never acceptable: committees cannot approve retrospectively, and journals will reject the paper.

17.5 Registration: putting the study on the public record

Registration means entering the key facts of a study (its design, population, interventions, primary and secondary outcomes, planned sample size and dates) in a public registry *before* the first participant is enrolled. Its purpose is to make the existence and the plan of every study visible, so that studies with disappointing results cannot quietly vanish and outcomes cannot be switched after the data are seen. These problems, publication bias and selective outcome reporting, are discussed in Chapter 18; registration is the main defence against them.

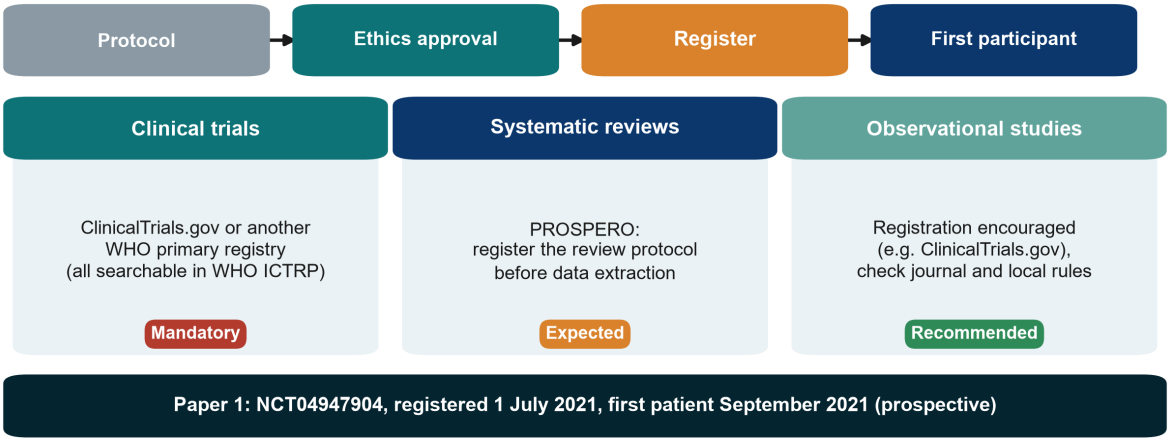


Figure 17.3: Registration comes after ethics approval and before the first participant. It is mandatory for clinical trials, expected for systematic reviews and recommended for observational studies. Paper 1 was registered on 1 July 2021, before its first patient in September 2021.

For **clinical trials**, registration is mandatory. Since 2005 the International Committee of Medical Journal Editors (ICMJE) has required prospective registration as a condition of publication (De Angelis et al. 2004; International Committee of Medical Journal Editors 2024), and many countries require it by law. Trials are registered in ClinicalTrials.gov or in another registry recognised as a primary registry by the World Health Organization. The WHO International Clinical Trials Registry Platform (ICTRP) is a search portal that searches all of these registries at once; it is not itself a registry. A trial registered after the first participant is called **retrospectively registered**, and many journals will not publish it.

For **systematic reviews**, registration of the review protocol in PROSPERO before data extraction is expected by many journals (Chapter 9). For **observational studies**, registration is encouraged rather than required: the Declaration of Helsinki asks for every study with human participants to be registered, some journals expect it, and it costs little. Check the target journal's policy and local rules.

The registration number then travels with the study: it goes in the abstract and the Methods of every paper. Paper 1 prints NCT04947904 in its abstract, which allows any reader to compare the registered primary outcome (fluid balance at day 5) with the one reported (Boulet et al. 2024).

Worked example: must these studies be registered?

Consider four studies and apply the rules above.

1. *A randomised trial comparing two dressings for pressure injuries.* This is a clinical trial: participants are allocated to an intervention to evaluate its effect on a health outcome.

Registration is **mandatory**, in a WHO primary registry or ClinicalTrials.gov, before the first patient.

2. *A systematic review of fluid strategies in sepsis.* Registration of the protocol in PROSPERO is **expected**, before data extraction begins.
3. *A retrospective cohort from ICU charts, like FLUID-ICU.* Registration is **recommended but not required**; check the target journal.
4. *An audit of hand-hygiene compliance against the hospital standard.* This is audit, not research. It is approved locally through quality structures and is not registered. If the team later wants to publish it as research, they should ask the ethics committee first.

17.6 Informed consent

Consent is the most visible expression of respect for a participant's autonomy. Its legal form is a signed document, but its substance is a conversation. Valid consent has four essential elements.

Table 17.2: The four elements of valid informed consent.

Element	What it means in practice
Informed	The person is told the purpose of the study, what will happen and for how long, the risks, burdens and possible benefits, the alternatives, how data will be protected, and whom to contact
Understood	Information is given in plain language, in a language the person reads fluently, with time to think and ask questions; understanding is checked, for example by asking the person to explain the study in their own words ("teach-back")
Voluntary	No pressure, no undue inducement; the person may refuse or withdraw at any time without giving a reason, and their care will not be affected
Documented	A signed and dated consent form, a copy for the participant, and re-consent if the study changes in a way that might affect their willingness to continue

The **participant information sheet** is where most consent problems begin. It is often written at the reading level of a medical journal, in the language of the researchers rather than of the patients. Good practice is to write as you would speak to a patient or relative: short sentences, everyday words ("drip" rather than "intravenous infusion"), "you" rather than "the participant", every medical term explained the first time it appears, and a length of a few pages. A useful structure is a series of questions the reader would ask: *Why is this study being done? Why have I been asked? Do I have to take part? What will happen if I take part? What are the risks and benefits? Will my information be kept confidential? What will happen to the results? Who has reviewed the study? Whom can I contact?* The sheet should say plainly what is research and what is normal care, and should give both a study contact and an independent contact (usually the ethics committee office) for complaints. It should then be translated into the languages patients speak, back-translated, and tested on two or three lay readers before submission.

The **consent form** lists short statements that the participant initials individually (that they have read the information sheet of a specified version and date, that participation is voluntary, that authorised people may inspect their records, what will happen to data if they withdraw), with any optional items, such as the use of anonymised data in future research, offered as separate yes/no choices that do not affect participation. If the participant cannot read, an impartial witness confirms that the

information was read accurately and consent was freely given.

Who takes consent matters. Ideally it is a trained member of the study team, delegated this task by the principal investigator, who is not the treating clinician under time pressure, so that the patient does not feel that their care depends on agreeing. Nurses and pharmacists are often the people patients ask afterwards "what did I sign?"; the whole team should understand the study well enough to answer.

Withdrawal is a right, not a failure. Participants may leave at any time without giving a reason and without any effect on their care. In Paper 1, two patients in the control group withdrew consent and were therefore not analysed, which is why the trial analysed 48 of the 50 patients randomised (Boulet et al. 2024). The protocol and consent form should say in advance what happens to data already collected when someone withdraws.

The test of informed consent

If the participant cannot explain the study back to you in their own words (what will happen, that they can say no, and whom to call), consent is not informed, whatever the form says.

17.7 Consent when the patient cannot decide

Much important research involves people who cannot consent at the moment they become eligible: patients in septic shock, sedated and ventilated patients, people with delirium, head injury or cardiac arrest. Excluding them all would leave the sickest patients without evidence-based treatment, which is itself unjust. The Declaration of Helsinki therefore allows research with people who lack capacity, but only when the research cannot be done with people who can consent, when it relates to the condition that causes the incapacity, and with additional safeguards (World Medical Association 2024; Council for International Organizations of Medical Sciences 2016). The figure below shows the usual sequence of decisions in an intensive care unit.

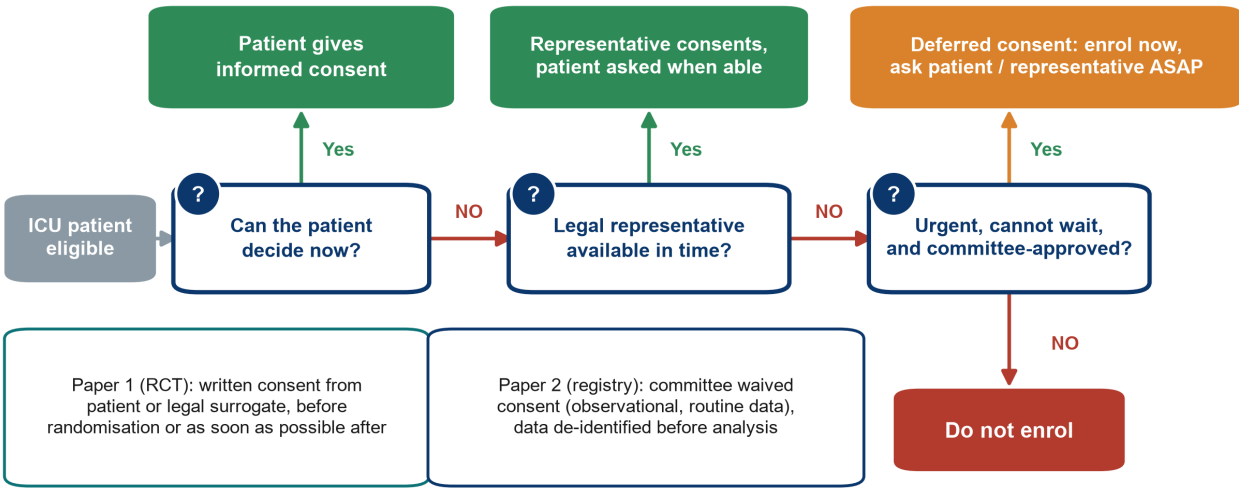


Figure 17.4: Consent for an eligible ICU patient: the patient consents if able; if not, a legally authorised representative consents and the patient is asked when able; if no representative can be reached in time and treatment cannot wait, deferred consent may be used only if the ethics committee approved it in advance. Paper 1 used written consent from patient or surrogate, before randomisation or as soon as possible after; Paper 2's committee waived consent for routine registry data.

1. Can the patient decide now? Capacity is specific to a decision and a moment. A patient with

pneumonia who is awake and oriented can usually consent for themselves, even in an ICU, and should be asked.

2. If not, is a legally authorised representative available in time? A **representative** (also called a surrogate, usually a close relative, as defined by local law) may consent on the patient's behalf. The representative is asked to decide on the basis of what the patient would have wanted, not what they themselves would prefer, so a version of the information sheet addressed to the representative ("your relative") is needed. When the patient regains capacity they must be told about the study and asked for their own consent to continue; they may withdraw.

3. If no representative can be reached in time and the intervention cannot wait, has the committee approved deferred consent? In true emergencies, where the treatment must begin within minutes or hours, some regulations allow the patient to be enrolled first and consent to be sought from the representative or the patient as soon as possible afterwards. This is **deferred consent** (sometimes called delayed consent or research without prior consent). It is legitimate only if the protocol explains why prior consent is impossible and the ethics committee has approved the procedure *in advance*. If neither condition holds, the patient is not enrolled.

The protocol and consent documents must also say what happens in the difficult cases: if the representative or patient later refuses (the patient leaves the study; whether data already collected may be used depends on the approved protocol and on what they agree to), and if the patient dies before consent could be sought. A continuing-consent form lets a recovered patient choose between continuing, stopping but allowing use of data collected so far, and stopping with removal of their data where this is still possible.

Paper 1 used exactly this pathway: written informed consent was obtained from patients or their legal surrogates "either before randomization or as soon as possible thereafter", under approval from a French regional ethics committee (number 2020-A01952-37) (Boulet et al. 2024).

A fourth route applies to a different kind of study. For **minimal-risk research using routine data**, where asking for consent is impracticable, the committee may grant a **waiver of consent**. Paper 2 analysed a prospectively collected sepsis registry of observational data from 20 hospitals; the institutional review board of each hospital approved the study and waived the requirement for individual consent "because of the observational nature of the study", and the data were anonymised and de-identified before analysis (Hyun et al. 2023). The criteria a committee usually applies are well established (Council for International Organizations of Medical Sciences 2016):

Table 17.3: Criteria for a waiver of consent, applied to an observational fluid-balance cohort.

Criterion for a waiver	How a FLUID-ICU-type cohort would meet it
No more than minimal risk	Observational; data already recorded for clinical care; no extra procedures
Consent impracticable	Most patients are sedated or in shock on admission; those who die early, often before a relative is reached, would be missed, biasing the results
Rights and welfare not harmed	Data pseudonymised; no care decision depends on the study
Participants informed when possible	A notice in the ICU and for relatives; recovered patients receive a leaflet and may ask for their data to be withdrawn
Permitted by law	National regulations allow a waiver for minimal-risk research on health records when approved by a committee

The second row deserves attention. A strong waiver request does not merely say that asking for

consent would be inconvenient; it explains that requiring consent would make the results *wrong*. If only patients who survive long enough to consent, or whose families arrive quickly, are included, the cohort systematically loses the sickest patients, a form of selection bias (Chapter 6).

Worked example: who consents for these three ICU patients?

Three patients become eligible for a randomised trial of fluid strategies. Walk each through the flowchart.

Mr A, 54 years, has pneumonia and is awake and oriented. He can decide now, so **Mr A gives his own consent**.

Mrs B, 71 years, is in septic shock, sedated and ventilated. Her daughter is at the bedside. Mrs B cannot decide now, but a representative is available. If local law recognises the daughter as a legally authorised representative, **the daughter consents**, on the basis of what she believes her mother would want. When Mrs B regains capacity, she is told about the study and asked whether she wishes to continue; she may withdraw.

Mr C, 38 years, develops shock at 03:00 and no family can be reached. The treatment under study must start now. **Deferred consent** is possible only if the protocol describes it and the committee approved it in advance; if so, Mr C is enrolled and consent is sought from his family or from him as soon as possible. If the committee did not approve deferred consent, Mr C is not enrolled.

Note what is *not* on this list: the treating doctor cannot consent on a patient's behalf. The doctor is not the patient's representative, and has an obvious interest in the study.

17.8 Confidentiality and data protection

Participants entrust researchers with information about their health, often without being able to check how it is used. Protecting it is a legal duty under data-protection law and an ethical one under the Declaration of Helsinki. The key principle is that confidentiality is protected **by design**, not by good intentions.

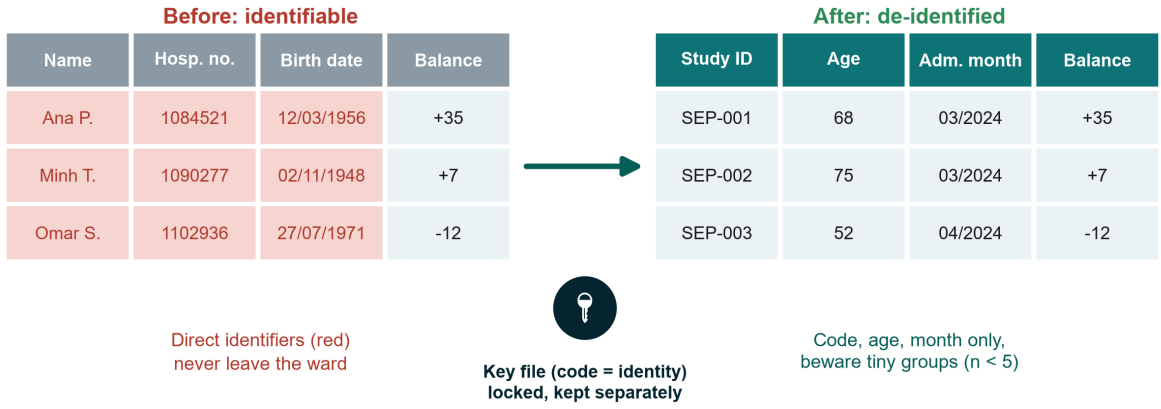


Figure 17.5: De-identification before analysis: direct identifiers such as name, hospital number and exact birth date stay on the ward; the analysis file carries only a study code, age, admission month and the study variables. The identifying file linking codes to identities is locked and kept separately. Small groups need care because they can identify people even without names.

The figure above shows the essential step. **Direct identifiers** (name, hospital number, exact date of birth, address, telephone number, photographs) are recorded only where they are needed and

never travel with the research data. Each participant receives a **study code** (for example SEP-001), and the analysis file holds only the code and the variables listed in the data dictionary, with dates coarsened where possible: age rather than birth date, admission month rather than exact date. The **identifying file** that links codes to identities is kept separately, password-protected, on an institution-approved server, with access limited to one or two named people. Paper copies of case report forms are kept in a locked cabinet; the database has individual log-ins and an audit trail.

Three further points are often missed. First, removing names is not always enough. **Indirect identification** is possible when a combination of characteristics is rare: "the only 19-year-old on ECMO in March" is identifiable without a name. Results for very small groups (say, fewer than five people) should be reported with care. Second, the terms differ: **pseudonymised** data can be re-identified by someone holding the key, and remain personal data under most laws; **anonymised** data cannot reasonably be re-identified by anyone. Third, the practical rules matter as much as the principles: no patient data on personal phones, messaging apps, private e-mail, USB sticks carried in pockets, or public cloud and artificial-intelligence tools; storage only on encrypted, institution-approved systems; and a stated retention period after which data are destroyed securely.

Paper 2 states that patient information was anonymised and de-identified before analysis (Hyun et al. 2023). That single sentence answers one of the first questions reviewers and editors ask about registry studies.

17.9 Worked example: the ethics paragraph of a cohort paper

Everything in this chapter so far eventually becomes a few sentences in the Methods and Declarations of a paper (Chapter 19). FLUID-ICU is simulated, so its own paper simply states: "This analysis used a simulated teaching dataset; no ethics approval or consent was required." But suppose it were a real multicentre cohort.

Worked example: four questions, four sentences

The ethics paragraph of an observational cohort answers four questions.

- **Approval.** Who approved the study, with what number and date, and was it before data collection? *"The study was approved by the research ethics committees of the four participating hospitals (approval numbers and dates) before data collection began."*
- **Consent.** Who consented, or why was consent waived? *"The committees waived individual consent because data were collected during routine care, the study carried no more than minimal risk, and consent was impracticable for patients admitted in shock."*
- **Confidentiality.** How were data protected? *"Patients were identified by study codes only; the identifying file was kept in each ICU, and data were de-identified before central analysis."*
- **Registration and sharing.** Was the study registered, and can the data be obtained? *"The study was registered at (registry, number). De-identified data and the data dictionary are available from the corresponding author under a data-use agreement, subject to committee approval."*

Paper 1 places its committee, approval number, consent procedure and funding inside the Methods; Paper 2 gives its IRB approval number and the consent waiver, with the reason, in its Declarations. Both forms are acceptable; follow the journal's instructions.

17.10 Feasibility: five honest questions

A study that is ethically sound and scientifically elegant can still fail because it cannot be done with the people, time, money and patients available. Feasibility failures are not harmless: participants who were enrolled in a trial that never reaches its sample size have borne burden for no answer. The committee will ask about feasibility, and so should you.

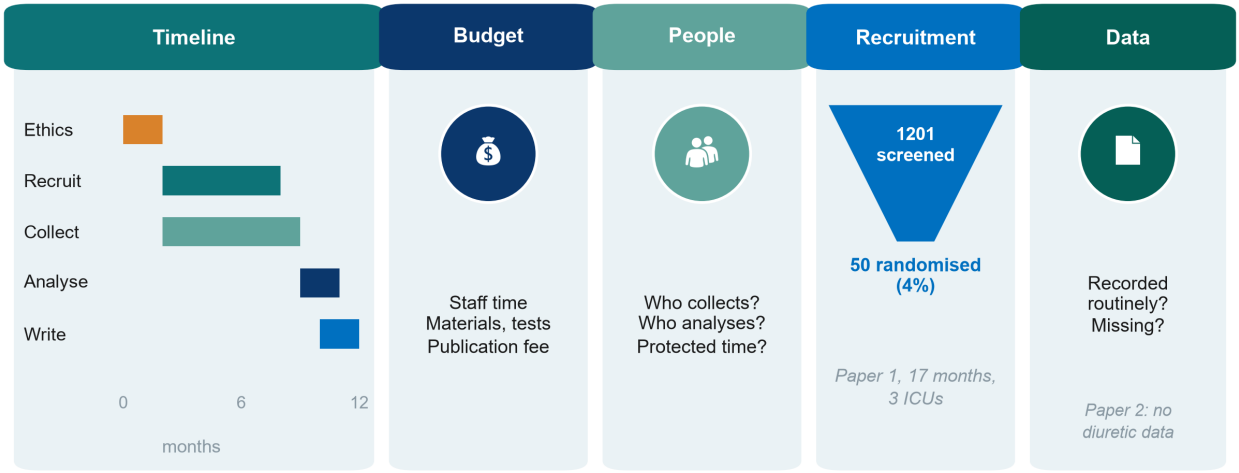


Figure 17.6: Five feasibility checks: a realistic timeline including ethics review; a budget with named headings; people to collect and analyse data, with protected time; recruitment based on real numbers (Paper 1 screened 1,201 patients in three ICUs over 17 months to randomise 50, about 4%); and data that are actually recorded (Paper 2 had no diuretic data).

Timeline. Ethics review alone often takes one to three months, recruitment almost always takes longer than planned, and analysis and writing need months, not weeks. Draw the timeline as a simple chart, as in the figure above.

Budget. Even an unfunded study has costs. Name the headings, even if the amounts are small: staff time for data collection, materials and tests, data management and statistical support, ethics fees, open-access publication charges, and a contingency.

People. Who screens and collects data, including on night shifts and out of hours? Who manages and analyses the data? Does anyone have protected time, or does the study depend on goodwill after long shifts?

Recruitment. This is where most studies fail. The number of patients you *need* (from the sample-size calculation, Chapter 12) must be compared with the number you can *get*. Count how many patients meeting your criteria your unit actually saw last year, then assume you will enrol only a fraction. Paper 1 screened 1,201 patients in three ICUs over 17 months to randomise 50, about 4% (Boulet et al. 2024). Most patients who look eligible turn out not to meet the criteria, are missed, or decline.

Data. Is the variable you need recorded routinely, and recorded well? A cohort built on routine data can only study what was recorded. Paper 2 could not examine whether diuretics modified the effect of fluid accumulation, because the registry never recorded them, and says so among its limitations (Hyun et al. 2023). A short pilot of the case report form on a handful of patients (Chapter 10) quickly shows which fields are missing in practice.

Worked example: a feasibility paragraph

"The unit admitted about 180 adults with septic shock last year. Allowing for exclusions and missed patients, we expect to enrol 120 eligible patients in 12 months, which meets the required sample size of 110. Two nurses per shift have been trained on the case report form, and data completeness is checked weekly. Budget headings: staff time for data entry, printing, statistical support and an open-access fee."
Notice that the expected recruitment comes from the unit's admissions, not from the sample-size calculation, and that the paragraph names who will do the work.

! Common mistake: consent planned in one vague sentence

"Informed consent will be obtained" is the most common weakness in ethics applications for ICU studies. The committee needs to know *who* consents (patient, representative, deferred, or a waiver requested), *who* takes consent, *when* and *where*, with which documents, and what happens when the patient regains capacity. Write the pathway out in full.

Summary

Research is ethically different from care because part of what is done serves knowledge rather than the individual patient, and so it needs independent safeguards. A study is ethical only if it is worth doing, well designed, fair in whom it includes, favourable in its balance of risks and benefits, independently reviewed, based on valid consent, and respectful of participants throughout. The protocol is the written, version-controlled plan of the study, approved before the first participant and changed only through formal amendments; SPIRIT describes its contents for trials. The Declaration of Helsinki (2024) states the principles, Good Clinical Practice, ICH E6(R3), describes how to run a trial so that participants are safe and data reliable, and a research ethics committee reviews every study, including retrospective ones, before data collection begins. Trials must be registered before the first participant; systematic reviews should be; observational studies are encouraged to be. Valid consent is informed, understood, voluntary and documented. When patients cannot decide, a legal representative may consent, deferred consent may be used in emergencies if approved in advance, and committees may waive consent for minimal-risk research on routine data. Confidentiality is protected by design through study codes, a separate identifying file and secure systems, and an honest feasibility check, based on real admission numbers, protects participants from studies that cannot finish.

Further reading

- The Declaration of Helsinki in its 2024 revision (World Medical Association 2024) is short and readable; every clinical researcher should read the full text once. The CIOMS guidelines (Council for International Organizations of Medical Sciences 2016) give more practical detail on consent waivers and research with people who lack capacity.
- Emanuel, Wendler and Grady's paper on what makes clinical research ethical (Emanuel et al. 2000) remains the clearest short framework for thinking about any study.
- The SPIRIT 2025 statement (Chan et al. 2025) and its explanation document are the best guide to writing a protocol, even for non-trial studies.

- ICH E6(R3) (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use 2025) explains the quality-by-design approach to running trials; Hulley and colleagues (Hulley et al. 2013) cover ethics and protocol writing for clinical researchers in an accessible way.

CHAPTER 18

Research integrity, authorship and conflicts of interest

Clinicians act on published research. A guideline panel pools trials into a meta-analysis; a pharmacist changes a dilution protocol because of a cohort study; a nurse argues for a new fluid chart because a paper showed that hidden fluids add up. All of this depends on trust: that the patients existed, that the numbers were measured and not invented, that every outcome was reported and not only the convenient ones, that the people named as authors did the work and stand behind it, and that readers know who paid for the study and who might gain from its conclusions. **Research integrity** is the set of practices that make that trust justified.

This chapter is not mainly about catching cheats. Serious fraud is rare. Most integrity problems start as small shortcuts under pressure: a missing value "filled in", an awkward outlier "cleaned", a head of department added to the author list, a paragraph copied into an introduction, a chatbot asked to "write the discussion". The aim is to recognise the slope early, and to build habits and systems that keep honest people on solid ground. As in the rest of Part IV, examples come from fluids in septic shock: Paper 1, a randomised pilot trial (Boulet et al. 2024), Paper 2, a registry cohort (Hyun et al. 2023), and FLUID-ICU, a cohort of 620 patients **simulated for teaching**.

In this chapter you will learn

- the three forms of research misconduct (fabrication, falsification and plagiarism) and how they differ from honest error;
- how duplicate publication, salami slicing and selective reporting distort the evidence, and how registration and pre-specification protect against them;
- who qualifies as an author under the ICMJE criteria, what gift and ghost authorship are, and how CRediT describes each contributor's role;
- what counts as a conflict of interest, how to declare it, and what a funding statement must say;
- how to plan and state data sharing;
- what artificial-intelligence tools may and may not be used for in research and writing.

18.1 What research integrity means

Integrity in research rests on a few simple values: **honesty** in proposing, performing and reporting research; **accountability** for what one does and publishes; **fairness** in giving credit and in dealing with colleagues and participants; and **transparency** about methods, data, funding and interests. Institutions, journals and bodies such as the Committee on Publication Ethics (COPE) translate these values into rules and procedures (Committee on Publication Ethics 2017), but the rules only work when researchers understand why they exist.

How common are serious breaches? In a systematic review of anonymous surveys, about 2% of scientists admitted having fabricated, falsified or modified data at least once, and around a third admitted other questionable practices; when asked about colleagues, the figures were considerably higher (Fanelli 2009). Whatever the true figure, even rare misconduct can do great harm when the work is influential, because a single fabricated trial can shift a meta-analysis and a guideline.

18.2 Fabrication, falsification and plagiarism

The three classic forms of research misconduct are usually abbreviated **FFP**.

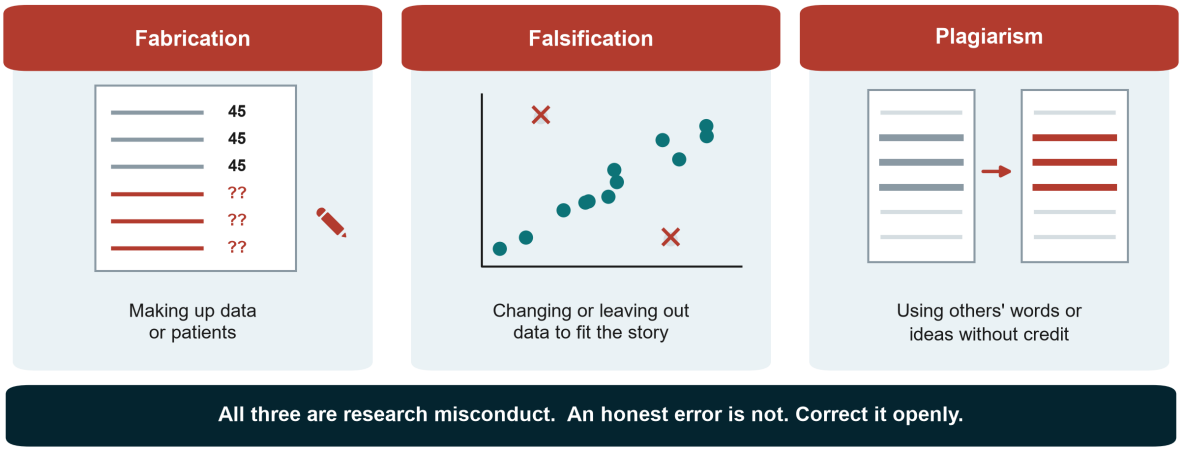


Figure 18.1: The three forms of research misconduct: fabrication (making up data or patients), falsification (changing or leaving out data to fit the story) and plagiarism (using others' words or ideas without credit). An honest error is not misconduct, but it must be corrected openly.

Fabrication is making up data, results or patients. The clinical version is rarely a whole invented study; it is far more often a gap filled from memory. Imagine that three patients' day-3 fluid charts have been lost, and a resident who "remembers roughly" what their balances were enters values so that the dataset is complete. Those values are invented, however well meant. The honest course is to record them as **missing**, report how many were missing, and handle them in the analysis (Chapter 15).

Falsification is manipulating research materials, equipment or processes, or changing or omitting data or results, so that the research record no longer represents what happened. Suppose that, after seeing the results, a team removes two patients with "strange" fluid values because they weaken the association, and the paper does not mention it. That is falsification. The protection is to decide in advance, in the data-quality plan (Chapter 10), what counts as an implausible value and what will be done about it: check the value against the source document; correct it if it was a transcription error; and if exclusion is justified, report the exclusion and show the results with and

without those patients as a sensitivity analysis. Pre-specified rules protect you from accidental falsification as much as from the deliberate kind.

Plagiarism is using other people's words, ideas, data or figures without appropriate credit. Its commonest form in clinical writing is copying sentences from a published paper or review into one's own introduction. Copying word for word *with* a citation but *without* quotation marks is still plagiarism, because it presents another author's wording as one's own. In clinical writing, direct quotation is rare; the norm is to paraphrase in your own words and cite the source. Journals routinely run submissions through similarity-checking software, which finds copied passages easily. **Self-plagiarism**, re-using large parts of one's own published text without acknowledgement, raises the same concern when it misleads readers about what is new; re-using a standard methods description from one's own earlier paper is usually acceptable if it is cited.

Honest error is not misconduct. Everyone makes mistakes: a mistyped number, a wrong figure call-out, a coding error discovered after publication. The difference lies in what happens next. An honest error is corrected openly, through a correction (erratum) to the published paper, or through retraction if the error invalidates the conclusions. Concealing a known error, or refusing to correct it, turns an error into an integrity problem.

- **Missing data are reported as missing**

  Never invented, never quietly removed. Missing values are a normal part of clinical research; how many there were, and how they were handled, is information the reader needs.

18.3 Distorting the record without inventing anything

Some practices distort the evidence even when every number is genuine. They matter because doctors and guideline panels rely on the published literature as a whole, not on single papers.

Duplicate (redundant) publication means publishing the same study, with the same or substantially overlapping results, in more than one place without saying so. Its main harm is that the same patients may be counted twice in a meta-analysis, exaggerating both the precision and, if the duplicated study was positive, the apparent effect. Legitimate secondary publications exist: a long-term follow-up, a pre-specified secondary analysis, a translation for a different readership. They are acceptable when they are clearly cross-referenced to the main paper and add something new, and when the editors know.

Salami slicing means cutting one study into the thinnest publishable pieces to multiply the number of papers. A cohort of ventilated patients published as three papers in three journals, one on mortality, one on length of stay and one on ventilator days, none mentioning the others, is the classic example. Readers never see the whole picture, the multiplicity of analyses is hidden, and patients can again be counted more than once. The remedy is one main paper reporting all pre-specified outcomes, with any genuine secondary paper cross-referencing it.

Selective reporting means publishing only the results that turned out "significant" or interesting. Its two forms are **publication bias**, where whole studies with disappointing results are never published, and **selective outcome reporting** (including **outcome switching**), where a study is published but the primary outcome is changed, or unfavourable outcomes are omitted, after the data have been seen. An empirical comparison of trial protocols with their publications found that around half of efficacy outcomes and two thirds of harm outcomes were incompletely reported, and that statistically significant outcomes were more likely to be reported fully (Chan et al. 2004). If only

positive results appear, the literature systematically overstates benefit and understates harm.

The defences are those described in Chapter 17: register the study before the first participant, pre-specify **one** primary outcome and the analysis plan, and report all of it. Paper 1 is a good example. It registered its protocol and statistical analysis plan at ClinicalTrials.gov on 1 July 2021, before recruitment began in September 2021, and its abstract reports plainly that the restrictive strategy "did not reduce the overall fluid balance at day 5" (Boulet et al. 2024). A negative pilot trial published with an honest abstract is exactly what the evidence base needs.

Closely related are the **questionable research practices** that sit between good practice and misconduct: trying many analyses and reporting the one that gives $p < 0.05$ ("p-hacking"); presenting a hypothesis formed after seeing the results as if it had been planned ("HARKing", hypothesising after the results are known); adding covariates, subgroups or patients until the result looks better; or stopping recruitment when the result happens to be favourable. Each one inflates the chance of a false-positive finding (Chapter 12). Again, pre-specification is the protection: analyses planned in the statistical analysis plan are confirmatory, anything else is labelled exploratory.

18.4 Authorship: who is an author?

Authorship confers credit, which matters for careers, and **responsibility**, which matters for patients. An author is someone who can stand behind the work and answer for it. The most widely used definition is that of the International Committee of Medical Journal Editors (ICMJE), which requires **all four** of the following criteria for every author (International Committee of Medical Journal Editors 2024).

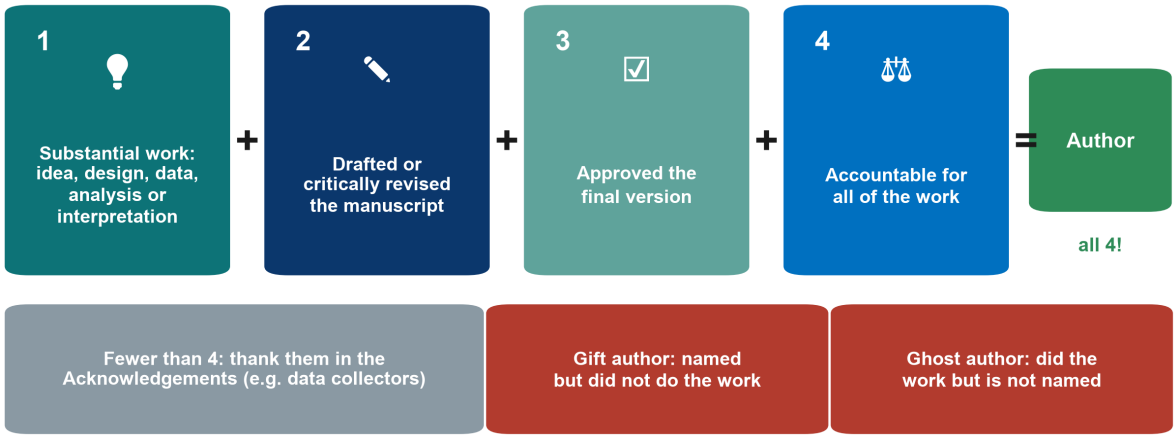


Figure 18.2: The four ICMJE criteria for authorship: a substantial contribution to the idea, design, data, analysis or interpretation; drafting or critically revising the manuscript; approving the final version; and agreeing to be accountable for all of the work. All four are needed; people who meet fewer are thanked in the Acknowledgements.

1. **Substantial contribution** to the conception or design of the work, or to the acquisition, analysis or interpretation of data.
2. **Drafting** the work or **revising it critically** for important intellectual content.
3. **Final approval** of the version to be published.
4. **Agreement to be accountable** for all aspects of the work, in ensuring that questions about the accuracy or integrity of any part are appropriately investigated and resolved.

Notice what these criteria exclude. Collecting data alone, providing patients, supervising the department in general terms, lending equipment, obtaining funding alone or providing language editing does not, by itself, make someone an author. Such contributors are thanked by name in the **Acknowledgements**, with their permission. Paper 1's acknowledgements, for instance, thank named people for help drafting the protocol, regulatory support, data collection and editorial assistance (Boulet et al. 2024). Conversely, someone who meets the first criterion must be given the opportunity to meet the others; it is not acceptable to exclude a contributor by never sending them the draft.

Two forms of authorship misconduct are named in the figure. A **gift (honorary) author** is named but did not do the work, typically a senior colleague added for seniority, friendship or the hope of easier acceptance. A **ghost author** did substantial work but is not named, typically a junior researcher, a statistician or a professional writer paid by a sponsor. Both mislead readers about who is responsible for the paper.

Group authorship is used by large networks. The named authors must each meet all four criteria, and the members of the study group who contributed are listed, for example in the Acknowledgements or an appendix. Paper 2 is published "on behalf of" the Korean Sepsis Alliance Investigators, and lists the steering committee and participating centres in its Acknowledgements, which is the correct way to credit a large collaboration (Hyun et al. 2023).

Author order is not defined by ICMJE; conventions differ between fields and countries. In clinical research, the first author usually did most of the work and wrote the first draft, and the last author is often the senior researcher who supervised it, which is legitimate only if that person meets all four criteria. Because order matters to people, disagreements are common and painful. The practical solution is simple: agree the provisional author list, order and roles **in writing at the protocol stage**, and revisit the agreement before submission, when contributions are known.

18.5 CRediT: saying who did what

Lists of names do not tell readers what each person did. The **Contributor Roles Taxonomy (CRediT)** provides 14 standard roles for describing contributions (Brand et al. 2015): conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing – original draft; and writing – review and editing. Most journals now ask for a CRediT statement. It does not replace the ICMJE criteria (it describes contributions; it does not decide who is an author) but it makes gift and ghost authorship harder to hide.

Paper 1 gives a model CRediT statement. The first author is credited with writing the original draft; the trial statistician with methodology and formal analysis; several investigators with investigation and validation; senior authors with review and editing and supervision; and the senior investigator with conceptualization, supervision and funding acquisition (Boulet et al. 2024). A reader can see at once who analysed the data and who conceived the study. Paper 2 describes contributions in free text (who conceived the study, who collected data, who analysed, who wrote the first draft) and states that all authors read and approved the final manuscript (Hyun et al. 2023); this meets the ICMJE requirements, but the CRediT format is what most journals now request.

Worked example: an author list for an observational ICU study

A team plans a cohort study of fluid balance and 28-day mortality in one ICU. They agree the following at the protocol stage.

Table 18.1: Planned author contributions for an observational ICU study.

Person	CRediT roles	Author?
Principal investigator (intensivist)	Conceptualization, Methodology, Writing – original draft	Yes, if all four criteria met
Two ICU nurses	Investigation, Data curation; one also Writing – review and editing	The nurse who also revises the draft critically, approves it and accepts accountability: yes. Data collection alone: Acknowledgements
Clinical pharmacist	Investigation, Writing – review and editing	Yes, if all four criteria met
Statistician	Formal analysis, Methodology	Yes, if they also revise, approve and accept accountability
Head of department	Supervision	Only if all four criteria are met; otherwise Acknowledgements

The table makes explicit, early, what each person must do to be an author, and invites contributors to take on the writing and checking roles that authorship requires.

18.6 Conflicts of interest and funding

A **conflict of interest** (or competing interest) exists when a person's judgement about the primary interest, the validity of the research and the welfare of participants, could be influenced, or could reasonably be seen to be influenced, by a secondary interest. The definition depends on perception: it does not matter whether the researcher believes they are unaffected, only whether a reasonable reader would want to know.

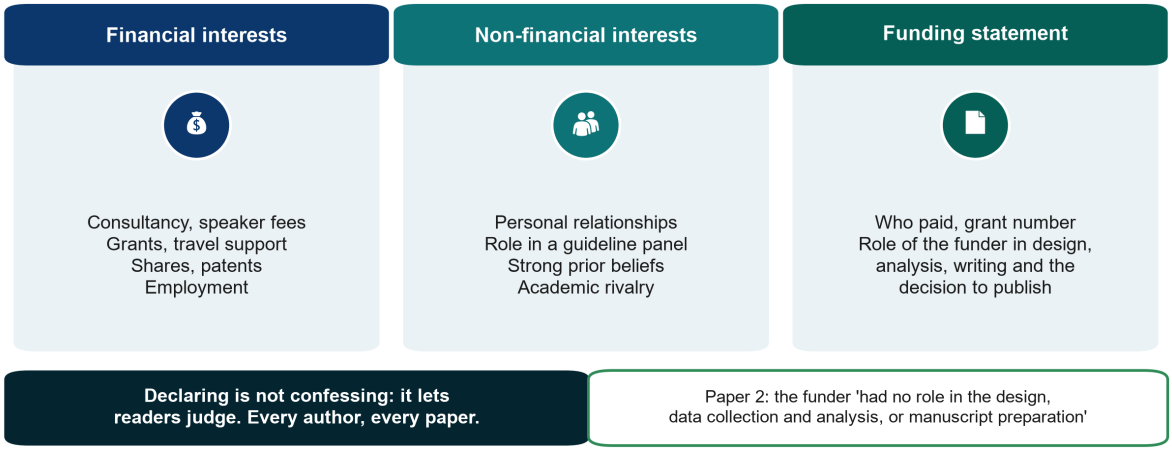


Figure 18.3: Financial interests (consultancy, speaker fees, grants, travel support, shares, patents, employment), non-financial interests (personal relationships, a role in a guideline panel, strong prior beliefs, academic rivalry) and the funding statement (who paid, grant number and the funder's role). Declaring is not confessing: it lets readers judge.

Financial interests are the most obvious: consultancy and speaker fees, research grants, travel support, shares, patents and employment, from any organisation that could gain or lose from the results. **Non-financial interests** are subtler but real: personal relationships, membership of a guideline panel that has taken a position, strong prior beliefs or a reputation built on one answer, and academic rivalry.

Having an interest is common and is not wrong. A clinician who has spoken at an industry-sponsored meeting can still do excellent research. What is wrong is **hiding** an interest. A declared interest is information; a hidden one, discovered later, destroys trust in the paper and its authors. Every author completes the journal's disclosure form (most journals use the ICMJE disclosure form, which typically covers the previous 36 months), and the paper carries a competing-interests statement, even if it says "The authors declare no competing interests", as Paper 1 does (Boulet et al. 2024). When in doubt, declare. If a drug company paid your travel to a congress last year, and the paper concerns a treatment area in which that company is active, a reader could reasonably see it as relevant.

The **funding statement** names who paid for the study, with grant numbers, and describes the **role of the funder** in the design, data collection, analysis, interpretation, writing and the decision to publish. The last part is an ICMJE requirement that is often forgotten. Paper 2 states it clearly: the work was funded by a national disease-control agency research programme, and "the funding body had no role in the design of the study, data collection and analysis, or manuscript preparation" (Hyun et al. 2023). Paper 1 states an internal grant from its university hospital (Boulet et al. 2024). When a funder did have a role, for example a company that supplied the drug and reviewed the manuscript, the statement must say exactly what that role was.

18.7 Data sharing

Readers increasingly expect to see the data and code behind the results. Shared data allow others to check analyses, combine studies in individual-participant-data meta-analyses and answer new questions without exposing new patients to research. Since 2018, the ICMJE has required trial reports to include a **data-sharing statement**, and trials must describe their data-sharing plan when they register (Taichman et al. 2017). Many journals now ask the same of observational studies.

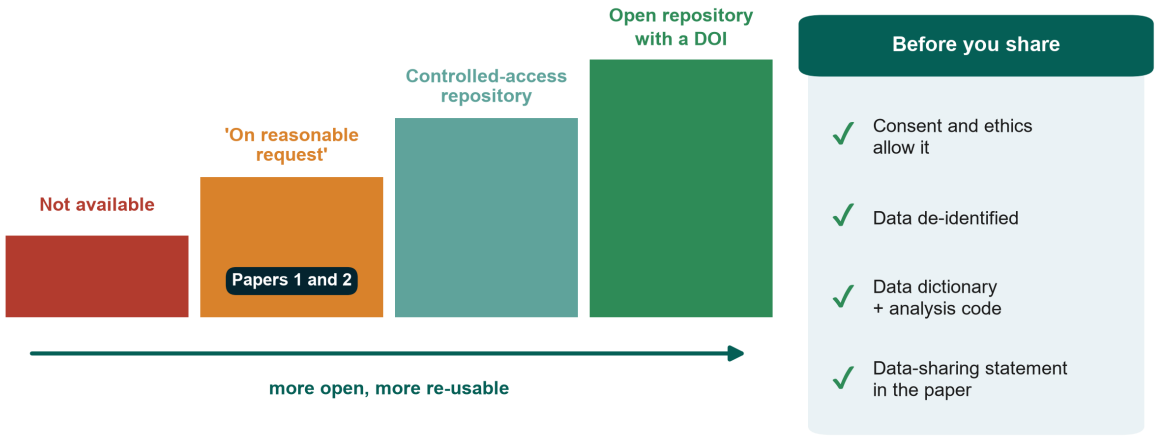


Figure 18.4: The data-sharing ladder, from not available, through "available on reasonable request" (used by both Papers 1 and 2), to a controlled-access repository and an open repository with a DOI. Before sharing: consent and ethics approval must allow it, data are de-identified, the data dictionary and analysis code accompany the data, and the paper contains a data-sharing statement.

The figure shows a ladder of openness. At the bottom, data are **not available**. Next comes **"available on reasonable request"** from the corresponding author, the weakest promise and the commonest statement; both Paper 1 and Paper 2 use it (Boulet et al. 2024; Hyun et al. 2023). Studies of such statements have found that requests often go unanswered, because authors move, files are lost and nobody is accountable. Higher up, a **controlled-access repository** holds the data and grants access to researchers who apply with a protocol and sign a data-use agreement, which suits sensitive clinical data. At the top, an **open repository with a DOI** makes fully de-identified data available to anyone. The FAIR principles summarise the goal: data should be **findable, accessible, interoperable and reusable** (Wilkinson et al. 2016).

Data sharing must be planned at the protocol stage, because the decisive step happens before the first participant: **the consent form and ethics approval must allow it**. A sentence asking permission for de-identified data to be shared for future research approved by an ethics committee, offered as an optional item on the consent form, is what makes sharing possible later. Shared data must be de-identified (Chapter 17), and they are usable only if the **data dictionary** and **analysis code** come with them (Chapter 10). A data-sharing statement then tells readers what is available, where, when and under what conditions.

Worked example: a data-sharing statement

For an observational fluid-balance cohort whose consent form included an optional data-sharing item:
"De-identified individual participant data, the data dictionary and the analysis code will be available from (repository name, DOI) from six months after publication, to researchers whose proposed use has been approved by an ethics committee, under a data-use agreement."

Compare this with “data are available on reasonable request”: the statement names what, where, when, to whom and on what conditions.

18.8 Artificial intelligence in research and writing

Generative artificial-intelligence (AI) tools, such as large language model chatbots, can draft text, translate, summarise papers, suggest code and produce images. They can save time, and they can also produce fluent, confident text that is wrong. Journals and publication-ethics bodies have converged on a small set of rules (International Committee of Medical Journal Editors 2024; Committee on Publication Ethics 2023; Flanagin et al. 2023).

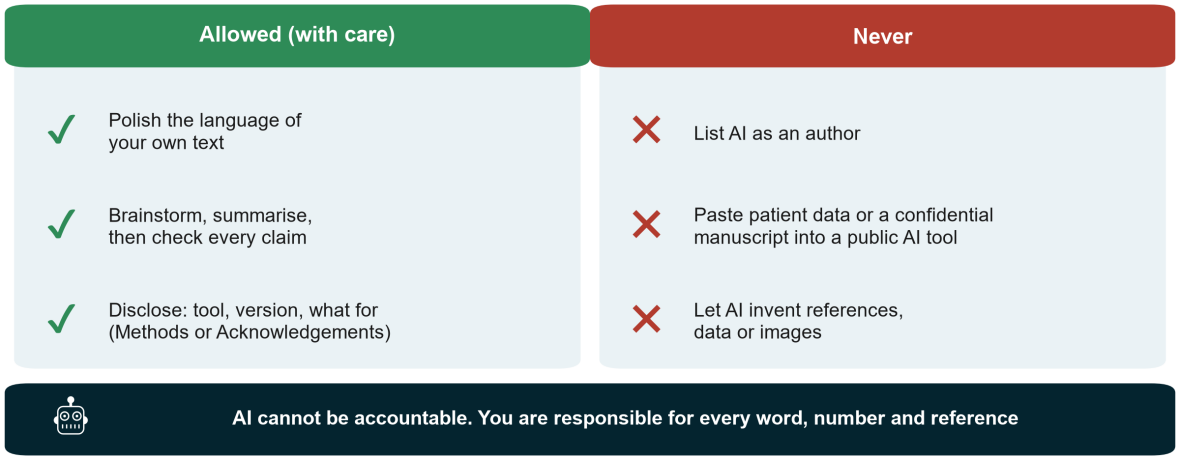


Figure 18.5: What AI tools may and may not be used for. Allowed with care: polishing the language of your own text, brainstorming and summarising with every claim checked, and disclosure of the tool, version and purpose. Never: listing AI as an author, pasting patient data or a confidential manuscript into a public tool, or letting AI invent references, data or images. The authors remain responsible for every word, number and reference.

AI tools cannot be authors. Authorship requires accountability (the fourth ICMJE criterion), and a software tool cannot take responsibility for the work, answer questions about it or consent to publication. **Authors who use AI must disclose it**: for help with writing, usually in the Acknowledgements or a dedicated statement; for use in data collection, analysis or figure production, in the Methods, naming the tool, the version and what it was used for. **Authors remain fully responsible** for every word, number and reference, including anything the tool produced.

Three dangers deserve special emphasis. First, **confidentiality**. Never paste identifiable patient data into a public AI tool: it is a breach of confidentiality and usually of data-protection law, because the data leave the institution’s control. The same applies to peer review: a manuscript you are reviewing is confidential, and uploading it to a public tool breaches that confidentiality. Second, **invented content**. Language models can generate references that look entirely real (plausible authors, journal, year and title) but do not exist, and can misstate numbers or findings from real papers. Every citation must be checked against the original source, for example in PubMed, and never cited without being read. Third, **images**. AI-generated or AI-altered images are not acceptable as research data.

Policies are changing quickly, and they differ between journals. Before submission, read the target journal’s own AI policy in its instructions for authors.

! Common mistake: asking a chatbot to write the Discussion

A colleague asks a chatbot to "write the discussion" of a fluid-balance paper and submits the result unread. In one step they have produced text for which no author can be accountable, very likely included references that do not exist or do not say what is claimed, and failed to disclose the use. Even if every sentence happened to be correct, the paper would misrepresent who did the intellectual work.

18.9 Worked example: green, amber or red?

The best way to learn where the lines fall is to classify real-looking situations. For each scenario below, decide whether it is **green** (acceptable as described), **amber** (acceptable only if something is fixed or disclosed) or **red** (misconduct or a serious breach).

Worked example: six integrity scenarios

Table 18.2: Six integrity scenarios classified.

#	Scenario	Verdict and reason	What to do instead
1	Three patients' day-3 fluid charts are lost; a resident enters the balances he "remembers roughly"	**Red: fabrication.** Remembered values are invented data	Record as missing, report the number, handle in the analysis; prevent with source checks
2	After seeing the results, two patients with "strange" values are removed because they weaken the association; the paper does not say so	**Red: falsification.** Selective removal after seeing results changes the findings	Pre-specified rules for implausible values; check against source; report any exclusion with a sensitivity analysis
3	The head of department, who took no part, asks to be last author "because it is my department"	**Red: gift authorship.** Seniority is not a contribution	Acknowledge, or invite a genuine contribution (critical revision) so all four criteria are met
4a	A nurse uses an AI tool to improve the English of the Discussion she wrote, checks every sentence and states this in the Acknowledgements	**Green.** Language editing of one's own text, checked and disclosed, is accepted	Check the journal's AI policy; name tool, version and purpose
4b	A colleague uploads the study spreadsheet, with names and hospital numbers, to a free public chatbot to "run the statistics"	**Red: breach of confidentiality,** and an analysis no one can account for or reproduce	De-identify; use institution-approved software; a named person runs and checks the analysis
5	A cohort is published as three papers (mortality, length of stay, ventilator days) in three journals, none mentioning the others	**Amber to red: salami slicing** and redundant publication	One main paper with all pre-specified outcomes; secondary papers cross-referenced and adding something new
6	Two paragraphs of a review are copied word for word into the Introduction, with the review cited but no quotation marks	**Red** (amber only if minor and fixed before submission): **plagiarism**	Paraphrase in your own words and cite

A seventh case is a useful test of the authorship criteria: a colleague lent the team an ultrasound machine and asks to be an author. Providing equipment is not authorship; the right place is the Acknowledgements.

18.10 When things go wrong

If you suspect misconduct in your own team, or in a paper you are reviewing or reading, the first step is to check the facts quietly and keep records; many apparent problems have innocent explanations. Concerns about a published paper go to the journal's editor, who will follow COPE procedures, which typically involve asking the authors for an explanation and, if necessary, referring the matter to their institution (Committee on Publication Ethics 2017). Concerns within an institution go to its research-integrity office or designated officer. Good institutions protect people who raise concerns in good faith.

The outcomes for a published paper range from a **correction** (an error that does not invalidate the conclusions), through an **expression of concern** (while an investigation is under way), to a **retraction** (when the findings are unreliable because of serious error or misconduct). Retraction is not only a sanction: researchers who discover a serious error in their own work and ask for retraction are doing exactly what integrity requires.

! Common mistake: thinking integrity is someone else's job

Many integrity problems in clinical research are system failures: no data-quality plan, no agreed author list, no secure storage, no one checking the tables against the text. A team that builds these into the protocol makes honest behaviour the easy default.

Summary

Research integrity means honesty, accountability, fairness and transparency at every stage of a study. Fabrication (inventing data), falsification (changing or selectively removing data) and plagiarism (using others' words or ideas without credit) are research misconduct; honest error is not, but it must be corrected openly. Missing data are reported as missing. Duplicate publication, salami slicing and selective reporting distort the evidence base even when every number is genuine; prospective registration, one pre-specified primary outcome and complete reporting are the main defences. Every author must meet all four ICMJE criteria (substantial contribution, drafting or critical revision, final approval and accountability); others are thanked in the Acknowledgements, and gift and ghost authorship are both misconduct. CRediT describes each contributor's role, and the author list is best agreed in writing at the protocol stage. Financial and non-financial interests must be declared, and the funding statement must describe the funder's role. Data sharing is planned before the first participant, because consent must allow it, and works only when data are de-identified and accompanied by a dictionary and code. AI tools cannot be authors, and their use must be disclosed. Patient data and confidential manuscripts must never be entered into public tools, and authors remain responsible for every word, number and reference.

Further reading

- The ICMJE Recommendations (International Committee of Medical Journal Editors 2024) are the reference for authorship, conflicts of interest, registration, data sharing and the use of AI in medical journals; they are free online and updated regularly.
- COPE's core practices (Committee on Publication Ethics 2017) and its position statement on authorship and AI tools (Committee on Publication Ethics 2023) explain how editors handle integrity problems and what they expect of authors.

- Fanelli's systematic review (Fanelli 2009) gives a sobering view of how often misconduct and questionable practices occur, and Chan and colleagues (Chan et al. 2004) show how selective outcome reporting distorts trial evidence.
- The FAIR principles (Wilkinson et al. 2016) and the ICMJE data-sharing statement requirement (Taichman et al. 2017) explain what good data sharing looks like.

CHAPTER 19

Writing the paper

A study that is never written up has, from the point of view of patients and other clinicians, never happened. And a study that is written up badly can be worse than unpublished: an exaggerated title, a result reported without its uncertainty or a Discussion that claims causation from an observational design can change practice in the wrong direction. Writing a research paper is therefore part of the research, not an administrative task after it. Fortunately, it is a learnable craft with a stable structure. Clinical papers follow a predictable shape, each section has a clear job, and international reporting guidelines list what readers need to find.

This chapter walks through the paper section by section, in the order in which it is best *written* rather than the order in which it is read. It draws heavily on two published papers about fluids in septic shock, which provide real examples of both good and poor writing: Paper 1, a small multicentre randomised pilot trial of a restrictive strategy for non-resuscitative fluids (Boulet et al. 2024), and Paper 2, a propensity-matched cohort from a national sepsis registry (Hyun et al. 2023). Both are good papers in well-regarded journals, and both contain errors that a careful co-author could have caught. For a paper of our own we use FLUID-ICU, a multicentre cohort of 620 adults with sepsis that is **simulated for teaching**; all its numbers are invented for this book, and the worked examples show how its primary question would be written up. FLUID-ICU and its analysis are described fully in Chapter 16.

In this chapter you will learn

- the IMRaD structure and the job of each section;
- why papers are best written in a different order from the one in which they are read;
- how to write Methods that another team could repeat, including the statistical-analysis paragraph;
- how to report Results (participants first, effect sizes with confidence intervals, no interpretation) and build tables and figures that stand alone;
- how to write an Introduction as a three-paragraph funnel and a Discussion in six moves;
- how to calibrate language to the study design, and how to avoid spin;
- how to write a structured abstract, a title and keywords, and how to use reporting guidelines such as CONSORT and STROBE.

19.1 The shape of a paper: IMRaD

Almost every original clinical research paper follows the **IMRaD** structure: Introduction, Methods, Results and Discussion, framed by the title and abstract at the front and closed by references and declarations at the back. The structure exists because it answers, in order, the four questions every reader asks: *Why did you do this? What did you do? What did you find? What does it mean?*

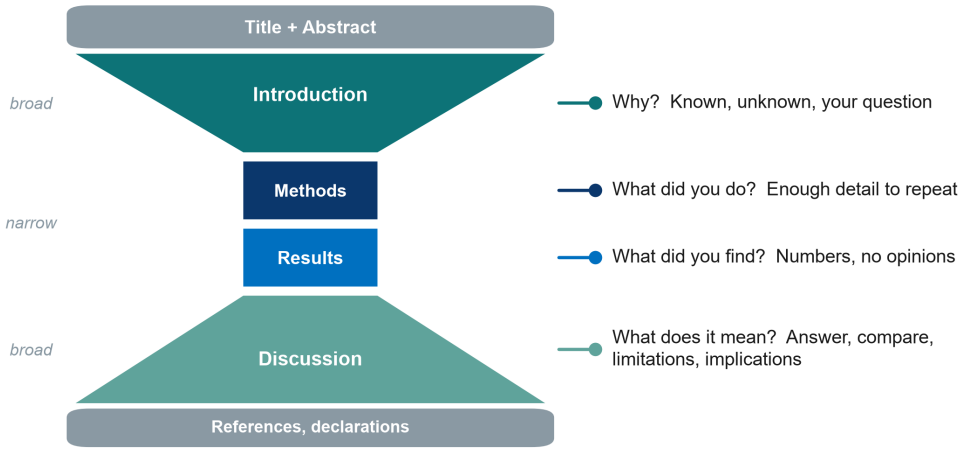


Figure 19.1: The IMRaD hourglass: the Introduction narrows from what is known to the study question; the Methods and Results are narrow and specific; the Discussion widens back out to what the findings mean.

The figure above shows why IMRaD is often drawn as an hourglass. The **Introduction** starts broad (the clinical problem) and narrows to one question. The **Methods** and **Results** are narrow: they describe exactly what was done to whom and exactly what was found, without opinions. The **Discussion** starts from the specific answer and widens again: how it compares with other studies, what its limitations are, and what it means for practice and research.

The hourglass also tells readers where to look. The study question (often in PICO form) is in the last paragraph of the Introduction. The design and primary outcome are in the Methods, the main effect is in the first paragraphs of the Results, and the limitations are towards the end of the Discussion. Many readers go first to the abstract and the conclusion, but the answer to the question “can I believe this?” is in the Methods.

19.2 Write in a different order from the one readers read

Readers meet the paper as title, abstract, Introduction, Methods, Results, Discussion. Writers who try to write in that order usually stall, often for months, on the Introduction, because they are trying to set up a story whose ending they have not yet written.

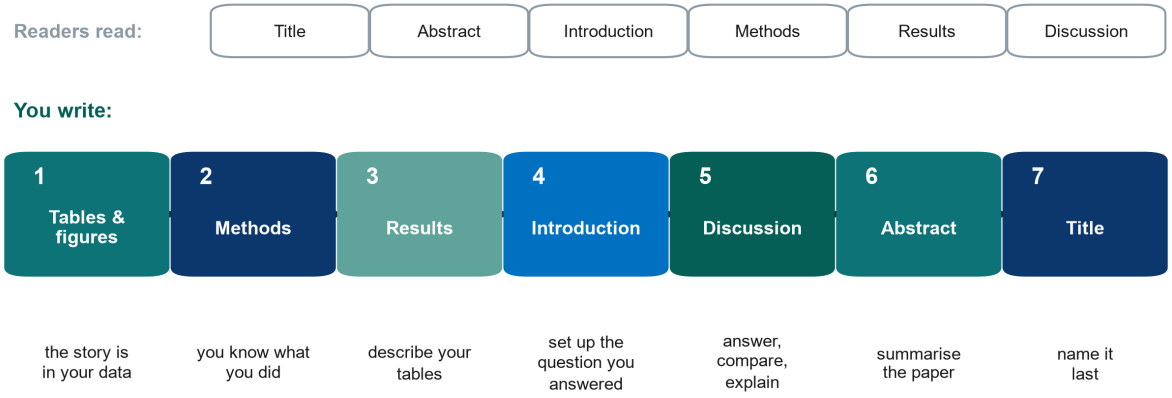


Figure 19.2: Readers read title, abstract, Introduction, Methods, Results and Discussion; writers do better to start with the tables and figures, then write the Methods, Results, Introduction and Discussion, and leave the abstract and title until last.

A more productive order, shown in the figure above, starts from what already exists.

1. **Tables and figures.** The story is in the data. Draft Table 1 (participant characteristics), the main outcome table and the key figure first; deciding what goes in them forces you to decide what the paper is about.
2. **Methods.** You know what you did, and most of it is already in the protocol and statistical analysis plan (Chapters 15 and 17). The Methods can be drafted before the data are even analysed.
3. **Results.** Describe the tables in words, in a logical order.
4. **Introduction.** Now set up exactly the question you answered, no more and no less.
5. **Discussion.** Answer, compare, explain, acknowledge limitations, draw implications.
6. **Abstract.** Summarise a paper that is finished.
7. **Title.** Name it last.

The logic is that each section is written from something that already exists, and the abstract and title summarise a finished paper, so they cannot promise more than it delivers. Before starting, two decisions help everything else: **choose the target journal** (its instructions for authors set the word limit, structure and reference style; Chapter 20) and **choose the reporting guideline** that matches the design (later in this chapter), which becomes your writing checklist.

It also helps to write down, before any drafting, the paper's **main message in one sentence**, including the key number. For FLUID-ICU's primary question it might be: *"In adults with sepsis, a positive fluid balance by ICU day 3 was modestly associated with higher 28-day mortality after adjustment for severity (adjusted OR 1.62, 95% CI 1.02 to 2.59)."* Every section should serve that sentence.

19.3 Methods: enough detail for someone to repeat the study

The Methods answer two questions: *what exactly was done?* and *can I trust it?* The test is simple: **could a competent colleague repeat the study from your Methods alone?** If not, add detail. The Methods are written in the past tense and follow the order of the relevant reporting checklist, usually under subheadings.

Table 19.1: The contents of the Methods section.

Subsection	What it contains
Study design and setting	The design named explicitly; where (type and number of sites), when (dates of recruitment and follow-up); registration number; where the protocol can be found
Participants	Eligibility criteria; how participants were identified and recruited (consecutive? convenience?)
Intervention or exposure, and comparator	Exactly what was done or measured, by whom, how much, how often, for how long
Outcomes	One primary outcome, then secondary and safety outcomes, each with its definition, unit and time point
Data sources and bias	Where data came from (case report form, registry, charts) and what was done to limit bias, including confounding
Study size	The sample-size calculation with every assumption and its source; or, for a fixed cohort, the statement that all eligible patients were included
Statistical analysis	Descriptive statistics; the test or model for every comparison; the effect measure with its 95% confidence interval; adjustment variables; missing data; subgroup and sensitivity analyses; multiplicity; software and version
Ethics	Committee name and approval number; consent procedure or waiver (Chapter 17)

The two papers show good and poor practice side by side. Paper 1 places its ethics committee, approval number, consent procedure and funding inside the Methods, answering the commonest editorial queries before they are asked. It describes the intervention well enough to replicate it (staff training before inclusion, a daily check of adherence by a clinical pharmacist, dilution charts in a supplement, rules for giving more fluid if needed). It separates one primary outcome from a list of secondary and safety outcomes, and it states its software and version (Boulet et al. 2024). Paper 2 defines precisely which inputs and outputs make up fluid balance, which is exactly where reviewers look for hidden assumptions, and cites the registry's cohort-profile paper instead of re-describing it (Hyun et al. 2023).

The same two papers also show what to avoid. Paper 1 is labelled a pilot but justifies its size with an efficacy calculation whose assumption (a control-group balance of 70 ml/kg) was not borne out. It names a Mann–Whitney test for the primary comparison but reports mean differences with confidence intervals, which that test does not produce. And it reports about 40 secondary p-values without a word on multiple comparisons. Paper 2 says patients were matched with "a caliper of 0.1" without saying 0.1 of what, does not say how missing covariates were handled or what the analysis was clustered by, and gives no statement about study size. None of these problems needed new data; each needed one or two more sentences.

19.3.1 The statistical-analysis paragraph

Reviewers often read the statistical paragraph first, because it tells them whether the results can be trusted. Its rule is simple: **name a method for every comparison you report, chosen before you saw the data.** Every number in the Results should be traceable to a sentence here. For each analysis, say how the data are summarised, which test or model was used, which effect measure is reported with its 95% confidence interval, what was adjusted for (and that the covariates were pre-specified), how missing data were handled, and how multiple comparisons were treated. End with the software and version.

Worked example: Methods for FLUID-ICU (simulated data)

Study design and setting. We analysed FLUID-ICU, a multicentre prospective cohort of consecutive adults with sepsis admitted to four intensive care units between January 2023 and December 2024. FLUID-ICU is a simulated teaching dataset; no ethics approval or consent was required.
Participants. All adults with sepsis enrolled in the cohort were eligible. Patients without a cumulative fluid balance on day 3 were excluded from the exposure analysis.
Variables. The exposure was a positive cumulative fluid balance by ICU day 3, defined as total fluid input minus total fluid output over ICU days 1 to 3 greater than 0 ml/kg, compared with a negative or zero balance. The primary outcome was death from any cause within 28 days of ICU admission. Confounders were pre-specified in the statistical analysis plan: age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease. Heart failure was pre-specified as a possible effect modifier.
Data sources and bias. Data were collected from ICU charts on a standard case report form. To address confounding by indication (sicker patients receive more fluid) we adjusted for markers of illness severity and comorbidity.
Study size. All eligible patients in the cohort were included; no separate sample-size calculation was performed.
Statistical analysis. Baseline characteristics are presented as mean (SD), median (IQR) or n (%), by fluid-balance group. Crude and adjusted odds ratios (ORs) for 28-day mortality were estimated by logistic regression. Survival to day 28 was described with Kaplan–Meier curves and analysed with Cox regression using the same covariates; the proportional-hazards assumption was checked. Effect modification by heart failure was tested with an interaction term. The main analysis used complete cases; missing lactate was handled by multiple imputation (20 imputed datasets) in a sensitivity analysis. Estimates are reported with 95% confidence intervals. Analyses used R (version stated).
Notice what each sentence does. The exposure definition is the sentence a reader needs to repeat the study. The study-size sentence does not invent a sample-size calculation for a fixed cohort. The bias sentence names the main threat (confounding by indication) and what was done about it.

19.4 Results: report, do not interpret

The Results say what was found, in numbers, and nothing more. Four rules cover most of what matters.

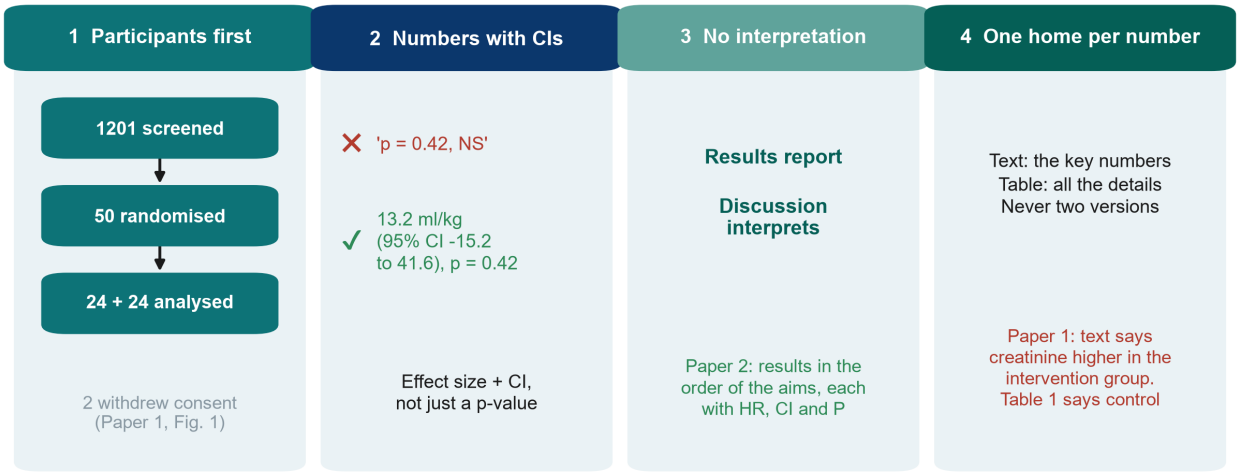


Figure 19.3: Four rules for the Results: participants first (Paper 1: 1,201 screened, 50 randomised, 24 + 24 analysed); numbers with confidence intervals rather than "p = 0.42, NS"; no interpretation, which belongs in the Discussion (Paper 2 reports each result in the order of its aims, with HR, CI and P); and one home per number, never two versions (Paper 1's text and Table 1 disagree about creatinine).

1. Participants first. Start with how many people were screened, eligible, included and analysed, and why people were lost at each step, ideally with a flow diagram. Paper 1 shows its flow chart in the main text: 1,201 patients screened, 50 randomised, two withdrawals, 48 analysed (Boulet et al. 2024). That 4% figure lets readers judge how representative the participants were. Paper 2 keeps its flow chart in the supplement, although 79% of the registry was excluded before matching, information that belongs in the main text (Hyun et al. 2023). Then describe the participants' baseline characteristics, referring to Table 1.

2. Numbers with confidence intervals. Report each outcome as an effect size with its 95% confidence interval, and add the exact p-value if you wish, never a p-value alone. "p = 0.42, NS" tells the reader almost nothing; "a difference of 13.2 ml/kg (95% CI −15.2 to 41.6; p = 0.42)" tells them the best estimate and the range of effects compatible with the data (Chapter 12). Give exact p-values (p = 0.044, not p < 0.05), except that very small values can be written p < 0.001. Every percentage should be traceable to its numerator and denominator. Call the analysis by its correct name: analysing 48 of 50 randomised patients is a *modified* intention-to-treat analysis, and Paper 1's Methods should have said so.

3. No interpretation. Words such as "because", "suggesting", "surprisingly" or "importantly" belong in the Discussion. Paper 2 is a good model here: it reports its results in the order of its aims (day 1, day 2, day 3), each with a hazard ratio, confidence interval and p-value, and no comment (Hyun et al. 2023). A useful check is to read each sentence of the Results and ask: is this something we found, or something we think?

4. One home for each number. Key numbers appear in the text; all the details are in the tables; and the two must agree. Paper 1's text says creatinine was higher in the intervention group while its Table 1 shows the opposite (Boulet et al. 2024). Text–table contradictions are a classic reason for rejection, because they make readers wonder what else is wrong.

The order within the Results usually mirrors the Methods: participants and their characteristics; the

primary outcome; secondary outcomes; harms (for trials); then subgroup and sensitivity analyses. For observational studies, STROBE asks for both the **crude** and the **adjusted** estimates, with a statement of what was adjusted for, because the difference between them is itself informative (Elm et al. 2007).

Worked example: Results for FLUID-ICU (simulated data)

Participants. Of 620 patients enrolled, 618 had a day-3 fluid balance and were included in the analysis; 527 had complete data for the adjusted model (91 without lactate or age). Mean age was 62.6 years (SD 15.2), 376 of 620 (60.6%) were men, and the median SOFA score was 7 (IQR 5 to 9).
Descriptive data. A positive balance by day 3 occurred in 373 of 618 patients (60.4%). These patients were sicker at baseline (median SOFA 8 [IQR 6 to 10] vs 6 [4 to 8]; vasopressors 205/373 [55.0%] vs 61/245 [24.9%]; mechanical ventilation 216/373 [57.9%] vs 97/245 [39.6%]) and less often had heart failure (43/373 [11.5%] vs 54/245 [22.0%]) (Table 1). Lactate was missing in 93 of 620 patients (15.0%).
Outcome data. By day 28, 134 of 373 patients (35.9%) with a positive balance and 45 of 245 (18.4%) with a negative balance had died (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4).
Main results. The crude OR for death with a positive balance was 2.49 (95% CI 1.70 to 3.70); after adjustment it was 1.62 (95% CI 1.02 to 2.59; p = 0.044; n = 527). The crude hazard ratio (HR) was 2.17 (95% CI 1.55 to 3.05) and the adjusted HR 1.46 (95% CI 1.00 to 2.13; p = 0.053).
Other analyses. With multiple imputation (n = 620) the adjusted OR was 1.70 (95% CI 1.09 to 2.64). The association differed by heart failure (p for interaction = 0.031): adjusted OR 4.39 (95% CI 1.65 to 12.38) with heart failure (n = 90) and 1.29 (95% CI 0.77 to 2.19) without (n = 437).
Notice what is *not* in these paragraphs: no “because”, no “significant”, no explanation of why the odds ratio shrank after adjustment. That explanation belongs in the Discussion.

! Common mistake: Methods, Results or Discussion?

Each sentence has one home. “We used logistic regression adjusted for SOFA and age” is Methods. “Of 640 patients screened, 600 were included” is Results. “This may be because sicker patients received more fluid” is Discussion, and is the classic sentence that slips into the Results. “Lactate was missing in 15% and was imputed” belongs half in each: *how* missing data were handled is Methods; *how many* were missing is Results.

19.5 Tables and figures that stand alone

Reviewers and clinicians read tables more carefully than text, and many readers look at nothing else. Every table and figure must therefore **stand alone**: a reader should understand it without reading the text.

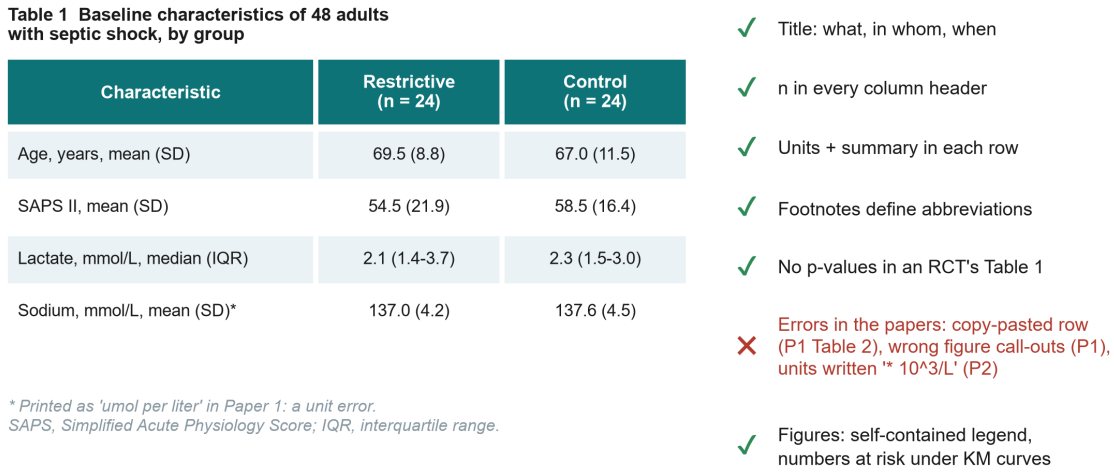


Figure 19.4: Anatomy of a stand-alone table: a title that says what, in whom and when; n in every column header; units and the summary statistic in each row; footnotes defining abbreviations; no p-values in a randomised trial's Table 1. The figure also lists real errors from the two papers, such as a copy-pasted row, wrong figure call-outs and incorrectly written units.

The rules shown in the figure are simple. The **title** says what is shown, in whom and when: "Table 2. Cumulative fluid volumes (ml/kg) during the first 5 and 7 ICU days, by group (n = 48)" rather than "Results". **Every column header** gives the group and its n. **Every row** gives the variable, its unit and the summary statistic, for example "Lactate, mmol/L, median (IQR)". **Footnotes** define abbreviations and state how many values were missing. Numbers are given with consistent, sensible precision (percentages to one decimal place, most continuous measures to one or two).

Two specific conventions are worth knowing. First, in a **randomised trial**, Table 1 should not contain p-values comparing baseline characteristics: any differences arose by chance, because allocation was random, so a significance test answers a question nobody asked (Schulz et al. 2010). Paper 1 gets this right (Boulet et al. 2024). Second, in an **observational study**, Table 1 often shows standardised mean differences instead, because the point is to show how different the groups were; Paper 2's Table 1 does this and adds footnotes on missing data (Hyun et al. 2023).

The papers also illustrate the errors that tables attract. Paper 1 prints sodium and potassium in "µmol per litre" when they are in mmol/L; repeats one row's estimate and confidence interval in the row beneath (a copy-paste error); refers twice to the wrong figure or panel in the text; and prints an interquartile range whose upper limit is below its lower limit (Boulet et al. 2024). Paper 2 writes white-cell units as "* 10^3/L" (Hyun et al. 2023). Each is small; together they tell the editor how carefully the rest was checked.

Worked example: from Table 1 to one sentence

A shortened Table 1 for FLUID-ICU (simulated data):

Table 19.2: Baseline characteristics of 618 adults with sepsis by cumulative fluid balance at ICU day 3 (FLUID-ICU, simulated data). IQR, interquartile range; SOFA, Sequential Organ Failure Assessment.

Characteristic	Negative balance (n = 245)	Positive balance (n = 373)
Age, years, mean	64.6	61.4
SOFA score, median (IQR)	6 (4 to 8)	8 (6 to 10)
Vasopressor, n (%)	61 (24.9%)	205 (55.0%)
Mechanical ventilation, n (%)	97 (39.6%)	216 (57.9%)
Lactate, mmol/L, median (IQR)	2.2 (1.4 to 3.0)	2.9 (1.9 to 4.0)
Heart failure, n (%)	54 (22.0%)	43 (11.5%)

The Results sentence that describes it: *"Patients with a positive balance were sicker at baseline (median SOFA 8 vs 6; vasopressors 55.0% vs 24.9%) and less often had heart failure (11.5% vs 22.0%)."* The sentence describes; it does not test, and it picks out the differences that matter for interpretation. Heart failure is less common in the positive-balance group because clinicians give less fluid to patients with heart failure, a reminder that treatment decisions shape observational data.

Figures show patterns; tables give exact values. Use a figure when the shape matters: a flow diagram of participants, Kaplan–Meier curves (with the numbers at risk shown beneath them), a forest plot of estimates with confidence intervals, or the distribution of a continuous variable. Show the data rather than hiding them: avoid three-dimensional effects, truncated axes and bar charts of means for continuous data, which conceal the distribution (Weissgerber et al. 2015). Each figure needs a self-contained legend saying what is shown, in whom, and what the symbols and abbreviations mean. Paper 2's timeline figure, which defines exactly what "day 1", "day 2" and "day 3" mean, is a good example of a figure doing what words could not (Hyun et al. 2023).

19.6 Reporting guidelines: your writing checklist

For each main study design there is an internationally agreed **reporting guideline**: a checklist of the items a reader needs in order to understand and appraise the study. All are collected, free, at the EQUATOR Network (Simera et al. 2010).

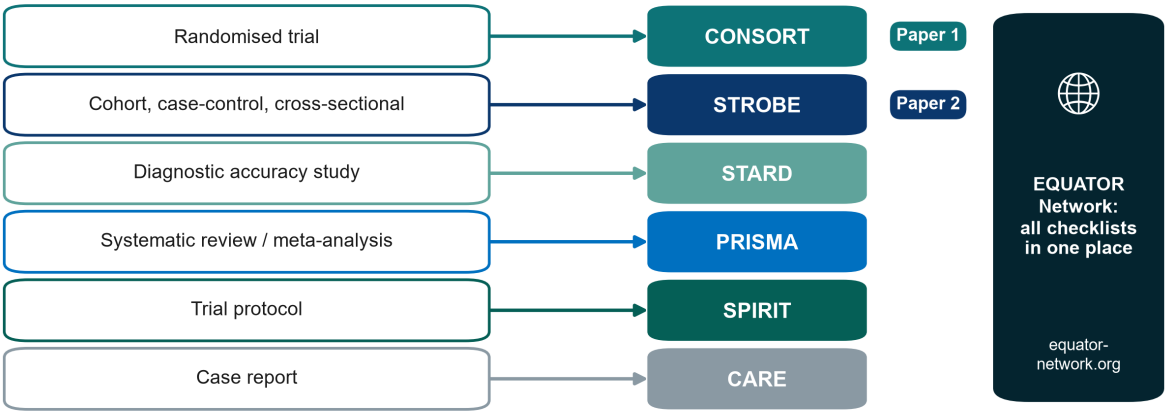


Figure 19.5: Reporting guidelines by design: CONSORT for randomised trials (Paper 1), STROBE for cohort, case-control and cross-sectional studies (Paper 2), STARD for diagnostic accuracy studies, PRISMA for systematic reviews, SPIRIT for trial protocols and CARE for case reports. All are available from the EQUATOR Network.

The main ones are **CONSORT** for randomised trials, updated as CONSORT 2025 (Schulz et al. 2010; Hopewell et al. 2025); **STROBE** for observational studies (Elm et al. 2007); **STARD** for diagnostic accuracy studies (Chapter 8); **PRISMA 2020** for systematic reviews (Chapter 9); **SPIRIT** for trial protocols (Chapter 17); and **CARE** for case reports. Many have extensions (for example, CONSORT for abstracts, cluster trials or pilot trials) and many have an "explanation and elaboration" paper that explains each item with examples (Vandenbroucke et al. 2007).

A common misunderstanding is that reporting guidelines tell you how to do a study. They tell you what to *report*. But reading the relevant guideline before you start is one of the best design checklists available, because every item you will need to report is something you will need to have planned.

Use the guideline as a writing checklist. FLUID-ICU is a cohort study, so its paper follows STROBE.

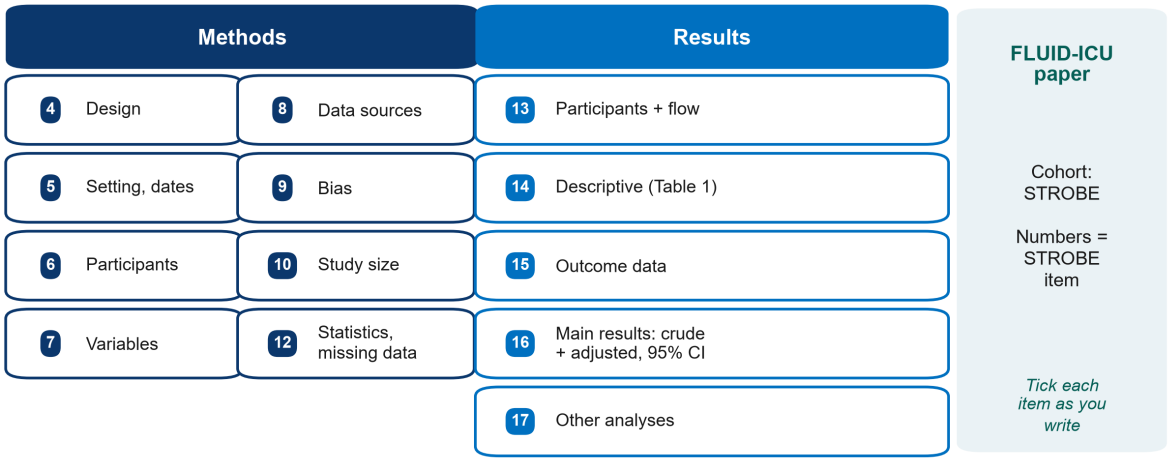


Figure 19.6: Where the STROBE items go in a cohort paper: Methods items 4 to 12 (design, setting and dates, participants, variables, data sources, bias, study size, statistics and missing data) and Results items 13 to 17 (participants and flow, descriptive data, outcome data, main results with crude and adjusted estimates and 95% CIs, other analyses). Tick each item as you write it.

The items most often forgotten in observational papers are item 9 (bias: say how confounding by indication was addressed), item 10 (study size: for a fixed dataset, say that all eligible patients were included), item 12 (how missing data were handled) and item 16 (give both crude and adjusted estimates, and the variables adjusted for). Many journals require the completed checklist at submission, with the page number on which each item appears. Interestingly, neither Paper 1 nor Paper 2 names its reporting guideline, though Paper 1 does include the CONSORT flow diagram; naming CONSORT or STROBE in the Methods and submitting the checklist is easy and expected.

19.7 Introduction: a funnel in three paragraphs

The Introduction is not a literature review. Its job is to show, as briefly as possible, why the question needed answering. The most effective structure is a funnel of three short paragraphs, together about 300 to 500 words.

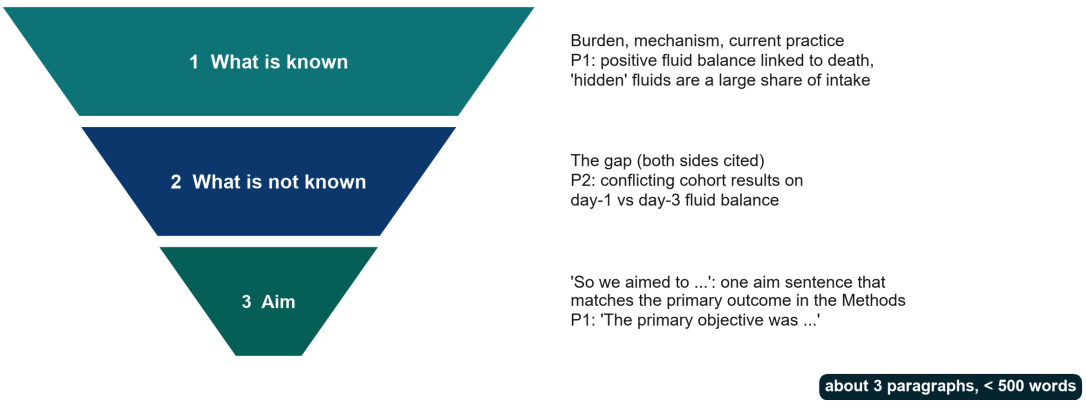


Figure 19.7: The Introduction funnel: what is known (burden, mechanism, current practice), what is not known (the gap, with both sides of any conflict cited), and the aim, in one sentence that matches the primary outcome in the Methods. About three paragraphs and under 500 words.

Paragraph 1: what is known. The clinical problem, its size, the mechanism if relevant and current practice, with key references. **Paragraph 2: what is not known.** The gap: conflicting evidence, a

limitation of earlier studies, a population not yet studied. The strongest gaps cite both sides of a disagreement. **Paragraph 3: the aim.** "We therefore aimed to..." in one sentence that matches the primary outcome in the Methods, with the same exposure, outcome and population, and the same strength of verb that the design will allow.

Paper 1 is a textbook funnel in under 500 words (Boulet et al. 2024). Its first paragraph establishes that a positive fluid balance is linked to death and that "hidden" maintenance and dilution fluids make up more than half of intake. Its second paragraph names the gap: earlier trials restricted only resuscitation fluids. Its third begins "We designed a pilot randomized controlled trial..." and states that "the primary objective was to assess the fluid balance after the first 5 days", matching the primary outcome in the Methods exactly. Paper 2's second paragraph defines its gap by citing conflicting cohort studies on both sides, a model to copy; but its Introduction runs to about 550 words, and its aim repeats the causal word "impact" for an observational design (Hyun et al. 2023).

Worked example: the FLUID-ICU Introduction funnel (simulated study)

Paragraph 1 (known). "Intravenous fluids are central to early sepsis treatment, but fluid given beyond the resuscitation phase accumulates. A positive cumulative fluid balance has been associated with higher mortality in several observational studies, including a large propensity-matched cohort in which a positive balance on ICU day 2 was associated with higher 28-day mortality (HR 2.13) [Paper 2]."
Paragraph 2 (not known). "Because sicker patients receive more fluid, it remains uncertain how much of this association reflects illness severity (confounding by indication) rather than the fluid itself. Randomised trials of restrictive strategies, such as a recent pilot targeting non-resuscitative fluids [Paper 1], have been small and inconclusive."
Paragraph 3 (aim). "We therefore examined whether a positive cumulative fluid balance by ICU day 3 was associated with 28-day mortality in adults with sepsis, before and after adjustment for severity."
Three paragraphs, about 350 words in full. The gap in paragraph 2 is precisely the confounding problem of Chapter 6, which is why the study is worth doing; and the aim uses "associated", the verb the design allows.

19.8 Discussion: six moves

A Discussion is not a summary of other studies, and it is not a second Results section. A good Discussion makes six moves, in this order.

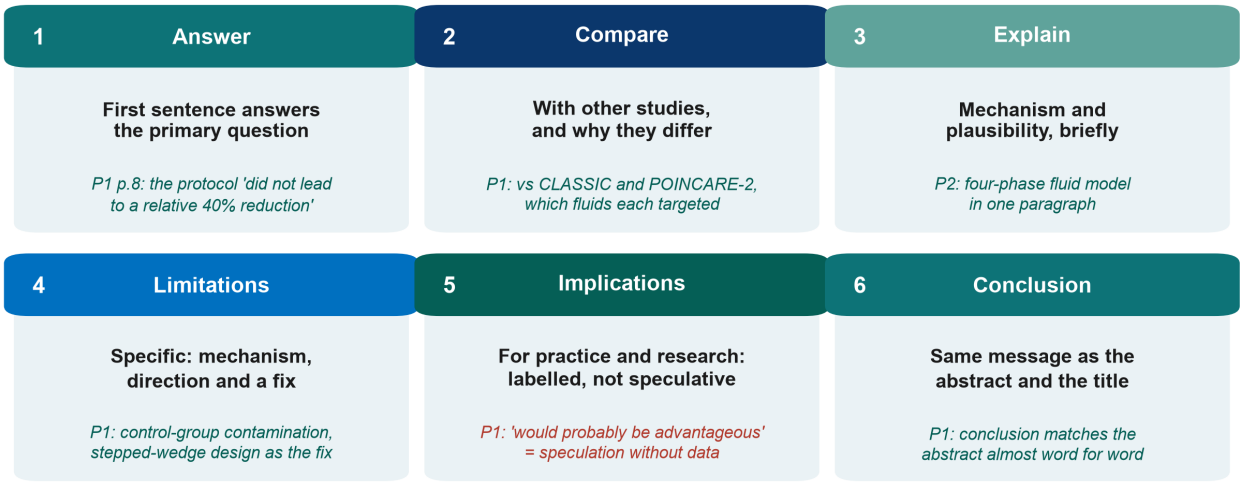


Figure 19.8: The six moves of a Discussion: answer the question in the first sentence; compare with other studies and explain why they differ; explain the mechanism briefly; state limitations specifically, with their direction and a fix; give implications for practice and research, labelled and not speculative; and conclude with the same message as the abstract and title. Examples from Papers 1 and 2 illustrate each move.

1. **Answer.** The first sentence answers the question posed at the end of the Introduction, with the key number. Paper 1's Discussion opens by stating that its protocol did not achieve the planned reduction in fluid balance (Boulet et al. 2024); the reader never has to wonder what was found.
2. **Compare.** How do the findings agree or disagree with previous studies, and *why* might they differ? Paper 1 compares itself with two earlier trials and explains the difference by what each targeted (resuscitation fluids versus maintenance and dilution fluids, use of diuretics). Paper 2 closes the loop by returning to the conflicting studies cited in its Introduction (Hyun et al. 2023). A comparison is an argument, not a list.
3. **Explain.** Possible mechanisms and plausibility, briefly. Paper 2 uses the four-phase model of fluid therapy (rescue, optimisation, stabilisation, de-escalation) in one paragraph.
4. **Limitations.** Specific limitations, each with its likely **direction** (does it make the effect look bigger or smaller?) and, ideally, what would fix it. Paper 1 does this well: it notes that its control group received far less fluid than in earlier trials, which narrowed the contrast between the groups and pushed the result towards no difference (unblinded clinicians, aware of the trial, may have restricted fluids in both arms); it explains that clinicians could not be blinded with individual randomisation; and it names the fix, a randomised stepped-wedge cluster design, together with its cost in centres and patients. Paper 2 lists genuine limitations but misses the three a reviewer would raise: survivor bias from requiring an ICU stay of at least three days, residual imbalance in lactate after matching, and confounding by indication.
5. **Implications.** For practice and for research, clearly labelled, and no stronger than the data. Paper 1's "would probably be advantageous" is speculation without data.
6. **Conclusion.** One or two sentences with the same message, and the same verb, as the abstract and the title. Paper 1's conclusion matches its abstract almost word for word.

Limitations are the move that first-time writers find hardest. “Our study has some limitations, such as its retrospective design” says nothing. A useful limitation names the specific problem, explains how it could affect the result and in which direction, and says what the reader should make of it.

The figure below condenses the six moves for FLUID-ICU’s primary question into one sentence each; the worked example that follows writes them out in full.

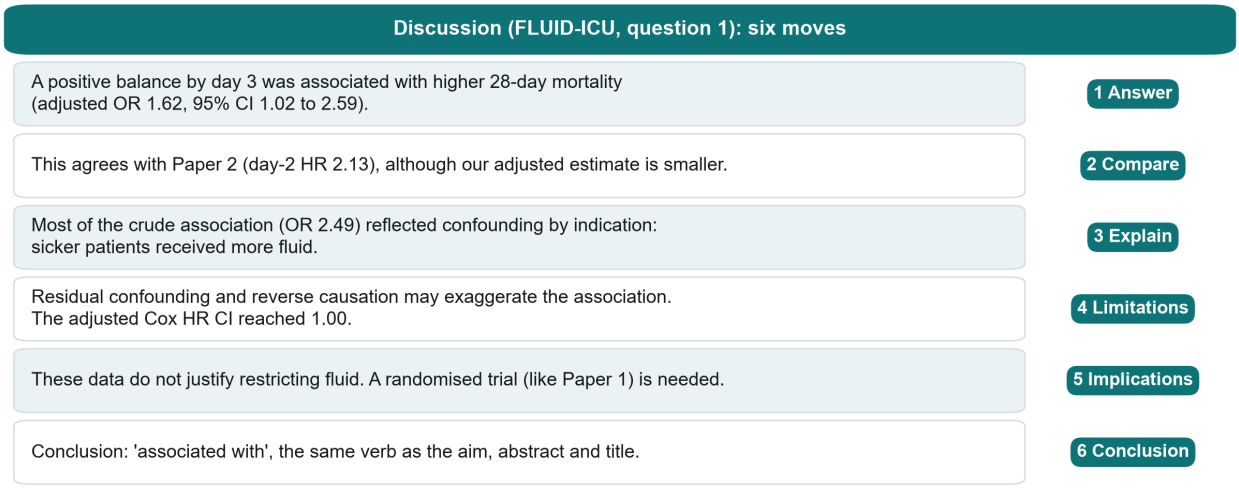


Figure 19.9: The six moves written out for FLUID-ICU’s primary question (simulated data): answer with the adjusted odds ratio; compare with Paper 2; explain the crude–adjusted gap as confounding by indication; limitations with their direction; implications (a trial is needed); and a conclusion using the same verb as the aim.

Worked example: the FLUID-ICU Discussion in six moves (simulated data)

1. **Answer.** “In this multicentre cohort, a positive fluid balance by ICU day 3 was associated with higher 28-day mortality after adjustment for severity (adjusted OR 1.62, 95% CI 1.02 to 2.59).”
2. **Compare.** “The direction agrees with Paper 2 (day-2 HR 2.13, 95% CI 1.48 to 3.06), although our adjusted estimate was smaller; differences in case mix, the timing of the balance and the adjustment set may explain this.”
3. **Explain.** “Much of the crude association (OR 2.49) disappeared after adjustment, consistent with confounding by indication: patients with a positive balance were sicker (median SOFA 8 vs 6; vasopressors 55.0% vs 24.9%). Fluid accumulation may contribute to organ oedema, but our data cannot show this.”
4. **Limitations.** “Residual confounding by unmeasured severity is likely and would exaggerate the association; reverse causation (patients who deteriorate accumulate fluid) acts in the same direction. Patients who died before day 3 had less time to accumulate fluid. The adjusted Cox HR (1.46, 95% CI 1.00 to 2.13) had a lower limit at 1. Lactate was missing in 15%, but multiple imputation gave a similar estimate (OR 1.70). The association appeared stronger in heart failure (adjusted OR 4.39 vs 1.29; p for interaction 0.031) but was based on 90 patients and should be treated as a hypothesis.”
5. **Implications.** “These findings do not justify restricting fluid in practice; a randomised trial of fluid strategies after the resuscitation phase is needed.”
6. **Conclusion.** “A positive cumulative fluid balance by ICU day 3 was modestly associated

with higher 28-day mortality; much of the crude association reflected severity of illness."

The third move deserves a closer look, because explaining the difference between crude and adjusted estimates is one of the most useful things an observational Discussion can do.

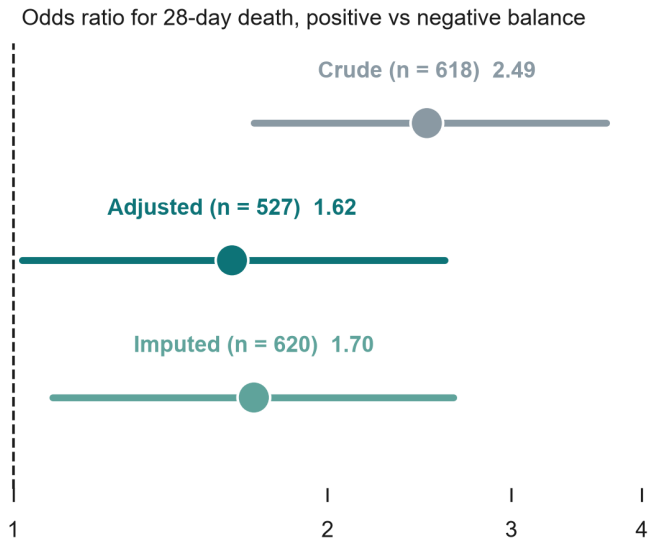


Figure 19.10: Odds ratios for 28-day death with a positive versus negative fluid balance in FLUID-ICU (simulated data): crude 2.49 (n = 618), adjusted 1.62 (n = 527) and adjusted after multiple imputation 1.70 (n = 620), with 95% confidence intervals. The drop from crude to adjusted reflects confounding by indication; the imputed estimate is close to the complete-case one.

The figure above shows that the crude odds ratio of 2.49 fell to 1.62 once severity was taken into account: sicker patients both received more fluid and died more often (Chapter 6). The imputed estimate (1.70) is close to the complete-case estimate, so missing lactate did not change the conclusion. And the adjusted confidence interval starts at 1.02, only just above 1, which is why "a modest association remained" is the calibrated way to describe it. Had the adjusted interval been 0.95 to 2.40, the honest sentence would have been "there was no clear evidence of an association after adjustment", and not "there was no effect".

19.9 Calibrated language: match the verb to the design

The strength of a verb is a claim. Over-interpretation, conclusions that go beyond what the data and design can support, is one of the commonest reasons reviewers reject papers (Bordage 2001). The remedy is to choose verbs that match the design.

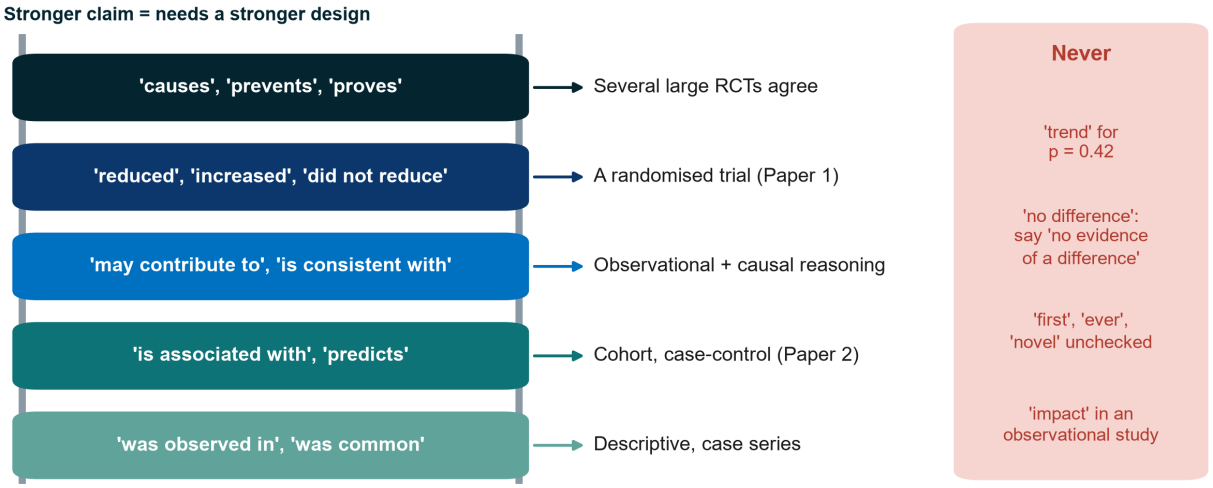


Figure 19.11: The verb ladder: "was observed in" for descriptive studies; "is associated with" or "predicts" for cohort and case-control studies (Paper 2); "may contribute to" for observational evidence with causal reasoning; "reduced" or "did not reduce" for a randomised trial (Paper 1); "causes" or "prevents" only when several large trials agree. Never: "trend" for p = 0.42, "no difference" instead of "no evidence of a difference", unchecked "first" or "novel", or "impact" in an observational study.

Read the ladder from the bottom. **Descriptive** studies and case series observe: "fluid overload was common". **Cohort and case-control** studies find that an exposure "is associated with" an outcome, or, in a prediction context, "predicts" it. Strong observational evidence combined with explicit causal reasoning (Chapter 6) may support "may contribute to" or "is consistent with". A **randomised trial** may say that an intervention "reduced" or "did not reduce" an outcome, because randomisation balances known and unknown confounders on average. "Causes", "prevents" and "proves" need several large, consistent trials.

Several specific phrases are traps.

- **"Trend" for a large p-value.** Paper 1's Discussion describes a difference with p = 0.42 as a "non-significant trend" and uses it to argue for a larger trial (Boulet et al. 2024). A p-value of 0.42 is not a trend towards anything. The honest sentence reports the estimate and its uncertainty: "The point estimate favoured the restrictive strategy, but the 95% CI was wide (−15.2 to 41.6 ml/kg) and included no difference; a larger trial would be needed to detect a smaller effect."
- **"No difference" from a small study.** With 48 patients, Paper 1 could not detect a difference in mortality. "No difference" claims too much; "no evidence of a difference; the study was not powered for mortality" is accurate. Absence of evidence is not evidence of absence (Altman and Bland 1995).
- **"Impact", "effect", "show" in observational studies.** Paper 2's title begins "Impact of a cumulative positive fluid balance..." and its Discussion says the day-1 balance "did not impact" mortality and that the results "show" a relationship, while its abstract correctly says "associated" (Hyun et al. 2023). The same paper thus claims two different strengths of evidence.
- **Superlatives.** "First", "ever", "novel" and "the most reliable" invite challenge. Paper 1 describes

its fluid balance as among the lowest "ever published"; Paper 2 calls its matched comparison "the most reliable". Protect such claims with "to our knowledge" and a citation, or drop them.

- **Ranking estimates without a test.** Paper 2 concludes that day 2 carried "the highest risk" (HR 2.13 vs 1.56 on day 3), but the estimates come from different matched cohorts, their confidence intervals overlap, and no formal comparison was made.

These are all forms of **spin**: reporting that makes results look more favourable than they are. Spin is common in the reports of trials with non-significant primary outcomes, where it appears in the abstract, the Discussion and the conclusions (Boutron et al. 2010). A good habit is a final "verb check": search the manuscript for *cause, impact, effect, show, prove, trend, significant, first* and *novel*, and justify or replace every one.

Worked example: fix the verb (FLUID-ICU, simulated data)

Table 19.3: Calibrating verbs to an observational design.

Draft sentence	Problem	Calibrated version
"A positive balance caused higher mortality (OR 1.62)."	Causal verb for a cohort	"A positive balance was associated with higher mortality (adjusted OR 1.62, 95% CI 1.02 to 2.59)."
"Fluid overload increases AKI (OR 1.27 per 10 ml/kg)."	Causal verb; reverse causation possible	"Each 10 ml/kg of positive balance was associated with higher odds of AKI (adjusted OR 1.27, 95% CI 1.16 to 1.40)."
"There was a trend towards death (Cox HR 1.46)."	"Trend"	"The adjusted HR was 1.46 (95% CI 1.00 to 2.13); the interval reached 1."
"Lactate predicts death (OR 1.39 per mmol/L)."	Acceptable in a prediction sense	If a causal role is implied: "Higher lactate was associated with death."

19.10 The structured abstract: the shop window

Most readers never go beyond the abstract, and many decide from it whether to read the paper at all. It must therefore stand alone. Most clinical journals ask for a **structured abstract** with headings (Background, Methods, Results, Conclusions) of about 250 to 300 words; trials add the registration number (Hopewell et al. 2008).

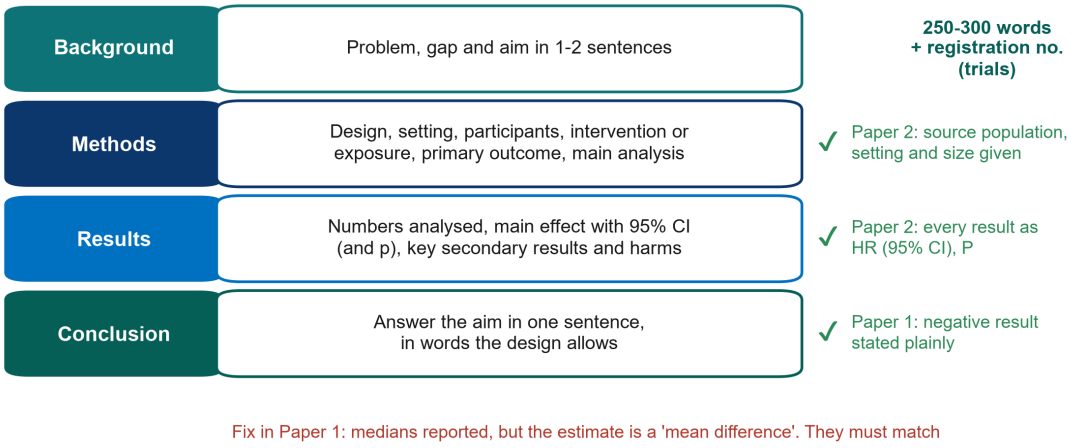


Figure 19.12: The structured abstract: Background (problem, gap and aim in one or two sentences); Methods (design, setting, participants, intervention or exposure, primary outcome, main analysis); Results (numbers analysed, main effect with 95% CI and p, key secondary results and harms); Conclusion (the answer in words the design allows). Paper 2 gives the source population, setting and size, and every result with its CI; Paper 1 states its negative result plainly but mislabels medians as a “mean difference”.

Good practice from the two papers is shown in the figure. Paper 2’s abstract gives the source population, setting and size (11,981 patients from 20 hospitals) and every result as a hazard ratio with 95% confidence interval and exact p-value (Hyun et al. 2023). Paper 1’s abstract conclusion reports its negative result plainly, without spin, and gives the registration number (Boulet et al. 2024). But Paper 1’s abstract also shows what to fix: it reports medians, labels the estimate a “mean difference”, and gives a p-value of 0.0506 to four decimal places with a confidence interval touching zero, which reads as an attempt to look “nearly significant”. The summary statistic, the estimate and the test must match.

Write the abstract last, copy the numbers from the Results (an abstract must contain no number that is not in the paper), and check that each one appears identically in the abstract, text and tables. Report the primary outcome first, even if it is negative. The conclusion must use the same verb as the Discussion’s conclusion and the title.

Worked example: abstract for FLUID-ICU (simulated data)

Background: A positive fluid balance is associated with death in sepsis, but sicker patients receive more fluid. We examined the association after adjustment for severity.
Methods: Multicentre prospective cohort (simulated teaching data) of 620 adults with sepsis in four ICUs, 2023 to 2024. Exposure: positive cumulative fluid balance by ICU day 3. Outcome: 28-day mortality. Logistic and Cox regression adjusted for age, SOFA, lactate, vasopressors, mechanical ventilation and chronic kidney disease; multiple imputation for missing lactate.
Results: Of 618 patients analysed, 373 (60.4%) had a positive balance; 134/373 (35.9%) vs 45/245 (18.4%) died. The crude OR was 2.49 (95% CI 1.70 to 3.70) and the adjusted OR 1.62 (95% CI 1.02 to 2.59); with imputation 1.70 (1.09 to 2.64). The adjusted HR was 1.46

(95% CI 1.00 to 2.13).
Conclusions: A positive fluid balance by day 3 was modestly associated with 28-day mortality; much of the crude association reflected illness severity. A randomised trial is needed before fluid strategies are changed.
About 190 words. Every number is copied from the Results; “modestly associated” matches an adjusted interval whose lower limit is just above 1.

19.11 Title and keywords

The title is the most-read sentence of the paper and the one that decides whether a search result is opened. Build it from the PICO: the design, the intervention or exposure (and comparator), the population and the outcome. Aim for about 10 to 16 words, state the design (usually after a colon), and avoid value-laden words such as “impact”, “optimised”, “novel” and “first”.

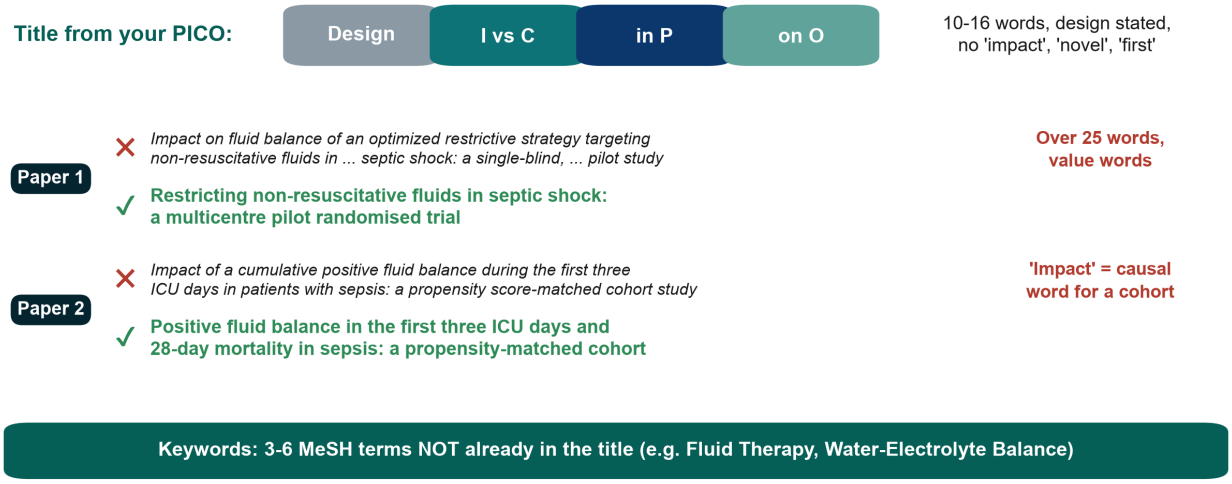


Figure 19.13: A title built from the PICO (design, intervention versus comparator, population, outcome), 10 to 16 words, with no “impact”, “novel” or “first”. Paper 1’s 27-word title contains value words; a shorter descriptive version is shown. Paper 2’s title uses the causal word “Impact” for a cohort; a calibrated version is shown. Keywords should be three to six MeSH terms not already in the title.

Paper 1’s title names its design, which is good, but runs to 27 words and contains the value words “impact” and “optimized”: “Restricting non-resuscitative fluids in septic shock: a multicentre pilot randomised trial” says the same in 13 words (Boulet et al. 2024). Paper 2’s title uses “Impact” for a propensity-matched cohort; “Positive cumulative fluid balance during the first three ICU days and 28-day mortality in sepsis: a propensity score-matched cohort study” is calibrated to the design (Hyun et al. 2023). For observational studies, “X and Y” or “Association between X and Y” is the safe form. The FLUID-ICU paper might be titled “Positive fluid balance by day 3 and 28-day mortality in adults with sepsis: a multicentre cohort study”.

Keywords help indexing and searching. Choose three to six terms, ideally Medical Subject Headings (MeSH; Chapter 3), that are **not** already in the title, so the paper is found by more searches. Paper 1’s keywords repeat words from its title; terms such as *Fluid Therapy* and *Water-Electrolyte Balance* would widen its reach. For FLUID-ICU: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Mortality; Cohort Studies.

19.12 Style, numbers and references

Clinical writing is plain writing. Use short sentences, one idea per sentence and the active voice where it is clearer ("we excluded patients who..." rather than "patients were excluded who..."). Define each abbreviation at first use, and use few. Keep terms consistent: if the exposure is "positive cumulative fluid balance by day 3" in the Methods, do not call it "fluid overload" in the Discussion. Replace adjectives with numbers: not "a strong association" but "OR 2.49 (95% CI 1.70 to 3.70)".

References should be managed with reference-management software from the start, in the journal's style (often NLM/Vancouver: numbered in order of citation). Cite original studies rather than only reviews, cite only what you have read, and check every citation against its source. Paper 2 cites the Sepsis-3 definitions paper as the source of its covariate list, almost certainly the wrong reference number (Hyun et al. 2023); a reference manager and a final check would have prevented it. Never cite a reference suggested by an AI tool without finding and reading it (Chapter 18).

19.13 The final read

Before submission, a co-author who did not write the section should read the whole manuscript with one job: to check it. Almost all of Paper 1's errors (the text–table contradiction, the copy-pasted row, the unit error, two wrong figure call-outs and the reversed interquartile range) would have been caught by a colleague reading the tables against the text (Boulet et al. 2024).

A final-read checklist

- Every number in the text and abstract matches the tables, and every percentage has its numerator and denominator.
- Every figure and table call-out points to the right figure, table or panel.
- Units, decimal places and abbreviations are consistent and correct.
- Every result has an effect size and 95% confidence interval; p-values are exact.
- The Results contain no interpretation; the Discussion makes the six moves.
- The verbs match the design throughout: title, aim, abstract, Discussion and conclusion agree.
- The reporting checklist is completed with page numbers.
- Every reference has been checked against its source.

Summary

A clinical paper follows the IMRaD hourglass: the Introduction narrows to one question, the Methods and Results report exactly what was done and found, and the Discussion widens to meaning. Papers are best written starting from the tables and figures, then the Methods, Results, Introduction and Discussion, with the abstract and title last. The Methods must allow another team to repeat the study, and the statistical paragraph must name a pre-specified method for every comparison reported. The Results start with the participants, give effect sizes with confidence intervals, contain no interpretation, and give each number one home; observational studies report both crude and adjusted estimates. Tables and figures must stand alone. The Introduction is a three-paragraph funnel (known, not known, aim) and the Discussion makes six moves (answer, compare, explain, limitations with direction, implications, conclusion). Verbs must match the design ("associated with" for observational studies; never "trend" for a large p-value; "no evidence of a difference" rather than "no difference"), because over-interpretation and spin are among the commonest reasons for rejection. The structured abstract and the title summarise the finished paper, with identical numbers and the same calibrated verb. Reporting guidelines such as CONSORT and STROBE are the writing checklist, and a final read by a co-author catches most errors before reviewers do.

Further reading

- The STROBE statement and its explanation and elaboration paper (Elm et al. 2007; Vandenbroucke et al. 2007), together with CONSORT (Schulz et al. 2010; Hopewell et al. 2025), show item by item what readers need, with examples of good reporting.
- Boutron and colleagues (Boutron et al. 2010) document how spin distorts the reporting of trials with non-significant results; Altman and Bland's one-page note (Altman and Bland 1995) is the classic statement of why "no difference" is the wrong conclusion from a small study.
- Weissgerber and colleagues (Weissgerber et al. 2015) explain why bar charts of means hide the data and what to show instead.
- Lang's *How to Write, Publish, and Present in the Health Sciences* (Lang 2010) is a practical, section-by-section guide to scientific writing for clinicians.

CHAPTER 20

Getting published

A finished manuscript is not yet a contribution to knowledge. It becomes one when it has passed through peer review, been published where the people who need it can find and read it, and been shared with colleagues and, where possible, with patients. This last stage of research has its own rules and its own traps: choosing a journal, deciding who will be able to read and re-use the work, avoiding journals that take fees without providing real peer review, assembling a complete submission, answering reviewers, surviving rejection, checking proofs and presenting the work at conferences. None of it is difficult once it is understood, but each step has cost many first-time authors months of delay.

This chapter follows a manuscript from the decision about where to send it to the day it is published and beyond. The examples are again from fluids in septic shock: Paper 1, a randomised pilot trial published in *Critical Care* (Boulet et al. 2024), Paper 2, a propensity-matched registry cohort published in *Annals of Intensive Care* (Hyun et al. 2023), and FLUID-ICU, a cohort of 620 adults with sepsis that is **simulated for teaching**. A FLUID-ICU paper would never actually be submitted, because its data are invented; we use it to show what each document in the process would look like for a typical observational study.

In this chapter you will learn

- how to choose a journal by fit, trustworthiness, rules and cost;
- what open access, article-processing charges and Creative Commons licences mean for authors and readers;
- how to recognise a predatory journal, and what checks to make before submitting;
- what preprints are, and their benefits and risks;
- what goes into a submission package, including the cover letter and declarations;
- how peer review works, how to respond to reviewers point by point, and how to handle rejection;
- what to do after acceptance (proofs, sharing, corrections), and how to write a conference abstract and design a poster.

20.1 Choosing a journal

Choose the target journal **before** you write, not after. Its instructions for authors (often called the *Guide for Authors*) set the word limit, the structure of the abstract, the number of tables and figures, the reference style, the reporting checklist and the registration rules, and rewriting a finished paper to fit them is wasted effort. Four checks help.

Table 20.1: Four checks for choosing a journal.

Check	Questions to ask
Fit	Who needs this result: intensivists, ICU nurses, pharmacists, general physicians? Does the journal publish studies of this design and topic? Read the abstracts of three recent issues.
Trust	Is the journal indexed in MEDLINE/PubMed, the Directory of Open Access Journals (DOAJ), Scopus or Web of Science? Does it describe its peer-review process, its editors and its policies on corrections and retractions?
Rules	What are the word, table and figure limits? Which reporting checklist is required? Must trials be registered? What is the policy on preprints, data sharing and AI?
Cost and speed	Is there an article-processing charge, and are waivers or institutional agreements available? Which licence will apply? How long does the journal take to make decisions?

Fit matters most. A good journal for a paper is one whose readers will use the result. A study of nurse-led fluid charting may have more impact in a critical-care nursing journal read by the people who will change the charts than in a high-profile general journal that rejects it. A useful rule is to start with a journal your own unit reads regularly. Look at whether the journal has recently published similar designs; a journal that never publishes observational studies is unlikely to start with yours.

Metrics such as the Journal Impact Factor (issued only by Clarivate for journals in Web of Science) describe the average citation rate of a journal's articles, not the quality of any single paper. They can help distinguish established journals from obscure ones, but they should not drive the choice, and any "impact factor" from an organisation other than Clarivate should be treated with suspicion (see predatory journals below).

Speed varies widely and is not in itself a sign of quality. Paper 1 went from submission to acceptance in about seven weeks (10 September to 31 October 2024); Paper 2 took about seven months (15 January to 24 August 2023) (Boulet et al. 2024; Hyun et al. 2023). Many journals publish their median times to first decision on their websites.

It is sensible to choose a first-choice journal and one or two alternatives at the outset, so that a rejection leads quickly to resubmission rather than to a new search.

20.2 Open access and licences

Traditionally, journals were funded by subscriptions, and only readers whose institutions paid could read the articles. **Open access** (OA) means that the article is free for anyone to read online. This matters especially for clinicians in settings where institutional subscriptions are limited.

There are several routes. In **gold open access**, the journal makes the final published article free immediately, usually funded by an **article-processing charge** (APC) paid by the authors, their institution or their funder. Both Paper 1 and Paper 2 are gold open access. In **green open access**, the article is published in a subscription journal, and the authors deposit their accepted manuscript (the peer-reviewed version before the publisher's typesetting) in an institutional or subject repository, often after an embargo period. Some journals, often run by societies or universities, are free both to read and to publish in (sometimes called **diamond** open access). APCs can be substantial; ask about waivers for authors from lower-income settings, discounts for society members, and agreements between publishers and your institution. Budget for the fee in the protocol (Chapter 17).

Open-access articles carry a **licence** that states what others may do with them. Most use Creative Commons (CC) licences.

Table 20.2: Common Creative Commons licences.

Licence	What others may do	Example
CC BY	Share, adapt and re-use, including commercially, with attribution	Paper 2 (Hyun et al. 2023)
CC BY-NC	Share and adapt for non-commercial purposes, with attribution	
CC BY-NC-ND	Share unchanged copies for non-commercial purposes, with attribution; no adaptations	Paper 1 (Boulet et al. 2024)

The licence matters to you as both author and reader. As an author, read it before you sign: it determines whether others may translate your paper, use your figures in teaching materials or include them in a book. As a reader, check it before re-using material. Suppose you want to redraw Paper 1's flow diagram for a departmental teaching presentation. Its CC BY-NC-ND licence allows sharing but not adaptation, so a redrawn version needs permission; citing and linking to the original is always allowed. Paper 2's CC BY licence would allow adaptation with attribution.

20.3 Predatory journals

Predatory journals are publications that charge fees while providing little or no genuine peer review, editing, indexing or long-term archiving. They exploit the pressure to publish. Papers sent to them disappear into journals that few people read or trust, may be cited as evidence of poor judgement, and usually cannot be withdrawn and submitted elsewhere, because the journal claims the work is already published. A group of researchers and publishers proposed a consensus definition: "entities that prioritize self-interest at the expense of scholarship", characterised by false or misleading information, deviation from best editorial and publication practice, lack of transparency and aggressive, indiscriminate solicitation (Grudniewicz et al. 2019).

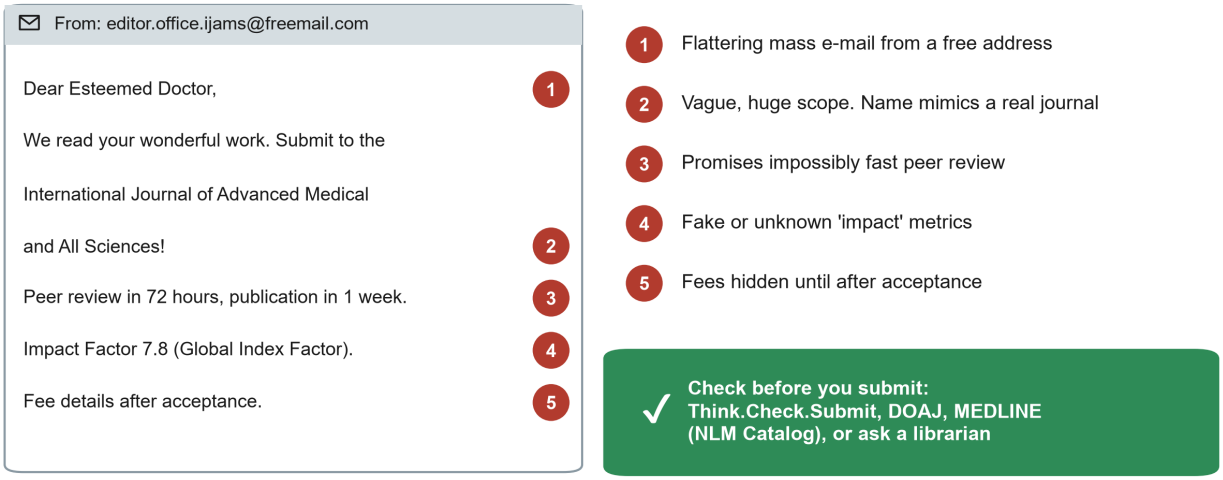


Figure 20.1: An invented invitation from a predatory journal, with its five red flags: a flattering mass e-mail from a free address; a vague, huge scope and a name that imitates a real journal; impossibly fast peer review; fake or unknown "impact" metrics; and fees hidden until after acceptance. Check before submitting: Think.Check.Submit, DOAJ, the NLM Catalog for MEDLINE indexing, or a librarian.

The figure shows a typical invitation. Its red flags are worth learning, because almost every clinician who has published even once receives such e-mails.

1. **Flattering mass e-mail**, often from a free e-mail address rather than a publisher's domain, addressed to "Dear Esteemed Doctor" and praising "your wonderful work" without naming it.
2. **A vague, enormous scope** ("medicine and all sciences") and a name that imitates a well-known journal.
3. **Impossibly fast peer review**: genuine review cannot be completed in 72 hours.
4. **Fake or unknown metrics**: an "impact factor" from an organisation other than Clarivate, or indexing claims that refer to services that are not quality indexes.
5. **Fees hidden until after acceptance.**

Other warning signs include an editorial board whose members cannot be verified at their stated institutions, no physical address, many spelling and grammatical errors, submission by replying to an e-mail rather than through an editorial system, and pressure tactics ("special discount this week only"). A particularly deceptive variant is the **hijacked journal**: a fake website that copies the name of a genuine journal. Always reach a journal through its publisher's site or the NLM Catalog, not through a link in an e-mail.

Not every invitation is predatory, and a new journal without an impact factor may be perfectly sound. The answer is to check, using a few free tools:

- the **Think. Check. Submit.** checklist, which walks through the questions to ask about any journal;
- the **Directory of Open Access Journals (DOAJ)**, which lists open-access journals that meet quality criteria;
- the **NLM Catalog**, filtered for journals "currently indexed in MEDLINE", to confirm indexing claims at the source;
- and a hospital or university **librarian**, who is often the quickest check of all.

Worked example: which journal would you trust?

Four invented journal profiles, and the verdict for each:

Table 20.3: Four journal profiles and verdicts.

Profile	Verdict and reason
J1: indexing in MEDLINE confirmed in the NLM Catalog; editors verifiable at universities; APC shown on the website; median nine weeks to first decision; requires CONSORT or STROBE checklists	**Trust.** Everything can be verified, and the policies are transparent
J2: claims indexing in "Google Scholar and Index Copernicus"; promises acceptance in 72 hours; no fee information; editorial board photographs without affiliations	**Avoid.** These are not quality indexes; the speed is implausible; fees and editors are hidden
J3: open access, listed in DOAJ, small society journal, no impact factor yet, clear peer-review and retraction policy	**Trust, or check further.** DOAJ listing and clear policies are good signs; no impact factor is normal for a new or small journal. Check that it publishes work like yours
J4: name differs by one word from a well-known journal; website address different from the real journal's; it e-mailed you first	**Avoid.** A hijacked or look-alike journal. Reach the real journal through its publisher or the NLM Catalog

20.4 Preprints

A **preprint** is a complete manuscript posted on a public server before, or at the same time as, it is submitted to a journal. For health research the best-known server is medRxiv. Preprints are screened only for basic suitability, not peer-reviewed.

Their benefits are speed (results become visible months before publication), a public timestamp establishing priority, and the chance of early comments that improve the paper. They became prominent during the COVID-19 pandemic, when waiting months for journal publication was not acceptable. Their risks follow from the absence of peer review: colleagues, guideline developers and the media may treat a preprint as established fact, and a preprint with serious flaws can spread before anyone checks it.

Most major journals accept submissions that have been posted as preprints, but policies vary, so check the target journal's policy first. Declare the preprint in the cover letter, label it clearly as not peer-reviewed in any presentation or discussion, and once the paper is published, update the preprint record with a link to the published version.

20.5 The submission package

Most journals use an online submission system that asks for each element of the submission separately. The editorial office checks that the package is complete before any scientific review, and missing items send the manuscript straight back to the authors, sometimes weeks later.

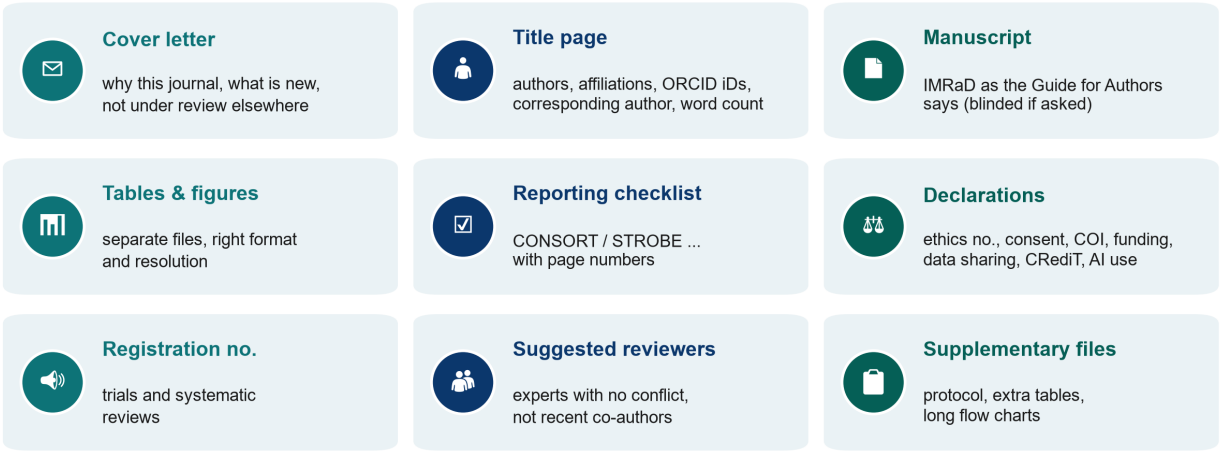


Figure 20.2: The submission package: cover letter, title page, manuscript, tables and figures, reporting checklist, declarations, registration number, suggested reviewers and supplementary files.

Cover letter. One page, addressed to the editor by name, that sells the paper (described below). **Title page.** The title, all authors with their affiliations and ORCID iDs, the corresponding author's contact details, word counts for the abstract and main text, and the numbers of tables and figures. An **ORCID iD** is a free, permanent identifier that distinguishes you from other researchers with similar names and links all your work; register at orcid.org before your first submission. **Manuscript.** Formatted as the instructions for authors require: usually double-spaced with line numbers, headings as specified and, if the journal uses double-blind review, with all identifying information removed. **Tables and figures.** In the required files, formats and resolution (often at least 300 dots per inch for images). **Reporting checklist.** The completed CONSORT, STROBE or other checklist, with the page number on which each item appears (Chapter 19). **Declarations.** Ethics approval (committee and number) and consent; competing interests (each author's disclosure form); funding and the funder's role; data availability; author contributions (CRediT); use of AI; and acknowledgements (Chapters 17 and 18). Paper 1 has a complete declarations block that matches, item by item, what a submission system asks for (Boulet et al. 2024). **Registration number**, for trials and systematic reviews. **Suggested reviewers.** Many journals ask for names of experts without conflicts of interest: not collaborators or co-authors from the past few years, not colleagues from your own institution. The editor decides whether to use them. You may also ask that particular people be excluded, with a reason. **Supplementary files.** The protocol, extra tables, detailed flow charts and analysis code.

Some requirements are journal-specific and are found only in the instructions for authors, such as a graphical abstract (Paper 1 has one), a short "key points" box or a plain-language summary.

Read the instructions twice.

20.5.1 The cover letter

The cover letter is the editor's first impression of the paper. It should fit on one page, on institutional letterhead, and answer the editor's implicit questions: what was asked, what was found, why it matters, and why it belongs in *this* journal. It also contains a set of standard statements.

Worked example: a cover letter for an observational cohort

Dear Dr [Editor's surname],
Please consider our manuscript "Positive fluid balance by day 3 and 28-day mortality in adults with sepsis: a multicentre cohort study" for publication as an Original Research article in [Journal].
A positive fluid balance has been associated with death in sepsis, but because sicker patients receive more fluid it is unclear how much of this association reflects severity of illness.
In a multicentre cohort of 618 adults with sepsis, a positive cumulative fluid balance by ICU day 3 was modestly associated with higher 28-day mortality after adjustment for severity (adjusted odds ratio 1.62, 95% CI 1.02 to 2.59), while most of the crude association reflected illness severity.
These findings are directly relevant to your readers in critical care and to the fluid-stewardship questions recently discussed in your journal.
We confirm that this manuscript is original, has not been published and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission. The study was approved by [committee] (number), and is reported according to STROBE; the completed checklist is attached. The authors declare no competing interests.
Yours sincerely, [corresponding author, on behalf of all authors]
Notice that the main finding carries its number and confidence interval and uses the same calibrated verb as the abstract. (FLUID-ICU itself is simulated and would not be submitted; this letter shows the form for a real study of the same kind.)

! Common mistake: submitting to two journals at once

Submitting the same manuscript to more than one journal at the same time, to save time, is **duplicate submission** and is treated as misconduct (Chapter 18). The cover letter's statement that the work "is not under consideration elsewhere" is a formal declaration. Submit to one journal at a time; if rejected, revise and submit to the next.

20.6 Peer review

Peer review is the evaluation of a manuscript by independent experts before publication. It is imperfect (reviewers miss errors, disagree with one another and occasionally are biased), but it remains the main quality filter of the medical literature, and most papers are improved by it.

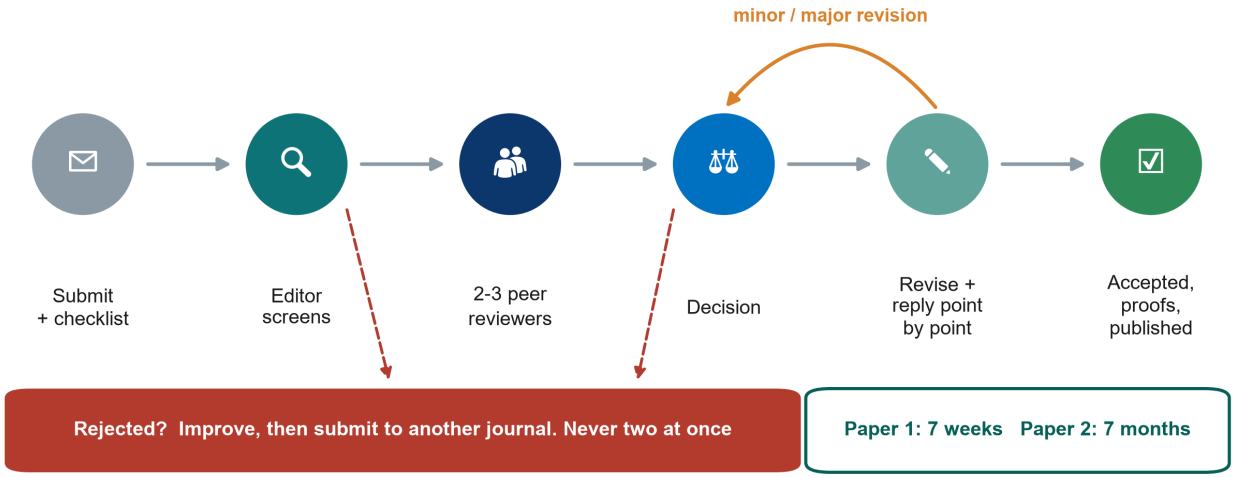


Figure 20.3: The peer-review journey: submission with checklist; screening by the editor (where many papers are desk-rejected); review by two or three peer reviewers; the editor's decision; revision with a point-by-point reply, often more than once; and acceptance, proofs and publication. If rejected, improve the paper and submit to another journal, never two at once. Paper 1 took seven weeks from submission to acceptance and Paper 2 seven months.

The figure follows the usual path. After submission, an **editor screens** the manuscript. Many papers are rejected at this stage without external review (**desk rejection**), usually because the topic does not fit the journal, the question is not of interest to its readers, or the reporting is so poor that review would be wasted. Papers that pass are sent to **two or three peer reviewers**, who comment on the importance of the question, the validity of the methods, the accuracy of the results and whether the conclusions follow. Some journals also use a statistical reviewer. The editor then makes a **decision**: accept (rare at first submission), **minor revision**, **major revision** or **reject**. After revision, the manuscript may go back to the same reviewers, sometimes for several rounds, before acceptance.

Journals use different models of review. In **single-blind** review the reviewers know who the authors are, but not vice versa; in **double-blind** review neither knows the other (which is why some journals require an anonymised manuscript); in **open** review identities are known and reviews may even be published with the paper.

It helps to understand what reviewers look for, and the commonest reasons they give for rejection. In a classic study of reviewers' comments, the leading reasons for rejection were statistics that were inappropriate or insufficiently described, over-interpretation of results, inaccurate or inconsistent data, inadequate description of the methods and poor tables and figures; among the leading reasons for acceptance was that the interpretation took the study's limitations into account (Bordage 2001). Almost all of these are under the authors' control and are covered in Chapter 19.

When appraising either of our two papers as a reviewer would, the same short list of questions applies: Is the question clear (can you write the PICO in one line)? Is the design right for it? Who was included and left out? Were the groups comparable at baseline, and how was that achieved? Who knew which group patients were in? Was there one pre-specified primary outcome? What is

the main result as an effect size with a 95% confidence interval? Do the words match the design? Are the main limitations stated honestly? Would this change practice? These are the questions of Chapter 4, and they are the questions your own reviewers will ask. Annotated as if by reviewers, Paper 1 would probably have received a minor revision (reconcile medians and "mean differences", correct text–table contradictions and unit errors, remove the "non-significant trend", justify blocks of two with unblinded clinicians) and Paper 2 a major revision (replace causal wording, discuss survivor bias and residual imbalance in lactate, name STROBE and move the flow diagram into the main text). Both were published, and both still contain errors a careful reviewer would have caught: publication is not the same as correctness.

20.7 Responding to reviewers

A request for revision is good news: the editor is interested and is telling you what is needed for acceptance. The response matters almost as much as the revision itself. Reviewers are usually unpaid volunteers, and a respectful, complete, well-organised response makes their second reading easy.

Reviewer comment	Our response	Change made
R1.1 Please do not call p = 0.42 a 'non-significant trend'.	Thank you, we agree. We now report the estimate with its 95% CI and removed 'trend'. Accept	Discussion, p.10, lines 4-6
R1.2 How was allocation concealed with blocks of 2?	Allocation was central and computer-generated. We added block size as a limitation. Clarify	Methods p.3, limitations p.10
R2.1 Please add 90-day mortality.	We respectfully disagree: 90-day status was not a pre-specified outcome and was not collected. Rebut	Noted as a limitation, p.10

Illustrative responses, built on reviewer points from the annotated Paper 1

Figure 20.4: A point-by-point response table with three illustrative rows built on reviewer points for Paper 1: accepting a request to stop calling p = 0.42 a "trend", clarifying allocation concealment with blocks of two, and politely rebutting a request for an outcome that was never collected. Each row gives the comment, the response and exactly where the manuscript changed.

The standard format is a **point-by-point table**, as in the figure: every comment copied verbatim and numbered (R1.1, R1.2 for reviewer 1; R2.1 for reviewer 2; E.1 for the editor), the authors' response, and exactly where the manuscript changed (page and line in the revised version), often with the new text quoted. A short covering letter to the editor thanks the reviewers once, summarises the main changes and explains how changes are marked (tracked changes or highlighting).

A few rules of tone and method make the difference.

- **Wait a day** before replying. First reactions to criticism are rarely the best ones.
- **Answer every comment.** Never ignore one, even a minor or mistaken one.
- **Agree where you can**, and say concretely what you changed. "Done" with no detail is not a response.
- **Disagree politely, with evidence.** When a request is wrong or impossible (a rebuttal), thank

the reviewer, explain why with a reason or a reference, and where possible still clarify the text or add a limitation so that the next reader is not confused in the same way. Never argue about the reviewer.

- **Do not accept a request that makes the paper wrong** just to please a reviewer, for example an unplanned subgroup analysis that would be underpowered and misleading, or a causal claim the design cannot support.
- **Re-check consistency** after revision: abstract, text and tables must still agree.

Worked example: four responses for an observational cohort

Table 20.4: Model responses to reviewers (fictional numbers for illustration).

#	Reviewer comment	Response	Change made
R1.1	"The title says your intervention *improves* outcomes, but this is an observational study."	"Thank you. We agree that our design supports association, not causation. We have changed the title to '... and ...: a cohort study' and replaced 'improves' with 'was associated with' throughout."	Title p1; Abstract; Discussion p14
R1.2	"Please report the main result with a confidence interval, not only the p-value."	"Thank you. The main result is now reported as RR 0.78 (95% CI 0.61 to 0.99) in the abstract, Results and Table 2."	Abstract; Results p9; Table 2
R1.3	"Text and Table 2 give different numbers of patients analysed (212 vs 208)."	"Thank you for spotting this. 212 patients were included and 4 were excluded from the adjusted model because of missing lactate; the flow diagram and text now show 212 to 208, with reasons."	Figure 1; Results p8
R2.1	"Please add 1-year mortality."	"We agree that 1-year mortality is important. Our protocol, approved by the ethics committee, followed patients to day 28 only, so these data were not collected. We now mention this as a limitation and suggest it for future studies."	Limitations p15

The first three accept and fix; the fourth is a polite rebuttal. It does not invent data that do not exist, gives a reason (the approved protocol) and still improves the paper by naming the limitation.

20.8 Rejection and resubmission

Rejection is a normal part of publishing, not a verdict on the researcher. Many good papers are rejected once or twice before acceptance, and high-profile journals reject the large majority of what they receive.

Desk rejection usually comes within days and usually means a problem of fit or of reporting. Re-target: choose a journal whose readers need the result, and check the reporting against the guideline. **Rejection after review** comes with reviewers' comments, which are free expert advice. Read them, wait a day, and then fix what they found before sending the paper elsewhere. Never resubmit an unchanged manuscript: the field is small, and the next journal may well invite the same reviewers, who will not be pleased to see their comments ignored.

When moving to the next journal, **reformat** the manuscript to its instructions for authors. Some publishers offer to **transfer** a rejected manuscript, with its reviews, to a sister journal, which can save months. An **appeal** against a rejection is appropriate only when the decision rests on a clear factual error (for example, a reviewer misread the design); disagreement with the editor's judgement of priority is not grounds for appeal.

After a rejection

Read the reviews for things to fix, not for reasons to be angry. The next reviewer may be the same person.

20.9 After acceptance: proofs, sharing and corrections

Acceptance is followed by copy-editing and typesetting, and then the publisher sends **proofs**, the paper as it will appear, often with a deadline of only a few days.

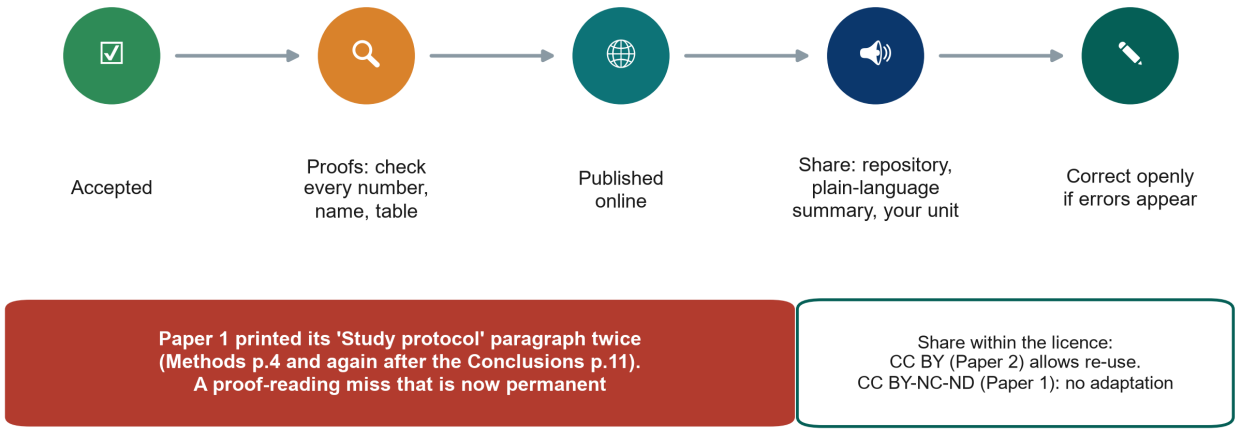


Figure 20.5: After acceptance: proofs, publication online, sharing and open correction of any errors. Paper 1 printed its "Study protocol" paragraph twice, once in the Methods and again after the Conclusions, a proof-reading miss that is now permanent. Share within the licence: CC BY (Paper 2) allows re-use; CC BY-NC-ND (Paper 1) allows no adaptation.

Read the proofs as carefully as the data. This is the last chance to correct errors before they become permanent. Check every number against your final tables, every author name and affiliation, every table and figure, and every equation and symbol, because typesetting can introduce errors that were not in the manuscript. Proofs are for corrections, not for new content or rewriting. Paper

1 prints its entire “Study protocol” paragraph twice, in the Methods and again after the Conclusions, a duplication that should have been caught at proof stage (Boulet et al. 2024). Paper 2’s author affiliations have inconsistent capitalisation, the kind of small error that proofs exist to catch (Hyun et al. 2023).

After publication, **share the work** as the licence and the journal’s policy allow: deposit it in your institutional repository, present it in your department, write a short plain-language summary for patients and colleagues, and update the trial registry with the results and the publication reference. If the press is interested, keep to what the design allows (calibrated language, Chapter 19). The Declaration of Helsinki asks researchers to make the results of all studies public, including negative and inconclusive ones (World Medical Association 2024).

If you find an **error after publication**, contact the journal. A **correction** (erratum) is the honest response to an error that does not change the conclusions; a **retraction** is for errors serious enough to invalidate them, or for misconduct (Chapter 18). Authors who correct their own errors openly earn trust, not lose it.

20.10 Conference abstracts and posters

A conference presentation is often the first public output of a study, and it is a good way to receive early feedback. But it is not a substitute for a paper: many conference abstracts are never followed by a full publication, and the evidence they contain is effectively lost.

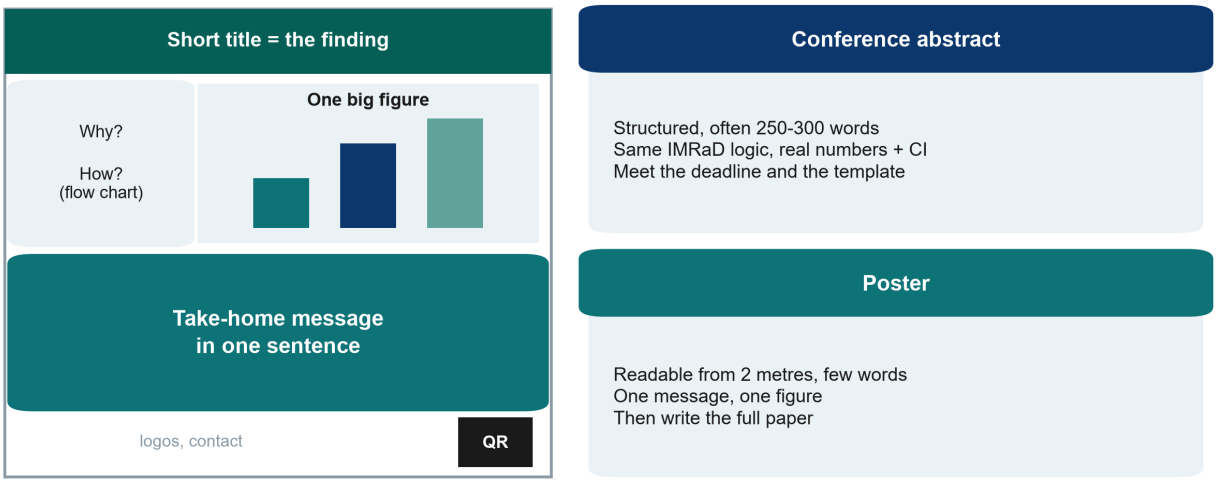


Figure 20.6: A conference abstract (structured, often 250 to 300 words, the same IMRaD logic, real numbers with confidence intervals, meeting the deadline and template) and a poster (a short title that states the finding, a brief why and how, one big figure, a one-sentence take-home message and a QR code), readable from two metres.

A **conference abstract** follows the same logic as a paper’s abstract (Background, Methods, Results, Conclusions) within a strict word limit, often 250 to 300 words, and usually without tables or references. It must contain **real results** with numbers and confidence intervals: an abstract that says “results will be presented” is usually rejected, so submit only when the main analysis is complete or nearly so. Follow the conference template and deadline exactly. A useful word budget for a 300-word abstract is about 40 to 50 words of background, 70 to 80 of methods, 90 to 110 of results and 30 to 40 of conclusions.

A **poster** is a billboard, not a paper pinned to a wall. Passers-by give it about a minute from a metre

or two away. A good poster has a short title that states the finding; a few words of background and methods, often as a flow chart; **one big figure** showing the primary result (for example, estimates with confidence intervals or Kaplan–Meier curves); a take-home message of one sentence in a prominent box; and a QR code linking to the abstract or paper. The text must be readable from about two metres (as a guide, a title of at least 85 point and body text of at least 24 point), with no more than a few hundred words in total, two or three colours plus grey, and dark text on a light background. The numbers on the poster must match the abstract. Prepare a one-minute spoken summary for the people who stop.

Presenting an abstract at a conference does not normally prevent later publication of the full paper, because most journals do not regard abstracts as prior publication; but check the target journal's policy. And then write the full paper.

Worked example: a conference abstract for FLUID-ICU (simulated data)

Title: Positive fluid balance by day 3 and 28-day mortality in sepsis: a multicentre cohort study
Background: Fluid accumulation in sepsis is associated with death, but sicker patients receive more fluid. We examined the association after adjustment for severity.
Methods: Prospective cohort (simulated teaching data) of adults with sepsis in four ICUs. Exposure: positive cumulative fluid balance by ICU day 3. Outcome: 28-day mortality. Logistic regression adjusted for age, SOFA, lactate, vasopressors, mechanical ventilation and chronic kidney disease.
Results: Of 618 patients, 373 (60.4%) had a positive balance; 35.9% vs 18.4% died by day 28. The crude OR was 2.49 (95% CI 1.70 to 3.70) and the adjusted OR 1.62 (95% CI 1.02 to 2.59).
Conclusions: A positive day-3 fluid balance was modestly associated with 28-day mortality; much of the crude association reflected severity. A randomised trial is needed.
The poster's single figure would be the three odds ratios (crude, adjusted and imputed) with their confidence intervals, and its take-home box would carry the conclusion sentence.

Summary

Getting published starts with choosing the journal before writing, by fit (whose readers need the result), trust (genuine indexing and transparent policies), rules (the instructions for authors) and cost and speed. Open access makes work free to read, through article-processing charges (gold) or repository deposit (green); the Creative Commons licence determines what others may do with it. Predatory journals take fees without real peer review; flattering mass e-mails, impossible speed, fake metrics and hidden fees are warning signs, and Think.Check.Submit, DOAJ, the NLM Catalog and librarians are the checks. Preprints offer speed and openness but are not peer-reviewed. A complete submission package (cover letter, title page with ORCID iDs, formatted manuscript, tables and figures, reporting checklist, declarations, registration number, suggested reviewers, supplements) avoids delays, and the manuscript goes to one journal at a time. Peer review leads to acceptance, revision or rejection; revisions are answered point by point, thanking reviewers, answering every comment, showing exactly what changed and disagreeing only politely and with evidence. Rejection is normal: fix what reviewers found and re-target. Proofs must be read as carefully as the data, results shared within the licence, and errors corrected openly. Conference abstracts need real numbers, posters need one message and one figure, and both should be followed by the full paper.

Further reading

- The ICMJE Recommendations (International Committee of Medical Journal Editors 2024) describe the responsibilities of authors, reviewers and editors throughout the publication process, including duplicate submission, preprints and corrections.
- Grudniewicz and colleagues (Grudniewicz et al. 2019) set out the consensus definition of predatory journals and why they matter.
- Bordage's analysis of reviewers' comments (Bordage 2001) shows which weaknesses lead to rejection and which strengths to acceptance; nearly all are within the authors' control.
- Lang (Lang 2010) gives practical guidance on submission, responding to reviewers, conference abstracts and posters.

APPENDIX A

Glossary: English–Vietnamese

Key terms used in this book, in alphabetical order, with the Vietnamese term and a plain-language definition in both languages. The definitions are written for clinicians new to research; they are not formal textbook definitions. Abbreviations widely used in Vietnamese practice (PICO, RCT, CRF, OR, RR, NNT, GCP …) are kept in English.

Các thuật ngữ chính dùng trong sách, xếp theo thứ tự chữ cái tiếng Anh, kèm thuật ngữ tiếng Việt và định nghĩa bằng ngôn ngữ đơn giản.

Term · Thuật ngữ	Definition · Định nghĩa
T	
2×2 table Bảng 2×2	A table of exposure (yes/no) by outcome (yes/no) used to calculate risks, RR, OR and NNT. *Bảng phơi nhiễm (có/không) theo kết cục (có/không), dùng để tính nguy cơ, RR, OR và NNT.*
A	
Adjusted estimate Ước lượng đã hiệu chỉnh	An effect estimate that accounts for other measured factors (confounders) in the model. *Ước lượng tác động đã tính đến các yếu tố khác đo được (yếu tố gây nhiễu) trong mô hình.*
AI disclosure Công khai việc sử dụng trí tuệ nhân tạo (AI)	Stating how AI tools were used in a study or manuscript; AI cannot be an author. *Nêu rõ cách sử dụng công cụ AI trong nghiên cứu hoặc bản thảo; AI không thể là tác giả.*
Aim Mục tiêu tổng quát	The broad purpose of the study in one sentence. *Mục đích chung của nghiên cứu, viết trong một câu.*
Alternative hypothesis Giả thuyết đối (H1)	What we accept if the data are incompatible with H0: usually that the groups differ, in either direction. *Điều được chấp nhận khi dữ liệu không phù hợp với H0: thường là các nhóm khác nhau, theo bất kỳ chiều nào.*
Analytical study Nghiên cứu phân tích	A study that compares groups to test whether an exposure or treatment is linked to an outcome. *Nghiên cứu so sánh các nhóm để kiểm định mối liên quan giữa phơi nhiễm hoặc điều trị và kết cục.*
Article processing charge (APC) Phí xử lý bài báo (APC)	A fee that some open-access journals charge authors when a paper is accepted. *Khoản phí một số tạp chí truy cập mở thu của tác giả khi bài báo được chấp nhận.*
Audit trail Nhật ký thay đổi dữ liệu (audit trail)	An automatic record of who changed which value, when and why. *Bản ghi tự động cho biết ai đã thay đổi giá trị nào, khi nào và vì sao.*

Term · Thuật ngữ	Definition · Định nghĩa
Authorship (ICMJE criteria) Quyền tác giả (tiêu chí ICMJE)	Authors must contribute substantially, draft or revise the work, approve the final version and take responsibility for it. *Tác giả phải đóng góp đáng kể, viết hoặc chỉnh sửa bản thảo, phê duyệt bản cuối và chịu trách nhiệm về công trình.*
B	
Bias Sai lệch (sai số hệ thống)	A systematic error in design or conduct that pushes results away from the truth. *Sai số có hệ thống trong thiết kế hoặc thực hiện, làm kết quả lệch khỏi sự thật.*
Binary variable Biến nhị phân	A variable with two categories, e.g. died / alive. *Biến có hai giá trị, ví dụ tử vong / sống.*
Blinding (masking) Làm mù	Keeping patients, staff or assessors unaware of which group a patient is in. *Giữ cho người bệnh, nhân viên hoặc người đánh giá không biết người bệnh thuộc nhóm nào.*
Boolean operators (AND, OR, NOT) Toán tử Boolean (AND, OR, NOT)	Words that combine search terms: AND narrows, OR widens, NOT excludes. *Các từ dùng để kết hợp từ khóa: AND thu hẹp, OR mở rộng, NOT loại trừ.*
Box plot Biểu đồ hộp	Shows the median, IQR and range of a variable; good for skewed data and for comparing groups. *Thể hiện trung vị, khoảng tứ phân vị và khoảng giá trị của một biến; phù hợp cho dữ liệu lệch và để so sánh các nhóm.*
Bradford Hill criteria Tiêu chí Bradford Hill	Points to weigh when judging whether an association is causal: strength, consistency, temporality, dose–response, plausibility and others. *Các điểm cần cân nhắc khi đánh giá một mối liên quan có phải quan hệ nhân quả hay không: độ mạnh, tính nhất quán, trình tự thời gian, quan hệ liều–đáp ứng, tính hợp lý và các tiêu chí khác.*
C	
Calibrated language Ngôn ngữ chừng mực, tương xứng với bằng chứng	Words that match the strength of the evidence: “associated with” for observational studies; “caused” only where the design supports it. *Từ ngữ tương xứng với mức độ bằng chứng: “có liên quan với” cho nghiên cứu quan sát; chỉ dùng “gây ra” khi thiết kế nghiên cứu cho phép.*
Calibration and discrimination Hiệu chuẩn và khả năng phân biệt	Calibration: predicted risks match observed risks. Discrimination: the model ranks patients who have the event above those who do not (AUC). *Hiệu chuẩn: nguy cơ dự báo khớp với nguy cơ quan sát được. Khả năng phân biệt: mô hình xếp người có biến cố cao hơn người không có biến cố (AUC).*
Case report / case series Báo cáo ca bệnh / loạt ca bệnh	A description of one or a few patients with an unusual condition or response. *Mô tả một hoặc một số người bệnh có tình trạng hoặc đáp ứng bất thường.*
Case report form (CRF) Phiếu thu thập dữ liệu (CRF)	The form on which study data are recorded for each participant. *Biểu mẫu dùng để ghi dữ liệu nghiên cứu của từng người tham gia.*
Case-control study Nghiên cứu bệnh–chứng	People with the outcome (cases) and without it (controls) are compared for past exposures. *So sánh người có kết cục (nhóm bệnh) và không có kết cục (nhóm chứng) về các phơi nhiễm trong quá khứ.*
Censoring Dữ liệu có thông tin khuyết (censored)	When a patient’s follow-up ends before the event happens (study ends, transfer, loss to follow-up). *Khi việc theo dõi người bệnh kết thúc trước khi biến cố xảy ra (nghiên cứu kết thúc, chuyển viện, mất theo dõi).*
Chi-square test Kiểm định Chi bình phương (χ^2)	Compares proportions of a categorical outcome between groups. *So sánh tỷ lệ của một biến phân loại giữa các nhóm.*

Term · Thuật ngữ	Definition · Định nghĩa
Citation style (NLM/Vancouver, APA) Kiểu trích dẫn (NLM/Vancouver, APA)	The format for citing and listing references. Most medical journals use a numbered style such as NLM/Vancouver (in Zotero: 'NLM/Vancouver: Citing Medicine 2nd edition (citation-sequence)'); APA ('APA Style 7th edition') gives author and year. *Quy cách trích dẫn và liệt kê tài liệu tham khảo. Đa số tạp chí y khoa dùng kiểu đánh số như NLM/Vancouver (trong Zotero: 'NLM/Vancouver: Citing Medicine 2nd edition (citation-sequence)'); APA ('APA Style 7th edition') ghi tác giả và năm.*
Cleaning log Nhật ký làm sạch dữ liệu	A record of every problem found and every change made, with the reason (e.g. "ID 0412: weight 7 kg → missing; implausible"). *Bản ghi mọi vấn đề phát hiện được và mọi thay đổi đã thực hiện, kèm lý do (ví dụ "ID 0412: cân nặng 7 kg → để trống; giá trị vô lý").*
Clinical audit Đánh giá thực hành lâm sàng (audit lâm sàng)	Measuring current practice against an agreed standard to see whether the standard is met. *Đo lường thực hành hiện tại so với một tiêu chuẩn đã thống nhất để xem tiêu chuẩn đó có được đáp ứng hay không.*
Cluster randomised trial Thử nghiệm ngẫu nhiên theo cụm	Groups such as wards or hospitals, not individual patients, are randomised. *Phân bổ ngẫu nhiên theo nhóm (ví dụ khoa phòng hoặc bệnh viện) thay vì theo từng người bệnh.*
Cohort study (prospective / retrospective) Nghiên cứu thuần tập (tiến cứu / hồi cứu)	People grouped by exposure are followed over time to see who develops the outcome; data may be collected forwards or from existing records. *Người tham gia được chia nhóm theo phơi nhiễm và theo dõi theo thời gian để xem ai xuất hiện kết cục; dữ liệu có thể thu thập về phía trước hoặc từ hồ sơ có sẵn.*
Comparison (comparator) Nhóm so sánh (nhóm chứng)	What the intervention is compared with, e.g. usual care or placebo. *Đối tượng hoặc phương pháp dùng để so sánh với can thiệp, ví dụ chăm sóc thường quy hoặc giả dược.*
Complete-case analysis Phân tích các trường hợp đầy đủ	Analysing only patients with no missing values; simple, but it can be biased and wastes data. *Chỉ phân tích những người bệnh không thiếu giá trị nào; đơn giản nhưng có thể sai lệch và lãng phí dữ liệu.*
Conference abstract and poster Tóm tắt và poster hội nghị	A short summary submitted to a scientific meeting and presented as a talk or a poster. *Bản tóm tắt ngắn gửi đến hội nghị khoa học, được trình bày dưới dạng báo cáo miệng hoặc poster.*
Confidence interval (95% CI) Khoảng tin cậy 95% (KTC 95%)	The range of values for the true effect that is compatible with the data; narrow = precise. *Khoảng giá trị của tác động thực phù hợp với dữ liệu; khoảng hẹp nghĩa là ước lượng chính xác.*
Confidentiality Bảo mật thông tin	Protecting participants' personal information from people who should not see it. *Bảo vệ thông tin cá nhân của người tham gia khỏi những người không có quyền tiếp cận.*
Conflict of interest Xung đột lợi ích	A financial or personal interest that could influence, or appear to influence, the research. *Lợi ích tài chính hoặc cá nhân có thể ảnh hưởng, hoặc bị xem là ảnh hưởng, đến nghiên cứu.*
Confounding by indication Nhiễu do chỉ định	Sicker patients receive more of a treatment, so the treatment looks harmful because of the illness, not the treatment. *Người bệnh nặng hơn được điều trị nhiều hơn, nên điều trị có vẻ gây hại nhưng thực ra do mức độ bệnh.*
Confounding (confounder) Nhiễu (yếu tố gây nhiễu)	A mixing of effects: a third factor linked to both exposure and outcome distorts their apparent relationship. *Sự pha trộn tác động: một yếu tố thứ ba liên quan đến cả phơi nhiễm và kết cục làm sai lệch mối liên quan quan sát được.*

Term · Thuật ngữ	Definition · Định nghĩa
Continuous variable Biến liên tục	A measured number that can take any value in a range, e.g. fluid balance in ml/kg. *Số đo có thể nhận bất kỳ giá trị nào trong một khoảng, ví dụ cân bằng dịch tính bằng ml/kg.*
Correlation coefficient (r) Hệ số tương quan (r)	A number from −1 to +1 showing how closely two continuous variables follow a straight line; correlation is not agreement and not causation. *Con số từ −1 đến +1 cho biết hai biến liên tục bám theo một đường thẳng chặt chẽ đến đâu; tương quan không phải là sự phù hợp và không phải là quan hệ nhân quả.*
Course peer review (individual) Bình duyệt chéo giữa các học viên	The after-course homework in which each participant reviews one classmate's Results section (reviewer assigned at random in Session 7) and then responds to the review of their own Results section. *Bài tập về nhà sau khóa học, trong đó mỗi học viên bình duyệt phần Kết quả của một bạn cùng lớp (người bình duyệt được phân công ngẫu nhiên ở Buổi 7) và sau đó phản hồi các nhận xét về phần Kết quả của mình.*
Cover letter Thư gửi ban biên tập (cover letter)	A one-page letter to the editor saying what the paper found, why it matters to the journal's readers, and the required declarations. *Thư một trang gửi biên tập viên, nêu phát hiện chính, tầm quan trọng với độc giả của tạp chí và các cam kết bắt buộc.*
Cox regression (proportional hazards) Hồi quy Cox (nguy cơ tỷ lệ)	A model for time-to-event outcomes that gives adjusted hazard ratios; it assumes the HR stays constant over time. *Mô hình cho kết cục dạng thời gian đến biến cố, cho ra tỷ số nguy cơ tức thời đã hiệu chỉnh; giả định HR không đổi theo thời gian.*
CRediT (contributor roles) Phân loại vai trò đóng góp CRediT	A standard list of 14 contributor roles (e.g. conceptualisation, data curation, writing) used to state who did what. *Danh mục chuẩn gồm 14 vai trò đóng góp (ví dụ lên ý tưởng, quản lý dữ liệu, viết bản thảo) dùng để nêu rõ ai đã làm gì.*
Critical appraisal Thẩm định phê bình (đánh giá y văn)	Systematically judging whether a study's methods and results are trustworthy and useful for your patients. *Đánh giá có hệ thống xem phương pháp và kết quả của một nghiên cứu có đáng tin cậy và hữu ích cho người bệnh của mình hay không.*
Crossover trial Thử nghiệm bắt chéo	Each participant receives both treatments in turn, separated by a washout period; suits stable, chronic conditions. *Mỗi người tham gia lần lượt nhận cả hai phương pháp điều trị, cách nhau một giai đoạn thải trừ; phù hợp với bệnh mạn tính, ổn định.*
Cross-sectional study Nghiên cứu cắt ngang	Exposure and outcome are measured at the same point in time, like a snapshot. *Phơi nhiễm và kết cục được đo cùng một thời điểm, giống như một "bức ảnh chụp nhanh".*
Crude vs adjusted estimate Ước lượng thô và ước lượng đã hiệu chỉnh	The crude estimate ignores other factors; the adjusted estimate accounts for measured confounders. A large gap between them signals confounding. *Ước lượng thô không tính đến các yếu tố khác; ước lượng đã hiệu chỉnh tính đến các yếu tố gây nhiễu đã đo. Chênh lệch lớn giữa hai ước lượng là dấu hiệu của nhiễu.*
D	
Data cleaning Làm sạch dữ liệu	Finding and fixing errors in a dataset (duplicates, impossible values, inconsistent codes, wrong units) before analysis, in a documented and repeatable way. *Phát hiện và sửa lỗi trong bộ dữ liệu (bản ghi trùng, giá trị vô lý, mã hóa không thống nhất, sai đơn vị) trước khi phân tích, theo cách có ghi chép và lặp lại được.*
Data dictionary Từ điển dữ liệu	A table defining every variable: name, meaning, type, unit or codes, allowed values. *Bảng định nghĩa từng biến: tên, ý nghĩa, loại biến, đơn vị hoặc mã hóa, giá trị cho phép.*

Term · Thuật ngữ	Definition · Định nghĩa
Data validation (range and consistency checks) Kiểm tra tính hợp lệ của dữ liệu (kiểm tra khoảng giá trị và tính nhất quán)	Checks that catch impossible or contradictory values, e.g. age 200, or discharge before admission. *Các bước kiểm tra phát hiện giá trị vô lý hoặc mâu thuẫn, ví dụ tuổi 200, hoặc ngày ra viện trước ngày nhập viện.*
Data-sharing statement Tuyên bố chia sẻ dữ liệu	A statement on whether, how and with whom the study data will be shared. *Tuyên bố về việc dữ liệu nghiên cứu có được chia sẻ hay không, bằng cách nào và với ai.*
Declaration of Helsinki Tuyên bố Helsinki	The World Medical Association's ethical principles for medical research involving people. *Các nguyên tắc đạo đức của Hiệp hội Y khoa Thế giới cho nghiên cứu y học trên con người.*
Deferred consent Đồng ý trì hoãn	In emergencies approved by an ethics committee, consent is sought after enrolment, as soon as possible. *Trong tình huống cấp cứu được hội đồng đạo đức chấp thuận, sự đồng ý được xin sau khi đưa vào nghiên cứu, càng sớm càng tốt.*
De-identification Loại bỏ thông tin định danh	Removing names and other identifiers and replacing them with a study code. *Xóa tên và các thông tin nhận dạng khác, thay bằng mã số nghiên cứu.*
Derived variable Biến dẫn xuất	A variable calculated from others, e.g. cumulative fluid balance (ml/kg) from daily fluid in, fluid out and body weight. *Biến được tính từ các biến khác, ví dụ cân bằng dịch tích lũy (ml/kg) từ lượng dịch vào, dịch ra hằng ngày và cân nặng.*
Descriptive study Nghiên cứu mô tả	A study that describes who has a condition, where and when, without comparing groups to test a cause. *Nghiên cứu mô tả ai mắc bệnh, ở đâu, khi nào, không so sánh các nhóm để kiểm định nguyên nhân.*
Dissemination Phổ biến kết quả nghiên cứu	Sharing results with colleagues, patients and policymakers through publication, presentations and local feedback. *Chia sẻ kết quả với đồng nghiệp, người bệnh và nhà hoạch định chính sách qua công bố, báo cáo và phản hồi tại đơn vị.*
Double data entry Nhập liệu kép	Two people enter the same data independently and the differences are checked. *Hai người nhập cùng một bộ dữ liệu một cách độc lập và các điểm khác biệt được kiểm tra lại.*
Duplicate and salami publication Xuất bản trùng lặp và xuất bản "cắt lát"	Publishing the same results twice, or slicing one study into many small papers. *Công bố cùng một kết quả hai lần, hoặc chia nhỏ một nghiên cứu thành nhiều bài báo nhỏ.*
Duplicate record Bản ghi trùng lặp	The same patient entered more than once; duplicates must be found and removed so that each patient is counted once. *Cùng một người bệnh được nhập nhiều lần; cần phát hiện và loại bỏ để mỗi người bệnh chỉ được tính một lần.*
Dynamite plot Biểu đồ cột kèm thanh sai số ("dynamite plot")	A bar showing only a mean with an error bar; it hides the data, so prefer dot plots or box plots. *Biểu đồ cột chỉ thể hiện giá trị trung bình kèm thanh sai số; che giấu dữ liệu thực, nên dùng biểu đồ chấm hoặc biểu đồ hộp thay thế.*
E	
Effect modification (interaction) Thay đổi tác động (tương tác)	The effect of an exposure differs between subgroups (e.g. it works in younger but not older patients); it should be reported, not "adjusted away". *Tác động của một phơi nhiễm khác nhau giữa các phân nhóm (ví dụ có hiệu quả ở người trẻ nhưng không ở người cao tuổi); cần được báo cáo, không "hiệu chỉnh để loại bỏ".*

Term · Thuật ngữ	Definition · Định nghĩa
Electronic data capture (EDC) Thu thập dữ liệu điện tử (EDC)	Entering study data directly into a secure system (e.g. REDCap, KoBoToolbox) with built-in checks. *Nhập dữ liệu nghiên cứu trực tiếp vào hệ thống bảo mật (ví dụ REDCap, KoBoToolbox) có sẵn các bước kiểm tra.*
Ethics committee Hội đồng đạo đức trong nghiên cứu y sinh học	An independent group that reviews and approves research to protect participants. *Hội đồng độc lập xem xét và phê duyệt nghiên cứu nhằm bảo vệ người tham gia.*
Evidence-based medicine Y học dựa trên bằng chứng	Combining the best research evidence with clinical expertise and patient values to make care decisions. *Kết hợp bằng chứng nghiên cứu tốt nhất với kinh nghiệm lâm sàng và giá trị, mong muốn của người bệnh để ra quyết định chăm sóc.*
Exposure Phơi nhiễm (yếu tố tiếp xúc)	A factor people have or receive that is not assigned by the researcher, e.g. a large fluid balance. *Yếu tố mà người bệnh có hoặc nhận được nhưng không do nhà nghiên cứu chỉ định, ví dụ cân bằng dịch dương nhiều.*
External validation Thẩm định ngoài	Testing a model on patients from a different time or place than those used to build it. *Kiểm tra mô hình trên người bệnh ở thời điểm hoặc địa điểm khác với dữ liệu dùng để xây dựng mô hình.*
External validity (generalisability) Giá trị ngoại suy (tính khái quát hóa)	Whether the result applies to other patients and settings, such as your own unit. *Kết quả có áp dụng được cho người bệnh và bối cảnh khác hay không, ví dụ đơn vị của bạn.*
F	
Factorial trial Thử nghiệm giai thừa	Tests two or more interventions at once by randomising to every combination (e.g. A, B, A + B, neither). *Đánh giá cùng lúc hai hay nhiều can thiệp bằng cách phân bổ ngẫu nhiên vào mọi tổ hợp (ví dụ A, B, A + B, không can thiệp).*
Feasibility Tính khả thi	Whether the study can realistically be done with the available time, money, staff and patients. *Khả năng thực hiện được nghiên cứu với thời gian, kinh phí, nhân lực và người bệnh hiện có.*
FINER criteria Tiêu chí FINER	A check that a question is Feasible, Interesting, Novel, Ethical and Relevant. *Bộ tiêu chí kiểm tra câu hỏi có Khả thi, Thú vị, Mới, Đạo đức và Phù hợp (thiết thực) hay không.*
Forest plot Biểu đồ rừng (forest plot)	A chart showing effect estimates and their CIs as squares and lines, with a vertical line at no effect. *Biểu đồ trình bày các ước lượng tác động và khoảng tin cậy bằng ô vuông và đoạn thẳng, có một đường dọc tại giá trị không tác động.*
Funding statement Tuyên bố nguồn tài trợ	Declaring who paid for the study and what role the funder had. *Nêu rõ ai tài trợ cho nghiên cứu và nhà tài trợ có vai trò gì.*
G	
Gantt chart Biểu đồ Gantt	A timeline chart showing each study task as a bar across months. *Biểu đồ tiến độ thể hiện mỗi công việc của nghiên cứu bằng một thanh ngang theo các tháng.*
Gift and ghost authorship Tác giả “danh dự” và tác giả “ma”	Gift: listed without contributing. Ghost: contributed substantially but not listed. Both are unacceptable. *Tác giả “danh dự”: có tên nhưng không đóng góp. Tác giả “ma”: đóng góp đáng kể nhưng không có tên. Cả hai đều không được chấp nhận.*
Good Clinical Practice (GCP) Thực hành tốt thử nghiệm lâm sàng (GCP)	The international standard for designing, running and reporting clinical trials so participants are protected and data are reliable. *Tiêu chuẩn quốc tế về thiết kế, thực hiện và báo cáo thử nghiệm lâm sàng nhằm bảo vệ người tham gia và đảm bảo dữ liệu tin cậy.*

Term · Thuật ngữ	Definition · Định nghĩa
Grey literature Tài liệu xám	Research not published in journals: reports, theses, conference abstracts, trial registries. *Tài liệu nghiên cứu không đăng trên tạp chí: báo cáo, luận văn, tóm tắt hội nghị, cơ sở dữ liệu đăng ký thử nghiệm.*
H	
Hazard ratio (HR) Tỷ số nguy cơ tức thời (HR)	Compares the rate at which an event (e.g. death) happens over time in two groups; 1 means no difference. *So sánh tốc độ xảy ra một biến cố (ví dụ tử vong) theo thời gian giữa hai nhóm; bằng 1 nghĩa là không khác biệt.*
Heterogeneity (I^2) Tính không đồng nhất (I^2)	How much study results differ beyond chance; an I^2 above about 50–75% means substantial heterogeneity. *Mức độ kết quả các nghiên cứu khác nhau vượt quá yếu tố ngẫu nhiên; I^2 trên khoảng 50–75% là không đồng nhất đáng kể.*
Hypothesis Giả thuyết nghiên cứu	A testable prediction about the result, naming the groups and the outcome. *Dự đoán có thể kiểm định về kết quả, nêu rõ các nhóm và kết cục.*
I	
Implausible value Giá trị vô lý (ngoài ngưỡng có thể)	A value outside the possible range (e.g. age 200, weight 7 kg in an adult); checked against source records, then corrected or set to missing. *Giá trị nằm ngoài ngưỡng có thể xảy ra (ví dụ tuổi 200, cân nặng 7 kg ở người lớn); cần đối chiếu hồ sơ gốc, sau đó sửa lại hoặc để trống.*
IMRaD Cấu trúc IMRaD	The standard structure of a research paper: Introduction, Methods, Results and Discussion. *Cấu trúc chuẩn của bài báo khoa học: Đặt vấn đề, Phương pháp, Kết quả và Bàn luận.*
Incidence Tỷ lệ mới mắc	The proportion of people free of a condition who develop it over a period (new cases). *Tỷ lệ người chưa mắc bệnh xuất hiện bệnh trong một khoảng thời gian (ca mới).*
Information bias (recall, observer) Sai lệch thông tin (nhớ lại, người quan sát)	Bias from measuring exposure or outcome differently between groups, e.g. cases remember past exposures better. *Sai lệch do đo phơi nhiễm hoặc kết cục khác nhau giữa các nhóm, ví dụ nhóm bệnh nhớ phơi nhiễm trong quá khứ rõ hơn.*
Informed consent Sự đồng ý tham gia nghiên cứu sau khi được cung cấp thông tin	A person's voluntary agreement to take part after understanding the purpose, risks, benefits and their right to withdraw. *Sự đồng ý tự nguyện tham gia nghiên cứu sau khi hiểu rõ mục đích, nguy cơ, lợi ích và quyền rút khỏi nghiên cứu.*
Interaction term Số hạng tương tác	A term added to a regression model to test whether the effect of one variable differs across levels of another (e.g. fluid balance × chronic kidney disease). *Thành phần được thêm vào mô hình hồi quy để kiểm định xem tác động của một biến có khác nhau theo các mức của biến khác hay không (ví dụ cân bằng dịch × bệnh thận mạn).*
Internal validity Giá trị nội tại	Whether the result is true for the people studied, i.e. free of important bias and confounding. *Kết quả có đúng với chính những người được nghiên cứu hay không, tức là không bị sai lệch và nhiễu đáng kể.*
Interquartile range (IQR) Khoảng tứ phân vị (IQR)	The range of the middle 50% of values (25th to 75th percentile); used with the median. *Khoảng chứa 50% giá trị ở giữa (từ bách phân vị 25 đến 75); dùng kèm trung vị.*
Interrupted time series Chuỗi thời gian ngắt quãng	Many measurements before and after a change, to see whether the level or trend shifts beyond the existing trend. *Nhiều lần đo trước và sau một thay đổi, để xem mức hoặc xu hướng có thay đổi vượt ra ngoài xu hướng sẵn có hay không.*

Term · Thuật ngữ	Definition · Định nghĩa
Intervention Can thiệp	A treatment or action deliberately given, e.g. a restrictive fluid strategy. *Phương pháp điều trị hoặc hành động được chủ động áp dụng, ví dụ chiến lược hạn chế dịch.*
K	
Kaplan–Meier curve Đường cong Kaplan–Meier	A step-shaped graph of the proportion of patients still free of the event over time, allowing for censoring. *Đồ thị dạng bậc thang thể hiện tỷ lệ người bệnh chưa xảy ra biến cố theo thời gian, có tính đến dữ liệu có thông tin khuyết.*
Keywords Từ khóa	3–6 terms (ideally MeSH terms) chosen so that others can find your paper. *3–6 thuật ngữ (nên dùng thuật ngữ MeSH) giúp người khác tìm thấy bài báo của bạn.*
L	
Legal representative (surrogate) Người đại diện hợp pháp	A person allowed by law to give consent for a patient who cannot decide, e.g. a sedated ICU patient. *Người được pháp luật cho phép đồng ý thay cho người bệnh không thể tự quyết định, ví dụ người bệnh ICU đang dùng an thần.*
Levels of evidence (evidence pyramid) Mức độ bằng chứng (tháp bằng chứng)	A ranking of study types by how well they protect against bias; systematic reviews of trials at the top, expert opinion at the bottom. *Thứ bậc các loại nghiên cứu theo khả năng hạn chế sai lệch; tổng quan hệ thống các thử nghiệm ở đỉnh, ý kiến chuyên gia ở đáy.*
Licence (CC BY) Giấy phép sử dụng (CC BY)	The terms under which others may reuse a published article; CC BY allows reuse with attribution. *Điều khoản cho phép người khác sử dụng lại bài báo đã xuất bản; CC BY cho phép sử dụng lại với điều kiện ghi nguồn.*
Likelihood ratio (LR) Tỷ số khả dĩ (LR)	How much a test result changes the odds of disease; LR+ above 10 or LR− below 0.1 change decisions strongly. *Mức độ một kết quả xét nghiệm làm thay đổi khả năng mắc bệnh; LR+ trên 10 hoặc LR− dưới 0,1 làm thay đổi quyết định rõ rệt.*
Limitation Hạn chế của nghiên cứu	A weakness in design or conduct that may affect the results; good papers state them honestly. *Điểm yếu trong thiết kế hoặc thực hiện có thể ảnh hưởng đến kết quả; bài báo tốt nêu rõ một cách trung thực.*
Literature review Tổng quan tài liệu	A structured summary of what published studies already show about the question. *Bản tóm tắt có hệ thống những gì các nghiên cứu đã công bố cho thấy về câu hỏi.*
Log transformation Biến đổi logarit	Taking the logarithm of a skewed variable (e.g. ICU length of stay) so that a linear model fits better; results are then reported as ratios. *Lấy logarit của một biến lệch (ví dụ thời gian nằm ICU) để mô hình tuyến tính phù hợp hơn; kết quả khi đó được báo cáo dưới dạng tỷ số.*
Log-rank test Kiểm định log-rank	Compares Kaplan–Meier curves between groups. *So sánh các đường cong Kaplan–Meier giữa các nhóm.*
M	
Machine learning Học máy	Computer methods that learn patterns from data to make predictions; judged like any prediction model (TRIPOD+AI). *Các phương pháp máy tính học quy luật từ dữ liệu để đưa ra dự báo; được đánh giá như mọi mô hình dự báo (TRIPOD+AI).*
Major / minor revision Yêu cầu sửa chữa lớn / nhỏ	Editorial decisions asking for substantial or small changes before a paper can be accepted. *Quyết định của tạp chí yêu cầu chỉnh sửa nhiều hoặc ít trước khi bài báo có thể được chấp nhận.*

Term · Thuật ngữ	Definition · Định nghĩa
Matching Ghép cặp	Choosing comparison participants with similar values of key factors (e.g. age, severity) to reduce confounding. *Chọn người so sánh có đặc điểm chính tương tự (ví dụ tuổi, mức độ nặng) để giảm nhiễu.*
MCAR / MAR / MNAR Thiếu hoàn toàn ngẫu nhiên / thiếu ngẫu nhiên / thiếu không ngẫu nhiên	Why data are missing: completely at random (unrelated to anything); at random (explained by data that were recorded); not at random (related to the missing value itself). *Lý do dữ liệu bị thiếu: hoàn toàn ngẫu nhiên (không liên quan đến yếu tố nào); ngẫu nhiên (giải thích được bằng dữ liệu đã ghi nhận); không ngẫu nhiên (liên quan đến chính giá trị bị thiếu).*
Mean Trung bình	The sum of values divided by their number; pulled by extreme values. *Tổng các giá trị chia cho số giá trị; bị ảnh hưởng bởi các giá trị cực đoan.*
Median Trung vị	The middle value when data are ordered; not affected by a few extreme values. *Giá trị ở giữa khi sắp xếp dữ liệu theo thứ tự; ít bị ảnh hưởng bởi vài giá trị cực đoan.*
MeSH (Medical Subject Headings) Tiêu đề chủ đề y khoa (MeSH)	PubMed's controlled vocabulary: standard terms that group articles on the same topic, whatever words the authors used. *Bộ từ vựng chuẩn của PubMed, gom các bài báo cùng chủ đề dù tác giả dùng từ ngữ khác nhau.*
Meta-analysis Phân tích gộp	Statistical combination of the results of several studies into one overall estimate. *Kỹ thuật thống kê gộp kết quả của nhiều nghiên cứu thành một ước lượng chung.*
Missing data Dữ liệu bị thiếu	Values that should have been recorded but were not; they can bias results and reduce power. *Giá trị lẽ ra phải được ghi nhận nhưng không có; có thể gây sai lệch kết quả và giảm lực thống kê.*
Multiple comparisons So sánh bội (nhiều phép so sánh)	Doing many tests increases the chance of a false-positive finding; pre-specify one primary outcome or adjust (e.g. Bonferroni). *Thực hiện nhiều kiểm định làm tăng khả năng có kết quả dương tính giả; cần xác định trước một kết cục chính hoặc hiệu chỉnh (ví dụ Bonferroni).*
Multiple imputation Quy gán đa bội (multiple imputation)	Filling in missing values several times from a model that uses other variables, then combining the results so the uncertainty is reflected. *Điền các giá trị bị thiếu nhiều lần dựa trên mô hình sử dụng các biến khác, rồi gộp kết quả để phản ánh độ không chắc chắn.*
N	
Nominal variable Biến định danh	Categories with no natural order, e.g. blood group. *Biến phân loại không có thứ tự tự nhiên, ví dụ nhóm máu.*
Non-parametric test (e.g. Mann–Whitney) Kiểm định phi tham số (ví dụ Mann–Whitney)	Compares groups using ranks; used for skewed or ordinal data. *So sánh các nhóm dựa trên thứ hạng; dùng cho dữ liệu lệch hoặc biến thứ hạng.*
Normal distribution Phân phối chuẩn	A symmetric, bell-shaped spread of values around the mean. *Phân bố các giá trị đối xứng hình chuông quanh giá trị trung bình.*
Null hypothesis Giả thuyết không (H0)	The starting assumption that there is no difference or no effect. *Giả định ban đầu rằng không có khác biệt hoặc không có tác động.*
Number needed to treat (NNT) Số người cần điều trị (NNT)	How many patients must be treated to prevent one extra bad outcome; NNT = 1 ÷ risk difference. *Số người bệnh cần được điều trị để ngăn ngừa thêm một kết cục xấu; NNT = 1 ÷ hiệu số nguy cơ.*
O	

Term · Thuật ngữ	Definition · Định nghĩa
Objective Mục tiêu cụ thể	A specific, measurable step towards the aim, e.g. “to compare 28-day mortality between groups”. *Bước cụ thể, đo lường được để đạt mục tiêu tổng quát, ví dụ “so sánh tỷ lệ tử vong 28 ngày giữa hai nhóm”.*
Odds ratio (OR) Tỷ số chênh (OR)	Odds of the outcome in one group divided by odds in the other; close to the RR when the outcome is rare. *Số chênh của kết cục ở nhóm này chia cho số chênh ở nhóm kia; gần bằng RR khi kết cục hiếm gặp.*
One-sided / two-sided test Kiểm định một phía / hai phía	A two-sided test looks for a difference in either direction and is the default. A one-sided test looks in one direction only and needs a strong reason written before the data are seen. *Kiểm định hai phía tìm khác biệt theo cả hai chiều và là lựa chọn mặc định. Kiểm định một phía chỉ xét một chiều và cần lý do vững chắc, viết ra trước khi xem dữ liệu.*
Open access Truy cập mở	Articles free for anyone to read online, often paid for by an author fee. *Bài báo được mọi người đọc miễn phí trên mạng, thường do tác giả trả phí xuất bản.*
ORCID Mã định danh ORCID	A free, permanent identifier that links a researcher to their publications. *Mã định danh miễn phí, cố định, liên kết nhà nghiên cứu với các công trình của họ.*
Ordinal variable Biến thứ hạng	Categories with a natural order but unequal steps, e.g. pressure injury stage 1–4. *Biến phân loại có thứ tự nhưng khoảng cách giữa các bậc không bằng nhau, ví dụ loét tì đè độ 1–4.*
Outcome Kết cục	The result that is measured to answer the question, e.g. death by day 28. *Kết quả được đo lường để trả lời câu hỏi, ví dụ tử vong trong 28 ngày.*
Overfitting (events per variable) Quá khớp (số biến cố trên mỗi biến)	A model with too many variables for the number of outcome events fits noise and fails on new patients; stepwise selection makes this worse. *Mô hình có quá nhiều biến so với số biến cố sẽ khớp cả nhiễu ngẫu nhiên và không dùng được cho người bệnh mới; chọn biến từng bước (stepwise) làm tình trạng này tệ hơn.*
P	
Paired vs unpaired data Dữ liệu ghép cặp và không ghép cặp	Paired data come from the same patients measured twice (or from matched pairs) and need paired tests. *Dữ liệu ghép cặp là cùng người bệnh được đo hai lần (hoặc các cặp được ghép) và cần dùng kiểm định ghép cặp.*
Paraphrasing Diễn giải bằng lời của mình	Expressing someone else’s idea in your own words and sentence structure, with a citation. *Trình bày ý tưởng của người khác bằng từ ngữ và cấu trúc câu của chính mình, kèm trích dẫn nguồn.*
Participant flow diagram Sơ đồ dòng người tham gia	A diagram showing how many patients were screened, excluded (with reasons) and analysed. *Sơ đồ cho biết số người bệnh được sàng lọc, bị loại (kèm lý do) và được đưa vào phân tích.*
Peer review Bình duyệt (phản biện đồng cấp)	Evaluation of a manuscript by independent experts before a journal decides to publish it. *Việc các chuyên gia độc lập đánh giá bản thảo trước khi tạp chí quyết định đăng.*
PICO(T) PICO(T)	A frame for a clinical question: Population, Intervention (or exposure), Comparison, Outcome, (Time). *Khung xây dựng câu hỏi lâm sàng: Đối tượng (P), Can thiệp hoặc phơi nhiễm (I), So sánh (C), Kết cục (O), (T, Thời gian).*

Term · Thuật ngữ	Definition · Định nghĩa
Pilot / feasibility study Nghiên cứu thí điểm / nghiên cứu khả thi	A small study to test whether a larger study can be done (recruitment, procedures), not to test whether the treatment works. *Nghiên cứu nhỏ để kiểm tra liệu một nghiên cứu lớn hơn có thực hiện được không (tuyển chọn, quy trình), không nhằm đánh giá hiệu quả điều trị.*
Plagiarism Đạo văn	Presenting someone else's words, ideas or data as your own, without proper citation. *Trình bày câu chữ, ý tưởng hoặc dữ liệu của người khác như của mình mà không trích dẫn đúng.*
Point-by-point response Phản hồi theo từng ý	Answering every reviewer comment separately: the comment, your reply, and the change made (with page and line). *Trả lời riêng từng nhận xét của người phản biện: nhận xét, câu trả lời và thay đổi đã thực hiện (kèm số trang và dòng).*
Pooled estimate Ước lượng gộp	The single combined effect from a meta-analysis, shown as the diamond on a forest plot. *Ước lượng tác động chung từ phân tích gộp, thể hiện bằng hình thoi trên biểu đồ rừng.*
Population Đối tượng nghiên cứu (quần thể)	The patients the question is about, e.g. adults with septic shock in the ICU. *Nhóm người bệnh mà câu hỏi hướng đến, ví dụ người lớn bị sốc nhiễm khuẩn tại ICU.*
Positive / negative predictive value (PPV, NPV) Giá trị tiên đoán dương / âm (PPV, NPV)	The chance that a positive (or negative) test result is correct; unlike sensitivity and specificity, they depend on prevalence. *Xác suất kết quả xét nghiệm dương (hoặc âm) là đúng; khác với độ nhạy và độ đặc hiệu, chúng phụ thuộc vào tỷ lệ hiện mắc.*
Power Lực thống kê (power)	The chance that a study detects a real effect of a given size; usually set at 80% or 90%. *Khả năng nghiên cứu phát hiện được một tác động có thật với độ lớn nhất định; thường đặt ở mức 80% hoặc 90%.*
Predatory journal Tạp chí săn mồi	A journal that charges authors but skips real peer review and editorial checks. *Tạp chí thu phí tác giả nhưng bỏ qua bình duyệt thực chất và kiểm tra biên tập.*
Prediction model Mô hình dự báo (tiên lượng)	A model that estimates an individual patient's risk of an outcome from several predictors. *Mô hình ước tính nguy cơ xảy ra một kết cục cho từng người bệnh dựa trên nhiều yếu tố dự báo.*
Preprint Bản tiền ấn phẩm (preprint)	A manuscript posted publicly (e.g. on medRxiv) before peer review. *Bản thảo được đăng công khai (ví dụ trên medRxiv) trước khi được bình duyệt.*
Prevalence Tỷ lệ hiện mắc	The proportion of people who have a condition at one point in time (all existing cases). *Tỷ lệ người đang mắc bệnh tại một thời điểm (tất cả các ca hiện có).*
Primary vs secondary research Nghiên cứu sơ cấp và nghiên cứu thứ cấp	Primary research collects new data; secondary research analyses existing data or studies (e.g. a records review or a systematic review). *Nghiên cứu sơ cấp thu thập dữ liệu mới; nghiên cứu thứ cấp phân tích dữ liệu hoặc nghiên cứu có sẵn (ví dụ hồi cứu hồ sơ bệnh án, tổng quan hệ thống).*
PRISMA flow diagram Sơ đồ PRISMA	The diagram in a systematic review showing how many records were found, screened, excluded and included. *Sơ đồ trong tổng quan hệ thống cho biết số tài liệu được tìm thấy, sàng lọc, loại trừ và đưa vào.*
Problem statement Đặt vấn đề	A short description of the clinical problem, how big it is, and what is not yet known. *Mô tả ngắn gọn vấn đề lâm sàng, mức độ quan trọng và những điều còn chưa biết.*
Proofs Bản in thử	The typeset version of an accepted paper, which the authors check for errors before publication. *Bản dàn trang của bài báo đã được chấp nhận, tác giả kiểm tra lỗi trước khi xuất bản.*

Term · Thuật ngữ	Definition · Định nghĩa
Propensity score matching Ghép cặp theo điểm xu hướng	Pairing exposed and unexposed patients who had a similar probability of exposure (based on measured factors), to make groups comparable in an observational study, as in Paper 2. *Ghép cặp người có và không có phơi nhiễm có xác suất phơi nhiễm tương tự (dựa trên các yếu tố đã đo), để các nhóm so sánh được trong nghiên cứu quan sát, như trong Bài báo 2.*
Protocol Đề cương nghiên cứu	The full written plan of a study: why, who, what, how and how it will be analysed. *Bản kế hoạch đầy đủ của nghiên cứu: lý do, đối tượng, nội dung, phương pháp và cách phân tích.*
P-value Giá trị p	If there were truly no effect, the probability of a result at least as extreme as the one observed. *Nếu thực sự không có tác động, đây là xác suất thu được kết quả ít nhất cực đoan như kết quả quan sát.*
Q	
Qualitative study (interviews, focus groups) Nghiên cứu định tính (phỏng vấn sâu, thảo luận nhóm tập trung)	Explores why and how (experiences, beliefs, barriers) through words rather than numbers. *Tìm hiểu "tại sao" và "như thế nào" (trải nghiệm, niềm tin, rào cản) qua lời nói thay vì con số.*
Quality improvement (QI) and PDSA cycle Cải tiến chất lượng (QI) và chu trình PDSA	Making and testing local changes to improve care, often in small Plan–Do–Study–Act cycles. *Thực hiện và thử nghiệm các thay đổi tại đơn vị để cải thiện chăm sóc, thường theo các chu trình nhỏ Lập kế hoạch – Thực hiện – Nghiên cứu – Hành động (PDSA).*
Quantitative / qualitative / mixed-methods research Nghiên cứu định lượng / định tính / phương pháp hỗn hợp	Quantitative research measures and counts; qualitative research explores experiences and meanings through words (interviews, observation); mixed methods combines both. *Nghiên cứu định lượng đo lường và đếm; nghiên cứu định tính tìm hiểu trải nghiệm và ý nghĩa qua lời nói (phỏng vấn, quan sát); phương pháp hỗn hợp kết hợp cả hai.*
Quasi-experimental (before–after) study Nghiên cứu bán thực nghiệm (trước–sau)	An intervention is introduced without randomisation and outcomes are compared before and after; common in QI. *Can thiệp được áp dụng không qua phân bổ ngẫu nhiên và kết cục được so sánh trước và sau; thường gặp trong cải tiến chất lượng.*
R	
R^2 Hệ số xác định (R^2)	The proportion of the variation in the outcome that a linear model explains. *Tỷ lệ biến thiên của kết cục được mô hình tuyến tính giải thích.*
Randomisation Phân bổ ngẫu nhiên	Allocating participants by chance so that groups are similar in both measured and unmeasured factors. *Chia nhóm người tham gia một cách ngẫu nhiên để các nhóm tương đồng về cả yếu tố đo được và không đo được.*
Randomised controlled trial (RCT) Thử nghiệm lâm sàng ngẫu nhiên có đối chứng (RCT)	Participants are allocated by chance to the intervention or the comparator, then followed for the outcome. *Người tham gia được phân bổ ngẫu nhiên vào nhóm can thiệp hoặc nhóm chứng, sau đó được theo dõi kết cục.*
Raw data vs clean data Dữ liệu thô và dữ liệu đã làm sạch	Raw data are kept exactly as collected and never edited; clean data are a separate, corrected copy produced by a documented script. *Dữ liệu thô được giữ đúng như khi thu thập và không bao giờ bị chỉnh sửa; dữ liệu sạch là một bản sao riêng đã được sửa lỗi bằng một quy trình (script) có ghi chép.*
Reference manager Phần mềm quản lý tài liệu tham khảo	Software (e.g. Zotero, Mendeley, EndNote) that stores references and inserts citations in the chosen style. *Phần mềm (ví dụ Zotero, Mendeley, EndNote) lưu trữ tài liệu tham khảo và chèn trích dẫn theo kiểu đã chọn.*

Term · Thuật ngữ	Definition · Định nghĩa
Regression (linear / logistic) Hồi quy (tuyến tính / logistic)	A model that relates an outcome to several factors at once; linear for continuous outcomes, logistic for binary outcomes. *Mô hình liên hệ một kết cục với nhiều yếu tố cùng lúc; hồi quy tuyến tính cho kết cục liên tục, hồi quy logistic cho kết cục nhị phân.*
Relative risk (RR) Nguy cơ tương đối (RR)	Risk in one group divided by risk in the other; 1 means no difference. *Nguy cơ ở nhóm này chia cho nguy cơ ở nhóm kia; bằng 1 nghĩa là không khác biệt.*
Reporting guideline (CONSORT, STROBE, STARD, PRISMA) Hướng dẫn báo cáo (CONSORT, STROBE, STARD, PRISMA)	A checklist of what to report for each study type: trials, observational studies, diagnostic studies, systematic reviews. *Bảng kiểm các nội dung cần báo cáo cho từng loại nghiên cứu: thử nghiệm, nghiên cứu quan sát, nghiên cứu chẩn đoán, tổng quan hệ thống.*
Research cycle Chu trình nghiên cứu	The repeating path from a clinical problem to a question, a study, results, publication and changed practice, which in turn raises new questions. *Con đường lặp lại từ vấn đề lâm sàng đến câu hỏi, nghiên cứu, kết quả, công bố và thay đổi thực hành, rồi từ đó nảy sinh câu hỏi mới.*
Research misconduct (fabrication, falsification and plagiarism, or FFP) Hành vi sai trái trong nghiên cứu (bịa đặt, làm sai lệch, đạo văn)	Inventing data, changing data or results, or using others' work or words without credit. *Bịa ra dữ liệu, sửa đổi dữ liệu hoặc kết quả, hoặc sử dụng công trình, câu chữ của người khác mà không ghi nguồn.*
Research question Câu hỏi nghiên cứu	The single, specific question a study is designed to answer. *Câu hỏi duy nhất, cụ thể mà nghiên cứu được thiết kế để trả lời.*
Response to reviewers Thư phản hồi ý kiến phản biện	A polite, point-by-point reply to each reviewer comment, saying what was changed and where. *Thư trả lời lịch sự, theo từng ý, cho mỗi nhận xét của người phản biện, nêu rõ đã sửa gì và ở đâu.*
Reverse causation Quan hệ nhân quả ngược	The outcome (or early disease) causes the exposure, not the other way round. *Kết cục (hoặc bệnh ở giai đoạn sớm) gây ra phơi nhiễm, chứ không phải ngược lại.*
Risk Nguy cơ	The number with the outcome divided by the number in the group, e.g. 30 deaths / 150 patients = 0.20. *Số người có kết cục chia cho tổng số người trong nhóm, ví dụ 30 tử vong / 150 người bệnh = 0,20.*
Risk difference (RD) Hiệu số nguy cơ (RD)	Risk in one group minus risk in the other (absolute difference). *Nguy cơ ở nhóm này trừ nguy cơ ở nhóm kia (chênh lệch tuyệt đối).*
Risk-of-bias tools Công cụ đánh giá nguy cơ sai lệch	Structured checklists for judging bias, chosen by design: RoB 2 (RCTs), ROBINS-I (non-randomised studies of interventions), Newcastle–Ottawa (cohort, case-control), QUADAS-2 (diagnostic accuracy), AMSTAR 2 (systematic reviews). *Bảng kiểm có cấu trúc để đánh giá sai lệch, chọn theo thiết kế: RoB 2 (RCT), ROBINS-I (nghiên cứu can thiệp không ngẫu nhiên), Newcastle–Ottawa (thuần tập, bệnh–chứng), QUADAS-2 (độ chính xác chẩn đoán), AMSTAR 2 (tổng quan hệ thống).*
ROC curve and AUC Đường cong ROC và diện tích dưới đường cong (AUC)	A plot of sensitivity against 1 − specificity at every cut-off; the AUC (0.5 = chance, 1 = perfect) summarises how well a test separates sick from well. *Đồ thị độ nhạy theo (1 − độ đặc hiệu) tại mọi ngưỡng cắt; AUC (0,5 = ngẫu nhiên, 1 = hoàn hảo) tóm tắt khả năng phân biệt người có bệnh và không có bệnh.*
S	
Sample Mẫu	The group of people actually studied, chosen to represent a larger population. *Nhóm người thực sự được nghiên cứu, được chọn để đại diện cho một quần thể lớn hơn.*

Term · Thuật ngữ	Definition · Định nghĩa
Sample size calculation Tính cỡ mẫu	Working out how many participants are needed to answer the question with enough power or precision. *Xác định số người tham gia cần thiết để trả lời câu hỏi với đủ lực thống kê hoặc độ chính xác.*
Search strategy Chiến lược tìm kiếm	The planned combination of terms, operators and filters used to search a database. *Sự kết hợp có kế hoạch giữa từ khóa, toán tử và bộ lọc để tìm kiếm trong cơ sở dữ liệu.*
Selection bias Sai lệch chọn lựa	Bias from the way participants are chosen or lost, so the groups are not comparable. *Sai lệch do cách chọn hoặc mất người tham gia, khiến các nhóm không còn so sánh được.*
Sensitivity Độ nhạy	The proportion of people with the disease whom a test correctly identifies as positive. *Tỷ lệ người có bệnh được xét nghiệm phát hiện đúng là dương tính.*
Sensitivity analysis Phân tích độ nhạy	Repeating the analysis under different assumptions to see whether the conclusion changes. *Lặp lại phân tích với các giả định khác nhau để xem kết luận có thay đổi hay không.*
Sentinel value (missing-value code) Giá trị quy ước cho dữ liệu thiếu (ví dụ 999)	A number such as 999 or −99 used to mean “missing”; it must be recoded as missing before analysis or it will distort the results. *Một con số như 999 hoặc −99 được dùng để biểu thị “thiếu dữ liệu”; phải mã hóa lại thành giá trị thiếu trước khi phân tích, nếu không sẽ làm sai lệch kết quả.*
Skewed distribution Phân phối lệch	An asymmetric spread with a long tail on one side, e.g. length of stay. *Phân bố không đối xứng, có một đuôi dài về một phía, ví dụ thời gian nằm viện.*
Specificity Độ đặc hiệu	The proportion of people without the disease whom a test correctly identifies as negative. *Tỷ lệ người không có bệnh được xét nghiệm xác định đúng là âm tính.*
Spin Trình bày sai lệch kết quả (“spin”)	Reporting that makes results look more favourable than they are, e.g. stressing secondary outcomes when the primary outcome was negative. *Cách trình bày khiến kết quả có vẻ tích cực hơn thực tế, ví dụ nhấn mạnh kết cục phụ khi kết cục chính không có khác biệt.*
SPIRIT Hướng dẫn SPIRIT	A guideline listing what a trial protocol should contain. *Hướng dẫn liệt kê các nội dung cần có trong đề cương thử nghiệm lâm sàng.*
Standard deviation (SD) Độ lệch chuẩn (SD)	How spread out values are around the mean; used with symmetric (normal) data. *Mức độ phân tán của các giá trị quanh trung bình; dùng cho dữ liệu đối xứng (phân phối chuẩn).*
Statistical analysis plan (SAP) Kế hoạch phân tích thống kê (SAP)	A detailed, dated plan of all analyses, written before the data are examined. *Kế hoạch chi tiết, có ghi ngày, cho tất cả các phân tích, được viết trước khi xem dữ liệu.*
Statistical vs clinical significance Ý nghĩa thống kê và ý nghĩa lâm sàng	A result can be unlikely to be due to chance (statistical) yet too small to matter to patients (clinical). *Một kết quả có thể ít khả năng do ngẫu nhiên (ý nghĩa thống kê) nhưng quá nhỏ để có giá trị với người bệnh (ý nghĩa lâm sàng).*
STROBE checklist Bảng kiểm STROBE	The 22-item checklist for reporting observational studies. *Bảng kiểm 22 mục để báo cáo nghiên cứu quan sát.*
Structured abstract Tóm tắt có cấu trúc	A short summary of a paper under fixed headings, usually Background, Methods, Results and Conclusions. *Bản tóm tắt ngắn của bài báo theo các mục cố định, thường là Đặt vấn đề, Phương pháp, Kết quả và Kết luận.*

Term · Thuật ngữ	Definition · Định nghĩa
Study registration (ClinicalTrials.gov, WHO ICTRP, PROSPERO) Đăng ký nghiên cứu (ClinicalTrials.gov, WHO ICTRP, PROSPERO)	Publicly recording a study's design and primary outcome before it starts; ICMJE journals require it for trials before the first participant is enrolled. *Công bố công khai thiết kế và kết cục chính của nghiên cứu trước khi bắt đầu; các tạp chí theo ICMJE yêu cầu thử nghiệm lâm sàng phải đăng ký trước khi tuyển người tham gia đầu tiên.*
Superiority vs non-inferiority trial Thử nghiệm vượt trội và thử nghiệm không kém hơn	A superiority trial asks whether a treatment is better; a non-inferiority trial asks whether it is not worse by more than a pre-set margin (useful if it is cheaper, simpler or safer). *Thử nghiệm vượt trội hỏi liệu điều trị có tốt hơn không; thử nghiệm không kém hơn hỏi liệu điều trị có kém hơn quá một ngưỡng định trước hay không (hữu ích khi điều trị rẻ hơn, đơn giản hơn hoặc an toàn hơn).*
Systematic review Tổng quan hệ thống	A review that finds, appraises and summarises all studies on a question using a pre-set, reproducible method. *Tổng quan tìm kiếm, thẩm định và tổng hợp tất cả nghiên cứu về một câu hỏi theo phương pháp định trước, có thể lặp lại.*
T	
Table 1 Bảng 1 (đặc điểm nền)	The table that describes the baseline characteristics of the participants, usually by group. *Bảng mô tả các đặc điểm ban đầu của người tham gia, thường theo từng nhóm.*
t-test Kiểm định t	Compares the means of a continuous, roughly normal outcome between two groups. *So sánh trung bình của một biến liên tục, xấp xỉ phân phối chuẩn, giữa hai nhóm.*
Type I error Sai lầm loại I	Concluding there is an effect when there is none (a false alarm). *Kết luận có tác động trong khi thực tế không có (báo động giả).*
Type II error Sai lầm loại II	Missing a real effect, often because the study is too small (a missed fire). *Bỏ sót một tác động có thật, thường do cỡ mẫu quá nhỏ (bỏ lỡ đám cháy thật).*
V	
Variable Biến số	Any characteristic recorded for each patient, e.g. age, sex, fluid balance, death. *Bất kỳ đặc điểm nào được ghi nhận cho từng người bệnh, ví dụ tuổi, giới, cân bằng dịch, tử vong.*
Vulnerable participants Đối tượng dễ bị tổn thương	People less able to protect their own interests, e.g. critically ill, children, people with impaired capacity. *Những người ít có khả năng tự bảo vệ quyền lợi, ví dụ người bệnh nặng, trẻ em, người suy giảm năng lực hành vi.*

APPENDIX B

Checklists and tools

This appendix collects four practical tools used throughout the book: a simplified ten-question checklist for appraising a published study, a guide to choosing a formal risk-of-bias tool, a checklist to work through before submitting a manuscript, and the reporting guidelines that match each study design.

B.1 The ten-question appraisal checklist

This short checklist is a simplified way of appraising any primary study, whatever its design (Chapter 4). Answer each question **Yes**, **No** or **Can't tell**, and note **where in the paper** you found the answer (section, page, table or line). "Can't tell" is a valid answer: it often means the study was poorly reported. There is no total score; questions 2, 4, 5 and 7 matter most when deciding whether to trust the result. Finish with a one-sentence verdict and compare the paper with the reporting guideline for its design.

You do not need to read every word. Read the abstract, then Table 1, the result for the primary outcome, and the first and last paragraphs of the Discussion; go to the Methods for the details.

Table B.1: The ten-question appraisal checklist.

#	Question	Yes	No	Can't tell
1	**Question:** Can you write the study's PICO in one line?	☐	☐	☐
2	**Design:** What is the design, and is it the right one for the question?	☐	☐	☐
3	**Participants:** Who was included, and who was left out? Would your patients have been included?	☐	☐	☐
4	**Comparison groups:** Were the groups similar at the start (Table 1)? How was this achieved (randomisation, matching, adjustment)?	☐	☐	☐
5	**Bias:** Who knew which group patients were in (blinding)? Could that change the outcome?	☐	☐	☐
6	**Outcome:** Was there one pre-specified primary outcome, measured the same way in both groups?	☐	☐	☐
7	**Numbers:** What is the main result as an effect size with a 95% confidence interval? Is the interval narrow or wide?	☐	☐	☐
8	**Words vs design:** Do the conclusions match the design ("associated with" vs "caused"; no "trend" for p = 0.42)?	☐	☐	☐
9	**Limitations:** Are the main limitations stated honestly? Name one the authors missed.	☐	☐	☐
10	**Use:** Would this change practice in your unit? Why, or why not?	☐	☐	☐

B.2 Choosing a risk-of-bias tool

When evidence is appraised formally, for example in a systematic review (Chapter 9), each study is judged with a validated tool that matches its design. Identify the design first (Chapter 5), choose the tool from the table, and appraise each outcome you care about rather than the paper as a whole. Judge each domain as low risk, some concerns or high risk; the overall judgement is usually set by the worst domain. Finally ask whether the conclusions are supported by the data.

Table B.2: Which risk-of-bias tool for which study design.

Study design	Tool	What it checks	Where to find it
Randomised controlled trial	RoB 2	Bias arising from randomisation, deviations from intended interventions, missing outcome data, measurement of the outcome and selection of the reported result	Cochrane (riskofbias.info)
Non-randomised study of an intervention	ROBINS-I	Confounding, selection of participants, classification of interventions, deviations, missing data, measurement and reporting, judged against an ideal “target trial”	Cochrane (riskofbias.info)
Cohort and case-control studies of exposures	Newcastle–Ottawa Scale, or ROBINS-E	Selection of groups, comparability (confounding) and ascertainment of outcome or exposure; ROBINS-E applies the ROBINS-I domains to exposures	Ottawa Hospital Research Institute; riskofbias.info
Diagnostic accuracy study	QUADAS-2	Patient selection, index test, reference standard, flow and timing (risk of bias and applicability)	University of Bristol
Systematic review or meta-analysis	AMSTAR 2	16 items, including a protocol registered in advance, a comprehensive search, risk-of-bias assessment of the included studies, appropriate meta-analysis and publication bias	amstar.ca
Qualitative study	CASP qualitative checklist	Aims, method and design, recruitment, data collection, the researcher’s relationship with participants, ethics, rigour of analysis, findings and value	Critical Appraisal Skills Programme

For prediction models use PROBAST; for cross-sectional studies and case series, the critical appraisal checklists of the Joanna Briggs Institute are widely used.

Reporting guidelines are not appraisal tools

Reporting guidelines such as CONSORT and STROBE tell authors what to **report**. Risk-of-bias tools judge how **trustworthy** a result is. A study can report everything a guideline asks and still be at high risk of bias. But a study that does not report the essentials cannot be judged at all.

B.3 Reporting guidelines by study design

Every main study design has a reporting guideline, developed by consensus and available free from the EQUATOR Network (equator-network.org). Use the guideline as a writing checklist from the first draft, not only at submission (Chapters 19 and 20).

Table B.3: Reporting guidelines for the main study designs.

Study design	Reporting guideline	Its most visible requirement
Randomised controlled trial	CONSORT	Participant flow diagram; primary outcome with effect size and confidence interval
Trial protocol	SPIRIT	Schedule of enrolment, interventions and assessments
Observational study (cohort, case-control, cross-sectional)	STROBE	Methods items on variables, bias and statistical methods; crude and adjusted estimates
Diagnostic accuracy study	STARD	Flow of participants; cross-tabulation of index test against reference standard
Systematic review and meta-analysis	PRISMA 2020	PRISMA flow diagram; risk-of-bias assessment
Prediction model (including machine learning)	TRIPOD+AI	Model development and validation; calibration and discrimination
Case report	CARE	Timeline of the case
Quality-improvement report	SQUIRE 2.0	Context, intervention and measures of improvement
Qualitative research	COREQ or SRQR	Research team, reflexivity, data collection and analysis

B.4 Writing and submission checklist

Work through this list before pressing “Submit”, and ask a co-author who did not write a section to check it against the tables.

Title and keywords

- ☐ The title states the population, the intervention or exposure, the outcome and the design; no abbreviations; within the journal’s limit
- ☐ 3–6 keywords (MeSH terms where possible) that do not simply repeat the title

Abstract

- ☐ Structured as the journal requires and within the word limit
- ☐ Primary outcome reported first, with effect size and 95% confidence interval; numbers identical to the Results
- ☐ Conclusion matches the design (no spin, no causal words for observational data); registration number given for trials

Introduction and Methods

- ☐ The Introduction is a funnel: what is known, what is unknown, the aim; it ends with the aim or hypothesis
- ☐ The Methods allow replication: design, setting, dates, eligibility, intervention or exposure, outcomes
- ☐ Sample size, statistical methods, handling of missing data and software are described
- ☐ Ethics approval (committee and number) and the consent procedure are stated

Results and numbers

- ☐ Participant flow comes first (flow diagram with numbers and reasons at each step)
- ☐ Every percentage has its numerator and denominator, and the denominators add up
- ☐ Effect sizes with 95% confidence intervals; exact p-values ($p = 0.042$ rather than $p < 0.05$; $p < 0.001$ is acceptable)
- ☐ Decimal places consistent, units given, each number in one place (text or table)
- ☐ No interpretation in the Results; all pre-specified outcomes reported

Tables and figures

- ☐ Each stands alone: an informative title, units, n per column, footnotes for abbreviations
- ☐ Numbers in tables match the text and the abstract
- ☐ Figures show the data (no 3-D effects, no truncated axes) at the resolution the journal requires

Discussion and conclusion

- ☐ Six moves: answer, compare, explain, limitations, implications, conclusion
- ☐ Limitations are specific, each with its likely direction of bias
- ☐ Language calibrated to the design: “associated with” for observational studies; no “trend” for a non-significant result

Reporting guideline, declarations and references

- ☐ The correct reporting guideline was used, its checklist completed with page numbers and its flow diagram included

- ☐ Ethics, consent, registration, funding and competing interests declared
- ☐ Data-availability statement, disclosure of any use of AI tools, and author contributions (CRediT)
- ☐ All authors meet the authorship criteria and approved the final version
- ☐ References in the journal's style; every citation in the list and vice versa; each reference checked against the original, none retracted

Journal, cover letter and submission package

- ☐ Scope, readership and article type fit; word, table and figure limits respected
- ☐ The journal is indexed and not predatory; fees and licence are known
- ☐ A one-page cover letter to the named editor: the question, the main finding, why this journal
- ☐ Statements of originality and of no simultaneous submission; any preprint disclosed
- ☐ Title page, manuscript (blinded if required), figures, supplementary files, checklist and suggested reviewers ready

After acceptance

- ☐ Proofs checked within the deadline: author names, affiliations, numbers, tables and figures
- ☐ Only corrections at proof stage, no new content; licence and open-access choice confirmed
- ☐ The paper shared with colleagues, the institution and, where allowed, a repository; the trial registry updated with results

APPENDIX C

The FLUID-ICU data dictionary

Simulated data

FLUID-ICU is a **simulated** cohort created for teaching: 620 adults with sepsis in four ICUs (ICU A–D), January 2023 to December 2024. It describes no real patient, hospital or country. Chapter 16 analyses it in full.

Main exposure: cumulative fluid balance at day 3 (`cum_fluid_balance_d3_ml_kg`, `positive_balance_d3`). **Primary outcome:** death by day 28 (`died_28d`, with `time_to_event_28d`). **Pre-specified confounders:** age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease. **Pre-specified effect modifier:** heart failure. DERIVED variables are calculated by the cleaning script and are not in the raw file; values outside the allowed range were set to missing and queried.

Table C.1: The FLUID-ICU data dictionary (clean file, 32 variables).

#	Variable	Label	Type	Units / codes	Allowed range
1	`study_id`	Study identifier	Character (ID)	FLU-001 ... FLU-620	Unique
2	`site`	Recruiting ICU	Categorical (4 levels)	ICU A / ICU B / ICU C / ICU D	4 levels
3	`admission_date`	Date of ICU admission (enrolment)	Date	YYYY-MM-DD	2023-01-01 to 2024-12-31
4	`age`	Age	Numeric (integer)	Years	18 to 110
5	`sex`	Sex	Categorical (2 levels)	Female / Male	2 levels
6	`weight_kg`	Body weight on admission	Numeric	kg (1 decimal)	30 to 250
7	`diabetes`	Diabetes mellitus	Binary	No / Yes	No, Yes
8	`ckd`	Chronic kidney disease	Binary	No / Yes	No, Yes
9	`heart_failure`	Chronic heart failure	Binary	No / Yes	No, Yes
10	`hypertension`	Hypertension	Binary	No / Yes	No, Yes
11	`infection_source`	Source of infection	Categorical (5 levels)	Respiratory / Urinary / Abdominal / Skin/soft tissue / Other	5 levels
12	`sofa`	SOFA score on ICU admission	Numeric (integer score)	Points	0 to 24

#	Variable	Label	Type	Units / codes	Allowed range
13	apache_ii	APACHE II score (first 24 h)	Numeric (integer score)	Points	0 to 71
14	lactate_mmol_l	Blood lactate on admission	Numeric	mmol/L (1 decimal)	0.1 to 30
15	map_mmhg	Mean arterial pressure on admission	Numeric (integer)	mmHg	20 to 200
16	vasopressor	Vasopressor on admission	Binary	No / Yes	No, Yes
17	mech_vent	Invasive mechanical ventilation on admission	Binary	No / Yes	No, Yes
18	fluid_in_d1_ml	Total fluid input, ICU day 1	Numeric (integer)	ml	0 to 20000
19	fluid_out_d1_ml	Total fluid output, ICU day 1	Numeric (integer)	ml	0 to 20000
20	fluid_in_d2_ml	Total fluid input, ICU day 2	Numeric (integer)	ml	0 to 20000
21	fluid_out_d2_ml	Total fluid output, ICU day 2	Numeric (integer)	ml	0 to 20000
22	fluid_in_d3_ml	Total fluid input, ICU day 3	Numeric (integer)	ml	0 to 20000
23	fluid_out_d3_ml	Total fluid output, ICU day 3	Numeric (integer)	ml	0 to 20000
24	cum_fluid_balance_d3_ml_kg	Cumulative fluid balance, ICU days 1-3	Numeric (DERIVED)	ml/kg (1 decimal)	-150 to 250
25	positive_balance_d3	Positive cumulative fluid balance at day 3	Binary (DERIVED)	No / Yes	No, Yes
26	aki_stage	Acute kidney injury, maximum KDIGO stage, ICU days 1-7	Ordinal (integer)	0 = none, 1, 2, 3	0 to 3
27	aki_any	Any acute kidney injury (KDIGO stage >= 1)	Binary (DERIVED)	No / Yes	No, Yes
28	rrt	Renal replacement therapy by day 28	Binary	No / Yes	No, Yes
29	vent_free_days_28	Ventilator-free days to day 28	Numeric (integer)	Days	0 to 28
30	icu_los_days	ICU length of stay	Numeric	Days (1 decimal)	> 0
31	died_28d	Death from any cause by day 28	Binary	No / Yes	No, Yes
32	time_to_event_28d	Time from ICU admission to death or censoring	Numeric (integer)	Days	1 to 28

C.1 Notes on the variables

- `study_id`: Raw column 'Study ID'. Raw file has 3 IDs entered twice (exact duplicate rows); removed in cleaning.
- `site`: Raw 'Site' typed several ways (icu a, ICU-A, ICUA); standardised. ICU C reported lactate in mg/dL.
- `admission_date`: Raw 'Admission Date' mixes YYYY-MM-DD, DD/MM/YYYY and DD-Mon-YYYY.
- `age`: Raw 'Age'. One impossible value (210) set to missing; query to site.
- `sex`: Raw 'Sex' has 8-9 spellings (M, m, male, F, f, female, ' Female').
- `weight_kg`: Raw 'Weight (kg)'. One impossible value (7) set to missing. Used to compute balance in ml/kg.
- `diabetes`: Raw codes Yes/No/Y/N/yes/no/1/0 (ICU B sends 1/0).
- `ckd`: Pre-specified confounder. Mixed Yes/No coding in raw file.
- `heart_failure`: Pre-specified effect modifier (Q1). Mixed coding in raw file.
- `hypertension`: Mixed coding in raw file.
- `infection_source`: Ordered as listed for tables.
- `sofa`: Severity; pre-specified confounder. Median 7 (IQR 5-9).
- `apache_ii`: Raw 'APACHE II'; 6 values coded -99 = not recorded.
- `lactate_mmol_l`: ~15% missing, more often in less severe patients (MAR). ICU C values in raw file are mg/dL (x 9.01): divided by 9.01 in cleaning.
- `map_mmhg`: Raw 'MAP (mmHg)'; 4 values coded 999.
- `vasopressor`: Pre-specified confounder. Mixed coding in raw file.
- `mech_vent`: Raw 'Mech vent'. Pre-specified confounder.
- `fluid_in_d1_ml`: All intravenous and enteral fluid incl. drugs and nutrition. Raw 'Fluid in D1 (ml)'.
- `fluid_out_d1_ml`: Urine, drains, ultrafiltration; insensible losses not included.
- `fluid_out_d2_ml`: One negative value (impossible) in raw file set to missing.
- `cum_fluid_balance_d3_ml_kg`: = (sum of fluid in d1-3 - sum of fluid out d1-3) / weight_kg. Not in the raw file. Missing for 2 patients (weight or output missing). Main exposure (Q2, Q3: per 10 ml/kg).
- `positive_balance_d3`: Yes if cum_fluid_balance_d3_ml_kg > 0. Main exposure for Q1 and Q5.
- `aki_stage`: Raw 'AKI stage (KDIGO)'.
- `aki_any`: Yes if aki_stage >= 1. Outcome for Q2.
- `rrt`: Mixed coding in raw file.
- `vent_free_days_28`: Days alive and free of invasive ventilation to day 28; 0 if died; 28 if never ventilated and alive. Raw 'VFD 28'. Outcome for Q3.

- `icu_los_days`: Right-skewed. For patients who died, stay ends at death. Outcome for Q5 (survivors).
- `died_28d`: Primary outcome (Q1, Q4). Mixed coding in raw file.
- `time_to_event_28d`: Death day if died; 28 (censored) if alive at day 28. Use with died_28d for Kaplan-Meier and Cox. Raw 'Time to event 28d (days)'.

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